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Balaji

M.S. CHOUHAN

ELEMENTARY PROBLEMS IN
**ORGANIC
CHEMISTRY**
for **JEE**

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Elementary Problems in
ORGANIC
CHEMISTRY
for **JEE**

& All Other Engg. Entrance Examinations

by:

M.S. Chouhan

Director
Vibrant Academy, Kota

Strictly follows the sequence prescribed by NCERT



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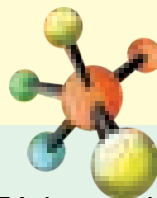
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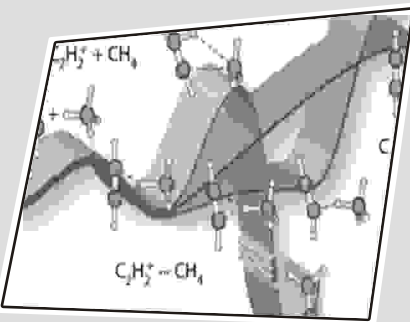
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Organic Chemistry: Some Basic Principles and Techniques

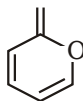


Only One Answer is Correct

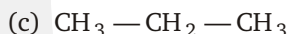
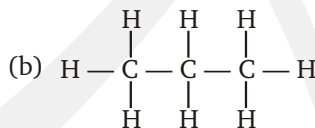
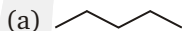
[A] Classification and Nomenclature of Organic Compounds

Functional Groups Degree of Unsaturation and Double Bond-equivalent

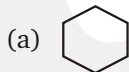
- How many rings and/or pi bonds are present in a hydrocarbon with the formula C_6H_{10} ?
(a) 0 (b) 1 (c) 2 (d) 3
- How many pi (π) electrons are there in the following molecule?



- (a) 4 (b) 6 (c) 8 (d) 10
- Which of the following is condensed structural formula?

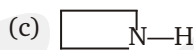
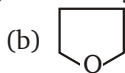
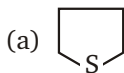


- Which of the following is homocyclic compound?



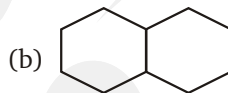
(d) Both (a) & (b)

5. Which of the following is heterocyclic compound?



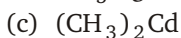
(d) All of these

6. Which of the following is bicyclo compound?



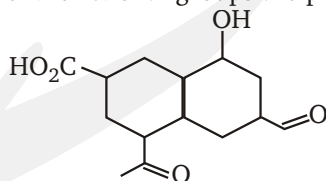
(d) All of these

7. Which of the following is organometallic compound?



(d) All of these

8. How many different functional groups are present in given compound?

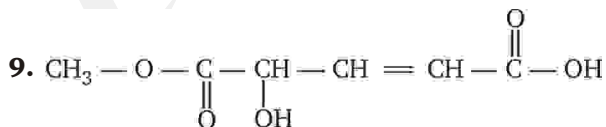


(a) 3

(b) 4

(c) 1

(d) 5



How many functional groups are present in above compound?

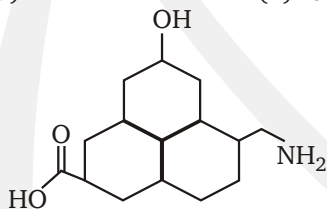
(a) 2

(b) 3

(c) 4

(d) 5

10.



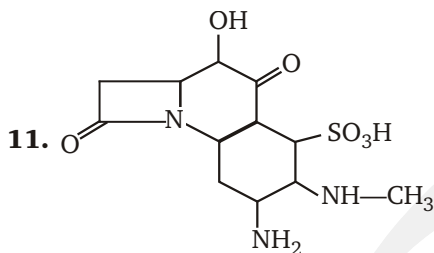
Functional groups present in above compound are:

(a) alcohol, aldehyde & carboxylic acid

(b) alcohol, amine & aldehyde

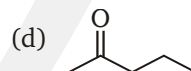
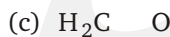
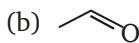
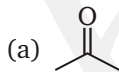
(c) amine, alcohol & carboxylic acid

(d) amine, amide & aldehyde



Which functional group is present in above compound?

- (a) Carboxylic acid (b) Aldehyde (c) Thiol (d) Alcohol
12. Which of the following is pair of homologues?
- (a) CH_3OH , $\text{CH}_3-\text{O}-\text{CH}_3$ (b) CH_3OH , CH_3SH
 (c) $\text{CH}_3-\text{O}-\text{CH}_3$, $\text{CH}_3-\text{CH}_2-\text{OH}$ (d) CH_3OH , $\text{CH}_3\text{CH}_2\text{OH}$
13. What is the homologue of $\text{CH}_3-\text{CH}_2-\text{OH}$?
- (a) $\text{CH}_3-\text{CH}_2-\text{O}-\text{CH}_3$ (b) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{OH}$
 (c) CH_3OCH_3 (d) $\text{CH}_3-\text{CH}_2-\text{O}-\text{CH}_2-\text{CH}_3$
14. In homologous series, the molecular mass of two consecutive members differ by:
- (a) 12 (b) 13 (c) 14 (d) 15
15. All the members of homologous series has:
- (a) same I.U.P.A.C. (b) same functional group
 (c) same boiling point (d) same number of carbons
16. Which of the following is smallest homologue of ketone?



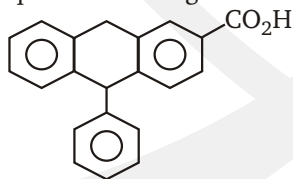
17. In homologous series, successive members differ by :

- (a) CH_3- (b) CH_3-CH_2- (c) $-\text{CH}_2-$ (d) $-\text{H}$

18. Degree of unsaturation for is:

- (a) 6 (b) 7 (c) 8 (d) 9

19. Double bond equivalent or degree of unsaturation for the compound given below is:



- (a) 12 (b) 13 (c) 14 (d) 15

20. Double bond equivalent of C_4H_6 is:

- (a) 0 (b) 1 (c) 2 (d) 3

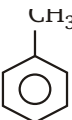
21. Double bond equivalent of $C_5H_{12}O$ is:

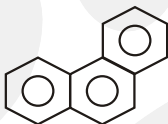
- (a) 0 (b) 1 (c) 2 (d) 3

22. Double bond equivalent of C_8H_8 is:

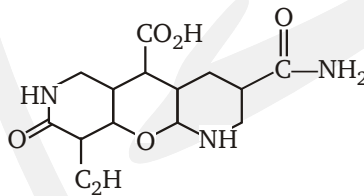
- (a) 4 (b) 5 (c) 6 (d) 7

23. Which of the following compounds has DBE equal to benzene?

- (a)  (b)  (c)  (d) All of these

24. Difference of DBE in  and  is:

- (a) 0 (b) 1 (c) 2 (d) 3

25.  ; Number of functional groups and degree of unsaturation (respectively) in the above compound is:

- (a) 7, 8 (b) 6, 8 (c) 6, 6 (d) 5, 6

26. Number of σ -bond and π -bond in anthracene are respectively:

- (a) 7, 26 (b) 24, 8 (c) 26, 7 (d) 26, 6

27. Which functional group can be presented in compound having molecular formula $C_4H_{10}O$?

- (a) Aldehyde (b) Ketone (c) Acid (d) Alcohol

28. Which functional group is present in compound having molecular formula C_4H_8O ?

- (a) Aldehyde (b) Ketone
(c) Unsaturated alcohol (d) All of these

29. Which functional group is present in compound having molecular formula C_4H_8 ?

- (a) Alkane (b) Alkene (c) Alkyne (d) Alcohol

30. Which functional group is present in compound having molecular formula $C_2H_4O_2$?

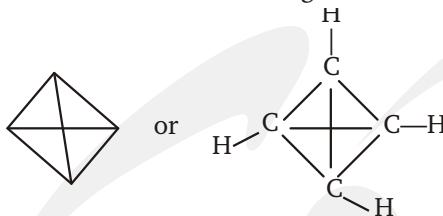
- (a) Carboxylic acid (b) Ester (c) Amine (d) Both (a) and (b)

31. How many nodal planes are there in $3d$ -orbital ?

- (a) 0 (b) 1 (c) 2 (d) 3

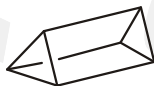


32. Double bond equivalent for tetrahedral molecule given below is:



- (a) 2 (b) 3 (c) 4 (d) 5

33.

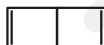


Prismane

Double bond equivalent for prismane is:

- (a) 2 (b) 3 (c) 4 (d) 4

34. Double bond equivalent for Dewar benzene is:



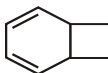
- (a) 2 (b) 3 (c) 4 (d) 5

Hybridization and Bonding in Organic Molecules

1. How many σ and π bonds present in $\text{N} \equiv \text{C}-\text{C} \equiv \text{N}$?

- (a) 4, 4 (b) 4, 5 (c) 4, 3 (d) 2, 1

2. Number of sp^2 sp^2 π -bonds present in given compound is:



- (a) 1 (b) 2 (c) 3 (d) 4

3. How many π -bonds are present in the given compound?



- (a) 6 (b) 12 (c) 18 (d) 24

4. What is the hybridization of carbon in compound $\text{H}-\text{C} \equiv \text{N}$?

- (a) sp^3 (b) sp^2 (c) sp (d) All of these

5. What is the hybridization of $\text{C}_1, \text{C}_2, \text{C}_3$ carbon in the following compound?



- (a) sp^3, sp^3, sp^3 (b) sp^3, sp^2, sp^2 (c) sp^3, sp^2, sp (d) sp, sp^2, sp^2

6. What is the hybridization of carbon in the following compounds?



- (a) sp^2, sp^2 (b) sp, sp (c) sp^3, sp^3 (d) sp^3, sp^2

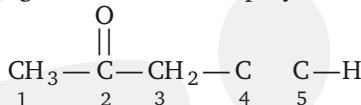
7. In which of the following carbon is sp^2 hybridized state ?

- (a) $\text{CH}_3 - \text{Cl}$ (b) $\text{NH}_2 - \overset{\text{O}}{\parallel} \text{C} - \text{NH}_2$ (c) $\text{H} - \text{C}$ (d) $\text{CH}_3 - \text{NH}_2$

8. What type of bonding is most important in $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$?

- (a) ionic (b) hydrogen (c) covalent (d) polar

9. Which carbon(s) in the following molecule is(are) sp hybridized?

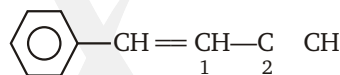


- (a) carbon 1 (b) carbon 2 (c) carbons 1, 3 (d) carbon 4 (e) carbons 4, 5

10. Which of the following has the strongest van der Waals' interaction between molecules of the same kind?

- (a) $\text{CH}_3 - \underset{\text{CH}_3}{\underset{|}{\text{CH}}} - \text{CH}_2\text{CH}_3$ (b) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$
 (c) $\text{CH}_3 - \underset{\text{CH}_3}{\underset{|}{\underset{\text{CH}_3}{|}{\text{C}}}} - \text{CH}_3$ (d) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ (e) $\text{CH}_3 - \underset{\text{CH}_3}{\underset{|}{\text{CH}}} - \text{CH}_3$

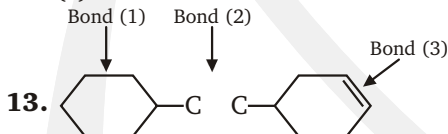
11. What is the hybridization of the carbon atoms numbered 1 and 2 respectively in the following structure?



- (a) sp^3, sp^2 (b) sp^2, sp^2 (c) sp, sp (d) sp^2, sp (e) sp, sp^2

12. The carbon-carbon triple bond of an alkyne is composed of

- (a) three bonds (b) two bonds and one bond
 (c) one bond and two bonds (d) three bonds

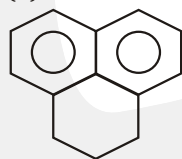


13.

Rank the indicated bonds in each compounds in order of increasing bond strength :

- (a) $1 < 2 < 3$ (b) $3 < 1 < 2$ (c) $1 < 3 < 2$ (d) $2 < 3 < 1$

14.

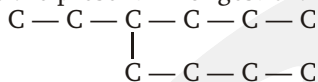


How many π -electrons are contained in given molecule?

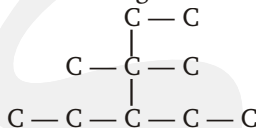
- (a) 6 (b) 8 (c) 10 (d) 12

IUPAC Nomenclature of Alkanes

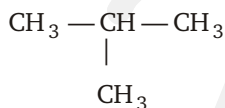
1. How many carbons are present in longest chain of given compound?



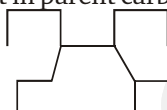
- (a) 6 (b) 7 (c) 8 (d) 9
2. How many carbons are present in longest carbon chain of given compound ?



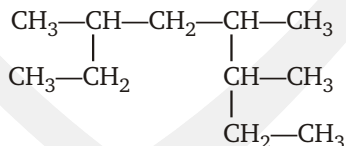
- (a) 4 (b) 5 (c) 6 (d) 7
3. How many carbons are present in nonane ?
- (a) 7 (b) 8 (c) 9 (d) 10
4. What is the common name for the following structure?



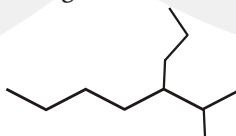
- (a) isobutane (b) *t*-butane (c) *n*-butane (d) *sec*-butane
5. How many carbon atoms are present in parent carbon chain in the given following compound?



- (a) 6C (b) 7C (c) 8C (d) 9C
6. Which of the following alphabet have a five membered chain as a longest chain?
A, E, F, H, I, K, M, T, V, W, X, Y, Z
- (a) A, E, H (b) E, M, W (c) K, M, X (d) F, W, Z
7. Give the complete IUPAC name for the structure below. Be careful to identify longest carbon chain:

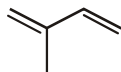


- (a) 3,5,6-trimethyl octane (b) 3,4,6-trimethyl octane
(c) 1,2,3,6-tetramethyl heptane (d) none of these
8. Give the IUPAC name for the following structure:



- (a) 2-methyl-3-ethylheptane (b) 3-ethyl-2-methylheptane
 (c) 5-isopropyloctane (d) 4-isopropyloctane
 (e) 2-methyl-3-propylheptane

9. Give the IUPAC name for the following compound:



- (a) isoprene (b) 3-methyl-1,3-butadiene
 (c) 2-methyl-1,3-butadiene (d) 2-methyl-2,3-butadiene (e) isobutadiene

10. ; How many methyl —CH_3 group are present in given alkane?

- (a) 4 (b) 5 (c) 6 (d) 7

11. ; How many 2° carbon present in above compound?

- (a) 4 (b) 5 (c) 6 (d) 7

12. The structure shown $\text{CH}_3\text{CH}_2\overset{\text{CH}_3}{\underset{\text{CH}_2\text{CH}_3}{\text{CH}}}\text{CHCH}_2\text{CH}_2\overset{\text{CH}_2\text{CH}_3}{\underset{\text{CH}_2\text{CH}_3}{\text{CH}}}\text{CH}_3$

- (a) 1° , 2° , and 3° carbon atoms (b) 1° and 2° carbon atoms only
 (c) 1° and 3° carbon atoms only (d) 2° and 3° carbon atoms only

13. How many 1° carbon atom will be present in a simplest hydrocarbon having two 3° and one 2° carbon atom ?

- (a) 3 (b) 4 (c) 5 (d) 6

14. 1° , 2° , 3° and 4° carbon atoms are present in :

- (a) 2, 2, 3-trimethyl pentane (b) 2, 3, 4-trimethyl pentane
 (c) both (a) and (b) (d) none of these

15. ; Correct IUPAC name of the following compound is:

- (a) 4-(1,1-dimethylethyl)-5-ethyloctane (b) 5-ethyl-4-(1,1-dimethylethyl) octane
 (c) 5-(1,1-dimethylethyl)-4-ethyloctane (d) 4-ethyl-5-(1,1-dimethylethyl) octane

IUPAC Nomenclature of Alkynes/Alkenes

1. Which of the following name will be incorrect ?



2. Which pair of names refers to the same compound ?

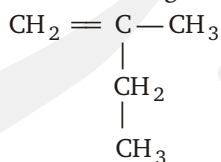
(a) Ethyne and Acetylene

(b) Ethyne and Ethene

(c) Ethane and Acetylene

(d) Ethane and Ethene

3. What is the IUPAC name for the following compound?



(a) 2-methyl-1-butene

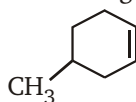
(b) isopentene

(c) 2-methylbutene

(d) 2-ethylpropene

(e) 3-methyl-3-butene

4. What is the IUPAC name for the following compound?



(a) 5-methylcyclohexene

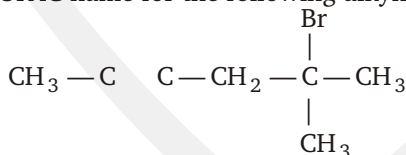
(b) 4-methylcyclohexene

(c) 1-methyl-3-cyclohexene

(d) 1-methyl-4-cyclohexene

(e) methylcyclohexene

5. What is the IUPAC name for the following alkyne?



(a) 5-bromo-2-heptyne

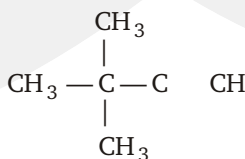
(b) 3-bromo-5-heptyne

(c) 2-bromo-2-methyl-4-hexyne

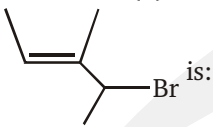
(d) 5-bromo-5,5-dimethylhexyne

(e) 5-bromo-5-methyl-2-hexyne

6. What is the common name for the following alkyne?



- (a) neoheptyne (b) trimethylpropyne
(c) trimethylacetylene (d) *tert*-butylacetylene (e) isopropylacetylene

7. The IUPAC name of  is:

- (a) 2-bromo-3-methylbut-3-ene (b) 4-bromo-3-methylpent-2-ene
(c) 2-bromo-3-methylpent-3-ene (d) 4-bromo-2,3-dimethylbut-2-ene

8. The IUPAC name of the compound $\text{CH}_3\text{CH}=\text{CHCH}=\text{CHC}\equiv\text{CH}_3$ is :

- (a) Octa-4,6-dien-2-yne (b) Octa-2,4-dien-6-yne
(c) Oct-2-yne-4,6-diene (d) Oct-6-yne-2,4-diene

9. The molecular formula of the first member of the family of alkenynes and its name is given by the set :

- (a) C_3H_2 , alkene (b) C_5H_6 , pent-1-en-3-yne
(c) C_6H_8 , hex-1-en-5-yne (d) C_4H_4 , butenyne

10. The correct IUPAC name of $\text{H}_2\text{C}=\text{CH}-\underset{\text{CH}_3}{\text{CH}}-\text{CH}_2\text{C}\equiv\text{CH}$ is

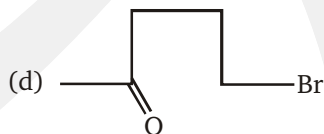
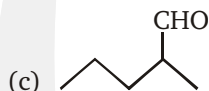
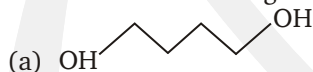
- (a) 3-methyl-1-hexen-5-yne (b) 4-methyl-5-hexen-1-yne
(c) 4-(ethenyl)1-pentyne (d) 3-(2-propenyl) butene-1

IUPAC Nomenclature of Alcohols/Aldehydes/Ketones/Acids

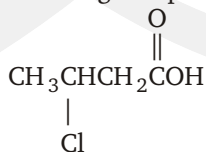
1. Which of the following compounds does not have the molecular formula $\text{C}_6\text{H}_{14}\text{O}$?

- (a) 1-hexanol (b) 2-hexanol
(c) 3-methyl-2-pentanol (d) 3-methyl-3-pentanol (e) cyclohexanol

2. Which of the following compound have 4 carbon atom in the parent chain ?

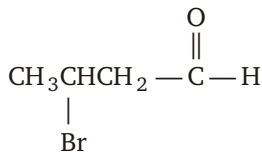


3. What is the common name of the following compound ?



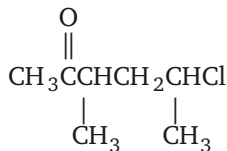
- (a) -chlorobutanoic acid (b) -chlorobutanoic acid
 (c) -chlorobutyric acid (d) -chlorobutyric acid (e) 3-chlorobutyric acid

4. What is the IUPAC name of the following compound ?



- (a) 2-bromobutanal (b) -bromobutanal
 (c) 3-bromobutanal (d) -bromobutyraldehyde (e) 3-bromobutanone

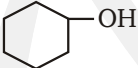
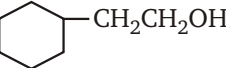
5. What is the IUPAC name of the following compound ?



- (a) 5-chloro-3-methylhexanone (b) 1-chloro-1,3-dimethyl-4-pentanone
 (c) 5-chloro-3,5-dimethyl-2-pentanone (d) 5-chloro-3-methyl-2-hexanone
 (e) 2-chloro-4-methyl-5-hexanone

6. The correct IUPAC name of $\text{CH}_3 - \text{CH}_2 - \underset{\text{CH}_2}{\overset{\parallel}{\text{C}}} - \text{COOH}$ is :

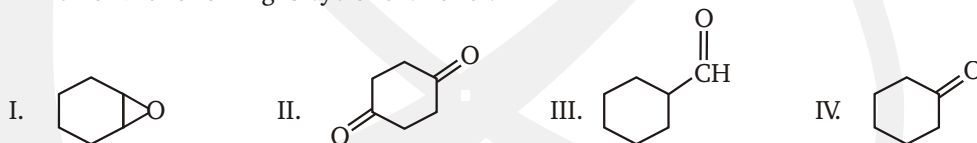
- (a) 2-Methyl butanoic acid (b) 2-Ethylprop-2-enoic acid
 (c) 2-Carboxy-1-butene (d) None of these

7. (i)  and (ii) 

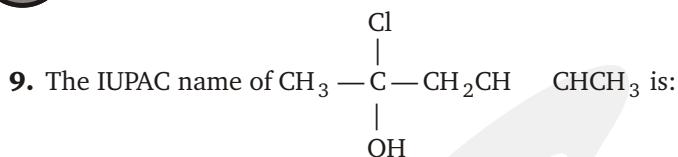
True statement for the above compounds is :

- (a) (i) is phenol while (ii) is alcohol (b) Both (i) and (ii) are primary alcohol
 (c) (i) is primary and (ii) is secondary alcohol (d) (i) is secondary and (ii) is primary alcohol

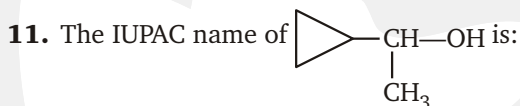
8. Which of the following is cyclohexanone ?



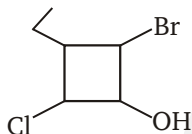
- (a) I (b) II (c) III (d) IV



- (a) 5-chloro-2-hydroxyhexene (b) 2-chloro-5-hydroxyhexene
 (c) 2-chloro-2-hydroxy-5-hexene (d) 2-chloro-4-hexenol-2
10. The IUPAC name of the *t*-butyl group is:
 (a) 2-methylpropyl (b) trimethylmethyl (c) 1, 1-dimethylethyl (d) 2-methyl-2-propyl



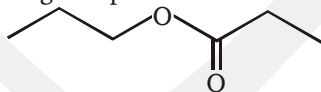
- (a) 1-hydroxy-1-cyclopropylethane (b) 1-hydroxyethylcyclopropane
 (c) cyclopropylethanol (d) 1-cyclopropylethanol
12. Correct IUPAC name of the given compounds is:



- (a) 2-bromo-3-ethyl-4-chlorocyclobutanol
 (b) 4-bromo-1-chloro-3-ethylcyclobutanol
 (c) 2-bromo-4-chloro-3-ethyl cyclohydroxybutane
 (d) 2-bromo-4-chloro-3-ethylcyclobutanol

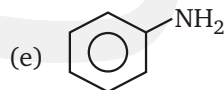
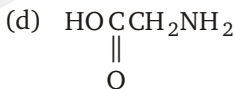
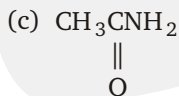
IUPAC Nomenclature of Acid Halides, Esters, Nitriles, Amines, Amides, Anhydrides

1. The IUPAC name of the following compound is:

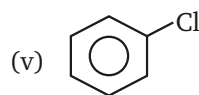
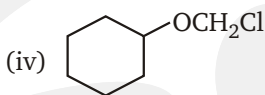
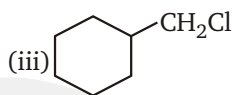
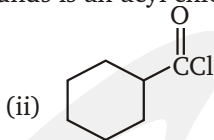
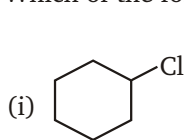


- (a) *n*-propyl ethanoate (b) ethyl propanoate
 (c) pentanoic anhydride (d) *n*-propyl propanoate

2. Which of the following compound is an amide ?



3. Which of the following compounds is an acyl chloride ?



(a) i

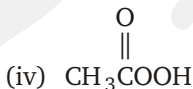
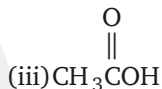
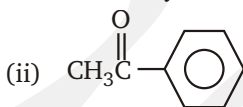
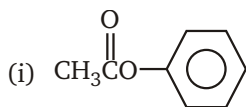
(b) ii

(c) iii

(d) iv

(e) v

4. Which of the following compounds is an anhydride ?



(a) i

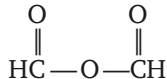
(b) ii

(c) iii

(d) iv

(e) v

5. What is the IUPAC name for the following compound ?



(a) Propanoic anhydride

(b) Methanoic anhydride

(c) Formyl formate

(d) Ethanoic anhydride

6. The IUPAC name of $\text{CH}_3 - \text{C}(=\text{O}) - \text{O} - \text{CH}_2 - \text{C}(=\text{O}) - \text{OH}$ is :

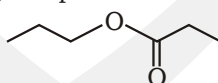
(a) 1-acetoxy acetic acid

(b) 2-acetoxy ethanoic acid

(c) 2-ethanoyloxy acetic acid

(d) 2-ethanoyloxyethanoic acid

7. The IUPAC name of the following compound is:



(a) *n*-propyl ethanoate

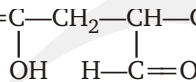
(b) ethyl propanoate

(c) pentanoic anhydride

(d) *n*-propyl propanoate

IUPAC Nomenclature of Poly Functional Compounds

1. The correct IUPAC name of the following compound $\text{O}=\text{C}-\text{CH}_2-\text{CH}-\text{CHO}$ is :



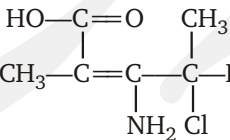
- (a) 3,3-Diformyl propanoic acid (b) 3-Formyl-4-oxo-butanoic acid
(c) 3,3-Dioxo propanoic acid (d) 3,3-Dicarbaldehyde propanoic acid

2. The correct IUPAC name of compound: $\text{CH}_3\text{—CH}_2\text{—}\underset{\text{O}}{\underset{\text{||}}{\text{C}}}\text{—}\underset{\text{H}}{\underset{|}{\text{CH}}}\text{—CHO}$ is :



- (a) 2-Cyano-3-oxopentanal (b) 2-Formyl-3-oxopentanenitrile
(c) 2-Cyano-1,3-pentanedione (d) 1,3-Dioxo-2-cyanopentane

3. The IUPAC name of compound $\text{CH}_3-\overset{\text{I}}{\underset{|}{\text{C}}}=\overset{\text{I}}{\underset{|}{\text{C}}}-\overset{\text{I}}{\underset{|}{\text{C}}}-\text{H}$ is :



- (a) 2-Amino-3-chloro-2-methylpent-2-enoic acid
(b) 3-Amino-4-chloro-2-methylpent-2-enoic acid
(c) 4-Amino-3-chloro-2-methylpent-2-enoic acid
(d) All of the above

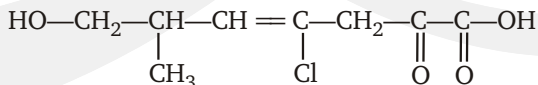
4. Which of the following compound is wrongly named ?

- (a) $\text{CH}_3\text{CH}_2\text{CH}_2\underset{\text{Cl}}{\text{CH}}\text{COOH}$; 2-Chloro pentanoic acid

- (b) $\text{CH}_3\text{C}(\text{CH}_3)\text{CH}_2\text{CH}_2\text{CH}_2\text{COOH}$; 2-Methyl hex-3-enoic acid

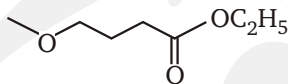
- (c) $\text{CH}_3\text{CH}_2\text{CH}=\text{CHCOCH}_3$; Hex-3-en-2-one
- (d) $\text{CH}_3-\underset{\text{CH}_3}{\underset{|}{\text{CH}}}\text{CH}_2\text{CH}_2\text{CHO}$; 4-Methyl pentanal

5. The suffix of the principal group, the prefixes for the other groups and the name of the parent in the structure



- (a) -oic acid, chloro, hydroxy, oxo, methyl, 4-heptene

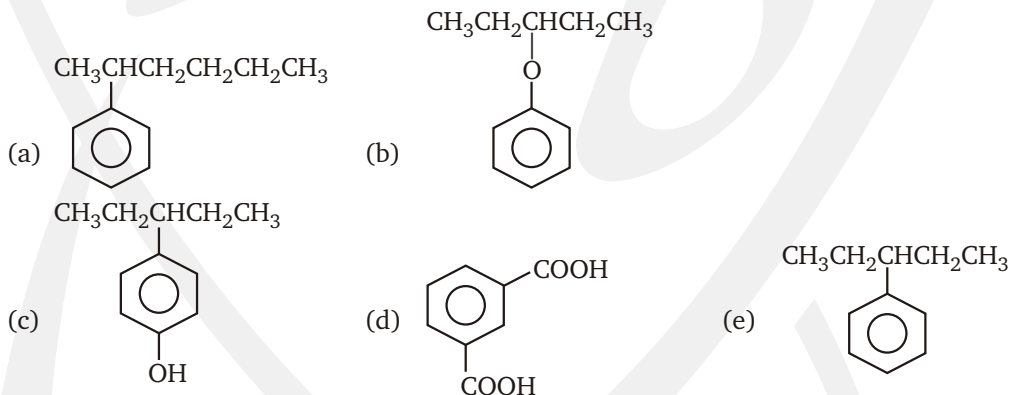
- (b) -oic acid, chloro, hydroxy, methyl, oxo, 4-heptene
 (c) -one, carboxy, chloro, methyl, hydroxy, 4-heptene
 (d) -one, carboxy, chloro, methyl, hydroxy, 4-heptene
 6. The IUPAC name of the following compound is:



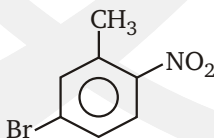
- (a) 3-methoxy ethylpropanoate (b) ethyl-4-methoxybutanoate
 (c) 1,4-diethoxybutane (d) 4-methoxy-ethylbutanoate
7. The IUPAC name of $\text{CH}_3 - \underset{\text{OH}}{\text{CH}} - \text{COOH}$ is:
- (a) 2-hydroxypropanoic acid (b) 1-hydroxypropanoic acid
 (c) 1-hydroxyethane carboxylic acid (d) 1-hydroxyethanoic acid

IUPAC Nomenclature of Aromatic Compounds & Bicyclic & Spiro Compounds

1. What is the structure of 3-phenylpentane ?

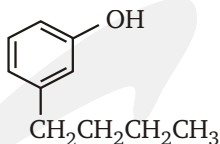


2. What is the IUPAC name of the following compound ?



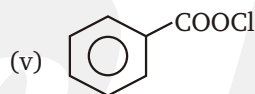
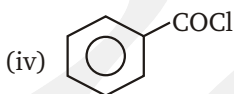
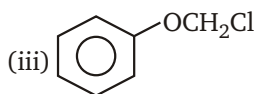
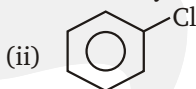
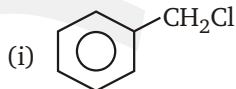
- (a) *o*-nitro-*m*-bromotoluene (b) 3-bromo-6-nitrotoluene
 (c) *m*-bromo-*o*-nitrotoluene (d) 5-bromo-2-nitrotoluene (e) 2-nitro-5-bromotoluene

3. What is the IUPAC name of the compound given below ?



- (a) 1-phenoxyethane (b) ethyl phenyl ether
(c) *m*-butylhydroxybenzene (d) *o*-butylhydroxybenzene (e) *m*-butylphenol

4. Which of the following compounds is benzoyl chloride ?

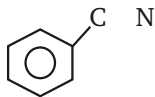


- (a) i (b) ii (c) iii (d) iv (e) v

5. The IUPAC name of is :

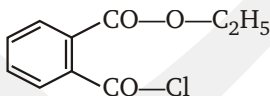
- (a) 2-Chlorocarbonyl ethylbenzoate (b) 2-Carboxyethyl benzoyl chloride
(c) Ethyl-2-(chlorocarbonyl) benzoate (d) Ethyl-1-(chlorocarbonyl) benzoate

6. What is the IUPAC name for the following compound ?



- (a) Benzenecarbonitrile (b) Benzenecyanide
(c) Phenylcyanide (d) Cyanophenyl

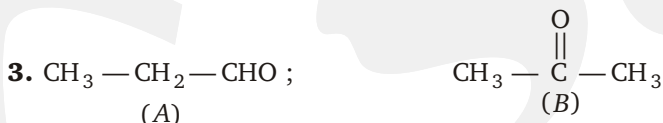
7. Write the IUPAC name of the following compound:



- (a) ethyl 2-(chlorocarbonyl) benzoate (b) ethyl-2-(chlorocarbonyl) hexanoate
(c) 2-(ethoxycarbonyl) benzoyl chloride (d) none of these

[B] Isomerism**Structural Isomerism**

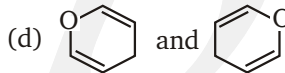
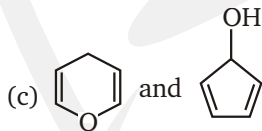
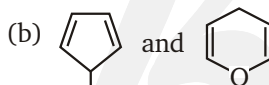
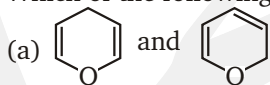
- How many dibromobenzenes are possible?
(a) 1 (b) 2 (c) 3 (d) 4
- The number of all the possible isomers of chlorobutane is:
(a) 2 (b) 4 (c) 6 (d) 5



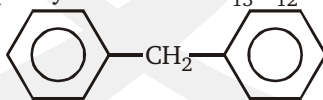
Relation between (A) and (B) is :

- (a) chain isomers (b) positional isomers
(c) functional isomers (d) metamers

4. Which of the following pairs of structures does not represent isomers?

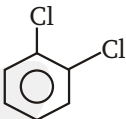


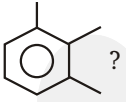
- How many distinct terminal alkynes are possible for a compound having molecular formula C_6H_{10} ?
(a) 3 (b) 4 (c) 5 (d) 6
- How many distinct terminal alkenes are possible for a compound having molecular formula C_6H_{12} ?
(a) 3 (b) 4 (c) 5 (d) 6
- How many distinct terminal alkynes are possible for a compound having molecular formula C_5H_8 ?
(a) 1 (b) 2 (c) 3 (d) 4 (e) 5
- How many distinct internal alkynes may exist with a molecular formula of C_6H_{10} ?
(a) 1 (b) 2 (c) 3 (d) 4 (e) 5
- The molecular formula of diphenyl methane is $\text{C}_{13}\text{H}_{12}$

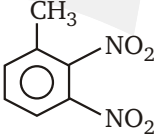


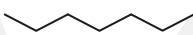
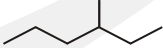
How many structural isomers are possible when one of the hydrogen is replaced by a chlorine atom ?

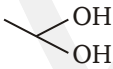
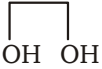
- (a) 4 (b) 6 (c) 7 (d) 8
10. The number of ether metamers represented by the molecular formula $\text{C}_4\text{H}_{10}\text{O}$ is:
(a) 1 (b) 2 (c) 3 (d) 4

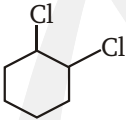
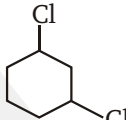
11. How many positional isomers are possible for  ?
 (a) 2 (b) 3 (c) 4 (d) 5

12. How many positional isomers are possible for  ?
 (a) 2 (b) 3 (c) 4 (d) 5

13.  ; Number of positional isomers for the compound given are :
 (a) 4 (b) 5 (c) 6 (d) 7

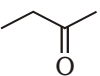
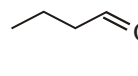
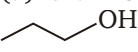
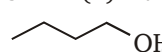
14.  (A)  (B) ; (A) and (B) are:
 (a) chain isomers (b) positional isomers (c) metamers (d) functional isomers

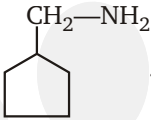
15.  (A)  (B) ; (A) and (B) are:
 (a) chain isomers (b) positional isomers (c) metamers (d) functional isomers

16.  (A)  (B) ; (A) and (B) are:
 (a) chain isomers (b) positional isomers (c) metamers (d) functional isomers

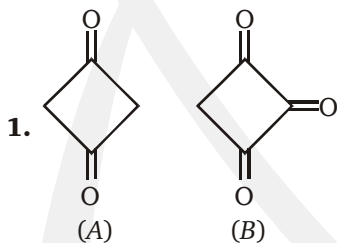
17. $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CO}_2\text{H}$ (A) $\text{CH}_3 - \underset{\text{CO}_2\text{H}}{\text{CH}} - \text{CH}_3$ (B) ; (A) and (B) are:
 (a) chain isomers (b) positional isomers (c) functional isomers (d) metamers

18.  (A)  (B) ; (A) and (B) are:
 (a) chain isomers (b) functional isomers (c) metamers (d) positional isomers

19.   ; (A) and (B) are:
 (A) (B)
 (a) chain isomers (b) functional isomers (c) metamers (d) positional isomers
20.   ; (A) and (B) are:
 (A) (B)
 (a) chain isomers (b) functional isomers (c) metamers (d) homologues
21. Which of the following pairs represent isomers?
 (a) $\text{CH}_3 - \text{CH}_3$, $\text{CH}_3 - \text{CH}_2 - \text{CH}_3$
 (b) $\text{H}_2\text{C} = \text{CH} - \text{OH}$, $\text{CH}_3 - \text{O} - \text{CH}_3$
 (c) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CH}_3$, $\text{CH}_3 - \underset{\text{CH}_3}{\text{CH}} - \text{CH}_3$
 (d) All of these
22. Which of the following represent a pair of chain isomers?
 (a) *n*-pentane, Iso-pentane (b) *n*-pentane, Neo-pentane
 (c) Iso-pentane, Neo-pentane (d) All of these
23. Total number of stereoisomers for 2,3-dichlorobutane is:
 (a) 3 (b) 4 (c) 5 (d) 6
24. $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \underset{\text{(A)}}{\text{C}} \equiv \text{N}$; $\text{CH}_3 - \underset{\text{CN}}{\text{CH}} - \text{CH}_3$ (B)
 Relation between (A) and (B) is:
 (a) chain isomers (b) positional isomers
 (c) functional isomers (d) metamers
25. Total number of stereoisomers possible for 2-Bromo-3-chlorobutane is:
 (a) 3 (b) 4 (c) 5 (d) 6
26. Molecule of $\text{C}_{13}\text{H}_{28}$ with the shortest possible parent carbon chain will contains:
 (a) 5 carbon atoms (b) 6 carbon atoms
 (c) 7 carbon atoms (d) 8 carbon atoms
27. $\text{CH}_3 - \text{O} - \text{CH}_2 - \text{CH}_2 - \text{CH}_3$; $\text{CH}_3 - \text{CH}_2 - \text{O} - \text{CH}_2 - \text{CH}_3$
 (A) (B)
 Relations between (A) and (B) is:
 (a) chain isomers (b) positional isomers
 (c) functional isomers (d) metamers
28. How many cyclic structural isomers are possible for C_5H_{10} ?
 (a) 4 (b) 5 (c) 6 (d) 7
29. How many structural isomers of C_9H_{18} may have a six-membered ring and are tri-substituted?
 (a) 5 (b) 6 (c) 7 (d) 8
30. How many structural isomers of C_6H_{12} may contain 3-membered ring?
 (a) 4 (b) 5 (c) 6 (d) 7
31. How many structural isomer of C_6H_{12} may contain 4-membered ring ?
 (a) 3 (b) 4 (c) 5 (d) 6

32. How many structural isomers are possible for molecular formula C_3H_6BrCl ?
 (a) 4 (b) 5 (c) 6 (d) 7
33. How many structural isomers are possible for C_8H_{16} that have a six-member ring ?
 (a) 4 (b) 5 (c) 6 (d) 7
34. How many structural isomers of C_4H_8 contain one π -bond?
 (a) 2 (b) 3 (c) 4 (d) 5
35. How many positional isomers are possible for  ?
 (a) 3 (b) 5 (c) 6 (d) 8
36. How many ketones are possible with molecular formula $C_6H_{12}O$?
 (a) 5 (b) 6 (c) 7 (d) 8
37. How many aldehydes are possible with molecular formula C_4H_8O ?
 (a) 1 (b) 2 (c) 3 (d) 4
38. There are 8 isomers that have the molecular formula $C_5H_{11}Br$. Among these, how many are tertiary alkyl bromides?
 (a) 0 (b) 1 (c) 2 (d) 3 (e) 8
39. What type of isomers $CH_3CH_2OCH_3$ and $CH_3CH_2CH_2OH$ are?
 (a) Constitutional (b) Symmetrical
 (c) Configurational (d) Conformational (e) Stereochemical

Tautomerism and Applications of Tautomerism



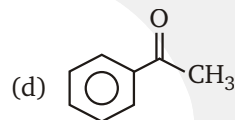
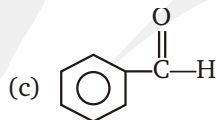
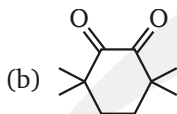
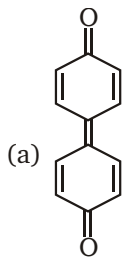
Enol content of (A) and (B) is:

- (a) B A (b) A B (c) A B (d) A B

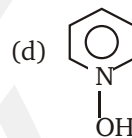
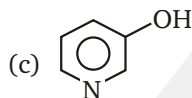
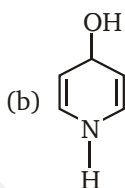
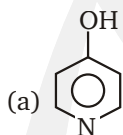
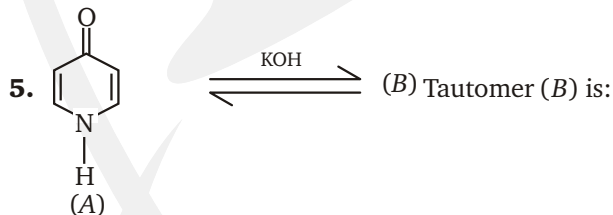
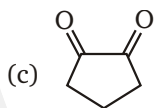
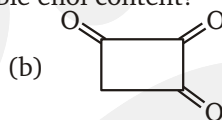
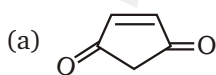
2. Which of the following compound will not show tautomerism ?



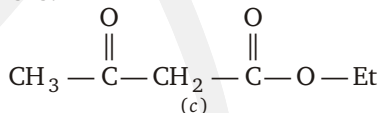
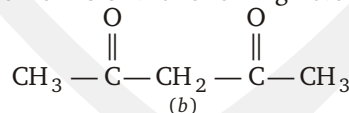
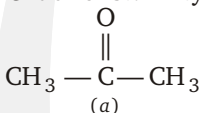
3. Which of the following compound will show tautomerism ?



4. Which of the following compound has negligible enol content?



6. Order of stability of enol forms of the following keto forms is:



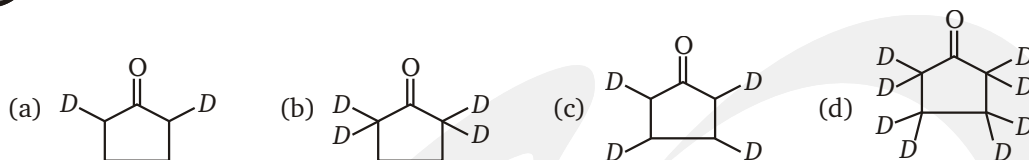
(a) $a > b > c$

(b) $b > a > c$

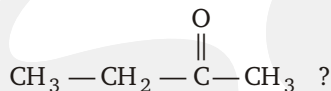
(c) $b > c > a$

(d) $a > b > c$

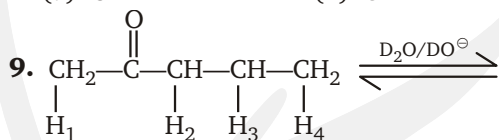




8. How many hydrogens are replaced by deuterium when given compound reacts with D_2O for long time



- (a) 3 (b) 5 (c) 8 (d) 10



Which hydrogen is not replaced by deuterium in the above process?

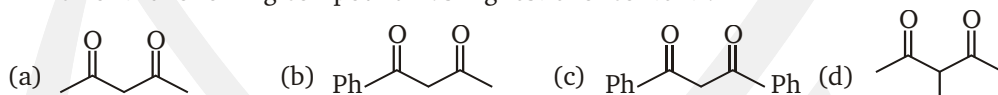
- (a) H_1 (b) H_3 (c) H_4 (d) (b) and (c) both



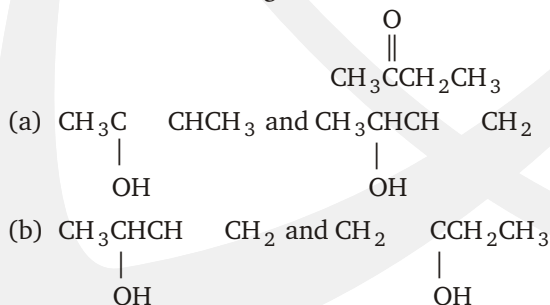
Stability order of enol forms is:

- (a) $i > ii > iii$ (b) $ii > i > iii$ (c) $iii > i > ii$ (d) $i > iii > ii$

11. Which of the following compound has highest enol content ?

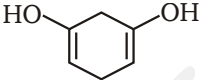
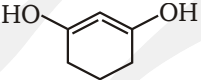
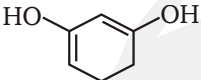
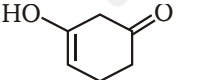
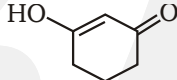


12. Which of the following are enol forms of 2-butanone ?



- (c) $\text{CH}_3\text{C}(\text{OH})\text{CHCH}_3$, $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$, CH_2 and CH_2 , CCH_2CH_3
 OH OH OH
- (d) $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{C}(\text{OH})\text{CHCH}_3$
 OH OH
- (e) $\text{CH}_3\text{C}(\text{OH})\text{CHCH}_3$ and CH_2 , CCH_2CH_3
 OH OH

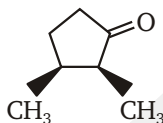
13. Which of the following is the most stable enol form of 1,3-cyclohexanedione ?

- (i)  (ii) 
 (iii)  (iv)  (v) 
 (a) i (b) ii (c) iii (d) iv (e) v

14. Which of the following compounds forms the most stable enol tautomer ?

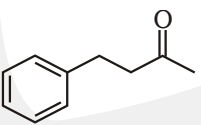
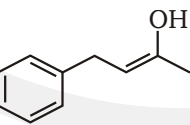
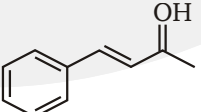
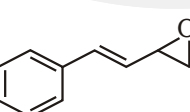
- (a) $\text{CH}_3\text{CH}(\text{O})$ (b) CH_3CCH_3
 O O
- (c) $\text{CH}_3\text{CCH}_2\text{CCH}_3$ (d) $\text{CH}_3\text{C}-\text{CCH}_3$ (e) $\text{CH}_3\text{C}-\text{COH}$
 O O O O

15. How many Hydrogens in the compound given below are replaced by deuterium when it is shaken in D_2O containing traces of hydroxide ?

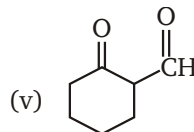
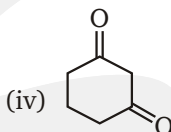
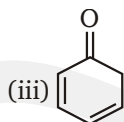
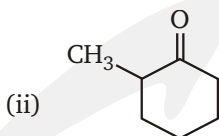
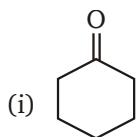


- (a) 1 (b) 2 (c) 3 (d) 6 (e) 7

16. Which of the following is the enol form of 4-phenyl-2-butanone?

- (a)  (b) 
 (c)  (d) 

17. When compared to the keto form, the enol form of which of the following compounds is most stable ?



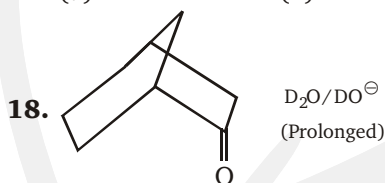
(a) i

(b) ii

(c) iii

(d) iv

(e) v



How many hydrogens are replaced by (D) in the above reaction?

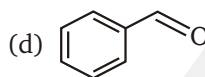
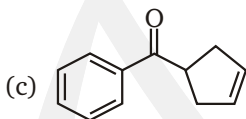
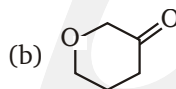
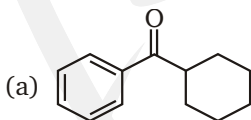
(a) 2

(b) 3

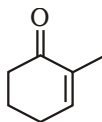
(c) 4

(d) 5

19. Which of the following will not form an enolate?



20. How many enolizable hydrogens are there in the following compound?



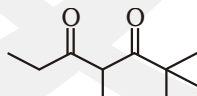
(a) 2

(b) 4

(c) 5

(d) 7

21. How many tautomers can you draw for the following diketone?



(a) 1

(b) 2

(c) 3

(d) 4

22. Keto and enol forms of a compound are related to each other as:

(a) resonance structures

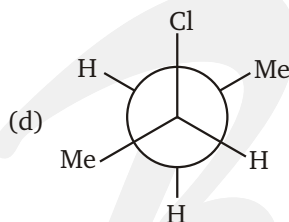
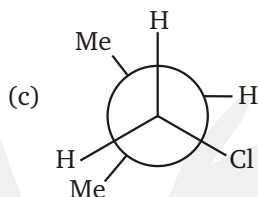
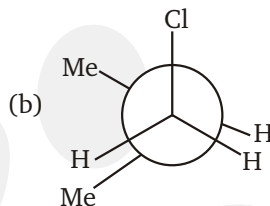
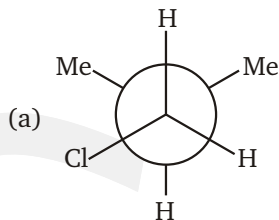
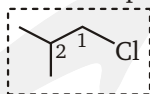
(b) conformations

(c) configurational isomers

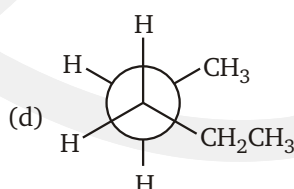
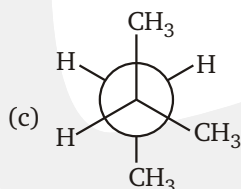
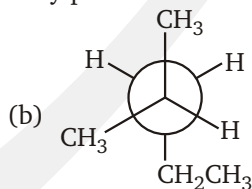
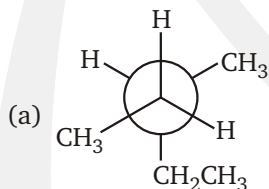
(d) constitutional isomers

Conformational Analysis

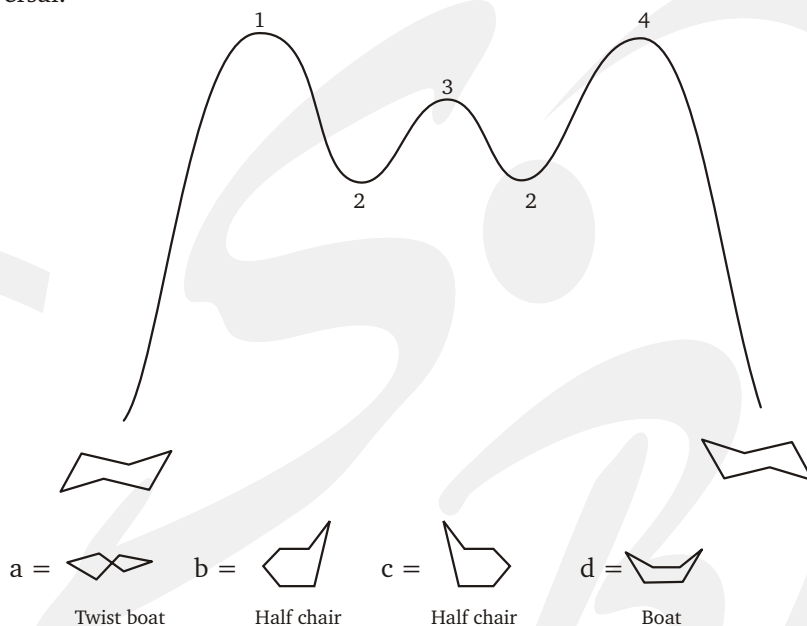
1. Which Newman projection formula does not represent the following compound?



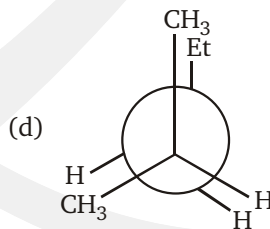
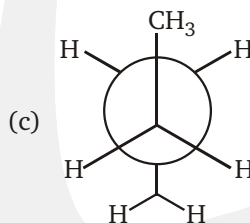
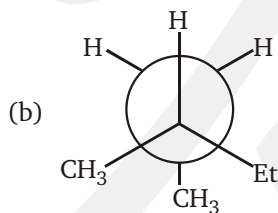
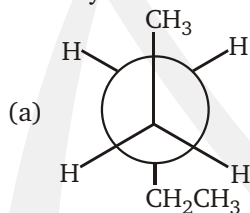
2. What are the total number of conformers (staggered and eclipsed) that you can draw for the following molecule?
- (a) 4 (b) 5 (c) 6 (d) 7
3. A “gauche” interaction costs 3.8 kJ/mol, a hydrogen–hydrogen eclipsing interaction costs 4.0 kJ/mol, a hydrogen–methyl eclipsing interaction costs 6.0 kJ/mol, and a methyl–methyl eclipsing interaction costs 11 kJ/mol. What is the relative energy cost of the most stable conformer of 2,3-dimethylbutane?
- (a) 0 kJ/mol (b) 7.6 kJ/mol (c) 8.6 kJ/mol (d) 11.4 kJ/mol
4. Choose the correct Newman projection for 2-methylpentane.

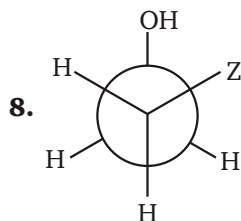


5. Place the following structures properly on the (abbreviated) energy surface for cyclohexane ring reversal.



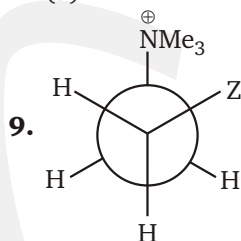
- (a) 1 b or c, 2 a, 3 d, 4 b or c (b) 1 d, 2 b or c, 3 d, 4 a
 (c) 1 d, 2 b or c, 3 a (d) 1 b or c, 2 d, 3 a, 4 b or c
6. Has how many eclipsing H-H interactions are present in boat conformation?
 (a) 1 (b) 2 (c) 4 (d) 6
7. Identify conformer of 2-methyl pentane:





Above gauche form is stable when Z is:

- (a) F (b) —OH (c) —OCH_3 (d) All of these



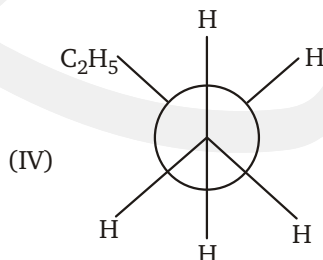
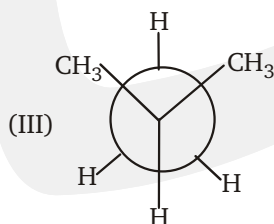
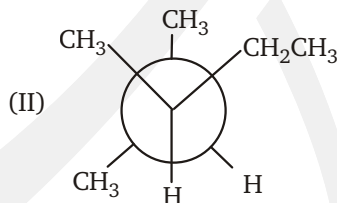
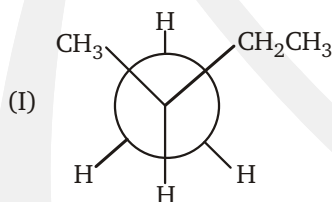
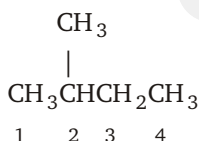
Above gauche form is stable when Z is:

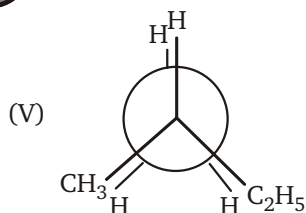
- (a) —CH_3 (b) —Et (c) —C(=O)OEt (d) —Ph

10. The eclipsed and staggered forms of ethane are said to differ in:

- (a) molecular formula (b) configuration
(c) conformation (d) constitution (e) structure

11. Which of the following is the staggered conformation for rotation about the $\text{C}_1\text{—C}_2$ bond in the following structure?



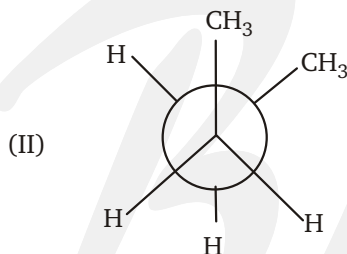
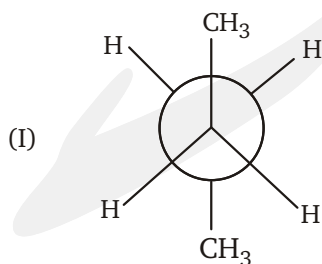


- (a) I (b) II (c) III (d) IV (e) V

12. Among the butane conformers, which occur at energy minima on a graph of potential energy versus dihedral angle?

- (a) gauche only (b) eclipsed and totally eclipsed
(c) gauche and anti (d) eclipsed only (e) anti only

13. Which of the following best explains the reason for the relative stabilities of the conformers shown?



- (a) I has more torsional strain. (b) I has more steric strain
(c) II has more torsional strain (d) II has more steric strain

14. Which of the following is the most stable conformation of bromocyclohexane?



- (a) I (b) II (c) III (d) IV

Geometrical Isomerism

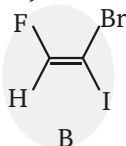
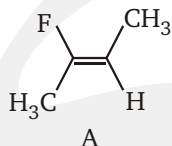
1. How many isomers are possible for dimethylcyclopentane (neglecting enantiomers)?

- (a) 3 (b) 4 (c) 5 (d) 6

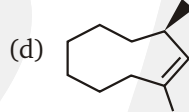
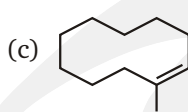
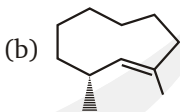
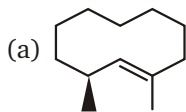
2. How many isomers are possible for dimethylcyclopropane (neglecting enantiomers)?

- (a) 1 (b) 2 (c) 3 (d) 4

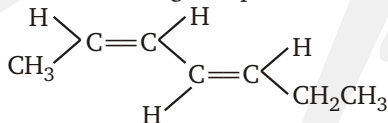
3. How many isomers are there of dimethylcyclohexane (neglecting enantiomers)?
 (a) 4 (b) 6 (c) 7 (d) 9
4. What is the relationship between *trans*-2-butene and *cis*-2-butene?
 (a) Unrelated compounds (b) Constitutional isomers
 (c) Enantiomers (d) Diastereomers
5. Determine the double bond stereochemistry (*E* or *Z*) for the following molecules.



- (a) A: *E*; B: *E* (b) A: *Z*; B: *Z* (c) A: *E*; B: *Z* (d) A: *Z*; B: *E*
6. Which of the following alkenes is (1*E*,3*S*)-1,3-dimethylcycloclodecene?



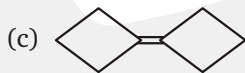
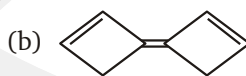
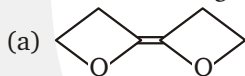
7. What is the IUPAC name for the following compound?



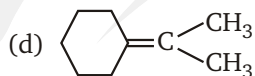
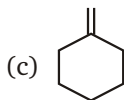
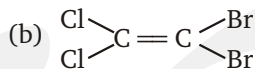
- (a) *cis*, *trans*-2, 4-heptadiene (b) 2*Z*, 4*E*-2,4-heptadiene
 (c) *cis*, *cis*-2,4-heptadiene (d) *trans*, *trans*-2,4-heptadiene
 (e) 2*E*, 4*E*-2,4-heptadiene
8. Geometrical isomerism results because the molecule:
 (a) rotates the plane of polarized light
 (b) has a plane of symmetry
 (c) has a centre of symmetry
 (d) has two dissimilar groups attached to both ends of double bond
9. (i) $\text{CH}_2 - \text{CH} = \text{CH}_2 - \text{CH} = \text{CH}_2$ (ii) $\text{CH}_2 - \text{CH} = \text{CH} - \text{CH}_3$
 (iii) $\text{CH}_3 - \text{CH} = \text{CH} - \text{CH} = \text{CH}_3$

The numbers of possible geometrical isomers for the above compounds respectively are:

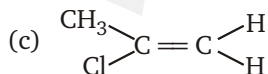
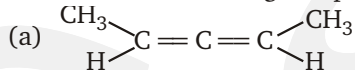
- (a) 0, 2, 4 (b) 2, 2, 4 (c) 0, 3, 3 (d) 0, 2, 3
10. Which of following compound will not show geometrical isomerism ?



11. Which of the following compound show *cis* and *trans* isomerism ?

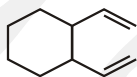


12. Which of the following compound not show geometrical isomerism ?



(d) All of these

13. How many geometrical isomer are possible for the given compound ?



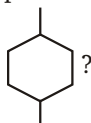
(a) 0

(b) 1

(c) 2

(d) 3

14. How many geometrical isomers are possible for



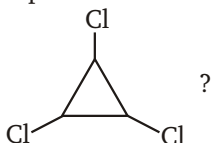
(a) 0

(b) 1

(c) 2

(d) 3

15. How many geometrical isomers are possible for



(a) 0

(b) 1

(c) 2

(d) 3

16. How many geometrical isomers are possible for



(a) 0

(b) 1

(c) 2

(d) 3

17. How many geometrical isomers are possible for



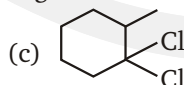
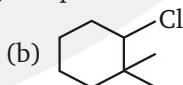
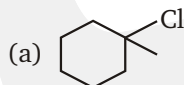
(a) 1

(b) 2

(c) 3

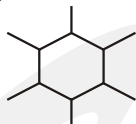
(d) 4

18. Which of the following compound will not show geometrical isomers ?



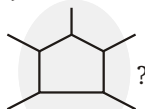
(d) All of these

19. How many geometrical isomers are possible for



- (a) 6 (b) 7 (c) 8 (d) 9

20. How many geometrical isomers are possible for



- (a) 3 (b) 4 (c) 5 (d) 6

21. How many geometrical isomers are possible for



- (a) 2 (b) 3 (c) 4 (d) 6

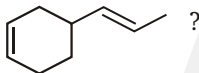
22. How many geometrical isomers are possible for $\text{CH}_3 - \text{CH} = \text{CH} - \text{CH} = \text{CH} - \text{Cl}$?

- (a) 2 (b) 3 (c) 4 (d) 5

23. How many geometrical isomers are possible for $\text{Ph} - \text{CH} = \text{CH} - \text{CH} = \text{CH} - \text{Ph}$?

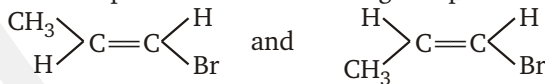
- (a) 2 (b) 3 (c) 4 (d) 5

24. How many geometrical isomers are possible for



- (a) 0 (b) 1 (c) 2 (d) 3

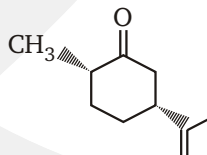
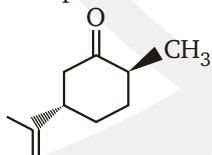
25. What is the relationship between the following compounds?



- (a) Configurational isomers (b) Conformational isomers
(c) Constitutional isomers (d) Structural isomers (e) Positional isomers

Optical Isomerism

1. Determine the relationship between the two molecules shown.

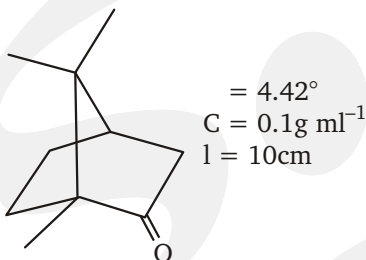


- (a) constitutional isomers (b) enantiomers
(c) diastereomers (d) identical molecules

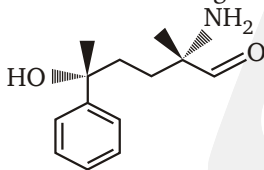
2. Which of the following diols exists as a pair of enantiomers?

- (a) *cis*-1,3-cyclohexanediol (b) *trans*-1,3-cyclohexanediol

- (c) *cis*-1,4-cyclohexanediol (d) *trans*-1,4-cyclohexanediol
3. Which of the 7 isomers of dichlorocyclohexane possess a plane of symmetry?
- (a) *cis*-1,2; *cis*-1, 3 (b) 1, 1; *cis*-1,2; *cis*-1, 3
- (c) *cis*-1,2; *cis*-1,3; *cis*-1,4 (d) 1,1; *cis*-1,2; *cis*-1,3; *cis*-1,4; *trans*-1,4
4. What is the specific rotation $^{20}[\alpha]_D$ of the following molecule?



- (a) 4.42 (b) 0.442 (c) 44.2 (d) 44.2
5. If a sample of 2-butanol has an enantiomeric excess of 60% of l-2-butanol, then per cent of each isomer in the mixture is:
- (a) 60% levorotatory and 40% dextrorotatory (b) 80% levorotatory and 20% dextrorotatory
- (c) 70% levorotatory and 30% dextrorotatory (d) 66% levorotatory and 34% dextrorotatory
6. How many stereocenters are there in the following molecule?

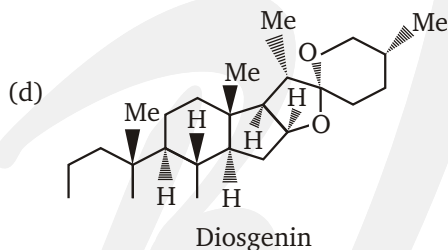
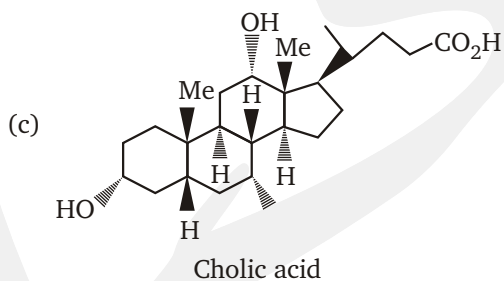
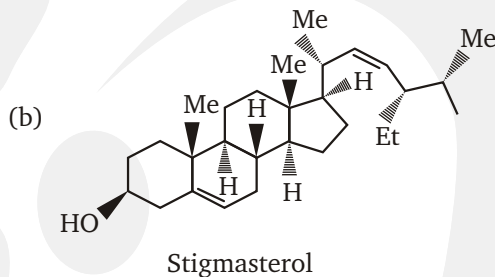
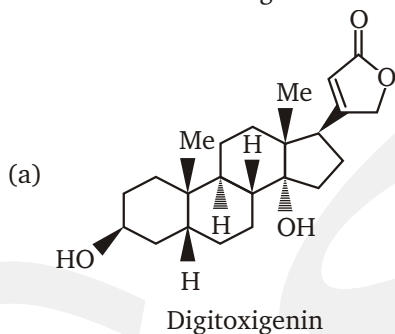


- (a) 0 (b) 2 (c) 3 (d) 1
7. Explain why the two stereoisomers of 1,4-dimethylcyclohexane are achiral?
- (a) Because they are identical
- (b) Because they are enantiomers
- (c) Because they both have the plane of symmetry
- (d) Because there are no stereocenters present in either molecule
8. Consider the following two stereoisomers. How are they different?

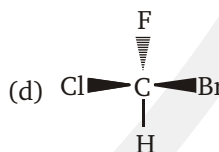
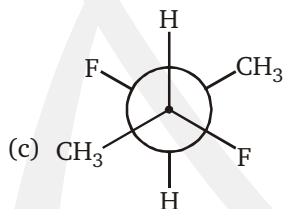


- (a) They have different melting points
- (b) They rotate plane-polarized light in opposite directions
- (c) They have different solubilities in water
- (d) They have different indices of refraction

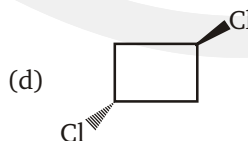
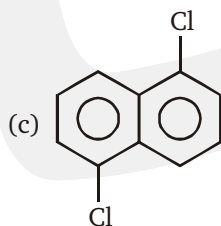
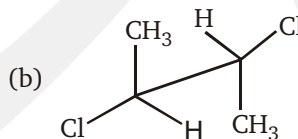
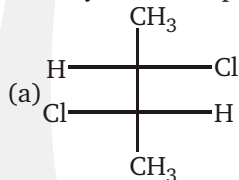
9. Which of the following steroid derivatives has 9 stereocenters?



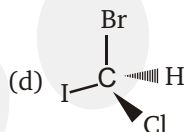
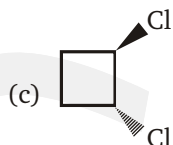
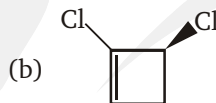
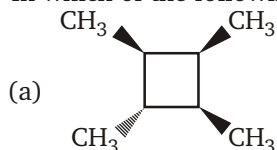
10. Which of the following compound is chiral ?



11. Identify meso compound:



12. In which of the following compound have plane of symmetry ?



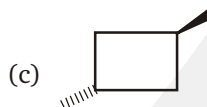
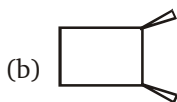
13. ; Relation between (A) and (B) is:

- (a) Diastereomers (b) Enantiomer (c) Identical (d) Structural isomer

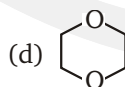
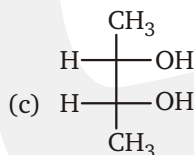
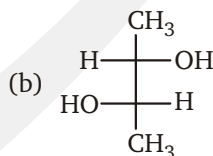
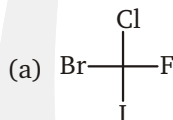
14. ; Number of plane of symmetry in above compound is:

- (a) 0 (b) 1 (c) 2 (d) 3

15. Which of the following is chiral compound ?



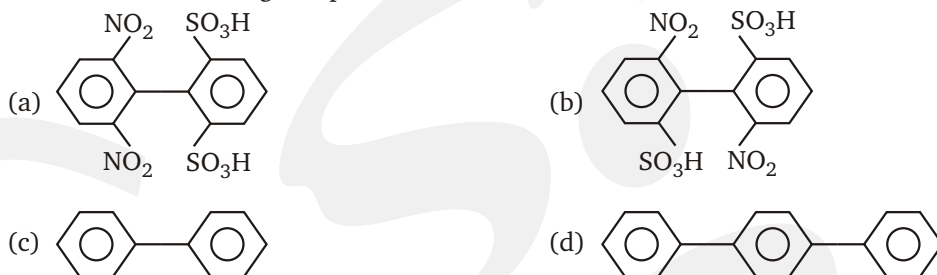
16. Which of the following compound is achiral although it contain chiral center ?



17. Which of the following compound is achiral ?



18. Which of the following compound is resolvable ?



19. How many planes of symmetry are present in $\begin{matrix} \text{H} & & \text{H} \\ & \diagdown & / \\ & \text{C} = \text{C} \\ & / & \diagdown \\ \text{H} & & \text{H} \end{matrix}$?

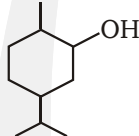
- (a) 1 (b) 2 (c) 3 (d) 4

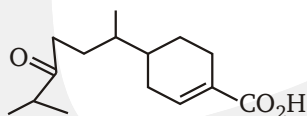
20. How many planes of symmetry are present in  (Benzene) ?

- (a) 3 (b) 5 (c) 7 (d) 9

21. $\begin{matrix} & \text{CO}_2\text{H} \\ & | \\ \text{H} & - & \text{C} & - & \text{OH} \\ | & & | \\ \text{H} & - & \text{C} & - & \text{OH} \\ & | \\ & \text{CO}_2\text{H} \end{matrix}$; Number of plane of symmetry in given compound is:

- (a) 1 (b) 2 (c) 3 (d) 4

22.  (a) (Number of chiral center)

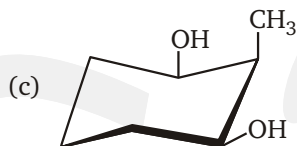
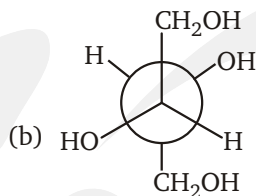
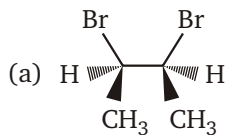


(b) (Number of chiral center)

sum of (a b ?) :

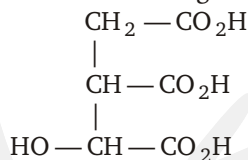
- (a) 3 (b) 4 (c) 5 (d) 6

23. Which of the following are meso compound ?



(d) All of these

24. For the following compound number of chiral centers and stereoisomers are:



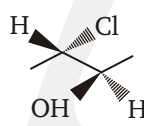
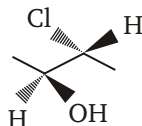
(a) 1, 2

(b) 2, 4

(c) 2, 3

(d) 3, 8

25. Relation between pair given below is:



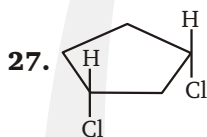
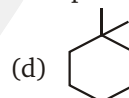
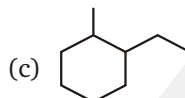
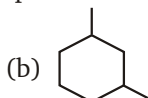
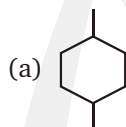
(a) enantiomers

(b) diastereomers

(c) identical

(d) structural isomer

26. Which of following compound has one of the stereoisomers as a meso compound ?



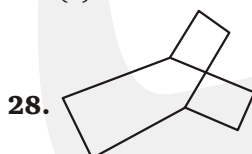
How many plane of symmetry is present in above compound ?

(a) 0

(b) 1

(c) 2

(d) 3



How many plane of symmetry is present in above compound ?

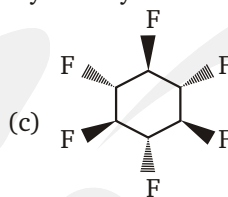
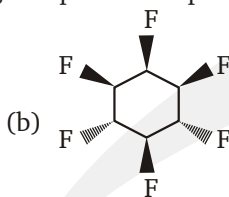
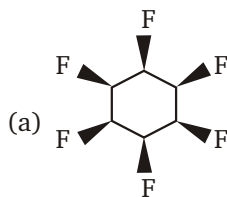
(a) 1

(b) 2

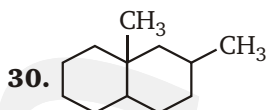
(c) 3

(d) 4

29. Which of the following compound has plane of symmetry and center of symmetry ?



(d) All of these



How many chiral center are present in above compound ?

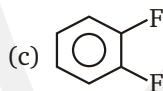
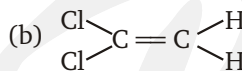
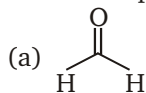
(a) 1

(b) 2

(c) 3

(d) 4

31. Which compound has plane of symmetry and C_2 axis of symmetry ?



(d) All of these

32. What is optical purity of 6 gm (d) and 4 gm (l) ?

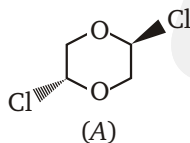
(a) 40%

(b) 20%

(c) 10%

(d) 60%

33. What is the optical rotation of (A) ?

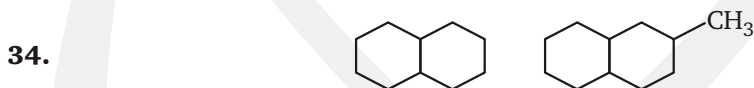


(a) 50

(b) 50

(c) 0

(d) 100



Number of chiral center present in above compounds are :

(a) 0, 1

(b) 0, 2

(c) 0, 3

(d) 1, 2



Number of chiral center present in above compounds are:

(a) 1, 2

(b) 2, 2

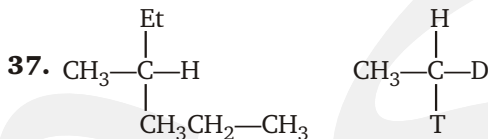
(c) 2, 1

(d) 2, 3



Number of chiral center present in above compounds are:

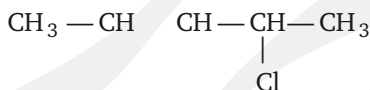
- (a) 1, 1 (b) 1, 2 (c) 2, 2 (d) 2, 1



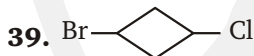
Number of chiral center present are:

- (a) 0, 0 (b) 1, 0 (c) 1, 1 (d) 2, 2

38. Number of stereo centre present are:

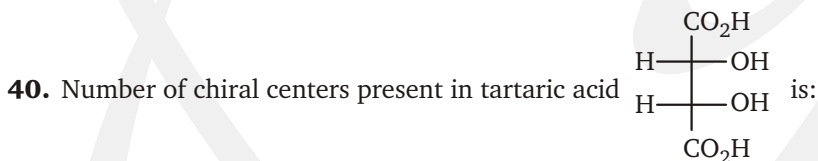


- (a) 2 (b) 3 (c) 4 (d) 5



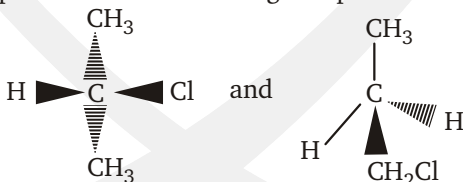
Number of stereocenters in the compound given are:

- (a) 0 (b) 1 (c) 2 (d) 3



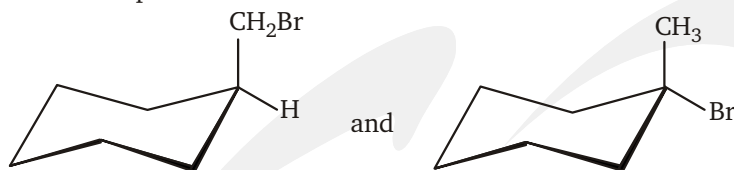
- (a) 1 (b) 2 (c) 3 (d) 4

41. What is the relationship between the following compounds?



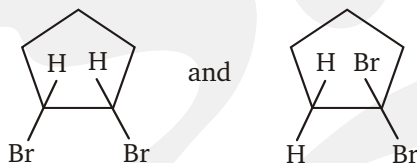
- (a) Constitutional isomers (b) Enantiomers
(c) Diastereomers (d) Conformational isomers
(e) Superimposable without bond rotation

42. What is the relationship between the structures shown below?



- (a) Not isomers (b) Constitutional isomers
(c) Conformational isomers (d) Configurational isomers (e) Enantiomers

43. What is the relationship between the structures shown below?

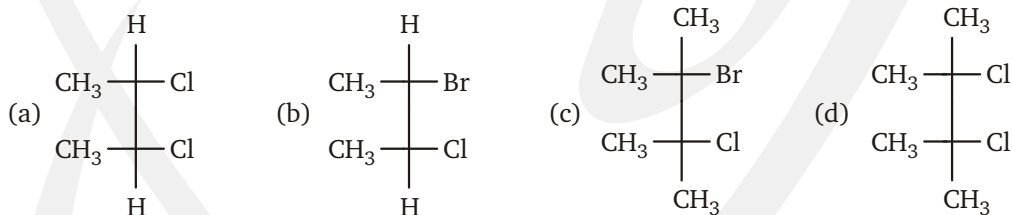


- (a) Diastereomers (b) Constitutional isomers
(c) Conformational isomers (d) Configurational isomers (e) Enantiomers

44. Which of the following compounds is never chiral?

- (a) 2,3-dibromobutane (b) 1,3-dibromobutane
(c) 1,2-dichlorobutane (d) 1,4-dibromobutane (e) 1-bromo-2-chlorobutane

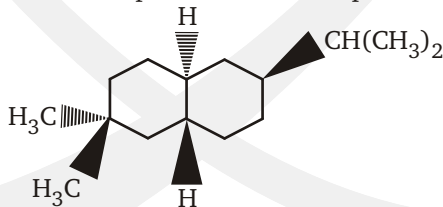
45. Which of the following compounds is chiral?



46. How many asymmetric centers are present in a molecule of 2,4,6-trimethylheptane?

- (a) 0 (b) 1 (c) 2 (d) 3 (e) 4

47. How many asymmetric centers are present in the compound shown below?

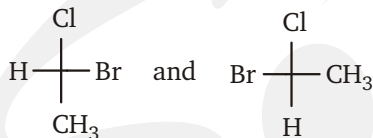


- (a) 1 (b) 2 (c) 3 (d) 4 (e) 5

48. A and B are stereoisomers. They are non-superimposable and are mirror images of each other. Which of the following best describes the relationship between A and B?

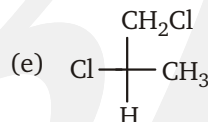
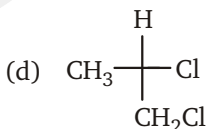
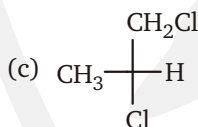
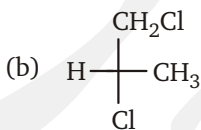
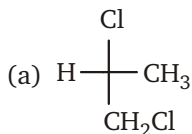
- (a) structural isomers (b) enantiomers
(c) conformational isomers (d) diastereomers (e) constitutional isomers

49. What is the relationship between the structures shown below?

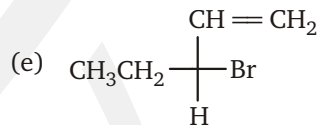
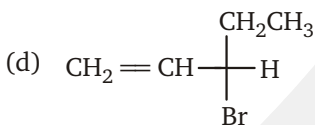
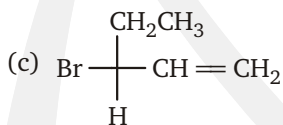
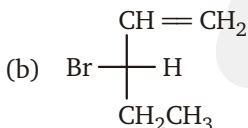
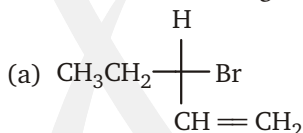


- (a) enantiomers (b) diastereomers
(c) configurational isomers (d) identical compounds (e) constitutional isomers

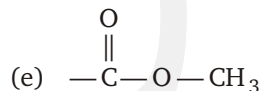
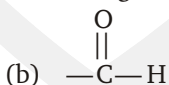
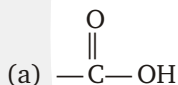
50. Which of the following compounds has an S configuration?



51. Which of the following compounds has an R configuration?



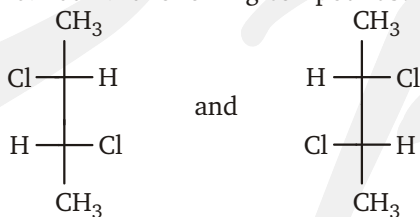
52. Which of the following groups has the highest priority using the Cahn, Ingold, Prelog rules?



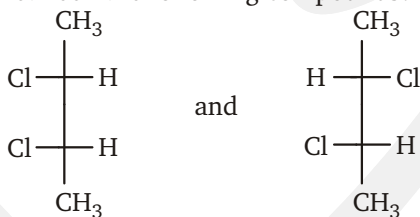
53. Which of the following is/are optically inactive?

- (a) A 50-50 mixture of R and S enantiomers (b) A meso compound
(c) Every achiral compound (d) A racemic mixture (e) All of the above

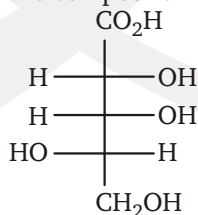
54. If (*S*)-glyceraldehyde has a specific rotation of -8.7° , what is the specific rotation of (*R*)-glyceraldehyde?
(a) -8.7° (b) $+8.7^\circ$
(c) 0.0° (d) cannot be determined from the information given
55. A mixture of equal amounts of two enantiomers.....
(a) is called a racemic mixture (b) is optically inactive
(c) implies that the enantiomers are meso forms (d) both (a) and (b)
56. The specific rotation of a pure substance is 1.68° . What is the specific rotation of a mixture containing 75% of this isomer and 25% of the ($-$) isomer?
(a) $+1.68^\circ$ (b) 0° (c) $+1.26^\circ$ (d) $+0.84^\circ$ (e) $+0.42^\circ$
57. ($-$)-Mandelic acid has a specific rotation of -158° . What would be the specific rotation of a solution which contains 40% ($-$)-mandelic acid and 60% ($+$)-mandelic acid?
(a) $+95^\circ$ (b) $+63^\circ$ (c) $+32^\circ$ (d) -32° (e) -63°
58. What is the relationship between the following compounds?



- (a) Enantiomers (b) Diastereomers
(c) Constitutional isomers (d) Conformational isomers (e) Identical compounds
59. What is the relationship between the following compounds?

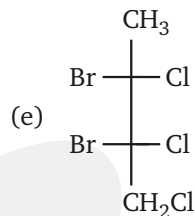
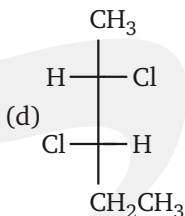
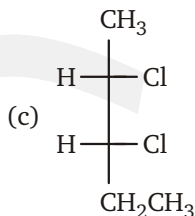
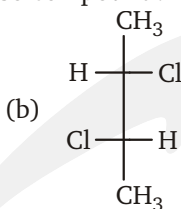
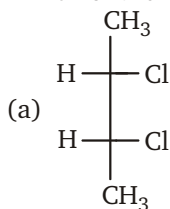


- (a) Enantiomers (b) Diastereomers
(c) Constitutional isomers (d) Conformational isomers (e) Identical compounds
60. How many diastereomers exist for the compound below?

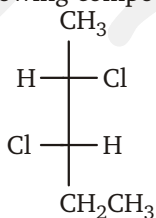


- (a) 2 (b) 4 (c) 6 (d) 7 (e) 8

61. Which of the following is a meso compound?

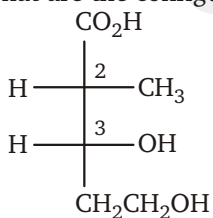


62. What is the configuration of the following compound?



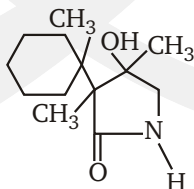
- (a) 2*S*,3*R* (b) 3*R*,3*S* (c) 2*S*,3*S* (d) 2*R*,3*R*
 (e) Can't do *R* and *S*; the compound is achiral

63. In the Fischer projection below, what are the configurations of the two asymmetric centers?



- (a) 2*R*,3*R* (b) 2*R*,3*S* (c) 2*S*,3*R* (d) 2*S*,3*S*
 (e) cannot be determined from structure

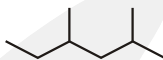
64. How many chiral carbon centers are in the following molecule ?



- (a) 0 (b) 1 (c) 2 (d) 3

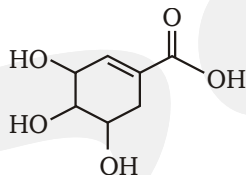
Miscellaneous Problems on Stereoisomers

1. How many stereoisomers are there for the following structure?



- (a) 1 (b) 2 (c) 3 (d) 4

2. Shikimic acid is an important natural product. How many stereoisomers are possible for shikimic acid?

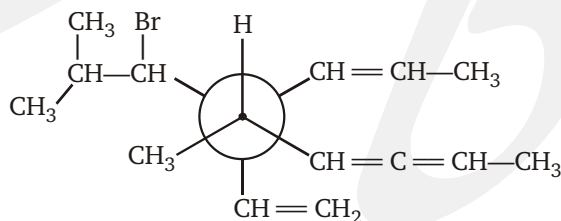


- (a) 16 (b) 8 (c) 4 (d) 2

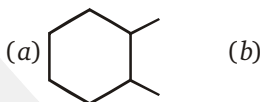
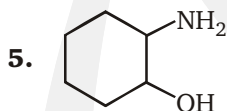
3. How many stereoisomers are there for 1-ethyl-3-methylcyclohexane?

- (a) 2 (b) 3 (c) 4 (d) 6

4. Total number of stereoisomer



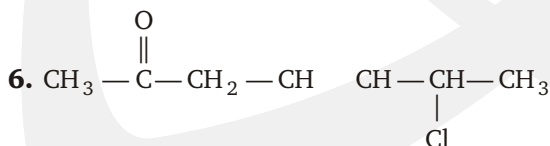
- (a) 2^3 (b) 2^4 (c) 2^5 (d) 2^6



(b)

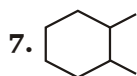
Number of stereoisomer (a b ?)

- (a) 5 (b) 6 (c) 7 (d) 8

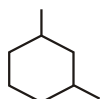


Total number of stereoisomers are:

- (a) 4 (b) 6 (c) 8 (d) 12



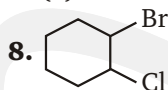
a(stereoisomers)



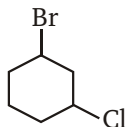
b (stereoisomers)

a and *b* are stereoisomers sum of (*a* *b* ?) :

- (a) 4 (b) 6 (c) 8 (d) 12



a



b

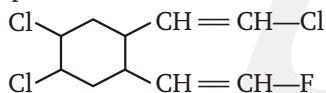
sum of stereoisomer (*a* *b* ?):

- (a) 4 (b) 6 (c) 8 (d) 12

9. How many isomer of C_6H_{12} containing cyclobutane ring ? (Including stereoisomeris)

- (a) 5 (b) 6 (c) 7 (d) 8

10. How many stereoisomers are possible for:



- (a) 2^4 (b) 2^5 (c) 2^6 (d) 2^7

11. How many stereoisomers exist for the compound given below?



- (a) 0 (b) 1 (c) 2 (d) 4 (e) 8

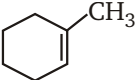
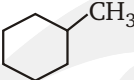
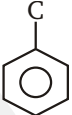
12. How many stereoisomers exist for the compound given below?



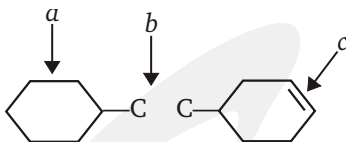
- (a) 0 (b) 1 (c) 2 (d) 3 (e) 4

[C] General Organic Chemistry

Inductive Effect & its Applications

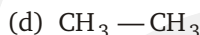
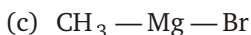
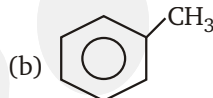
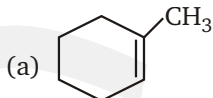
- Which of the following has strong +I effect ?
 (a) $\text{O}^\ominus$ (b) $-\text{Et}$ (c) $-\text{CH}_3$ (d) $-\text{NH}-\text{CH}_3$
- In which of the following compounds inductive effect is not present ?
 (a) $\text{CH}_3-\text{CH}_2-\text{Br}$ (b) $\text{CH}_3-\text{CH}=\text{CH}_2$
 (c) $\text{CH}_3-\text{C}(\text{CH}_3)=\text{CH}_2$ (d) $\text{CH}_3-\text{CH}_2-\text{CH}_3$
- In which of the following compounds I effect of methyl group is not present ?
 (a)  (b) 
 (c) $\text{H}_2\text{C}=\text{CH}-\text{CH}_3$ (d) $\text{CH}_3-\text{CH}=\text{O}$
- In which of the following compounds -I effect of methyl group is operating?
 (a) CH_3-Cl (b) CH_3-CH_3
 (c) CH_3MgBr (d) CH_3-NO_2
- In which of the following compounds (I) effect of alkene will operate?
 (a) $\text{CH}_3-\text{CH}=\text{CH}_2$ (b) $\text{CH}_3-\text{CH}_2-\text{CH}=\text{CH}_2$
 (c) $\text{HC}(\text{CH}_3)=\text{C}(\text{CH}_3)-\text{CH}_2\text{CH}_3$ (d) $\text{H}_2\text{C}=\text{CH}-\text{CH}=\text{CH}_2$
-  ; 
 (a) ; (b)
 In the compounds (a) and (b) inductive effect of phenyl group is:
 (a) I, I (b) I, I (c) I, I (d) I, -I
- $\text{CH}_3-\underset{4}{\text{CH}_2}-\underset{3}{\text{CH}_2}-\underset{2}{\text{CH}_2}-\underset{1}{\text{CH}_2}-\boxed{\text{Cl}}$
 Inductive effect of chlorine is minimum on:
 (a) C_1 (b) C_2 (c) C_3 (d) C_4
- Which of the following substituent/group is an electron donating group overall ?
 (a) $-\text{Br}$ (b) $-\text{C}(\text{O})\text{CH}_3$ (c) $-\text{OCH}_3$ (d) $-\text{N}^+(\text{CH}_3)_3$ (e) $-\text{CCl}_3$

9. Rank the indicated bond in the given compound in order of decreasing bond strength:

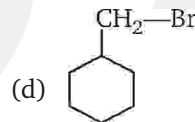
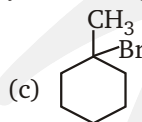
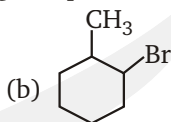
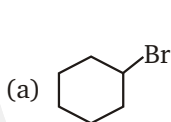


- (a) $a \ b \ c$ (b) $b \ a \ c$ (c) $b \ c \ a$ (d) $a \ c \ b$

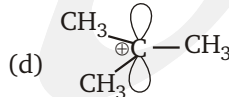
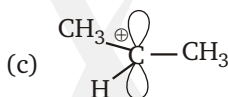
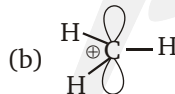
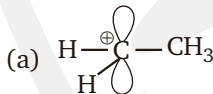
10. In which of the following compounds —CH₃ shows *I* effect ?



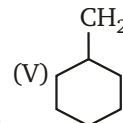
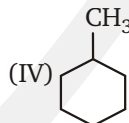
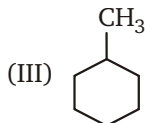
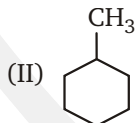
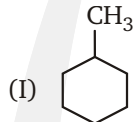
11. Which of the following compounds on heterolytic fission gives most stable carbocation ?



12. Which of the following represent methyl carbocation ?

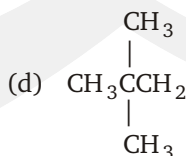
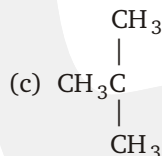


13. Which of the following is the most stable carbocation?

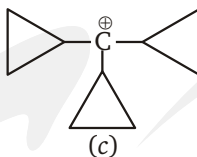
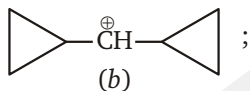
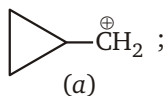


- (a) I (b) II (c) III (d) IV (e) V

14. Which of the following carbocations is the most stable?



15. Stability order of given carbocation is:



(a) b a c

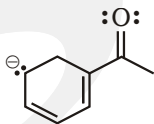
(b) a b c

(c) c b a

(d) c a b

Resonance

1. How many total resonance structures can be drawn for the following anion (including those without separation of charge)?



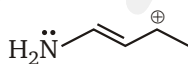
(a) 1

(b) 2

(c) 3

(d) 4

2. How many resonance structures can be drawn for the following molecule?



(a) 1

(b) 4

(c) 3

(d) 2

3. How many other major contributing resonance structures are possible for the following heterocycle?



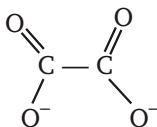
(a) 2

(b) 4

(c) 6

(d) 8

4. How many energetically equivalent resonance structures exist for the oxalate dianion?



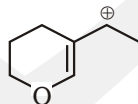
(a) 1

(b) 2

(c) 3

(d) 4

5. How many resonance structures can be drawn for the following carbocation?



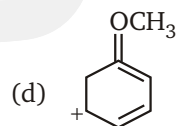
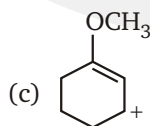
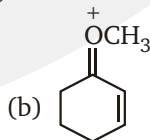
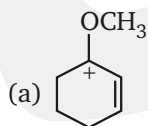
(a) 0

(b) 1

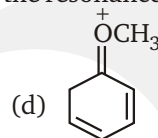
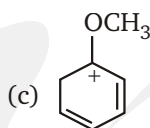
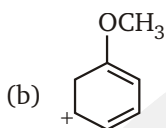
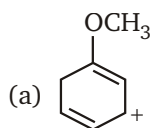
(c) 3

(d) 4

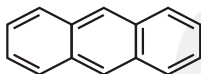
6. Which of the following structures is not a resonance structure of the others?



7. Which of the following resonating structures contributes maximum to the resonance hybrid?



8. How many resonating structures are there for anthracene?



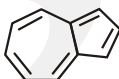
(a) 6

(b) 5

(c) 4

(d) 3

9. How many uncharged resonating structures are there for azulene?



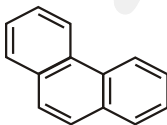
(a) 1

(b) 2

(c) 3

(d) 4

10. How many resonating structures are there for phenanthrene?



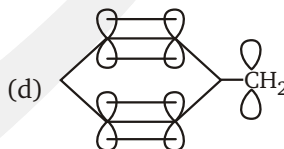
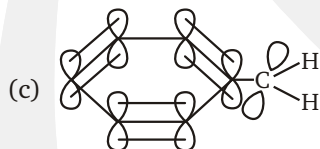
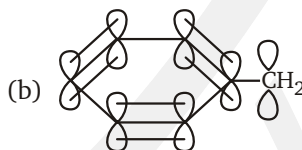
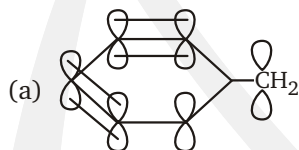
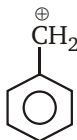
(a) 6

(b) 5

(c) 4

(d) 2

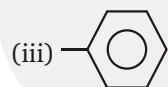
11. Draw the given carbocation in orbital phase.



12. Which of the following substrates is an electron withdrawing group ?

(i) $-\text{CH}_3$

(ii) $-\text{OH}$



(iv) $-\text{OCH}_3$

(v) $-\text{SO}_3\text{H}$

(a) i

(b) ii

(c) iii

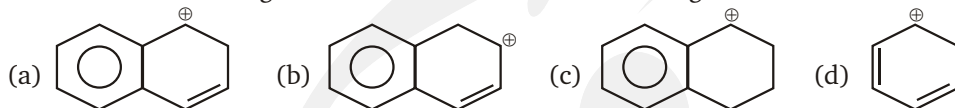
(d) iv

(e) v

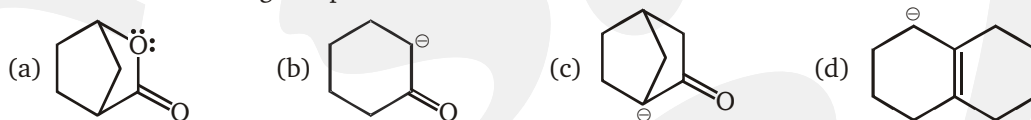
13. Which of the following has two equivalent resonating structure ?



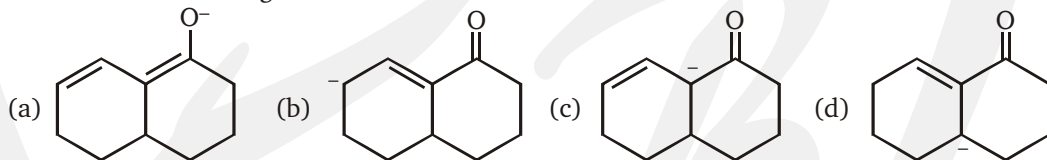
14. Which of the following has maximum number of resonating structure ?



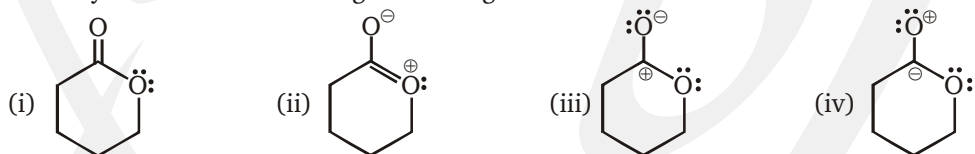
15. Which of the following compound is not resonance stabilized ?



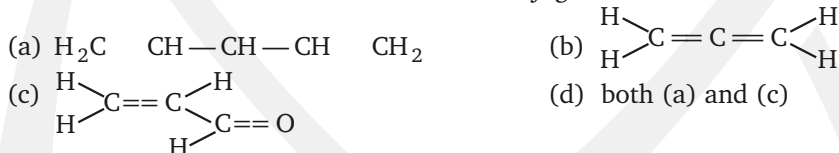
16. Which of the following is not a resonance structure of the others ?



17. Stability order of the following resonating structure will be:



18. Select the molecule in which bonds are conjugated ?

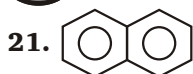


19. Which of the following is not a valid resonance structures ?



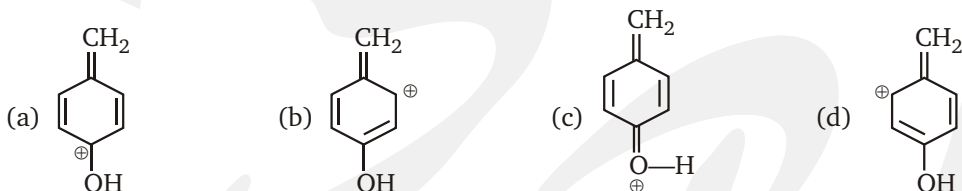
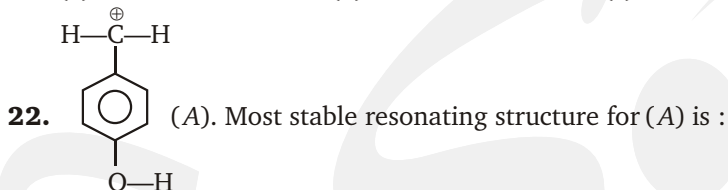
20. In which of the following compound resonance is not possible?





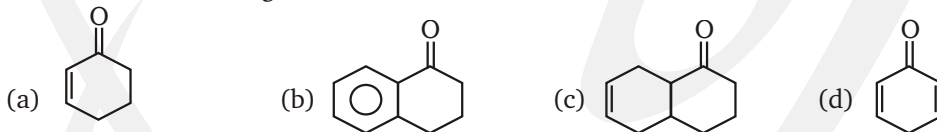
How many resonating structure are possible for Naphthalene (lacking formal charges) ?

- (a) 1 (b) 2 (c) 3 (d) 4



- (a) 1 (b) 2 (c) 3 (d) 4

24. Which of the following is not resonance stabilized ?



25. In which of the following compounds lone pair of oxygen is delocalized ?



26. Resonance hybrid of $\text{H}_2\text{C}=\text{CH}-\text{CH}_2^+$ is :



27. In which of the following resonating structures, extra bond will be formed ?



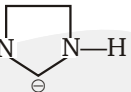
28. In which of the following compounds *d*-orbital resonance takes place ?

- (a) $\text{Cl}-\overset{\ominus}{\text{C}}\text{H}_2$ (b) $\text{CH}_3-\text{S}-\overset{\ominus}{\text{C}}\text{H}_3$ (c) $\text{Me}_3\overset{\oplus}{\text{P}}-\overset{\ominus}{\text{C}}\text{H}_2$ (d) All of these

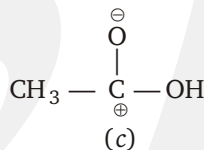
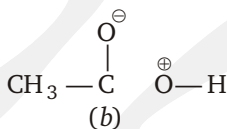
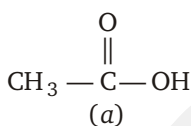
29. $\text{Cl}-\overset{\ominus}{\text{C}}\begin{matrix} \text{Cl} \\ \diagup \\ \text{Cl} \end{matrix}$; In above carbanion electron is delocalized in :

- (a) *p*-orbital (b) *d*-orbital (c) *s*-orbital (d) *f*-orbital

30. Which of the following carbanions is resonance stabilized ?

- (a)  (b)  (c)  (d) 

31. Compare relative stability of following resonating structure:



- (a) *a b c* (b) *a c b* (c) *b a c* (d) *c a b*

32. Which of the following is not the resonating structure of each other?



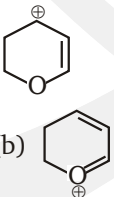
- (c) $\text{CH}_3-\text{C}^{\oplus}(\text{N}^{\ominus})-\text{N}^{\oplus}$ $\text{CH}_3-\text{N}^{\oplus}(\text{C}^{\ominus})-\text{C}^{\oplus}$ (d) All of these

33. Which of the following is not the resonating structure of given?

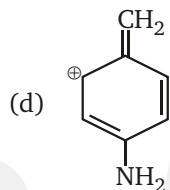
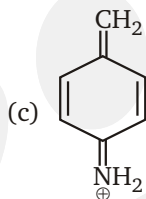
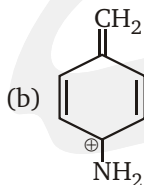
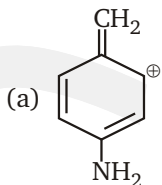
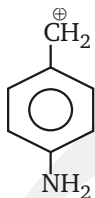


- (d) both (b) and (c)

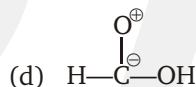
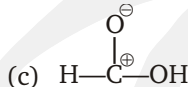
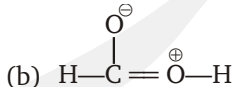
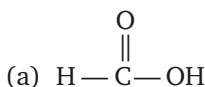
34. Most stable resonating structure of given cation is:

- (a)  (b)  (c)  (d) 

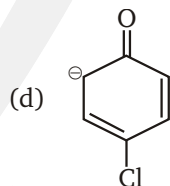
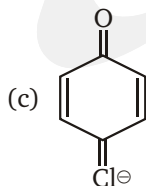
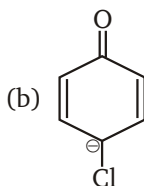
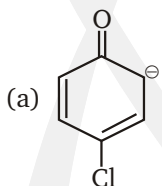
35. Most stable resonating structure of given cation is:



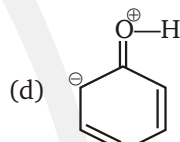
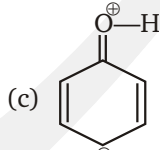
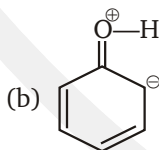
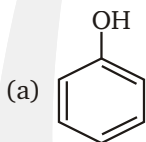
36. Most stable resonating structure of formic acid is:



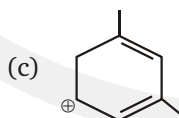
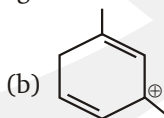
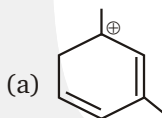
37. Most stable resonating structure of is:



38. Most unstable resonating structure of phenol is:

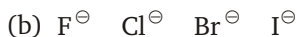
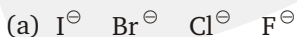


39. Most unstable resonating structure of given is:

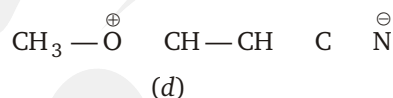
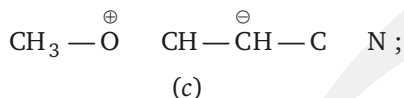
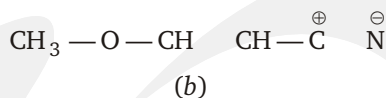
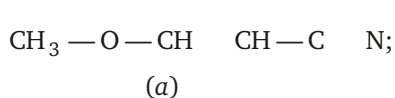


(d) All of these

40. Compare leaving group tendency of following halide ion :



41. Compare relative stability of following resonating structures :



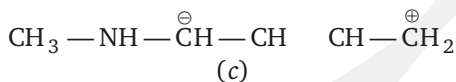
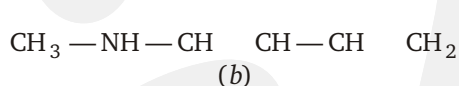
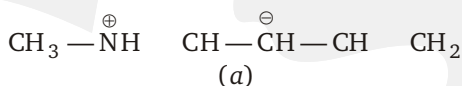
(a) a b c d

(b) a c d b

(c) a d c b

(d) c d b a

42. Compare relative stability of following resonating structures :



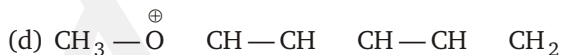
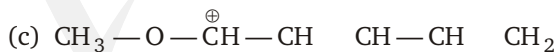
(a) a b c

(b) b a c

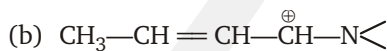
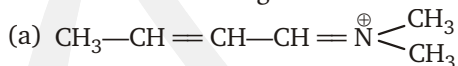
(c) b c a

(d) a c b

43. Most stable resonating structure is:



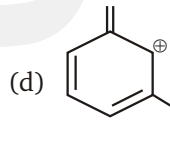
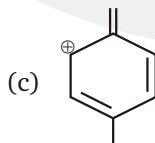
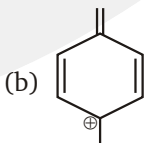
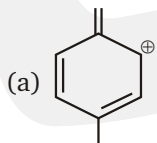
44. Most stable resonating structure is:



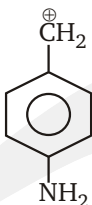
45. Most stable resonating structure of



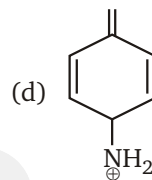
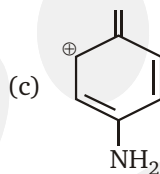
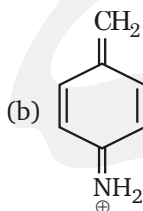
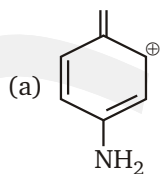
is:



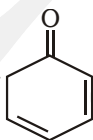
46. Most stable resonating structure of



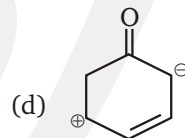
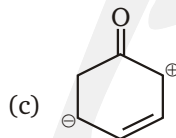
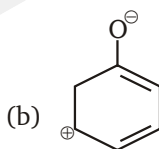
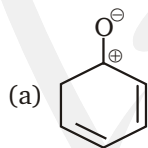
is:



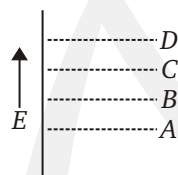
47. Most unstable resonating structure of



is :

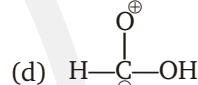
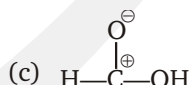
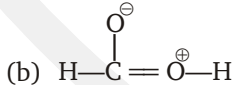
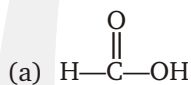


48.

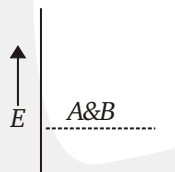


Formic acid

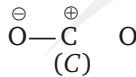
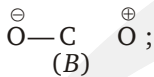
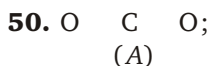
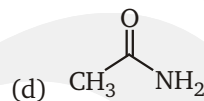
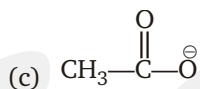
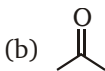
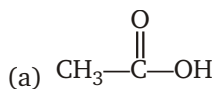
A, B, C, D are resonating structure of formic acid. Structure of B is :



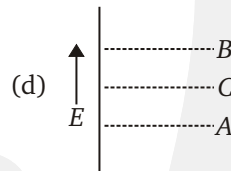
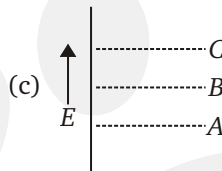
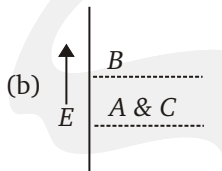
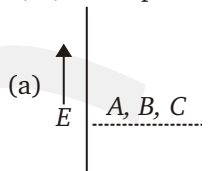
49.



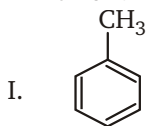
A and B are resonating structure of :



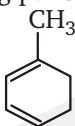
A, B, C is represent by:



51. Which of the following pairs are resonance structures?



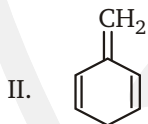
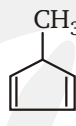
and



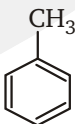
IV.



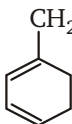
and



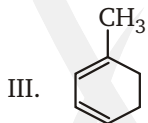
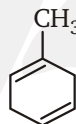
and



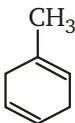
V.



and



and



(a) I

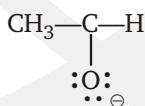
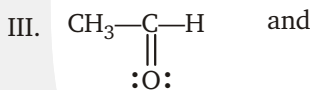
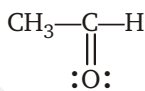
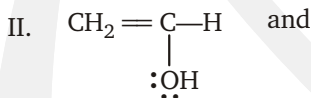
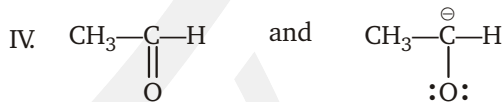
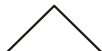
(b) II

(c) III

(d) IV

(e) V

52. Which of the following pairs are resonance structures?



(a) I

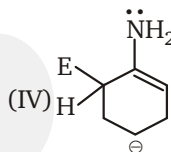
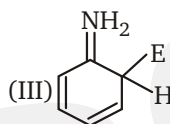
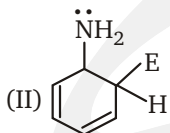
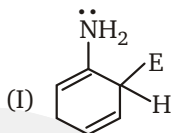
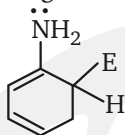
(b) II

(c) III

(d) IV

(e) V

53. Which of the following is not a resonating structure of the species shown?



(a) I

(b) II

(c) III

(d) IV

54. In which of the following does resonance delocalization of electron density not play a role?

(a) allyl cation

(b) benzyl anion

(c) CO_3^{2-} (d) O_3

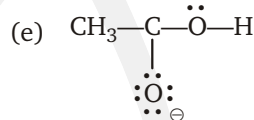
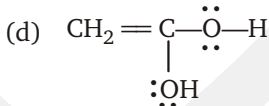
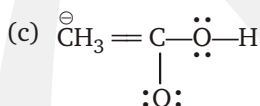
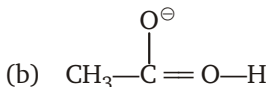
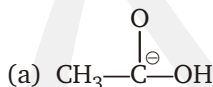
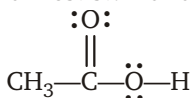
(e) cyclohexyl radical

55. Due to electron delocalization, one would predict that the carbon-oxygen bond in acetamide, CH_3CONH_2 , -----.

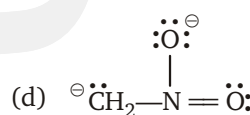
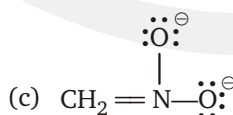
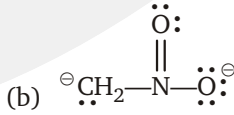
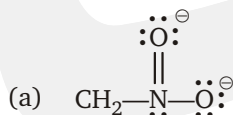
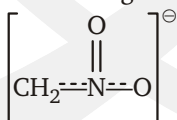
(a) is nonpolar

(b) has more double bond character than the carbon-oxygen bond of acetone, $(\text{CH}_3)_2\text{CO}$ (c) is longer than the carbon-oxygen bond of dimethyl ether, $(\text{CH}_3)_2\text{O}$ (d) is longer than the carbon-oxygen bond of acetone, $(\text{CH}_3)_2\text{CO}$ (e) is formed by overlapping sp^3 orbitals

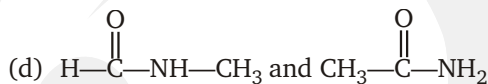
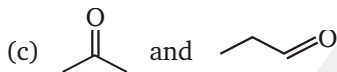
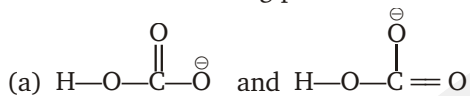
56. Which of the following is the most stable resonance contributor to acetic acid ?



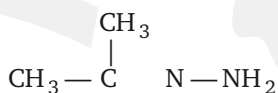
57. Which is the most important contributing resonating structure of the following ?



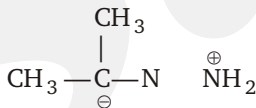
58. Which of the following pairs are resonating structure?



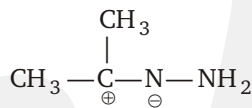
59. Rank the resonance structure in each group in order of increasing contribution to the resonance hybrid :



(i)



(ii)



(iii)

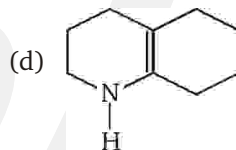
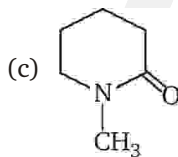
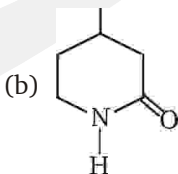
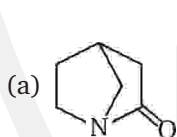
(a) $\text{i} < \text{ii} < \text{iii}$

(b) $\text{ii} < \text{i} < \text{iii}$

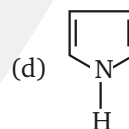
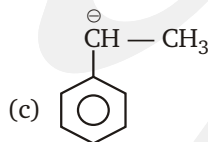
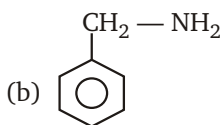
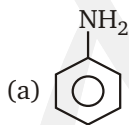
(c) $\text{iii} < \text{ii} < \text{i}$

(d) $\text{iii} < \text{i} < \text{ii}$

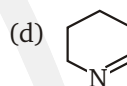
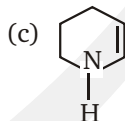
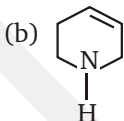
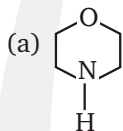
60. In which of the compound lone pair of nitrogen is not involved in resonance?



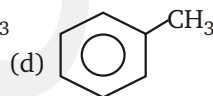
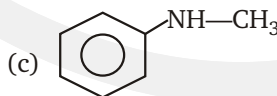
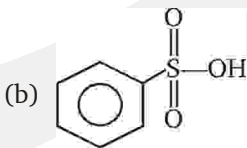
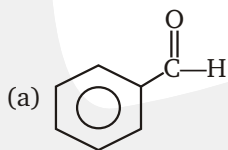
61. Which of the following compounds has localized lone pair of electrons?



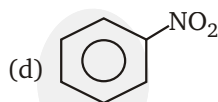
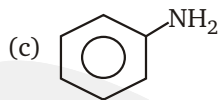
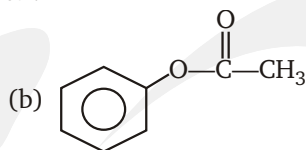
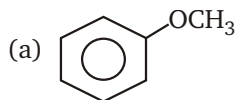
62. Which of the following compounds has delocalized lone pair of electrons?



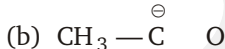
63. Which of the following group shows (+M) effect ?



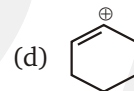
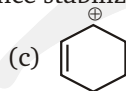
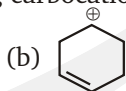
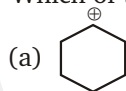
64. Which of the following group shows ($-M$) effect ?



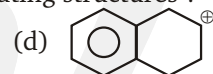
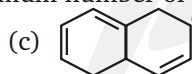
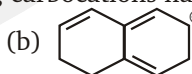
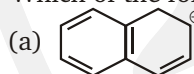
65. Which of the following carbanions is resonance stabilized?



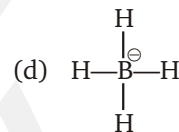
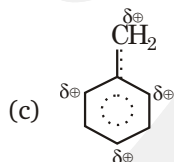
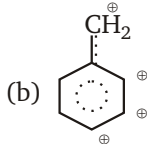
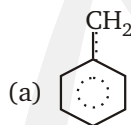
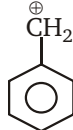
66. Which of the following carbocations is resonance stabilized?



67. Which of the following carbocations has maximum number of resonating structures ?



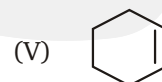
68. Resonance hybrid of given carbocation is:



69. Resonance hybrid of $\text{H}_2\text{C}^+ - \text{CH}^+ - \text{CH}_3$ is :



70. Which of the following is an allylic carbocation?



(a) I

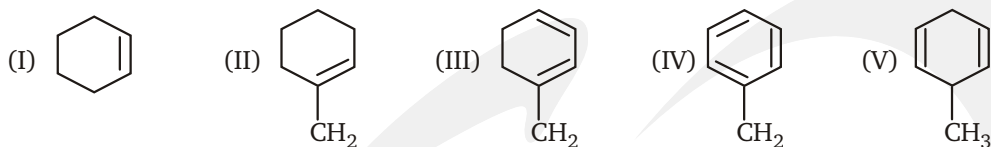
(b) II

(c) III

(d) IV

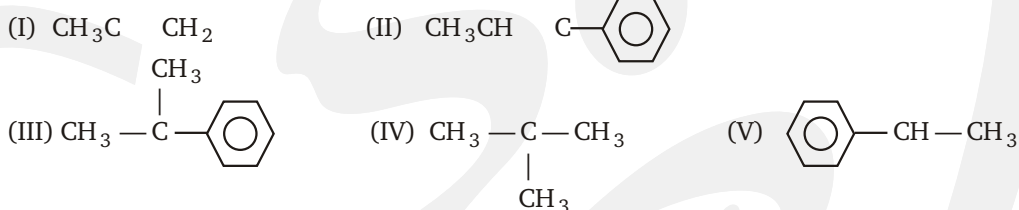
(e) V

71. Which of the following is a benzylic cation?



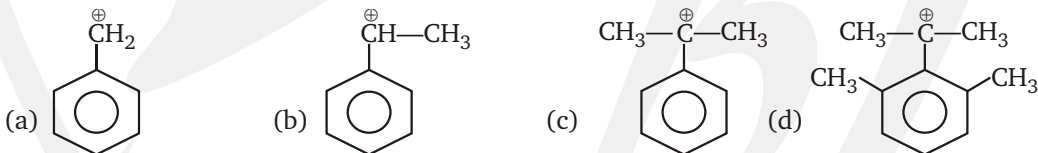
(a) I (b) II (c) III (d) IV (e) V

72. Which of the following is the most stable carbocation?

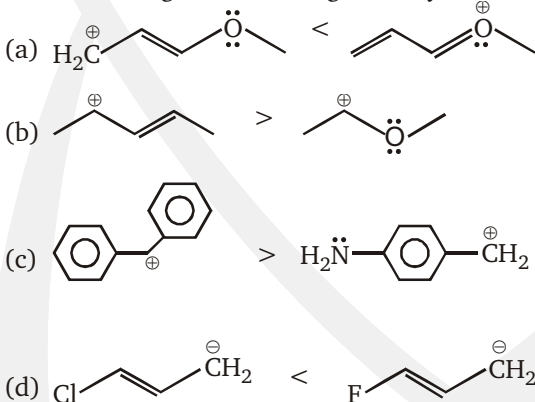


(a) I (b) II (c) III (d) IV (e) V

73. Which of the following carbocation is not resonance stabilized?



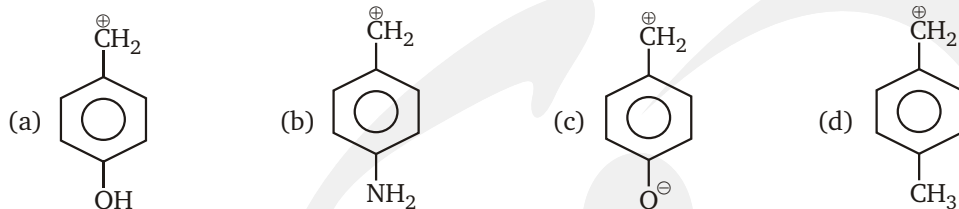
74. Which of the given following stability order is correct ?



75. Which of the following carbocation is most stable ?



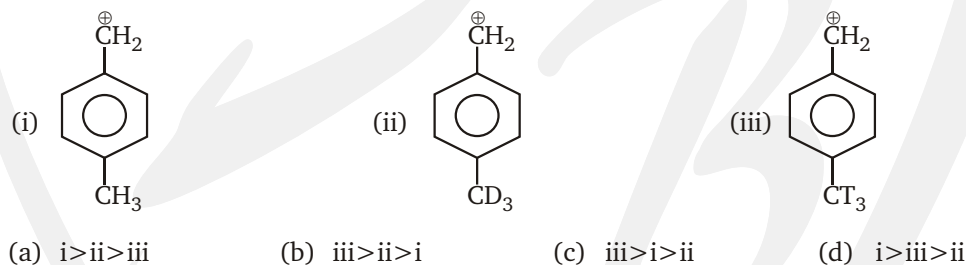
76. Which of the following carbocation is most stable?



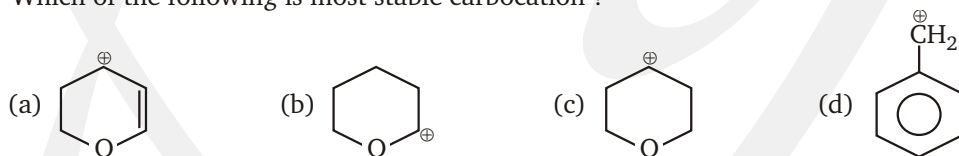
77. Which of the following carbocation is highly unstable?



78. Compare stability of the following carbocation?



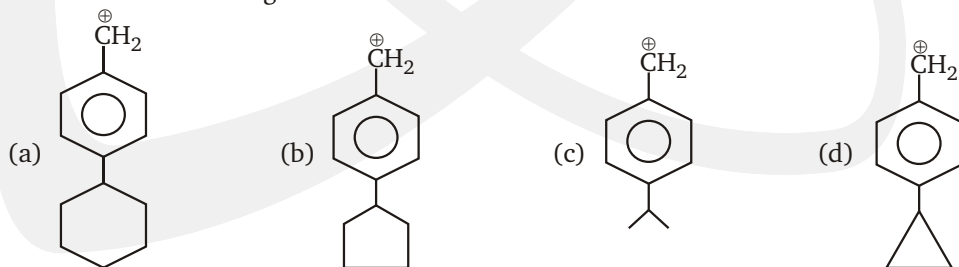
79. Which of the following is most stable carbocation?



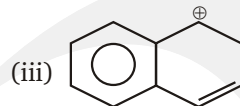
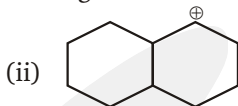
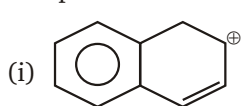
80. Which of the following carbocation is most stable?



81. Which of the following carbocation is most stable?



82. Compare relative stability of following carbocation :



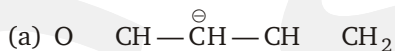
(a) $i > ii > iii$

(b) $iii > i > ii$

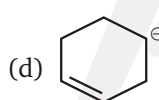
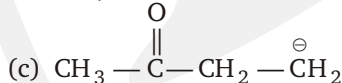
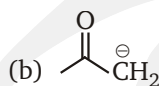
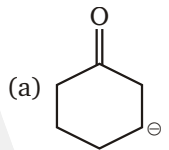
(c) $i > iii > ii$

(d) $iii > ii > i$

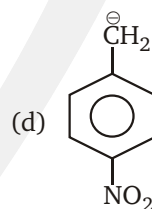
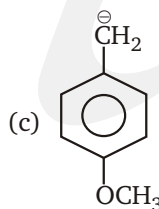
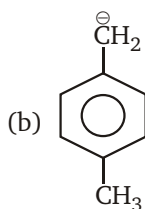
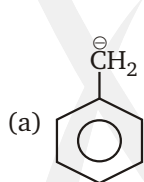
83. Most stable resonating structure of the given anion is:



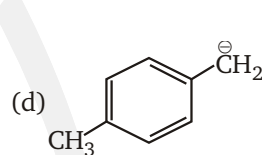
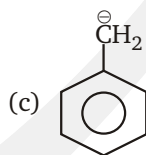
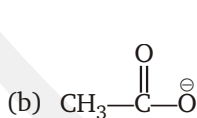
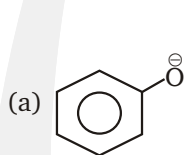
84. Which of the following carbanion is/are resonance stabilized ?



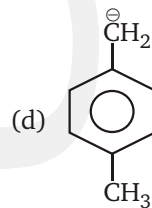
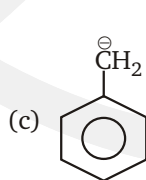
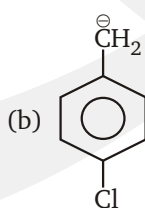
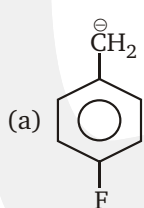
85. Which of the carbanion is most stable?



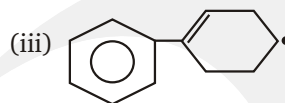
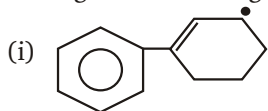
86. Which of the following is most stable carbanion ?



87. Which of the following carbanion is most stable ?



88. Arrange the following radical in their decreasing order of stability :



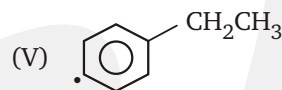
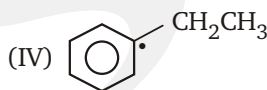
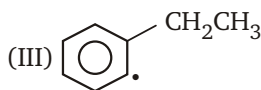
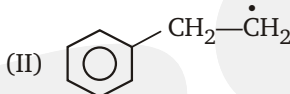
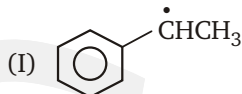
(a) ii > i > iii

(b) i > ii > iii

(c) i > iii > ii

(d) ii > iii > i

89. Which of the following is the most stable radical?



(a) I

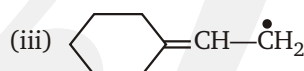
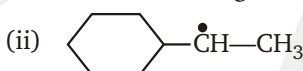
(b) II

(c) III

(d) IV

(e) V

90. Arrange the following radical in order of decreasing stability :



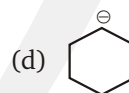
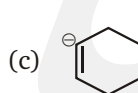
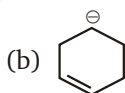
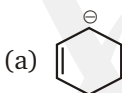
(a) ii > i > iii

(b) i > ii > iii

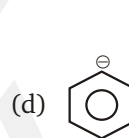
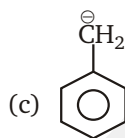
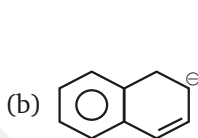
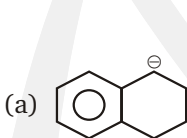
(c) iii > ii > i

(d) iii > i > ii

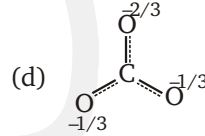
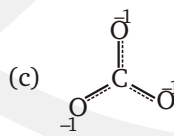
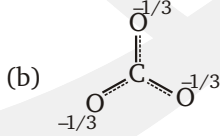
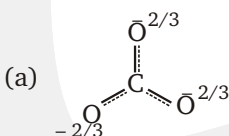
91. Which of the following carboanion is resonance stabilized ?



92. Which of the following carboanion is not resonance stabilized ?



93. Resonance hybrid of is:



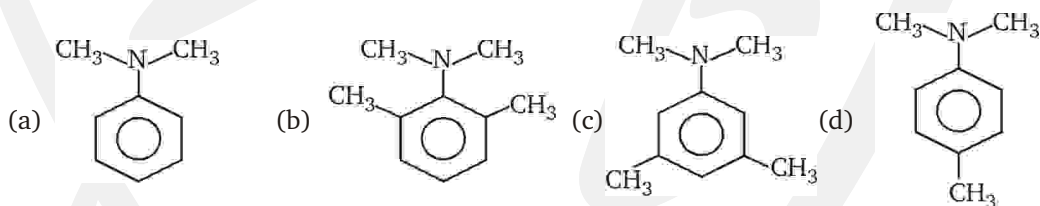
94. In which of the following carbanions electron will be delocalized in d -orbital?



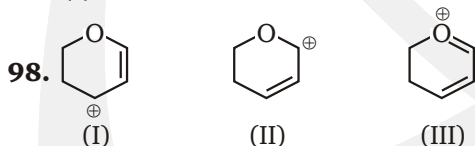
95. In which of the following compound Steric inhibition of resonance effect will operate?



96. In which of the following lone pair of nitrogen is not involved in resonance?



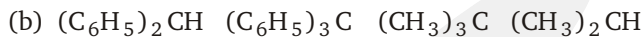
97. Most unstable resonating structure amongst those given below is :



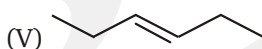
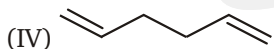
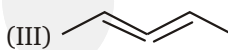
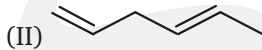
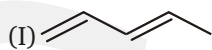
The most stable canonical structure among the given structure is:

- (a) I (b) II
(c) III (d) all are equally stable

99. The increasing order of stability of the following free radicals is:



100. Which of the following is a cumulated diene?



(a) I

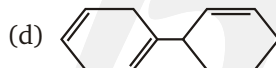
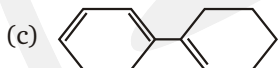
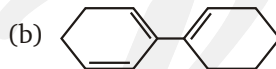
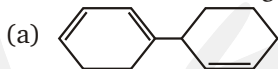
(b) II

(c) III

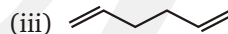
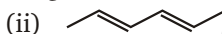
(d) IV

(e) V

101. Which of the following is most stable?



102. Arrange the following in decreasing order of stability :



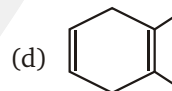
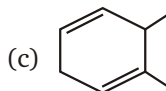
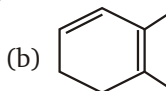
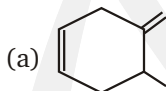
(a) $i > ii > iii$

(b) $ii > iii > i$

(c) $ii > i > iii$

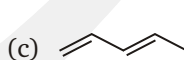
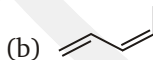
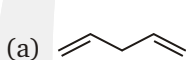
(d) $iii > ii > i$

103. Which of the following is most stable dienes?

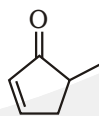


Hyper Conjugation

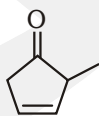
1. Which of the following is the most stable diene?



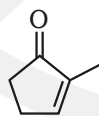
2. Which of the following is the most stable, and which is the next most stable isomer?



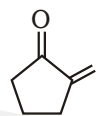
a



b



c



d

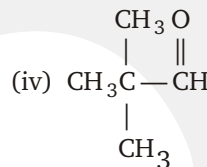
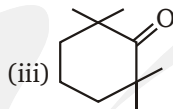
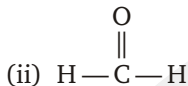
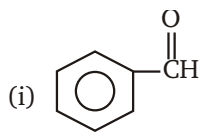
(a) c a

(b) a c

(c) d c

(d) b a

3. Which of the following compounds does not have an alpha hydrogen ?



(a) i

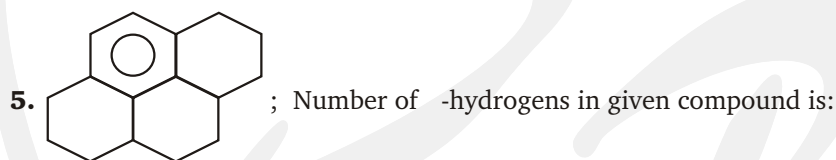
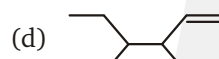
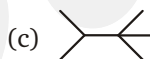
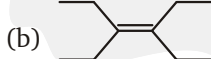
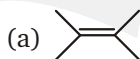
(b) ii

(c) iii

(d) iv

(e) all of the above

4. Which of the following has maximum number of α -hydrogen?



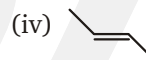
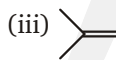
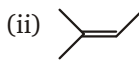
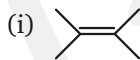
(a) 4

(b) 5

(c) 6

(d) 7

6. Sum of α -hydrogens in given alkene is:



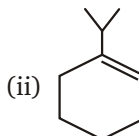
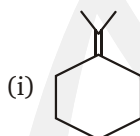
(a) 30

(b) 31

(c) 32

(d) 33

7. Sum of α -hydrogens in :



(i) (ii) ?

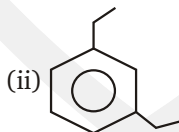
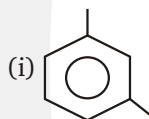
(a) 13

(b) 14

(c) 15

(d) 16

8. Sum of α -hydrogen in :



(i) (ii) ?

(a) 8

(b) 9

(c) 10

(d) 11

9. Sum of α -hydrogen in :

(i) 1-hexene ,

(ii) 2-hexene ,

(iii) 3-hexene

(i) (ii) (iii) ?

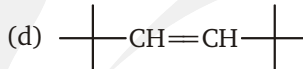
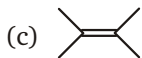
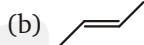
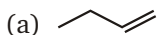
(a) 8

(b) 10

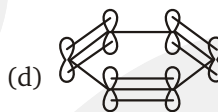
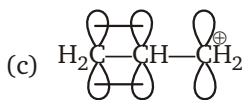
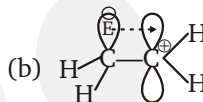
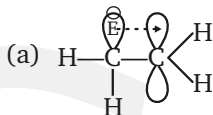
(c) 11

(d) 12

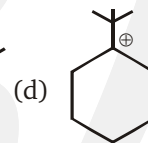
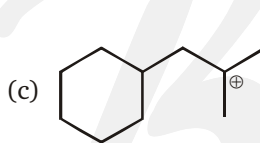
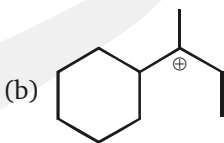
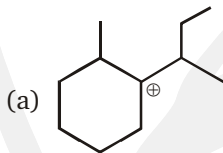
10. In which of the following alkene hyperconjugation does not take place?



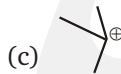
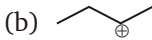
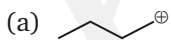
11. Which of the following represent hyperconjugation in carbocation?



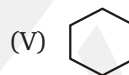
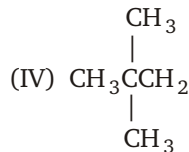
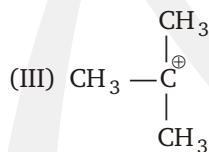
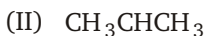
12. Which carbocation is most stable ?



13. Most stable structure of carbocation $C_4H_9^+$ is:



14. Which of the following carbocations is least stable?



(a) I

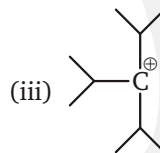
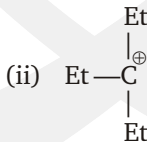
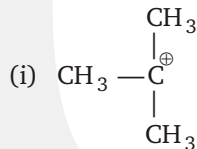
(b) II

(c) III

(d) IV

(e) V

15. Sum of β -hydrogens in given carbocation :



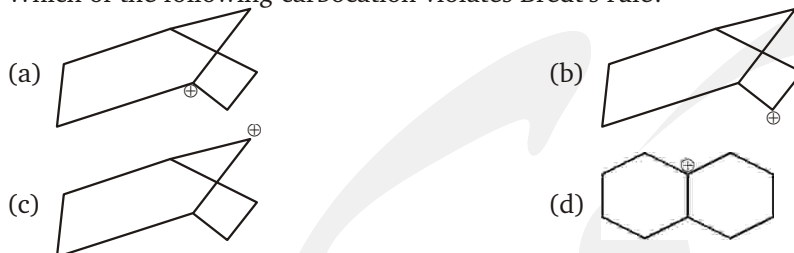
(a) 15

(b) 16

(c) 17

(d) 18

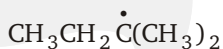
16. Which of the following carbocation violates Bredt's rule?



17. Rank the free radicals (I-III) shown below in order of decreasing stability (i.e., from most stable to least stable).



I



II



III

(a) I > III > II

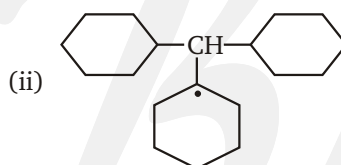
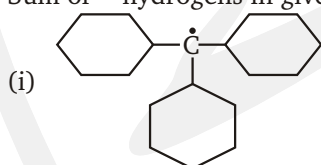
(b) II > III > I

(c) I > II > III

(d) II > I > III

(e) III > II > I

18. Sum of α -hydrogens in given radical:



(i) (ii) ?

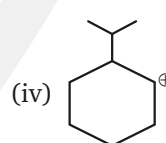
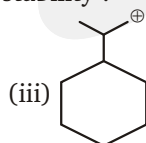
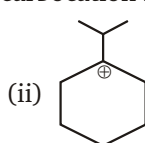
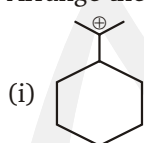
(a) 6

(b) 7

(c) 8

(d) 9

19. Arrange the following carbocation in order of stability :



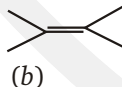
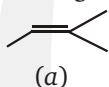
(a) i > ii > iii > iv

(b) i > ii > iv > iii

(c) ii > i > iv > iii

(d) ii > i > iii > iv

20. Arrange the following in decreasing order of number of α hydrogens:



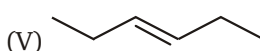
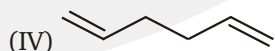
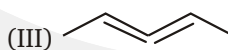
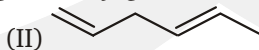
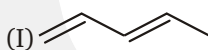
(a) a b c d

(b) b a c d

(c) c a b d

(d) d a b c

21. Which of the following is a conjugated diene ?



(a) I

(b) II

(c) III

(d) IV

(e) V

22. Which of the following is most stable alkene ?



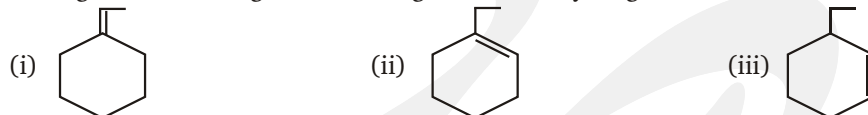
23. Which of the following is most stable alkene ?



24. Which of the following is least stable alkene ?



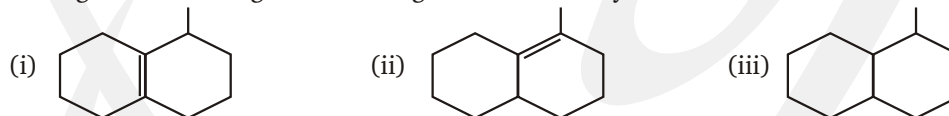
25. Arrange the following in decreasing order of $\Delta H_{\text{hydrogen}}$:



26. Which of the following is most stable?



27. Arrange the following in decreasing order of stability :



28. Which of the following is least stable alkene?



29. Which of the following alkene is more stable than 1-butene?



Heat of Hydrogenation, Heat of Combustion and Bond Length

1. Consider the bond dissociation energies listed below in kcal/mole.

$\text{CH}_3\text{—Br}$	70
$\text{CH}_3\text{CH}_2\text{—Br}$	68
$(\text{CH}_3)_2\text{CH—Br}$	68
$(\text{CH}_3)_3\text{C—Br}$	65

These data show that the carbon-bromine bond is weakest when bromine is bound to a

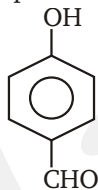
- (a) methyl carbon (b) primary carbon
(c) secondary carbon (d) tertiary carbon (e) quaternary carbon

2. On the basis of bond dissociation energies given below (in kcal/mole), estimate the H° for the propagation step

$(CH_3)_2CH \cdot$	Cl_2	$(CH_3)_2CHCl$	$Cl \cdot$
$CH_3CH_2CH_2 \cdot$			98
$(CH_3)_2CH \cdot$			95
$Cl-Cl$			58
$H-Cl$			103
$CH_3CH_2CH_2-Cl$			81
$(CH_3)_2CH-Cl$			80

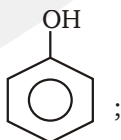
- (a) -30 kcal/mole (b) +22 kcal/mole
(c) -40 kcal/mole (d) +45 kcal/mole (e) -45 kcal/mole

3. Compare bond length in the given compounds across (carbon-oxygen):



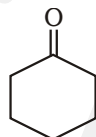
(a)

(a) a b c



(b)

(b) b c a



(c)

(c) b a c

(d) a c b

4. Compare bond length in the given compounds:



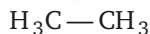
(a)

(a) a b c



(b)

(b) c a b

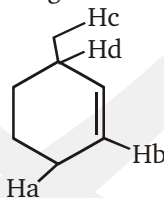


(c)

(c) c b a

(d) a c b

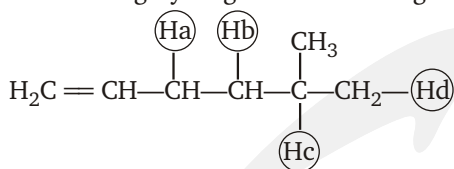
5. Arrange the proton shown, in decreasing order of their bond energy ?



- (a) $H_a > H_b > H_c > H_d$
(c) $H_b > H_c > H_d > H_a$

- (b) $H_b > H_c > H_a > H_d$
(d) $H_d > H_a > H_c > H_b$

6. Arrange the following hydrogen in decreasing order of bond energy :



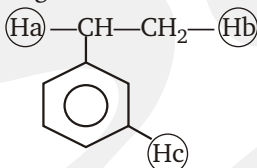
(a) Ha Hc Hb Hd

(b) Ha Hd Hb Hc

(c) Hd>Hc>Hb>Ha

(d) Hd>Hb>Hc>Ha

7. Arrange the following in decreasing order of bond energy :



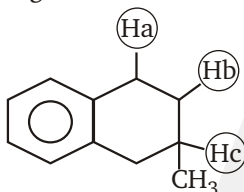
(a) Ha>Hb>Hc

(b) Hb>Ha>Hc

(c) Hc>Ha>Hb

(d) Hc>Hb>Ha

8. Arrange the following in decreasing order of bond energy :



(a) Ha>Hc>Hb

(b) Ha>Hb>Hc

(c) Hb>Ha>Hc

(d) Hb>Hc>Ha

9. Which of the following compounds contains the longest carbon-carbon single bond?

(a) allene

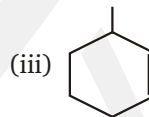
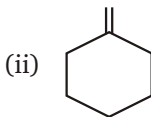
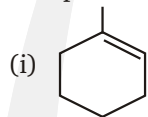
(b) 1,3-butadiyne

(c) 1,3-butadiene

(d) propyne

(e) ethane

10. Compare heat of hydrogenation of the following :



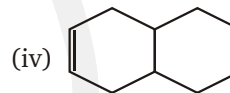
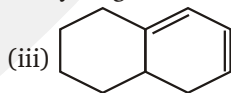
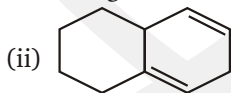
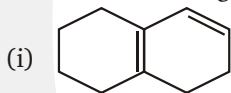
(a) i>ii>iii

(b) iii>ii>i

(c) ii>i>iii

(d) ii>iii>i

11. Rank the following in decreasing order of heat of hydrogenation :



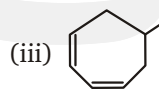
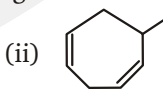
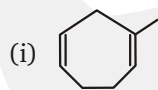
(a) iii>i>ii>iv

(b) i>iii>ii>iv

(c) iv>ii>iii>i

(d) iv>iii>ii>i

12. Rank the following in decreasing order of heat of hydrogenation :



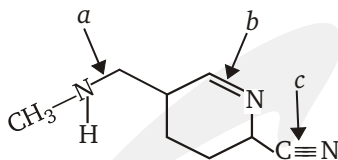
(a) i>ii>iii

(b) ii>iii>i

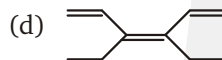
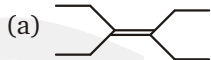
(c) i>iii>ii

(d) ii>i>iii

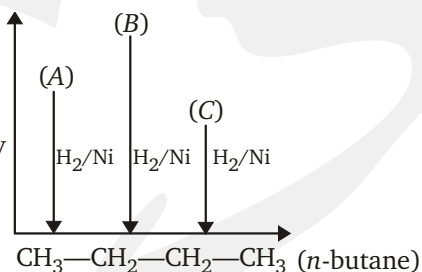
13. Rank the indicated bond in the given compound in order of decreasing bond strength:

(a) $a \ b \ c$ (b) $c \ b \ a$ (c) $b \ c \ a$ (d) $c \ a \ b$

14. Which of the following has minimum heat of hydrogenation?



15. Energy



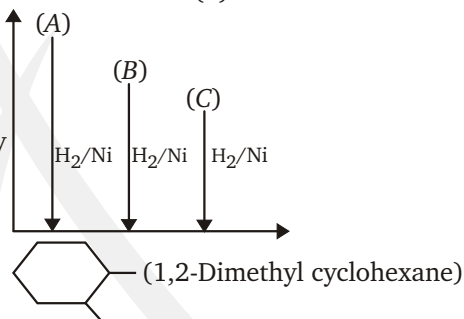
Alkene (C) is:

(a) *cis*-2-butene(b) *trans*-2-butene

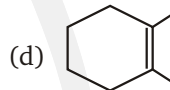
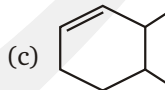
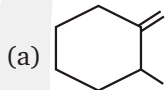
(c) 1-butene

(d) ethene

16. Energy



Structure of (A) is:



17. Which of the following has largest heat of combustion ?

(a) Ethyl cyclopentane

(b) Ethyl cyclohexane

(c) Ethyl cyclobutane

(d) Ethyl cycloheptane

18. Compare the heats of combustion for the compounds given:

(i) 2,3-dimethyl-2-butene

(ii) 2-methyl-2-butene

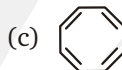
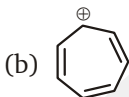
(iii) 1,3-butadiene

(iv) 1,2-butadiene

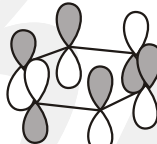
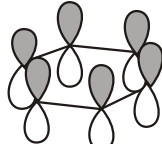
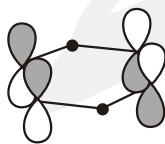
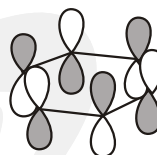
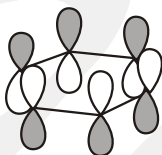
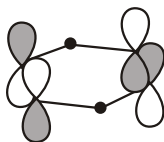
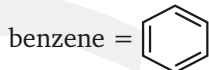
(a) $\text{iii} > \text{iv} > \text{ii} > \text{i}$ (b) $\text{iii} > \text{iv} > \text{i} > \text{ii}$ (c) $\text{i} > \text{ii} > \text{iv} > \text{iii}$ (d) $\text{i} > \text{ii} > \text{iii} > \text{iv}$

Aromaticity

1. Which of the following is antiaromatic?



2. Based upon your understanding of the basic principles of molecular orbitals it should be possible to answer the following question. Consider the following molecular orbitals of benzene. Which of these molecular orbitals has the lowest energy?



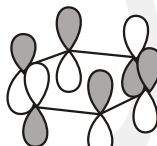
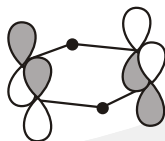
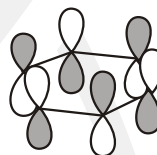
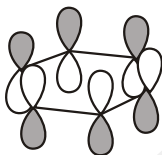
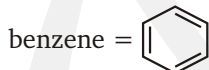
(a) orbital II

(b) orbital III

(c) orbital IV

(d) orbital V

3. Based upon your understanding of the basic principles of molecular orbitals it should be possible to answer the following question. Consider the following molecular orbitals of benzene. Which of these molecular orbitals has the highest energy?



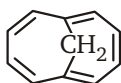
(a) orbital I

(b) orbital II

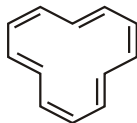
(c) orbital III

(d) orbital IV

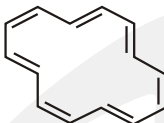
4. Which of the following structures are aromatic according to Hückel's rule?



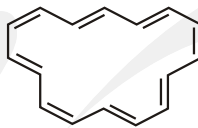
a



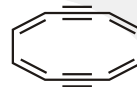
b



c



d



e

(a) a and c

(b) a, c, and d

(c) b and d

(d) b, d, and e

5. Which of the following cations is aromatic?



a



b



c



d

(a) b, d

(b) d

(c) a, b, d

(d) b, c, d

6. Benzene can be reduced to cyclohexane when it is treated with hydrogen in the presence of a nickel catalyst under conditions of high temperature and pressure. Predict the signs of H and S for this process.

(a) H 0, S 0

(b) H 0, S 0

(c) H 0, S 0

(d) H 0, S 0

(e) H 0, S 0

7.



Cyclononatetraenyl cation

(A)



Cyclononatetraenide anion

(B)

Number of π -electrons in (A) and (B) respectively are :

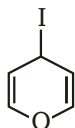
(a) 8, 8

(b) 8, 10

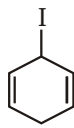
(c) 10, 8

(d) 10, 10

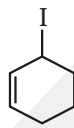
8. Compare the stability of carbocation which are formed when react with AgNO_3 ?



(A)



(B)



(C)

(a) A B C

(b) A C B

(c) B C A

(d) B A C

9. The order of stability of the following carbanion is:



(I)



(II)



(III)



(IV)

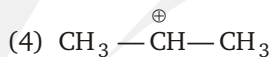
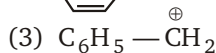
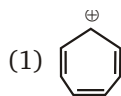
(a) I II III IV

(b) I III II IV

(c) IV III II I

(d) III IV I II

10. Decreasing order of stability of given carbocations is as:



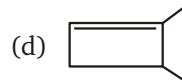
(a) 3 2 4 1

(b) 1 3 4 2

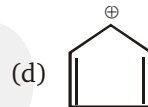
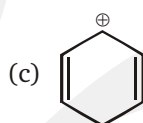
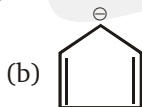
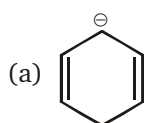
(c) 1 3 2 4

(d) 3 2 1 4

11. Which of the following is least stable dienes?



12. Which of the following is aromatic compound ?

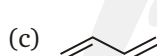
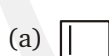


13. Which of the following is Aromatic ?



(d) All of these

14. Which of the following is non-Aromatic ?

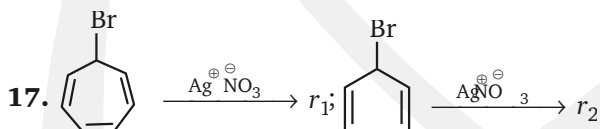


(d) All of these

15. + 2 Na ; Product of the reaction is:



16. + 2 Fe³⁺ ; Product of the reaction is:



Rate of reaction r_1 and r_2 is:

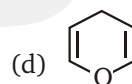
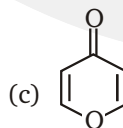
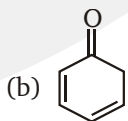
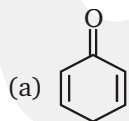
(a) $r_1 > r_2$

(b) $r_2 > r_1$

(c) $r_1 > r_2$

(d) $r_2 > r_1$

18. Which of the following is aromatic ?



19. Which of the following is Anti-aromatic ?

- (a)  (b)  (c)  (d) All of these

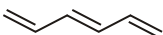
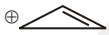
20. Which of the following statements is incorrect about benzene?

- (a) All of the carbon atoms are sp hybridized
 (b) It has delocalized electrons
 (c) The carbon-carbon bond lengths are all the same
 (d) The carbon-hydrogen bond lengths are all the same
 (e) All twelve atoms lie in the same plane

21. Aromatic molecules contain.....-electrons.

- (a) no (b) $4n + 2$ (with n an integer)
 (c) $4n + 2$ (where $n = 0.5$) (d) $4n$ (with n an integer) (e) unpaired

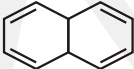
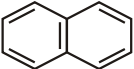
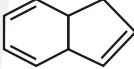
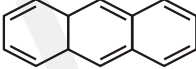
22. Which of the following is aromatic ?

- (i)  (ii) 
 (iii)  (iv)  (v) 
 (a) i (b) ii (c) iii (d) iv (e) v

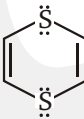
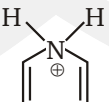
23. Which of the structures below would be aromatic ?

- (i)  (ii)  (iii)  (iv) 
 (a) i and iv (b) i, iii and iv (c) iii and iv (d) ii

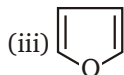
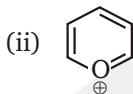
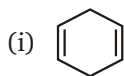
24. Which of the following is aromatic ?

- (i)  (ii) 
 (iii)  (iv)  (v) 
 (a) i (b) ii (c) iii (d) iv (e) v

25. Which of the following is aromatic ?

- (i)  (ii) 
 (iii)  (iv)  (v) 
 (a) i (b) ii (c) iii (d) iv (e) v

26. Which of the following is antiaromatic ?



(a) iii

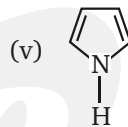
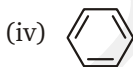
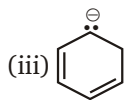
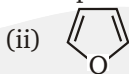
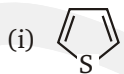
(b) i

(c) ii

(d) v

(e) iv

27. Which of the following aromatic compounds is the most stable ?



(a) i

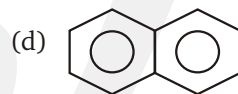
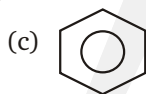
(b) ii

(c) iii

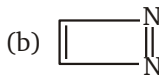
(d) iv

(e) v

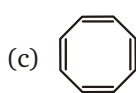
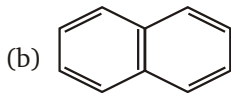
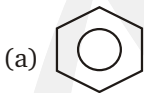
28. Which of the following compound will undergo dimerization ?



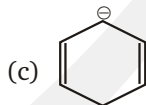
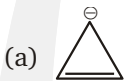
29. Which of the following compound will not undergo dimerization ?



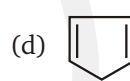
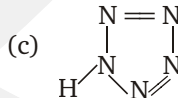
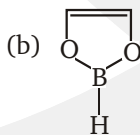
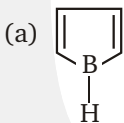
30. Which of the following is non-planar compound ?



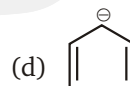
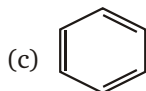
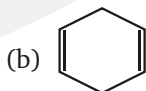
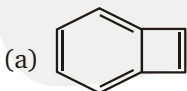
31. Which of the following carbanion is aromatic ?



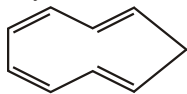
32. Which of the following is anti-aromatic ?



33. Which of the following is non-aromatic ?

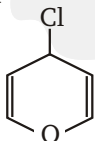


34. To convert cyclononatetraene in aromatic ion, the process needed is :



- (a) addition of one more π -bond
- (b) addition of two more π -bond
- (c) loss of H^+ from sp^3 -hybridized carbon
- (d) loss of H^+ from sp^2 -hybridized carbon

35. To convert given compound into aromatic compound, best reagent is:



- (a) $AlCl_3$
- (b) HCl
- (c) H_2SO_4
- (d) Na

36. To convert given compound into aromatic best reagent is:



- (a) H^-
- (b) H^+
- (c) $NaOH$
- (d) $NaHCO_3$

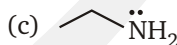
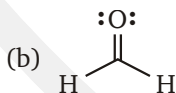
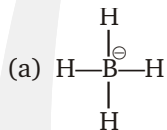
37. To convert given compound into aromatic best reagent is:



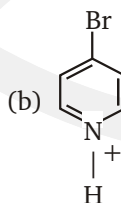
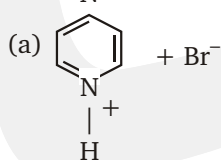
- (a) HCl
- (b) $NaOH$
- (c) NaH
- (d) $NaHCO_3$

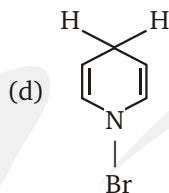
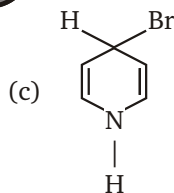
Acidic and Basic Strength

1. Which of the following compounds can behave as an acid as well as a base?



2. What are the product(s) of the following acid-base reaction?





3. Using pK_a values which is the strongest acid (HA is any acid).

pK_a	HA
3.7	HCOOH
4.2	C_6H_5COOH
4.75	CH_3COOH
15.5	CH_3OH

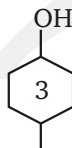
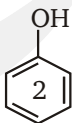
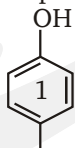
(a) Acetic acid

(b) Benzoic acid

(c) Methanol

(d) Formic acid

4. Rank the following compounds in order from most basic to least basic.



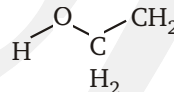
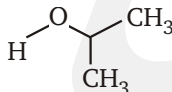
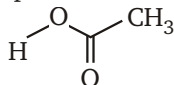
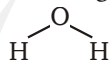
(a) 2 3 4 1

(b) 4 3 1 2

(c) 3 1 2 4

(d) 1 4 3 2

5. Rank the following compounds in order from most acidic to least acidic.



(a)

(b)

(c)

(d)

(a) c d a b

(c) b a d c

(b) d a b c

(d) a b d c

6. Which is the strongest base?

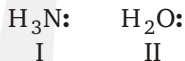
(a) LiOH

(b) CH_3Li

(c) $LiNH_2$

(d) LiF

7. Consider ammonia (I) and water (II). Which of the following statements is true?



(a) I is more basic and more nucleophilic than II

(b) I is less basic and less nucleophilic than II

(c) I is more basic but less nucleophilic than II

(d) I is less basic but more nucleophilic than II

8. Which of the following is the most acidic?

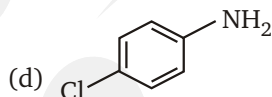
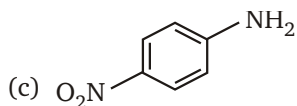
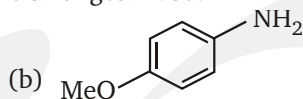
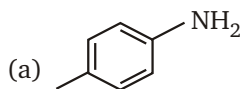
(a) 1-butyne

(b) 2-butyne

(c) 1-butene

(d) butane

9. Which of the following aniline derivatives is the strongest base?



10. Which of the following compounds is most basic?

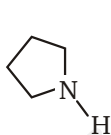
(a) cyclohexylamine

(b) aniline

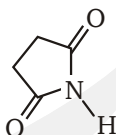
(c) *p*-methoxyaniline

(d) *p*-nitroaniline

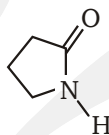
11. Which of the following compounds has the most basic nitrogen?



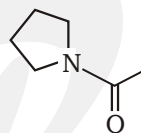
a



b



c



d

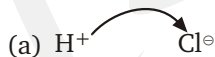
(a) a

(b) b

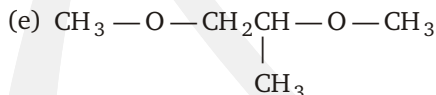
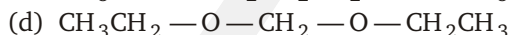
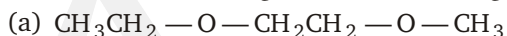
(c) c

(d) d

12. Which of the following is correct representation of curved arrow ?



13. Which of the following would have the highest boiling point?



14. An alcohol, an aldehyde and a carboxylic acid of comparable mass will have their boiling points in the order:

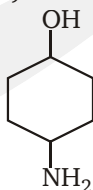
(a) alcohol < carboxylic acid < aldehyde

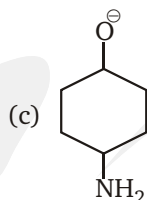
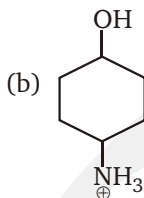
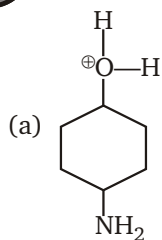
(b) aldehyde < alcohol < carboxylic acid

(c) alcohol < aldehyde < carboxylic acid

(d) carboxylic acid < aldehyde < alcohol

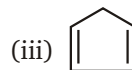
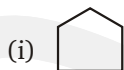
15. What is the conjugate acid of (A)?





(d) both (a) and (b)

16. Rank the following in decreasing order of acidity :



(a) $\text{iii} > \text{i} > \text{ii}$

(b) $\text{i} > \text{ii} > \text{iii}$

(c) $\text{ii} > \text{i} > \text{iii}$

(d) $\text{iii} > \text{ii} > \text{i}$

17. Compare acidic strength :



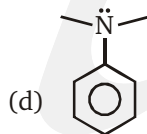
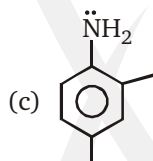
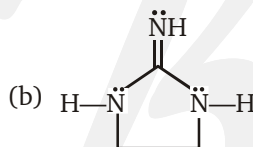
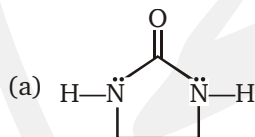
(a) $\text{i} > \text{ii} > \text{iii}$

(b) $\text{ii} > \text{i} > \text{iii}$

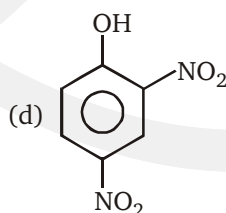
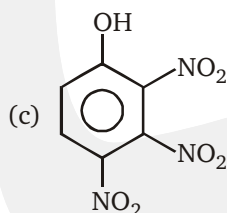
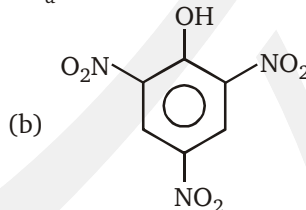
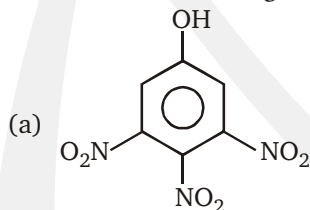
(c) $\text{iii} > \text{i} > \text{ii}$

(d) $\text{ii} > \text{iii} > \text{i}$

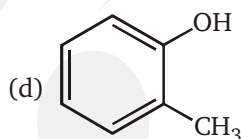
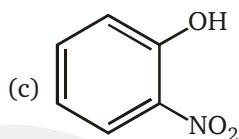
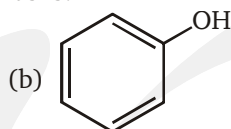
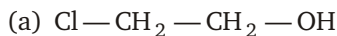
18. Which of the following is strongest base ?



19. Which of the following compound have highest K_a value ?



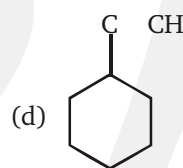
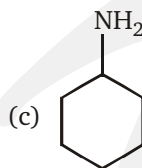
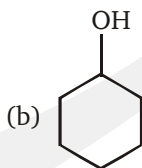
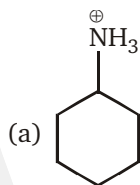
20. The most acidic among the following compounds is:



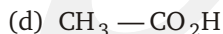
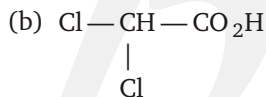
21. The strongest bronsted base among the following compounds is:



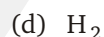
22. Which of the following is lowest pK_a value ?



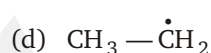
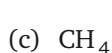
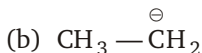
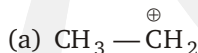
23. Which of following is lowest pK_a value ?



24. Conjugate base of $\text{HO}^\ominus$



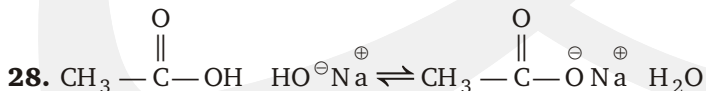
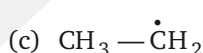
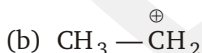
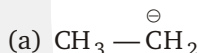
25. Conjugate base of CH_3-CH_3 is:



26. Conjugate acid of H_2O is:

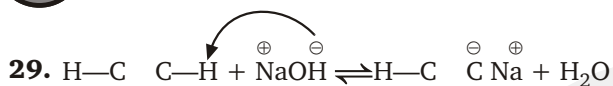


27. Conjugate acid of $\text{H}_2\text{C}=\text{CH}_2$ is:



in the above reaction:





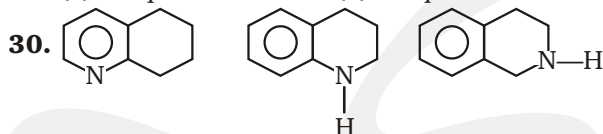
For the above reaction:

(a) $K_{\text{eq}} > 1$

(b) $K_{\text{eq}} < 1$

(c) $K_{\text{eq}} = 1$

(d) $K_{\text{eq}} = 0$



(a)

(b)

(c)

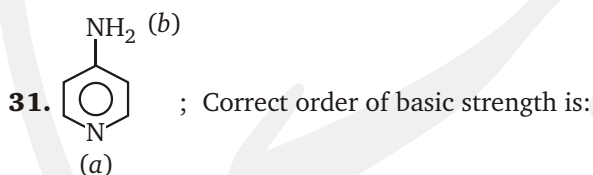
Compare basic strength:

(a) a b c

(b) b a c

(c) c a b

(d) c b a

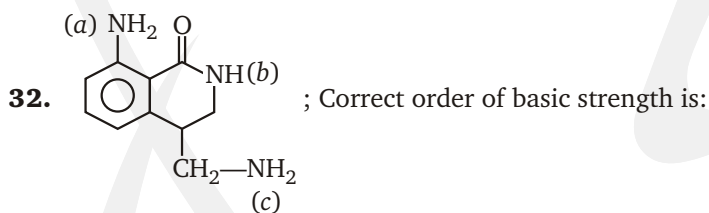


(a) a b

(b) b a

(c) a b

(d) b a

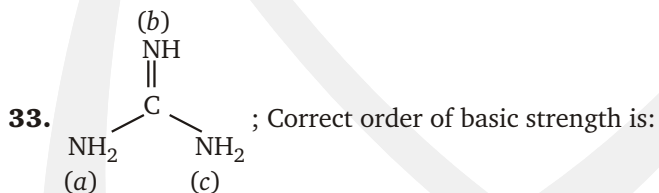


(a) a b c

(b) b a c

(c) c a b

(d) c b a

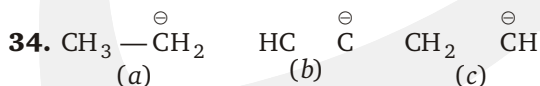


(a) b a c

(b) a c b

(c) b a c

(d) c a b



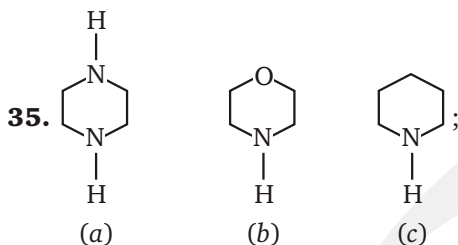
Compare basic strength:

(a) b c a

(b) c b a

(c) a b c

(d) a c b



Compare basic strength

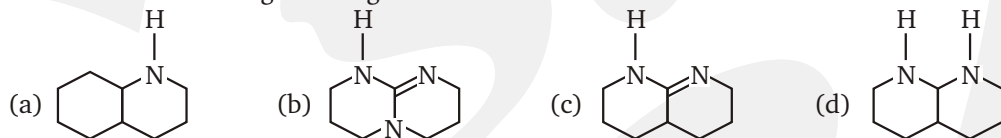
(a) $a > b > c$

(b) $c > a > b$

(c) $b > a > c$

(d) $c > b > a$

36. Which of the following is strongest base ?



37. Correct order of basic strength is: $\text{CH}_3 - \text{C} \equiv \text{N}$ (a)

$\text{CH}_3 - \text{CH} = \text{NH}$ (b)

$\text{CH}_3 - \text{CH}_2 - \text{NH}_2$ (c)

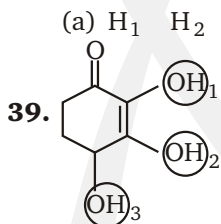
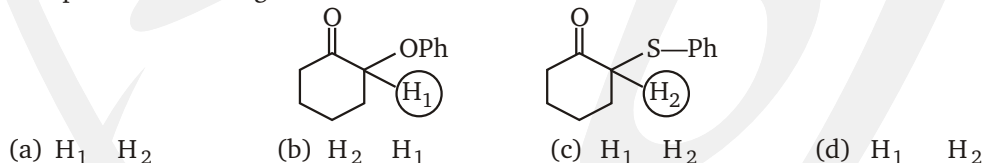
(a) $a > b > c$

(b) $c > b > a$

(c) $b > a > c$

(d) $a > b > c$

38. Compare acidic strength of H-atoms marked.



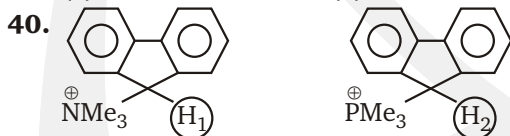
Compare acidic strength of hydroxylic H-atoms marked.

(a) $1 > 2 > 3$

(b) $2 > 1 > 3$

(c) $3 > 2 > 1$

(d) $3 > 1 > 2$



Compare acidic strength of H-atoms marked.

(a) $H_1 > H_2$

(b) $H_2 > H_1$

(c) $H_1 > H_2$

(d) Cannot compare

41. $\text{CH}_3 - \text{CO}_2\text{H}$; $\text{CH}_3 - \text{OH}$; $\text{CH}_3 - \text{C} \equiv \text{CH}$; $\text{CH}_3 - \text{NH}_2$
 (a) (b) (c) (d)

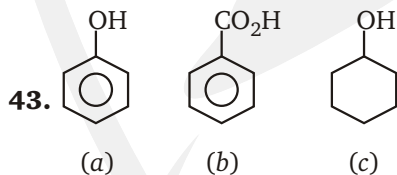
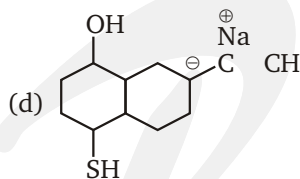
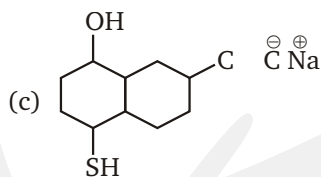
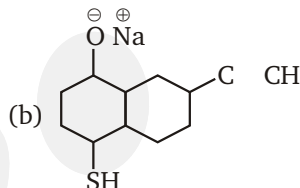
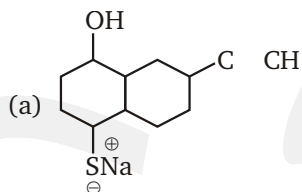
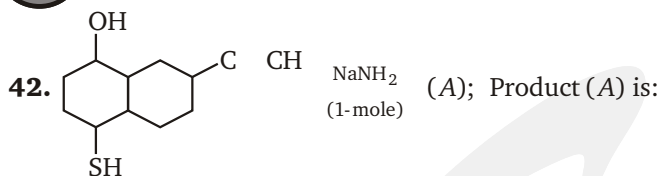
Compare acidic strength of compounds a d :

(a) $a > b > c > d$

(b) $b > c > a > d$

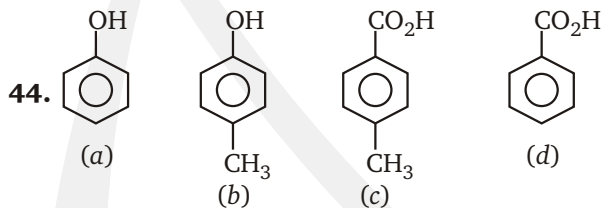
(c) $b > a > c > d$

(d) $a > c > b > d$



Correct order of acidic strength of the compounds given is :

- (a) a b c (b) b c a (c) b a c (d) a c b



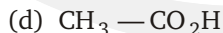
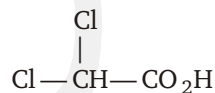
Correct order of acidic strength of the compounds given is :

- (a) a c d b (b) d a c b (c) d c a b (d) d c b a

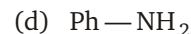
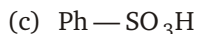
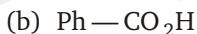
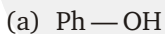
45. Which of the following has highest pK_a value ?



(b)



46. Which of the following has lowest pK_a value ?



47. Which of the following contains least acidic hydrogen ?

- (a) $R-\overset{\oplus}{N}H_3$ (b) $R-CO_2H$ (c) $R-OH$ (d) $R-SO_3H$

48. Which of the following has highest pK_a value ?

- (a) CH_3-SO_3H (b) $CH_3-\overset{\overset{O}{\parallel}}{C}-OH$ (c) CH_3-OH (d) H_2O

49. Which of the following ions is most acidic?

- (a) $C_6H_5NH_3^+$ (b) $C_6H_5CH_2NH_3^+$
(c) $(CH_3)_2CHNH_3^+$ (d) $CH_3CH_2NH_3^+$ (e) $CH_3NH_3^+$

50. Which of the following is the weakest base?

- (a) Phenolate ion ($C_6H_5O^-$) (b) Ethoxide ion ($CH_3CH_2O^-$)
(c) Hydroxide ion (HO^-) (d) Acetate ion ($CH_3CO_2^-$) (e) Methoxide ion (CH_3O^-)

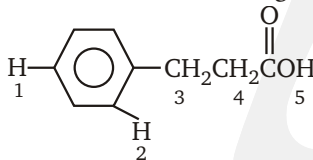
51. Which of the following is strongest acid?

- (a) CH_3NH_2 (b) CH_3OH (c) CH_3SH (d) CH_3OCH_3 (e) CH_3Cl

52. Which of the following is strongest base?

- (a) $HOMgBr$ (b) H_2O (c) CH_4 (d) CH_3OH (e) CH_3MgBr

53. Which of the labeled hydrogen atoms in the following structure is the most acidic ?

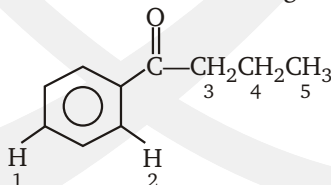


- (a) 1 (b) 2 (c) 3 (d) 4 (e) 5

54. Which of the following underlined alpha hydrogen are the most acidic ?

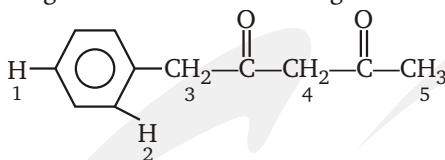
- (a) $CH_3-\overset{\overset{O}{\parallel}}{C}-\underline{CH_3}$ (b) $CH_3-\overset{\overset{O}{\parallel}}{C}-\underline{CH_2}-\overset{\overset{O}{\parallel}}{C}-CH_3$
(c) $CH_3-\overset{\overset{O}{\parallel}}{C}-\underline{CH_2}-\overset{\overset{O}{\parallel}}{C}-OCH_3$ (d) $\underline{CH_3}-\overset{\overset{O}{\parallel}}{C}-CH_3$ (e) $\underline{CH_3}-\overset{\overset{O}{\parallel}}{C}-O-CH_3$

55. Which of the labeled hydrogen atoms in the following structure is the most acidic ?



- (a) 1 (b) 2 (c) 3 (d) 4 (e) 5

56. Which of the labeled hydrogen atom in the following structure is the most acidic ?



- (a) 1 (b) 2 (c) 3 (d) 4 (e) 5

57. Rank the following compound in order of increasing acidic strength :

- (i) H_3O^+ (ii) H_2O (iii) HO^-

- (a) $i < ii < iii$ (b) $ii < i < iii$ (c) $iii < ii < i$ (d) $iii < i < ii$

58. Rank the following compound in order of increasing acidic strength :

- (i) NH_3 (ii) H_2O (iii) H_2S

- (a) $i < ii < iii$ (b) $ii < i < iii$ (c) $i < iii < ii$ (d) $iii < ii < i$

59. Rank the following compound in order of increasing acidic strength :

- (i) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{OH}$ (ii) $\text{Cl} - \text{CH}_2 - \text{CH}_2 - \text{OH}$ (iii) $\text{CH}_3 - \text{CH}_2 - \text{OH}$

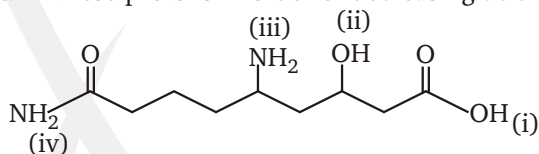
- (a) $i < ii < iii$ (b) $i < iii < ii$ (c) $ii < i < iii$ (d) $ii < iii < i$

60. Rank the following compound in order of increasing acidic strength :

- (i) CH_3OH (ii) $\text{CH}_3 - \text{NH}_2$ (iii) $\text{CH}_3 - \text{CH}_3$

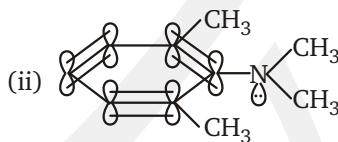
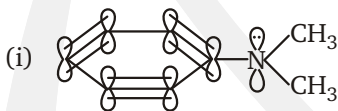
- (a) $i < ii < iii$ (b) $iii < ii < i$ (c) $i < iii < ii$ (d) $ii < i < iii$

61. Rank these protons in order of decreasing acidic strength :



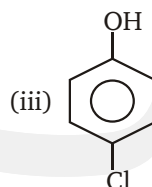
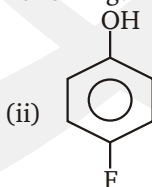
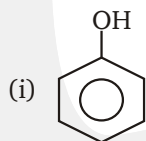
- (a) $iv > i > iii > ii$ (b) $i > ii > iv > iii$ (c) $i > iv > ii > iii$ (d) $i > iv > iii > ii$

62. Compare basic strength of the following:



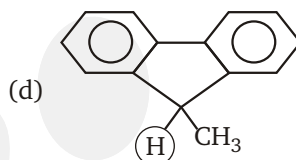
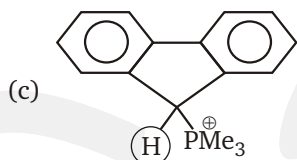
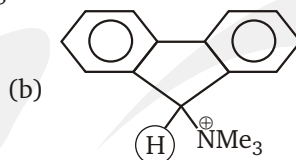
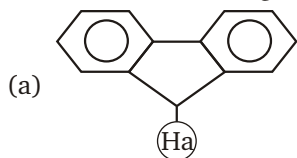
- (a) $i > ii$ (b) $i = ii$ (c) $ii > i$ (d) $i >> ii$

63. Compare acidic strength of the following:

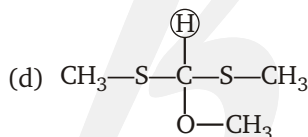
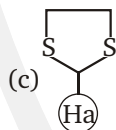
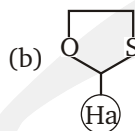
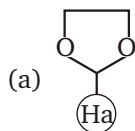


- (a) $i > ii > iii$ (b) $ii > iii > i$ (c) $iii > ii > i$ (d) $iii > i > ii$

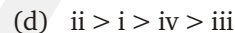
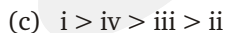
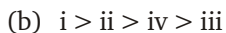
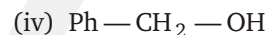
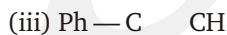
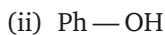
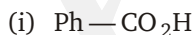
64. Which of the following has most acidic hydrogen?



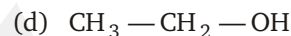
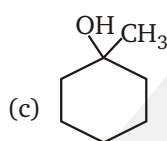
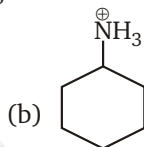
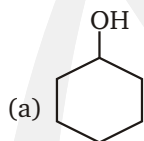
65. Which of the following has least acidic hydrogen?



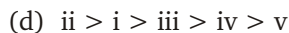
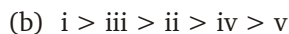
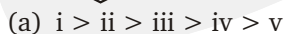
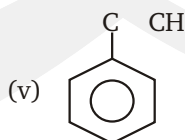
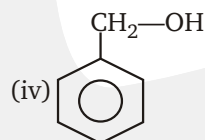
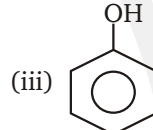
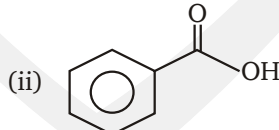
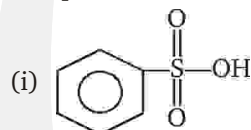
66. Compare acidic strength :



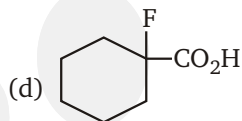
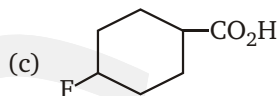
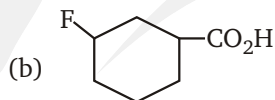
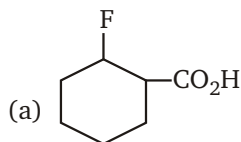
67. Which of the following is most acidic?



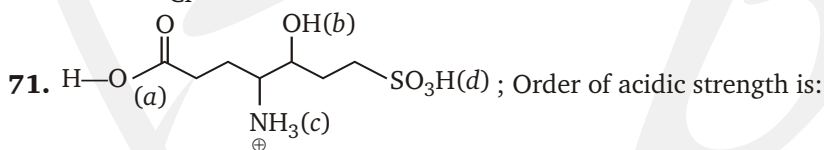
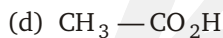
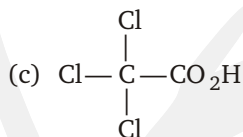
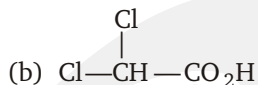
68. Compare acidic strength :



69. Which of the following has highest pK_a value?



70. Which of the following has highest K_a value ?



Order of acidic strength is:

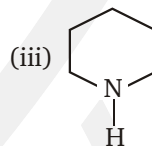
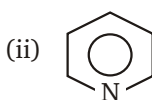
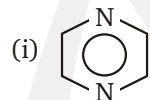
(a) $d > c > b > a$

(b) $d > a > b > c$

(c) $d > a > c > b$

(d) $a > d > c > b$

72. Compare basic strength :



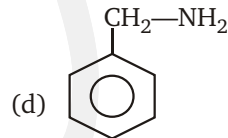
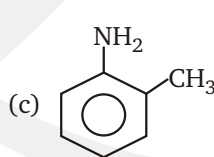
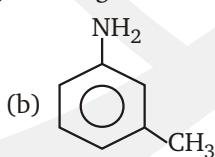
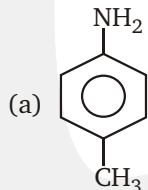
(a) $i > ii > iii$

(b) $i > iii > ii$

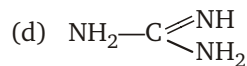
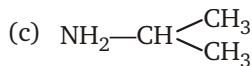
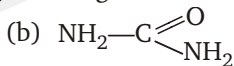
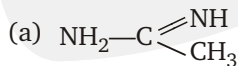
(c) $iii > i > ii$

(d) $iii > ii > i$

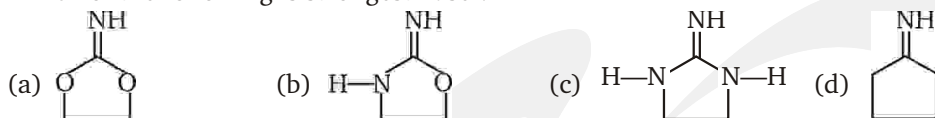
73. Which of the following is strongest base?



74. Which of the following is strong base?



75. Which of the following is strongest base ?



76. Which of the following is weakest base?



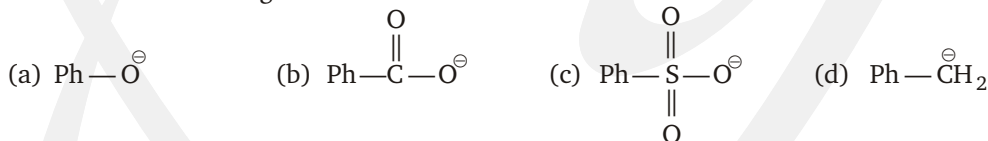
77. Which of the following compound is virtually non-basic?

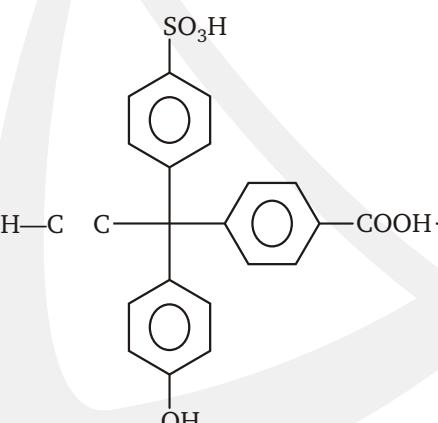


78. Which of the following is weakest base?



79. Which of the following is weakest base?

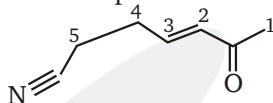


80.  $\xrightarrow{(x) \text{ NaHCO}_3}$; Total number of moles of NaHCO_3 consumed during given reaction will be (x).

So the value of x will be :

- (a) 1 (b) 2 (c) 3 (d) 4

81. Which of the following carbon will be deprotonated first on treatment with base?

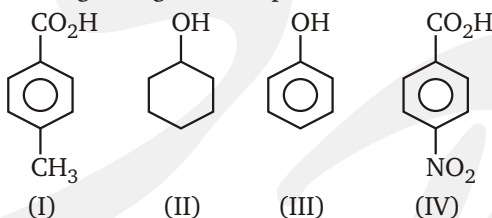


- (a) 4 (b) 5 (c) 2 (d) 3

82. When benzene sulfonic acid and *p*-nitrophenol are treated with NaHCO_3 , the gases released respectively are:

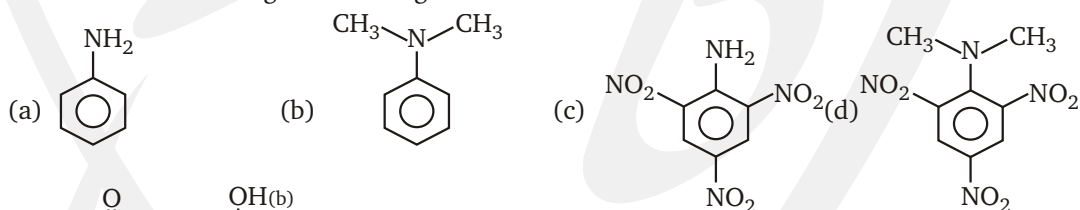
- (a) SO_2, NO_2 (b) SO_2, NO (c) SO_2, CO_2 (d) CO_2, CO_2

83. Correct order of acidic strength of given compounds will be:



- (a) IV II I III (b) IV III I II (c) IV I II III (d) IV I III II

84. Which of the following is the strongest base?



85. $\text{H}-\text{O}-\text{C}(=\text{O})-\text{CH}_2-\text{CH}_2-\text{CH}(\text{NH}_3^+)-\text{CH}(\text{OH})-\text{CH}_2-\text{CH}_2-\text{SO}_3\text{H}$; Order of acidic strengths is:

- (a) d c b a (b) d a b c (c) d a c b (d) a d c b

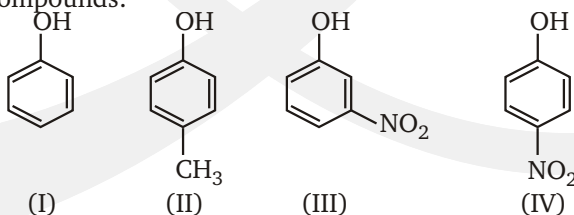
86. The strongest Bronsted base among the following compounds is:

- (a) NaCH_3 (b) NaOH (c) NaF (d) NaNH_2

87. Oxygen atom of the carbonyl group is most basic in:

- (a) Amide (b) Ester (c) Acid chloride

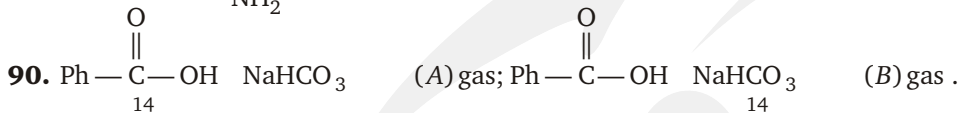
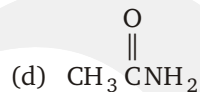
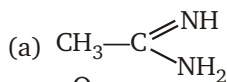
88. In the following compounds:



the order of acidity is:

- (a) III IV I II (b) I IV III II (c) II I III IV (d) IV III I II

89. Most basic compound is:



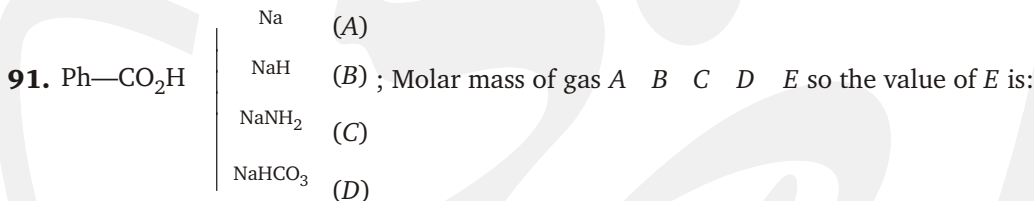
Difference between the molar mass of A and B gas is:

(a) 1

(b) 2

(c) 3

(d) 4



(a) 19

(b) 63

(c) 64

(d) 65

92. Which one among the following is the least basic?



Electrophile & Nucleophile and Type of Reaction & Solvent

1. $\text{C}_6\text{H}_{13}\text{Br}$ OH^- $\text{C}_6\text{H}_{13}\text{OH}$ Br^- . This reaction is an example of :

(a) nucleophilic addition

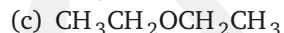
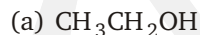
(b) nucleophilic substitution

(c) electrophilic addition

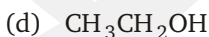
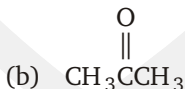
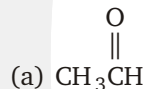
(d) electrophilic substitution

(e) Free radical substitution

2. Which of the following solvents is aprotic ?



3. Which of the following solvents is protic ?



4. Protic and aprotic solvents are very similar as solvents except for their :

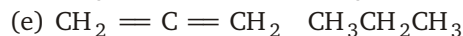
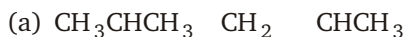
(a) polarity

(b) dielectric constant

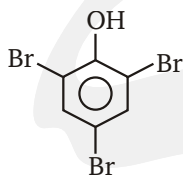
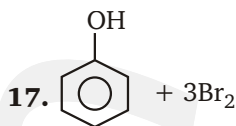
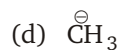
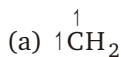
(c) ability to stabilize anions by hydrogen bonding

(d) ability to stabilize cations by hydrogen bonding

(e) ability to stabilize cations with unshared pairs of electrons



16. Which of the following is singlet carbene?



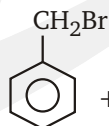
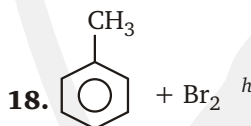
+ 3HBr This reaction is an example of :

(a) nucleophilic addition

(c) electrophilic addition

(b) nucleophilic substitution

(d) electrophilic substitution



+ HBr This reaction is an example of :

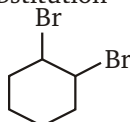
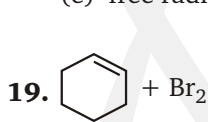
(a) nucleophilic addition

(c) electrophilic addition

(e) free radical substitution

(b) nucleophilic substitution

(d) electrophilic substitution



This reaction is an example of :

(a) nucleophilic addition

(c) electrophilic addition

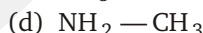
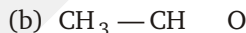
(e) free radical substitution

(b) nucleophilic substitution

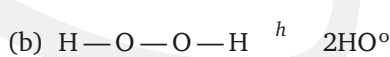
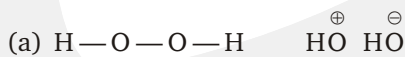
(d) electrophilic substitution

Miscellaneous Problems

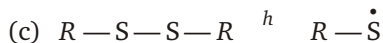
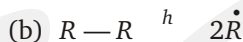
1. Which of the following compounds easily undergo homolytic bond fission ?



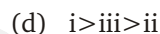
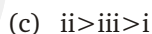
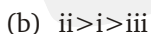
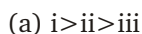
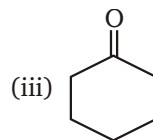
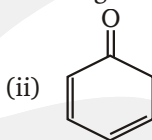
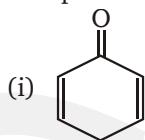
2. Which of the following cleavage is homolytic cleavage ?



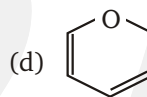
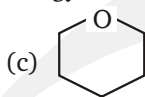
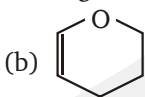
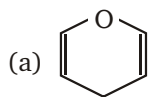
3. Which of the following cleavage is heterolytic cleavage?



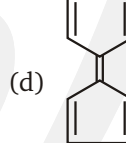
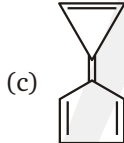
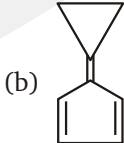
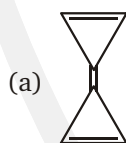
4. Compare resonance energy of following :



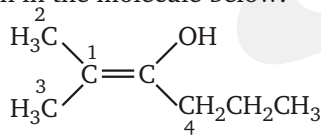
5. Which of the following has highest resonance energy?



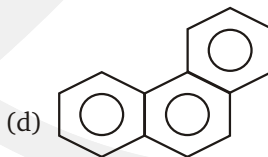
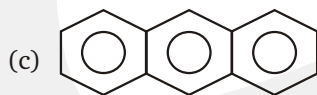
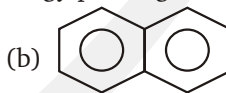
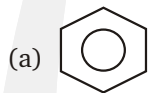
6. Which of the following compound has highest dipole moment ?

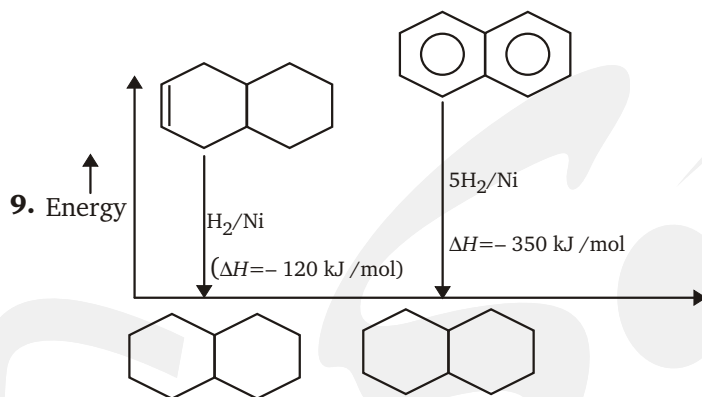


7. The most electron-rich carbon in the molecule below:



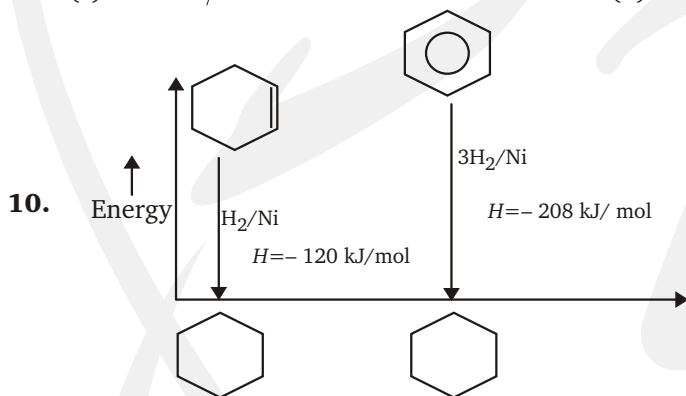
8. Which of the following has lowest resonance energy per ring ?





Resonance energy of naphthalene is:

- (a) -150 kJ/mol (b) 250 kJ/mol
(c) -350 kJ/mol (d) -450 kJ/mol



Resonance energy of benzene is:

- (a) -152 kJ/mol (b) -252 kJ/mol (c) -92 kJ/mol (d) -42 kJ/mol

Answers

[A] Classification and Nomenclature of Organic Compounds

Functional Groups Degree of Unsaturation and Double Bond-equivalent

1.	(c)	2.	(d)	3.	(c)	4.	(d)	5.	(d)	6.	(d)	7.	(d)	8.	(b)	9.	(c)	10.	(c)
11.	(d)	12.	(d)	13.	(b)	14.	(c)	15.	(b)	16.	(a)	17.	(c)	18.	(b)	19.	(c)	20.	(c)
21.	(a)	22.	(b)	23.	(d)	24.	(a)	25.	(b)	26.	(c)	27.	(d)	28.	(d)	29.	(b)	30.	(d)
31.	(c)	32.	(b)	33.	(c)	34.	(c)												

Hybridization and Bonding in Organic Molecules

1.	(c)	2.	(c)	3.	(c)	4.	(c)	5.	(c)	6.	(c)	7.	(b)	8.	(c)	9.	(e)	10.	(d)
11.	(d)	12.	(c)	13.	(c)	14.	(c)												

IUPAC Nomenclature of Alkanes

1.	(c)	2.	(c)	3.	(c)	4.	(a)	5.	(c)	6.	(b)	7.	(b)	8.	(d)	9.	(c)	10.	(b)
11.	(b)	12.	(a)	13.	(b)	14.	(a)	15.	(a)										

IUPAC Nomenclature of Alkynes/Alkenes

1.	(c)	2.	(a)	3.	(a)	4.	(b)	5.	(e)	6.	(d)	7.	(b)	8.	(a)	9.	(d)	10.	(a)
----	-----	----	-----	----	-----	----	-----	----	-----	----	-----	----	-----	----	-----	----	-----	-----	-----

IUPAC Nomenclature of Alcohols/Aldehydes/Ketones/Acids

1.	(e)	2.	(a)	3.	(c)	4.	(c)	5.	(d)	6.	(b)	7.	(d)	8.	(d)	9.	(d)	10.	(c)
11.	(d)	12.	(d)																

IUPAC Nomenclature of Acid Halides, Esters, Nitriles, Amines, Amides, Anhydrides

1.	(d)	2.	(c)	3.	(b)	4.	(e)	5.	(b)	6.	(d)	7.	(d)						
----	-----	----	-----	----	-----	----	-----	----	-----	----	-----	----	-----	--	--	--	--	--	--

IUPAC Nomenclature of Poly Functional Compounds

1.	(b)	2.	(b)	3.	(b)	4.	(b)	5.	(b)	6.	(b)	7.	(a)						
----	-----	----	-----	----	-----	----	-----	----	-----	----	-----	----	-----	--	--	--	--	--	--

IUPAC Nomenclature of Aromatic Compounds & Bicyclic & Spiro Compounds

1.	(e)	2.	(d)	3.	(e)	4.	(d)	5.	(c)	6.	(a)	7.	(a)						
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[B] Isomerism**Structural Isomerism**

1.	(c)	2.	(d)	3.	(c)	4.	(d)	5.	(b)	6.	(d)	7.	(b)	8.	(c)	9.	(a)	10.	(c)
11.	(b)	12.	(b)	13.	(c)	14.	(a)	15.	(b)	16.	(b)	17.	(a)	18.	(d)	19.	(b)	20.	(d)
21.	(c)	22.	(d)	23.	(a)	24.	(a)	25.	(b)	26.	(a)	27.	(d)	28.	(b)	29.	(b)	30.	(c)
31.	(b)	32.	(b)	33.	(b)	34.	(b)	35.	(a)	36.	(b)	37.	(b)	38.	(b)	39.	(a)		

Tautomerism and Applications of Tautomerism

1.	(a)	2.	(a)	3.	(d)	4.	(a)	5.	(a)	6.	(c)	7.	(b)	8.	(b)	9.	(d)	10.	(a)
11.	(c)	12.	(e)	13.	(e)	14.	(c)	15.	(c)	16.	(b)	17.	(c)	18.	(a)	19.	(d)	20.	(d)
21.	(c)	22.	(d)																

Conformational Analysis

1.	(d)	2.	(c)	3.	(b)	4.	(b)	5.	(a)	6.	(c)	7.	(d)	8.	(d)	9.	(c)	10.	(c)
11.	(a)	12.	(c)	13.	(d)	14.	(c)												

Geometrical Isomerism

1.	(c)	2.	(c)	3.	(c)	4.	(d)	5.	(d)	6.	(a)	7.	(a)	8.	(d)	9.	(d)	10.	(c)
11.	(a)	12.	(d)	13.	(c)	14.	(c)	15.	(c)	16.	(c)	17.	(b)	18.	(d)	19.	(c)	20.	(b)
21.	(a)	22.	(c)	23.	(b)	24.	(c)	25.	(a)										

Optical Isomerism

1.	(c)	2.	(b)	3.	(d)	4.	(c)	5.	(b)	6.	(b)	7.	(c)	8.	(b)	9.	(d)	10.	(d)
11.	(b)	12.	(a)	13.	(a)	14.	(b)	15.	(d)	16.	(c)	17.	(b)	18.	(b)	19.	(c)	20.	(c)
21.	(a)	22.	(c)	23.	(d)	24.	(b)	25.	(a)	26.	(b)	27.	(b)	28.	(d)	29.	(c)	30.	(c)
31.	(d)	32.	(b)	33.	(c)	34.	(c)	35.	(c)	36.	(b)	37.	(c)	38.	(b)	39.	(c)	40.	(b)
41.	(a)	42.	(b)	43.	(b)	44.	(d)	45.	(b)	46.	(a)	47.	(c)	48.	(b)	49.	(b)	50.	(b)
51.	(a)	52.	(d)	53.	(e)	54.	(b)	55.	(d)	56.	(d)	57.	(c)	58.	(a)	59.	(b)	60.	(c)
61.	(a)	62.	(c)	63.	(a)	64.	(c)												

Miscellaneous Problems on Stereoisomers

1.	(b)	2.	(b)	3.	(c)	4.	(c)	5.	(c)	6.	(a)	7.	(b)	8.	(c)	9.	(c)	10.	(c)
11.	(c)	12.	(d)																

[C] General Organic Chemistry**Inductive Effect & its Applications**

1.	(a)	2.	(d)	3.	(b)	4.	(c)	5.	(c)	6.	(b)	7.	(d)	8.	(c)	9.	(c)	10.	(c)
11.	(c)	12.	(b)	13.	(d)	14.	(c)	15.	(c)										

Resonance

1.	(d)	2.	(d)	3.	(c)	4.	(d)	5.	(c)	6.	(d)	7.	(d)	8.	(c)	9.	(b)	10.	(b)
11.	(b)	12.	(e)	13.	(b)	14.	(b)	15.	(c)	16.	(d)	17.	(a)	18.	(c)	19.	(d)	20.	(d)
21.	(c)	22.	(c)	23.	(c)	24.	(c)	25.	(c)	26.	(a)	27.	(c)	28.	(d)	29.	(b)	30.	(c)
31.	(a)	32.	(d)	33.	(d)	34.	(b)	35.	(c)	36.	(a)	37.	(c)	38.	(c)	39.	(c)	40.	(a)
41.	(c)	42.	(b)	43.	(d)	44.	(a)	45.	(b)	46.	(b)	47.	(c)	48.	(b)	49.	(c)	50.	(c)
51.	(c)	52.	(c)	53.	(d)	54.	(e)	55.	(d)	56.	(b)	57.	(c)	58.	(a)	59.	(c)	60.	(a)
61.	(b)	62.	(c)	63.	(c)	64.	(d)	65.	(c)	66.	(c)	67.	(a)	68.	(c)	69.	(b)	70.	(b)
71.	(d)	72.	(c)	73.	(d)	74.	(a)	75.	(a)	76.	(c)	77.	(d)	78.	(a)	79.	(a)	80.	(d)
81.	(d)	82.	(c)	83.	(b)	84.	(b)	85.	(d)	86.	(b)	87.	(b)	88.	(b)	89.	(a)	90.	(d)
91.	(a)	92.	(d)	93.	(a)	94.	(b)	95.	(c)	96.	(b)	97.	(c)	98.	(c)	99.	(d)	100.	(c)
101.	(c)	102.	(c)	103.	(c)														

Hyper Conjugation

1.	(c)	2.	(a)	3.	(e)	4.	(a)	5.	(c)	6.	(d)	7.	(c)	8.	(c)	9.	(c)	10.	(d)
11.	(b)	12.	(c)	13.	(c)	14.	(a)	15.	(d)	16.	(a)	17.	(b)	18.	(c)	19.	(b)	20.	(b)
21.	(a)	22.	(c)	23.	(a)	24.	(b)	25.	(a)	26.	(d)	27.	(b)	28.	(c)	29.	(d)		

Heat of Hydrogenation, Heat of Combustion and Bond Length

1.	(d)	2.	(a)	3.	(c)	4.	(b)	5.	(c)	6.	(d)	7.	(d)	8.	(d)	9.	(e)	10.	(b)
11.	(c)	12.	(d)	13.	(b)	14.	(a)	15.	(b)	16.	(a)	17.	(d)	18.	(c)				

Aromaticity

1.	(a)	2.	(d)	3.	(a)	4.	(a)	5.	(c)	6.	(e)	7.	(b)	8.	(a)	9.	(d)	10.	(c)
11.	(c)	12.	(b)	13.	(d)	14.	(d)	15.	(b)	16.	(b)	17.	(a)	18.	(c)	19.	(c)	20.	(a)
21.	(b)	22.	(c)	23.	(c)	24.	(b)	25.	(a)	26.	(e)	27.	(d)	28.	(b)	29.	(d)	30.	(c)
31.	(b)	32.	(a)	33.	(b)	34.	(c)	35.	(a)	36.	(b)	37.	(a)						

Acidic and Basic Strength

1.	(c)	2.	(a)	3.	(d)	4.	(c)	5.	(c)	6.	(b)	7.	(c)	8.	(a)	9.	(b)	10.	(a)
11.	(a)	12.	(b)	13.	(c)	14.	(b)	15.	(d)	16.	(d)	17.	(a)	18.	(b)	19.	(b)	20.	(c)
21.	(a)	22.	(a)	23.	(c)	24.	(c)	25.	(b)	26.	(a)	27.	(b)	28.	(a)	29.	(b)	30.	(c)
31.	(a)	32.	(c)	33.	(c)	34.	(d)	35.	(b)	36.	(b)	37.	(b)	38.	(b)	39.	(b)	40.	(b)
41.	(a)	42.	(a)	43.	(c)	44.	(c)	45.	(d)	46.	(c)	47.	(c)	48.	(c)	49.	(a)	50.	(d)
51.	(c)	52.	(e)	53.	(e)	54.	(b)	55.	(c)	56.	(d)	57.	(c)	58.	(a)	59.	(b)	60.	(b)
61.	(c)	62.	(c)	63.	(c)	64.	(c)	65.	(a)	66.	(b)	67.	(b)	68.	(a)	69.	(c)	70.	(c)
71.	(c)	72.	(c)	73.	(d)	74.	(d)	75.	(c)	76.	(b)	77.	(d)	78.	(c)	79.	(c)	80.	(b)
81.	(a)	82.	(d)	83.	(d)	84.	(d)	85.	(c)	86.	(a)	87.	(a)	88.	(d)	89.	(a)	90.	(b)
91.	(d)	92.	(d)																

Electrophile & Nucleophile and Type of Reaction & Solvent

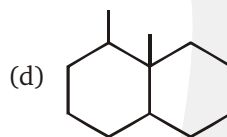
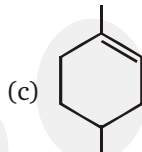
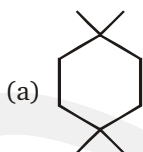
1.	(b)	2.	(c)	3.	(d)	4.	(c)	5.	(a)	6.	(d)	7.	(e)	8.	(e)	9.	(d)	10.	(a)
11.	(e)	12.	(a)	13.	(e)	14.	(e)	15.	(d)	16.	(c)	17.	(d)	18.	(e)	19.	(c)		

Miscellaneous Problems

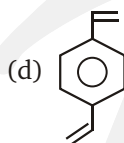
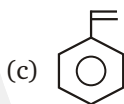
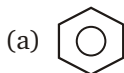
1.	(c)	2.	(b)	3.	(a)	4.	(b)	5.	(d)	6.	(c)	7.	(a)	8.	(c)	9.	(b)	10.	(a)
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One or More than One Answers is/are Correct

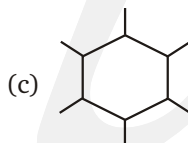
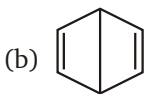
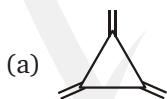
1. Which of the compound contains only 1, 2 and 4 carbon atoms ?



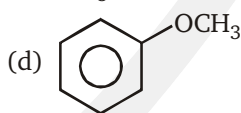
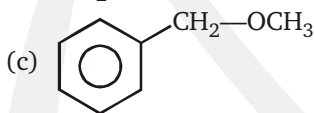
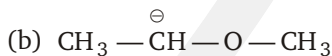
2. Which of the following compounds has maximum number of π -bonds?



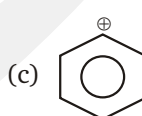
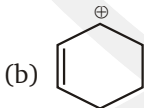
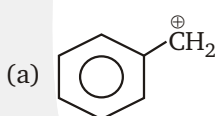
3. Which of the following is isomer of benzene ?



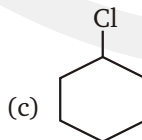
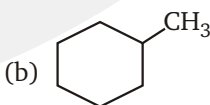
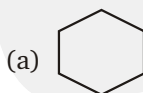
4. In which of the following compound M of $-\text{OCH}_3$ will operate ?



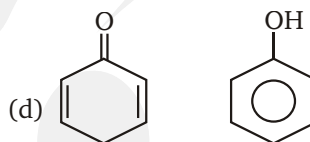
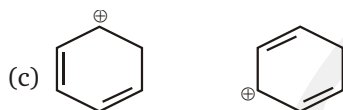
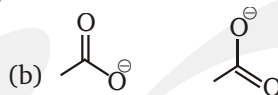
5. Which of the following carbocations is/are resonance stabilized?



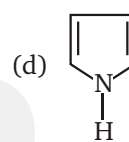
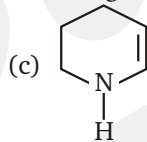
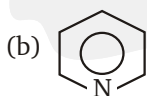
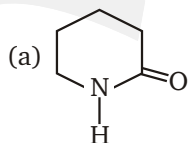
6. In which of the following compounds inductive effect is absent ?



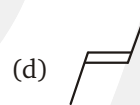
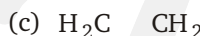
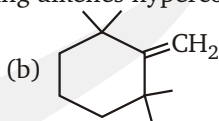
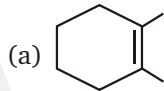
7. Which of the following pairs is/are resonating structure?



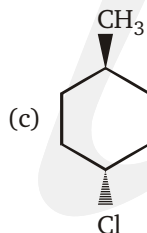
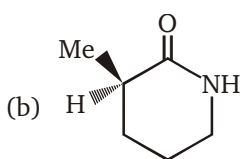
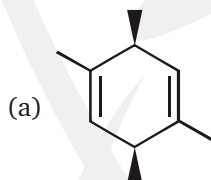
8. In which of the following compounds lone pair of nitrogen is involved in resonance?



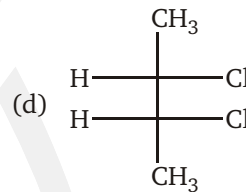
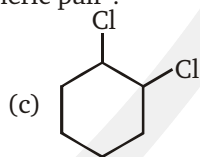
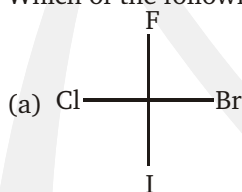
9. In which of the following alkenes hyperconjugation is not possible?



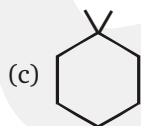
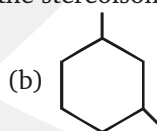
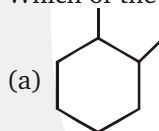
10. Which of the following compound is chiral ?



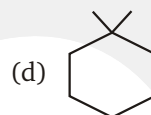
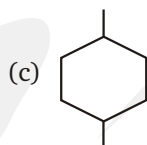
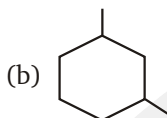
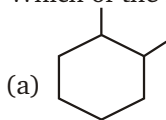
11. Which of the following can exist in diastereomeric pair ?



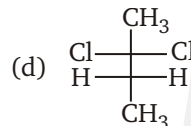
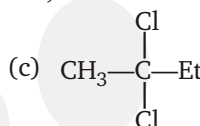
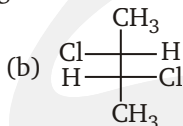
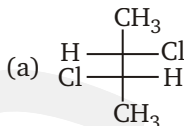
12. Which of the following compound has one of the stereoisomer as a meso compound ?



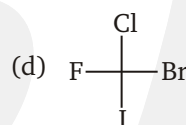
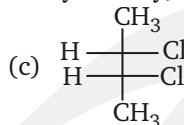
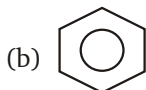
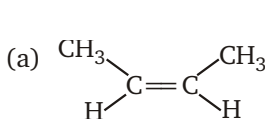
13. Which of the following has one of the isomer is meso?



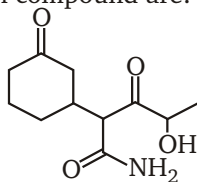
14. Which of the following is diastereomer of meso-2,3-dichloro butane?



15. Which of the following has/have P. O.S. (plane of symmetry)?



16. Functional groups present in given compound are:



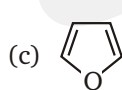
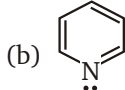
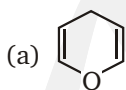
(a) alcohol

(b) ketone

(c) carboxylic acid

(d) amide

17. Which of the following is aromatic?



18. Phenol is less acidic than:

(a) acetic acid

(b) *p*-methoxy phenol

(c) *p*-nitrophenol

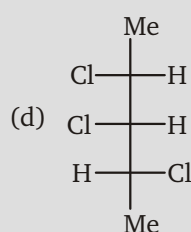
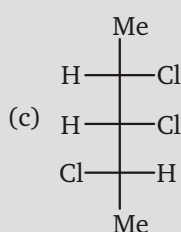
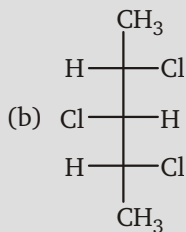
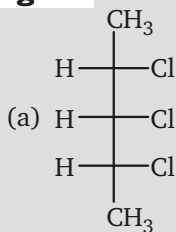
(d) ethanol

Answers

1.	(a, b)	2.	(b)	3.	(a, b)	4.	(a, d)	5.	(a, b, d)	6.	(a, b)
7.	(b, c)	8.	(a, c, d)	9.	(b, c)	10.	(a, b)	11.	(b, c, d)	12.	(a, b)
13.	(a, b)	14.	(a, b)	15.	(a, b, c)	16.	(a, b, d)	17.	(b, c)	18.	(a, c)

Comprehension Based Problems

Passage-1:



1. Pair of enantiomers:

- (a) 0 (b) 1 (c) 2 (d) 3

2. Pair of Diastereomers are:

- (a) 3 (b) 4 (c) 5 (d) 6

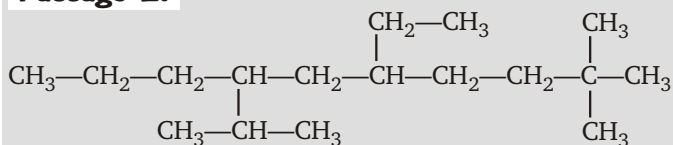
3. Number of meso isomers are:

- (a) 1 (b) 2 (c) 3 (d) 4

4. Number of chiral isomers are:

- (a) 1 (b) 2 (c) 3 (d) 4

Passage-2:



1. Number of 1 carbon in above compounds are:

- (a) 5 (b) 6 (c) 7 (d) 8

2. Number of 2 carbon in above compounds are:

- (a) 5 (b) 6 (c) 7 (d) 8

3. Number of 3 carbon are:

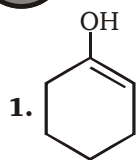
- (a) 1 (b) 2 (c) 3 (d) 4

4. Number of 4 carbon are:

- (a) 1 (b) 2 (c) 3 (d) 4

Passage-3:

Keto-enol tautomerism is a dynamic equilibrium which is depends on the stability of keto or enol tautomerism take place in dilute acidic or basic medium.



Prolonged treatment



How many hydrogen atoms replaced by deuterium ?

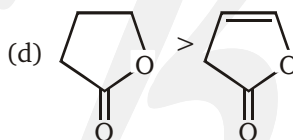
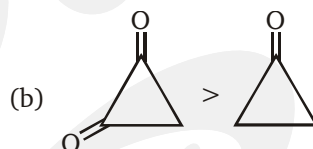
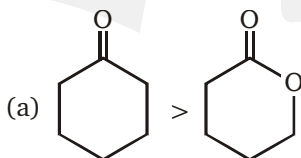
(a) 1

(b) 3

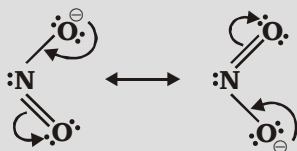
(c) 4

(d) 5

2. Which of the % enol content order is wrong ?

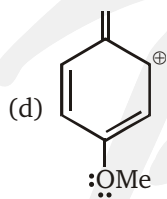
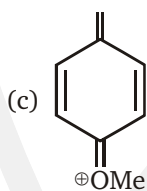
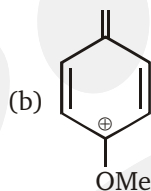
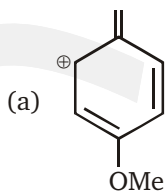
**Passage-4:**

The theory of resonance was developed primarily by Pauling in the 1930s. According to this theory, many molecules and ions are best described by writing two or more lewis structures and considering the real molecule or ion to be a composite of these structures. Individual lewis structure are called contributing structures. They are also sometimes referred to as resonance structures or resonance contributors. We show that the real molecule or ion is a resonance hybrid of the various contributing structures by interconnecting them with double headed arrows. Resonance structures are not in equilibrium with each other.

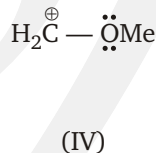
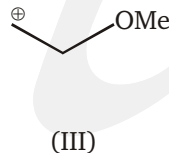
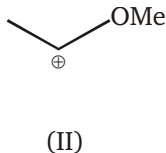
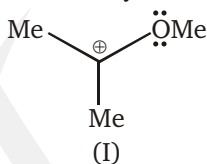


(equivalent contributing structures)

1. More stable resonating structure of the given cation is:

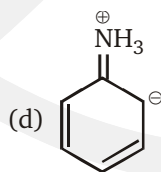
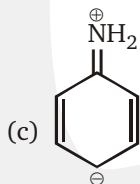
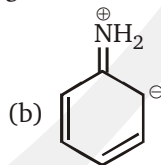
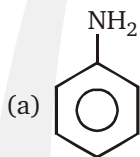


2. Find out the stability order of intermediate:



- (a) I > II > III > IV (b) I > II > IV > III (c) IV > I > II > III (d) III > I > II > IV

3. Which of the following is not a valid resonating structure ?



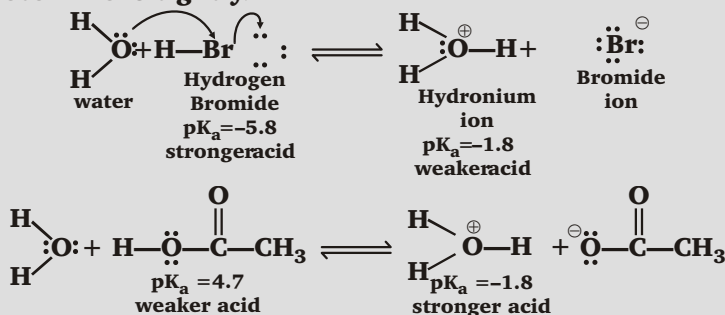
Passage-5:

In any proton transfer reaction:

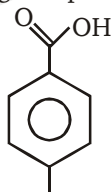
Acid + Base $\rightleftharpoons$ conjugate acid + conjugate base the reaction proceeds in the direction that converts the stronger acid and the stronger base to the weaker acid and the weaker base.

Stronger acid + Stronger base $\xrightleftharpoons{K>1}$ weaker acid + weaker base

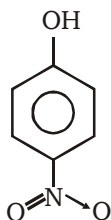
The reaction will be favourable when the stronger acid is on the left and the weaker acid is on the right. The equilibrium lies to the side of the acid that holds the proton more tightly.



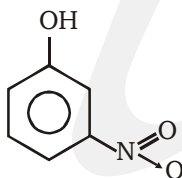
1. In the following compound;



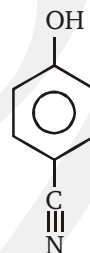
(I)



(II)



(III)

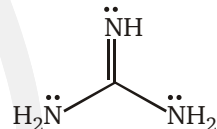
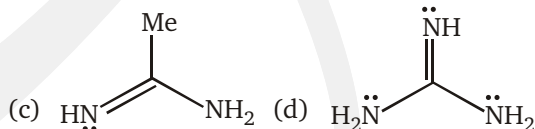
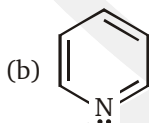
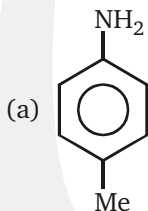


(IV)

the decreasing order of acidic strength:

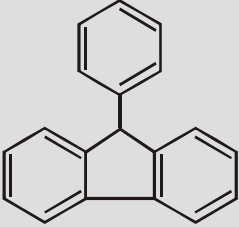
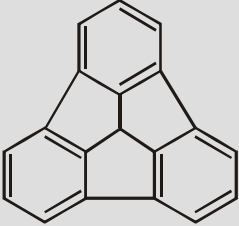
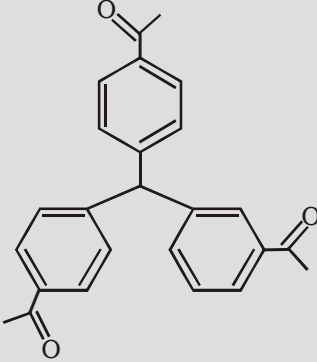
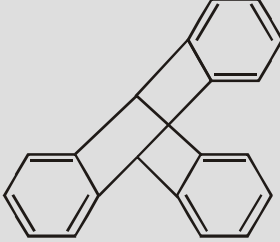
- (a) I > II > IV > III (b) II > I > IV > III (c) II > IV > I > III (d) II > IV > III > I

2. Which of the following compound is strongest base ?

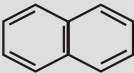
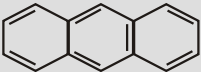
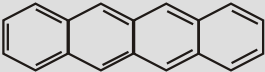


Matching Type Problems

1. Match the following:

Column-I		Column-II (Double bond equivalent value)	
(a)		(p)	11
(b)		(q)	12
(c)		(r)	13
(d)		(s)	14
		(t)	15

2. Match the column:

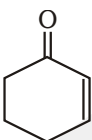
Column-I		Column-II	
(a)		(p)	Number of π -bonds is even
(b)		(q)	Number of π -bonds is odd
(c)		(r)	Number of π -bonds is odd
(d)		(s)	Number of π -bonds is even

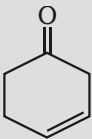
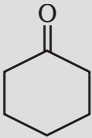
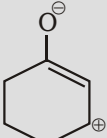
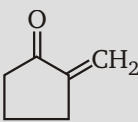
3. Match the column:

Column-I		Column-II	
(a)	C_4H_8	(p)	Number of structural isomer=2
(b)	C_5H_{12}	(q)	Number of structural isomer=3
(c)	C_6H_{14}	(r)	Number of structural isomer=4
(d)	$C_4H_{10}O$ (alcohol only)	(s)	Number of structural isomer=5

4. Match the column:

Column-I		Column-II	
(a)	Chain isomer	(p)	$CH_3 - CH_2 - OH, \quad CH_3 - CH_2 - CH_2 - OH$
(b)	Positional isomer	(q)	$CH_3 - CH_2 - OH, \quad CH_3 - O - CH_3$
(c)	Metamers	(r)	$CH_3 - \overset{\overset{O}{\parallel}}{C} - O - CH_2 - CH_3,$ $CH_3 - CH_2 - \overset{\overset{O}{\parallel}}{C} - O - CH_3$
(d)	Functional isomer	(s)	$CH_3 - CH_2 - CH_2 - NH_2, \quad CH_3 - \underset{\underset{NH_2}{ }}{CH} - CH_3$
(e)	Homologous	(t)	$CH_3 - CH_2 - CH_2 - CH_3, \quad \begin{array}{c} \diagup \quad \diagdown \\ CH_3 \end{array}$

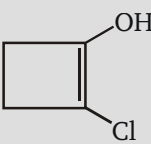
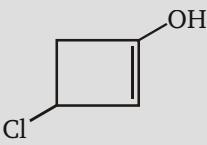
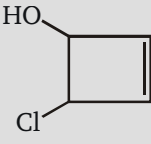
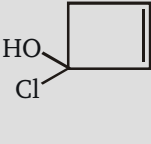
5. With reference to compound A  drawn below label each compound as an isomer resonating structure or neither :

Column-I		Column-II	
(a)		(p)	Isomers
(b)		(q)	Resonating structure
(c)		(r)	No relation
(d)		(s)	Homologous

6. Relation between molecules or ions in each pairs related :

Column-I		Column-II	
(a)	$\text{CH}_3 - \text{O} - \text{CH}_2 - \text{CH}_3, \text{CH}_3 - \underset{\text{H}}{\overset{\text{CH}_3}{\text{C}}} - \text{OH}$	(p)	Resonating structure
(b)	$\text{H}_2\text{C} = \text{CH} - \text{CH}^{\ominus} - \text{CH}_3, \text{CH}_2^{\ominus} - \text{CH} = \text{CH} - \text{CH}_3$	(q)	Isomers
(c)	 , $\text{CH}_3 - \underset{\text{H}}{\text{C}} - \underset{\text{H}}{\text{C}} - \text{CH}_3$	(r)	No relation
(d)	$\text{CH}_3 - \text{CH}_2 - \text{CH}_3, \text{CH}_3 - \text{CH}_2 - \text{CH}_2^{\ominus}$	(s)	Homologous

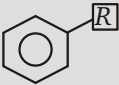
7. Match the column:

Column-I		Column-II	
(a)		(p)	Resonance stabilization in the compound
(b)		(q)	IUPAC name is 2-chlorocyclobutenol
(c)		(r)	IUPAC name is 3-chlorocyclobutenol
(d)		(s)	Hyperconjugation is possible
		(t)	Aromatic compound

8. Match the column:

Column-I (Various isoelectronic species)		Column-II (Relative stability)	
(a)	$\begin{array}{cccccc} & (a) & & & (b) & \\ \ominus & \text{C} & \text{O} & \oplus & \text{C} & \ominus \\ & & & & \text{—O} & \end{array}$	(p)	$a > b$
(b)	$\begin{array}{cccccc} \ominus & \oplus & & & \oplus & \ominus \\ \text{N} & \text{N} & \text{O} & \text{N} & \text{N} & \text{—O} \end{array}$	(q)	$b > a$
(c)	$\text{:}\overset{\ominus 2}{\text{N}}\text{—}\ddot{\text{N}}=\text{O} \longleftrightarrow \text{:}\overset{-1}{\text{N}}\equiv\ddot{\text{N}}\text{—}\overset{\ominus}{\text{O}}$	(r)	$a = b$
(d)	$\begin{array}{cccc} \ominus & & & \ominus \\ \text{O} & \text{—C} & \text{O} & \text{O} & \text{C} & \text{—O} \end{array}$		

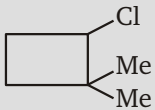
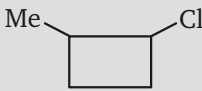
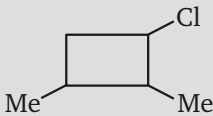
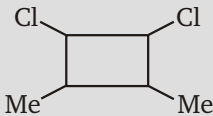
9. Match the column:

Column-I		Column-II	
(a)		(p)	R groups show I effect
(b)	$\boxed{R}-Mg-Br$	(q)	R groups show I effect
(c)	$\boxed{H_2C=CH}-R$	(r)	$(H_2C \quad CH-)$ show I effect
(d)	$\boxed{H_2C=CH}-C\equiv CH$	(s)	$(H_2C \quad CH-)$ show I effect

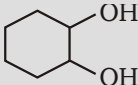
10. Match the column:

	Column-I		Column-II
(a)	2,3-dichlorobutane	(p)	Number of stereoisomers are odd numbers
(b)	2,4-dichloropentane	(q)	Number of stereoisomer are even numbers
(c)	1,2-dichlorocyclohexane	(r)	Number of meso are odd numbers
(d)	1,3-dichlorocyclohexane	(s)	Number of meso are even numbers

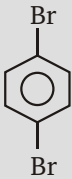
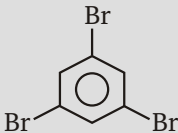
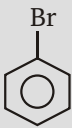
11. Match the column:

	Column-I		Column-II
(a)		(p)	Number of chiral center = 1
(b)		(q)	Number of chiral center = 2
(c)		(r)	Number of chiral center = 3
(d)		(s)	Number of chiral center = 4

12. Match the column:

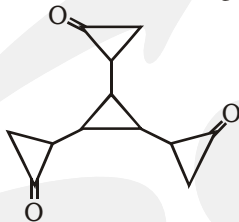
Column-I		Column-II	
(a)	$\text{CH}_3 - \text{CH} \quad \text{CH} - \text{CH}_3$	(p)	Number of stereoisomer are=2
(b)	$\text{CH}_3 - \text{CH} \quad \text{CH} - \text{CH} - \text{CH}_3$ OH	(q)	Number of stereoisomer are=3
(c)	$\text{CH}_3 - \text{CH} - \text{CH} \quad \text{CH} - \text{CH} - \text{CH}_3$ OH OH	(r)	Number of stereoisomer are=4
(d)		(s)	Number of stereoisomer are=6

13. Match the column:

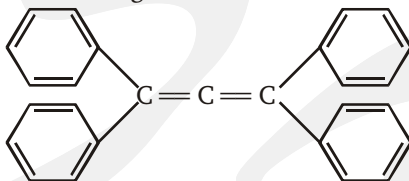
Column-I		Column-II	
(a)		(p)	C_2 axis of symmetry
(b)		(q)	C_3 axis of symmetry
(c)		(r)	Plane of symmetry
(d)		(s)	Center of symmetry

Multi Concept Subjective Problems

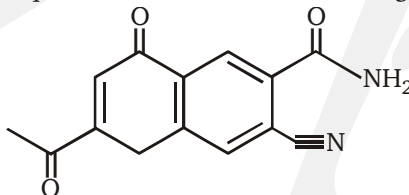
1. Find out the value of degree of unsaturation in the given compound.



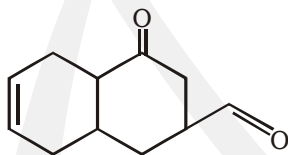
2. How many 2 carbon in the following ?



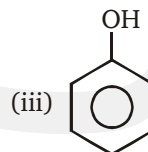
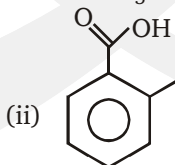
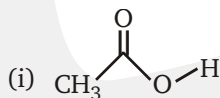
3. Find out the double bond equivalent (D. B. E) value of the given following compound.

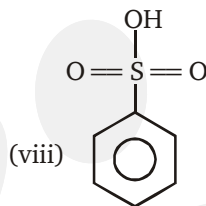
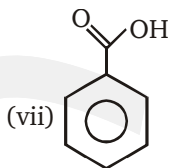
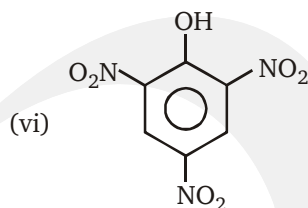
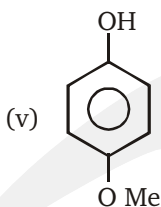
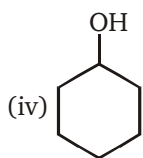


4. How many structural isomers are possible for $C_4H_{10}O$ (only alcohol) ?
5. The total number of functional groups present in the given following compound:

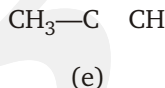
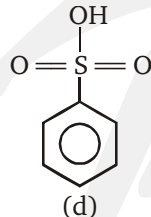
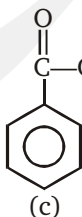
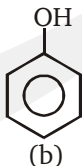
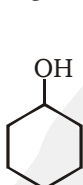


6. How many 5 members parent chain alkane are possible for C_7H_{16} ?
7. Total number of possible structural isomers of $C_5H_{11}Br$.
8. Total number of structural isomers of C_9H_{18} containing cyclohexane ring:
9. Provide the structure of all distinct terminal alkynes with a molecular formula of C_6H_{10} .
10. How many compound reacts with $NaHCO_3$?

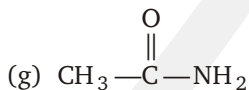
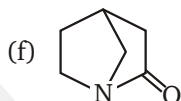
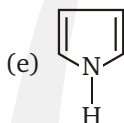
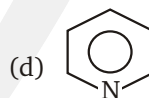
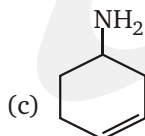
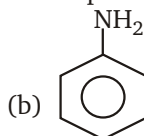
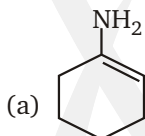




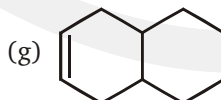
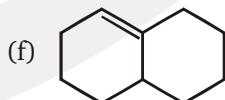
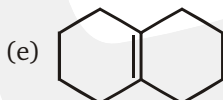
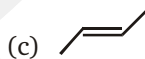
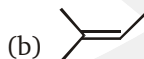
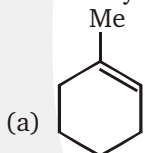
11. Which of the given following compound will react with NaHCO_3 or soluble in NaHCO_3 ?



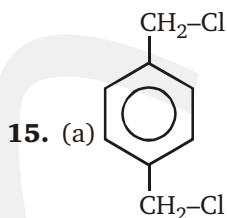
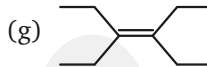
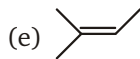
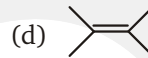
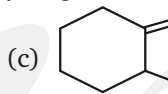
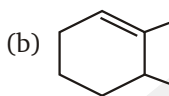
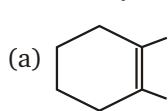
12. In how many compound lone pair of nitrogen is involved in resonance ?



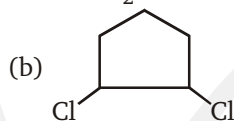
13. How many compound have odd number of π -hydrogens ?



14. How many compounds has odd number of π -hydrogens ?



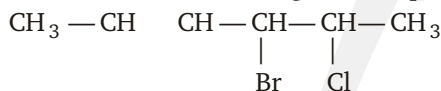
(x) (Number of plane of symmetry)



(y) (Number of meso isomer of 1,2-dichlorocyclopentane).

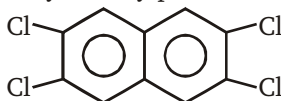
Sum of (x + y) ?

16. Find out the total number of stereocentre in the given compound:



17. Find the total number of isomers of C_7H_{14} (only 5-member ring):

18. Total number of plane of symmetry present in given compound is:

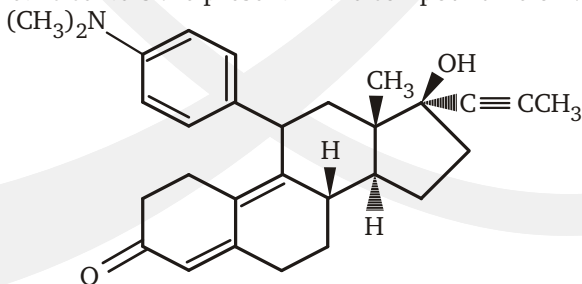


19. Total number of isomers for $\text{C}_4\text{H}_6\text{Br}_2$ containing cyclobutane ring are (including stereoisomer).

20. How many structural isomers are possible for $\text{C}_4\text{H}_9\text{Cl}$?

21. Total number of stereoisomer of 2, 3-dichlorobutane containing compound:

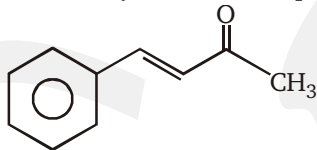
22. How many asymmetric centers are present in the compound below?



23. Total number of β -hydrogen in the given following compound is:

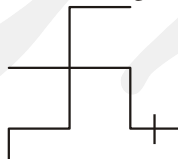


24. How many carbon atom present in the parent chain in the given following compound ?

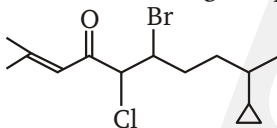


25. How many alkane will be formed by molecular formula C_6H_{14} ?

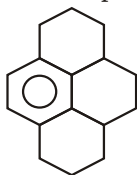
26. Total number of carbon in the parent chain of given compound is:



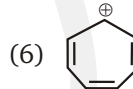
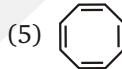
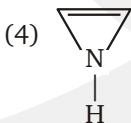
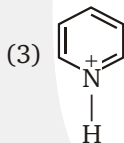
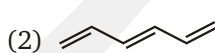
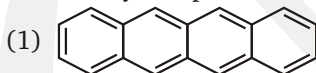
27. Total number of substituents present in following compound are:



28. Total number of β -hydrogen in the given compound is:



29. How many compounds are aromatic?

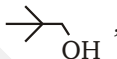
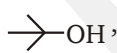


30. x Types of functional group
y Double bond equivalent

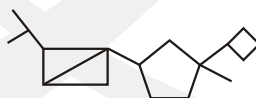
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$$\begin{array}{c}
 \text{CH}_3 \\
 | \\
 \text{CH}_3 - \text{C} - \text{CH}_3 \\
 | \\
 \text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{C} - \text{CH}_2 - \text{CH}_2 - \text{CH}_3 \\
 | \\
 \text{CH}_3 - \text{CH}_2 - \text{C} - \text{CH}_2 - \text{CH}_2 - \text{CH}_3 \\
 | \\
 \text{CH}_3
 \end{array}$$

(B)
gas

34. $\text{—}\text{N}\text{H}_2$, $\text{—}\text{N}\text{H—}$, $\text{—}\text{N}\text{H}_2$, $\text{—}\text{N}\text{H—}$



hen what is the

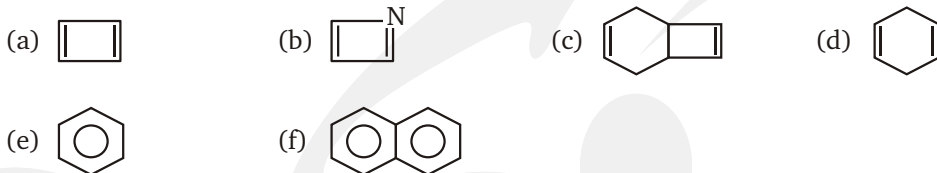
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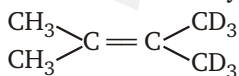
(ii) $\text{Ph}-\text{OH}$ NaNH_2 $B(\text{gas})$

Sum of mol. wt. of A B X, so the value of X 54 is.

37. Number of compounds undergo dimerization at room temperature?



38. X Number of α -hydrogen.



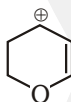
39. Y Number of resonating structure.



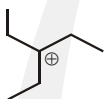
40. X Number of R.S. (Resonating structure)



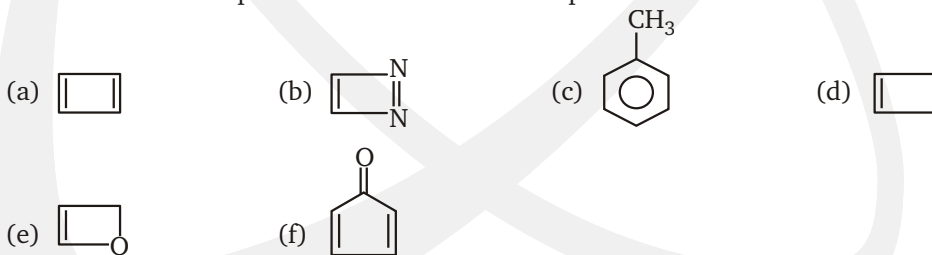
41. X Number of R.S. (Resonating structure)

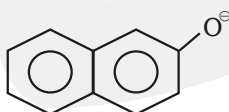


42. X Number of α -Hydrogen,



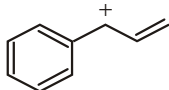
43. X Number of compound unstable at room temperature.



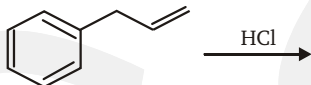
44.  x No. of resonating structure.

Unified Conceptual Problems

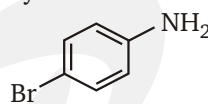
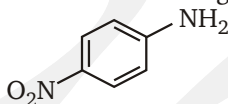
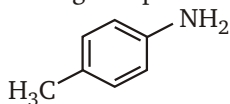
1. Draw the important resonance contributing forms for the structure shown below.



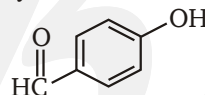
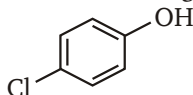
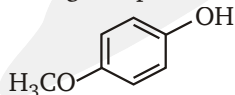
2. Provide the structure of the major organic product(s) in the reaction below:



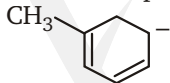
3. List the following compounds in order of decreasing basicity.



4. List the following compounds in order of decreasing acidity.



5. Draw the important resonance structures of:



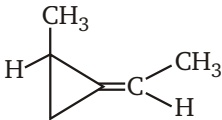
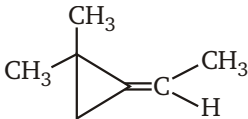
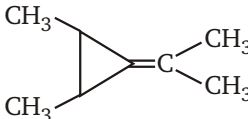
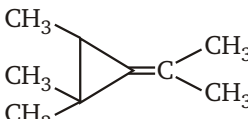
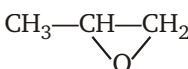
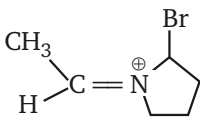
6. Provide 1, 2, 3 and 4 carbons and hydrogens in each structure:

Structures	1	2	3	4
	C H	C H	C H	C

7.

S.No.	Molecular Formula	Number of Structural Isomers
1.	C_4H_8	
2.	C_3H_6	
3.	C_6H_{12} (having cyclobutane ring)	
4.	C_4H_6 (cyclic)	
5.	C_7H_{16}	
6.	$C_4H_{10}O$	
7.	C_6H_{14}	
8.	C_3H_9N	
9.	C_6H_{12} (cyclopropane ring only)	
10.	$C_3H_6Cl_2$	
11.	C_8H_{10} (only one benzene ring)	
12.	C_7H_8O (only one benzene ring)	
13.	C_5H_{12}	
14.	C_7H_{14} (cyclobutane ring)	
15.	$C_5H_{10}O$ (only aldehyde and ketone)	
16.	C_5H_{10} (cyclic)	
17.	$C_5H_{11}Br$	
18.	$C_4H_{11}N$	
19.	$C_3H_5Br_3$	
20.	C_7H_7Cl (only one benzene ring)	
21.	$C_5H_{12}O$ (only ether)	

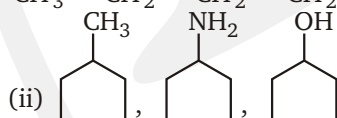
8. Find the number of Stereocenter and Chiral center in all structural isomers of given molecular formula:

S.No.	Molecular Formula	Stereocenter	Chiral Center
(i)	C_6H_{12} (open chain)		
(ii)	C_6H_{12} (excluding cyclopropane ring)		
(iii)	C_6H_{14}		
(iv)	$C_3H_6Cl_2$		
(v)	C_5H_{12}		
(vi)	C_5H_{10} (all cyclic isomers)		
(vii)	$C_5H_{11}Br$		
(viii)	C_4H_6 (all cyclic isomers)		
(ix)	C_7H_{14} (cyclopentane ring)		
(x)			
(xi)			
(xii)			
(xiii)			
(xiv)			
(xv)			

(xvi)			
(xvii)	2,4-Hexadiene		
(xviii)			

9. Rank the following groups of compounds in order of decreasing solubility in H_2O :

- (i) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{OH}$, $\text{Cl} - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{Cl}$,
 $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{OH}$, $\text{HO} - \text{CH}_2 - \text{CH}_2 - \text{OH}$



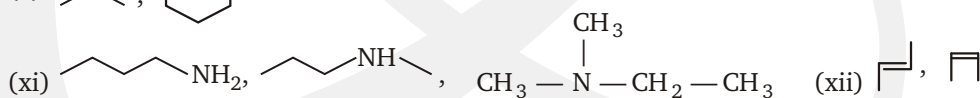
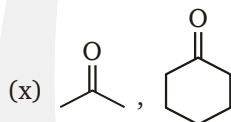
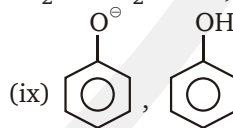
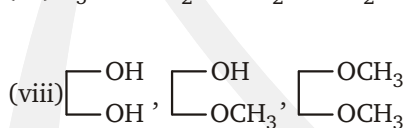
- (iii) $\text{CH}_3 - \text{F}$, $\text{CH}_3 - \text{Cl}$, $\text{CH}_3 - \text{Br}$

- (iv) $\text{CH}_3 - \text{O} - \text{CH}_3$, $\text{CH}_3 - \text{O} - \text{CH}_2 - \text{CH}_3$, $\text{CH}_3 - \text{CH}_2 - \text{O} - \text{CH}_2 - \text{CH}_3$

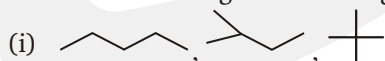
- (v) EtOH , $\text{CH}_3 - \text{CH}_2 - \text{CH}_3$

- (vi) $\text{CH}_3 - \text{O} - \text{CH}_3$, $\text{CH}_3 - \text{CH}_2 - \text{CH}_3$

- (vii) $\text{H}_3\text{C} - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{OH}$, $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{OH}$, $\text{CH}_3 - \text{CH}_2 - \text{OH}$

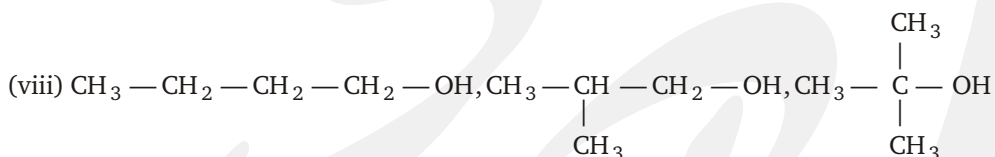
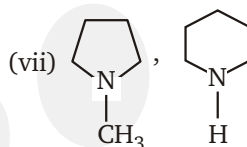
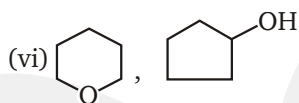
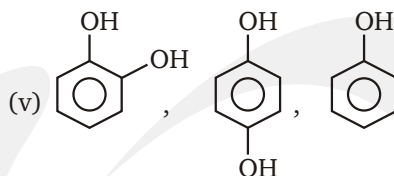
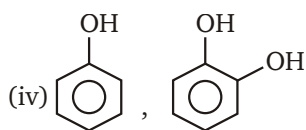


10. Rank the following in decreasing order of boiling point :



- (ii) EtOH , $\text{Et} - \text{NH}_2$, $\text{CH}_3 - \text{O} - \text{CH}_3$, $\text{CH}_3 - \text{CH}_2 - \text{CH}_3$

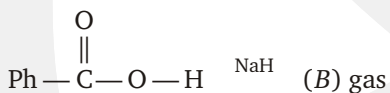
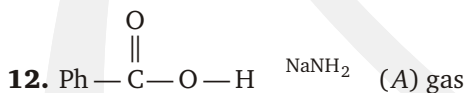
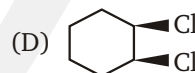
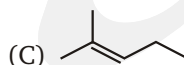
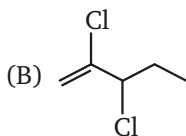
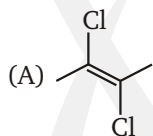
- (iii) $\text{CH}_3\text{CO}_2\text{H}$, $\text{CH}_3 - \text{CH}_2 - \text{OH}$, $\text{CH}_3 - \text{CH} = \text{O}$, $\text{CH}_3 - \text{CH}_2 - \text{CH}_3$



(ix) cyclooctane, cycloheptane, cyclohexane

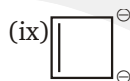
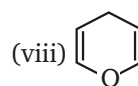
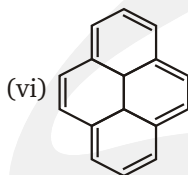
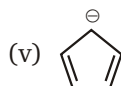
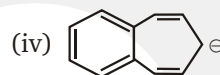
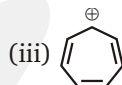
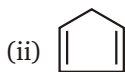


11. Amongst the following, the total number of compound will show geometrical isomerism:

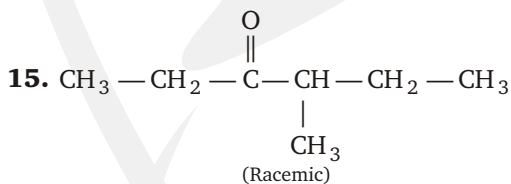
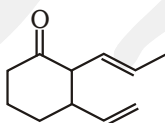


(A) and (B) are molecular mass of gas so the value of $A - B - 10$ is:

13. Amongst the following, the total number of compounds which will show aromatic character will be:



14. How many stereoisomers of given compound are possible when alkenyl substituent have a *cis*-configuration.

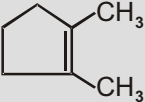
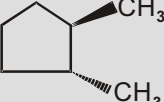
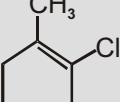
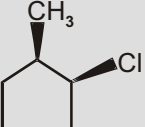
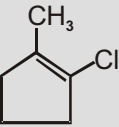
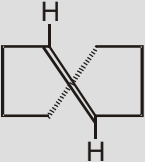
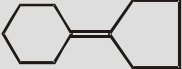
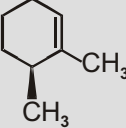
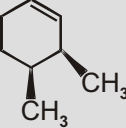
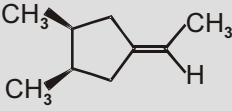
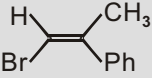
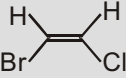
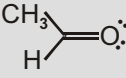
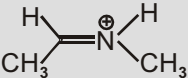


Total number of enol (including stereoisomer) possible of given compound is (x).
So the value of (x) is:

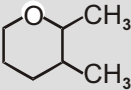
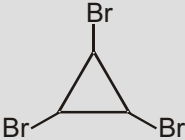
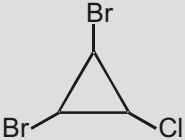
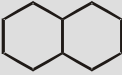
16. Identify which of the following compounds may show geometrical isomerism?

1.		2.		3.	
4.		5.		6.	
7.		8.		9.	
10.		11.		12.	
13.		14.		15.	
16.		17.		18.	
19.		20.		21.	
22.		23.		24.	

17. Identify which of the compound is *cis* or *trans* or none of these?

1.		2.		3.	
4.		5.		6.	
7.		8.		9.	
10.		11.		12.	
13.		14.		15.	
16.		17.		18.	
19.		20.		21.	

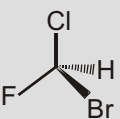
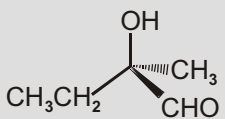
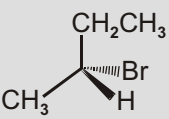
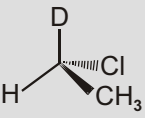
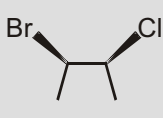
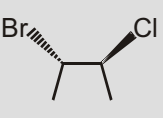
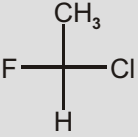
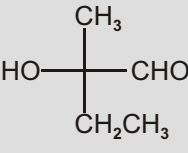
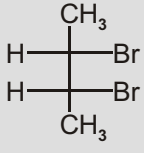
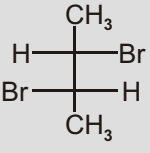
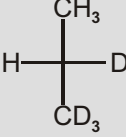
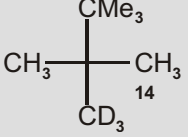
18.

	Compound	Total number of Geometrical Isomers possible
1.	$\text{CH}_3 - \text{CH} = \text{CH} - \text{CH}_2\text{CH}_3$	
2.	$\text{CH}_3 - \text{CH} = \text{CH} - \text{CH} = \text{CH}_2$	
3.	$\text{CH}_3 - \text{CH} = \text{CH} - \text{CH} = \text{CH} - \text{CH}_3$	
4.	$\text{CH}_3 - \text{CH} = \text{CH} - \text{CH} = \text{CH} - \text{Br}$	
5.	$\text{CH}_3 - \text{CH} = \text{CH} - \text{CH} = \text{CH} - \text{CH} = \text{CH}_2$	
6.	$\text{CH}_3 - (\text{CH} = \text{CH})_3 - \text{Ph}$	
7.	$\text{CH}_3(\text{CH} = \text{CH})_3\text{CH}_3$	
8.		
9.		
10.		
11.		
12.		
13.	$\text{CH}_3 - \text{CH} = \text{CH}_2$	

19. Count the total number of chiral center.

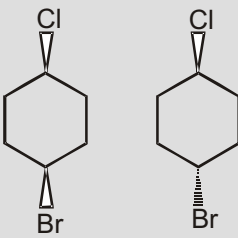
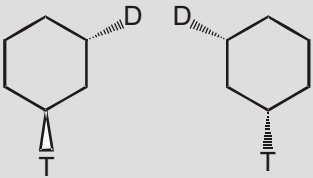
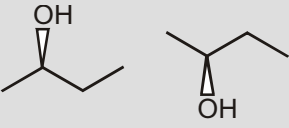
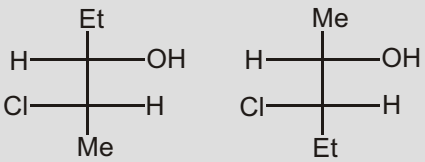
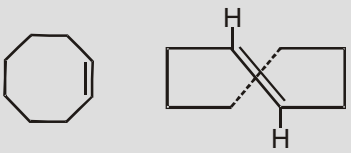
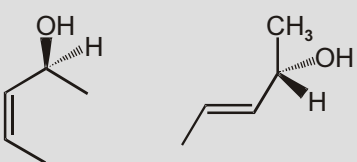
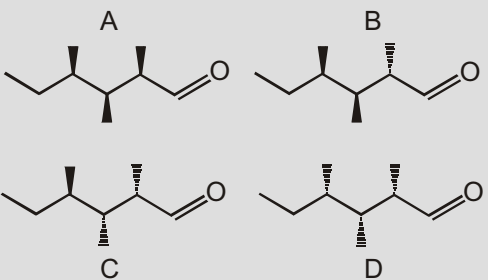
1.		2.		3.	
4.		5.		6.	
7.		8.		9.	
10.		11.		12.	
13.		14.		15.	
16.			Cholesterol		
17.			Estrone		

20. R-S System

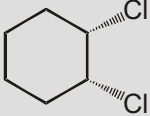
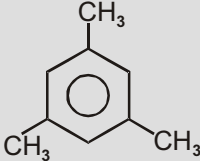
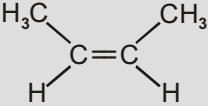
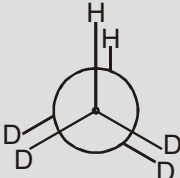
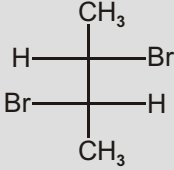
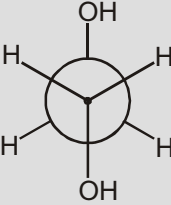
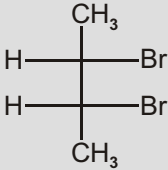
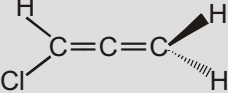
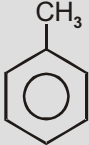
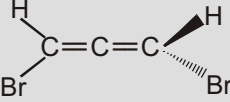
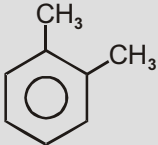
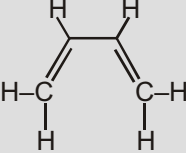
1.		2.		3.	
4.		5.		6.	
7.		8.		9.	
10.		11.		12.	
13.		14.			

21. Relationship between enantiomer, diastereomer and other :

	Identify the relation	Enantiomer	Diastereomer	Other
1.				
2.				
3.				
4.				
5.				
6.				
7.				

8.				
9.				
10.				
11.				
12.				
13.				
14.				

22. Identify the plane of symmetry (POS) in the given molecule.

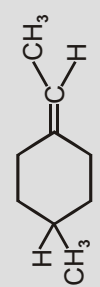
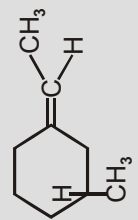
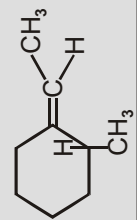
		POS			POS
1.			7.		
2.			8.		
3.			9.		
4.			10.		
5.			11.		
6.			12.		

		POS			POS
13.			15.		
14.					

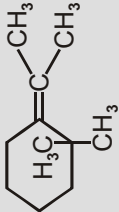
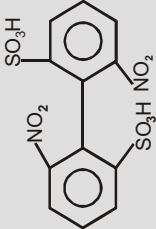
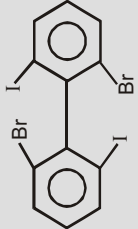
23. Identify the molecules having centre of symmetry

		C_i			C_i
1.			4.		
2.			5.		
3.			6.		

		C_i			C_i
7.			12.		
8.			13.		
9.			14.		
10.			15.		
11.					

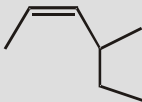
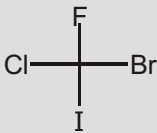
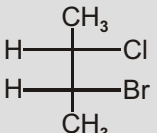
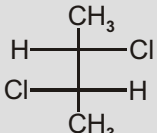
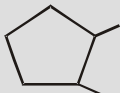
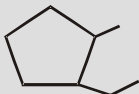
S. NO.	Compound	Chiral carbon	POS	COS	Chiral compound	Optically active	Optically isomerism	Geometrical isomerism
1.								
2.								
3.								
4.								
5.								
6.								
7.								
8.								

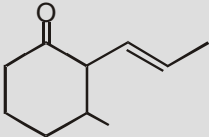
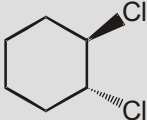
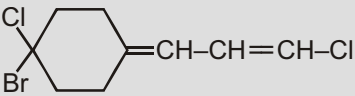
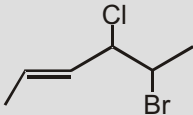
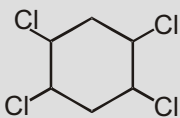
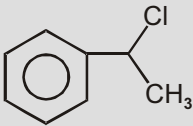
24.

S. NO.	Compound	Chiral carbon	POS	COS	Chiral compound	Optically active	Optically isomerism	Geometrical isomerism
1.								
2.								
3.								
4.								
5.								
6.								
7.								

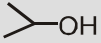
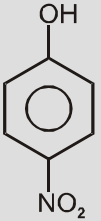
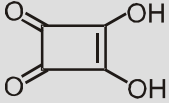
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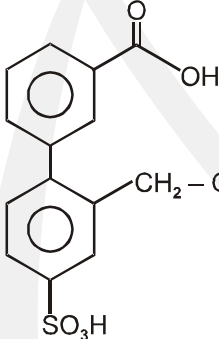
26. Calculate the number of isomeric molecules as per direction.

S. No.	Compound	Total no. of stereoisomers	Number of meso compd.	Number of optically active isomers
1.				
2.				
3.	$\text{CH}_3 - \text{CH} = \text{C} = \text{CH} - \text{CH}_3$			
4.				
5.				
6.				
7.	2,4-dichloropentane			
8.	2,3-dichloropentane			
9.	1,2-dichloropentane			
10.				
11.				

S. No.	Compound	Total no. of stereoisomers	Number of meso compd.	Number of optically active isomers
12.				
13.				
14.				
15.				
16.				
17.	$\begin{array}{c} \text{CH}_3 - \text{CH} = \text{C} = \text{CH} - \text{CH} = \\ \text{CH} - \text{CH}_3 \end{array}$			
18.				
19.	Tartaric acid			

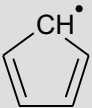
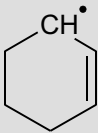
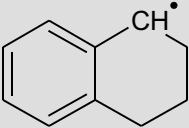
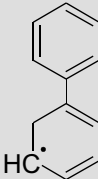
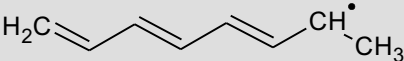
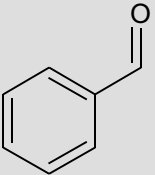
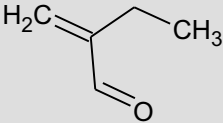
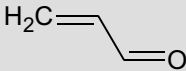
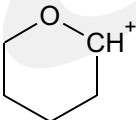
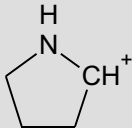
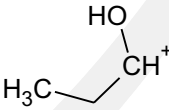
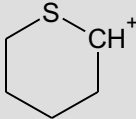
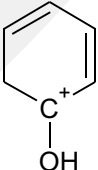
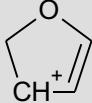
27. State different types of gases evolved in the following given blocks :

S.No.	Compound	Na	NaH	NaNH ₂	NaHCO ₃	CH ₃ — Li
a.	EtOH					
b.	H — C C — H					
c.						
d.	Ph — OH					
e.	Ph — CO ₂ H					
f.	Ph — SO ₃ H					
g.						
h.						

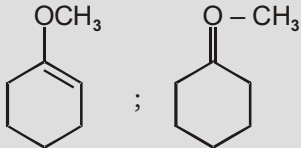
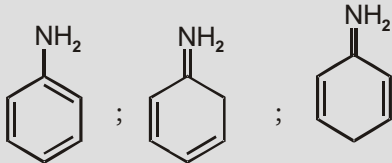
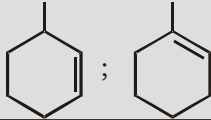
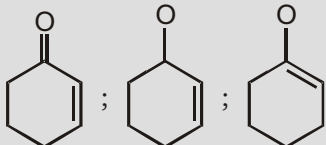
28.  gas (B) sum of molecular weight of (A + B)?

29.  OH $\xrightarrow{\text{Na}}$ x mole $\xrightarrow{\text{Al}}$ y mole $\xrightarrow{\text{NaH}}$ z mole Find sum of x + y + z?

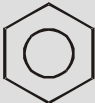
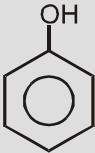
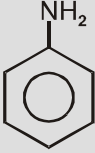
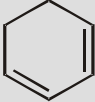
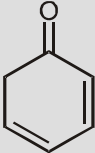
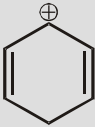
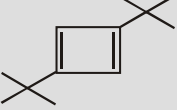
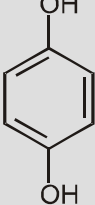
30. Draw the resonating structure and resonance hybrid of following species?

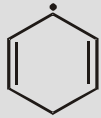
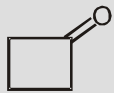
S. No.	Compounds	Resonating Hybrid	S. No.	Compounds	Resonating Hybrid
1.			2.		
3.			4.		
5.					
6.			7.		
8.			9.		
10.			11.		
12.			13.		
14.					

31.

S.No.	Resonating Structure	Relative Stability
1.		
2.		
3.	$\text{CH}_3 - \overset{\text{O}}{\parallel}{\text{C}} - \text{OH} , \text{CH}_3 - \overset{\text{O}}{\parallel}{\text{C}} - \text{OH} , \text{CH}_3 - \overset{\text{O}}{\parallel}{\text{C}} = \ddot{\text{O}} - \text{H}$	
4.		
5.	$\begin{array}{c} \text{O} \quad \text{C} \quad \text{O} \\ \parallel \quad \parallel \\ \text{O} = \text{C} - \text{O} \end{array} ; \begin{array}{c} \text{O} \quad \text{C} - \text{O} \\ \parallel \quad \parallel \\ \text{O} - \text{C}^{+2} - \text{O} \end{array}$	
6.	$\begin{array}{c} \text{O} \\ \parallel \\ \text{H} - \text{O} - \text{C} - \text{NH}_2 \end{array} ; \begin{array}{c} \text{O} \\ \parallel \\ \text{H} - \text{O} = \text{C} - \text{NH}_2 \end{array} ;$ $\begin{array}{c} \text{O} \\ \parallel \\ \text{H} - \text{O} - \text{C} = \text{NH}_2 \end{array} ; \begin{array}{c} \text{O} \\ \parallel \\ \text{H} - \text{O} - \text{C} - \text{NH}_2 \end{array}$	
7.		
8.		

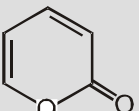
32.

S.No.	Compound	Huckel Rule ($4n + 2$) e	Planarity	Cyclic	Complete delocalisation	Nature
1.						
2.						
3.						
4.						
5.						
6.						
7.						
8.						

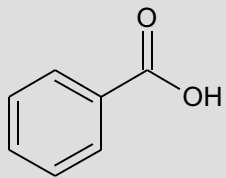
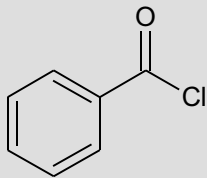
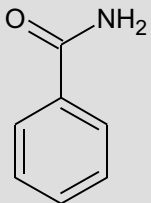
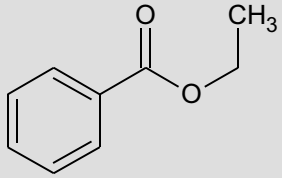
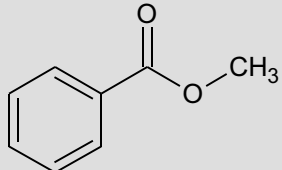
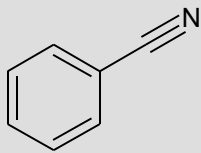
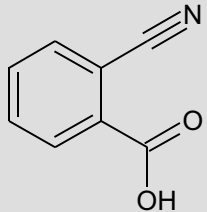
S.No.	Compound	Huckel Rule ($4n + 2$) e	Planarity	Cyclic	Complete delocalisation	Nature
9.						
10.						

33.

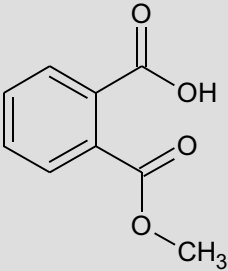
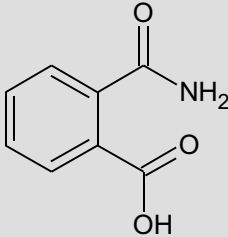
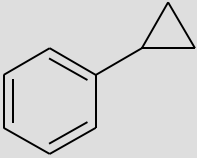
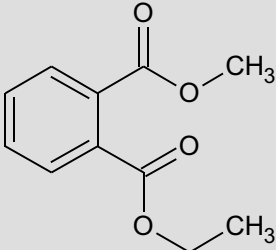
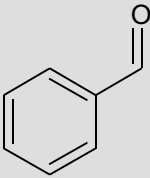
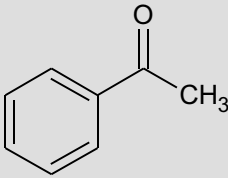
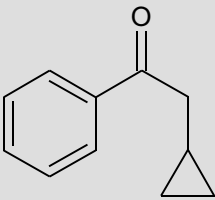
S.N.	Compounds	Huckel Rule ($4n + 2$) e	Planarity	Cyclic	Complete delocalisation	Nature
1.						
2.						
3.						
4.						
5.						
6.						
7.						
8.						
9.						
10.						

11.						
12.						

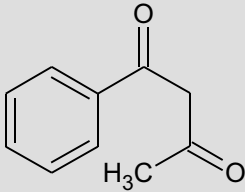
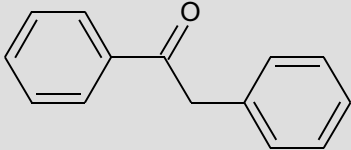
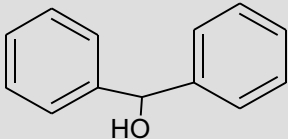
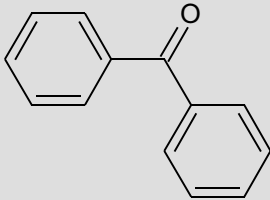
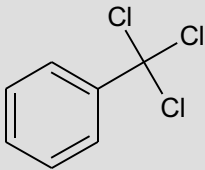
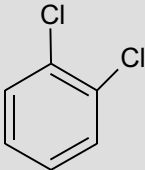
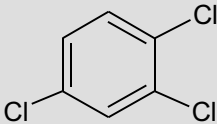
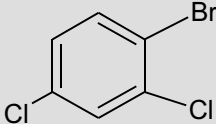
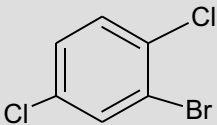
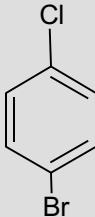
34. Write the IUPAC name of the compounds given in the blank box.

S. No.	Compounds	S. No.	Compounds
1.		2.	
3.		4.	
5.		6.	
7.			

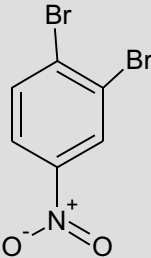
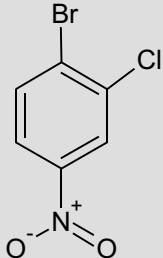
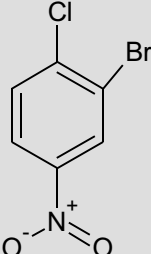
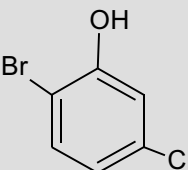
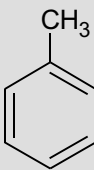
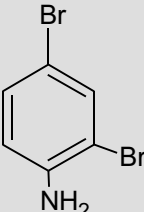
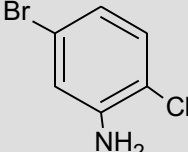
35. Write the IUPAC name of the compounds given in the blank box.

S.No.	Compounds	S.No.	Compounds
1.		2.	
3.		4.	
5.		6.	
7.			

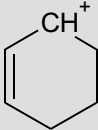
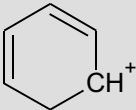
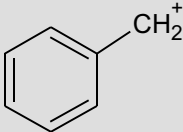
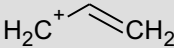
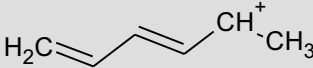
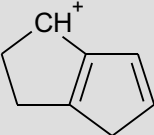
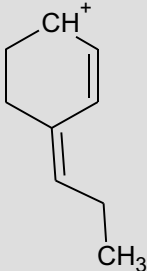
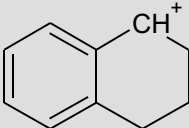
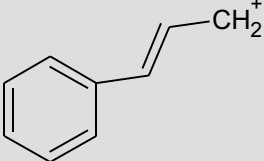
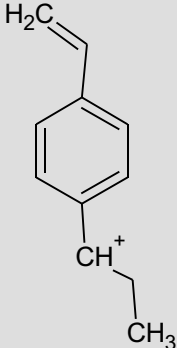
36. Write the IUPAC name of the compounds given in the blank box.

S.No.	Compounds	S.No.	Compounds
1.		2.	
3.		4.	
5.		6.	
7.		7.	
9.		10.	

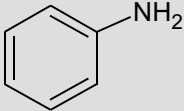
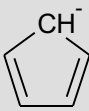
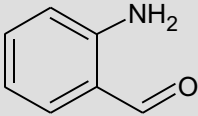
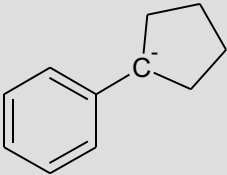
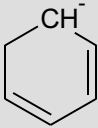
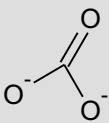
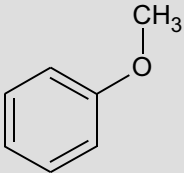
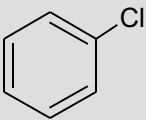
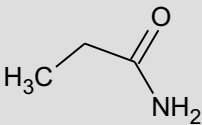
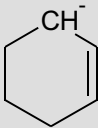
37. Write the IUPAC name of the compounds given in the blank box.

S. No.	Compound	S. No.	Compound
1.		2.	
3.		4.	
5.		6.	
7.			

38. Draw Resonance hybrid of following ?

1.		2.	
3.		4.	
5.		6.	
7.		8.	
9.		10.	
11.			

39. Draw the resonating structure and resonance hybrid of following species?

S. No.	Compounds	Resonating Structure	S. No.	Compounds	Resonating Structure
1.	$\text{H}_2\text{C}^-\text{CH}=\text{CH}_2$		2.	$\text{H}_3\text{C}-\text{C}(=\text{O})\text{O}^-$	
3.			4.		
5.			6.		
7.			8.		
9.			10.		
11.	$\text{H}_2\text{C}^-\text{CH}=\text{CH}-\text{CH}=\text{CH}-\text{CH}_2-\text{CH}_3$				
12.			13.		

S. No.	Compounds	Resonating Structure	S. No.	Compounds	Resonating Structure
14.			15.		
16.			17.		
18.					

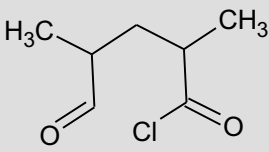
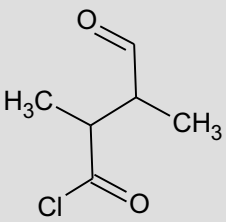
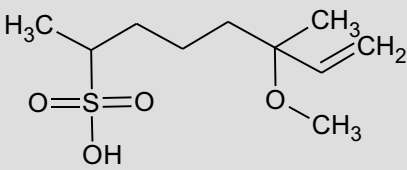
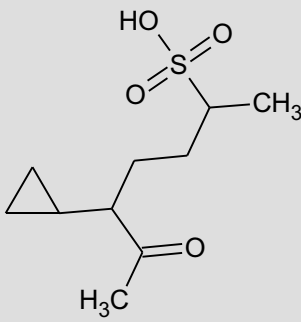
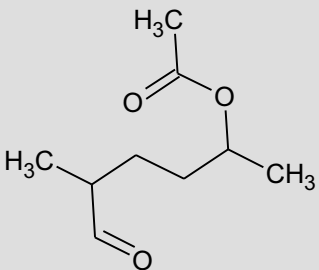
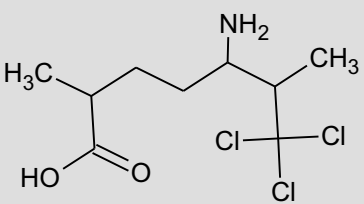
40. Write the IUPAC name of the compounds given in the blank box.

S. No.	Compounds	S. No.	Compounds
1.		2.	
3.		4.	

S. No.	Compounds	S. No.	Compounds
5.		6.	
7.		8.	
9.		10.	
11.		12.	
13.		14.	

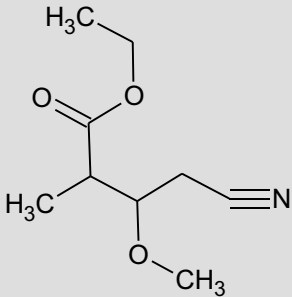
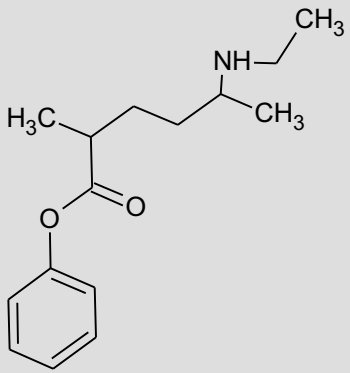
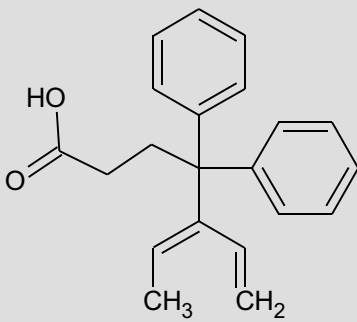
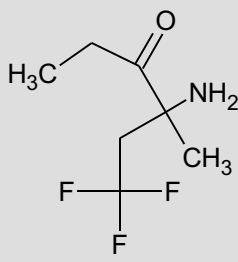
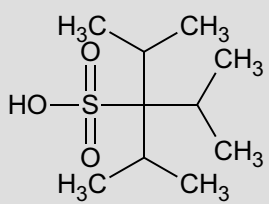
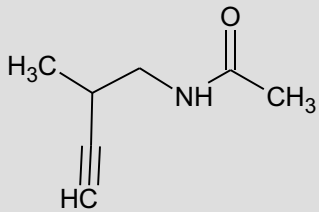
41. Write the IUPAC name of the compounds given in the blank box.

S. No.	Compounds	S. No.	Compounds
1.		2.	
3.		4.	
5.		6.	
7.		8.	

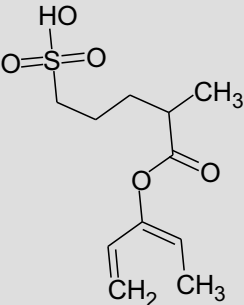
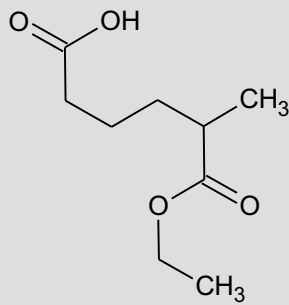
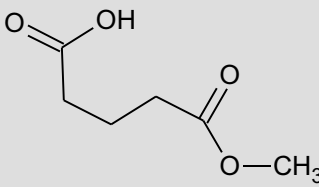
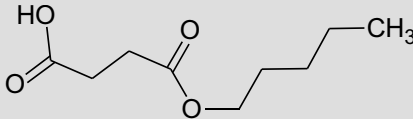
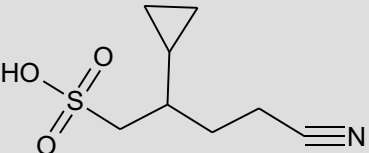
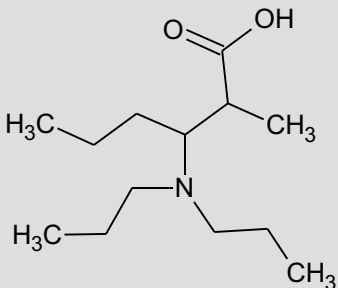
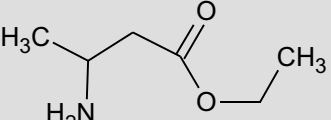
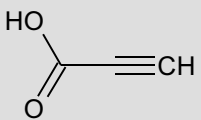
9.		10.	
11.		12.	
13.		14.	

42. Write the IUPAC name of the compounds given in the blank box.

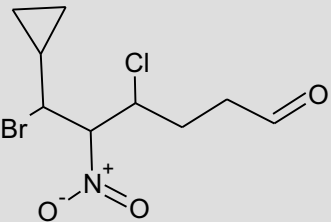
S. No.	Compounds	S. No.	Compounds
1.		2.	
3.		4.	
5.		6.	

S. No.	Compounds	S. No.	Compounds
7.		8.	
9.		10.	
11.		12.	

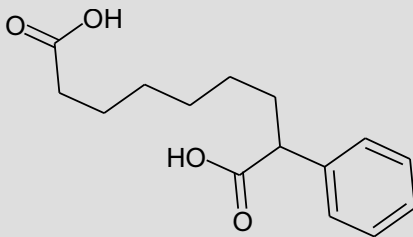
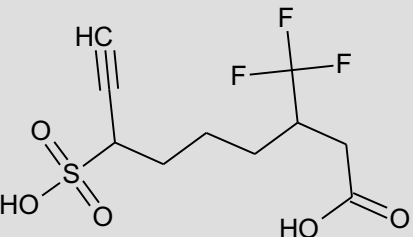
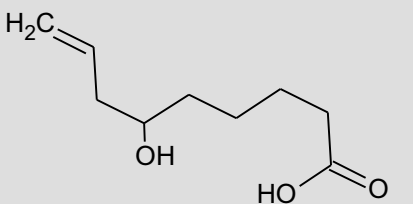
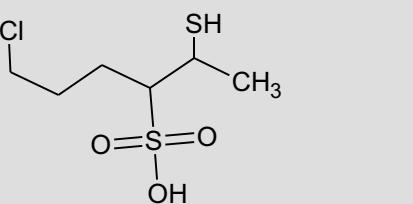
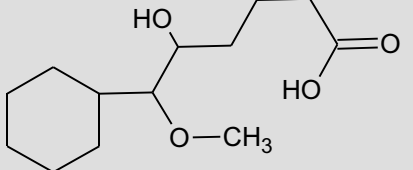
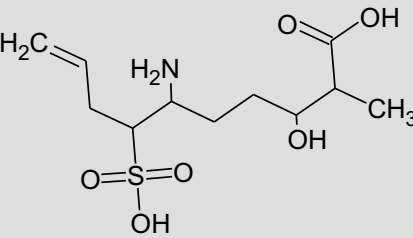
43. Write the IUPAC name of the compounds given in the blank box.

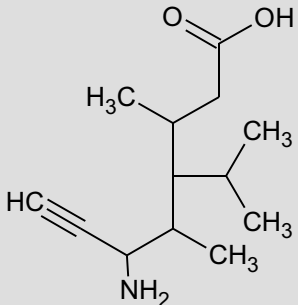
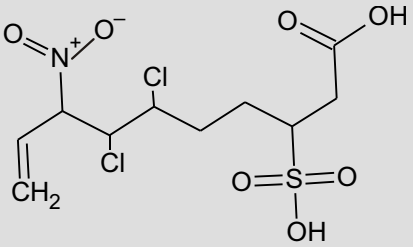
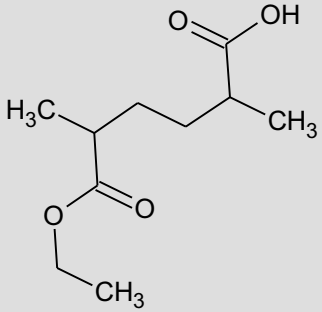
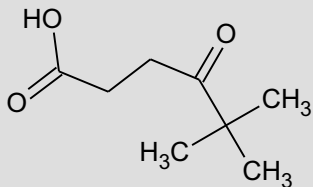
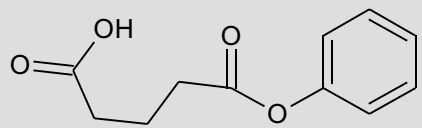
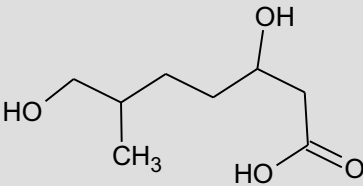
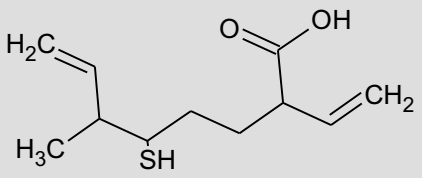
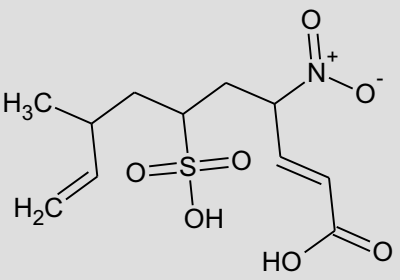
S. No.	Compounds	S. No.	Compounds
1.		2.	
3.		4.	
5.		6.	
7.		8.	

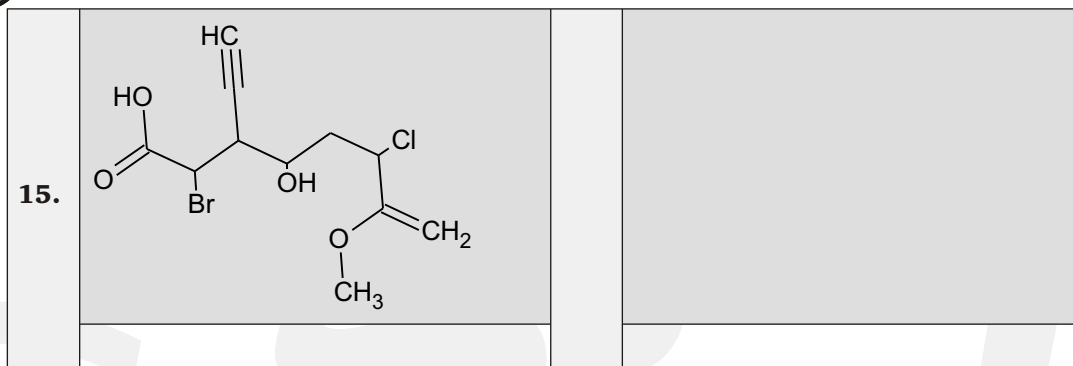
9.		10.	
11.		12.	
13.		14.	
15.		16.	

17.			
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44. Write the IUPAC name of the compounds given in the blank box.

S. No.	Compounds	S. No.	Compounds
1.		2.	
3.		4.	
5.		6.	

7.		8.	
9.		10.	
11.		12.	
13.		14.	

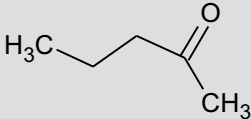
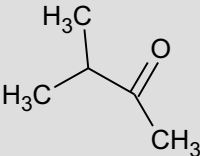
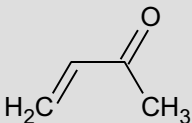
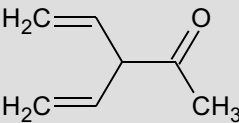
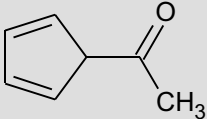
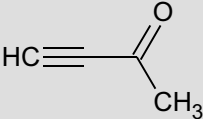
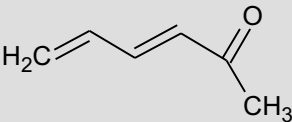
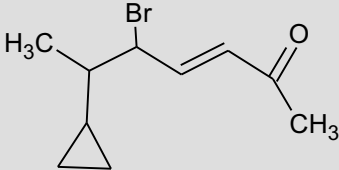
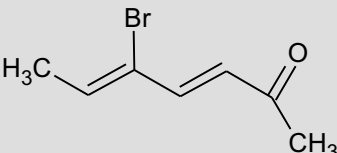


45. Write the IUPAC name of the compounds given in the blank box.

S. No.	Compounds	S. No.	Compounds
1.		2.	
3.		4.	
5.		6.	

S. No.	Compounds	S. No.	Compounds
7.		8.	
9.		10.	
11.		12.	
13.		14.	
15.			

46. Write the IUPAC name of the compounds given in the blank box.

S. No.	Compounds	S. No.	Compounds
1.		2.	
3.		4.	
5.		6.	
7.		8.	
9.		10.	

47. Write the IUPAC name of the compounds given in the blank box.

S. No.	Compounds	S. No.	Compounds
1.		2.	
3.		4.	
5.		6.	
7.		8.	

48. Write the IUPAC name of the compounds given in the blank box.

S. No.	Compounds	S. No.	Compounds
1.		2.	

3.		4.	
5.		6.	
7.			

49. Write the IUPAC name of the compounds given in the blank box.

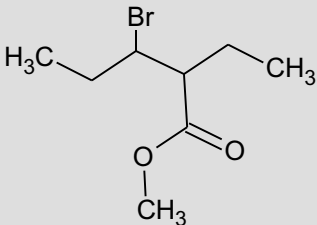
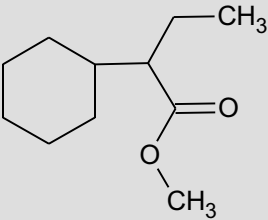
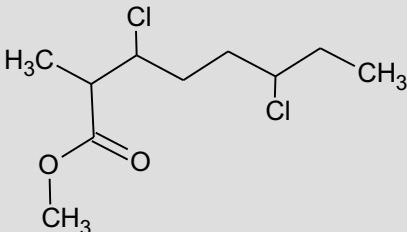
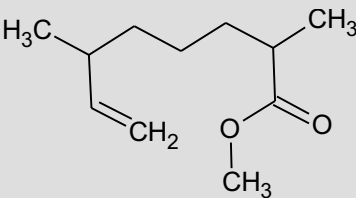
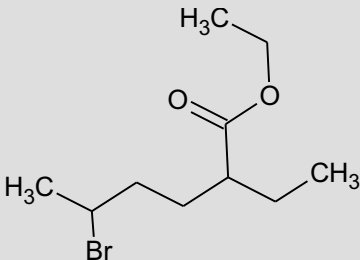
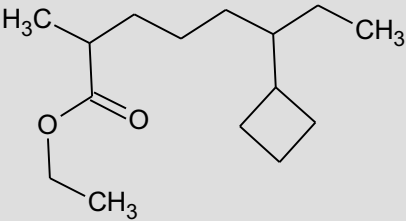
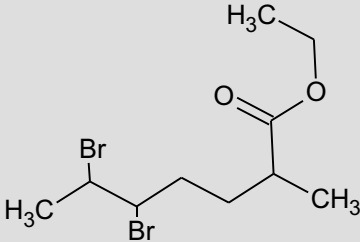
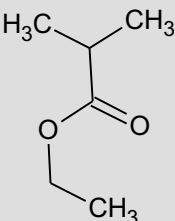
S. No.	Compounds	S. No.	Compounds
1.		2.	

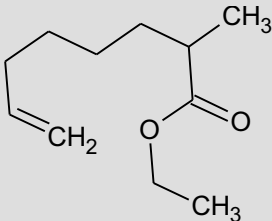
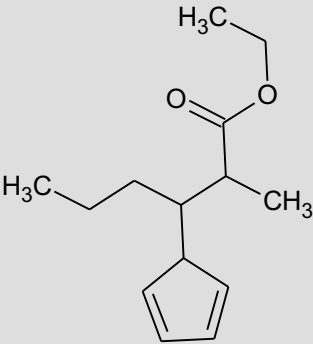
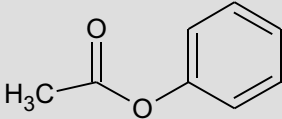
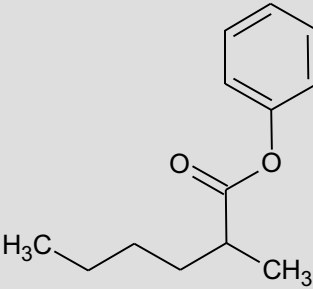
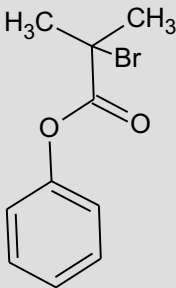
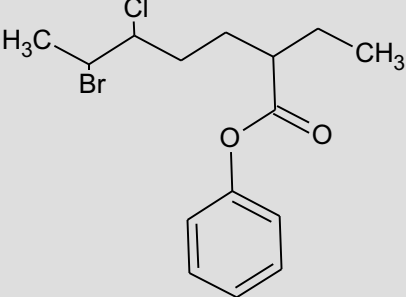
S. No.	Compounds	S. No.	Compounds
3.		4.	
5.		6.	
7.		8.	
9.		10.	

S. No.	Compounds	S. No.	Compounds
11.		12.	
13.		14.	
15.			

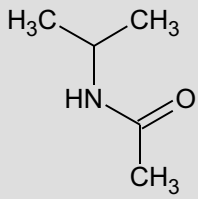
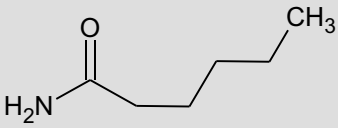
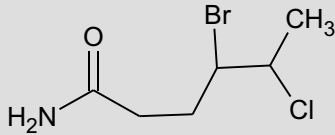
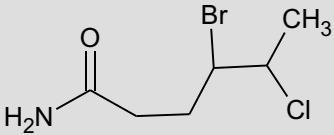
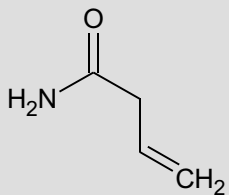
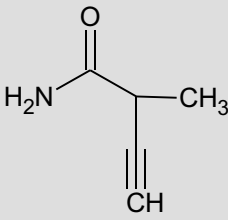
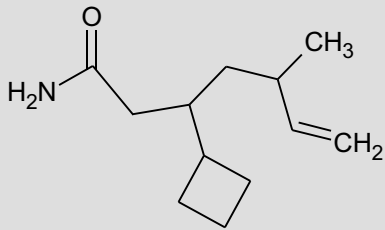
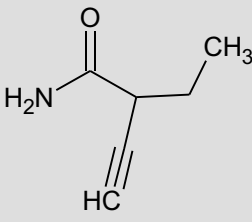
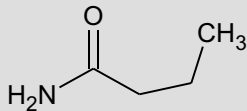
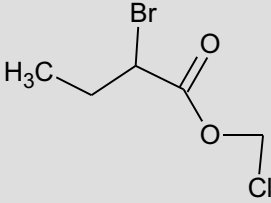
50. Write the IUPAC name of the compounds given in the blank box.

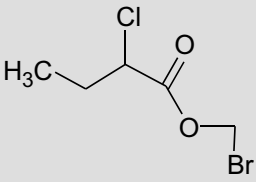
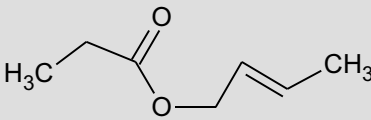
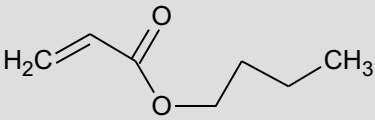
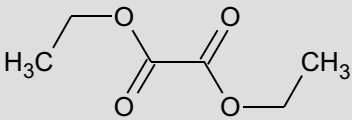
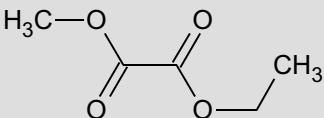
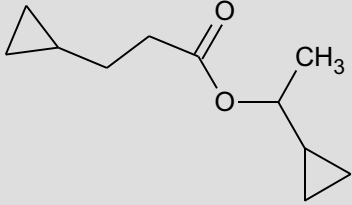
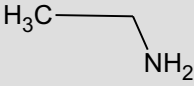
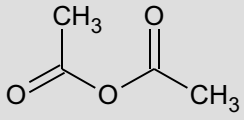
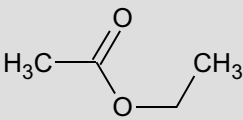
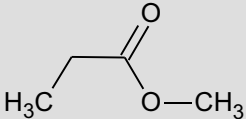
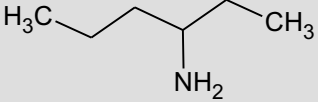
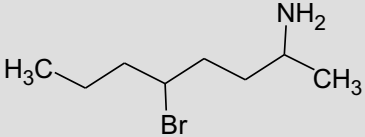
S. No.	Compounds	S. No.	Compounds
1.		2.	

S. No.	Compounds	S. No.	Compounds
3.		4.	
5.		6.	
7.		8.	
9.		10.	

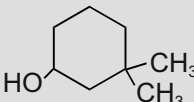
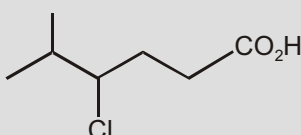
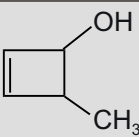
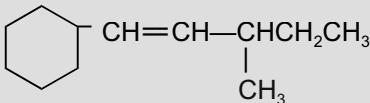
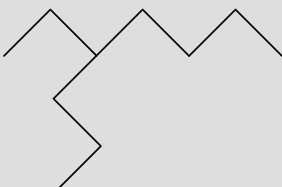
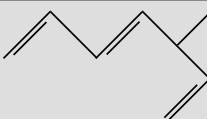
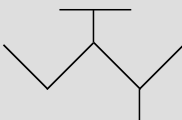
S. No.	Compounds	S. No.	Compounds
11.		12.	
13.		14.	
15.		16.	

51. Write the IUPAC name of the compounds given in the blank box.

S. No.	Compounds	S. No.	Compounds
1.		2.	
3.		4.	
5.		6.	
7.		8.	
9.		10.	

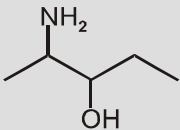
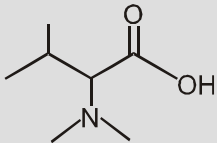
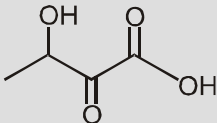
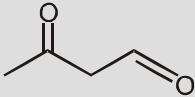
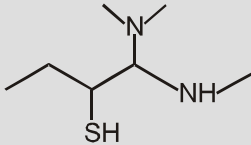
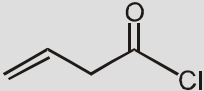
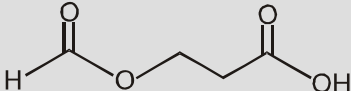
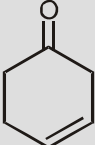
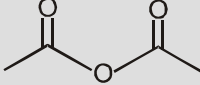
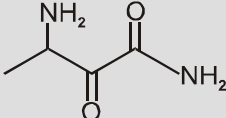
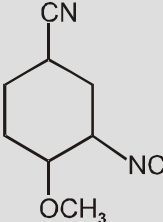
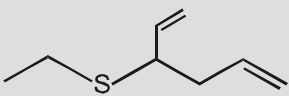
S. No.	Compounds	S. No.	Compounds
11.		12.	
13.		14.	
15.		16.	
17.		18.	
19.		20.	
21.		22.	

52. Count the number of primary, secondary, tertiary, quaternary carbon as well as hydrogen in given compound :

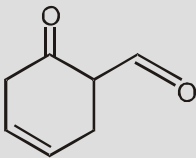
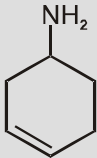
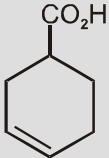
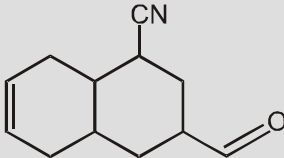
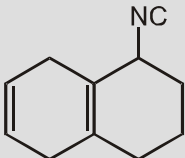
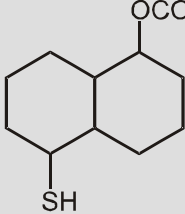
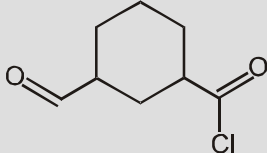
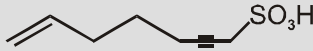
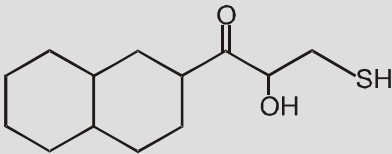
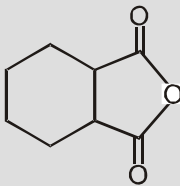
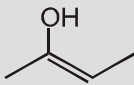
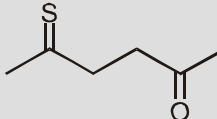
S. No.	Compounds	1°C	2°C	3°C	4°C	1°H	2°H	3°CH
1.	$ \begin{array}{c} \text{CH}_3 \\ \\ \text{CH}_3\text{CH}_2\text{CHCHCHCH}_2\text{CH}_2\text{CHCH}_3 \\ \qquad \qquad \\ \text{CH}_2\text{CH}_3 \quad \text{CH}_2\text{CH}_3 \end{array} $							
2.	$ \begin{array}{ccc} \text{CH}_2 & \text{CH} & \text{CH}_2 \\ & & \\ \text{OH} & \text{OH} & \text{OH} \end{array} $							
3.								
4.								
5.								
6.								
7.								
8.								
9.								

10.								
11.	 $\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$							

53. Circle the functional groups and give the name in the following structures ?

1.		2.	
3.		4.	
5.		6.	
7.		8.	
9.		10.	
11.		12.	
13.	HCOOCH_3	14.	CH_3COOH
15.			

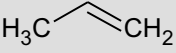
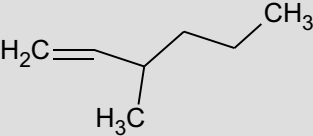
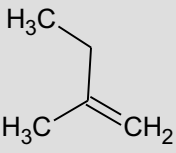
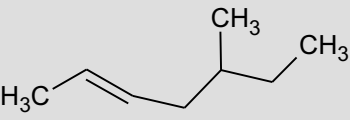
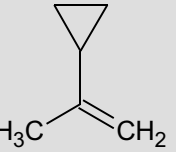
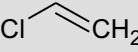
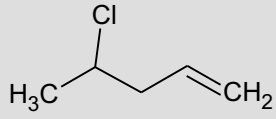
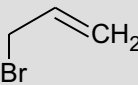
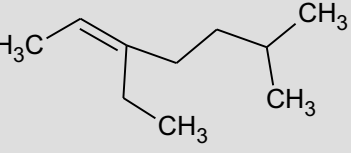
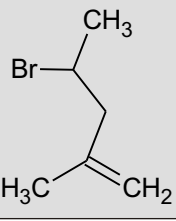
54. Circle the functional groups in the following compounds & find out the DBE value of each compound ?

1.		2.	
3.		4.	
5.		6.	
7.		8.	
9.		10.	
11.		12.	

55.

S. No.	Compounds	Name of Functional Group	DBE Value
1.			
2.			
3.			
4.			
5.			
6.			

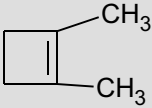
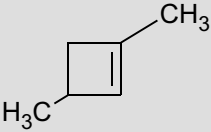
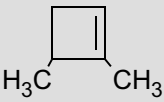
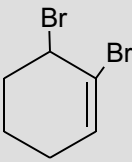
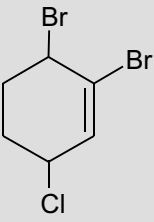
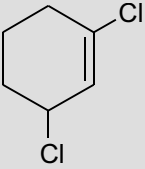
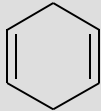
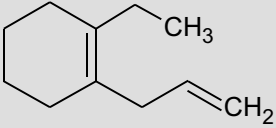
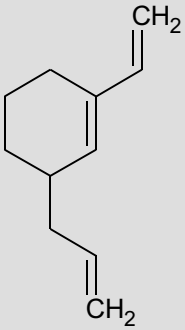
56. Write the IUPAC name of the compounds given in the blank box.

S. No.	Compounds	S. No.	Compounds
1.		2.	
3.		4.	
5.		6.	
7.		8.	
9.		10.	

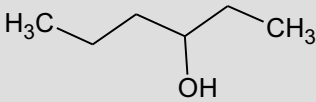
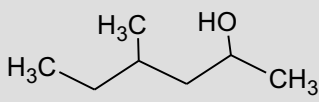
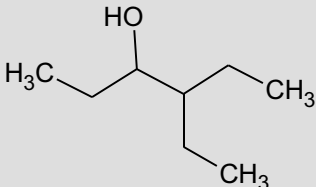
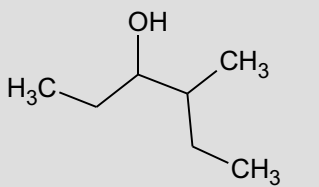
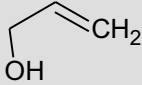
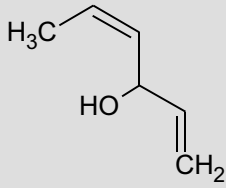
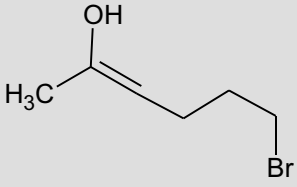
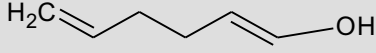
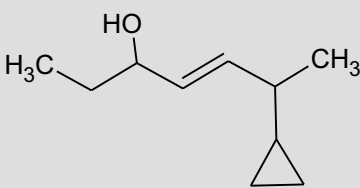
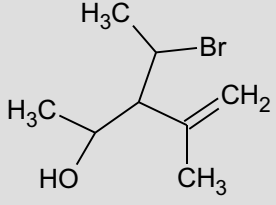
57. Write the IUPAC name of the compounds given in the blank box.

S. No.	Compounds	S. No.	Compounds
1.		2.	
3.		4.	
5.		6.	
7.		8.	
9.		10.	
11.			

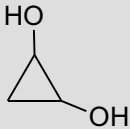
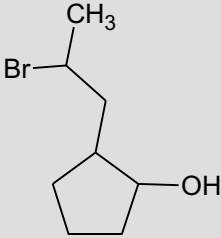
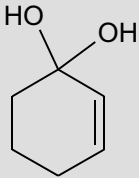
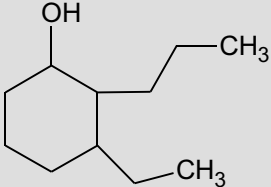
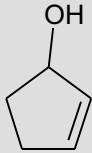
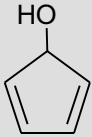
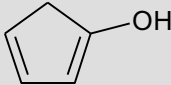
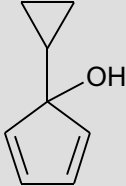
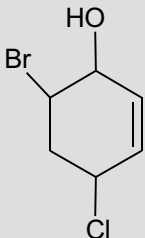
58. Write the IUPAC name of the compounds given in the blank box.

S.No.	Compounds	S.No.	Compounds
1.		2.	
3.		4.	
5.		6.	
7.		8.	
9.		10.	

59. Write the IUPAC name of the compounds given in the blank box.

S.No.	Compounds	S.No.	Compounds
1.		2.	
3.		4.	
5.		6.	
7.		8.	
9.		10.	

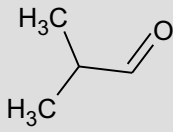
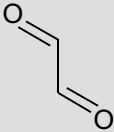
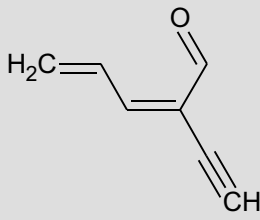
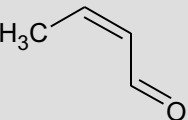
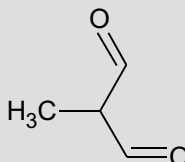
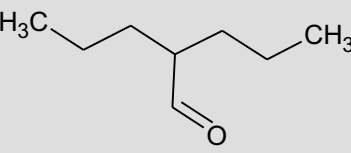
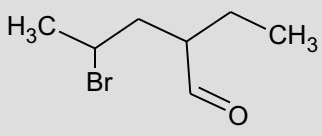
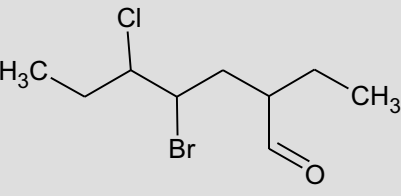
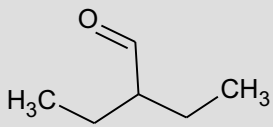
60. Write the IUPAC name of the compounds given in the blank box.

S. No.	Compounds	S. No.	Compounds
1.		2.	
3.		4.	
5.		6.	
7.		8.	
9.			

61. Write the IUPAC name of the compounds given in the blank box.

S. No.	Compounds	S. No.	Compounds
1.		2.	
3.		4.	
5.		6.	
7.		8.	

62. Write the IUPAC name of the compounds given in the blank box.

S. No.	Compounds	S. No.	Compounds
1.		2.	
3.		4.	
5.		6.	
7.		8.	
9.			

Answers

Comprehension Based Problems:

Passage-1:	1.	(b)	2.	(c)	3.	(b)	4.	(b)		
Passage-2:	1.	(c)	2.	(b)	3.	(c)	4.	(a)		
Passage-3:	1.	(c)	2.	(d)						
Passage-4:	1.	(c)	2.	(b)	3.	(d)				
Passage-5:	1.	(a)	2.	(d)						

Matching Type Problems:

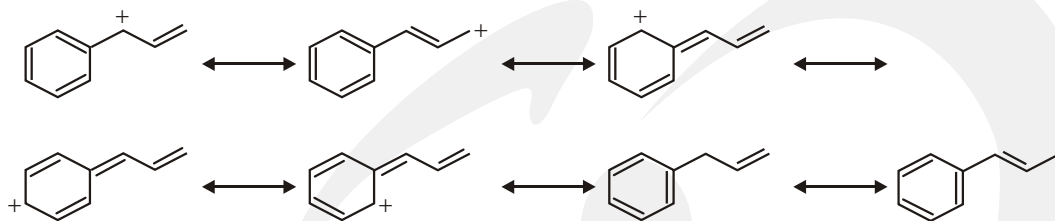
1. a r; b t; c t; d s
2. (a) q, s; (b) q, r; (c) q, r; (d) q, r
3. (a) s; (b) q; (c) s; (d) r
4. (a) t; (b) s; (c) r; (d) q; (e) p
5. (a) p; (b) r; (c) q; (d) p
6. (a) q; (b) p; (c) q; (d) r
7. (a) p,q,s; (b) p,r,s; (c) s; (d) s
8. (a) p, (b) q, (c) q, (d) r
9. (a) p, (b) q, (c) s, (d) r
10. (a) p, r; (b) p, r; (c) p, r; (d) p, r
11. (a) p; (b) q; (c) r; (d) s
12. (a) p; (b) r; (c) s; (d) q
13. (a) p, r, s; (b) p, q, r; (c) p, r; (d) p, q, r, s

Multi Concept Subjective Problems

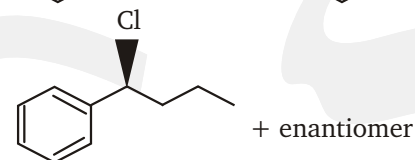
1.	7	2.	21	3.	11	4.	4	5.	3	6.	5	7.	8	8.	12	9.	5	10.	5
11.	2	12.	4	13.	3	14.	2	15.	4	16.	4	17.	8	18.	3	19.	6	20.	4
21.	3	22.	5	23.	6	24.	4	25.	5	26.	8	27.	4	28.	6	29.	5	30.	9
31.	8	32.	65	33.	7	34.	4	35.	17	36.	7	37.	2	38.	6	39.	5	40.	7
41.	3	42.	6	43.	3	44.	6												

Unified Conceptual Problems

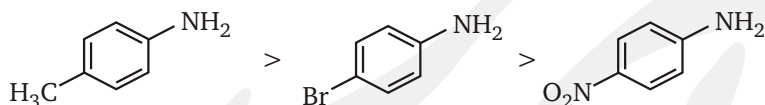
1.



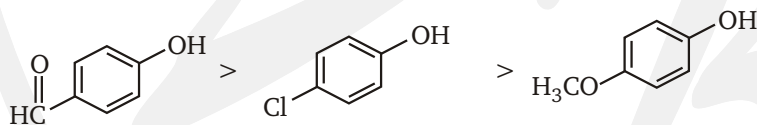
2.



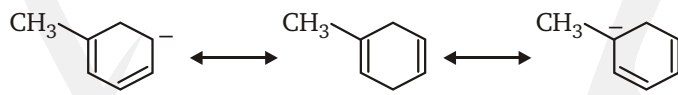
3.



4.



5.



6. Provide 1, 2, 3 and 4 carbons and hydrogens in each structure:

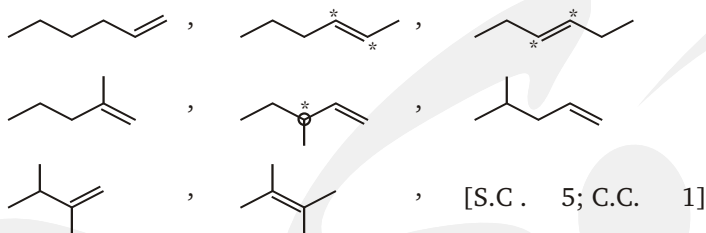
Structures	1		2		3		4
	C	H	C	H	C	H	C
	2	6	4	8	0	0	0
	3	9	2	4	1	1	0
	3	9	2	4	1	1	0
	4	12	1	2	0	0	1
	4	12	0	0	2	2	0

7.

S.No.	Molecular Formula	Number of Structural Isomers
1.	C_4H_8	5
2.	C_3H_6	2
3.	C_6H_{12} (having cyclobutane ring)	4
4.	C_4H_6 (cyclic)	5
5.	C_7H_{16}	9
6.	$C_4H_{10}O$	7
7.	C_6H_{14}	5
8.	C_3H_9N	4
9.	C_6H_{12} (cyclopropane ring only)	6
10.	$C_3H_6Cl_2$	4
11.	C_8H_{10} (only one benzene ring)	4
12.	C_7H_8O (only one benzene ring)	5
13.	C_5H_{12}	3
14.	C_7H_{14} (cyclobutane ring)	8
15.	$C_5H_{10}O$ (only aldehyde and ketone)	7
16.	C_5H_{10} (cyclic)	5
17.	$C_5H_{11}Br$	8
18.	$C_4H_{11}N$	8
19.	$C_3H_5Br_3$	5
20.	C_7H_7Cl (only one benzene ring)	4
21.	$C_5H_{12}O$ (only ether)	6

8. Short abbreviations * (S.C. = Stereocenter) (C.C. = Chiral center)

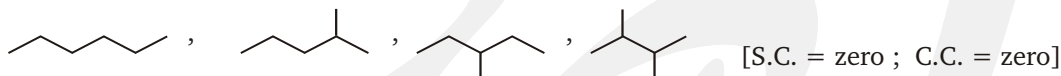
(i)



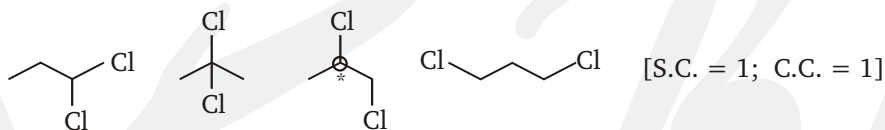
(ii)



(iii)



(iv)



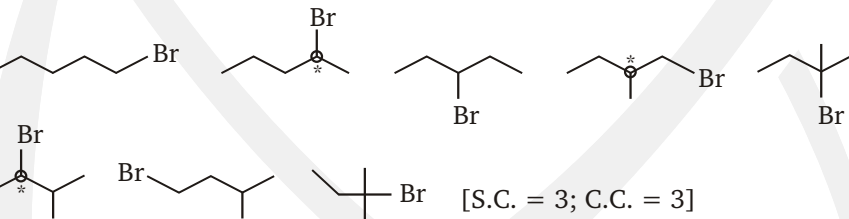
(v)



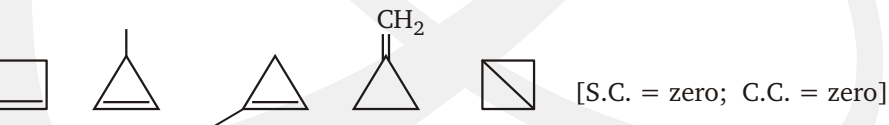
(vi)



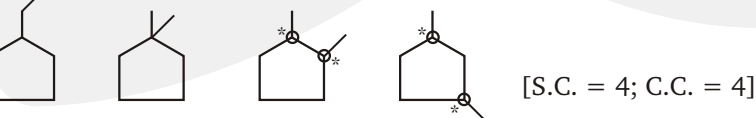
(vii)

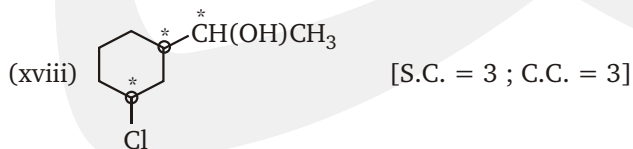
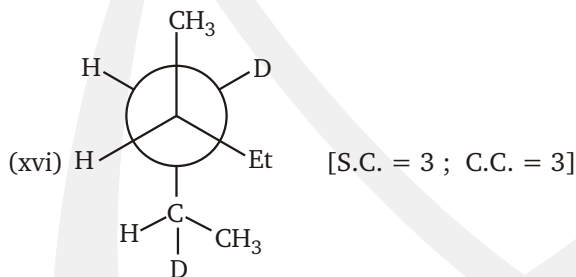
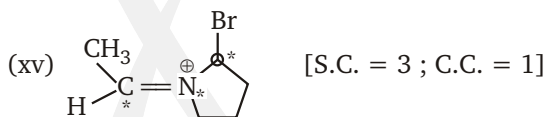
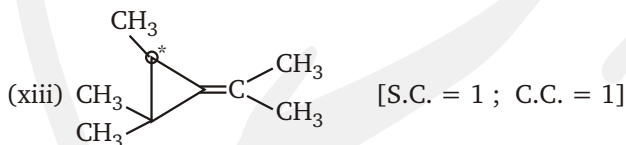
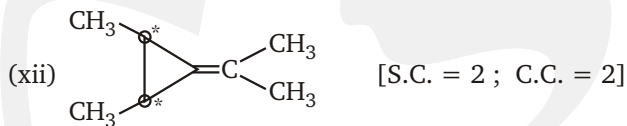
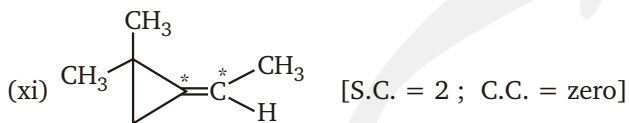
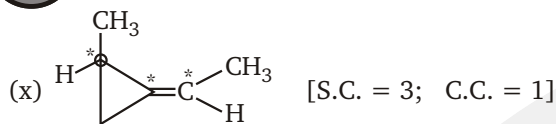


(viii)



(ix)





9.

- | | | | | |
|-------------|-------------|--------------|------------|---------|
| (i) 4 1 3 2 | (ii) 2 3 1 | (iii) 1 2 3 | (iv) 1 2 3 | (v) 1 2 |
| (vi) 1 2 | (vii) 3 2 1 | (viii) 1 2 3 | (ix) 1 2 | (x) 1 2 |
| (xi) 1 2 3 | (xii) 2 1 | | | |

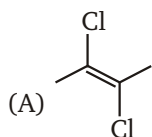
10.

- | | | | | |
|-----------|--------------|---------------|------------|-----------|
| (i) 1 2 3 | (ii) 1 2 3 4 | (iii) 1 2 3 4 | (iv) 2 1 | (v) 2 1 3 |
| (vi) 2 1 | (vii) 2 1 | (viii) 1 2 3 | (ix) 1 2 3 | (x) 2 1 |

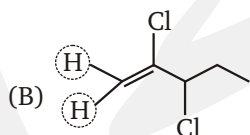
11. (3)

Strategy

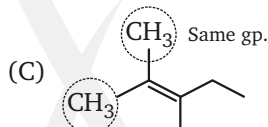
(1) Restricted rotation

(2) l_1 l_2 **G.I.**

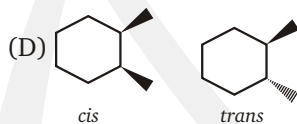
✓

 $(l_1 \quad l_2)$

✗

 $(l_1 \quad l_2)$

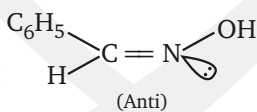
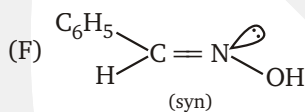
✗



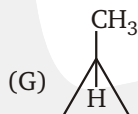
✓

 $(l_1 \quad l_2)$

✗



✓

 $(l_1 \quad l_2)$

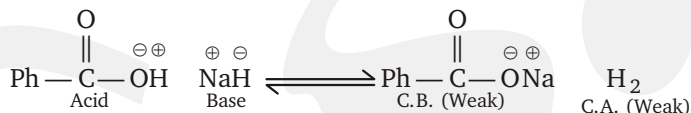
✗

12. (9)

Strategy Acid-Base reaction always **favour** the formation of **weaker acid and weaker base**.



(A) NH_3 gas



(B) H_2 gas

Mol. mass

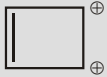
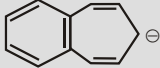
(A) NH_3 17

(B) H_2 2

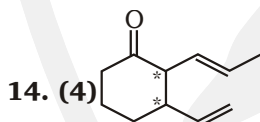
A B 19

A B 10 9

13.

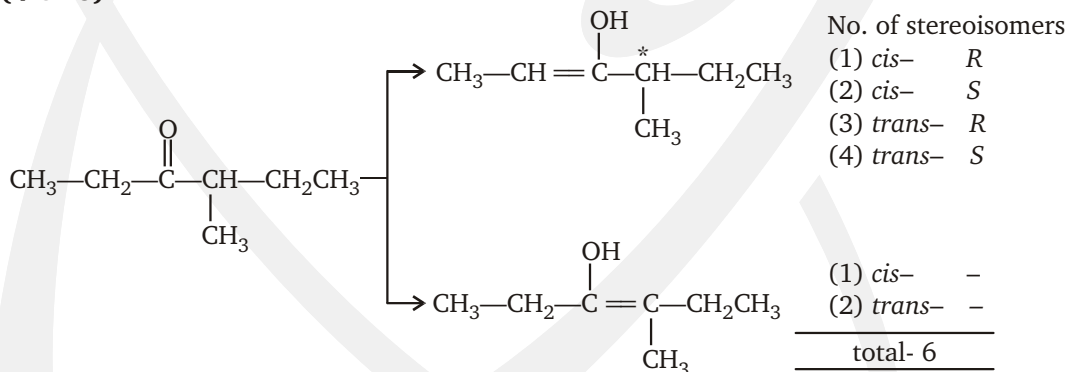
Compound	No. of e	Value of ' n '	Nature
	2	0	A
	4	1	N.A.
 Tropolium ion	6	1	A
	12	3	AA
	6	1	A

	14	3	A Perimeter Reso. Present
 (Tub shaped molecule)	8	2	N.A.
	5	1	N.A. (Hybrid state of one of the carbon is sp^3)
	6	1	A



Total stereoisomers of given compound $2^3 = 8$
but when alkenyl substituent have *cis* then number of S.I. 4.

15. (4 or 6)



Total enols possible are 6.

16. Compounds that show geometrical isomerism are:

1, 2, 4, 5, 6, 8, 11, 12, 13, 14, 15, 17, 20, 21, 22, 23

17.

- | | | | | |
|---------------|-----------------|-----------------|---------------|------------------|
| 1. <i>cis</i> | 2. <i>trans</i> | 3. <i>trans</i> | 4. none | 5. <i>cis</i> |
| 6. <i>cis</i> | 7. <i>trans</i> | 8. <i>trans</i> | 9. <i>cis</i> | 10. <i>trans</i> |

11. cis 12. none 13. none 14. cis 15. cis
 16. none 17. cis 18. none 19. cis 20. trans
 21. trans

18.

1.	2	2.	2	3.	3	4.	4	5.	4	6.	8	7.	6	8.	2	9.	2	10.	3
11.	2	12.	2	13.	0														

19.

1.	1	2.	2	3.	3	4.	1	5.	4	6.	1	7.	2	8.	3	9.	0	10.	1
11.	3	12.	1	13.	1	14.	1	15.	1	16.	8	17.	4						

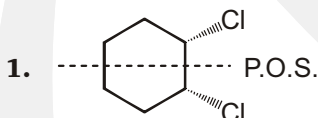
20.

1. S 2. R 3. R 4. R 5. 2-R, 3S
 6. 2-S, 3-S 7. 2-R, 3-R 8. 2-S, 3-R 9. R 10. R
 11. 2-S, 3R 12. 3-S, 3-S 13. S 14. R

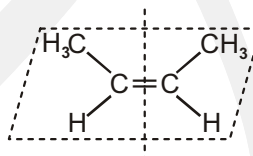
21.

1. Enantiomer 2. Enantiomer 3. Identical
 4. Identical 5. Enantiomer 6. Identical
 7. Diastereomer 8. Diastereomer 9. Diastereomers
 10. Enantiomer 11. Positional isomers 12. Diastereomers
 13. Diastereomers 14. A, B diastereomers
 A, C diastereomers
 A, D enantiomers
 B, C diastereomers

22.



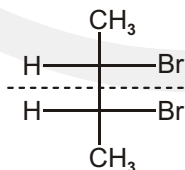
2.

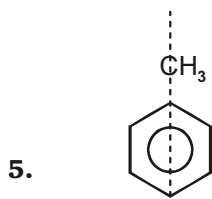


Vertical + molecular P.O.S.

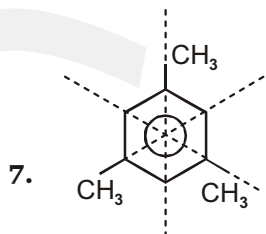
3. No. P.O.S.

4.

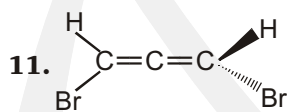
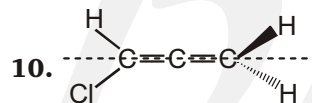
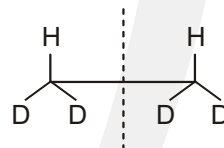
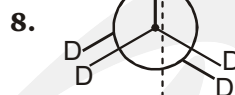
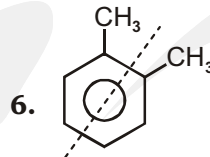
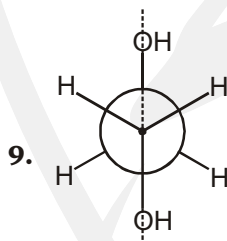




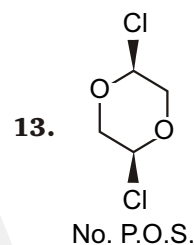
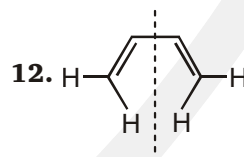
Molecular + vertical



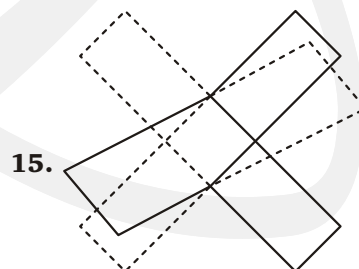
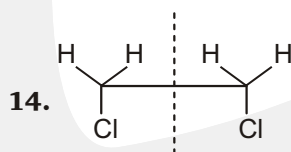
+ molecular P.O.S.



No. P.O.S.



No. P.O.S.



+ molecular P.O.S.

23. The molecules having centre of symmetry are

1, 3, 6, 7, 8, 12, 15

24.

Compound	Chiral carbon	POS	COS	Chiral compound	Optically active	Optical isomerism	G.I.
1.	×	✓	×	×	×	×	×
2.	×	×	×	✓	✓	✓	×
3.	×	✓	×	×	×	×	×
4.	×	✓	×	×	×	×	✓
5.	×	✓	×	×	×	×	×
6.	×	×	×	✓	✓	✓	×
7.	✓	×	×	✓	✓	✓	✓
8.	✓	×	×	✓	✓	✓	✓

25.

Compound	Chiral carbon	POS	COS	Chiral compound	Optically active	Optical isomerism	G.I.
1.	×	✓	×	×	×	×	×
2.	×	×	×	✓	✓	✓	×
3.	×	✓	×	×	×	×	×
4.	×	✓	×	×	×	×	✓
5.	×	✓	×	×	×	×	×
6.	×	×	×	✓	✓	✓	×
7.	×	×	×	✓	✓	✓	×

26.

Compound	Total number of stereoisomers	Number of meso compound	Number of optically active isomers
1.	2	0	0
2.	4	0	4
3.	2	0	2
4.	2	0	2

[illegible]

28. A B 90

29. 3

30.

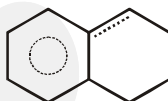
1.



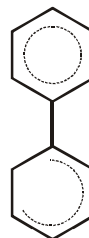
2.



3.



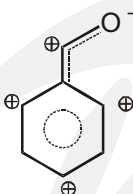
4.



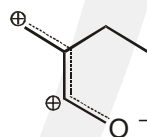
5.



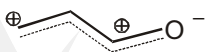
6.



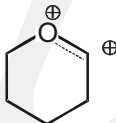
7.



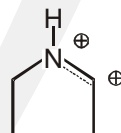
8.



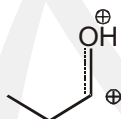
9.



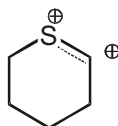
10.



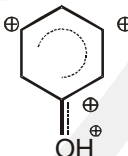
11.



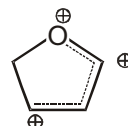
12.



13.



14.



31.

1. I > II

2. II > I

3. I > III > II

4. I > II > III

5. I > II > III > IV

6. I > III > II > IV

7. I > II

8. I > II > III

32.

1.	6	✓	✓	✓	Aromatic	2.	6	✓	✓	✓	Aromatic
3.	6	✓	✓	✓	Aromatic	4.			✓		Non-Aromatic
5.			✓		Non-Aromatic	6.			✓		Non-Aromatic

7.		✓	✓	✓	Anti-Aromatic	8.	6	✓	✓	✓	Aromatic
9.			✓		Not-Aromatic	10.			✓		Non-Aromatic

33.

1.			✓		Non-Aromatic	2.		✓	✓	✓	Anti-Aromatic
3.	2	✓	✓	✓	Aromatic	4.	2	✓	✓	✓	Aromatic
5.	6	✓	✓	✓	Aromatic	6.			✓		Non-Aromatic
7.	6	✓	✓	✓	Aromatic	8.		✓	✓	✓	Anti-Aromatic
9.		✓	✓	✓	Anti-Aromatic	10.	6	✓	✓	✓	Aromatic
11.		✓		✓	Non-Aromatic	12.	6	✓	✓	✓	Aromatic

34.

1. benzenecarboxylic acid or benzoic acid
2. benzenecarbonyl chloride or benzoyl chloride
3. benzenecarboxamide or benzamide
4. ethyl benzene carboxylate or ethyl benzoate
5. methyl benzene carboxylate or methyl benzoate
6. benzenecarbonitrile
7. 2-cyano benzenecarboxylic acid or 2-cyanobenzoic acid

35. 1. 2-(methoxycarbonyl)benzenecarboxylic acid

2. 2-carbamoyl benzenecarboxylic acid
3. cyclopropylbenzene
4. ethyl methyl benzene-1,2-dicarboxylate
5. benzene carbaldehyde or benzaldehyde
6. 1-phenylethanone
7. 2-cyclopropyl-1-phenylethanone

36. 1. 1-phenylbutane-1,3-dione

3. diphenylmethanol
5. (trichloromethyl)benzene
7. 1,2,4-trichlorobenzene
9. 2-bromo-1,4-dichlorobenzene

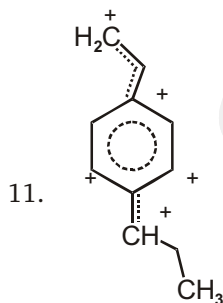
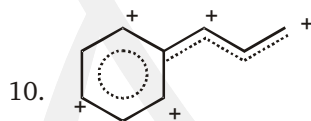
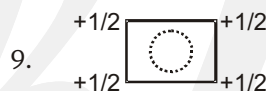
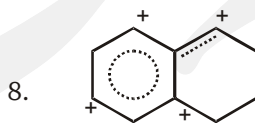
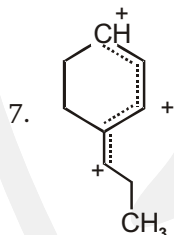
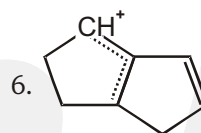
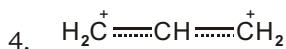
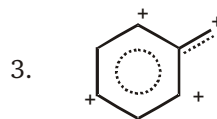
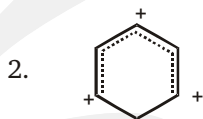
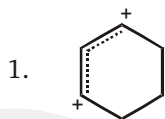
2. 1,2-diphenylethanone

4. diphenylmethanone
6. 1,2-dichlorobenzene
8. 1-bromo-2,4-dichlorobenzene
10. 1-bromo-4-chlorobenzene

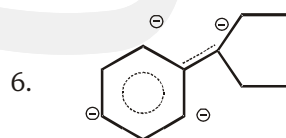
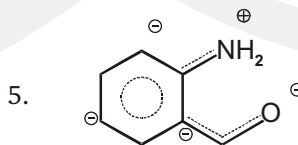
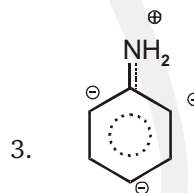
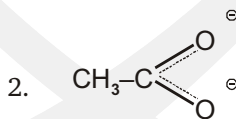
37. 1. 1,2-dibromo-4-nitrobenzene
 3. 2-bromo-1-chloro-4-nitrobenzene
 5. toluene or methylbenzene
 7. 5-bromo-2-chloro aniline

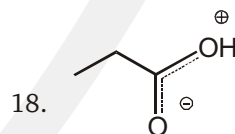
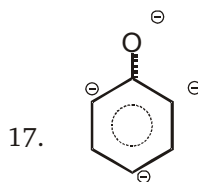
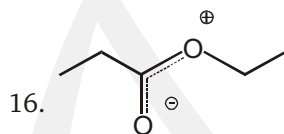
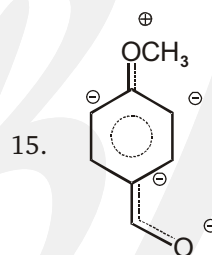
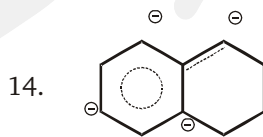
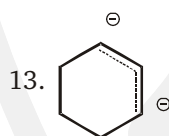
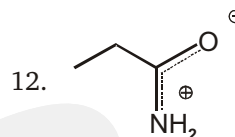
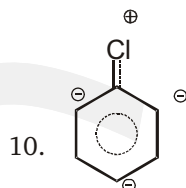
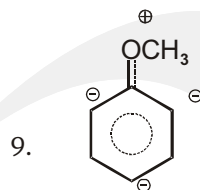
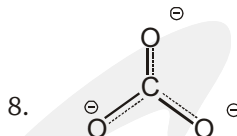
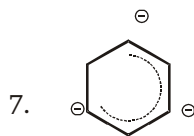
2. 1-bromo-2-chloro-4-nitrobenzene
 4. 2-bromo-5-chlorophenol
 6. 2,4-dibromo aniline

38.



39.





40.

- | | |
|--|--------------------------------|
| 1. 2-bromo-N-chloroacetamide | 2. N-bromo-2-chloro ethanamide |
| 3. N-propylbutanamide | 4. acetamide |
| 5. N-[but-2-enyl]pent-2-enamide | 6. 3-cyclopropylbut-3-enamide |
| 7. N-(2-bromopropyl)-3-chloropentanamide | 8. N-butylacetamide |
| 9. N-[penta-1,3-dienyl]but-3-ynamide | |
| 10. ethanoic cyclohexanecarboxylic anhydride | |
| 11. cyclobutylethanoic cyclohexanecarboxylic anhydride | |
| 12. hepta-3,5-dien-1-amine | |
| 13. hexa-1,4-dien-3-amine | |
| 14. 3-bromo-7-cyclobutylnon-5-en-4-amine | |

41.

1. 4-amino-2-methylpentanoic acid
2. 2-amino-3-chloro-2-methylbutanoic acid
3. 2-methyl-5-sulfohexanoic acid
4. 2-cyclopropyl-2,7-dimethyl-6-sulfo non-8-enoic acid
5. 7-aminooctane-2-sulfonic acid
6. 6-methylheptan-2-amine
7. 8-cyclohexyl-4-methyl-9-oxodec-2-enoic acid
8. 2,6dimethyl-7-oxo octanoyl chloride
9. 2,4-dimethyl-5-oxopentanoyl chloride
10. 2,3-dimethyl-4-oxobutanoyl chloride
11. 6-methoxy-6-methyloct-7-ene-2-sulfonic acid
12. 5-cyclopropyl-6-oxoheptane-2-sulfonic acid
13. 4-Formyl-1-methylpentylethanoate
14. 5-amino-7,7,7-trichloro-2, 6-dimethylheptanoic acid

42.

1. 2-carbamoylpentane-3- sulphonic acid
2. methyl 2-ethyl-6-methyl-7-oxoheptanoate
3. 6-cyclohexyl-3-methylheptan-2-one
4. ethyl-4-amino-4-cyclohexyl-2-methyl butanoate
5. 3-amino-2-methyl-7-sulfo octanoic acid
6. 6-ethyl 1-methyl 3-amino-2,5-dimethylhexanedioate
7. ethyl 4-cyano-3-methoxy-2-methylbutanoate
8. phenyl-5-(N-ethylamino)-2-methylhexanoate
9. 5-ethenyl-4,4-diphenylhept-5-enoic acid
10. 4-amino-6,6,6-trifluoro-4-methylhexan-3-one
11. 2,4-dimethyl-3-(1-methylethyl)pentane-3-sulfonic acid
12. N-(2-methylbut-3-ynyl)acetamide

43.

1. 4-(1-ethenylprop-1-enoxycarbonyl) pentane-1-sulfonic acid
2. 5-ethoxycarbonylhexanoic acid
3. 4-methoxycarbonylbutanoic acid
4. 3-pentoxycarbonylpropanoic acid
5. 4-cyano-2-cyclopropylbutane-1-sulfonic acid
6. 3-(N,N-dipropylamino)-2-methylhexanoic acid
7. ethyl 3-aminobutanoate
8. prop-2-ynoic acid
9. ethyl 5-(chlorocarbonyl)-6,6,6-trifluoro-4-methylhexanoate

10. ethyl 2-amino-3-chloro-4-cyclohexylhex-5-enoate
11. 3-ethynyldec-9-enoic acid
12. 4-(chlorocarbonyl)undec-1-en-10-yne-3-sulfonic acid
13. 6,6-dichloro-5-sulfo nonanoic acid
14. 7-(chlorocarbonyl)-3-methyldeca-1,9-diene-4-sulfonic acid
15. 9-carbamoyl-7-methylnona-1,8-diene-1-sulfonic acid
16. 2,4-dimethyl-3-(1-methylethyl)pentane
17. 6-bromo-4-chloro-6-cyclopropyl-5-nitrohexanal

44.

1. 2-phenylnonanedioic acid
2. 7-sulfo-3-(trifluoromethyl)non-8-ynoic acid
3. 6-hydroxy non-8-enoic acid
4. 6-chloro-2-merceptohexane-3-sulfonic acid
5. 6-cyclohexyl-5-hydroxy-6-methoxyhexanoic acid
6. 6-amino-3-hydroxy-2-methyl-7-sulfodec-9-enoic acid
7. 6-amino-3,5-dimethyl-4-(1-methylethyl)oct-7-ynoic acid
8. 6,7-dichloro-8-nitro-3-sulfodec-9-enoic acid
9. 5-(ethoxyformyl)-2-methylhexanoic acid
10. 5,5-dimethyl-4-oxohexanoic acid
11. 4-(Phenoxycarbonyl)butanoic acid
12. 3,7-dihydroxy-6-methylheptanoic acid
13. 2-ethenyl-6-methyl-5-mercepto oct-7-enoic acid
14. 8-methyl-4-nitro-6-sulfodeca-2,9-dienoic acid
15. 2-bromo-6-chloro-3-ethynyl-4-hydroxy-7-methoxyoct-7-enoic acid

45.

1. 2,4-diamino-5-methylhexan-3-ol
2. 3-amino-7-hydroxy-2-methyloctanedioic acid
3. 3-(chlorocarbonyl)-4-cyanopentanoic acid
4. 6-carbamoyl-5-formyloct-3-enoic acid
5. 4-bromo-5-(ethoxycarbonyl)-3-ethylheptanoic acid
6. 5-(acetyl-amino)-4-(trichloromethyl)heptanoic acid
7. 6-carbamoyl-5-(chlorocarbonyl)nona-4,7-dienoic acid
8. 4-sulfohexanoic acid
9. 3-cyano-4-ethylhex-4-ene-2-sulfonic acid
10. 2-(1-ethoxycarbonylpropyl) non-8-enoic acid
11. 2-cyano-4-propyloct-5-enoyl chloride
12. 2-oxopropanal
13. 5,10-dioxodec-6-ene-3-sulfonic acid

14. cyclohexanecarboxylic acid
15. cyclohexanecarbaldehyde

46.

- | | |
|---|---------------------------------|
| 1. pentan-2-one | 2. 3-methylbutan-2-one |
| 3. but-3-en-2-one | 4. 3-ethenylpent-4-en-2-one |
| 5. 1-(cyclopenta-2,4-dien-1-yl)ethanone | 6. but-3-yn-2-one |
| 7. ethenone | 8. hexa-3,5-dien-2-one |
| 9. 5-bromo-6-cyclopropylhept-3-en-2-one | 10. 5-bromohepta-3,5-dien-2-one |

47.

- | | |
|---|--|
| 1. 2-methylpropanoyl chloride | 2. 2-methylpentanoyl chloride |
| 3. 4-methyl-2-(methylethyl)pentanoyl chloride | 4. 2,6-dimethyloct-7-enoyl chloride |
| 5. 4,7-dichloro-2-ethylnonanoyl chloride | |
| 6. 6-cyclopropyl-2-methylheptanoyl chloride | 7. 6-cyclobutyl-2-ethylnonanoyl chloride |
| 8. 7-ethyl-2-methylnon-8-enoyl chloride | |

48.

- | | |
|--------------------------------------|-----------------------------------|
| 1. 2-methylbutanenitrile | 2. 2-methylpentanenitrile |
| 3. 2-propylpentanenitrile | 4. 2,5-dimethylhept-6-yenenitrile |
| 5. 2-cyclobutylpropanenitrile | 6. 2,7-dimethylnon-8-yenenitrile |
| 7. 4-cyclohexyl-2-ethylbutanenitrile | |

49.

- | | |
|--|---|
| 1. acetic acid | 2. 2-methylhex-5-enoic acid |
| 3. 2-ethyl-4-methylhept-6-enoic acid | 4. 3-bromo-2-ethylpentanoic acid |
| 5. 3-ethyl-2-methylpent-4-ynoic acid | 6. 5-cyclopropyl-2-methylheptanoic acid |
| 7. 3-(cyclopenta-2,4-dien-1-yl)-2-methylhexanoic acid | |
| 8. 4-methyl-2-(methylethyl)pentanoic acid | |
| 9. 5-bromo-2-(1-bromoethyl)-3-cyclopropylhexanoic acid | |
| 10. 2,4-dimethylhex-5-ynoic acid | 11. 2,3,6-trimethylheptanoic acid |
| 12. 8-butyl-2-methyl dec-9-enoic acid | 13. 2-ethylheptanoic acid |
| 14. 3-cyclohexyl-2-methylhexanoic acid | 15. 4-cyclobutyl-2-methylbutanoic acid |

50.

- | | |
|--|---|
| 1. methyl acetate or methyl ethanoate | 2. methyl 2-methylpentanoate |
| 3. methyl 3-bromo-2-ethylpentanoate | 4. methyl 2-cyclohexylbutanoate |
| 5. methyl 3,6-dichloro-2-methyloctanoate | 6. methyl 2,6-dimethyloct-7-enoate |
| 7. ethyl 5-bromo-2-ethylhexanoate | 8. ethyl 6-cyclobutyl-2-methyloctanoate |
| 9. ethyl 5,6-dibromo-2-methylheptanoate | 10. ethyl 2-methylpropanoate |

11. ethyl 2-methyloct-7-enoate
12. ethyl 3-(cyclopenta-2,4-dienyl)-2-methylhexanoate
13. phenyl ethanoate
14. phenyl 2-methylhexanoate
15. phenyl 2-bromo-2-methylpropanoate
16. phenyl 6-bromo-5-chloro-2-ethylheptanoate

51.

1. N-(1-methyl ethyl) ethanamide
2. hexanamide
3. 4-bromo-5-chlorohexanamide
4. 4-bromo-5-chlorohexanamide
5. but-3-enamide
6. 2-methylbut-3-ynamide
7. 3-cyclobutyl-5-methylhept-6-enamide
8. 2-ethylbut-3-ynamide
9. butanamide
10. chloromethyl 2-bromobutanoate
11. bromomethyl 2-chlorobutanoate
12. but-2-enyl propanoate
13. butyl prop-2-enoate
14. diethyl ethanedioate
15. ethyl methyl ethanedioate
16. 1-cyclopropylethyl 3-cyclopropylpropanoate
17. ethanamine
18. ethanoic anhydride
19. ethyl ethanoate
20. methyl propanoate
21. hexan-3-amine
22. 5-bromooctan-2-amine

52.

S. No.	1°C	2°C	3°C	4°C	1°H	2°H	3°H
1.	5	5	3	0	15	10	3
2.	2	1	0	0	4	1	0
3.	2	5	0	1	6	9	0
4.	3	3	1	0	6	5	1
5.	1	3	1	0	3	3	1
6.	2	8	2	0	6	14	2
7.	3	6	1	0	9	12	1
8.	3	4	1	0	7	4	1
9.	5	1	3	0	15	2	3
10.	0	4	0	0	0	4	0
11.	1	8	1	0	3	16	1

53.

- | | |
|-------------------------------------|--------------------------------|
| 1. 1° amine, alcohol | 2. 3° amine, carboxylic acid |
| 3. alcohol, ketone, carboxylic acid | 4. Ketone, Aldehyde |
| 5. Thio alcohol, 2° amine, 3° amine | 6. Alkene, Acid halide |
| 7. ester, carboxylic acid | 8. Ketone, alkene |
| 9. Ether | 10. Anhydride |
| 11. 1° amine, ketone, amide | 12. cyanide, isocyanide, ether |
| 13. Ester | 14. Carboxylic Acid |
| 15. Thio ether, alkene | |

54.

Functional Groups	DBE
1. Alkene, ketone, Aldehyde	4
2. Alkene, 1° amine	2
3. Alkene, Carboxylic acid	3
4. Alkene, cyanide, aldehyde	6
5. Alkene, isocyanide	6
6. Thioalcohol, ester	3
7. Aldehyde, Acid halide	3
8. Alkene, Alkyne, Sulphonic acid	5
9. Ketone, alcohol, Thio alcohol	3
10. Anhydride	4
11. Alkene, alcohol (enol)	1
12. Ketone, Thio ketone	2

55.

Name of functional Groups	DBE Value
1. Alcohol, ether, Thio ester	2
2. Alkene	4
3. Amide	3
4. Ether, anhydride	5
5. Alkene, Alcohol, Ketone	4
6. Ether, Carboxylic acid, alkene	10

56.

- | | |
|-------------------------------|--------------------------------|
| 1. prop-1-ene | 2. 3-methylhex-1-ene |
| 3. 2-methylbut-1-ene | 4. 5-methylhept-2-ene |
| 5. 2-cyclopropyl propene | 6. chloroethene |
| 7. 4-chloropent-1-ene | 8. 3-bromoprop-1-ene |
| 9. 3-ethyl-6-methylhept-2-ene | 10. 4-bromo-2-methylpent-1-ene |

57.

1. hexa-1,4-diene
2. hexa-1,5-diene
3. 2-methylpenta-1,4-diene
4. but-1-en-3-yne
5. pent-3-en-1-yne
6. oct-5-en-1-yne
7. 3-ethylhex-1-en-5-yne
8. 7-chlorodec-1-en-4-yne
9. 5-chlorocyclopenta-1,3-diene
10. cyclopenta-1,3-diene
11. 5-(methyl ethyl)cyclopenta-1,3-diene

58.

1. 1,2-dimethylcyclobutene
2. 1,3-dimethylcyclobutene
3. 1,4-dimethylcyclobutene
4. 1,6-dibromocyclohexene
5. 1,6-dibromo-3-chlorocyclohexene
6. 1,3-dichlorocyclohexene
7. cyclohexa-1,4-diene
8. 1-ethyl-2-(prop-2-enyl)cyclohexene
9. 1-ethenyl-3-(prop-2-enyl)cyclohexene
10. 3-methylcyclopropene

59.

1. hexan-3-ol
2. 4-methylhexan-2-ol
3. 4-ethylhexan-3-ol
4. 4-methylhexan-3-ol
5. prop-2-en-1-ol
6. hexa-1,4-dien-3-ol
7. 6-bromohex-2-en-2-ol
8. hexa-1,5-dien-1-ol
9. 6-cyclopropylhept-4-en-3-ol
10. 3-(1-bromoethyl)-4-methylpent-4-en-2-ol

60.

1. cyclopropane-1,2-diol
2. 2-(2-bromopropyl)cyclopentanol
3. cyclohex-2-ene-1,1-diol
4. 3-ethyl-2-propylcyclohexanol
5. cyclopent-2-en-1-ol
6. cyclopenta-2,4-dien-1-ol
7. cyclopenta-1,3-dien-1-ol
8. 1-cyclopropylcyclopenta-2,4-dien-1-ol
9. 6-bromo-4-chlorocyclohex-2-en-1-ol

61.

1. 4-ethyl-2-methylhex-5-enal
2. 3-bromo-2,5-dimethyloctanal
3. 2,6-dimethyloct-7-enal
4. 4-methyl-2-(2-methylpropyl)hexanal
5. 4-bromo-2,2,6-trimethylheptanal
6. 4-ethyl-2,3-dimethylhex-5-ynal
7. 2,4-dimethylhex-5-enal
8. 2-(1-bromopropyl)-4-chloro-3-cyclopropylhexanal

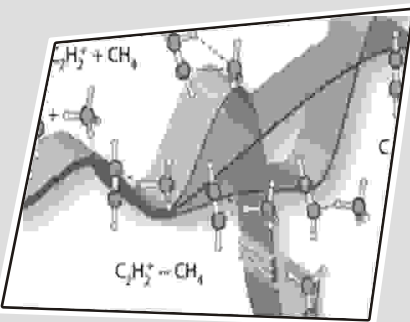
62.

1. 2-methylpropanal
2. ethanedial
3. 2-ethynylpenta-2,4-dienal
4. but-2-enal
5. methylpropanedial
6. 2-propylpentanal
7. 4-bromo-2-ethylpentanal
8. 4-bromo-5-chloro-2-ethylheptanal
9. 2-ethylbutanal

2

Hydrocarbons

(Aliphatic and Aromatic Hydrocarbons)

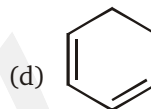
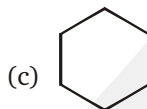
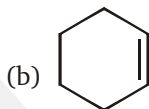
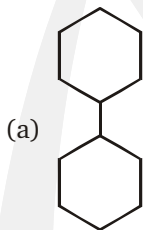


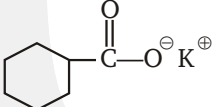
Only One Answer is Correct

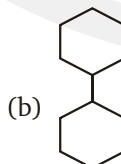
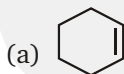
[A] Aliphatic Hydrocarbons

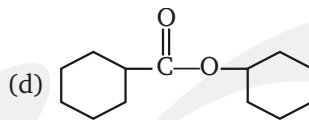
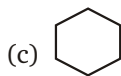
Alkanes

1.  $\xrightarrow[h\nu]{\text{Br}_2}$ (A) $\xrightarrow[\text{Dry ether}]{\text{Na}}$ (B). Product B is:



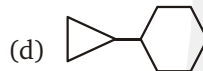
2.  $\xrightarrow{\text{electrolysis}}$ (A) ; Major-product (A) is :

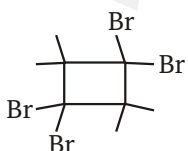


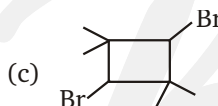
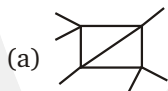


3. $\text{CH}_3\text{MgBr} + \text{D}_2\text{O}$ (A); Product (A) of the reaction is:
 (a) CH_4 (b) CH_3D (c) CH_2D_2 (d) CHD_3

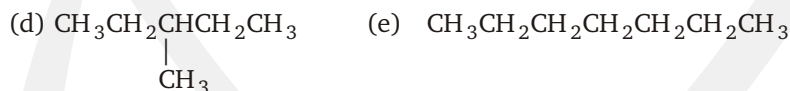
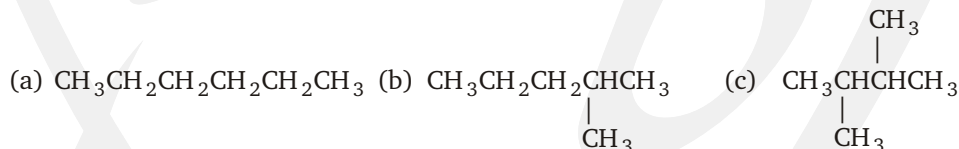
4. Which of the following alkane will be formed in high yield by Wurtz reaction ?



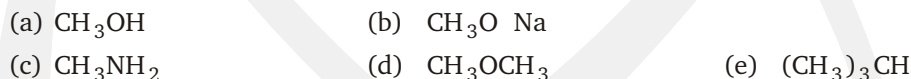
5.  $\xrightarrow[\text{dry ether}]{2\text{Na}}$ (A); Product A is:



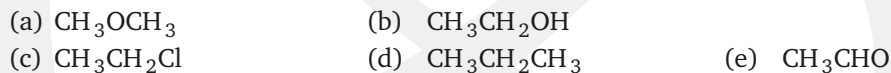
6. Which of the following has the lowest boiling point?



7. Which of the following has highest solubility in $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$?



8. Which of the following is most soluble in H_2O ?



9. During the free radical chlorination of methane, which of the following reactions has lowest collision frequency?



10. In the reaction $\text{Br}_2 + \text{CH}_3\text{Br} \xrightarrow{h\nu} \text{CH}_2\text{Br}_2 + \text{HBr}$ which of the following mechanism steps is productive, but relatively unlikely to occur?

- (a) $\text{Br}\cdot + \text{CH}_3\text{Br} \rightarrow \text{HBr} + \cdot\text{CH}_2\text{Br}$ (b) $\text{Br}\cdot + \cdot\text{CH}_2\text{Br} \rightarrow \text{CH}_2\text{Br}_2$
 (c) $\text{Br}\cdot + \text{Br}_2 \rightarrow \text{Br}_2 + \text{Br}\cdot$ (d) $\text{Br}\cdot + \cdot\text{CH}_3 \rightarrow \text{CH}_3\text{Br}$

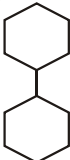
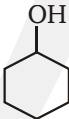
11. Which of the following reactions is a termination step in the free radical chlorination of methane?

- (a) $\text{Cl}_2 + \text{Cl}\cdot \rightarrow \text{Cl}\cdot + \text{Cl}_2$ (b) $\text{Cl}_2 \rightarrow 2\text{Cl}\cdot$
 (c) $\cdot\text{CH}_3 + \text{Cl}\cdot \rightarrow \text{CH}_3\text{Cl}$ (d) $\text{CH}_4 + \text{Cl}\cdot \rightarrow \text{HCl} + \cdot\text{CH}_3$

12. Which of the following is a chain propagation step in the free radical chlorination of methane?

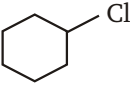
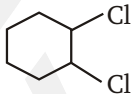
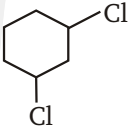
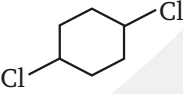
- (a) $\text{CH}_4 + \text{Cl}\cdot \rightarrow \cdot\text{CH}_3 + \text{HCl}$ (b) $\text{Cl}_2 \rightarrow 2\text{Cl}\cdot$
 (c) $\text{Cl}\cdot + \cdot\text{CH}_3 \rightarrow \text{CH}_3\text{Cl}$ (d) $\cdot\text{CH}_3 + \text{CH}_4 \rightarrow \text{CH}_4 + \cdot\text{CH}_3$

13. $2 \text{ } \text{Cyclohexyl-Br} \xrightarrow[\text{ether}]{\text{Mg}}$ (A); Product (A) is:

- (a)  (b)  (c)  (d) 

14. What is the major product of the following reaction?

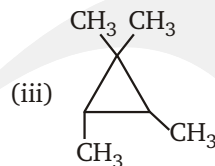
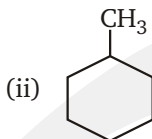
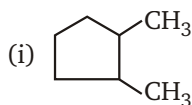


- (I)  (II)  (III) 
 (IV)  (V) 
 (a) I (b) II (c) III (d) IV (e) V

15. Compare the heats of combustion of the following compounds:

- (i)  (ii)  (iii) 
 (a) $i > ii > iii$ (b) $ii > i > iii$ (c) $iii > i > ii$ (d) $ii > iii > i$

16. Compare heats of combustion of the following compounds:

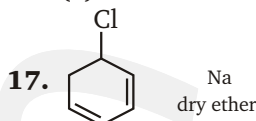


(a) $i > ii > iii$

(b) $ii > i > iii$

(c) $iii > i > ii$

(d) $iii > ii > i$



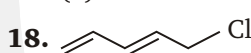
(x) Dimeric product (x) is :

(a) 1

(b) 2

(c) 3

(d) 4



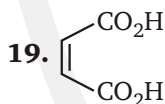
(x) . Products x is :

(a) 1

(b) 2

(c) 3

(d) 4



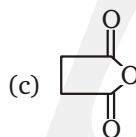
2NaOH

(A) electrolysis

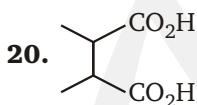
(B); Product (B) is:

(a) $\text{HC} \quad \text{CH}$

(b) $\text{H}_2\text{C} \quad \text{CH}$



(d)



2KOH

(A) electrolysis

(B) + (C) ; Products (B) and (C) are:

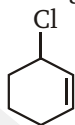
(a) geometrical isomers

(b) enantiomer

(c) positional isomer

(d) structural isomer

21. How many products will formed in the given reaction ?



$\xrightarrow[\text{dry ether}]{\text{Na}}$

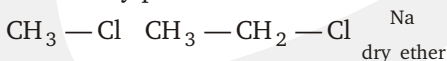
(a) 1

(b) 2

(c) 3

(d) 4

22. How many products will formed in the following reaction ?



(a) 1

(b) 2

(c) 3

(d) 4

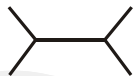
23. Which of the following correctly ranks the cycloalkanes in order of increasing ring strain per methylene?

(a) cyclopropane < cyclobutane < cyclohexane < cycloheptane

- (b) cyclohexane < cyclopentane < cyclobutane < cyclopropane
 (c) cyclopentane < cyclobutane < cyclopentane < cyclopropane
 (d) cyclopentane < cyclopropane < cyclobutane < cyclohexane
 (e) cyclopropane < cyclopentane < cyclobutane < cyclohexane

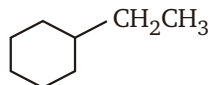
24. Cl—C—Cl bond angle in 1,1,2,2-tetrachloro ethene and tetrachloromethane is:

- (a) 120 and 109.5 (b) 90 and 109.5 (c) 109.5 and 90 (d) 109.5 and 120

25.  $\xrightarrow[h\nu]{\text{Cl}_2}$ (X) mono chloro product: (including stereoisomer)

- (a) 2 (b) 3 (c) 4 (d) 5

26. How many products are formed from the monochlorination of ethylcyclohexane? Ignore stereoisomers?



- (a) 6 (b) 8 (c) 5 (d) 9 (e) 11

27. How many distinct dichlorination products can result when isobutane is subjected to free radical chlorination?

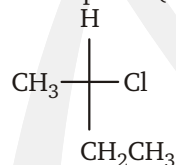
- (a) 1 (b) 2 (c) 3 (d) 4 (e) 6

28. How many distinct monochlorinated products, including stereoisomers, can result when the alkane below is heated in the presence of Cl_2 ?

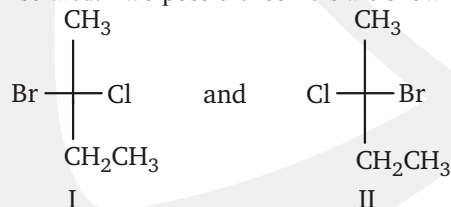


- (a) 2 (b) 4 (c) 5 (d) 6 (e) 8

29. A sample of (R)-2-chlorobutane,



reacts with Br_2 in the presence of light, and all the products having the formula $\text{C}_4\text{H}_8\text{BrCl}$ were isolated. Two possible isomers are shown below:

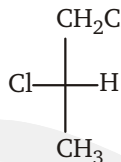


of these:

- (a) only I was formed
 (b) both I and II were formed in equal amounts

- (c) both I and II were formed in unequal amounts
 (d) only II was formed
 (e) neither I nor II was formed

30. How many dichlorinated products, including stereoisomers, can be isolated when (*S*)-2-chlorobutane reacts with $\text{Cl}_2/h\nu$?

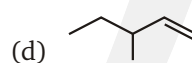
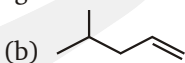
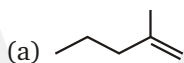


(*S*)-2-chlorobutane

- (a) 1 (b) 2 (c) 3 (d) 4 (e) 5

Alkenes

1. In which of the following alkenes will a hydrogen shift occur upon addition of HCl?



2. Predict the major product of the reaction of 1,4-pentadiene with excess HCl.

(a) 1,4-dichloropentane

(b) 1,5-dichloropentane

(c) 2,4-dichloropentane

(d) 4-chloro-1-pentene

3. Indicate the product resulting from the hydrogenation (excess H_2/Pd) of 2-hexyne.

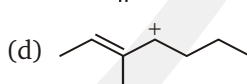
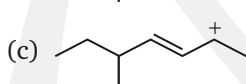
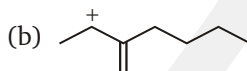
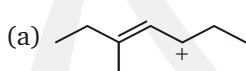
(a) (*Z*)-2-hexene

(b) (*E*)-2-hexene

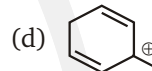
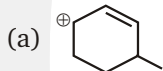
(c) hexane

(d) 1-hexene

4. Consider the following carbocations and their resonance hybrids. Which is the MOST stable carbocation?



5. Which of the following is the MOST stable carbocation?



6. In the addition of HBr to 1,3-butadiene, one product is called the kinetic product and the other is called the thermodynamic product. Which of the following descriptors applies to the kinetic product?

(a) 1,2 product

(b) 1,4 product

(c) most stable

(d) least stable

(e) formed slowest

(f) formed fastest

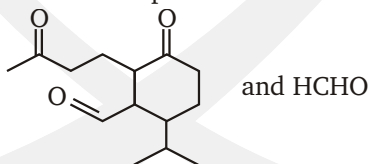
(g) formed at low temperature

(h) formed at high temperature

- (i) most substituted alkene
(k) bromide attack at more substituted site
(a) a, c, e, g, j, l
(c) a, d, f, g, j, k
- (j) least substituted alkene
(l) bromide attack at least substituted site
(b) b, c, f, g, i, l
(d) b, d, f, h, i, k
7. Which of the following statements regarding 1,2- or 1,4-conjugate addition of HBr to 1,3-butadiene is false?
- (a) The kinetic product, 3-bromo-1-butene, arises from the more stable carbocation intermediate.
(b) The thermodynamic product, 1-bromo-2-butene, is the more stable product.
(c) The kinetic product is favored by carrying out the reaction at high temperature.
(d) Formation of the two products arises from a common resonance stabilized carbocation intermediate.
8. What is the product of the following reaction?

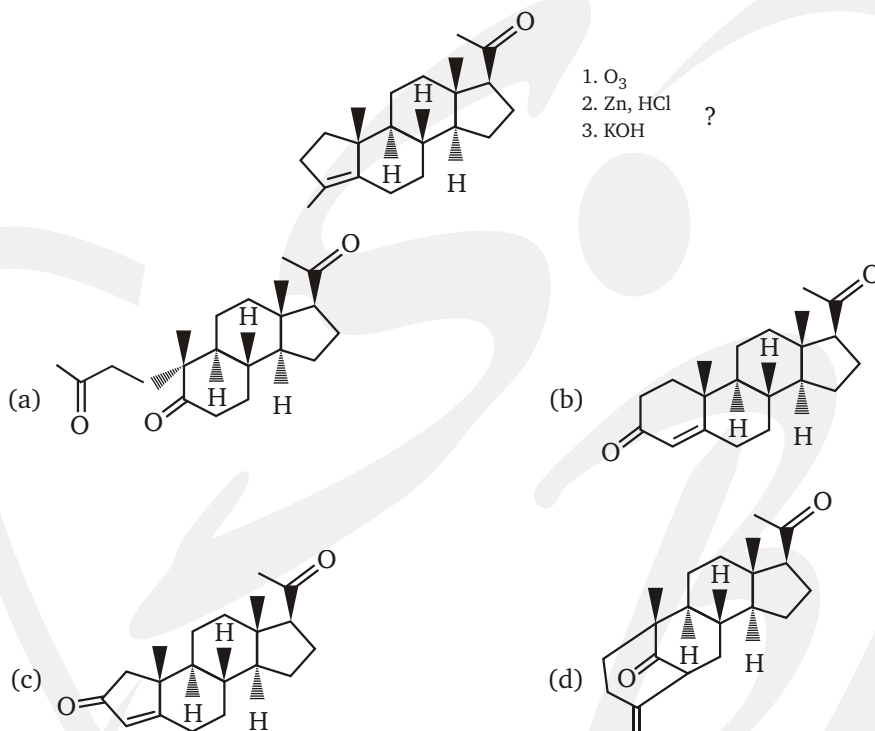


- (a)  +  + 
- (b)  + 
- (c)  +  + 
- (d)  +  + 
9. The molecular formula of compound A is $C_{15}H_{24}$. Ozonolysis gives the following compound and formaldehyde. Which could be compound A?

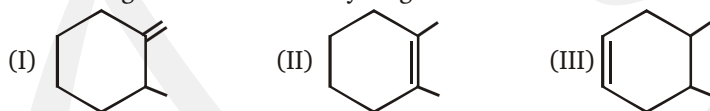


- [illegible]

10. What is the product of the following synthesis?

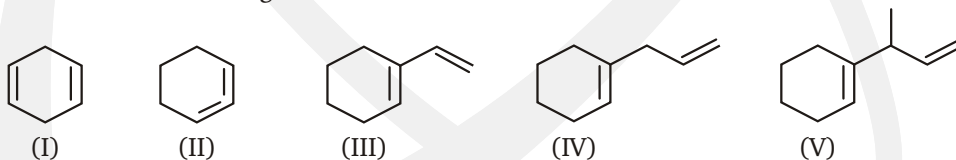


11. Decreasing order of heat of hydrogenation ?

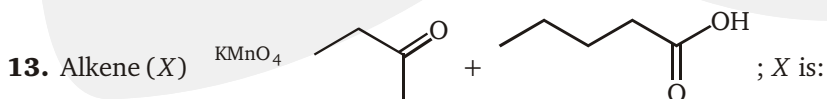


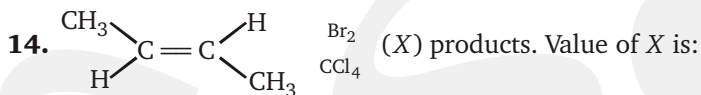
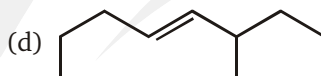
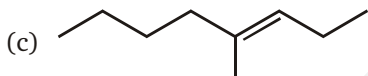
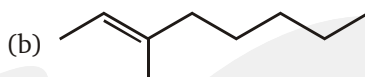
- (a) $I > III > II$ (b) $II > I > III$ (c) $III > I > II$ (d) $II > III > I$

12. Which of the following is the most stable diene?



- (a) I (b) II (c) III (d) IV (e) V



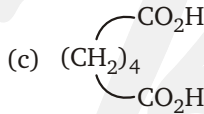
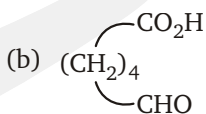
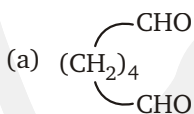
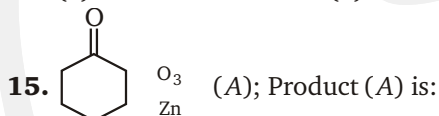


(a) 0

(b) 1

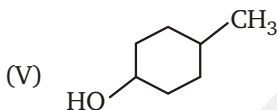
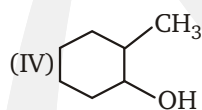
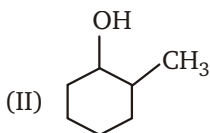
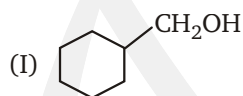
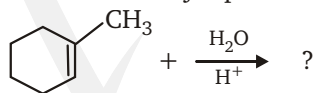
(c) 2

(d) 3



(d) All of these

16. What is the major product of the following reaction?



(a) I

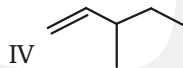
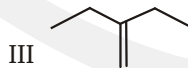
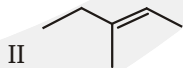
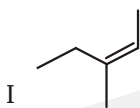
(b) II

(c) III

(d) IV

(e) V

17. Which of the following alkenes yield(s) 3-bromo-3-methylpentane as the major product upon addition of HBr?



(a) I and II only

(b) III only

(c) I, II, and III only

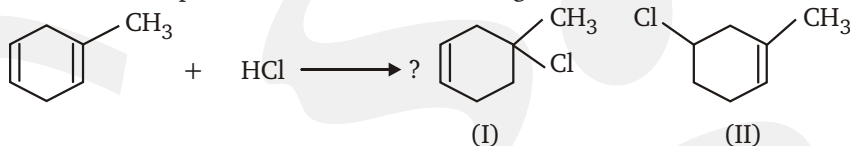
(d) all of them

(e) none of them

18. Which of the following alkenes reacts with HCl at the slowest rate?

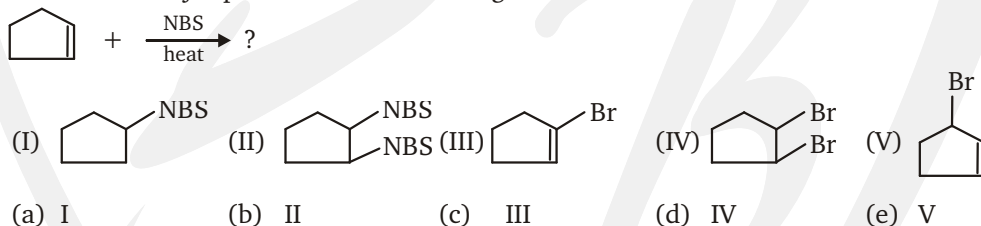
- (a) CH_3CHCH_2 (b) $\text{CH}_3\text{CH}_2\text{CCH}_2$ (c) CH_3CCH_2
 $\quad \quad \quad |$ $\quad \quad \quad |$ $\quad \quad \quad |$
 $\quad \quad \quad \text{CH}_3$ $\quad \quad \quad \text{CH}_3$ $\quad \quad \quad \text{CH}_3$
- (d) $\text{H} \quad \text{CH}_3$ (e) $\text{CH}_3\text{CH}_2 \quad \text{CH}_3$
 $\quad \quad \quad \backslash \quad /$ $\quad \quad \quad \backslash \quad /$
 $\quad \quad \quad \text{C} = \text{C}$ $\quad \quad \quad \text{C} = \text{C}$
 $\quad \quad \quad / \quad \backslash$ $\quad \quad \quad / \quad \backslash$
 $\quad \quad \quad \text{CH}_3 \quad \text{CH}_3$ $\quad \quad \quad \text{CH}_3 \quad \text{CH}_3$

19. What is/are the product(s) from the following reaction?

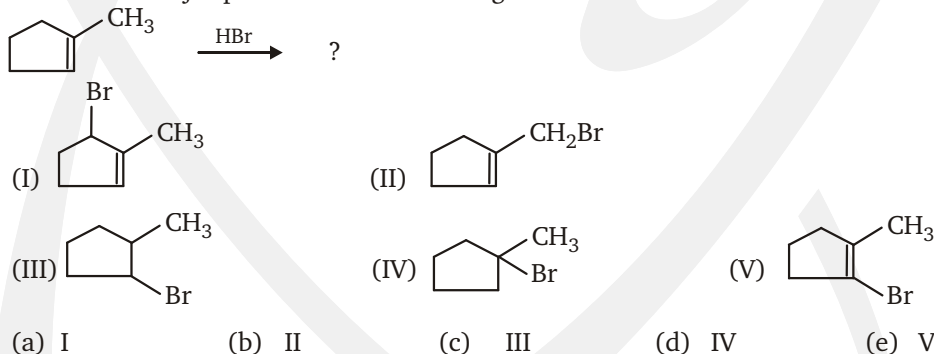


- (a) I only (b) II only (c) I is minor, II is major
 (d) I is major, II is minor (e) equal amounts of I and II

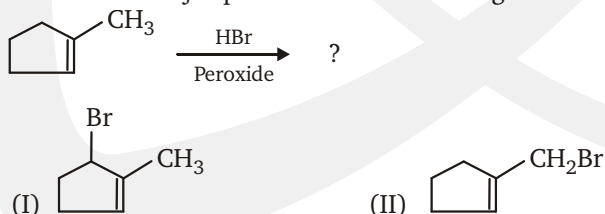
20. What is the major product of the following reaction?

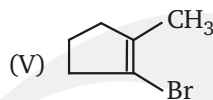
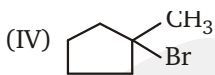
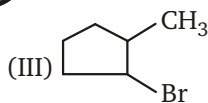


21. What is the major product of the following reaction?



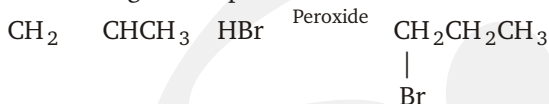
22. What is the major product of the following reaction?





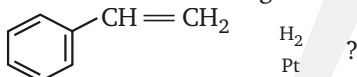
- (a) I (b) II (c) III (d) IV (e) V

23. Which of the following is a step in the mechanism of the reaction shown?



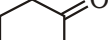
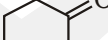
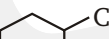
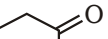
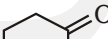
- (a) $\text{CH}_2\dot{\text{C}}\text{HCH}_3$ HBr
|
Br
- (b) $\text{CH}_2\text{CH}_2\text{CH}_3$ HBr
- (c) CH_2CHCH_3 HBr
|
Br
- (d) $\text{CH}_2\text{CH}_2\text{CH}_3^+$ HBr
- (e) CH_2CHCH_3 H
|
Br

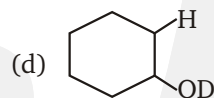
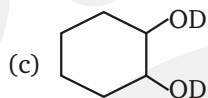
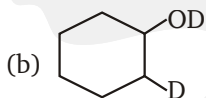
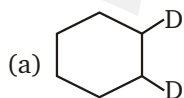
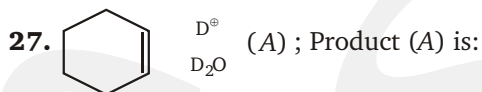
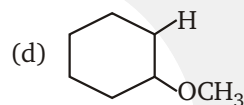
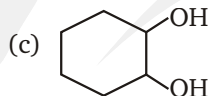
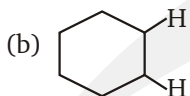
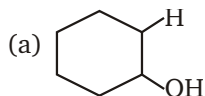
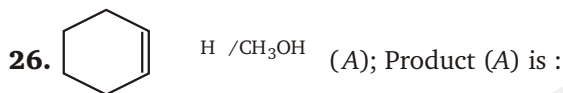
24. What is the major organic product of the following reaction ?



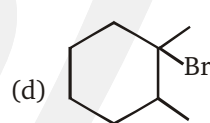
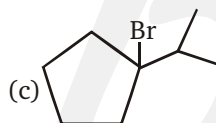
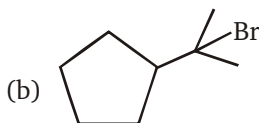
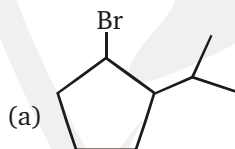
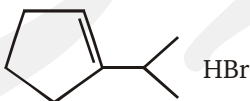
- (i)  C=Cc1ccccc1
- (ii)  CCc1ccccc1
- (iii)  CCc1ccccc1
- (iv)  C=Cc1ccccc1
- (v)  C=Cc1ccccc1
- (a) i (b) ii (c) iii (d) iv (e) v

25. What are the major organic products when the following molecule is treated with ozone, and then with Zn/H₂O?

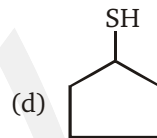
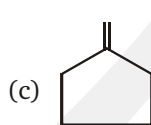
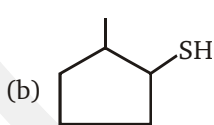
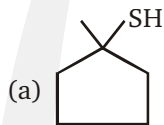
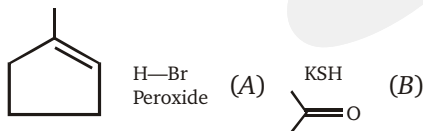
- (i)  , $\text{H}-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$ (ii)  , $\text{CO}_2, \text{H}_2\text{O}$
- (iii)  , CH_2O (iv)  , $\text{H}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH}$ (v)  , $\text{HO}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH}$
- (a) i (b) ii (c) iii (d) iv (e) v



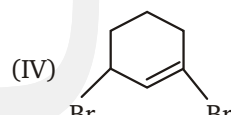
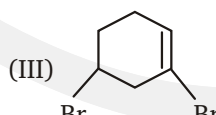
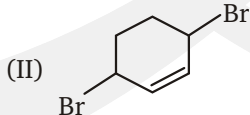
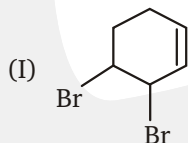
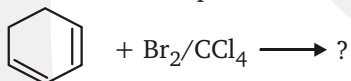
28. Which of the following product is not formed in the following reaction ?



29. Product (B) of the reaction is:



30. What is/are the product(s) from the following reaction?



(a) I and II

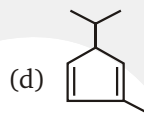
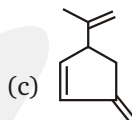
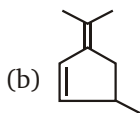
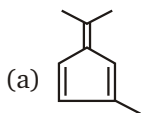
(b) I and III

(c) I and IV

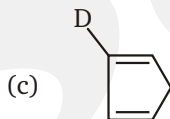
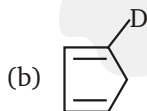
(d) II and III

(e) II and IV

31. Most acidic hydrogen is present in:

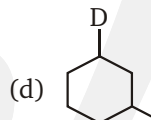
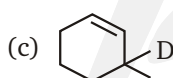
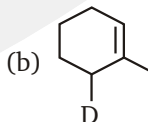
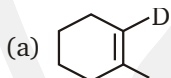


32. (1) NaH (2) H₂O *P*, Product formed in this reaction is:

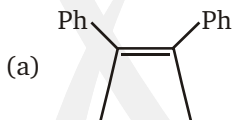


(d) All of these

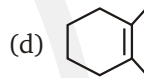
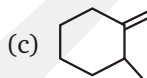
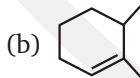
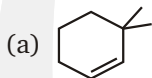
33. D_2O (A); Product (A) is:



34. Which of the following reactant is more reactive towards electrophilic addition reaction with HBr.



35. H^+ Product ; Product of the reaction is:



36. (A) CH_3COCl AlCl_3 (B) PCl_5 (C) 2NaNH_2 (D), If (A) is C_6H_6 , then:

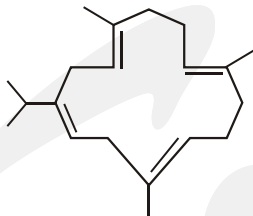
(a) D is a terminal alkyne

(b) (A) is an aromatic compound

(c) D has 8 carbon

(d) All the statements are true.

37. How many different products are obtained upon the ozonolysis of the following C_{20} -tetraene?



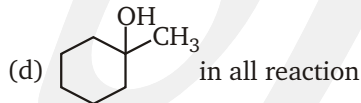
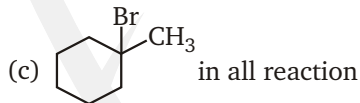
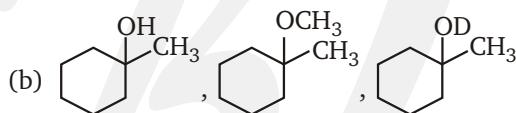
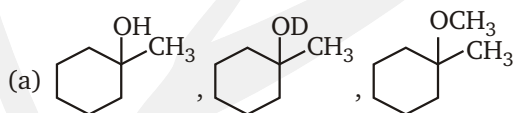
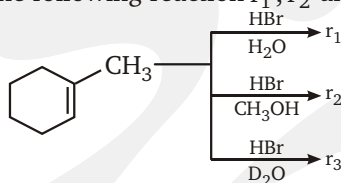
(a) 2

(b) 3

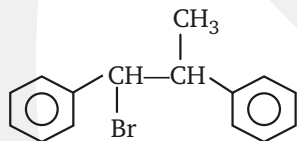
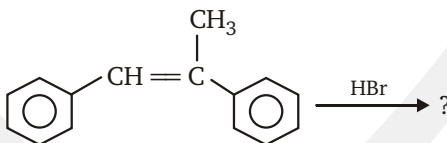
(c) 4

(d) 5

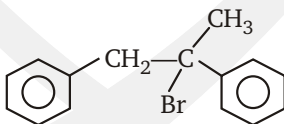
38. Major product obtained in the following reaction r_1 , r_2 and r_3 in respectively is:



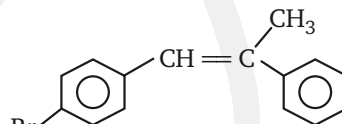
39. Which of the following is/are the major product(s) of the following reaction?



(I)



(II)



(III)

(a) I

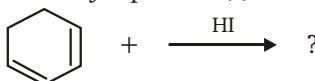
(b) I and III

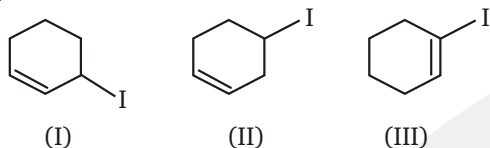
(c) II

(d) II and III

(e) III

40. Which of the following is/are the major product(s) of the following reaction?



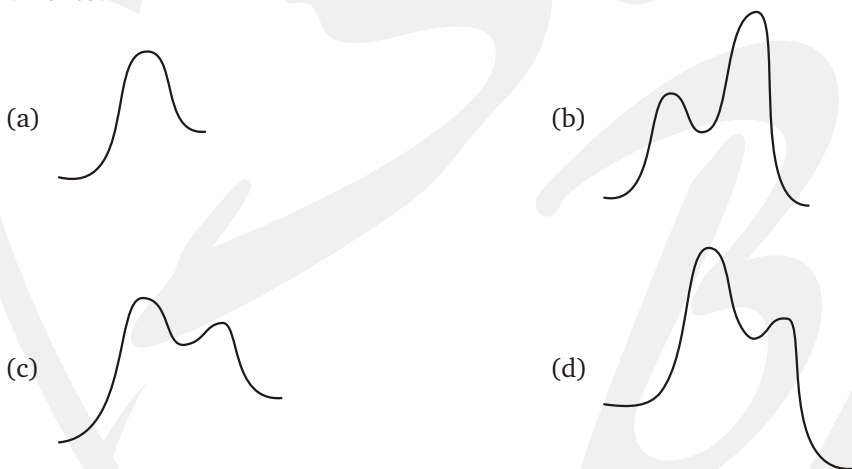


- (a) I (b) I and II (c) I and III (d) II (e) III

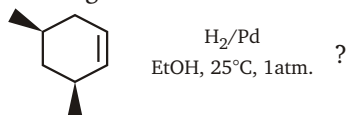
41. What compound results from the 1,4-addition of one equivalent of HBr to 1,3-butadiene?

- (a) 1-bromo-1-butene (b) 2-bromo-2-butene
(c) 4-bromo-1-butene (d) 3-bromo-1-butene (e) 1-bromo-2-butene

42. Which energy diagram best describes the electrophilic addition of hydrogen halides to alkenes?



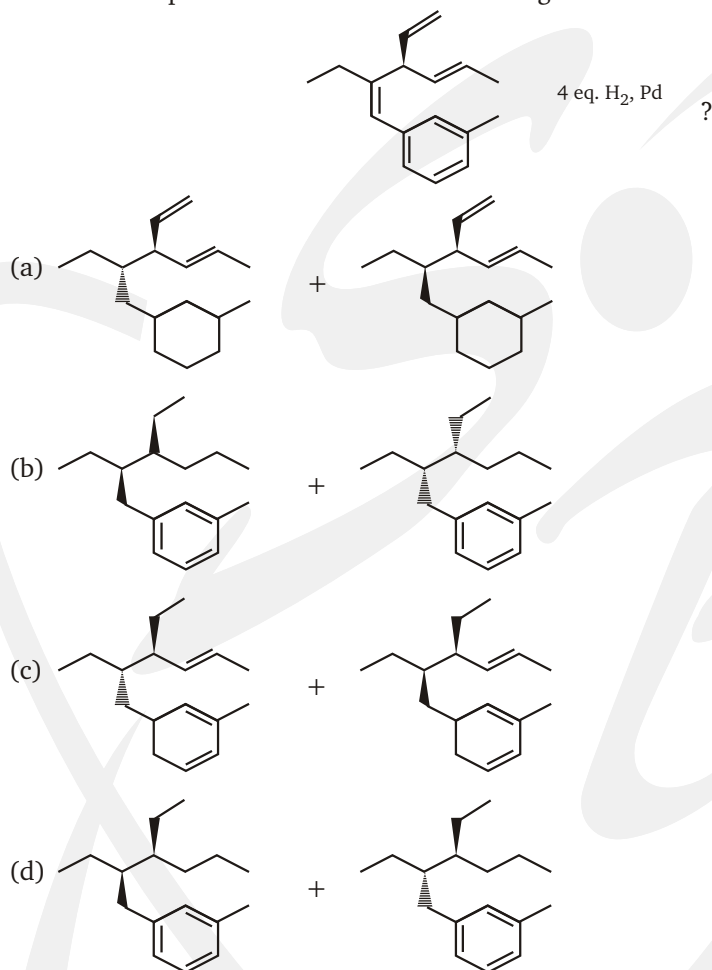
43. What is the product of the following reaction?



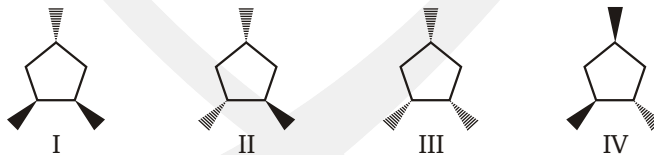
44. Which alkene will give only optically active product upon catalytic hydrogenation (H_2/Pd)?



45. What are the products formed in the following reaction?



46. Which of the following product(s) is(are) formed when 1,2,4-trimethylcyclopentene is hydrogenated?



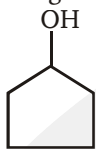
(a) I

(b) I and III

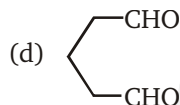
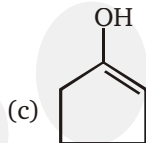
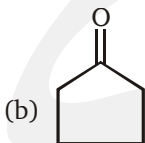
(c) II

(d) I and IV

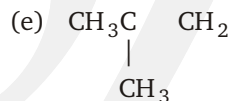
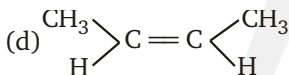
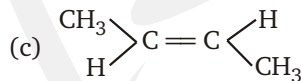
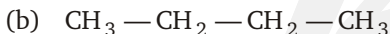
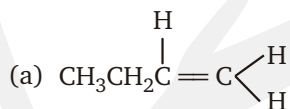
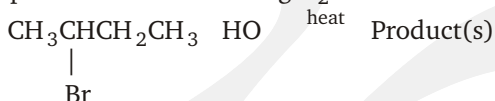
47. The major product of the following reaction is:



- (i) $\text{H}_3\text{PO}_4, 150^\circ\text{C}$
(ii) $\text{C}_6\text{H}_5\text{COOH}$



48. What is the major product of the following E_2 reaction?



49. Which of the alkyl chlorides listed below undergoes dehydrohalogenation in the presence of a strong base to give 2-pentene as the only alkene product?

(a) 1-chloropentane

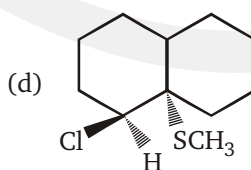
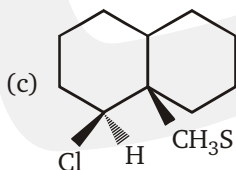
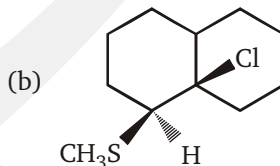
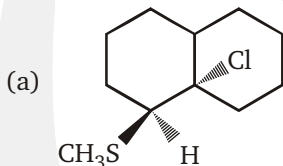
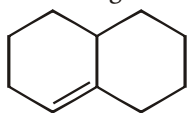
(b) 2-chloropentane

(c) 3-chloropentane

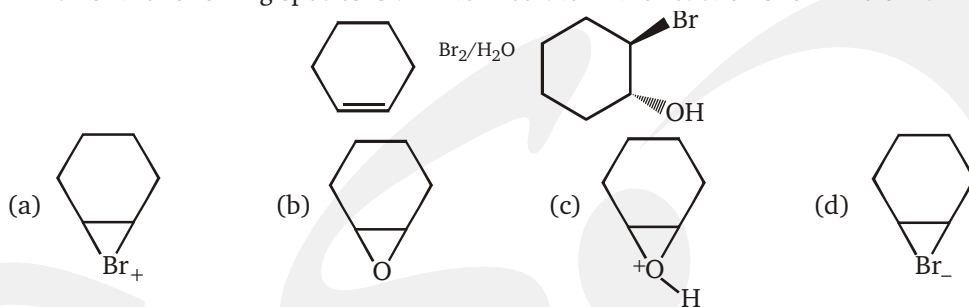
(d) 1-chloro-2-methylbutane

(e) 1-chloro-3-methylbutane

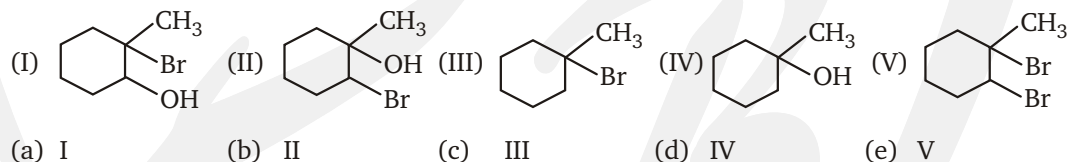
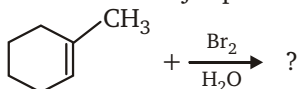
50. Give the major product of the following reaction :



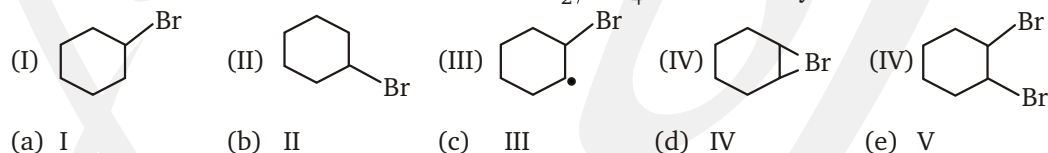
51. Which of the following species is an intermediate in the reaction shown below ?



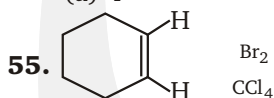
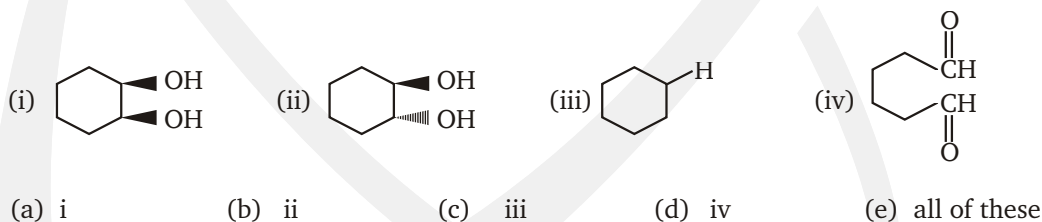
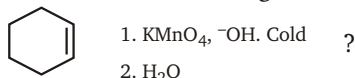
52. What is the major product of the following reaction?



53. Which reaction intermediate is formed when Br_2/CCl_4 reacts with cyclohexene?

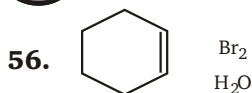


54. What is the major organic product of the following reaction ?



Product of above reaction is:

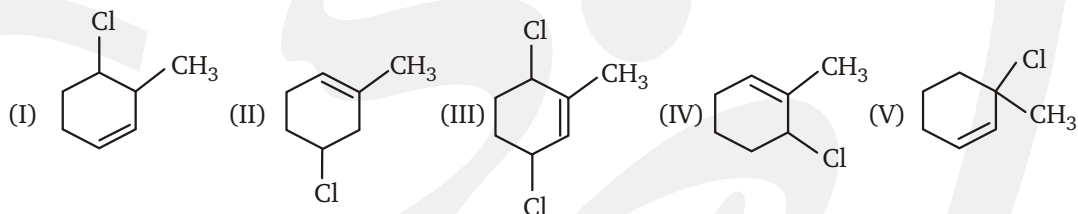
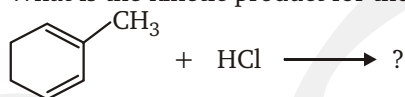
- (a) meso (b) racemic (c) diastereomer (d) structural isomer



Product of above reaction is:

- (a) meso (b) racemic (c) diastereomer (d) structural isomer

57. What is the kinetic product for the following reaction?



- (a) I (b) II (c) III (d) IV (e) V

58. How many Grignard reacts with alcohol to give *n*-butane?

(Excluding stereoisomer)

- (a) 1 (b) 2 (c) 3 (d) 4

59. Addition of Br_2 on *cis*-2 butene gives:

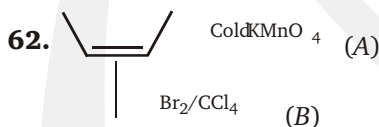
- (a) A racemic mixture of 2,3-dibromobutene (b) Meso form of 2,3-dibromobutane
(c) Dextro form of 2,3-dibromobutane (d) Laevo form of 2,3-dibromobutane

60. Classify the reaction below as an oxidation, a reduction, or neither.

cis-2-pentene $\longrightarrow$ pentane

- (a) oxidation (b) reduction (c) redox reaction

61. A $\xrightarrow[\text{CCl}_4]{\text{Br}_2}$ Meso 2,3-dibromobutane, identify reactant (A) :



Relation between (A) and (B) :

- (a) Diastereomers (b) Enantiomers
(c) Structural isomer (d) Different compound

Alkynes

1. Which of the following will produce 4-methyl-1-pentyne?

- (a) ethyne + 1) NaNH_2 2) 1-iodobutane
 (b) propyne + 1) NaNH_2 2) 1-iodopropane
 (c) ethyne + 1) NaNH_2 2) 1-iodo-2-methylpropane
 (d) propyne + 1) NaNH_2 2) 2-iodopropane

2. Predict the product of the following reaction sequence.

ethyne + 1) excess NaNH_2 2) excess $\text{I}-\text{CH}_2-(\text{CH}_2)_2-\text{CH}_3$

- (a) 6-iodo-1-hexyne (b) 1-hexyne
 (c) 5-decyne (d) 1-iodo-1-hexene

3. Which of the following anions is the strongest base?



i



ii



iii



iv

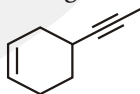
(a) i

(b) ii

(c) iii

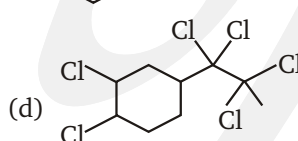
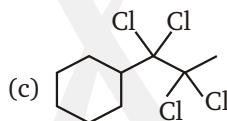
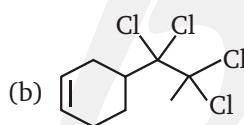
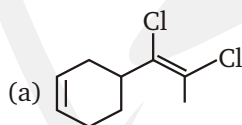
(d) iv

4. What is the product of the following reaction?



3 eq. Cl_2
 CCl_4

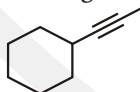
?



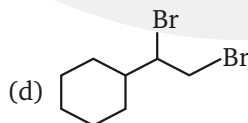
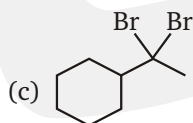
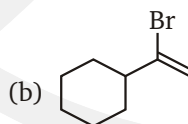
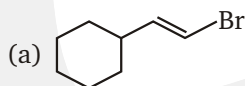
5. Predict the product of the reaction of 1-pentyne and excess HCl in the presence of hydrogen peroxide.

- (a) 1,1-dichloropentane (b) 1,2-dichloropentane
 (c) 2,2-dichloropentane (d) 2-chloro-1-pentene

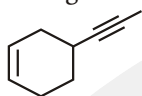
6. Identify the product of the following reaction.



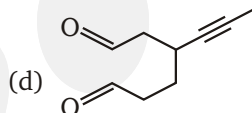
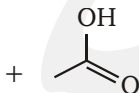
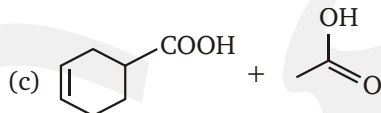
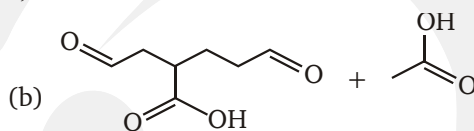
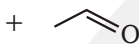
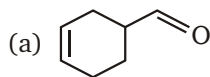
Excess HBr



7. What is the product of the following reaction?



1. O_3
2. Zn, HOAc ?



8. How many compounds are formed upon the ozonolysis of the following triyne?



(a) 2

(b) 3

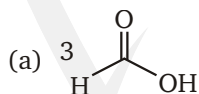
(c) 4

(d) 5

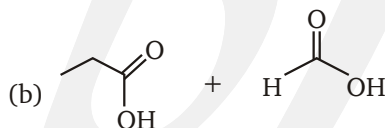
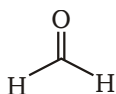
9. What is the product of the following reaction?



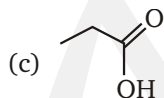
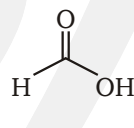
$\xrightarrow[2. \text{H}_3\text{O}^+, \text{H}_2\text{O}]{1. \text{KMnO}_4, \text{NaOH}, \text{H}_2\text{O, Heat}}$?



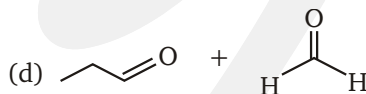
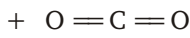
+



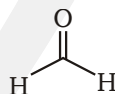
+



+



+



10. $\text{CH}_3-\text{C} \equiv \text{C}-\text{CH}_3 \xrightarrow[(2) \text{Br}_2]{(1) \text{H}_2/\text{Pd}/\text{CaCO}_3} \text{X} :$

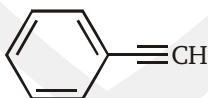
(a) (d)-2,3-Dibromobutane

(b) (l)-2,3-Dibromobutane

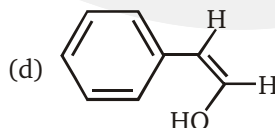
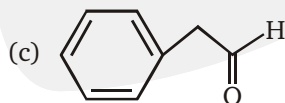
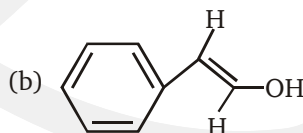
(c) (dl)-2,3-Dibromobutane

(d) meso-2,3-Dibromobutane

11. What is the major organic product of the following reaction ?



$\xrightarrow[\text{H}_2\text{SO}_4]{\text{Hg}^2}$

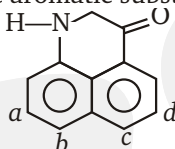


- For More Material Join: @JEEAdvanced_2024

[B] Aromatic Hydrocarbons

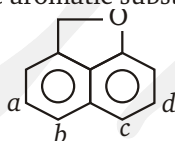
Preparation & Properties

1. Identify the site where electrophilic aromatic substitution takes place ?



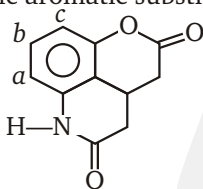
- (a) a (b) b (c) c (d) d

2. Identify the site where electrophilic aromatic substitution takes place ?



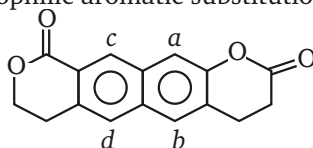
- (a) a (b) b (c) c (d) d

3. Identify the site where electrophilic aromatic substitution takes place ?

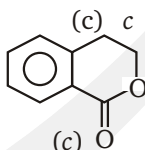
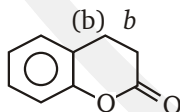
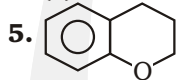


- (a) a, b (b) b, c (c) a, c (d) b

4. Identify the site where electrophilic aromatic substitution takes place ?



- (a) a (b) b (c) c (d) d



- (d) d

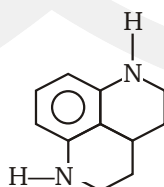
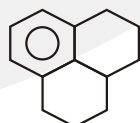
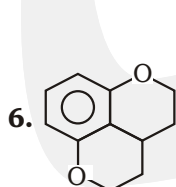
Compare rate of EAS:

- (a) a b c

- (b) b c a

- (c) a c b

- (d) b a c



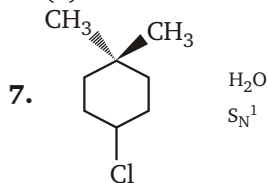
Compare rate of EAS

(a) $a > b > c$

(b) $a > c > b$

(c) $b > a > c$

(d) $c > a > b$



Total number of plane of symmetry present in carbocation formed (of given compound) in rate determining step will be:

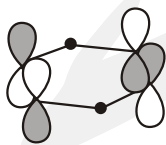
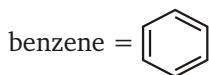
(a) 1

(b) 2

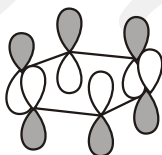
(c) 3

(d) 4

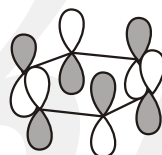
8. Based upon your understanding of the basic principles of molecular orbitals it should be possible to answer the following question. Consider the following six molecular orbitals of benzene. How many of these orbitals will contain electrons?



I



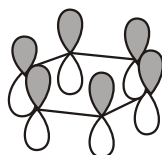
II



III



IV



V



VI

(a) 1

(b) 3

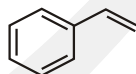
(c) 4

(d) 6

9. How would you synthesize the following compound?



?



(a) 1. MeCOCl , AlCl_3

2. Zn (Hg) HCl

(c) 1. $\text{CH}_3\text{CH}_2\text{Br}$, FeBr_3

2. Br_2 , light

3. $t\text{BuONa}$, $t\text{BuOH}$, heat

(b) 1. CH_3COOH , AlCl_3

3. H_2SO_4 , H_2O , heat

(d) 1. CH_3CHO , Mg in ether

2. NaBH_4 , MeOH

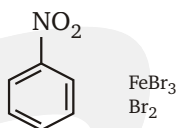
3. EtONa , EtOH

2. H_2 , Pd

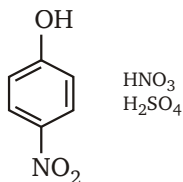
10. Which of the following represents the best reagent(s) for the electrophilic nitration of benzene?

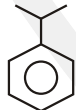


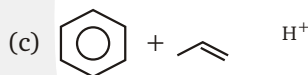
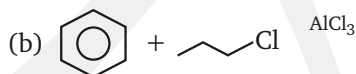
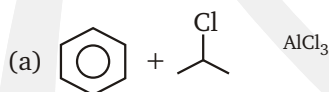
- (a) NO_2 (b) HNO_2 (c) HNO_3 (d) $\text{H}_2\text{SO}_4, \text{HNO}_3$
11. Predict the major product of the following reaction.



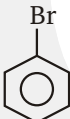
- (a) *o*-bromonitrobenzene and *p*-bromonitrobenzene
 (b) *m*-bromonitrobenzene
 (c) *o*-bromoaniline and *p*-bromoaniline
 (d) *m*-bromoaniline
12. Predict the major product of the following reaction.

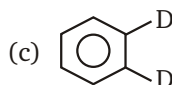
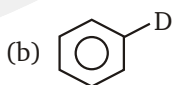
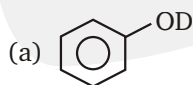


- (a) 2,4-dinitrophenol (b) 3,4-dinitrophenol
 (c) 2,4,6-trinitrophenol (d) 3,4,5-trinitrophenol
13. Cumene () is prepared by:

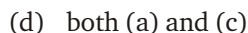
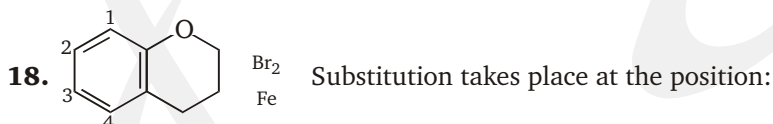
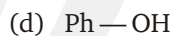
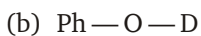
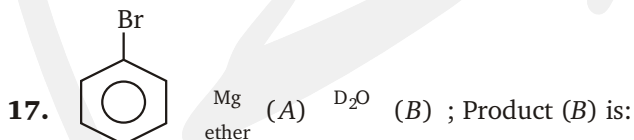
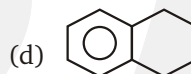
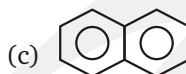
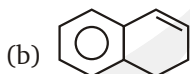
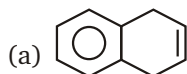
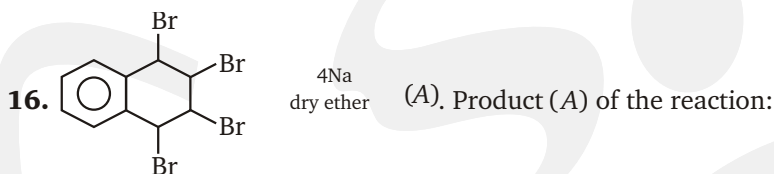
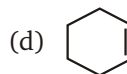
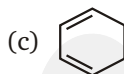
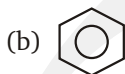
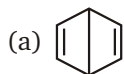
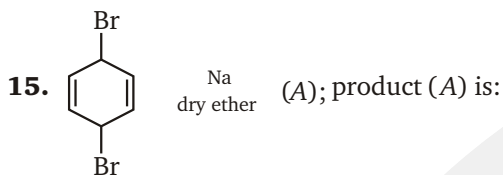


(d) All of these

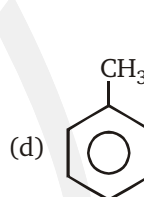
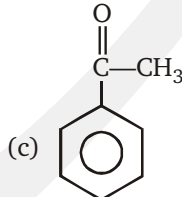
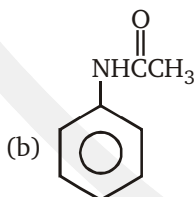
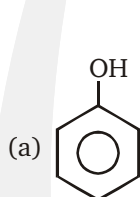
14.  $\xrightarrow[\text{ether}]{\text{Mg}}$ (A) $\xrightarrow{\text{D}-\text{O}-\text{D}}$ (B); Product (B) is:



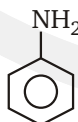
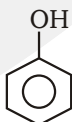
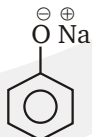
(d) none of these



19. Which of the following compounds is more reactive than benzene toward ring bromination?



20. Compare rate of EAS



(a)

(b)

(c)

(d)

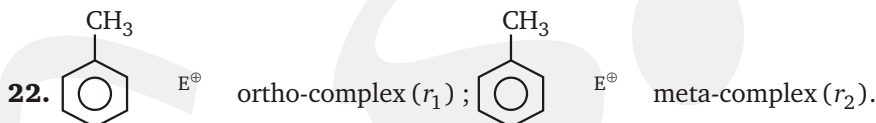
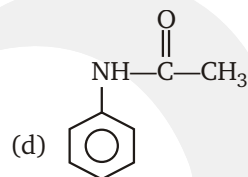
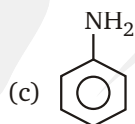
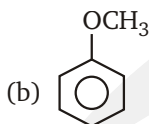
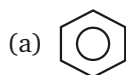
(a) a b c d

(b) a c b d

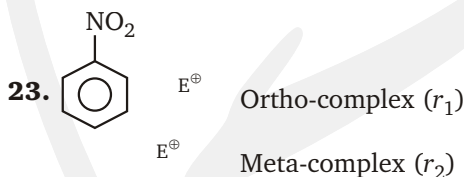
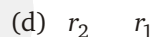
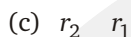
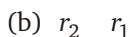
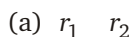
(c) c a b d

(d) c b a d

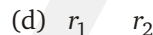
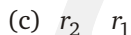
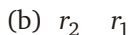
21. Which of the compound will not undergo Friedel Craft reaction ?



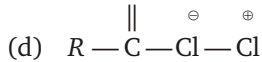
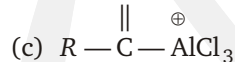
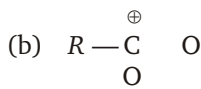
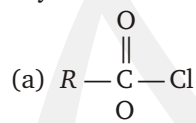
Rate of reaction is:



Rate of reaction is:



24. Which of the following is the electrophile that attacks the aromatic ring during Friedel-Crafts acylation ?



25. Which of the following substituents acts as a moderate activator and *o/p* director in electrophilic aromatic substitution reactions ?



26. In electrophilic aromatic substitution reactions a chlorine substituent :

(a) is a deactivator and a *m*-director

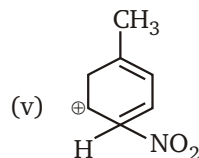
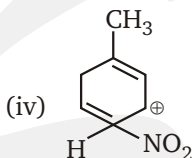
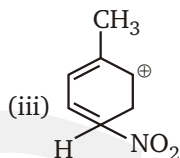
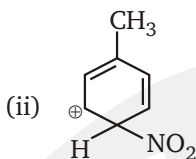
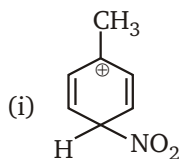
(b) is a deactivator and an *o,p*-director

(c) is an activator and a *m*-director

(d) is an activator and an *o,p*-director

(e) none of the above

27. Which of the following structures is the most important contributor to the resonance hybrid formed when toluene undergoes para-nitration ?



(a) i

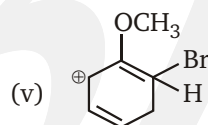
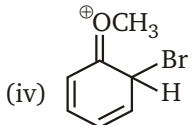
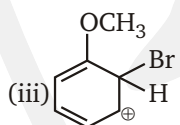
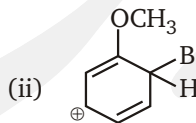
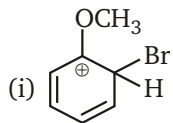
(b) ii

(c) iii

(d) iv

(e) v

28. Which of the following structure is the most important contributor to the resonance hybrid formed when anisole undergoes *o*-bromination ?



(a) i

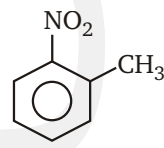
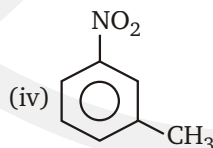
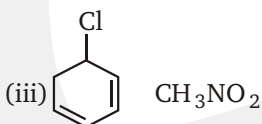
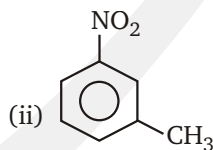
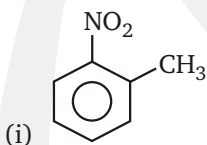
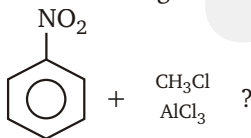
(b) ii

(c) iii

(d) iv

(e) v

29. What is/are the product(s) from the following reaction ?



(a) i

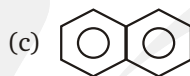
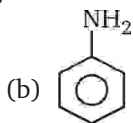
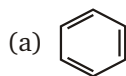
(b) ii

(c) iii

(d) iv

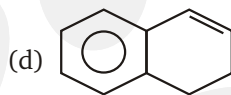
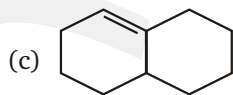
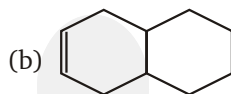
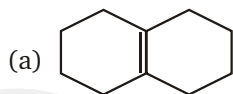
(e) no reaction

30. Which of the following is benzenoid aromatic compound?

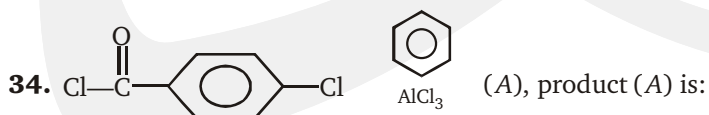
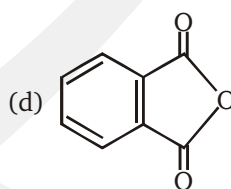
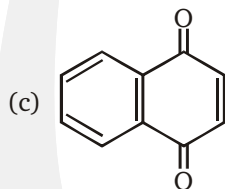
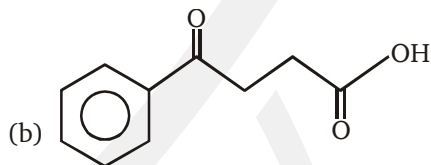
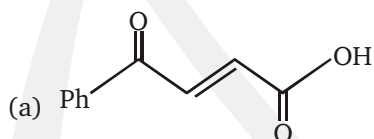
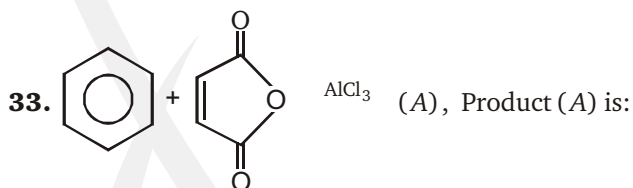
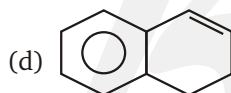
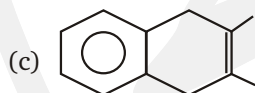
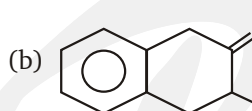
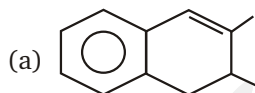


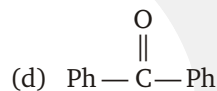
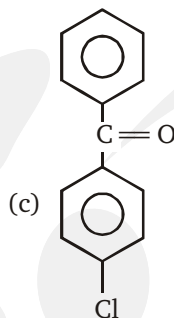
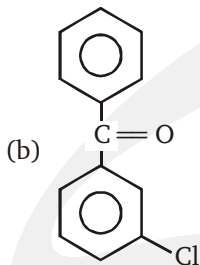
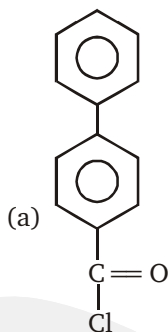
(d) All of these

31. Which of the following is most stable alkene?



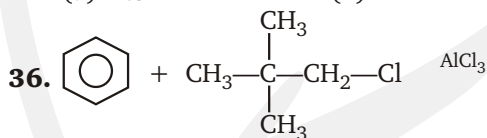
32. Which of the following is least stable alkene?



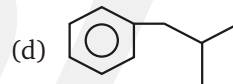
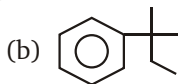


35. Compound A (C_6H_8) which is optically active, react with excess H_2 in Pd to give C_6H_{14} which is resolvable compound. How many moles of H_2 molecules are required for the hydrogenation of compound A ?

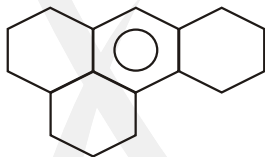
- (a) 1.5 (b) 2 (c) 3 (d) 4



Major product of the reaction is:



37.



Number of benzylic hydrogens in given compound is:

- (a) 7 (b) 8 (c) 9 (d) 10

38. What is the major organic product of the reaction between benzene and isobutyl chloride in the presence of AlCl_3 ?

- (a) *tert*-butylbenzene (b) isobutylbenzene
(c) *n*-butylbenzene (d) chlorobenzene (e) *sec*-butylbenzene

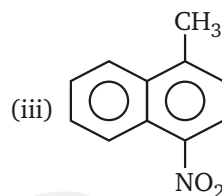
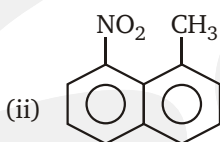
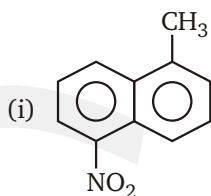
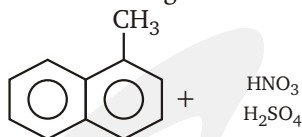
39. What is the major organic product that results when benzene is treated with 1-chlorobutane and AlCl_3 ?

- (a) Ethylbenzene (b) Isobutylbenzene
(c) Butylbenzene (d) *Sec*-butylbenzene (e) *Ter*butylbenzene

40. Which of the following aromatic rings is alkylated most rapidly by $\text{CH}_3\text{CH}_2\text{Cl}/\text{AlCl}_3$?

- (a) Benzene (b) Ethylbenzene
(c) Chlorobenzene (d) Benzenesulfonic acid (e) Nitrobenzene

41. What is the major product of the following reaction ?



(a) i

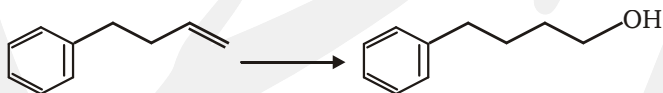
(b) ii

(c) iii

(d) a and b equally

(e) a and c equally

42. What reagents can best be used to accomplish the following transformation?



(a) 1. BH_3 , THF 2. HO^- , H_2O_2

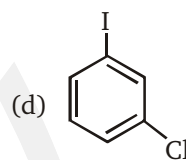
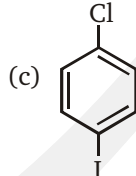
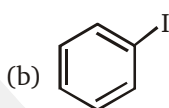
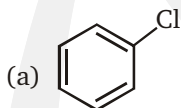
(b) H^+ , H_2O

(c) 1. $\text{Hg}(\text{OAc})_2$, H_2O /THF 2. NaBH_4

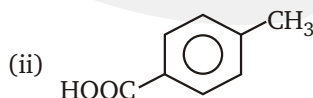
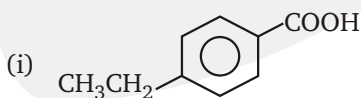
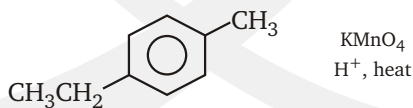
(d) 1. $\text{Hg}(\text{O}_2\text{CCF}_3)_2$, CH_3OH 2. NaBH_4

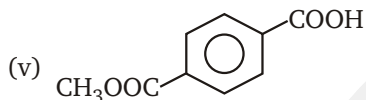
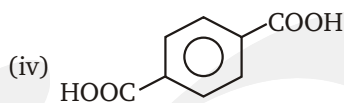
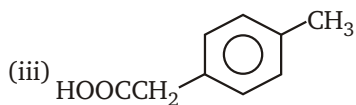
(e) NaOH , H_2O

43. The compound X in the reaction,



44. What is the major product of the following reaction ?





(a) i

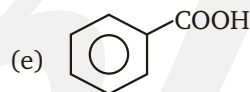
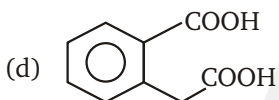
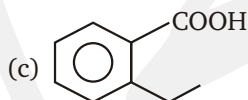
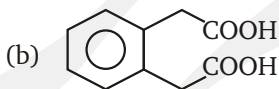
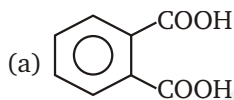
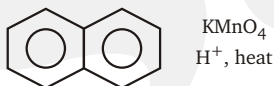
(b) ii

(c) iii

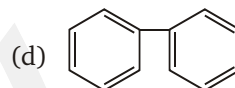
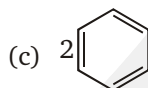
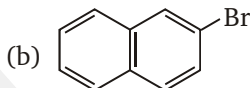
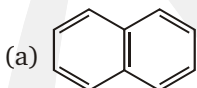
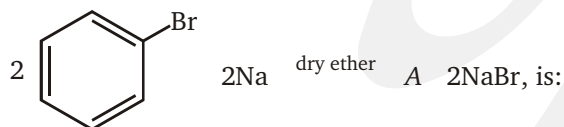
(d) iv

(e) v

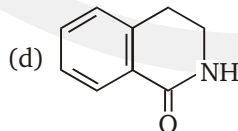
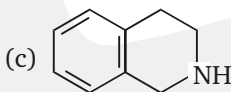
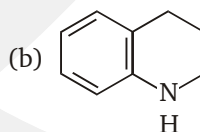
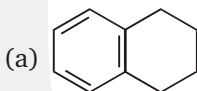
45. What is the major product of the following reaction ?



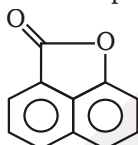
46. The product A in the reaction



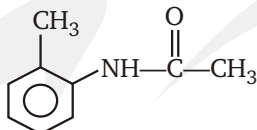
47. Which one of the following compounds undergoes bromination of its aromatic ring (electrophilic aromatic substitution) at the fastest rate ?



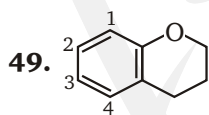
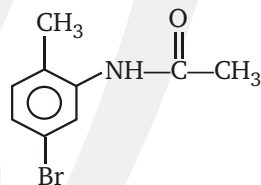
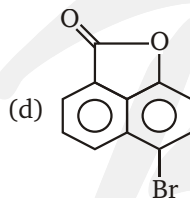
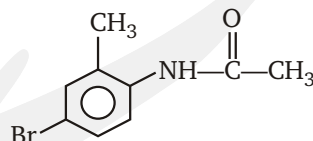
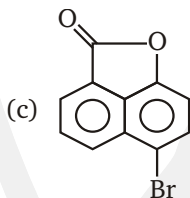
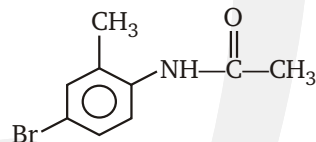
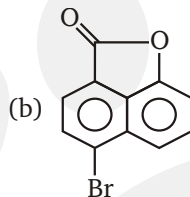
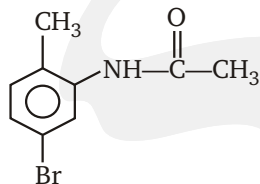
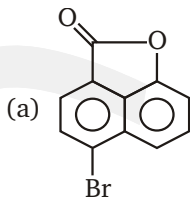
48. The compound *P*, *Q* were separately subjected to halogenation using Br_2/Fe . The major product formed in each case respectively is:



(P)

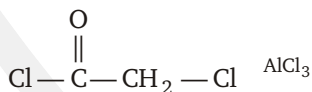
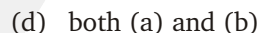


(Q)

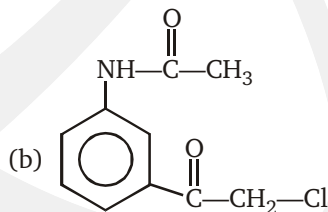
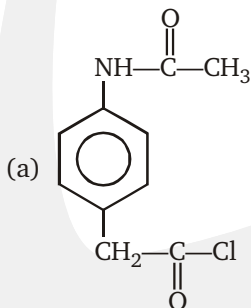


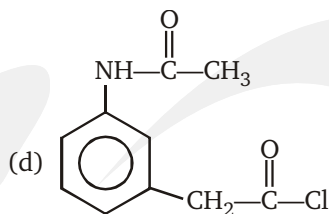
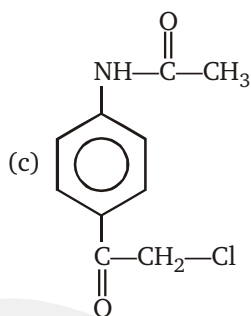
Br_2
 Fe

substitution take place at the position.

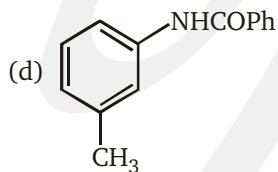
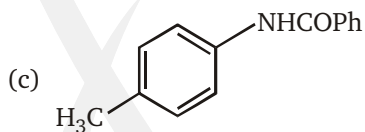
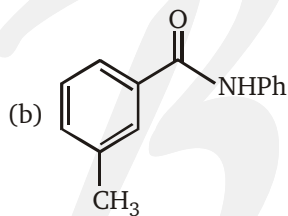
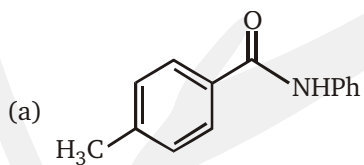
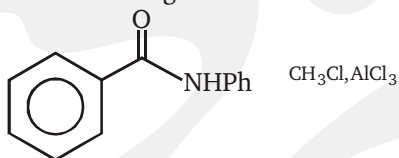


(A) (79–83%). Product (A) is:





51. Give the major product of the following reaction:



Answers

[A] Aliphatic Hydrocarbons

Alkanes

1.	(a)	2.	(b)	3.	(b)	4.	(a)	5.	(b)	6.	(c)	7.	(e)	8.	(b)	9.	(a)	10.	(b)
11.	(c)	12.	(a)	13.	(b)	14.	(b)	15.	(a)	16.	(c)	17.	(c)	18.	(c)	19.	(a)	20.	(a)
21.	(a)	22.	(c)	23.	(b)	24.	(a)	25.	(b)	26.	(a)	27.	(b)	28.	(e)	29.	(b)	30.	(e)

Alkenes

1.	(d)	2.	(c)	3.	(c)	4.	(a)	5.	(d)	6.	(c)	7.	(c)	8.	(c)	9.	(a)	10.	(b)
11.	(a)	12.	(c)	13.	(a)	14.	(b)	15.	(b)	16.	(c)	17.	(d)	18.	(a)	19.	(d)	20.	(e)
21.	(d)	22.	(c)	23.	(a)	24.	(c)	25.	(a)	26.	(d)	27.	(b)	28.	(d)	29.	(b)	30.	(a)
31.	(a)	32.	(d)	33.	(c)	34.	(b)	35.	(d)	36.	(d)	37.	(b)	38.	(b)	39.	(c)	40.	(a)
41.	(e)	42.	(d)	43.	(b)	44.	(b)	45.	(d)	46.	(b)	47.	(a)	48.	(c)	49.	(c)	50.	(a)
51.	(a)	52.	(b)	53.	(d)	54.	(b)	55.	(b)	56.	(b)	57.	(e)	58.	(b)	59.	(a)	60.	(b)
61.	(a)	62.	(d)																

Alkynes

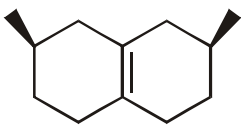
1.	(c)	2.	(c)	3.	(d)	4.	(d)	5.	(c)	6.	(c)	7.	(b)	8.	(c)	9.	(b)	10.	(c)
11.	(a)	12.	(b)	13.	(e)	14.	(e)	15.	(d)	16.	(a)	17.	(a)	18.	(c)	19.	(a)		

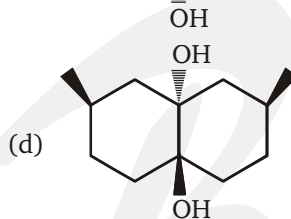
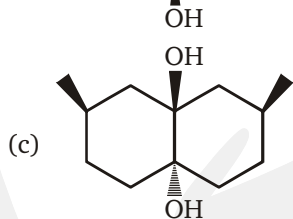
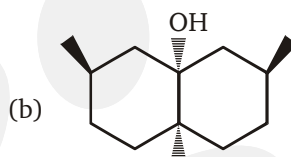
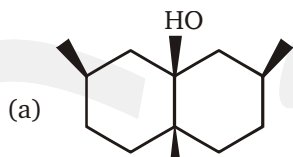
[B] Aromatic Hydrocarbons

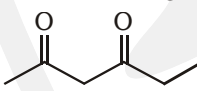
Preparation & Properties

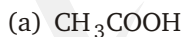
1.	(b)	2.	(c)	3.	(c)	4.	(a)	5.	(a)	6.	(d)	7.	(b)	8.	(b)	9.	(c)	10.	(d)
11.	(b)	12.	(a)	13.	(d)	14.	(b)	15.	(b)	16.	(c)	17.	(a)	18.	(d)	19.	(a)	20.	(b)
21.	(c)	22.	(a)	23.	(b)	24.	(b)	25.	(d)	26.	(b)	27.	(a)	28.	(d)	29.	(e)	30.	(d)
31.	(d)	32.	(b)	33.	(a)	34.	(c)	35.	(c)	36.	(b)	37.	(c)	38.	(a)	39.	(d)	40.	(b)
41.	(c)	42.	(a)	43.	(b)	44.	(d)	45.	(a)	46.	(d)	47.	(b)	48.	(c)	49.	(d)	50.	(c)
51.	(d)																		

One or More than One Answers is/are Correct

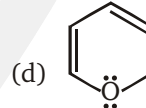
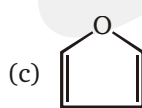
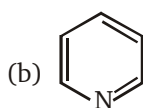
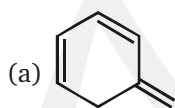
1.  cold KMnO_4 (dil) Identify products:



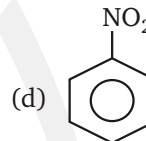
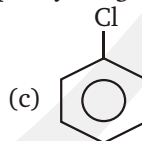
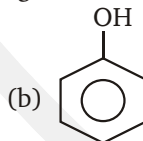
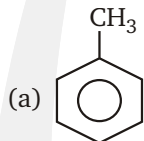
2.  O_3 / Zn Identify all possible product:

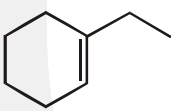


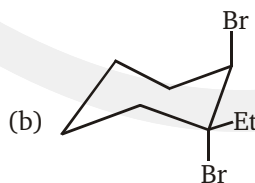
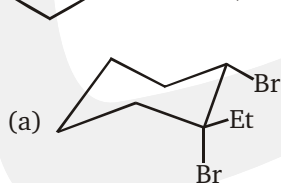
3. Which of the following compounds are aromatic ?

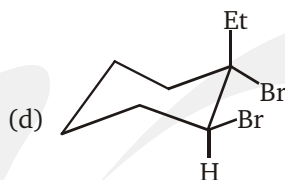
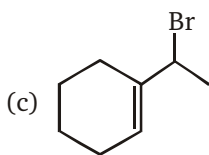


4. In which of the following electron density on phenyl ring is more than benzene?

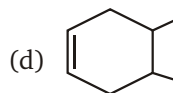
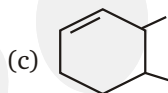
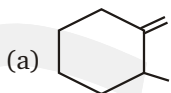


5.  Br_2 / CCl_4 Identify the products.

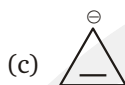
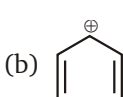
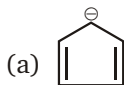




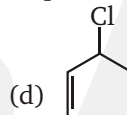
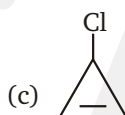
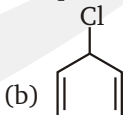
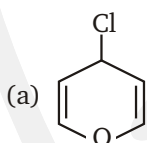
6. Heat of hydrogenation of is less than:



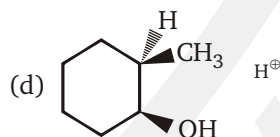
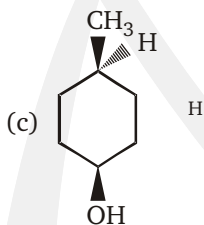
7. Which of the following is/are aromatic ?



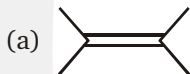
8. Which of the following compounds when react with AlCl_3 aromatic compound will form ?



9. In which of the following reaction formation of racemic mixture ?



10. Which of the following is most reactive toward bromination ?



Answers

1.	(a, b)	2.	(a, b, c, d)	3.	(b, c)	4.	(a, b)	5.	(b, d)	6.	(a, b, c, d)
7.	(a, d)	8.	(a, c)	9.	(c)	10.	(a)				

Comprehension Based Problems

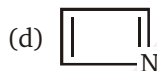
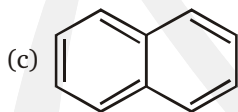
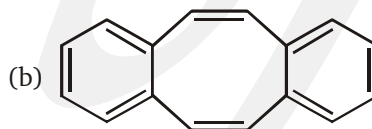
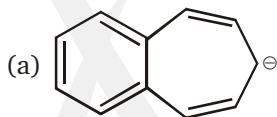
Passage-1:

We've now learned that benzene is unusually stable, and that this stability seems to be correlated with the overlap of its carbon $2p$ orbitals to form molecular orbitals. In 1931, Erich Huckel (1896-1980), a German chemist physicist, elucidated with molecular orbital arguments the criteria for this sort of stability, which has come to be called aromaticity.

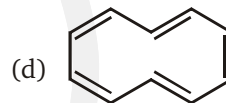
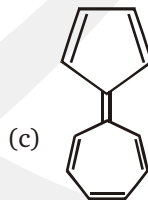
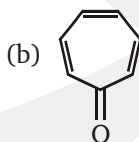
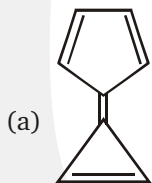
Criteria for aromaticity:

- Aromatic compounds contain one or more rings that have a cyclic arrangement of p -orbitals. Thus, aromaticity is a property of certain cyclic compounds.
- Every atom of an aromatic ring has a p -orbital.
- Aromatic rings are planar.
- The cyclic arrangements of p -orbitals in an aromatic compound must contain $4n + 2$ electrons, where n is any positive integer (0, 1, 2, ...). In other words, an aromatic ring must contain 2, 6, 10, ... electrons.

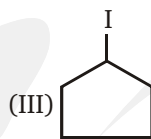
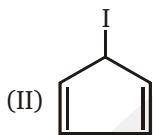
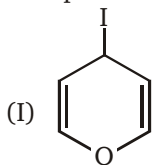
1. Which of the following compound is not anti-aromatic ?



2. Which of the following compound is non-aromatic ?



3. Compare rate of reaction with AgNO_3



(a) $\text{I} > \text{II} > \text{III}$

(b) $\text{I} > \text{III} > \text{II}$

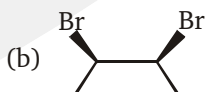
(c) $\text{II} > \text{III} > \text{I}$

(d) $\text{II} > \text{I} > \text{III}$

Passage-2:

In electrophilic addition reaction, alkene behave as a nucleophile or base. During reaction electrophile react with alkene in rate determining step to form carbocation according to advance markownikoff rule. In second step carbocation react with nucleophile to give product so overall addition reaction take place.

1. Br_2 Possible product:



2. Br_2 CCl_4 Number of product formed in the reaction:

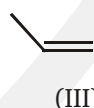
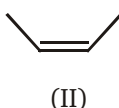
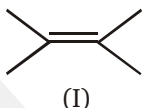
(a) 1

(b) 2

(c) 3

(d) 4

3. Correct order of rate of Bromination:



(a) $\text{I} > \text{II} > \text{III}$

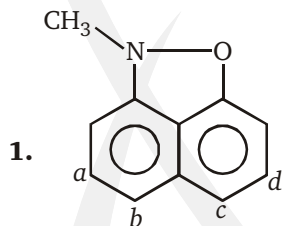
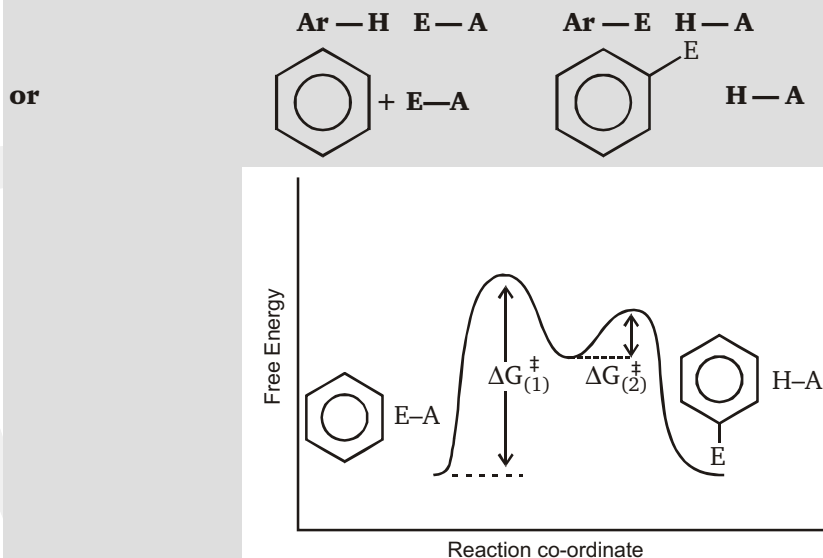
(b) $\text{II} > \text{I} > \text{III}$

(c) $\text{III} > \text{II} > \text{I}$

(d) $\text{II} > \text{III} > \text{I}$

Passage-3:

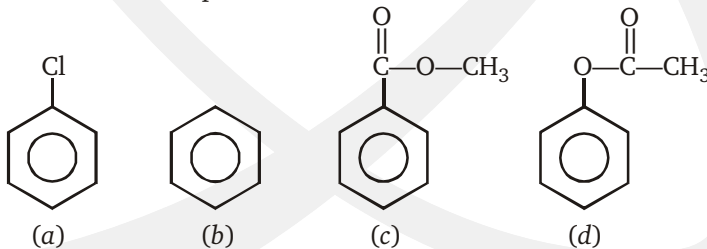
The most characteristic reactions of benzenoid arenes are the substitution reaction that occur when they react with electrophilic reagents. These reactions are of the general type shown below.



Identify the position where EAS (Electrophilic Aromatic Substitution) reaction takes place:

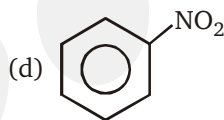
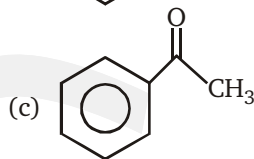
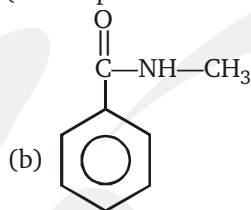
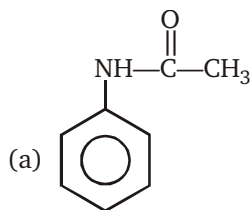
- (a) a (b) b (c) c (d) d

2. Compare rate of EAS (Electrophilic Aromatic Substitution):

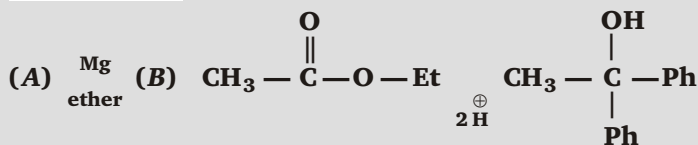


- (a) $a > b > c > d$ (b) $b > a > c > d$ (c) $d > b > c > a$ (d) $d > b > a > c$

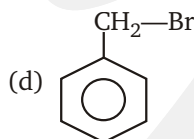
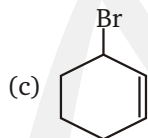
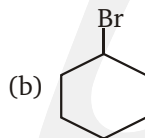
3. Which of the following is *o/p* director for EAS (Electrophilic Aromatic Substitution) ?



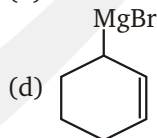
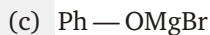
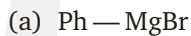
Passage-4:



1. Compound (A) is:

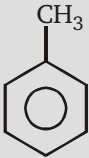
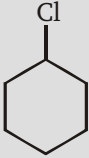


2. Compound (B) is:

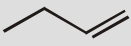
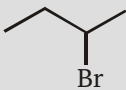
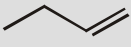
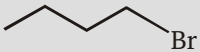
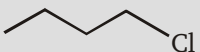
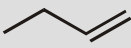
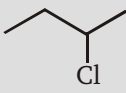
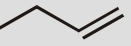
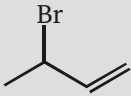


Matching Type Problems

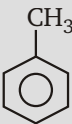
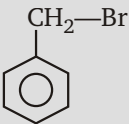
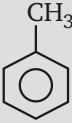
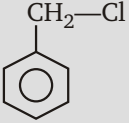
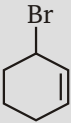
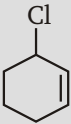
1. Match the column:

Column-I		Column-II	
(a)	 HBr Peroxide	(p)	Electrophilic-addition reaction
(b)	 HCl Peroxide	(q)	Free-radical-addition reaction
(c)	 NBS	(r)	Free-radical-substitution reaction
(d)	 aq. KOH	(s)	Formation of free-radical as a intermediate in major product formation
		(t)	One-step reaction

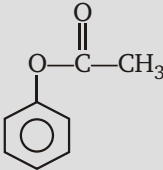
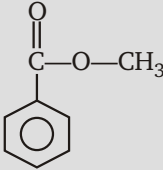
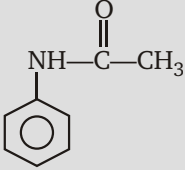
2. Match the column:

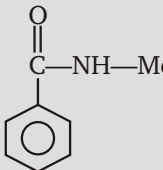
Column-I		Column-II	
(a)	 HBr Peroxide	(p)	
(b)	 HBr No Peroxide	(q)	
(c)	 HCl Peroxide	(r)	
(d)	 HCl CCl ₄	(s)	
(e)	 NBS	(t)	

3. Match the column:

Column-I		Column-II (Reagent used is)	
(a)	 	(p)	NBS
(b)	 	(q)	$\text{Br}_2/h\nu$
(c)	 	(r)	$\text{SO}_2\text{Cl}_2/h\nu$
(d)	 	(s)	$\text{Cl}_2/h\nu$

4. Match the column:

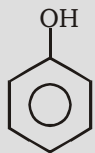
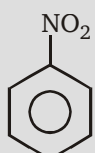
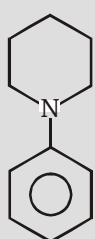
Column-I		Column-II	
(a)		(p)	<i>o/p</i> -directive
(b)		(q)	meta directive
(c)		(r)	rate of EAS is less than benzene

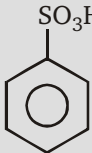
(d)		(s)	rate of EAS is more than benzene.
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5. Match the column:

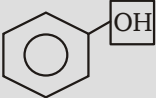
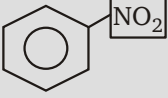
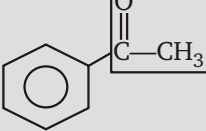
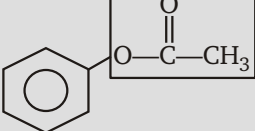
Column-I (Groups are attached with phenyl ring)		Column-II	
(a)	$\text{—}\overset{\oplus}{\text{N}}\text{R}_3$	(p)	shows <i>I</i> effect
(b)	$\text{—}\overset{\text{O}}{\parallel}\text{C—H}$	(q)	shows <i>H</i> effect
(c)	$\text{—}\ddot{\text{O}}\text{—Me}$	(r)	shows <i>M</i> effect
(d)	—CH_3	(s)	shows <i>I</i> effect
		(t)	shows <i>M</i> effect

6. Match the column:

Column-I		Column-II	
(a)		(p)	Group attached with benzene ring is <i>M</i> group.
(b)		(q)	Group attached with benzene ring is <i>M</i> group.
(c)		(r)	Group attached with benzene ring is <i>I</i> .

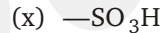
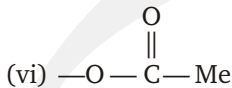
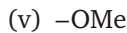
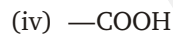
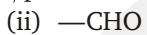
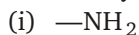
(d)		(s)	Group attached with benzene ring is +I.
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7. Match the column:

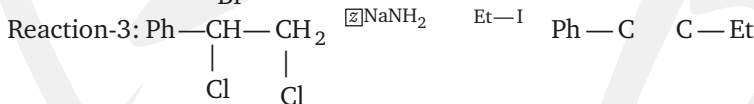
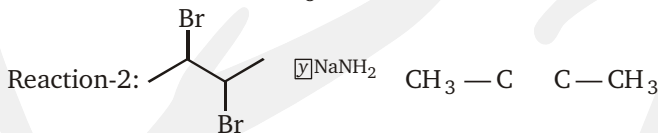
Column-I		Column-II (Group attached with phenyl ring is)	
(a)		(p)	<i>M</i>
(b)		(q)	<i>M</i>
(c)		(r)	<i>I</i>
(d)		(s)	<i>I</i>

Multi Concept Subjective Problems

1. How many groups are *o/p* director in the electrophilic aromatic substitution ?



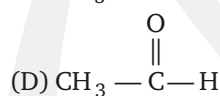
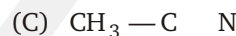
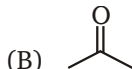
2. How many isomer of C_8H_{10} when reacts with hot alkaline KMnO_4 give di-carboxylic acid as a product ?



x, y, z are moles used.

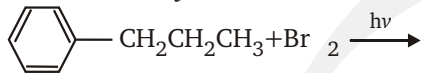
Sum of $[x \quad y \quad z]$

4. Amongst the following, the total number of compounds which will undergo addition reaction will be:

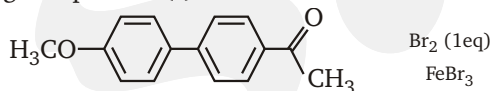


Unified Conceptual Problems

1. Predict the major monobromination product in the following reaction.



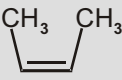
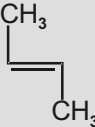
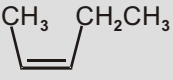
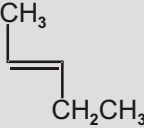
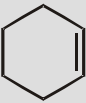
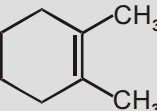
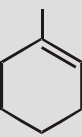
2. Provide the major organic product(s) of the reaction below.



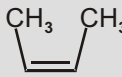
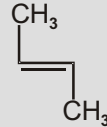
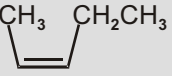
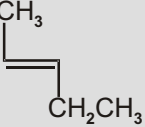
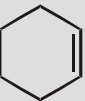
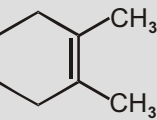
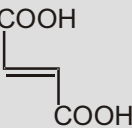
3. This questions gives you an idea about mono and dihalogenated products of an reactant.

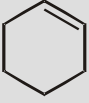
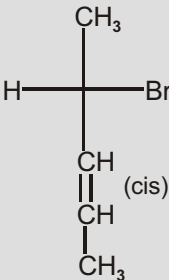
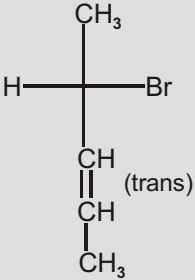
S.No.	Compound	Number of monochloro product	Number of monochloro product (excluding stereoisomer)
1.			
2.			
3.			
4.			
5.			
6.			
7.			
8.			
9.			

4. Identify product including stereo chemistry

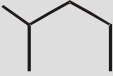
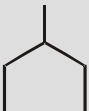
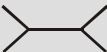
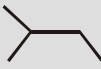
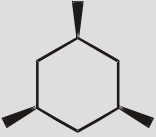
1.	 Ni/D_2	2.	 Ni/D_2
3.	 Ni/D_2	4.	 Ni/D_2
5.	 Ni/D_2	6.	 Ni/D_2
7.	$\text{CH}_3 - \text{CH} = \text{CH}_2$ Ni/D_2	8.	$\text{CH}_3 - \text{C} \quad \text{C} - \text{CH}_3$ Ni/D_2
9.	 Ni/D_2	10.	 Ni/D_2

5. Identify product including stereo chemistry

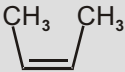
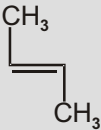
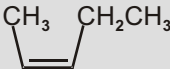
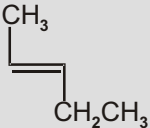
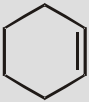
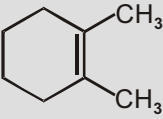
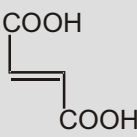
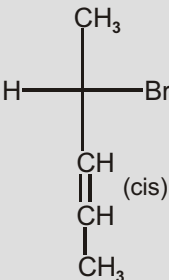
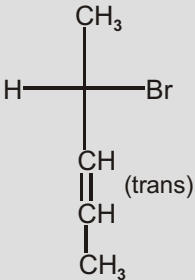
1.	 $\text{Br}_2, \text{CCl}_4$	2.	 $\text{Br}_2, \text{CCl}_4$
3.	 $\text{Br}_2, \text{CCl}_4$	4.	 $\text{Br}_2, \text{CCl}_4$
5.	 $\text{Br}_2, \text{CCl}_4$	6.	 $\text{Br}_2, \text{CCl}_4$
7.	$\text{CH}_3 - \text{CH} = \text{CH}_2$ $\text{Br}_2, \text{CCl}_4$	8.	 $\text{Br}_2, \text{CCl}_4$
9.	$\text{CH}_3 - \text{CH} \quad \text{C} - \text{CH}_3$ $\text{Br}_2, \text{CCl}_4$	10.	 $\text{Br}_2, \text{CCl}_4$

11.	$\text{CH}_2=\text{CH}-\text{CH}_2-\text{C}(\text{CH}_3)_2$	$\text{Br}_2, \text{CCl}_4$	
12.	 Br_2 H_2O		13.  $\text{Br}_2, \text{CCl}_4$
14.	 $\text{Br}_2, \text{CCl}_4$		

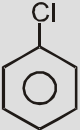
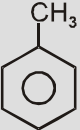
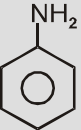
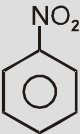
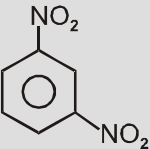
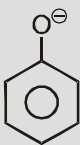
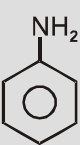
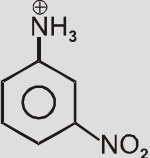
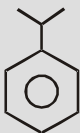
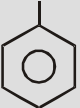
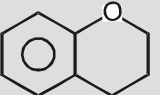
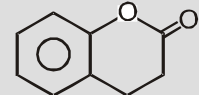
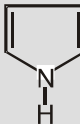
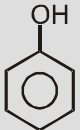
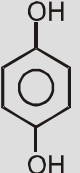
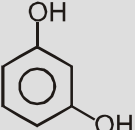
6. Identify monochlorination products (including stereo)

1.	$\text{CH}_3\text{CH}_2\text{CH}_3$	Cl_2/h	2.	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$	Cl_2/h
3.		Cl_2/h	4.		Cl_2/h
5.		Cl_2/h	6.		Cl_2/h
7.		Cl_2/h	8.		Cl_2/h
9.		Cl_2/h	10.		Cl_2/h

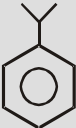
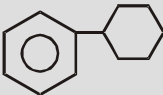
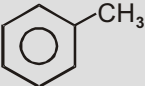
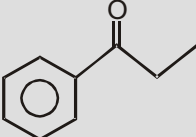
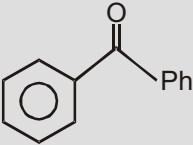
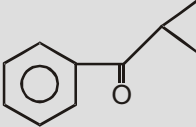
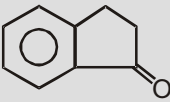
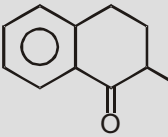
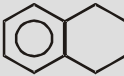
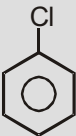
7. Identify product including stereo chemistry:

1.		Cold KMnO_4	2.		Cold KMnO_4
3.		Cold KMnO_4	4.		Cold KMnO_4
5.		Cold KMnO_4	6.		Cold KMnO_4
7.	$\text{CH}_3 - \text{CH} = \text{CH}_2$	Cold KMnO_4	8.		Cold KMnO_4
9.	$\text{CH}_3 - \text{C} \quad \text{C} - \text{CH}_3$	Cold KMnO_4	10.		Cold KMnO_4
11.	$\text{CH}_2 = \text{CH} - \text{CH}_2 - \text{C} \quad \text{CH}$	Cold KMnO_4	12.		Cold KMnO_4
13.		Cold KMnO_4			

8.

S.No.	Compare Rate of EAS
1.	   
2.	  
3.	   
4.	  
5.	  
6.	  
7.	  

9. Prepare the following by Friedal Craft reaction.

S. No.	Compounds	S. No.	Compounds
1.		2.	
3.		4.	
5.		6.	
7.		8.	
9.		10.	
11.		12.	

Answers

Comprehension Based Problems:

Passage-1:	1.	(c)	2.	(d)	3.	(b)				
Passage-2:	1.	(a)	2.	(b)	3.	(a)				
Passage-3:	1.	(b)	2.	(d)	3.	(a)				
Passage-4:	1.	(a)	2.	(a)						

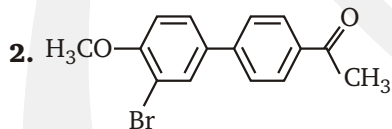
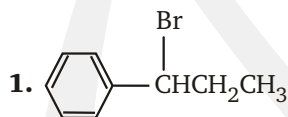
Matching Type Problems:

- (a) q,s; (b) p; (c) r,s; (d) t
- (a) q; (b) p; (c) s; (d) s; (e) t
- (a) p, q; (b) r, s; (c) p, q; (d) r, s
- (a) p, s; (b) q, r; (c) p, s; (d) q, r
- (a) s; (b) s,t; (c) r,s; (d) p,q
- (a) p, r; (b) q, r; (c) p, r; (d) q, r
- (a) p,r (b) q,r (c) q,r (d) p,r

Multi Concept Subjective Problems

1.	6	2.	3	3.	8	4.	5												
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Unified Conceptual Problems



- 3.
- | | | | |
|---------|---------|---------|----------|
| 1. 4, 3 | 2. 3, 2 | 3. 6, 4 | 4. 10, 6 |
| 5. 6, 4 | 6. 1, 1 | 7. 1, 1 | 8. 1, 1 |
| 9. 2, 2 | | | |

4. Ni / D₂ performs syn addition

- | | | |
|---------------------|--------------------|--------------------|
| 1. Meso | 2. Racemic mixture | 3. Racemic mixture |
| 4. Racemic mixture | 5. Meso | 6. Meso |
| 7. Racemic mixture | 8. Single product | 9. Racemic mixture |
| 10. Racemic mixture | | |

5.

- | | | | | |
|---------------------|---------------------|--------------|--------------------|--------------|
| 1. 2-racemic | 2. 1-meso | 3. 2-racemic | 4. 2-racemic | 5. 2-racemic |
| 6. 2-racemic | 7. 2-racemic | 8. 1-meso | 9. 2-diastereomers | |
| 10. 2-racemic | 11. 2-diastereomers | | 12. 2-racemic | |
| 13. 2-diastereomers | 14. 2-diastereomers | | | |

6.

- | | | | | |
|------|------|------|------|-------|
| 1. 2 | 2. 3 | 3. 4 | 4. 8 | 5. |
| 6. 2 | 7. 5 | 8. 3 | 9. 1 | 10. 5 |

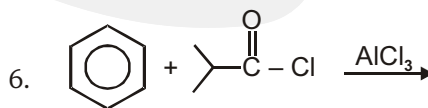
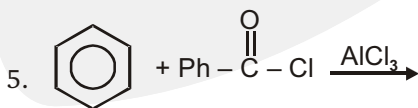
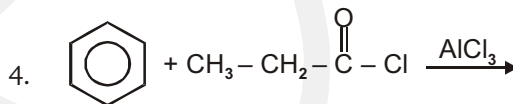
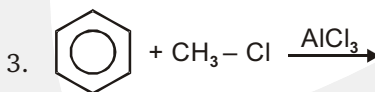
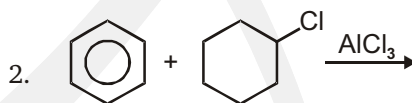
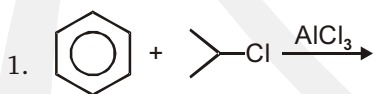
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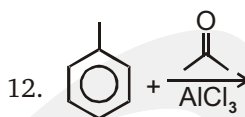
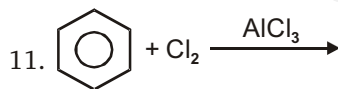
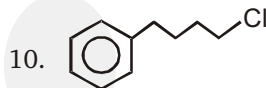
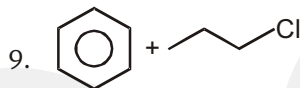
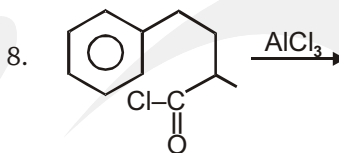
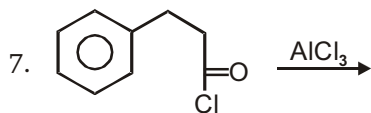
- | | | | |
|-----------------------|-----------------------|--------------|--------------|
| 1. 1-Meso | 2. 2-Racemic | 3. 2-Racemic | 4. 2-Racemic |
| 5. 1-Meso | 6. 1-Meso | 7. 2-Racemic | 8. 2-Racemic |
| 9. Left | 10. 1-Meso | 11. 1-Meso | |
| 12. 2-(Diastereomers) | 13. 2-(Diastereomers) | | |

8.

- | | | |
|------------|----------|------------|
| 1. 4 3 1 2 | 2. 1 2 3 | 3. a b d c |
| 4. 3 1 2 | 5. 2 3 1 | 6. a b c |
| 7. c b a | | |

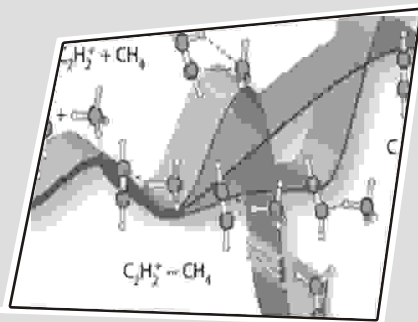
9.





3

Haloalkanes and Haloarenes



Only One Answer is Correct

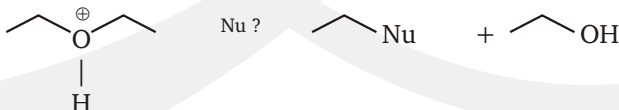
[A] Haloalkanes : Preparation and Properties

Bimolecular Nucleophilic Substitution (S_N2) Reactions

1. What is the K_{eq} of the following reaction?

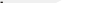


- (a) 1.7 (b) 0.17 (c) 17 (d) 0.017
2. Which halide ion is the best nucleophile in dimethyl sulfoxide solution?
(a) F (b) Cl (c) Br (d) I
3. Which is the best leaving group in a substitution reaction of an alkyl halide?
(a) Cl (b) Br (c) I (d) F
4. Which statement is true with respect to an S_N2 reaction?
(a) A good leaving group is a strong base
(b) A good leaving group is a weak base
(c) A leaving group must be negatively charged
(d) A leaving group must be a halide
5. Which of the following nucleophiles is going to promote an S_N2 reaction?



- (a) NaI (b) Water (c) NH_4Cl (d) MeONa

CCBr
 $\xrightarrow[2. \text{H}_2, \text{Pd}]{1. \text{NaS}}$
CC(C)=C

- (a)  (b)  (c)  (d) 

BrCC(C)(C)CCBr + HO^- (1 equivalent)

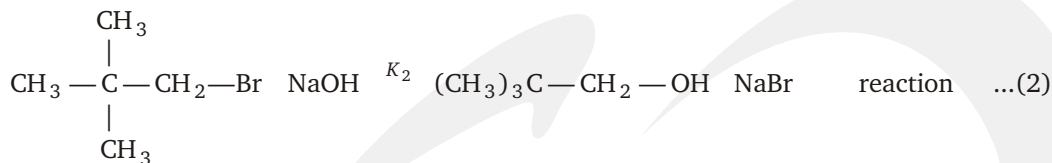
- (a)
- (b)
- (c)
- (d)

8.  NaI/acetone
(S_N2) condition

- (a) 
- (b) 
- (c) 
- (d) 

(a) $\text{CH}_3 - \text{Cl}$ (b) $\text{H}_2\text{C} = \text{CH} - \text{Cl}$ (c)  (d) $\text{CH}_3 - \text{F}$

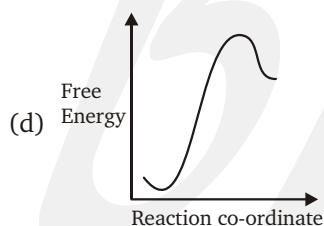
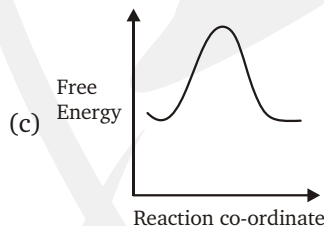
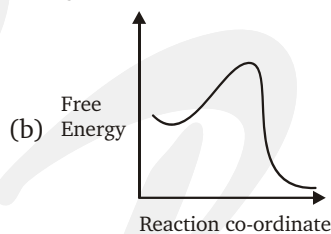
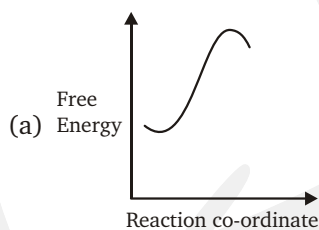
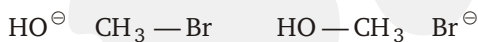
(a) $\text{CH}_3\text{CH}_2-\text{Cl}$ (b) $\text{CH}_3\text{CH}_2-\text{Br}$
(c) $\text{CH}_3\text{CH}_2-\text{I}$ (d) $\text{CH}_3\text{CH}_2-\text{F}$ (e) $\text{CH}_3\text{CH}_2-\text{OH}$



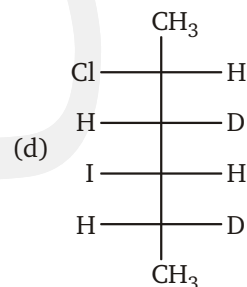
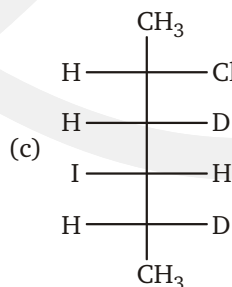
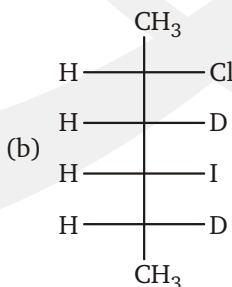
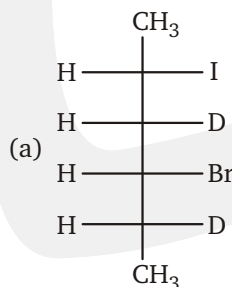
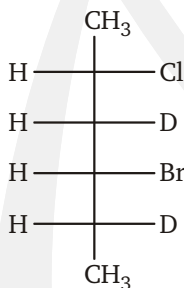
K_1 and K_2 are rate constant for above reactions. Correct relation is:

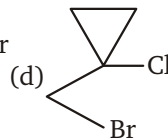
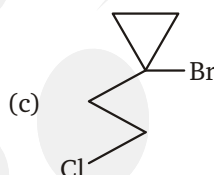
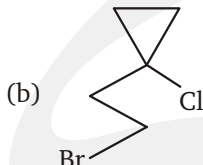
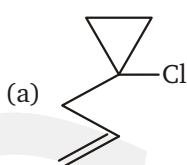
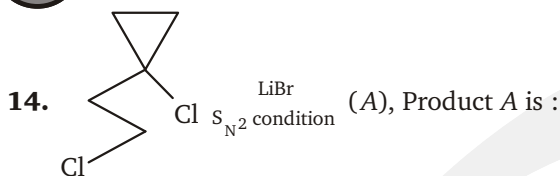
- (a) $K_1 < K_2$ (b) $K_1 > K_2$ (c) $K_1 = K_2$ (d) $K_1 \gg K_2$

12. Free energy profile for given reaction is



13. (A), Major product (A) is:





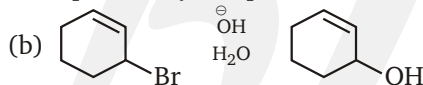
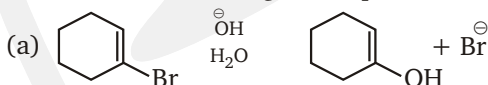
Which of the following solvents is most suitable for this reaction ?

- (a) H_2O (b) EtOH (c) Et_2O (d) None of these

16. Which of the following reactions is not possible ?

- (a) $\text{R} - \text{OH} \xrightarrow{\text{NaBr}} \text{R} - \text{Br} \xrightarrow{\text{NaOH}}$ (b) $\text{R} - \text{OH} \xrightarrow{\text{HBr}} \text{R} - \text{Br} \xrightarrow{\text{H}_2\text{O}}$
 (c) Both reactions are possible (d) Both reactions are not possible

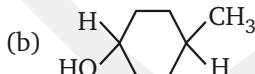
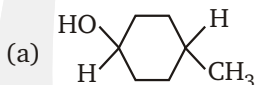
17. Which of the following nucleophilic substitutions is practically not possible ?



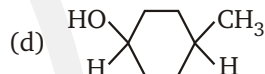
- (c) Both (a) and (b) (d) none of these

18. From each of the following pairs select the compound that will react faster with sodium iodide in acetone:

- (A) 2-Chloropropane or 2-bromopropane (B) 1-Bromobutane or 2-bromobutane
 I II I II
 (a) (A)-I, (B)-I (b) (A)-I, (B)-II (c) (A)-II, (B)-I (d) (A)-II, (B)-II



- (c) Both (a) and (b)



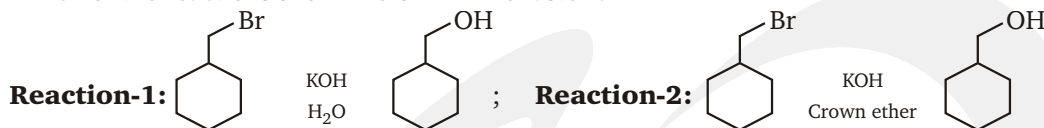
20. Correct order of nucleophilicity in polar protic solvent is:

- (a) $\text{I}^- \quad \text{Br}^- \quad \text{Cl}^- \quad \text{F}^-$ (b) $\text{F}^- \quad \text{Cl}^- \quad \text{Br}^- \quad \text{I}^-$
 (c) $\text{I}^- \quad \text{Cl}^- \quad \text{Br}^- \quad \text{F}^-$ (d) $\text{F}^- \quad \text{I}^- \quad \text{Br}^- \quad \text{Cl}^-$

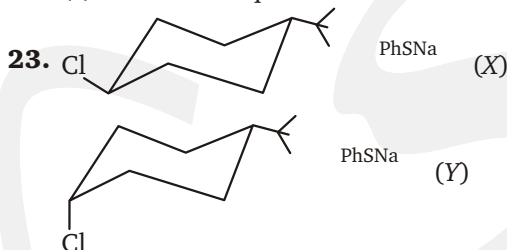
21. Correct order of nucleophilicity in polar aprotic solvent is:

- (a) $\text{I}^- \quad \text{Br}^- \quad \text{Cl}^- \quad \text{F}^-$ (b) $\text{F}^- \quad \text{Cl}^- \quad \text{Br}^- \quad \text{I}^-$
 (c) $\text{I}^- \quad \text{Cl}^- \quad \text{Br}^- \quad \text{F}^-$ (d) $\text{F}^- \quad \text{I}^- \quad \text{Br}^- \quad \text{Cl}^-$

22. Which of the reactions shown below will be faster ?

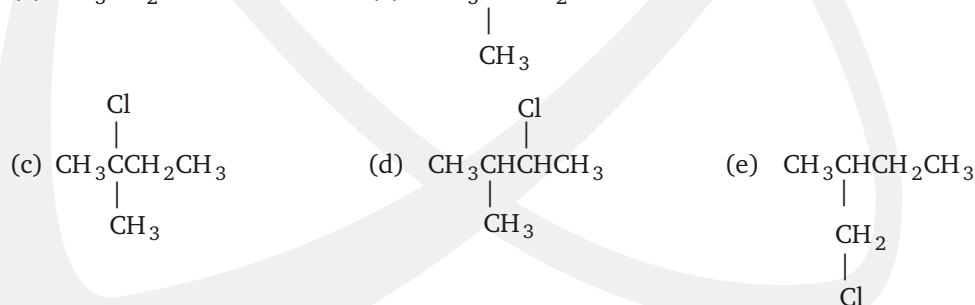


- (a) Reaction 1
(b) Reaction 2
(c) Both have equal rates of reaction
(d) Cannot be predicted



Relationship between (X) and (Y) is:

- (a) constitutional isomer
(b) enantiomer
(c) identical
(d) none of these
24. Without exception, $\text{S}_{\text{N}}2$ reaction at chiral carbon gives:
- (a) enantiomer of substrate
(b) product of opposite polarity
(c) diastereomeric mixture
(d) only one stereoisomer
25. The order of reactivities of the following alkyl halides for a $\text{S}_{\text{N}}2$ reaction is:
- (a) $\text{RF} > \text{RCl} > \text{RBr} > \text{RI}$
(b) $\text{RF} > \text{RBr} > \text{RCl} > \text{RI}$
(c) $\text{RCl} > \text{RBr} > \text{RF} > \text{RI}$
(d) $\text{RI} > \text{RBr} > \text{RCl} > \text{RF}$
26. $(\text{CH}_3)_3\text{C MgCl}$ on reaction with D_2O produces:
- (a) $(\text{CH}_3)_3\text{CD}$
(b) $(\text{CH}_3)_3\text{OD}$
(c) $(\text{CD}_3)_3\text{CD}$
(d) $(\text{CD}_3)_3\text{OD}$
27. When chloroform is hydrolysed with KOH , then the final product is:
- (a) HCOOH
(b) HCOOK
(c) CH_3OH
(d) None of these
28. Which of the following alkyl halides gives the slowest $\text{S}_{\text{N}}2$ reaction?



29. Assuming no other changes, what is the effect on rate of reaction if the concentration of the alkyl halide doubled in a $\text{S}_{\text{N}}2$ reaction?



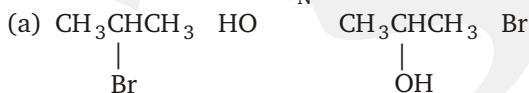
- (a) No change (b) Doubles the rate
(c) Triples the rate (d) Quadruples the rate (e) Rate is halved

30. Assuming no other changes, what is the effect on rate of reaction if the concentrations of both alkyl halide and the nucleophile are doubled ?



- (a) No change (b) Doubles the rate
(c) Triples the rate (d) Quadruples the rate (e) Rate is halved

31. Which of the following $\text{S}_{\text{N}}2$ reactions is the fastest ?



32. Which of the following $\text{S}_{\text{N}}2$ reactions is the slowest ?

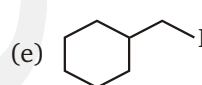
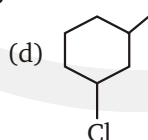
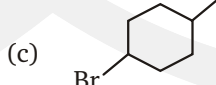
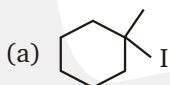


33. Which halide reacts most rapidly via an $\text{S}_{\text{N}}2$ mechanism?



(e) all primary halides react at the same rate in $\text{S}_{\text{N}}2$

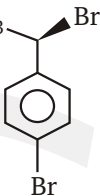
34. Which halide reacts most rapidly via an $\text{S}_{\text{N}}2$ mechanism ?

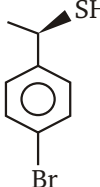
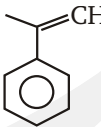
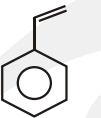
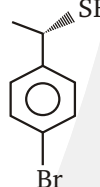


35. Assuming no other changes, what is the effect of doubling only the concentration of the alkyl halide in the following S_N1 reaction?

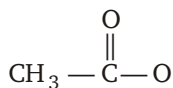
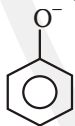


- (a) No change (b) Doubles the rate
(c) Triples the rate (d) Quadruples the rate (e) Rate is halved

36.  KSH (P) ; Major product (P) is:

- (a)  (b)  (c)  (d) 

37. Arrange the following in order of their nucleophilicity in protic solvent



(i)

(ii)

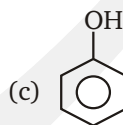
(iii)

- (a) i > ii > iii (b) iii > ii > i (c) ii > i > iii (d) None

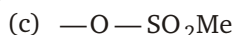
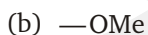
38. In which of the following protic solvents, S_N2 reaction will occur most rapidly?

- (a) CH_3O (1m) (b) CH_3O (2m)
(c) CH_3O (3m) (d) all have equal rate.

39. Rate of S_N2 reaction will be maximum in which of the following solvents?



40. For the following groups, the order of leaving group ability is:



(I)

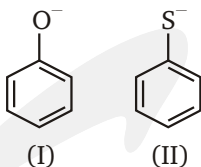
(II)

(III)

(IV)



41. Consider the following two anionic molecules. Which of the following statements is TRUE?

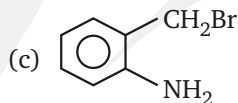
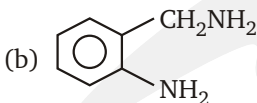
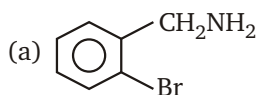
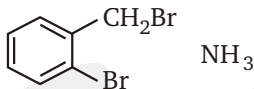


- (a) I is more basic and more nucleophilic than II
 (b) I is less basic and less nucleophilic than II
 (c) I is more basic but less nucleophilic than II
 (d) I is less basic but more nucleophilic than II
42. Which of the following is a secondary alkyl halide?
 (a) CH_3Br (b) $(\text{CH}_3)_3\text{CBr}$ (c) $(\text{CH}_3)_2\text{CHBr}$ (d) $(\text{CH}_3)_2\text{CHCH}_2\text{Br}$
43. Which of the following compounds will undergo an $\text{S}_{\text{N}}2$ reaction most readily?
 (a) $(\text{CH}_3)_3\text{CCH}_2\text{I}$ (b) $(\text{CH}_3)_3\text{CCl}$
 (c) $(\text{CH}_3)_2\text{CHI}$ (d) $(\text{CH}_3)_2\text{CHCH}_2\text{CH}_2\text{CH}_2\text{Cl}$
 (e) $(\text{CH}_3)_2\text{CHCH}_2\text{CH}_2\text{CH}_2\text{I}$
44. Which of the following $\text{S}_{\text{N}}2$ reactions is the fastest ?
 (a) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br} \xrightarrow{\text{OH}^-} \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$ Br^-
 (b) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br} \xrightarrow{\text{H}_2\text{O}} \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$ HBr
 (c) $\text{CH}_3\text{CH}_2\text{CHBrCH}_3 \xrightarrow{\text{OH}^-} \text{CH}_3\text{CH}_2\text{CHOHCH}_3$ Br^-
 (d) $\text{CH}_3\text{CH}_2\text{CHBrCH}_3 \xrightarrow{\text{H}_2\text{O}} \text{CH}_3\text{CH}_2\text{CHOHCH}_3$ HBr
45. Which of the following $\text{S}_{\text{N}}2$ reactions is the slowest ?
 (a) $\text{CH}_3\text{CH}_2\text{CHBrCH}_3 \xrightarrow{\text{OH}^-} \text{CH}_3\text{CH}_2\text{CHOHCH}_3$ Br^-
 (b) $\text{CH}_3\text{CH}_2\text{CHBrCH}_3 \xrightarrow{\text{H}_2\text{O}} \text{CH}_3\text{CH}_2\text{CHOHCH}_3$ HBr
 (c) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br} \xrightarrow{\text{OH}^-} \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$ Br^-
 (d) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br} \xrightarrow{\text{H}_2\text{O}} \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$ HBr
46. Which of the following bromides reacts readily via an $\text{S}_{\text{N}}2$ reaction with NaN_3 ?
 (a) $\text{C}_6\text{H}_5\text{Br}$ (b) $\text{CH}_3\text{CH}_2\text{CHBrCH}_3$
 (c) $(\text{C}_6\text{H}_5)_3\text{CBr}$ (d) $(\text{CH}_3)_3\text{CCH}_2\text{CH}_2\text{CH}_2\text{Br}$
 (e) 1-bromo-1-methylcyclohexane

47. (A), Product (A) is:

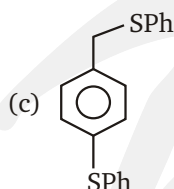
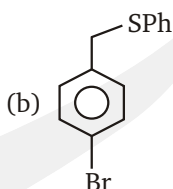
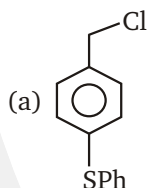
- (a) (b)
- (c) (d)

48. The major product in the given reaction:



(d) All of these

49. (A), Structure of product (A) is:

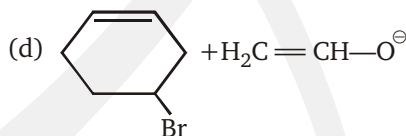
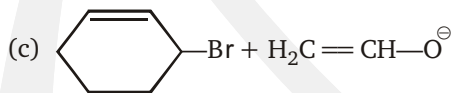
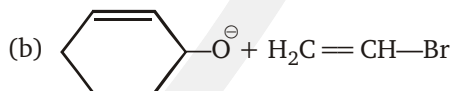
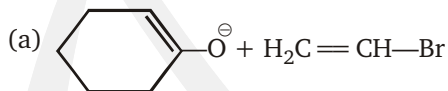
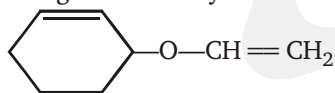


(d) None

50. The substances used for the preparation of ether by Williamson's synthesis are:

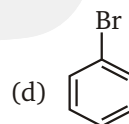
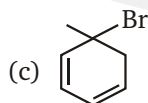
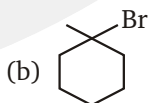
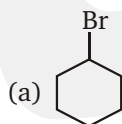


51. The best method for synthesis of given ether by Williamson's ether synthesis is:

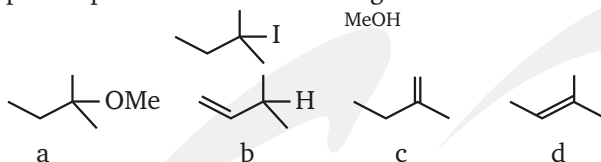


Unimolecular Nucleophilic Substitution ($\text{S}_{\text{N}}1$) Reactions

1. What is the most reactive alkyl halide for the following reaction?

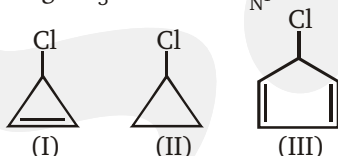


2. Give all of the expected products of the following reaction.



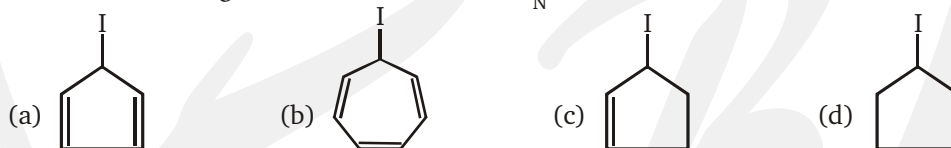
- (a) a (b) a and c (c) a, b, c and d (d) a, c, and d

3. Order of rate of reaction with AgNO_3 or rate of $\text{S}_{\text{N}}1$:

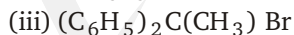


- (a) $\text{I} > \text{III} > \text{II}$ (b) $\text{II} > \text{III} > \text{I}$ (c) $\text{I} > \text{II} > \text{III}$ (d) $\text{III} > \text{I} > \text{II}$

4. Which of following is least reactive toward $\text{S}_{\text{N}}1$?

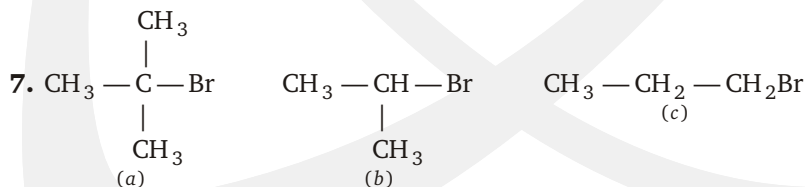
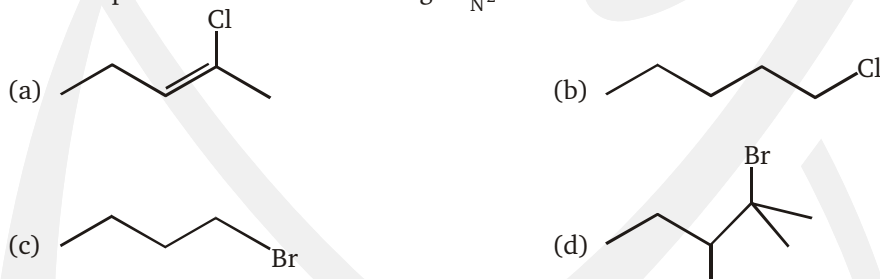


5. $\text{S}_{\text{N}}1$ reactivity of the following halides,



- (a) $\text{iv} > \text{i} > \text{ii} > \text{iii}$ (b) $\text{ii} > \text{i} > \text{iii} > \text{iv}$ (c) $\text{i} > \text{iii} > \text{ii} > \text{iv}$ (d) $\text{iii} > \text{ii} > \text{i} > \text{iv}$

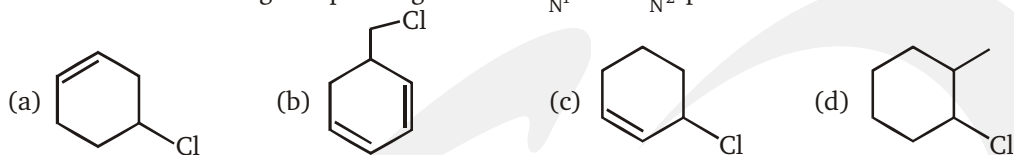
6. The compound which would undergo $\text{S}_{\text{N}}2$ reaction fastest is:



Reactivity toward $\text{S}_{\text{N}}1$:

- (a) $a > b > c$ (b) $b > a > c$ (c) $b > c > a$ (d) $c > b > a$

8. Which of the following compound give same S_N1 and S_N2 product ?



9. $\text{Ph}-\underset{\text{CH}_3}{\text{CH}}-\text{Cl} \xrightarrow{\text{CH}_3\text{CO}_2\text{H}}$, This reaction is known as:

- (a) methanolysis (b) ethanolysis (c) acetolysis (d) formalysis

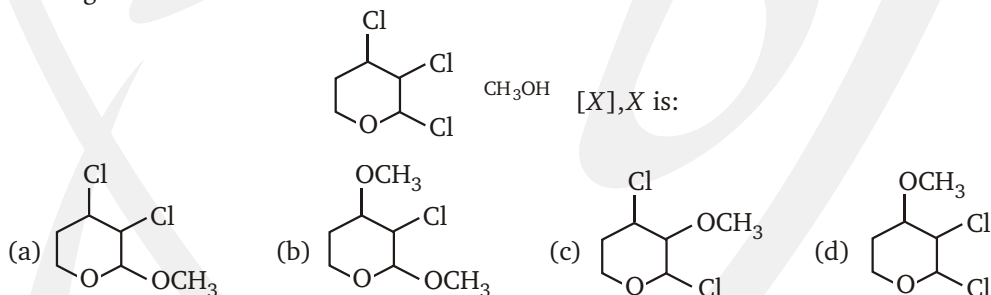
10. The rate of solvolysis of *tert*-butyl bromide will be maximum in which of the following solvents?

- (a) Et_2O (b) CCl_4 (c) EtOH (d) H_2O

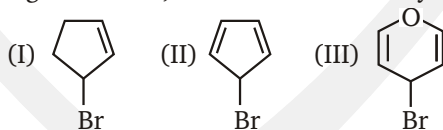
11. Which one of the following compounds will be most reactive for S_N1 reactions ?



12. In the given reaction:



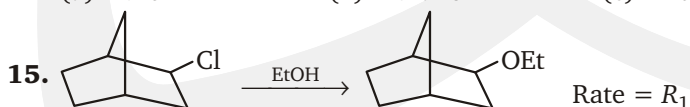
13. Among the bromides I-III given below, the order of reactivity S_N1 reaction is:

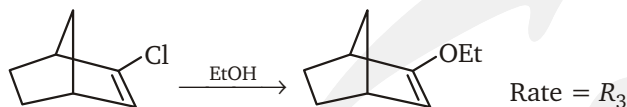
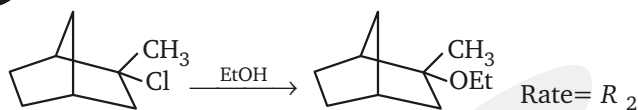


- (a) $\text{III} > \text{I} > \text{II}$ (b) $\text{III} > \text{II} > \text{I}$ (c) $\text{II} > \text{III} > \text{I}$ (d) $\text{II} > \text{I} > \text{III}$

14. When ethyl bromide is treated with moist Ag_2O , main product is:

- (a) Ether (b) Ethanol (c) Aldehyde (d) All of these





Which of the following is correct order?

- (a) R_1 R_2 R_3 (b) R_3 R_2 R_1 (c) R_2 R_1 R_3 (d) R_2 R_3 R_1

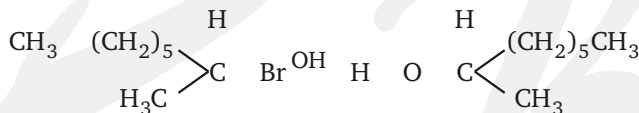
16. Which of the following compounds has the largest dipole moment?

- (a) CH_3Cl (b) CH_2Cl_2 (c) CHCl_3 (d) CCl_4

17. Which of the following will react with water?

- (a) CHCl_3 (b) Cl_3CCHO (c) CCl_4 (d) $\text{ClCH}_2\text{CH}_2\text{Cl}$

18. The reaction is described as:



- (a) $\text{S}_{\text{E}2}$ (b) $\text{S}_{\text{E}1}$ (c) $\text{S}_{\text{N}2}$ (d) $\text{S}_{\text{N}1}$

19. Assuming no other changes, what is the effect of doubling only the concentration of the nucleophile in the following reaction?



- (a) No change (b) Doubles the rate
(c) Triples the rate (d) Quadruples the rate (e) Rate is halved

20. Assuming no other changes, what is the effect of doubling both the concentration of the alkyl halide and the nucleophile in the following reaction?

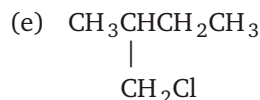
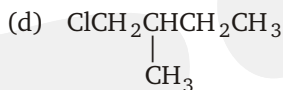
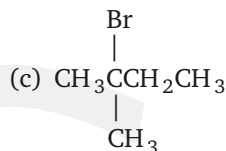
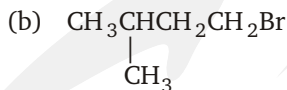
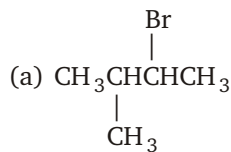


- (a) No change (b) Doubles the rate
(c) Triples the rate (d) Quadruples the rate (e) Rate is halved

21. Which of the following alkyl halides gives the fastest $\text{S}_{\text{N}1}$ reaction?

- (a) $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$ (b) $\text{CH}_3\text{CH}(\text{Br})\text{CH}_3$
(c) $\text{CH}_3\text{CH}(\text{I})\text{CH}_3$ (d) $\text{CH}_3\text{CH}_2\text{CH}_2\text{I}$ (e) $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$

22. Which of the following alkyl halides gives the fastest S_N1 reaction?



23. Which of the following factors has no effect on the rate of S_N1 reactions?

- (a) Nature of the alkyl halide
(b) Nature of the leaving group
(c) Concentration of the alkyl halide
(d) Concentration of the nucleophile
(e) Value of the rate constant

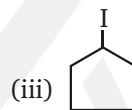
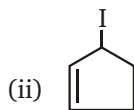
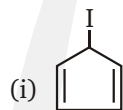
24. Which of the following is the rate law for S_N1 mechanisms?

- (a) $\text{Rate} = k [\text{Alkyl Halide}] [\text{Nucleophile}]$
(b) $\text{Rate} = k [\text{Nucleophile}]$
(c) $\text{Rate} = k [\text{Alkyl Halide}]$
(d) $\text{Rate} = k [\text{Alkyl Halide}] [\text{Nucleophile}] + k_2 [\text{Alkyl Halide}]$
(e) $\text{Rate} = k_1 [\text{Alkyl Halide}] + k_2 [\text{Nucleophile}]$

25. In the S_N1 hydrolysis mechanism of $(\text{CH}_3)_3\text{CBr}$, there are ----- elementary steps, ----- distinct transition states, and ----- distinct intermediates.

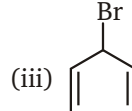
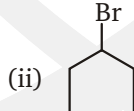
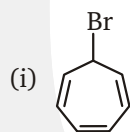
- (a) 2, 2, 2 (b) 2, 2, 3 (c) 2, 3, 2 (d) 3, 2, 3 (e) 3, 3, 2

26. Compare rate of reaction with Ag^+NO_3^- or rate of S_N1 reaction :



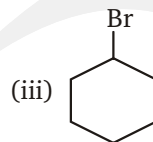
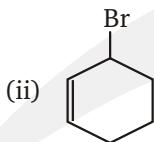
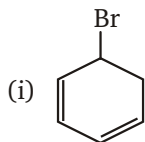
- (a) $i > ii > iii$ (b) $ii > iii > i$ (c) $ii > i > iii$ (d) $i > iii > ii$

27. Compare rate of reaction with Ag^+NO_3^- or rate of S_N1 reaction :



- (a) $i > iii > ii$ (b) $i > ii > iii$ (c) $iii > ii > i$ (d) $ii > i > iii$

28. Compare rate of reaction with $\text{Ag}^\oplus\text{NO}_3^\ominus$ or rate of $\text{S}_{\text{N}}1$ reaction :



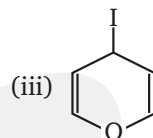
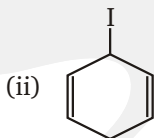
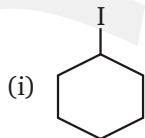
(a) $\text{i} > \text{ii} > \text{iii}$

(b) $\text{ii} > \text{iii} > \text{i}$

(c) $\text{iii} > \text{i} > \text{ii}$

(d) $\text{ii} > \text{i} > \text{iii}$

29. Compare rate of reaction with $\text{Ag}^\oplus\text{NO}_3^\ominus$ or rate of $\text{S}_{\text{N}}1$ reaction :



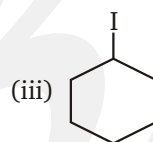
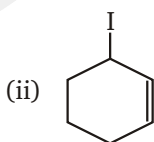
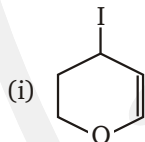
(a) $\text{i} > \text{ii} > \text{iii}$

(b) $\text{ii} > \text{i} > \text{iii}$

(c) $\text{iii} > \text{ii} > \text{i}$

(d) $\text{ii} > \text{iii} > \text{i}$

30. Compare rate of reaction with $\text{Ag}^\oplus\text{NO}_3^\ominus$ or rate of $\text{S}_{\text{N}}1$ reaction :



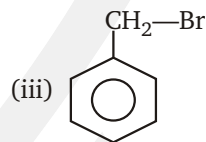
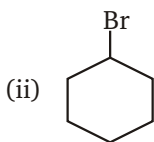
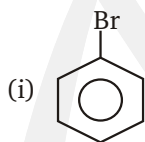
(a) $\text{i} > \text{ii} > \text{iii}$

(b) $\text{ii} > \text{iii} > \text{i}$

(c) $\text{ii} > \text{i} > \text{iii}$

(d) $\text{iii} > \text{ii} > \text{i}$

31. Compare rate of reaction with $\text{Ag}^\oplus\text{NO}_3^\ominus$ or rate of $\text{S}_{\text{N}}1$ reaction :



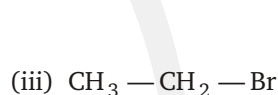
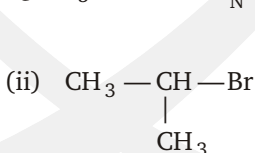
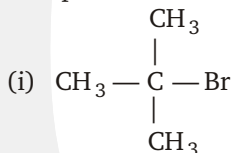
(a) $\text{i} > \text{ii} > \text{iii}$

(b) $\text{ii} > \text{iii} > \text{i}$

(c) $\text{iii} > \text{ii} > \text{i}$

(d) $\text{iii} > \text{i} > \text{ii}$

32. Compare rate of reaction with $\text{Ag}^\oplus\text{NO}_3^\ominus$ or rate of $\text{S}_{\text{N}}1$ reaction :



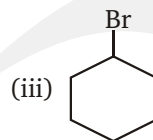
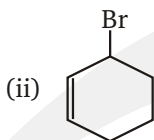
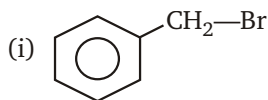
(a) $\text{i} > \text{ii} > \text{iii}$

(b) $\text{ii} > \text{iii} > \text{i}$

(c) $\text{iii} > \text{ii} > \text{i}$

(d) $\text{ii} > \text{i} > \text{iii}$

33. Compare rate of reaction with $\text{Ag}^\oplus\text{NO}_3^\ominus$ or rate of $\text{S}_{\text{N}}1$ reaction :



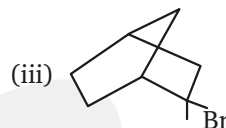
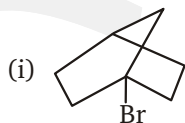
(a) $\text{i} > \text{ii} > \text{iii}$

(b) $\text{ii} > \text{iii} > \text{i}$

(c) $\text{ii} > \text{i} > \text{iii}$

(d) $\text{iii} > \text{i} > \text{ii}$

34. Compare rate of reaction with $\text{Ag}^\oplus\text{NO}_3^\ominus$ or rate of $\text{S}_{\text{N}}1$ reaction :



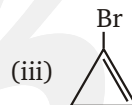
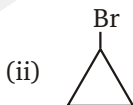
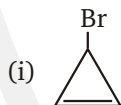
(a) $\text{i} > \text{ii} > \text{iii}$

(b) $\text{iii} > \text{ii} > \text{i}$

(c) $\text{ii} > \text{iii} > \text{i}$

(d) $\text{iii} > \text{i} > \text{ii}$

35. Compare rate of reaction with $\text{Ag}^\oplus\text{NO}_3^\ominus$ or rate of $\text{S}_{\text{N}}1$ reaction :



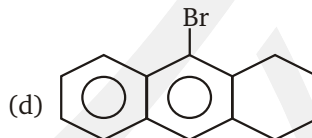
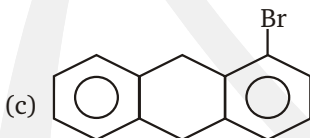
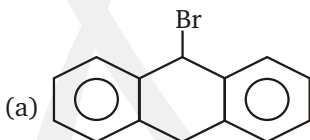
(a) $\text{i} > \text{iii} > \text{ii}$

(b) $\text{ii} > \text{iii} > \text{i}$

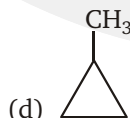
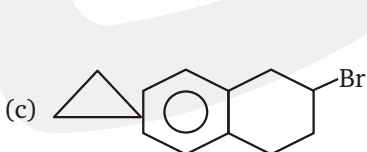
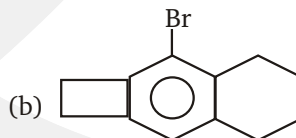
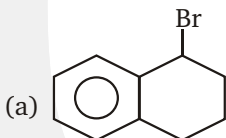
(c) $\text{i} > \text{ii} > \text{iii}$

(d) $\text{iii} > \text{i} > \text{ii}$

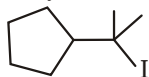
36. Which of the following is most reactive toward $\text{Ag}^\oplus\text{NO}_3^\ominus$ or rate of $\text{S}_{\text{N}}1$ reaction ?



37. Which of the following is most reactive toward $\text{Ag}^\oplus\text{NO}_3^\ominus$ or rate of $\text{S}_{\text{N}}1$ reaction ?

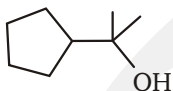


38. Which of the following is least likely to be found in the product mixture which results when the alkyl iodide below is heated in water?



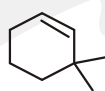
I

(a) I



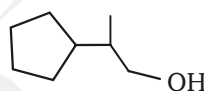
II

(b) II



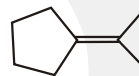
III

(c) III



IV

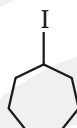
(d) IV



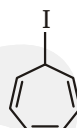
V

(e) V

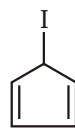
39. Compare the reactivity of these halide in S_N1 reactions



(I)



(II)



(III)

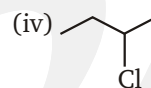
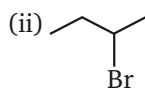
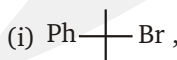
(a) II III I

(b) II I III

(c) I II III

(d) III II I

40. Arrange in order of increasing S_N1 reaction rate (protic solvent).



(a) iv ii iii i

(b) i ii iii iv

(c) iv i iii ii

(d) iv i ii iii

41. CHCl_3 aq. KOH (Exces) (A) + H_2O



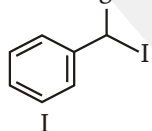
(a) A is acid, B is aldehyde

(b) (A) is acid, B is alcohol

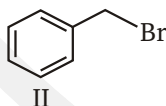
(c) A is acid, B is CO_2

(d) A is aldehyde, B is acid.

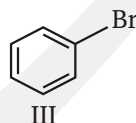
42. Rank the following in order of decreasing reactivity with methanol.



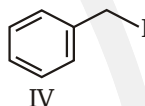
I



II



III



IV

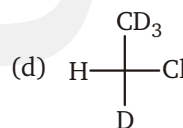
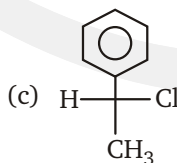
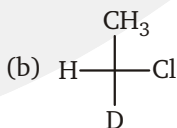
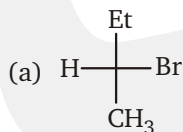
(a) III II IV I

(b) IV I II III

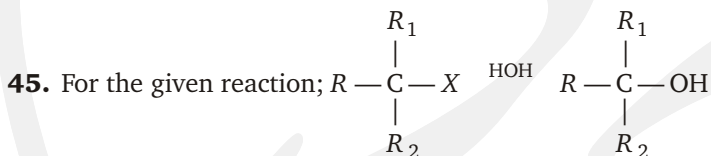
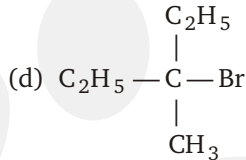
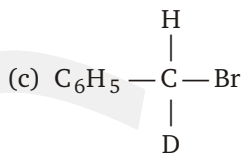
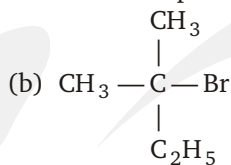
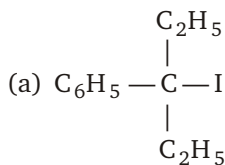
(c) I IV II III

(d) I IV III II

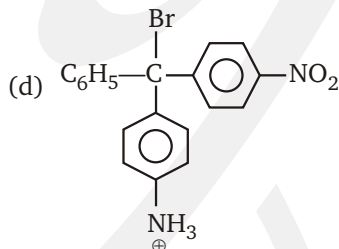
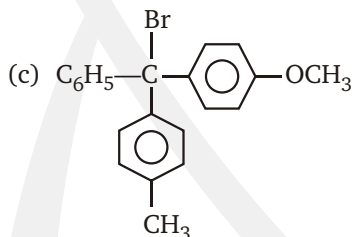
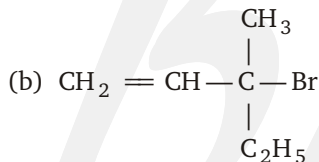
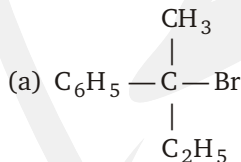
43. Under identical conditions, solvolysis of which of the following substrates would lead to maximum racemization?



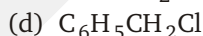
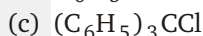
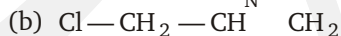
44. Which one of the following compounds will give enantiomeric pair on treatment with HOH ?



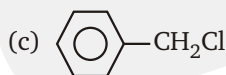
Which substrate will give maximum racemisation ?



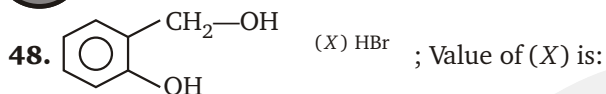
46. Which of the following compounds is most rapidly hydrolysed by $\text{S}_{\text{N}}1$ mechanism ?



47. Which will give white ppt. with AgNO_3 ?



(d) Both (a) and (c)



- (a) 1 (b) 2 (c) 3 (d) 4

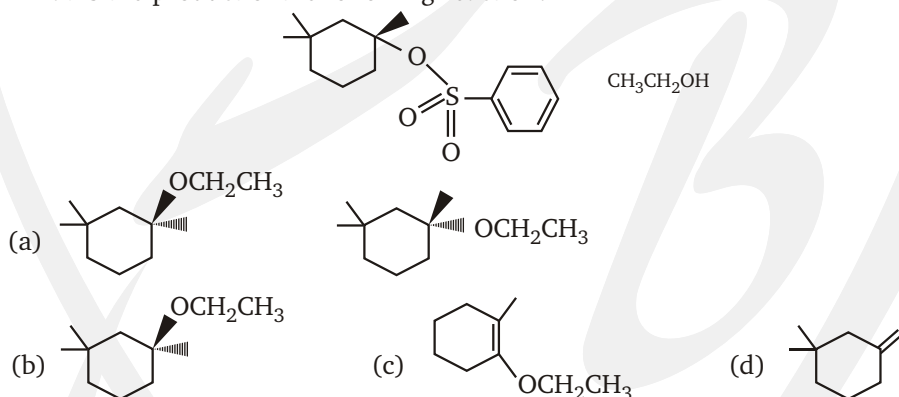
49. Aryl halides are less reactive towards nucleophilic substitution reaction as compared to alkyl halides due to:

- (a) The formation of less stable carbonium ion
(b) Resonance stabilization
(c) Longer carbon-halogen bond
(d) The inductive effect

50. The reaction products of: $\text{C}_6\text{H}_5\text{OCH}_3$ HI is :

- (a) $\text{C}_6\text{H}_5\text{OH}$ CH_3I (b) $\text{C}_6\text{H}_5\text{I}$ CH_3OH (c) $\text{C}_6\text{H}_5\text{CH}_3$ HOI (d) C_6H_6 CH_3OI

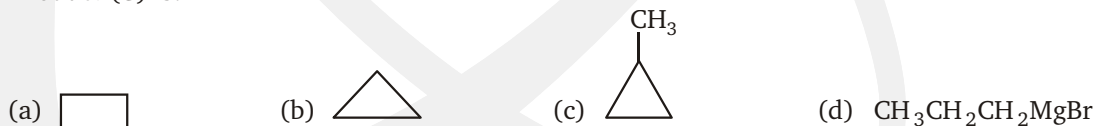
51. What is the product of the following reaction?



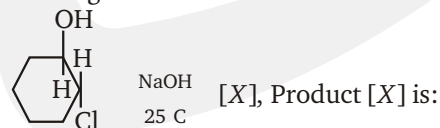
Neighbouring Group Participation and $\text{S}_\text{N}2$

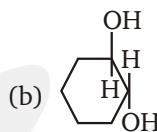
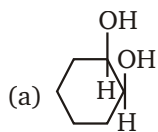


Product (C) is:

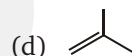
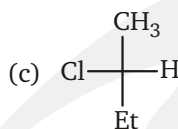
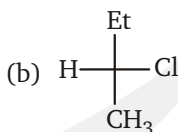
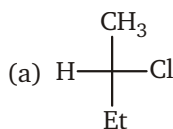
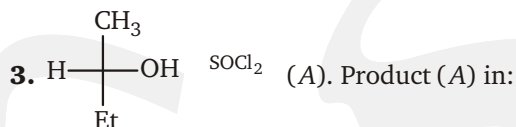
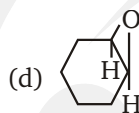


2. In the given reaction





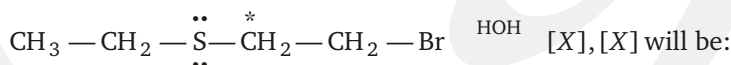
(c) Mixture of (a) and (b)



4. The reaction of SOCl_2 on alkanols to form alkyl chlorides gives good yields because:

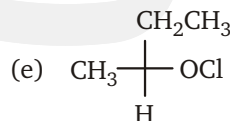
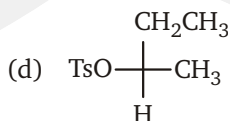
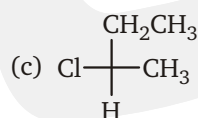
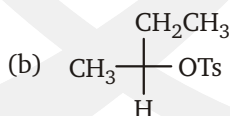
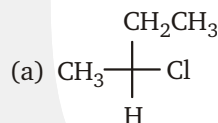
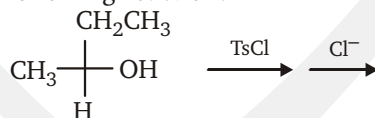
- (a) alkyl chlorides are immiscible with SOCl_2
- (b) the other products of the reaction are gaseous and escape out
- (c) alcohol and SOCl_2 are soluble in water
- (d) the reaction does not occur via intermediate formation of an alkyl chloro sulphite

5. In the given reaction:

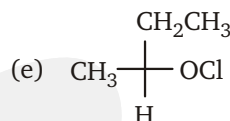
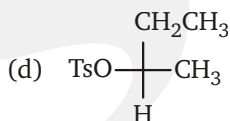
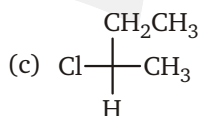
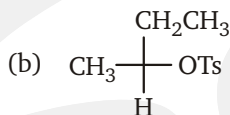
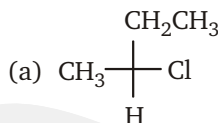
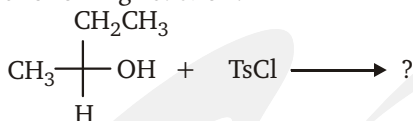


- (a) $\text{CH}_3 - \text{CH}_2 - \text{S}^* - \text{CH}_2 - \text{CH}_2 - \text{OH}$
- (b) $\text{CH}_3 - \text{CH}_2 - \text{S} - \text{CH}_2 - \text{CH}_2^* - \text{OH}$
- (c) 1 : 1 mixture of (a) and (b)
- (d) 2 : 1 mixture of (a) and (b)

6. What is the product of the following reaction?



7. What is the product of the following reaction?



Unimolecular Elimination Reactions (E_1)

1. How many alkenes are formed by E_2 elimination of HBr from 2-bromo-2,3-dimethylhexane using a strong base such as sodium methoxide?

- (a) 1 (b) 2 (c) 3 (d) 4

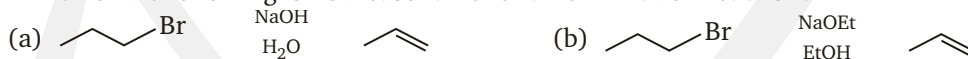
2. How many alkenes are formed by E_2 elimination of HBr from 3-bromo-3,4-dimethylhexane using a strong base such as sodium methoxide?

- (a) 2 (b) 3 (c) 4 (d) 5

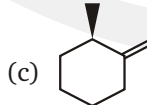
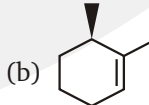
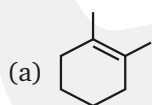
3. How many alkenes are formed by E_1 elimination of HCl from 3-chloro-3,4-dimethylheptane in methanol?

- (a) 2 (b) 3 (c) 4 (d) 5

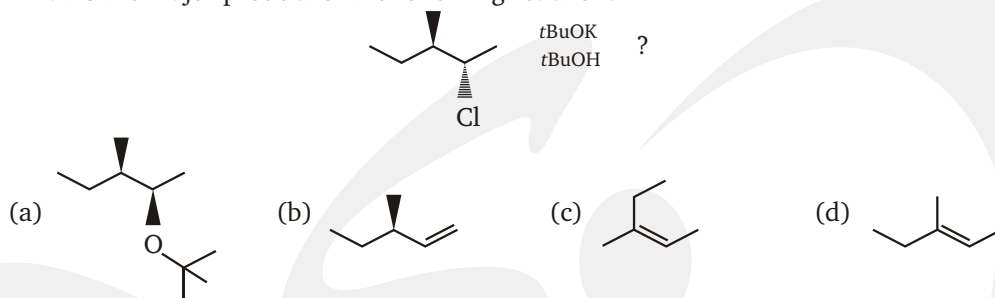
4. Which of the following is not reasonable for an elimination reaction?



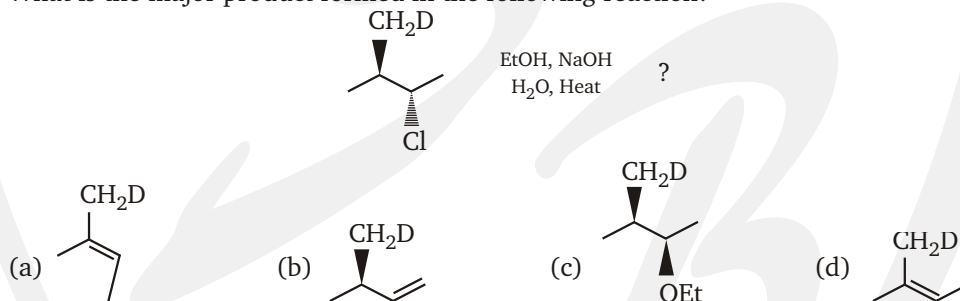
5. What is the major product of the following reaction?



6. What is the major product of the following reaction?



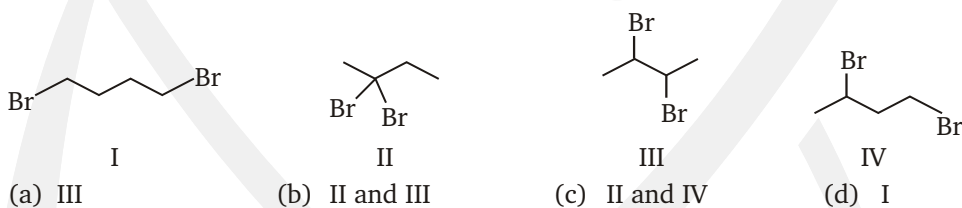
7. What is the major product formed in the following reaction?



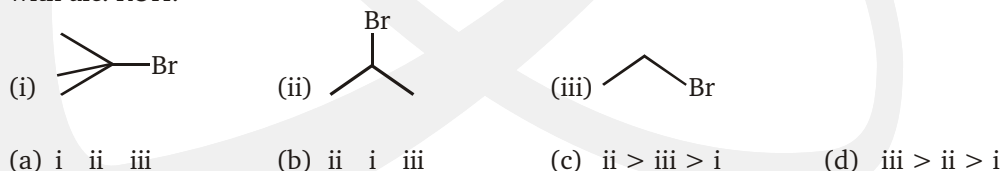
8. The major product of the reaction of 1,2-dibromobutane and excess NaNH_2 is

- (a) 2-bromo-1-butene
 (b) 1-bromo-1-butene
 (c) 1-butyne
 (d) 1,2-butadiene

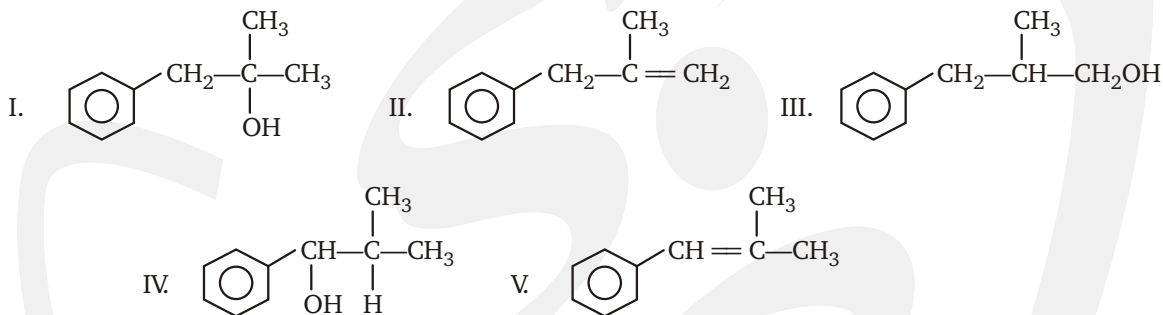
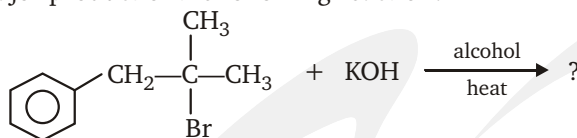
9. Which of the following dibromides would yield 2-butyne upon elimination?



10. Identify the rate of dehydrohalogenation in the following compound when allowed to react with alc. KOH:

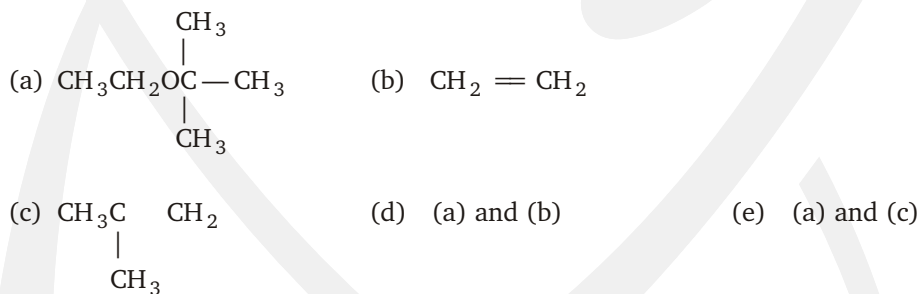
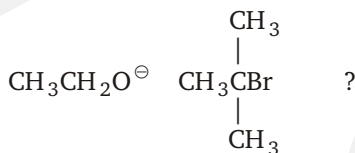


11. What is the major product of the following reaction?

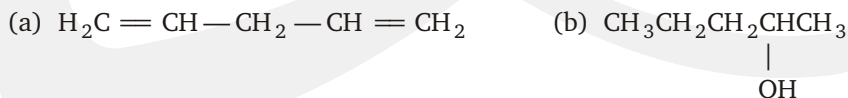
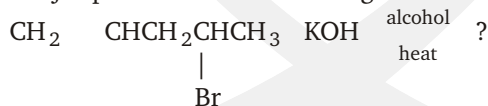


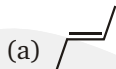
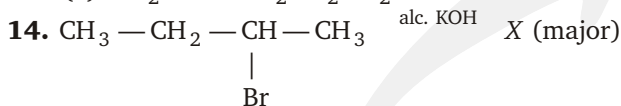
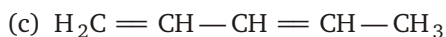
- (a) I (b) II (c) III (d) IV (e) V

12. What is/are the product(s) of the following reaction?

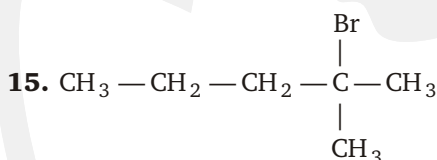
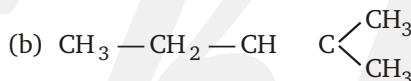
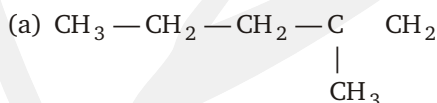


13. What is the major product of the following reaction?



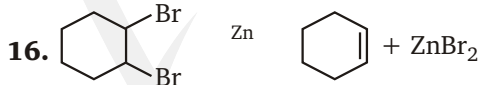


(d) None of these

Identify all possible product obtained by E_2 reaction :

(c) Both (a) and (b)

(d) None of these



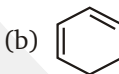
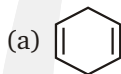
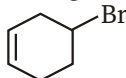
This reaction is an example of :

(a) -elimination

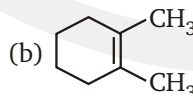
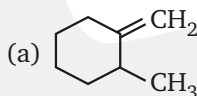
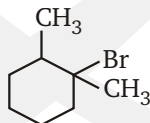
(b) -elimination

(c) -elimination

(d) None of these

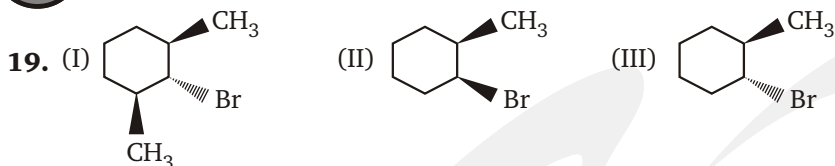
17. The major product obtained when following substrate is subjected to E_2 reaction will be:

(c) Both (a) and (b) (d) None of these

18. The major product obtained when following substrate is subjected to E_2 reaction under the treatment of potassium *tert*-butoxide will be:

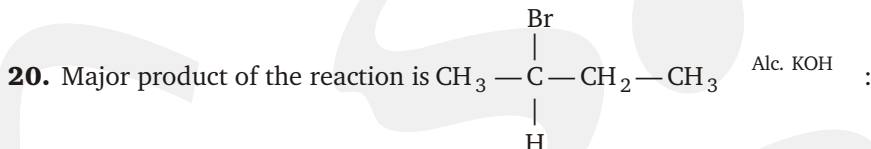
(c) Both in equal proportions

(d) None of these



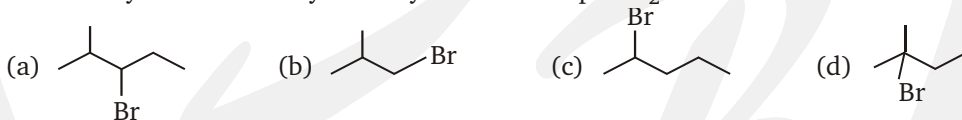
Ease of $\text{-dehydrobromination}$ among these substrates under the treatment of strong base will be in the order as:

- (a) $\text{I} > \text{II} > \text{III}$ (b) $\text{III} > \text{II} > \text{I}$ (c) $\text{II} > \text{I} > \text{III}$ (d) $\text{II} > \text{III} > \text{I}$

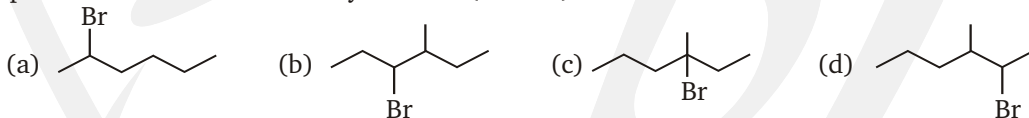


- (a) Butene-1 (b) Butene-2 (c) Butane (d) Butyne-1

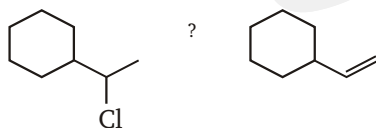
21. Which alkyl bromide will yield only one alkene upon E_2 elimination ?



22. Which alkyl bromide will yield 3-methyl-1-hexene as the major product upon treatment with potassium *t*-butoxide in *t*-butyl alcohol (solvent) ?



23. In order to accomplish the following conversion, what reagent and conditions would be required ?



- (a) Cold sodium hydroxide (b) Hot conc. sodium hydroxide
(c) Potassium *t*-butoxide and heat (d) Hot water

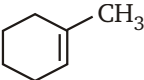
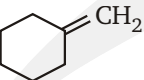
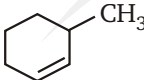


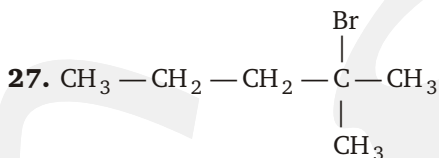
- (a) I (b) II (c) III (d) All with same ease

25. In dehydrohalogenations, the base (alcoholic KOH) abstracts:

- (a) The halides ion
(b) The proton present on the carbon next to the carbon to which the halogen is attached
(c) The proton present on the carbon to which the halogen is attached
(d) The proton on the β -carbon

26. The major product obtained when this substrate is subjected to E_2 reaction will be:

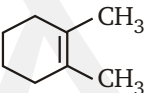
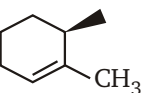
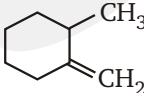
- (a)  (b)  (c)  (d) Both (a) and (c)



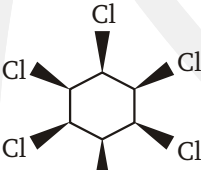
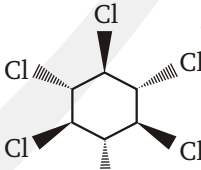
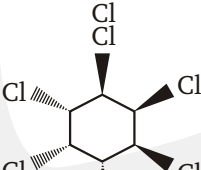
Major product obtained when this alkyl halide is subjected to E_2 reaction under the treatment of potassium *tert*-butoxide will be:

- (a) $\text{CH}_3 - \text{CH}_2 - \text{CH} = \text{C}(\text{CH}_3)_2$ (b) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{C}(\text{CH}_3) = \text{CH}_2$
(c) Both (a) and (b) (d) *Trans* isomer

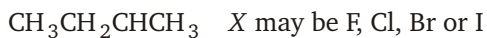
28. The major product obtained when following substrate is subjected to E_2 reaction will be:

- (a)  (b)  (c)  (d) None of these

29. Which of the following isomeric hexachlorocyclohexanes is least reactive in dehydrochlorination on treatment with strong base?

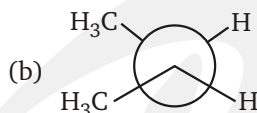
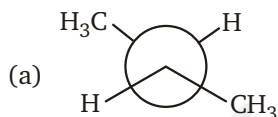
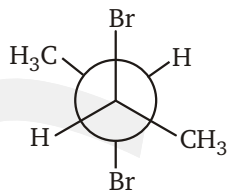
- (a)  (b) 
(c)  (d) All three are equally reactive

30. Correct order of yield of Hofmann alkene in following reaction will be:



- (a) $\text{F} > \text{Cl} > \text{Br} > \text{I}$ (b) $\text{I} > \text{Br} > \text{Cl} > \text{F}$ (c) $\text{Cl} > \text{F} > \text{Br} > \text{I}$ (d) $\text{I} > \text{Br} > \text{F} > \text{Cl}$

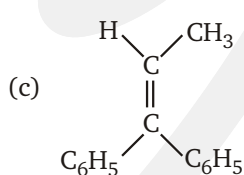
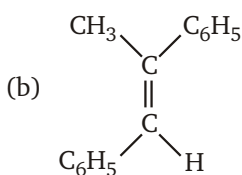
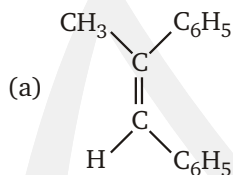
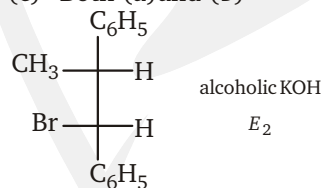
31. ? Product is:



- (c) Both (a) and (b)

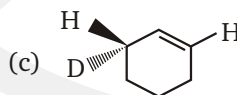
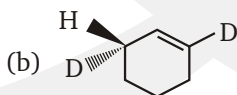
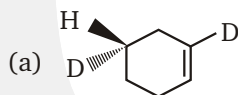
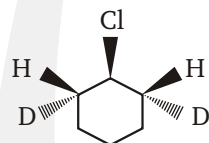
- (d) None of these

32. A



- (d) None is correct

33. A. The structure of 'A' is:

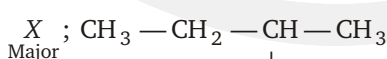


34. $\text{CH}_3 - \text{CH}_2 - \text{CH}(\text{Br}) - \text{CH}_3$



alc. KOH

X
Major



EtONa

Y
Major

Product (X) and (Y) respectively is:

- (a) 1-butene, *trans*-2-butene (b) 1-butene, *cis*-2-butene
(c) *cis*-2-butene, 1-butene (d) *trans*-2-butene, 1-butene

35. Debromination of meso-dibromobutane gives mainly:

- (a) *n*-butane (b) 1-butene
(c) *cis*-2-butene (d) *trans*-2-butene

36. Identify the set of reagents/reaction-conditions 'X' and 'Y' in the following set of transformation:

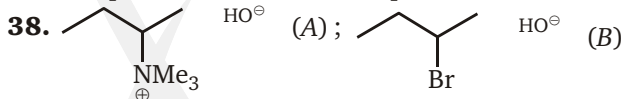


- (a) X dilute aqueous NaOH, 20 °C; Y HBr / CH₃COOH, 20 °C
(b) X conc. alcoholic NaOH, 80 °C; Y HBr / CH₃COOH
(c) X dilute aqueous NaOH, 20 °C; Y Br₂ / CHCl₃, 20 °C
(d) X conc. alcoholic NaOH, 80 °C; Y Br₂ / CHCl₃

37. Under certain conditions alkyl halides reacts with base to give an alkene and HCl (elimination reaction). For example, $\text{RCH}_2\text{CH}_2\text{Cl} \xrightarrow{\text{R}} \text{CH}=\text{CH}_2 + \text{HCl}$

The extent of these reaction depends on the structure of alkyl halides (e.g., primary, secondary and tertiary). The relative extent to which such reactions take place is in the order (of halo alkanes):

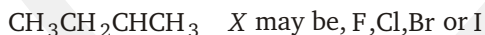
- (a) $p < s < t$ (b) $p > s > t$ (c) $p > s < t$ (d) $p < s > t$



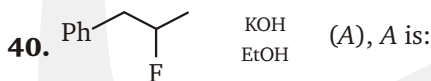
Relation between (A) and (B) is:

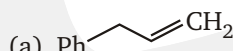
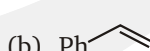
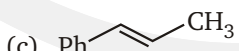
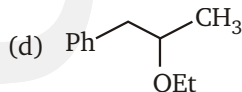
- (a) positional isomer (b) functional isomer
(c) chain isomer (d) mesomers

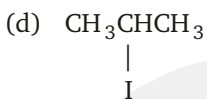
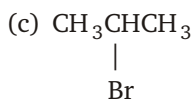
39. Correct order of yield of Hofmann alkene in following reaction will be:



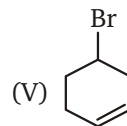
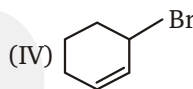
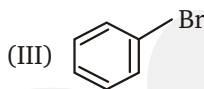
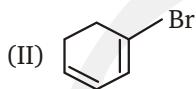
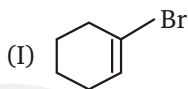
- (a) $\text{F} > \text{Cl} > \text{Br} > \text{I}$ (b) $\text{I} > \text{Br} > \text{Cl} > \text{F}$ (c) $\text{Cl} > \text{F} > \text{Br} > \text{I}$ (d) $\text{I} > \text{Br} > \text{F} > \text{Cl}$



- (a)  (b)  (c)  (d) 

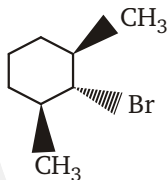


48. Which of the following alkyl halides forms the most stable carbocation when it undergoes an E_1 reaction?



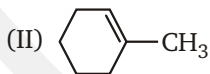
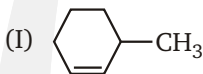
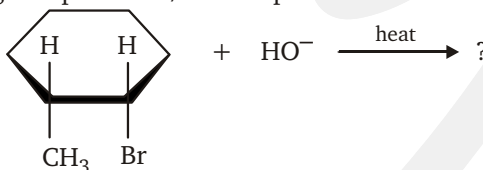
- (a) I (b) II (c) III (d) IV (e) V

49. Why is the alkyl halide below not capable of undergoing an E_2 reaction upon treatment with sodium ethoxide?



- (a) Br is too poor a leaving group
 (b) the substrate is too hindered
 (c) too much angle strain would be present in the alkene product
 (d) sodium ethoxide is a poor base to use in E_2 reactions
 (e) the C—H and C—Br bonds which need to break cannot achieve an anti-periplanar orientation

50. Which of the following compounds is/are the products of the reaction:

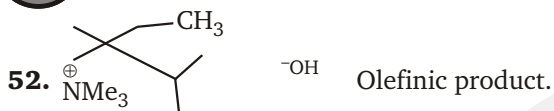


- (a) I only (b) II only
 (c) I and II are of equal yield (d) I is major, II is minor (e) I is minor, II is major

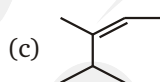
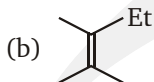
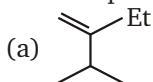
51. Which of the following statements correctly describe(s) E_1 reactions of alkyl halides (RX)?

- I. Rate = k [base]
 II. Rate = k [base][RX]
 III. Rate = k [RX]
 IV. The reactions occur in two distinct steps
 V. Rearrangements are sometimes seen

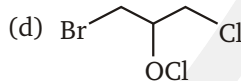
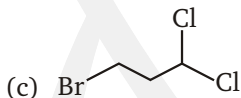
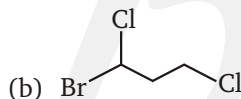
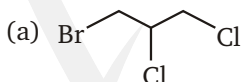
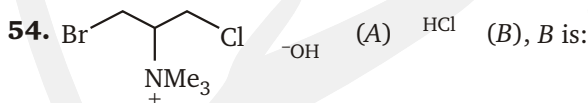
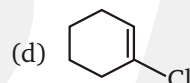
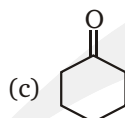
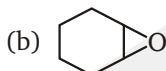
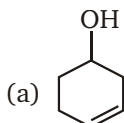
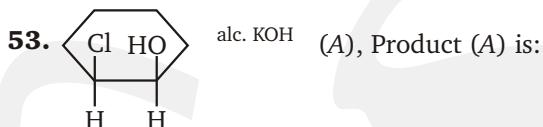
- (a) II and IV (b) III and V
 (c) I, IV and V (d) I only (e) III, IV and V



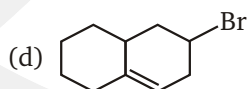
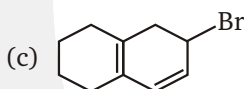
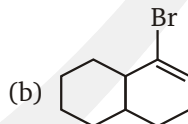
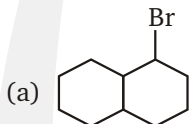
Olefinic product formed in the reaction is:

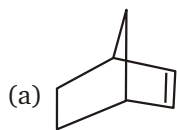


(d) None of these

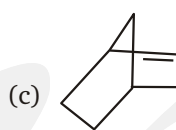


55. Which of the following undergo fastest dehydrobromination?

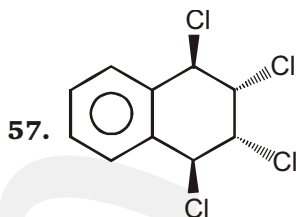




(b) No-reaction



(d) None



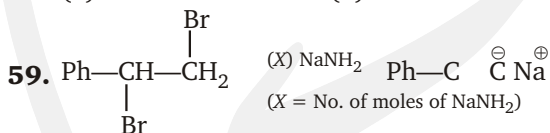
(X) alc. KOH

Moles of alc. KOH consumed are:

- (a) 0 (b) 1 (c) 2 (d) 3

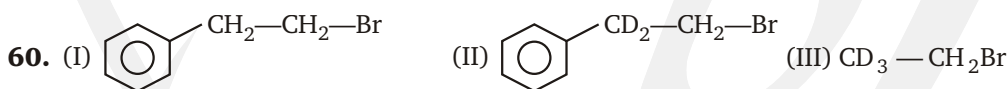
58. $\text{Ph}-\text{Br} \xrightarrow{\text{Mg}}$ solvent used in the above reaction is:

- (a) ethane (b) ethanol (c) diethyl ether (d) acid



Value of X is:

- (a) 1 (b) 2 (c) 3 (d) 4

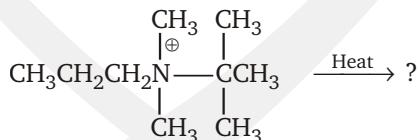


Ease of $-\text{dehydrobromination}$ among these halides under the treatment of strong base will be in the order as:

- (a) $\text{I} > \text{II} > \text{III}$ (b) $\text{III} > \text{II} > \text{I}$ (c) $\text{III} > \text{I} > \text{II}$ (d) $\text{II} > \text{III} > \text{I}$

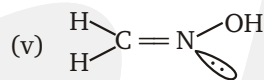
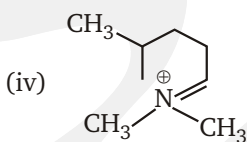
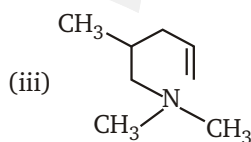
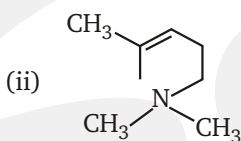
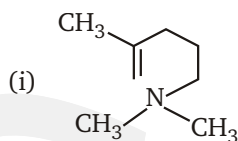
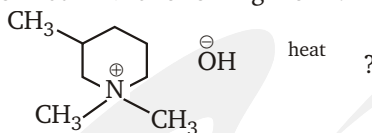
Elimination Reaction E_2 /Hoffmann Exhaustive Methylation and Elimination

1. What is the major alkene formed in the following Hofmann elimination ?



- (a) $\text{CH}_3\text{CH}=\text{CH}_2$ (b) $\text{CH}_3-\text{C}(\text{CH}_3)=\text{CH}_2$
(c) $\text{CH}_3\text{CH}_2\text{CH}=\text{CH}_2$ (d) $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}=\text{CH}_2$ (e) $\text{CH}_3\text{CH}_2\text{C}(\text{CH}_3)=\text{CH}_2$

2. What is the major alkene formed in the following Hofmann elimination ?



(a) i

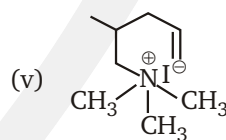
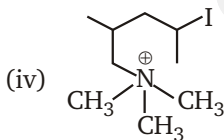
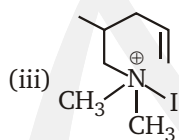
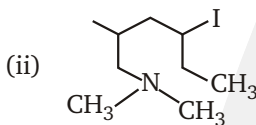
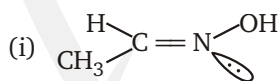
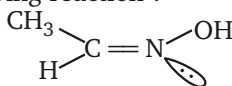
(b) ii

(c) iii

(d) iv

(e) v

3. What is the product of the following reaction ?



(a) i

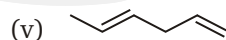
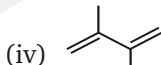
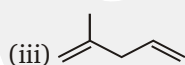
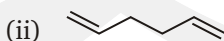
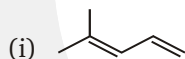
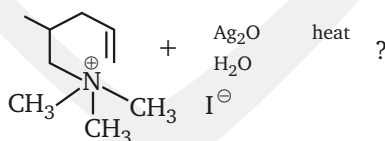
(b) ii

(c) iii

(d) iv

(e) v

4. What is the major alkene formed in the following reaction ?



(a) i

(b) ii

(c) iii

(d) iv

(e) v

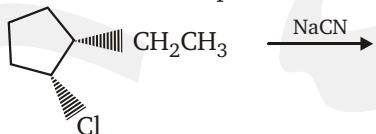
5. *R*-2-chloro butane KSH
acetone

- (a) S_{N^1} (b) S_{N^2} (c) E_2 (d) E_1

6. Predict the two most likely mechanisms for the reaction of 2-iodohexane with sodium ethoxide.

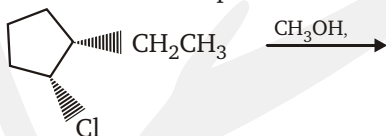
- (a) S_{N^2} and S_{N^1} (b) E_1 and E_2
(c) S_{N^2} and E_2 (d) E_1 and S_{N^1} (e) E_2 and S_{N^1}

7. What mechanism predominates in the reaction below?



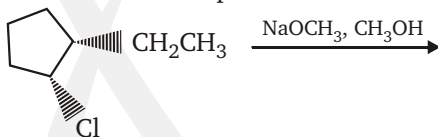
- (a) S_{N^2} (b) S_{N^1} without rearrangement
(c) S_{N^1} with rearrangement (d) E_2 (e) E_1

8. What mechanism predominates in the reaction below?

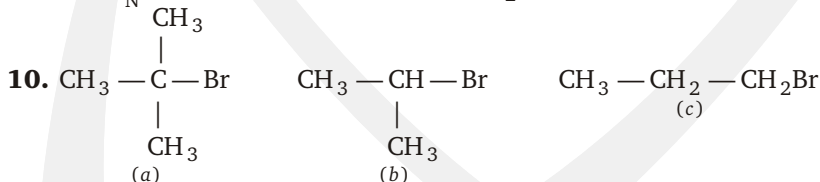


- (a) S_{N^2} (b) S_{N^1} without rearrangement
(c) S_{N^1} with rearrangement (d) E_2 (e) E_1

9. What mechanism predominates in the reaction below?



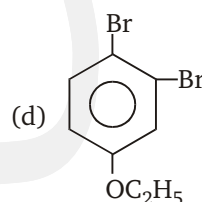
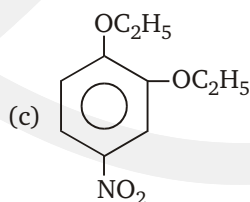
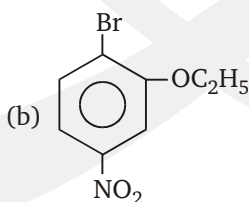
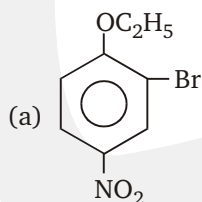
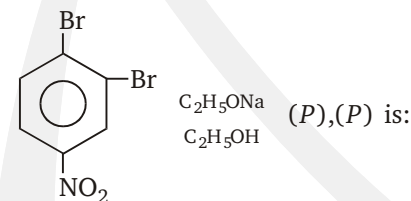
- (a) S_{N^2} (b) S_{N^1} without rearrangement
(c) S_{N^1} with rearrangement (d) E_2 (e) E_1

Reactivity toward E_2 :

- (a) $a > b > c$ (b) $b > a > c$ (c) $b > c > a$ (d) $c > b > a$

[B] Haloarenes

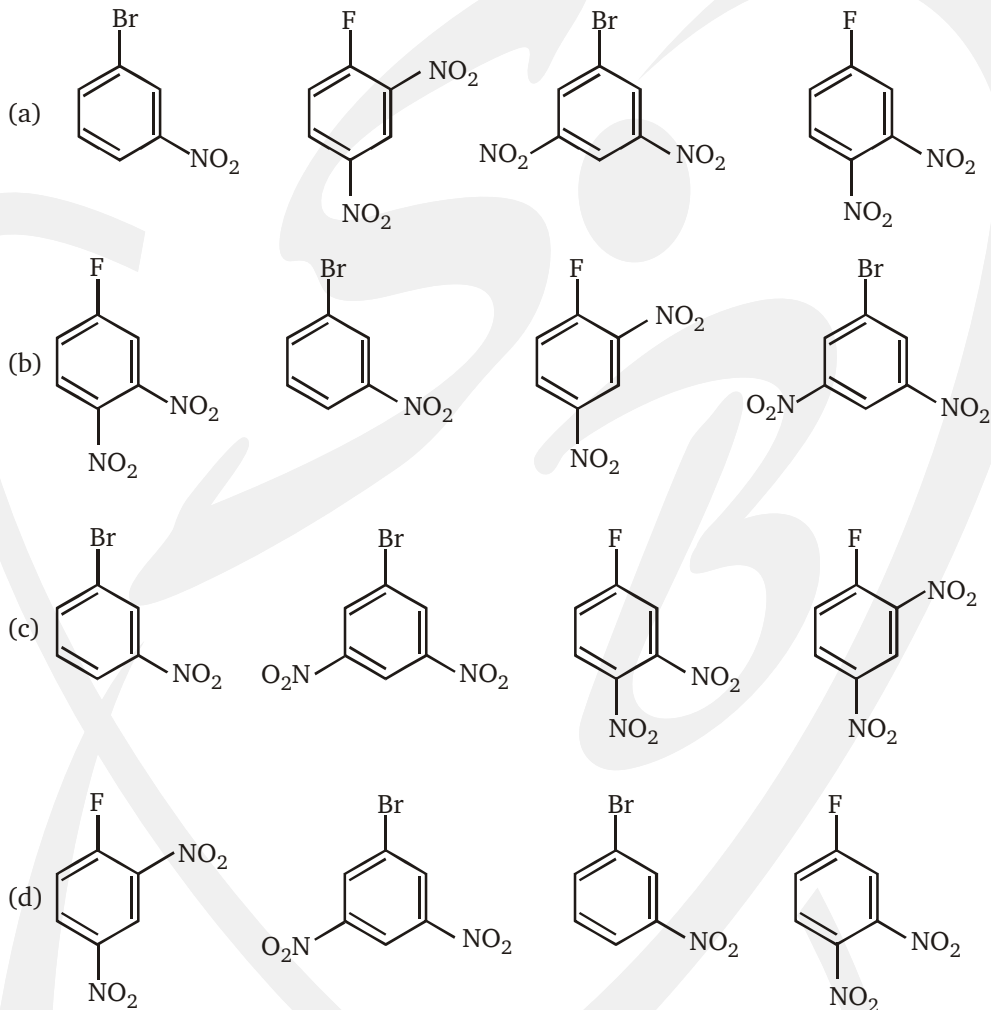
- Which of the following is used as refrigerant?
 (a) CO_2 (b) CHCl_3 (c) SiC (d) CF_2Cl_2
- CCl_4 is used as fire extinguisher because:
 (a) high melting point (b) due to covalent bond
 (c) low boiling point (d) gives incombustible vapour
- CCl_2F_2 destroy ozone by liberating:
 (a) Cl_2 (b) F_2 (c) NO_2 (d) All of these
- Freons are:
 (a) fluoro derivatives of ethylene (b) $\text{CCl}_4 + \text{CS}_2$
 (c) chloro-fluoro derivatives of alkanes (d) chloro derivative of alkanes
- Dipole moment is highest in:
 (a) CHCl_3 (b) CH_4 (c) CHF_3 (d) CCl_4
- Boiling point of alkyl iodide, bromide, chloride and methane are in this order:
 (a) Iodide > Bromide > Chloride > CH_4 (b) CH_4 > Chloride > Bromide > Iodide
 (c) CH_4 > Iodide > Bromide > Chloride (d) Chloride > Bromide > Iodide > CH_4
- What compound is obtained when ethanol is distilled with potassium bromide and concentrated sulphuric acid?
 (a) Ethyl hydrogen sulphate (b) Ethyl bromide
 (c) Ethylene bromide (d) Acetylene dibromide
- Which one of the following compounds does not react with bromine?
 (a) Ethylamine (b) Propane (c) Phenol (d) Chloroform
- In the given reaction :



10. Which of the following correctly ranks the aryl halides in decreasing order of reactivity toward sodium methoxide in methanol ?

Slowest rate

Fastest rate



11. The reduction of benzoyl chloride with H_2 / $\text{Pd} - \text{BaSO}_4$ gives:

- (a) $\text{C}_6\text{H}_5\text{CHO}$ (b) $\text{C}_6\text{H}_5\text{CH}_2\text{OH}$ (c) $\text{C}_6\text{H}_5\text{COOH}$ (d) $\text{C}_6\text{H}_5\text{CH}_2\text{CN}$

12. In chlorobenzene, the $-\text{Cl}$ group:

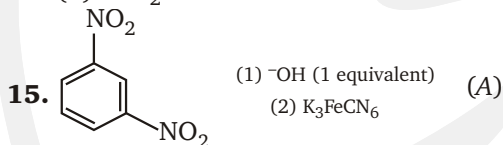
- (a) Activates the benzene ring more via resonance effect than deactivating it via inductive effect
 (b) Deactivates the benzene ring more via inductive effect than activating in via resonance effect
 (c) Activates the benzene ring via resonance effect and deactivates it via inductive effect. Both these effects are more evenly matched
 (d) None of these

13. The chemical name of DDT is:

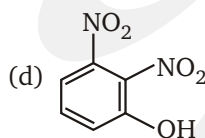
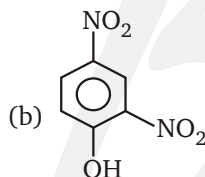
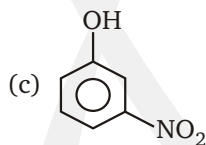
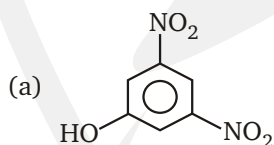
- (a) Dichlorodinitrotoluene
- (b) Dichlorodimethyltoluene
- (c) *p,p*-dichlorodiphenyltrichloroethane
- (d) None of these

14. Replacement of Cl of chlorobenzene to give phenol requires drastic conditions but chlorine of 2,4-dinitrochlorobenzene is readily replaced because:

- (a) NO_2 makes the electron rich ring at ortho and para positions
- (b) NO_2 withdraws electrons at metaposition
- (c) NO_2 donate electrons at *m*-position
- (d) NO_2 withdraws electrons at ortho and para positions



Organic product (A) is:



Answers

[A] Haloalkanes

30. Biomolecular Nucleophilic Substitution (S_N2) Reactions

1.	(a)	2.	(a)	3.	(c)	4.	(b)	5.	(a)	6.	(c)	7.	(b)	8.	(c)	9.	(a)	10.	(c)
11.	(b)	12.	(b)	13.	(c)	14.	(b)	15.	(c)	16.	(a)	17.	(a)	18.	(c)	19.	(c)	20.	(a)
21.	(b)	22.	(b)	23.	(d)	24.	(a)	25.	(d)	26.	(a)	27.	(b)	28.	(c)	29.	(b)	30.	(d)
31.	(b)	32.	(e)	33.	(d)	34.	(e)	35.	(b)	36.	(d)	37.	(c)	38.	(c)	39.	(d)	40.	(b)
41.	(c)	42.	(c)	43.	(e)	44.	(a)	45.	(b)	46.	(d)	47.	(c)	48.	(a)	49.	(b)	50.	(c)
51.	(c)																		

Unimolecular Nucleophilic Substitution (S_N1) Reactions

1.	(c)	2.	(d)	3.	(c)	4.	(a)	5.	(d)	6.	(c)	7.	(a)	8.	(c)	9.	(c)	10.	(d)
11.	(a)	12.	(a)	13.	(a)	14.	(b)	15.	(c)	16.	(a)	17.	(b)	18.	(c)	19.	(a)	20.	(b)
21.	(c)	22.	(c)	23.	(d)	24.	(c)	25.	(e)	26.	(b)	27.	(b)	28.	(a)	29.	(c)	30.	(a)
31.	(c)	32.	(a)	33.	(a)	34.	(b)	35.	(c)	36.	(a)	37.	(d)	38.	(d)	39.	(b)	40.	(a)
41.	(c)	42.	(c)	43.	(c)	44.	(c)	45.	(c)	46.	(c)	47.	(d)	48.	(a)	49.	(b)	50.	(a)
51.	(a)																		

Neighbouring Group Participation and S_N1

1.	(b)	2.	(d)	3.	(a)	4.	(b)	5.	(c)	6.	(c)	7.	(b)						
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Unimolecular Elimination Reactions (E_1)

1.	(b)	2.	(d)	3.	(d)	4.	(d)	5.	(a)	6.	(b)	7.	(a)	8.	(c)	9.	(b)	10.	(a)
11.	(e)	12.	(c)	13.	(c)	14.	(a)	15.	(c)	16.	(b)	17.	(b)	18.	(a)	19.	(d)	20.	(b)
21.	(b)	22.	(d)	23.	(c)	24.	(a)	25.	(b)	26.	(a)	27.	(b)	28.	(a)	29.	(b)	30.	(a)
31.	(a)	32.	(a)	33.	(c)	34.	(d)	35.	(d)	36.	(b)	37.	(a)	38.	(a)	39.	(a)	40.	(c)
41.	(a)	42.	(b)	43.	(b)	44.	(c)	45.	(e)	46.	(b)	47.	(d)	48.	(d)	49.	(e)	50.	(e)
51.	(e)	52.	(a)	53.	(c)	54.	(c)	55.	(c)	56.	(a)	57.	(c)	58.	(c)	59.	(c)	60.	(a)

Elimination Reaction E_2 / Hoffmann Exhaustive Methylation and Elimination

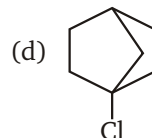
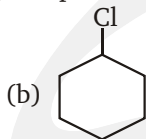
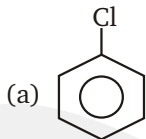
1.	(b)	2.	(c)	3.	(e)	4.	(c)	5.	(b)	6.	(c)	7.	(a)	8.	(e)	9.	(d)	10.	(a)
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[B] Haloarenes

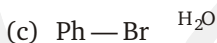
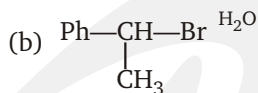
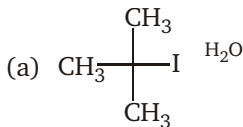
1.	(d)	2.	(d)	3.	(b)	4.	(c)	5.	(c)	6.	(a)	7.	(b)	8.	(d)	9.	(a)	10.	(c)
11.	(a)	12.	(c)	13.	(c)	14.	(d)	15.	(b)										

One or More than One Answers is/are Correct

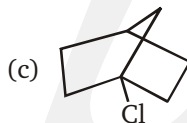
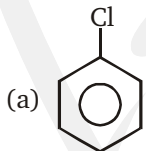
1. Which of the following compounds will not undergo S_N2 reaction?



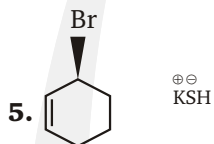
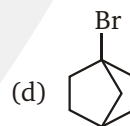
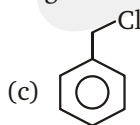
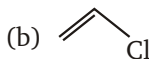
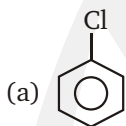
2. In which of the following reactions S_N1 reaction takes place?



3. In which of the following reaction neither S_N1 nor S_N2 take place ?



4. Which of the following compounds will not undergo nucleophilic substitution reaction?



Correct statement regarding given reaction will be:

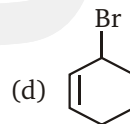
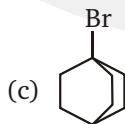
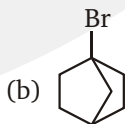
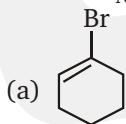
(a) above reaction is S_N2

(b) inversion of configuration takes place during given reaction

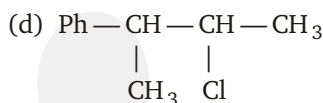
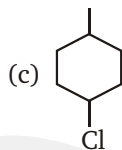
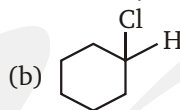
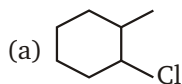
(c) bimolecular reaction

(d) second order reaction

6. Rate of S_N2 will be negligible in:



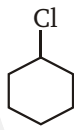
7. S_N1 and S_N2 product are same in (excluding stereoisomer):



8. $\text{CH}_3-\text{CH}_2-\text{O}-\text{CH}_3 \xrightarrow{\text{PCl}_5}$

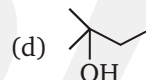
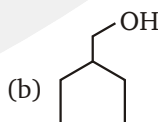
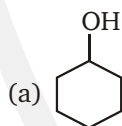
Products are:



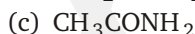
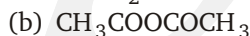
9.  $\xrightarrow{\text{moist Ag}_2\text{O}}$

S_N1

Product; product of the reaction is:



10. Which of the following will give yellow precipitate with I_2 / NaOH ?



11. Given rate of reaction is observed $\text{CH}_3-\overset{\text{CH}_3}{\underset{\text{CH}_3}{\text{C}}}-\text{Br}$ $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{Br}$ $\text{CH}_3-\text{CH}_2-\text{Br}$, for:

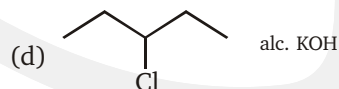
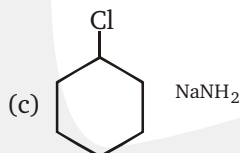
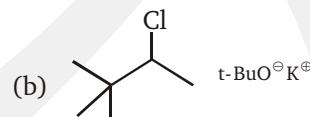
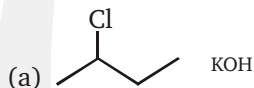
(a) E_1 reaction

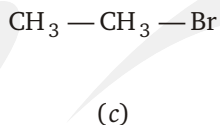
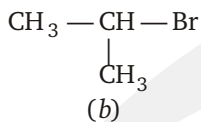
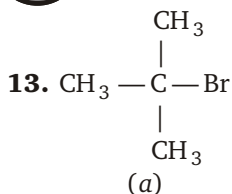
(b) E_2 reaction

(c) S_N2 reaction

(d) S_N1 reaction

12. In which of the following reaction Diastereomer isomer will formed ?





Correct statement is:

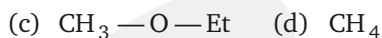
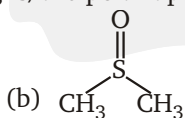
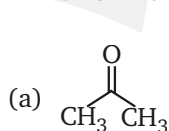
(a) rate of $S_{\text{N}}1$ a b c

(b) rate of $S_{\text{N}}2$ a b c

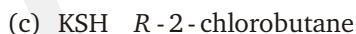
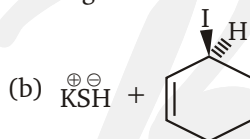
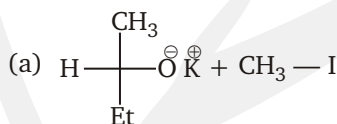
(c) rate of E_2 a b c

(d) rate of E_1 a b c

14. Which of the following is/are polar aprotic solvent?



15. In which of the following reactions inversion of configuration will take place?



16. Which of the following is/are bimolecular reaction?

(a) $S_{\text{N}}1$

(b) $S_{\text{N}}2$

(c) E_1

(d) E_2

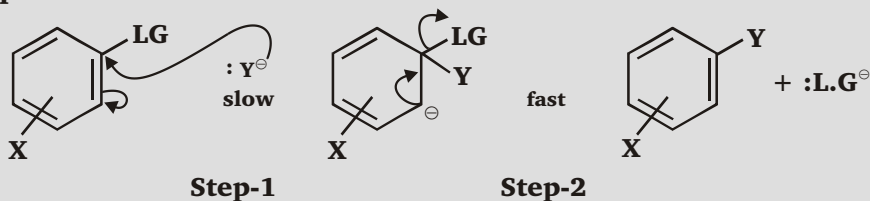
Answers

1.	(a, c, d)	2.	(a, b)	3.	(a, b, c)	4.	(a, b, d)	5.	(a, b, c, d)	6.	(a, b, c)
7.	(b, c)	8.	(a, b)	9.	(a)	10.	(a, d)	11.	(a, b, d)	12.	(a, d)
13.	(a, c, d)	14.	(a, b, c)	15.	(a, c)	16.	(b, d)				

Comprehension Based Problems

Passage-1:

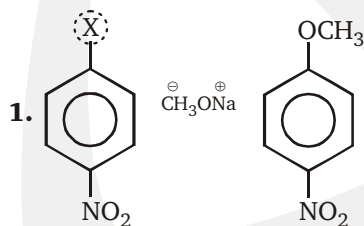
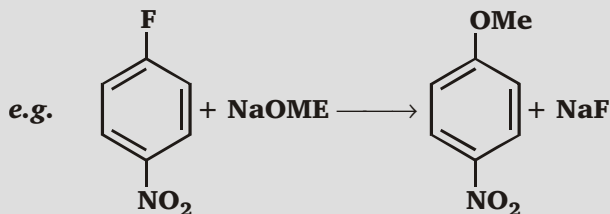
Nucleophilic aromatic substitution:



The species that adds to the aromatic ring in step-1 needs to be a nucleophile such as methoxide (CH_3O^-) instead of an electrophile such as nitronium ion (NO_2^+).

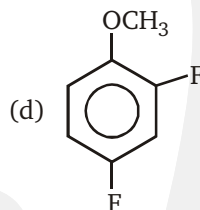
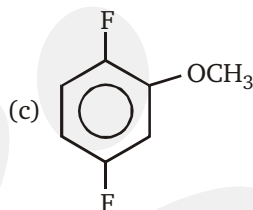
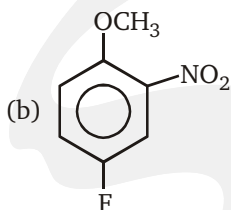
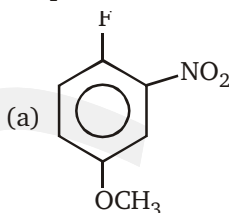
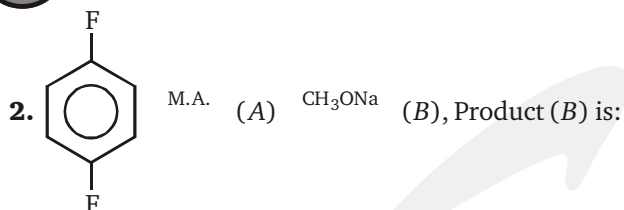
Step-1 will be facilitated by electron withdrawing rather than electron donating substituents. Substituents that are deactivating in electrophilic aromatic substitution are activating in nucleophilic aromatic substitution. The effect is most pronounced for strongly electron-withdrawing group ortho and/or para to the leaving group.

-In order of step-2 to occur, the pair of electrons bonding the leaving group to the ring are lost with the leaving group, rather than becoming part of the ring's system.

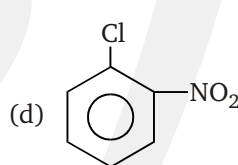
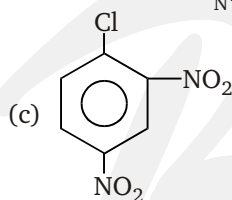
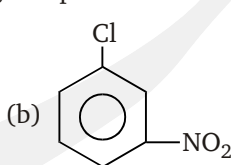
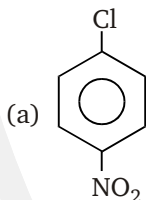


Above reaction has maximum rate when:

- (a) X I (b) X Br (c) X Cl (d) X F

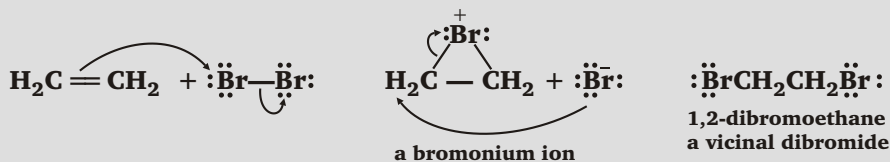


3. Which of the following compounds is most reactive towards ArS_{N}^2 reaction ?



Passage-2:

The halogens, Br_2 and Cl_2 , add to alkenes. When the electrons of the alkene approach a molecule of Br_2 or Cl_2 , one of the halogen atoms accepts them and releases the shared electrons to the other halogen atom. Therefore, in an electrophilic addition reaction, Br_2 behaves as if it were Br^+ and Br^- , and Cl_2 behaves as if it were Cl^+ and Cl^- .

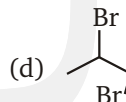
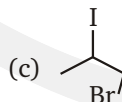
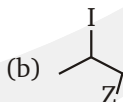
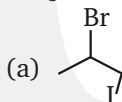


1. *cis*-2-butene $\xrightarrow[\text{CCl}_4]{\text{Br}_2}$

How many products will be obtained in this reaction ?

- (a) 1 (b) 2 (c) 3 (d) 4

2. $\text{CH}_3-\text{CH}=\text{CH}_2 \xrightarrow{\text{Br}}$ Product (A) is :

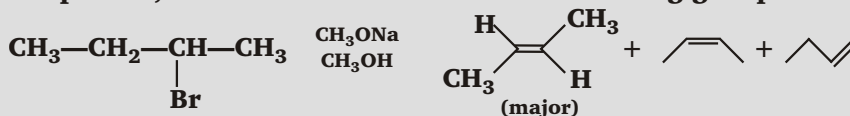
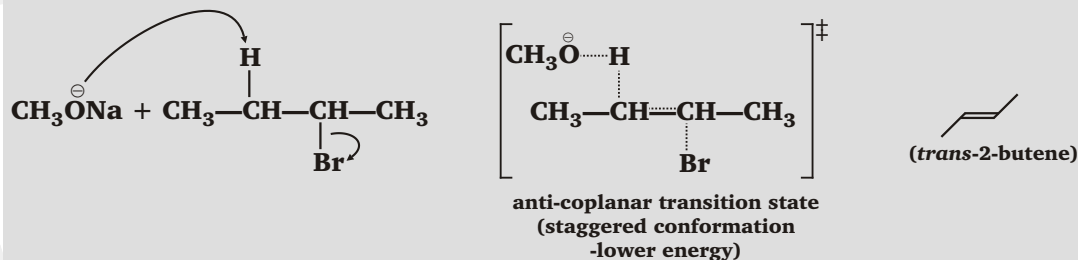


3. (x) isomer of butene $\xrightarrow[\text{CCl}_4]{\text{Br}_2}$ (y) products (including stereoisomer) sum of x + y :

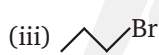
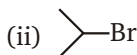
- (a) 8 (b) 9 (c) 10 (d) 11

Passage-3:**E₂ reaction Elimination bimolecular**

In the general mechanism of the E₂ reaction a strong base abstract a proton on a carbon atom adjacent to the one of the leaving group. As the base abstracts a proton, a double bond forms and the leaving group leaves.

**Mechanism:**

1. Give the order of reaction rate with alc. KOH:



(a) (i) (ii) (iii) (b) (ii) (i) (iii) (c) (iii) (ii) (i) (d) (iii) (i) (ii)

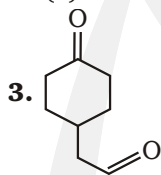
2. How many isomer of C₄H₉Br when reacts with NaNH₂, diastereomers will form? (excluding stereoisomer)

(a) 0

(b) 1

(c) 2

(d) 3



Sum of x y :

(a) 4

(b) 6

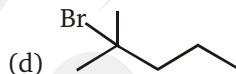
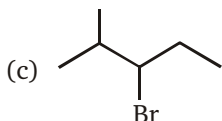
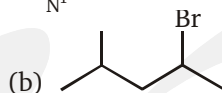
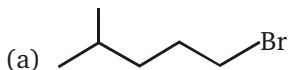
(c) 3

(d) 2

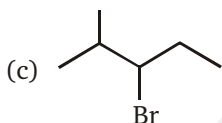
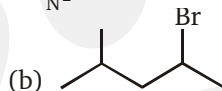
Passage-4:

These problems differ from those in earlier chapters in that they directly test your knowledge of core material rather than using a descriptive passage to extend the material or introduce new ideas. The number of factors that contribute to nucleophilic substitution can be daunting. The really major ones, though, are few and readily applied to specific by using the S_N¹ and S_N² mechanisms to guide your analysis.

1. Which compound undergoes substitution by the S_N1 mechanism at the fastest rate ?



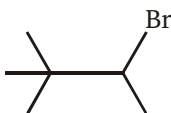
2. Which compound undergoes substitution by the S_N2 mechanism at the fastest rate ?



3. Which reaction takes place at the fastest rate ?



4. Identify the mechanism most responsible for the major product in the following reaction.



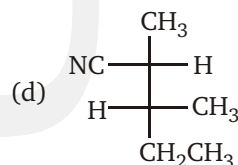
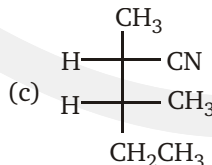
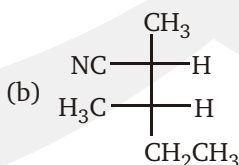
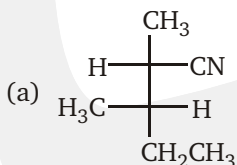
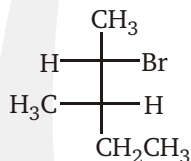
(a) S_N1

(b) S_N2

(c) E_1

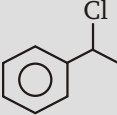
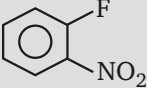
(d) E_2

5. NaCN ; Which is the major product of the reaction shown ?



Matching Type Problems

1. Match the Column:

Column-I		Column-II	
(a)		(p)	Nucleophilic-aromatic substitution reaction
(b)		(q)	Elimination reaction
(c)		(r)	S_N1 reaction
(d)		(s)	S_N2 reaction

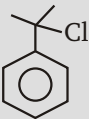
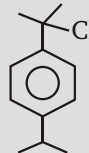
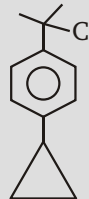
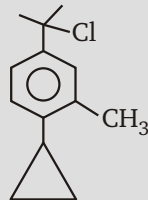
2. Match the column:

Column-I (Primary alkyl bromide)		Column-II (S_N2 relative rate)	
(a)	$\text{CH}_3 - \text{CH}_2 - \text{Br}$	(p)	10^5
(b)	$\text{Me} - \text{CH}_2 - \text{CH}_2 - \text{Br}$	(q)	10^2
(c)	$\text{Me} - \underset{\text{Me}}{\text{CH}} - \text{CH}_2 - \text{Br}$	(r)	0.8
(d)	$\text{Me} - \underset{\text{Me}}{\overset{\text{Me}}{\text{C}}} - \text{CH}_2 - \text{Br}$	(s)	1

3. Match the column:

	Substrate		E_2 elimination		S_N2 -substitution
(a)	$\text{CH}_3 - \text{CH}_2 - \text{Br}$	(p)	1	(w)	0
(b)	$(\text{CH}_3)_2\text{CH} - \text{Br}$	(q)	80	(x)	20
(c)	$(\text{CH}_3)_3\text{CBr}$	(r)	100	(y)	90

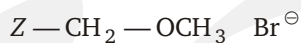
4. Match the column:

Column-I (Compound)		Column-II (Relative solvolysis rate in aqueous acetone)	
(a)		(p)	172
(b)		(q)	157
(c)		(r)	178
(d)		(s)	1

5. Match the column:

Column-I (Reaction)		Column-II (Relative rate of reaction)	
(a)	$\text{HO}^\oplus \quad \text{R} - \text{CH}_2 - \text{I}$	(p)	1
(b)	$\text{HO}^\ominus \quad \text{R} - \text{CH}_2 - \text{Br}$	(q)	200
(c)	$\text{HO}^\ominus \quad \text{R} - \text{CH}_2 - \text{Cl}$	(r)	10,000
(d)	$\text{HO}^\ominus \quad \text{R} - \text{CH}_2 - \text{F}$	(s)	30,000

6. Match the column-I with column-II for given S_N2 reaction and select the correct answer from the codes given below:



Column-I		Column-II (Relative reactivity)	
(a)	H —	(p)	0.1
(b)	CH ₃ —	(q)	3
(c)	C ₂ H ₅ —	(r)	1
(d)	$\begin{array}{c} CH_3 \\ \diagup \\ CH \\ \diagdown \\ CH_3 \end{array} -$	(s)	100

7. Match the column-I with column-II (no. of structural isomers produced in $-E_2$ elimination) and select the correct answer:

Column-I		Column-II	
(a)		(p)	Three
(b)		(q)	Zero
(c)		(r)	One
(d)		(s)	Two

8. Match the column:

Column-I		Column-II	
(a)		(p)	Undergo S_NAr (Nucleophilic aromatic substitution) with CH_3O^-
(b)		(q)	Undergo S_{N1} reaction with H_2O
(c)		(r)	Undergo S_{N2} reaction with aq. KOH
(d)		(s)	Undergo E_2 reaction with alc. KOH
		(t)	Undergo E_1 reaction with EtOH on strong heating

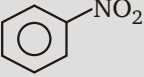
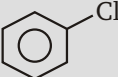
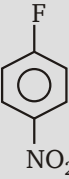
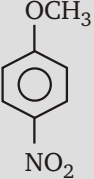
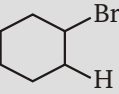
9. Match the column:

Column-I E_2 reaction (elimination bimolecules)		Column-II No. of possible products (including stereoisomerism)	
(a)	alc. KOH	(p)	0
(b)	alc. KOH	(q)	1
(c)	alc. KOH	(r)	2
(d)	alc. KOH	(s)	3

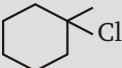
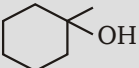
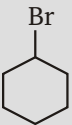
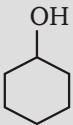
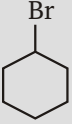
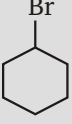
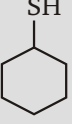
10. Match the column:

Column-I		Column-II	
(a)	 aq. KOH	(p)	Hard base (strong base)
(b)	 alc. KOH	(q)	Soft base (weak base)
(c)	 aq. KOH	(r)	S_{N2} reaction
(d)	 alc. KOH	(s)	E_2 reaction

11. Match the column:

Column-I		Column-II	
(a)	 HNO_3 H_2SO_4 	(p)	Substitution reaction
(b)	 Cl_2 AlCl_3 	(q)	Addition reaction
(c)	 $\text{CH}_3\text{O}^\oplus$ 	(r)	Carbocation is intermediate
(d)	 HBr CCl_4 	(s)	Carbanion is intermediate

12. Match the Column:

Column-I			Column-II	
(a)	 $\xrightarrow{\text{H}_2\text{O}}$ 		(p)	Substitution reaction
(b)	 $\xrightarrow{\text{Moist Ag}_2\text{O}}$ 		(q)	Elimination reaction
(c)	 $\xrightarrow{\text{NaNH}_2}$ 		(r)	Carbocation is intermediate
(d)	 $\xrightarrow{\text{KSH}}$ 		(s)	Carbanion is intermediate

A to Z (Learning Chart)

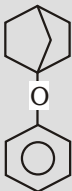
Reaction scheme for the identification of compounds A through Z:

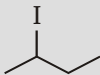
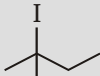
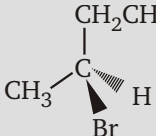
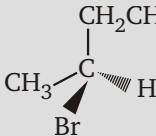
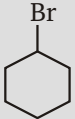
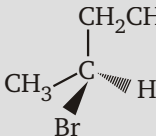
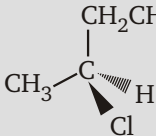
Starting materials: c1ccc(cc1)Cl (1-chloro-4-iodobenzene) and c1ccccc1O (cyclohexanol).

Reactions:

- c1ccc(cc1)Cl + MgBr → c1ccc(cc1)[Mg]Br (Grignard reagent)
- c1ccc(cc1)[Mg]Br + c1ccccc1O → c1ccc(cc1)O (4-phenylphenol) → c1ccc(cc1)Cl (1-chloro-4-iodobenzene) (S)
- c1ccc(cc1)Cl + D3O+ → c1ccc(cc1)[O+]D3 (T)
- c1ccc(cc1)[O+]D3 + Cu → c1ccc(cc1)Cu (300°C) → c1ccc(cc1)Cl (U)
- c1ccc(cc1)Cl + CH3CO2Et → c1ccc(cc1)O (V)
- c1ccc(cc1)O + NaOH / CaO → c1ccc(cc1)O (W)
- c1ccc(cc1)O + SOCl2 → c1ccc(cc1)Cl (X)
- c1ccc(cc1)Cl + NH3 → c1ccc(cc1)O (Y)
- c1ccc(cc1)O + KOBr → c1ccc(cc1)O (Z)
- c1ccc(cc1)O + Na / dry Ether → c1ccc(cc1)O (B)
- c1ccc(cc1)O + HBr / CCl4 → c1ccc(cc1)O (C)
- c1ccc(cc1)O + Br2 / CCl4 → c1ccc(cc1)O (D + E)
- c1ccc(cc1)O + Mg / dry Ether → c1ccc(cc1)O (F)
- c1ccc(cc1)O + Br2 / H2O → c1ccc(cc1)O (G)
- c1ccc(cc1)O + Na / Ether → c1ccc(cc1)O (H)
- c1ccc(cc1)O + CH3MgBr → c1ccc(cc1)O (I)
- c1ccc(cc1)O + CO2 → c1ccc(cc1)O (J)
- c1ccc(cc1)O + H+ → c1ccc(cc1)O (K)
- c1ccc(cc1)O + CH3-Li (excess) → c1ccc(cc1)O (L)
- c1ccc(cc1)O + Al(OH)3 / H2O → c1ccc(cc1)O (M)
- c1ccc(cc1)O + CH3MgBr / H+ → c1ccc(cc1)O (N)
- c1ccc(cc1)O + CH3MgBr / H+ → c1ccc(cc1)O (P)
- c1ccc(cc1)O + CH3MgBr / H+ → c1ccc(cc1)O (Q + R)

2. Determine whether a substitution reaction will take place by an S_N1 or an S_N2 pathway.

S.No.		S_N1	S_N2	No reaction
1.	$\begin{array}{c} \text{CH}_3 \\ \\ \text{CH}_3 - \text{C} - \text{Br} \\ \\ \text{CH}_3 \end{array} \quad \text{H}_2\text{O}$			
2.	$\begin{array}{c} \text{Br} \\ \\ \text{CH}_3 - \text{CH} - \text{CH}_3 \end{array} \quad \text{HO}^-$			
3.	$\begin{array}{c} \text{Br} \\ \\ \text{CH}_3 - \text{CH} - \text{CH}_3 \end{array} \quad \text{I}^-$			
4.	$\text{CH}_3 - \text{Br} \quad \text{HO}^-$			
5.	$\text{Ph} - \text{O} - \text{CH}_3 \quad \text{H}_3\text{O}^+$			
6.	$\text{Ph} - \text{O} - \text{C(CH}_3)_3 \quad \text{H}_3\text{O}^+$			
7.	 H_3O^+			
8.	$\text{Ph} - \text{O} - \text{CH}_2\text{CH}_3 \quad \text{H}_3\text{O}^+$			
9.	 HI			
10.	$\text{Ph} - \text{O} - \text{CH}_3 \quad \text{HI}$			
11.	$\begin{array}{c} \text{CH}_3 \\ \\ \text{CH}_3 - \text{C} - \text{OH} \\ \\ \text{CH}_3 \end{array} \quad \text{H}-\text{I}$			
12.	 H_3O^+			

13.	$\text{Ph} - \text{O} - \text{Ph} \quad \text{H}_3\text{O}^+$			
14.	 H_3O^+			
15.	 OH^-			
16.	 H_2O			
17.	 $+ \text{CH}_3\text{O}^-$ high concentration			
18.	 $+ \text{CH}_3\text{OH}$			
19.	 Moist Ag_2O			
20.	$\text{CH}_3\text{CH}_2\text{CH}(\text{Cl})\text{CH}_3 \quad \text{HO}^-$ high concentration			
21.	$\text{CH}_3\text{CH}_2\text{CH}(\text{I})\text{CH}_3 \quad \text{H}_2\text{O}$			
22.	 $+ \text{CH}_3\text{CH}_2\text{CH}_2\text{O}^-$ high concentration			
23.	 $+ \text{EtOH}$			

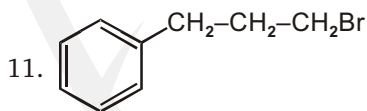
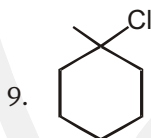
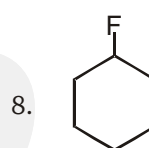
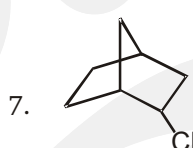
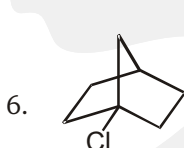
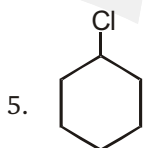
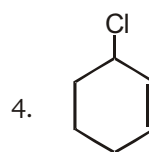
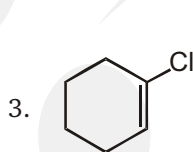
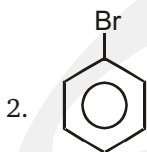
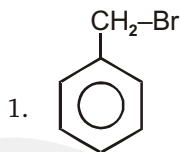
24.	$+ \text{CH}_3\text{O}^-$ high concentration			
25.	$+ \text{CH}_3\text{OH}$			

3.

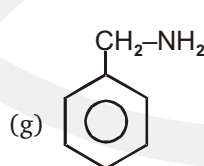
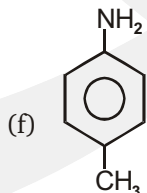
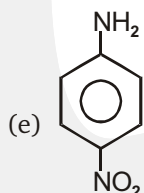
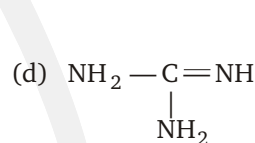
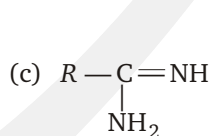
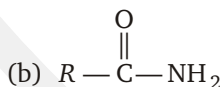
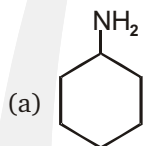
S.No.	Compound	$\text{S}_{\text{N}}1$ E_1 (Total products) (including Stereoisomer)
(i)	 EtOH	
(ii)	 EtOH	
(iii)		
(iv)		
(v)		
(vi)		
(vii)		
(viii)		
(ix)		

Unified Conceptual Problems

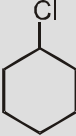
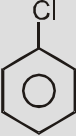
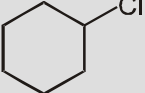
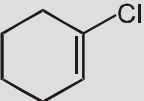
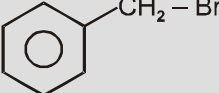
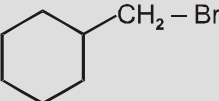
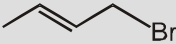
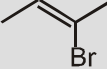
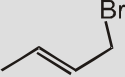
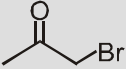
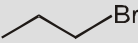
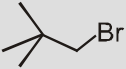
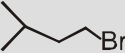
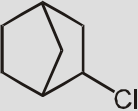
1. X The number of compounds do not show S_N2 .



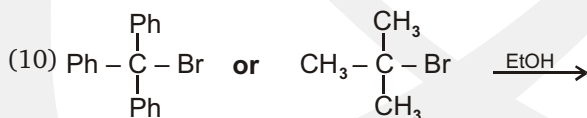
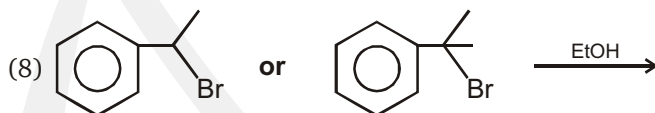
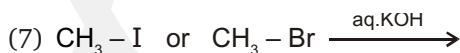
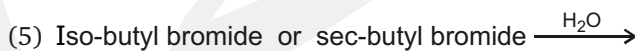
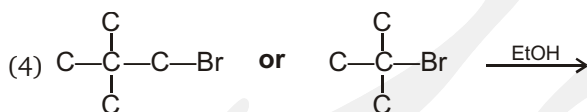
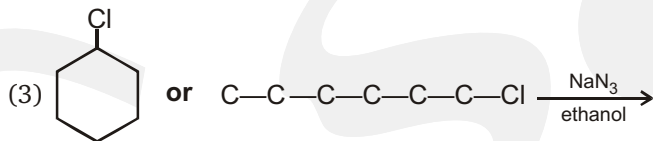
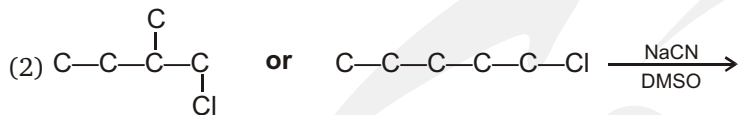
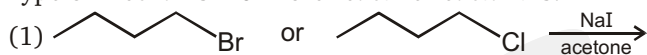
2. X number of compound which are more basic than



3.

S.No.	Compound		Compare rate of S_N2	Compare rate of S_N1
1.				
2.				
3.				
4.				
5.				
6.				
7.				
8.				
9.				
10.	$\text{CH}_3 - \text{O} - \text{CH}_2 - \text{Br}$	$\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{Br}$		
11.				
12.				

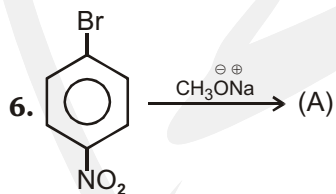
4. Type of mechanism & more reactive reactant is:



5. Identify the missing structure in each reaction ?

S.No.		
1.	<chem>Clc1ccc([N+](=O)[O-])cc1</chem> $\xrightarrow[160^{\circ}\text{C}]{15\%\text{NaOH}}$	
2.	<chem>O=[N+]([O-])c1c(Cl)cc([N+](=O)[O-])cc1[N+](=O)[O-]</chem> $\xrightarrow{\text{H}_2\text{O}}$ (A) $\longrightarrow$ acid	
3.	<chem>Brc1cc([N+](=O)[O-])cc([N+](=O)[O-])c1</chem> $\xrightarrow{\text{CH}_3\text{O}^-}$	
4.	<chem>Brc1cc(Br)cc([N+](=O)[O-])c1</chem> $\xrightarrow{\text{CH}_3\text{O}^-}$	
5.	<chem>Brc1cc([N+](=O)[O-])cc([N+](=O)[O-])c1</chem> $\xrightarrow{\text{CH}_3\text{—CH}_2\text{—NH}_2}$ (A) $\xrightarrow{\text{HO}^-}$ (B)	
6.	<chem>Fc1ccc([N+](=O)[O-])cc1</chem> $\xrightarrow{\text{HO}^-}$ (A) $\xrightarrow{\text{HO}^-}$ <chem>Oc1ccc([N+](=O)[O-])cc1</chem> Stable intermediate	
7.	<chem>Fc1ccccc1[N+](=O)[O-]</chem> $\xrightarrow{\text{KOH/}}$	

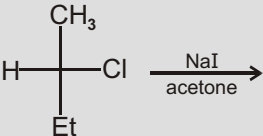
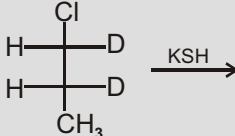
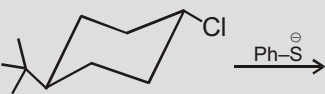
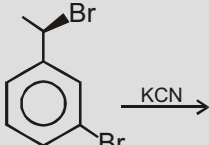
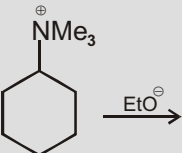
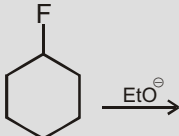
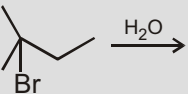
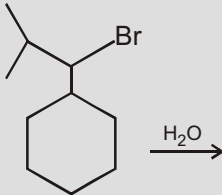
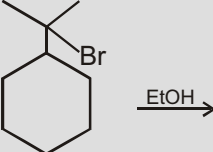
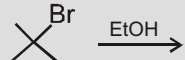
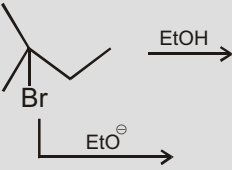
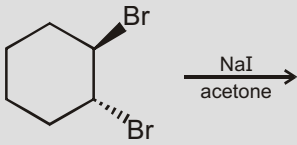
8.	 <chem>O=[N+]([O-])c1ccc(Br)cc1Br >> [KOH]</chem>	
9.	 <chem>Cc1ccc(Cl)cc1 >> [KNH2, liq. NH3] (A) (major)</chem>	
10.	 <chem>COc1ccccc1Cl >> [KNH2, liq. NH3] (B) (major)</chem>	

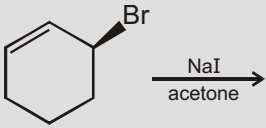
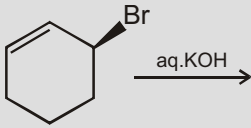
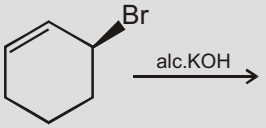
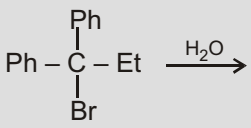
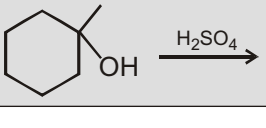
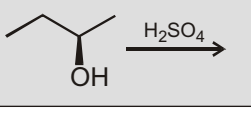
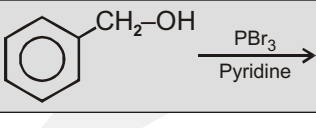


Energy profile diagram of given reaction is:

7. Identify major product and type of reaction (S_N1 / S_N2 / E_1 / E_2).

S.No.	Reaction	S.No.	Reaction
1.	 <chem>ClC1CCCCC1 >> [alc. KOH]</chem>	2.	 <chem>CCC(Br)C >> [NaNH2]</chem>
3.	 <chem>CC(C)C(Br)C >> [t-BuO- K+]</chem>	4.	 <chem>c1ccccc1CCl >> [CH3O-]</chem>

5.		6.	
7.		8.	
9.		10.	
11.		12.	
13.		14.	
15.		16.	

17.	 <chem>C1CCC(CC1)Br >> [NaI][acetone]</chem>	18.	 <chem>C1CCC(CC1)Br >> [aq.KOH]</chem>
19.	 <chem>C1C=CCCC1Br >> [alc.KOH]</chem>	20.	 <chem>CC(Br)(c1ccccc1)c2ccccc2 >> [H2O]</chem>
21.	 <chem>CC1(O)CCCCC1 >> [H2SO4]</chem>	22.	 <chem>CCC(O)C >> [H2SO4]</chem>
23.	 <chem>c1ccccc1CO >> [PBr3][Pyridine]</chem>		

Answers

Comprehension Based Problems:

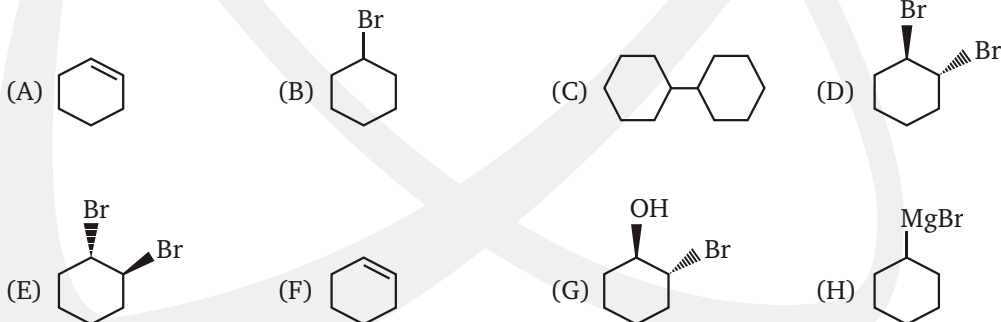
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Passage-2:	1.	(b)	2.	(a)	3.	(c)				
Passage-3:	1.	(a)	2.	(b)	3.	(b)				
Passage-4:	1.	(d)	2.	(a)	3.	(c)	4.	(d)	5.	(b)

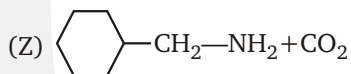
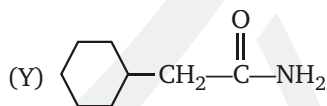
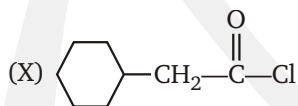
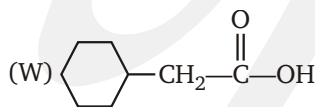
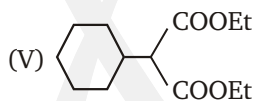
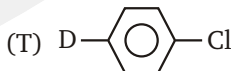
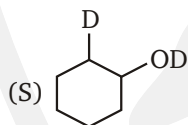
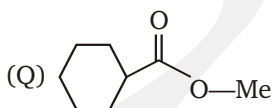
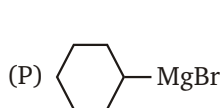
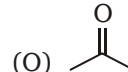
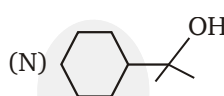
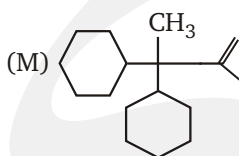
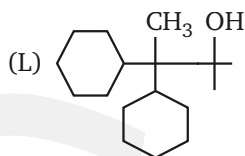
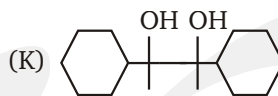
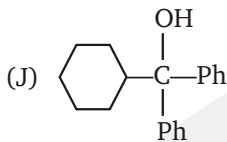
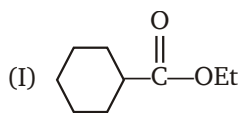
Matching Type Problems:

- (a) r, s; (b) q, r, s; (c) p; (d) q, r
- (a) s; (b) r; (c) q; (d) p
- (a) p, y; (b) q, x; (c) r, w
- (a) s; (b) q; (c) p; (d) r
- (a) s; (b) r; (c) q; (d) p
- (a) s; (b) q; (c) r; (d) p
- (a) s; (b) r; (c) p; (d) q
- (a) q, r, s, t; (b) p; (c) p; (d) p, q, r, s, t
- (a) s; (b) r; (c) q; (d) p
- (a) q, r; (b) p, s; (c) q, r; (d) p, s
- (a) p, r; (b) p, r; (c) p, s; (d) q, r
- (a) p, r; (b) p, r; (c) q; (d) p

Multi Concept Subjective Problems:

1.





2.

S.No.	S_{N^1}	S_{N^2}	No reaction	S.No.	S_{N^1}	S_{N^2}	No reaction
1.	✓			2.		✓	
3.		✓		4.		✓	
5.		✓		6.	✓		
7.			✓	8.		✓	
9.		✓		10.		✓	
11.	✓			12.	✓		
13.			✓	14.			✓
15.		✓		16.	✓		
17.		✓		18.	✓		
19.	✓			20.		✓	
21.	✓			22.		✓	
23.	✓			24.		✓	
25.	✓						

3.

- (i) S_{N^1} 3; E_1 2 1
 (iii) S_{N^1} 3; E_1 3
 (v) S_{N^1} 4; E_1 3
 (vii) S_{N^1} 2; E_1 4
 (ix) S_{N^1} 2; E_1 5

- (ii) S_{N^1} 1; E_1 2
 (iv) S_{N^1} 2; E_1 3
 (vi) S_{N^1} 4; E_1 7
 (viii) S_{N^1} 1; E_1 2

Unified Conceptual Problems:

1. x 5(2, 3, 6, 8, 9)

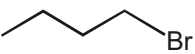
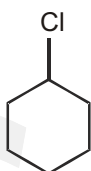
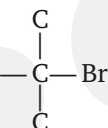
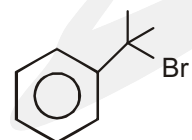
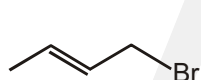
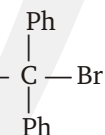
2. x 5(a, c, d, f, g)

3.

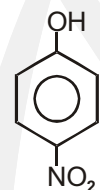
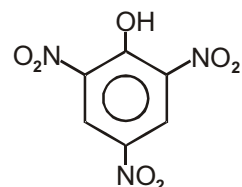
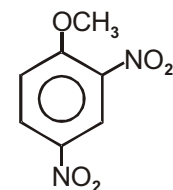
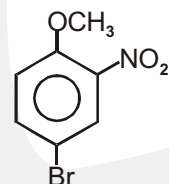
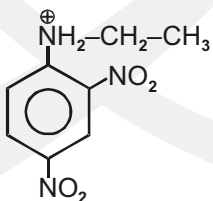
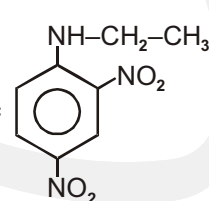
1. $a > b$; $a > b$ 2. $a > b$; $b > a$ 3. $b > a$; $b > a$
 4. $a > b$; $b > a$ 5. $a > b$; $a > b$ 6. $a > b$; $a > b$

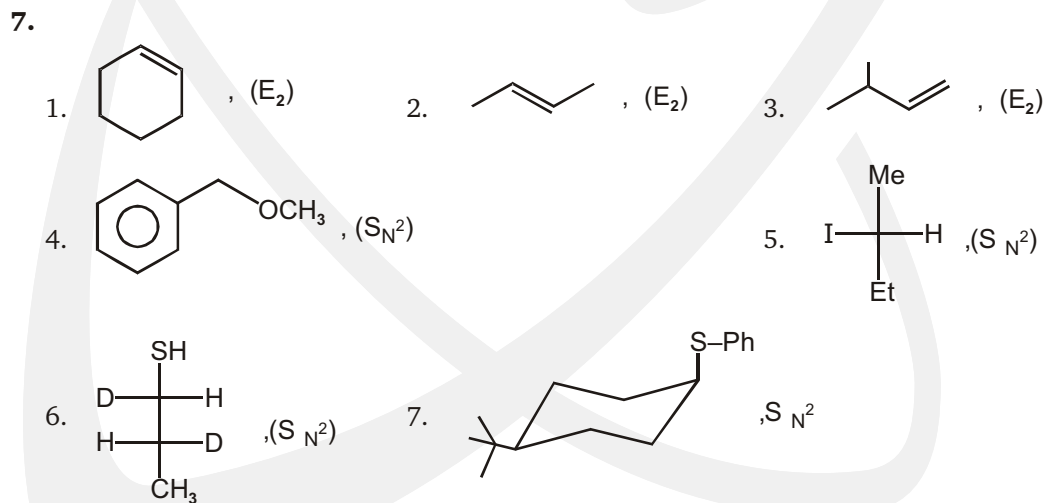
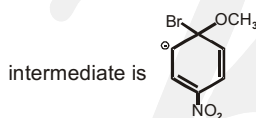
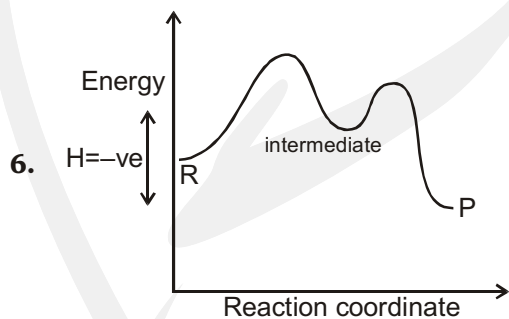
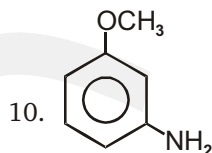
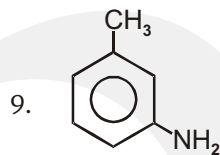
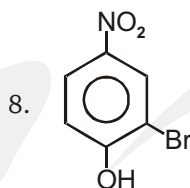
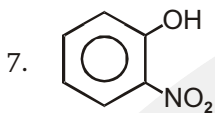
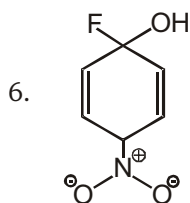
7. $a > b$; $a > b$ 8. $b > a$; $b > a$ 9. $a > b$; $b > a$ 10. $a > b$; $a > b$ 11. $b > a$; $b > a$ 12. $b > a$; $b > a$

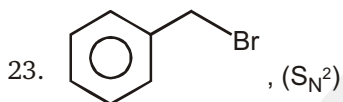
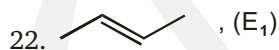
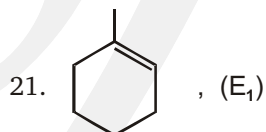
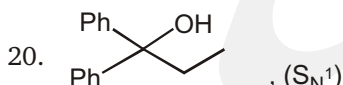
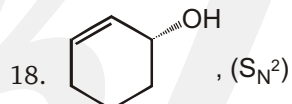
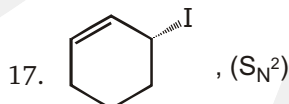
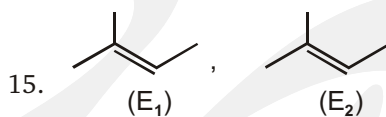
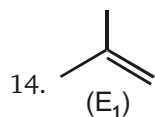
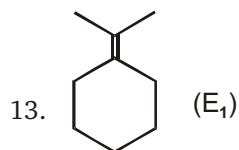
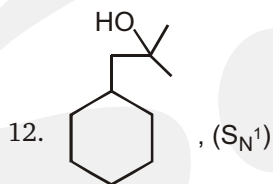
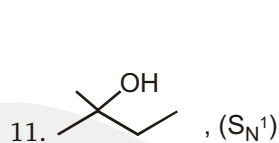
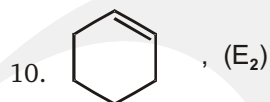
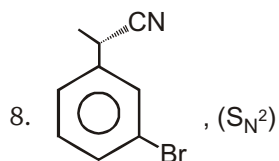
4.

1. S_N2 , 2. S_N2 , $C-C-C-C-C-Cl$ 3. S_N1 , 4. S_N1 , 5. S_N1 , $C-C-C-Br$
6. S_N2 , 7. S_N2 , CH_3I 8. S_N1 , 9. S_N2 , 10. S_N1 , 

5.

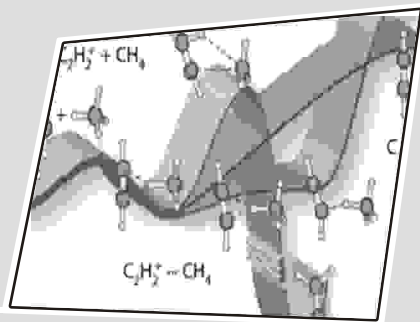
1. 2. 
(picric acid)3. 4. 5. A = 
 $NH_2-CH_2-CH_3$, B = 
 $NH-CH_2-CH_3$





4

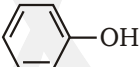
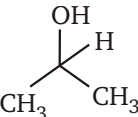
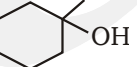
Alcohols, Phenols and Ethers



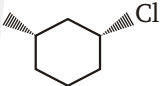
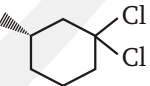
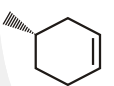
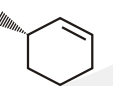
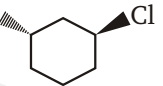
Only One Answer is Correct

[A] Alcohols

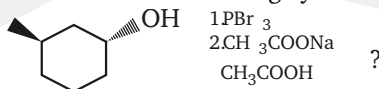
1. Which of the following is a secondary alcohol?

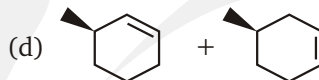
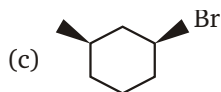
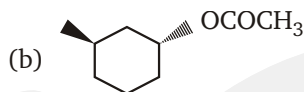
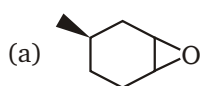
- (a)  (b)  (c)  (d) CH_3OH

2. What is the product of the following reaction?

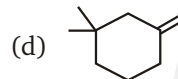
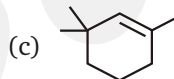
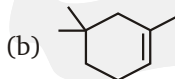
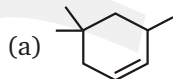
- (a)  + enantiomer
- (b)  ?
- (c)  + 
- (d)  optically active

3. What is the major product formed in the following synthesis?

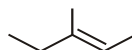




4. Which is not a likely product formed when the following alcohol is treated with sulfuric acid?

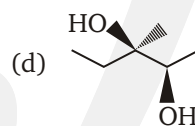
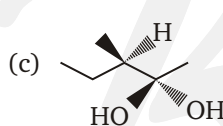
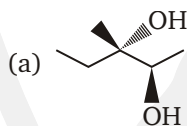


5. What are the products of the following reaction?

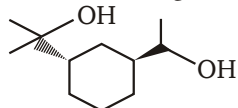


KMnO_4
Cold
 $\text{NaOH, H}_2\text{O}$

Product Enantiomer

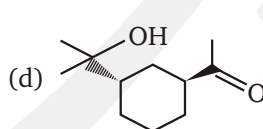
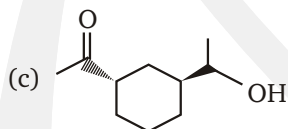
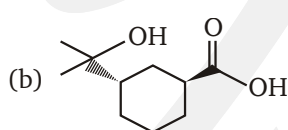
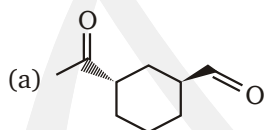


6. What is the product of the following reaction?



$\text{CrO}_3, \text{HCl, pyridine}$

?



7. How many compounds with molecular formula $\text{C}_5\text{H}_{10}\text{O}$ may be reduced with NaBH_4 to a primary alcohol?

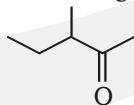
(a) 2

(b) 3

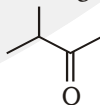
(c) 4

(d) 5

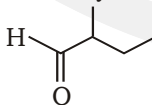
8. Which of the following compounds gives 3-methyl-2-butanol upon reduction with LiAlH_4 ?



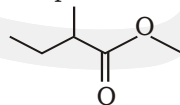
a



b



c



d

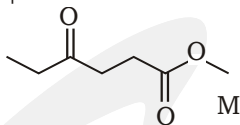
(a) a

(b) b

(c) c

(d) d

9. Which of the following products is formed in good yield by the reduction of methyl 4-oxo-hexanoate (M) with LiAlH_4 in ether?

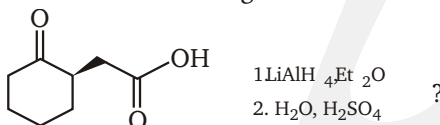


- (a) 6-Methoxy-3-hexanone (b) 1,4-Hexanediol
(c) Methyl 4-hydroxyhexanoate (d) 4-Hydroxyhexanal
10. What is the major product of the following reaction?



- (a) + enantiomer (b)
(c) (d) None of these

11. What are the major products of the following reaction?



- (a) + enantiomer (b) +
(c) + (d) +

12. Predict the product of the following reaction. $\text{CH}_3\text{COOH} \xrightarrow{\text{NaBH}_4}$

- (a) CH_3CHO (b) $\text{CH}_3\text{CH}_2\text{OH}$ (c) CH_3CH_3 (d) no reaction

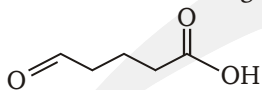
13. Which of the following compounds does not give a tertiary alcohol upon reaction with methylmagnesium bromide?

- (a) 3-methylpentanal (b) ethyl benzoate
(c) 4,4-dimethylcyclohexanone (d) 4-heptanone

14. Which of the following compounds gives a secondary alcohol upon reaction with methylmagnesium bromide? Assume the usual acid workup.

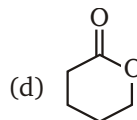
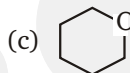
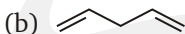
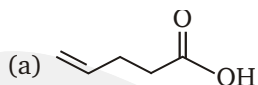
- (a) butyl formate (b) 3-pentanone (c) pentanal (d) methyl butanoate

15. Which of the following compounds is not oxidized by pyridinium chlorochromate (PCC)?
 (a) 2-methyl-2-butanol (b) 1-pentanol
 (c) 2-pentanol (d) 1,3-propanediol
16. What is the major product of the following reaction?



1. NaBH_4 , MeOH
 2. Dilute acid, heat

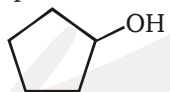
?



17. Which of the following is an allylic alcohol?



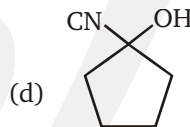
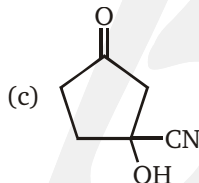
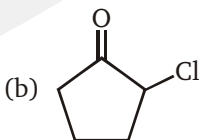
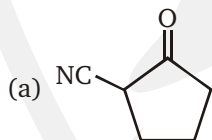
18. Give the major product of the following reaction:



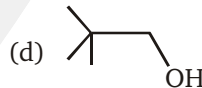
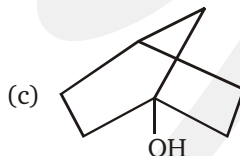
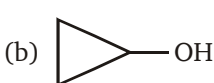
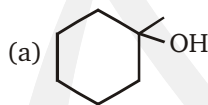
$\text{CrO}_3/\text{H}_2\text{SO}_4$

NaCN/HCl

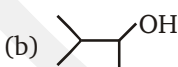
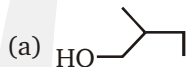
Major product



19. Identify which of the compound having 2° alcohol?

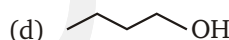
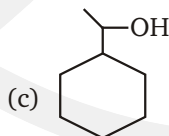
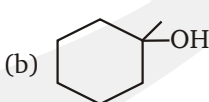
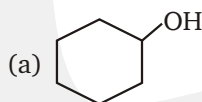


20. Identify 2° alcohol in the given following alcohols:

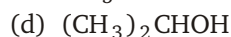
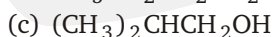


(d) None of these

21. Which of the following is /are 1° alcohol?

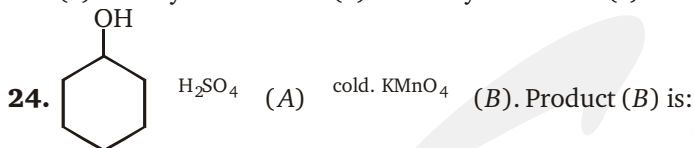


22. Which of the following is sec-butyl alcohol?

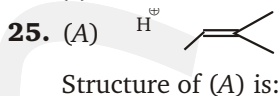


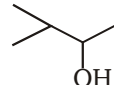
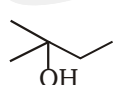
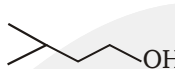
23. The isomeric alcohol which has a chiral carbon atom is:

- (a) *n*-butyl alcohol (b) iso-butyl alcohol (c) sec-butyl alcohol (d) tert-butyl alcohol

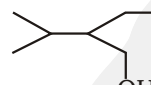
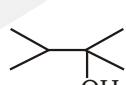


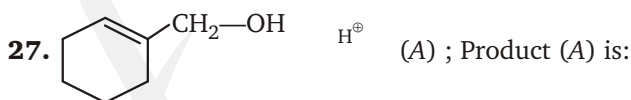
- (a) diastereomer (b) meso
(c) racemic (d) geometrical isomer

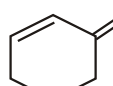
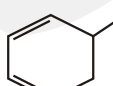
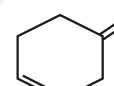


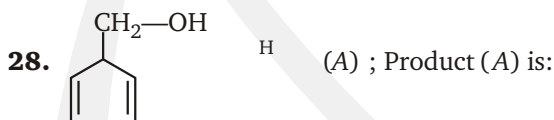
- (a)  (b)  (c)  (d) all of these

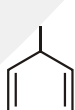
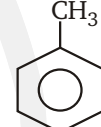


- (a)  (b)  (c)  (d) all of these



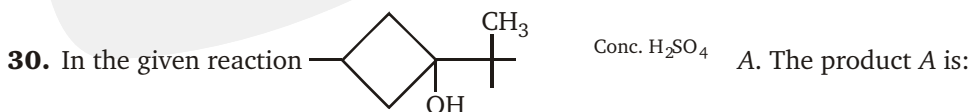
- (a)  (b)  (c)  (d) 

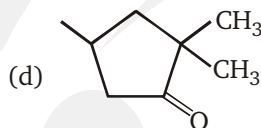
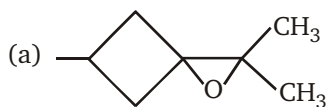


- (a)  (b)  (c)  (d) 

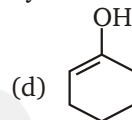
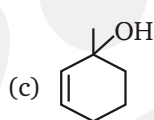
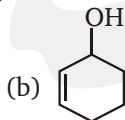
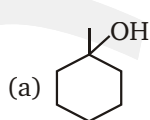
29. Best method to convert alcohol to halide is:

- (a) $\text{ROH} \xrightarrow{\text{SOCl}_2} \text{R}-\text{Cl}$ (b) $\text{R}-\text{OH} \xrightarrow{\text{PCl}_3} \text{R}-\text{Cl}$
(c) $\text{R}-\text{OH} \xrightarrow{\text{PCl}_5} \text{R}-\text{Cl}$ (d) $\text{R}-\text{OH} \xrightarrow{\text{HCl}} \text{R}-\text{Cl}$

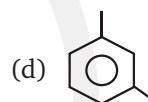
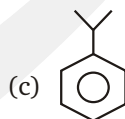
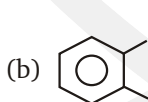
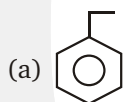
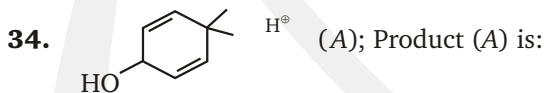
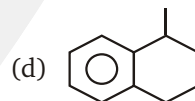
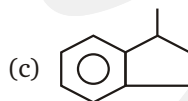
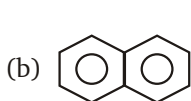
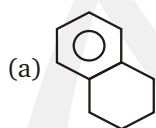
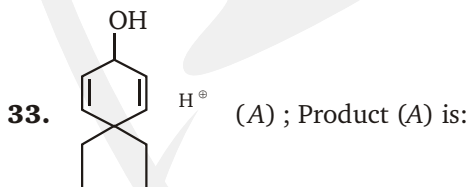
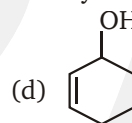
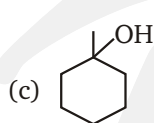
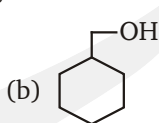
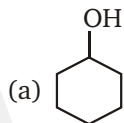




31. Which of the following is most reactive towards the acid-catalysed dehydration?



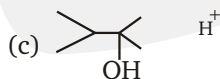
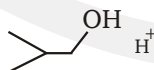
32. Which of the following alcohols is least reactive towards acid-catalysed dehydration?



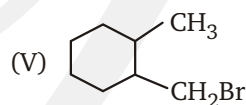
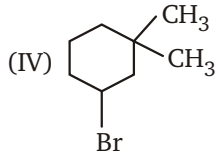
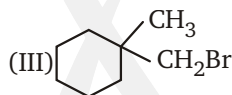
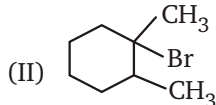
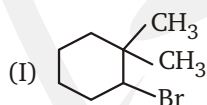
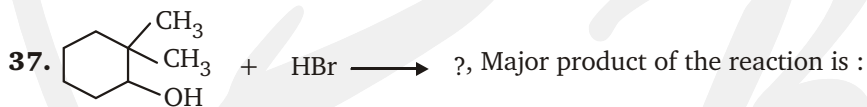
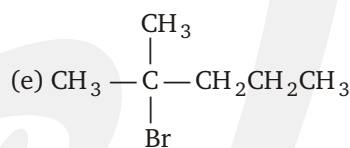
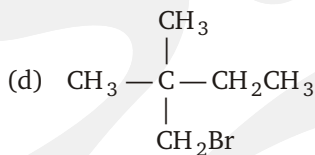
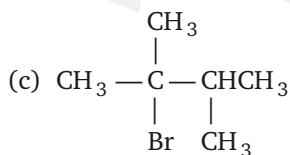
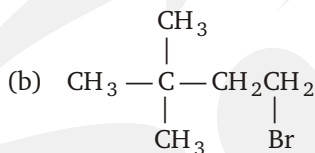
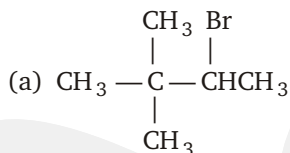
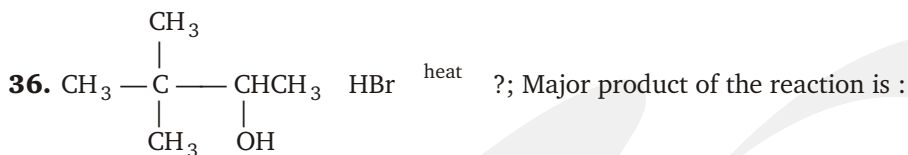
35. In which of the following reaction pair of diastereomers will form ?

(a)

(b)



(d) All of these



(a) I

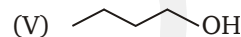
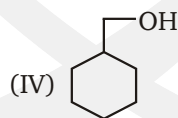
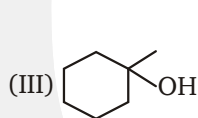
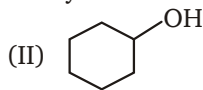
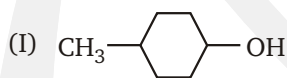
(b) II

(c) III

(d) IV

(e) V

38. Which of the following alcohols dehydrates most readily ?



(a) I

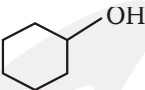
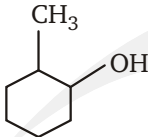
(b) II

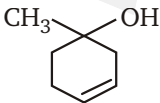
(c) III

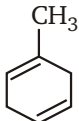
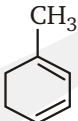
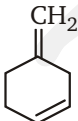
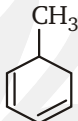
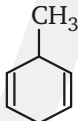
(d) IV

(e) V

39. Which of the following alcohols gives a rearranged carbocation when dehydrated?

- $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$ (I) $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$ (II)  (III)  (IV) $\text{H}_3\text{C}-\text{C}(\text{OH})(\text{CH}_3)_2$ (V)
- (a) I (b) II (c) III (d) IV (e) V

40.  $\xrightarrow[\text{heat}]{\text{H}_2\text{SO}_4}$? , Major product of the reaction is :

- (a)  (b)  (c)  (d)  (e) 

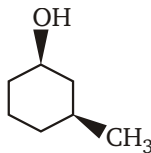
41. Which of the following reagents can be used to oxidize 1° alcohols to aldehydes ?

- (a) KMnO_4 (b) MnO_2 (c) $\text{K}_2\text{Cr}_2\text{O}_7$ (d) H_2O_2 (e) PCC

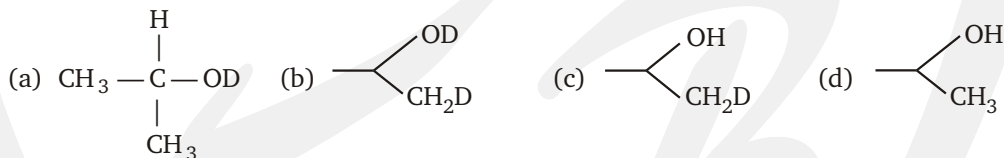
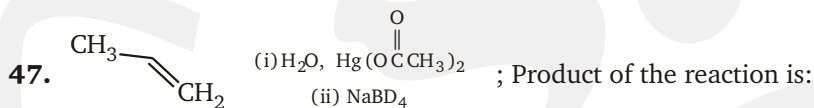
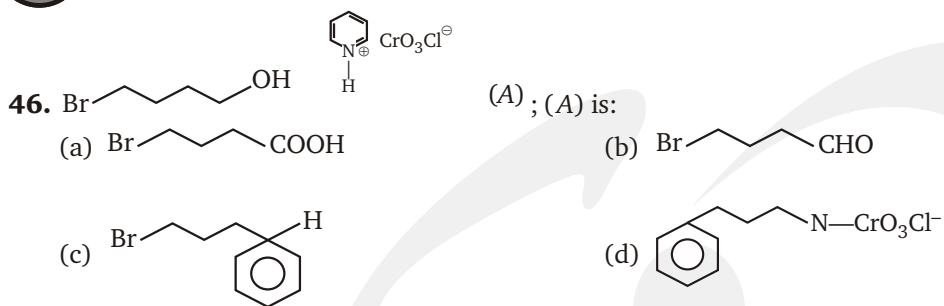
42. In the chromic acid oxidation of alcohols, the chromium is:

- (a) reduced from Cr^{6+} to Cr^{3+} (b) oxidized from Cr^{6+} to Cr^{3+}
 (c) reduced from Cr^{3+} to Cr^{6+} (d) oxidized from Cr^{3+} to Cr^{6+}
 (e) none of the above

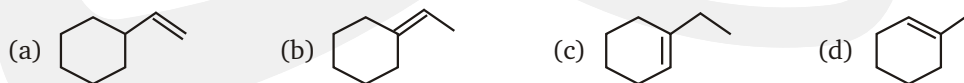
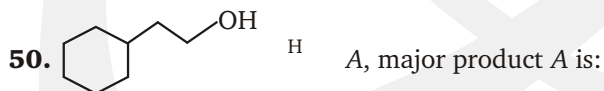
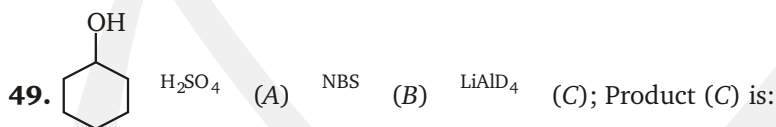
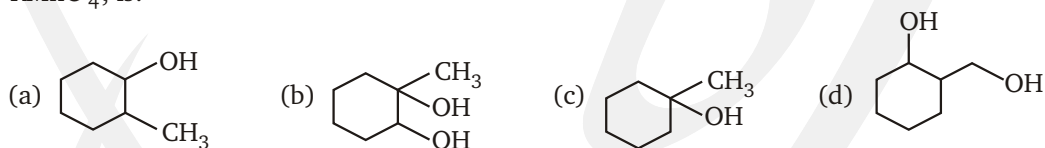
43. When the optically active alcohol given below is oxidized by H_2CrO_4 , the major product is



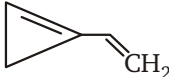
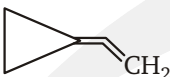
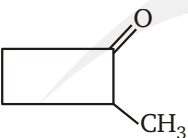
- (a) a ketone which is meso and optically inactive
 (b) an optically active ketone
 (c) a racemic mixture of ketones which is optically inactive
 (d) an aldehyde which is meso and optically active
 (e) an acyclic carboxylic acid
44. Which of the following compound contains most acidic hydrogen ?
- (a) $\text{HC}=\text{CH}$ (b) CH_3OH (c) NH_3 (d) $\text{H}_2\text{C}=\text{CH}_2$
45. Which of the following compound will evolve H_2 gas with Na ?
- (a) $\text{H}_2\text{C}=\text{CH}_2$ (b) $\text{R}-\text{O}-\text{R}$ (c) $\text{R}-\text{CH}_3$ (d) $\text{R}-\text{O}-\text{H}$

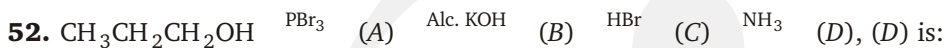


48. The alcohol, which would not generate a carbonyl compound on oxidation with acidified KMnO_4 , is:

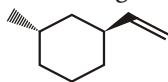
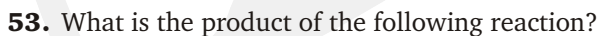




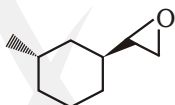
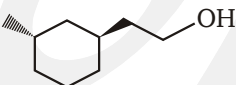
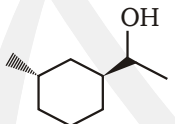
- (a)  (b)  (c)  (d) None of these

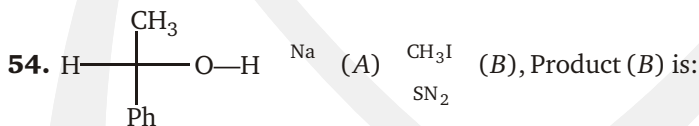


- (a) $\text{CH}_3 - \text{CH}_2 - \text{CH} - \text{NH}_2$ (b) $\text{CH}_3 - \underset{\text{NH}_2}{\text{CH}} - \text{CH}_3$
 (c) $\text{CH}_3 - \underset{\text{CH}_3}{\overset{\text{H}}{\text{C}}} - \text{NH}_2$ (d) $\text{CH}_3 - \underset{\text{CH}_3}{\overset{\text{CH}_3}{\text{C}}} - \text{NH}_2$

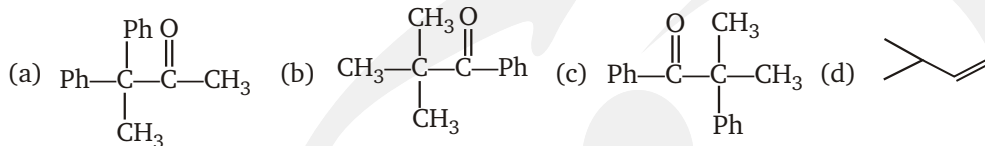
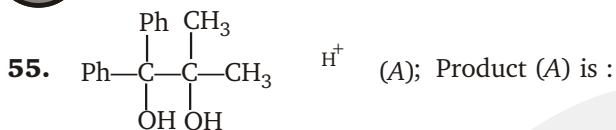


1. B_2H_6 , THF
 2. NaOH , H_2O_2 ?

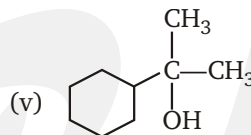
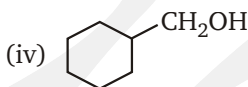
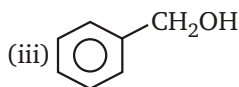
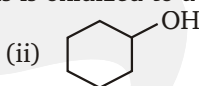
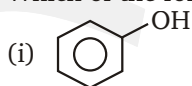
- (a)  (b) 
 (c)  (d) 



- (a) $\text{H} - \underset{\text{OCH}_3}{\overset{\text{CH}_3}{\text{C}}} - \text{Ph}$ (b) $\text{H} - \underset{\text{Ph}}{\overset{\text{CH}_3}{\text{C}}} - \text{O} - \text{CH}_3$
 (c) $\text{H} - \underset{\text{CH}_3}{\overset{\text{Ph}}{\text{C}}} - \text{O} - \text{CH}_3$ (d) $\text{CH}_3\text{O} - \underset{\text{CH}_3}{\overset{\text{Ph}}{\text{C}}} - \text{H}$



56. Which of the following alcohols is oxidized to a ketone by chromic acid ?



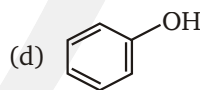
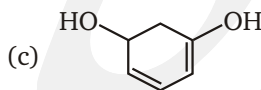
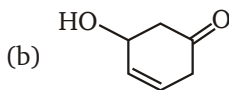
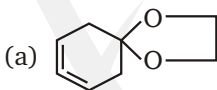
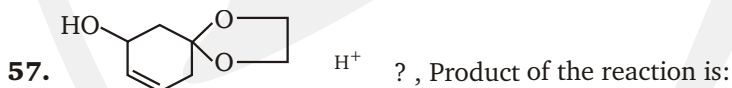
(a) i

(b) ii

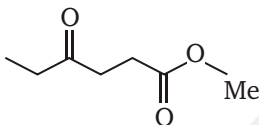
(c) iii

(d) iv

(e) v



58. Which of the following products is formed in good yields by the reduction of methyl 4-oxo-hexanoate (M) with NaBH_4 ?

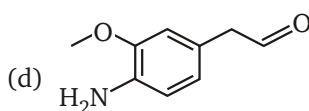
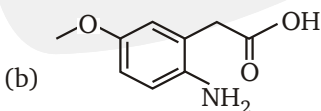
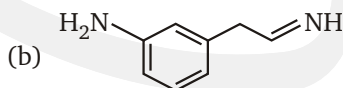
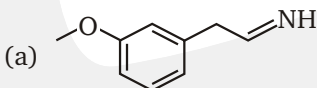
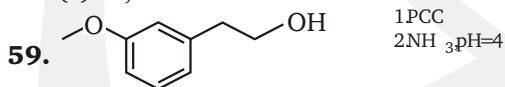


(a) Methyl 4-hydroxyhexanoate

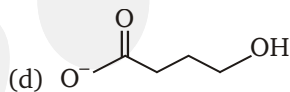
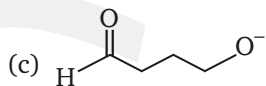
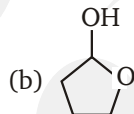
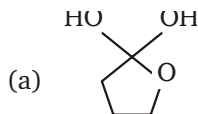
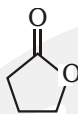
(b) 6-Hydroxy-3-hexanone

(c) 1,4-Hexanediol

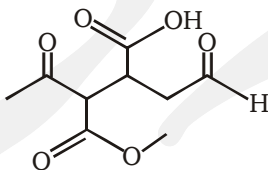
(d) 4-Hydroxyhexanal



60. What is the product formed when the following lactone is treated with aqueous sodium hydroxide solution?

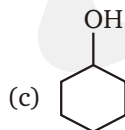
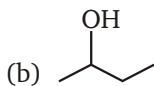
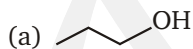


61.

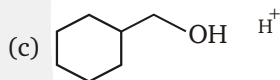
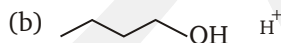
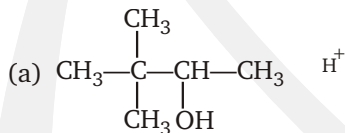


In the above given compound how many functional groups are reduced by LAH (Lithium aluminium hydride) and SBH (Sodium borohydride) respectively ?

- (a) 4, 4 (b) 4, 3 (c) 3, 4 (d) 4, 2
62. Which of the following alcohol will give turbidity of alkyl chloride immediately at room temperature when reacts with Lucas reagent?



63. In which of the reaction rearrangement take place with change in carbon skeleton ?

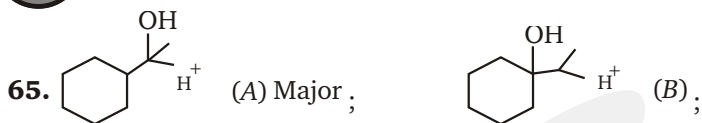


(d) All of these

64. (A); (B); (C)

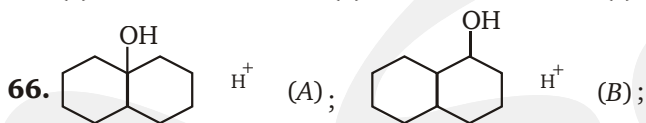
Sum of -hydrogen in (A+B+C=?)

- (a) 17 (b) 18 (c) 19 (d) 20



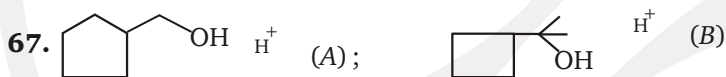
Sum of α -hydrogen in (A B ?)

- (a) 18 (b) 19 (c) 20 (d) 21



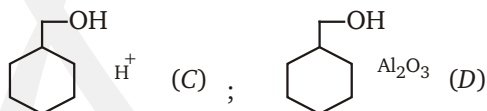
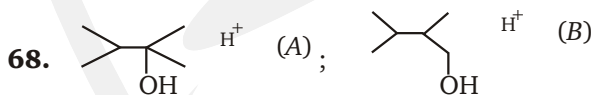
Sum of α -hydrogen in (A B ?)

- (a) 15 (b) 16 (c) 17 (d) 18



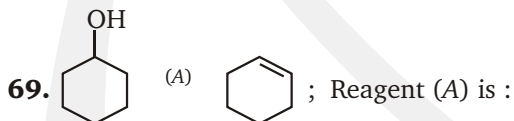
Sum of α -hydrogen in (A+B =?)

- (a) 12 (b) 13 (c) 14 (d) 15



Sum of α -hydrogen in A+B+C+D is :

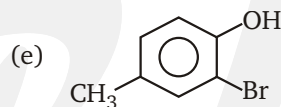
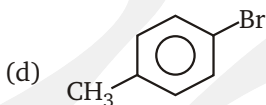
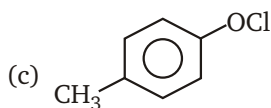
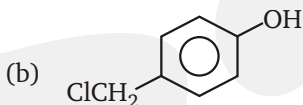
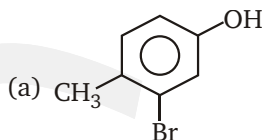
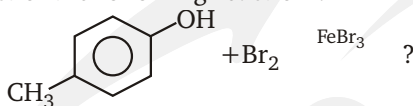
- (a) 34 (b) 35 (c) 36 (d) 37



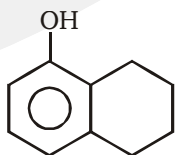
- (a) H_3O^+ (b) $\text{H} /$ (c) HO^- (d) NH_2^-

[B] Phenols

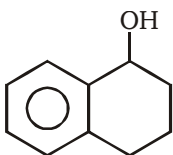
1. What is the major product of the following reaction ?



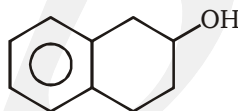
2. Compare the rate of acid catalyst dehydration:



I



II



III

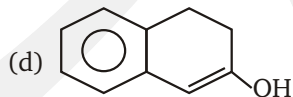
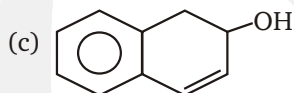
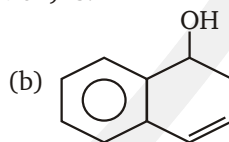
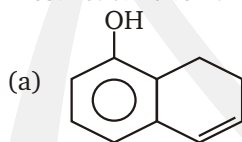
(a) I > II > III

(b) II > III > I

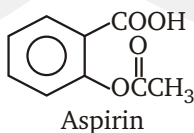
(c) III > II > I

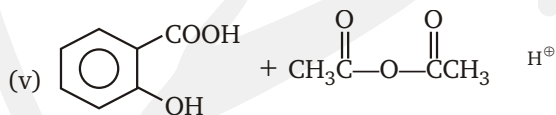
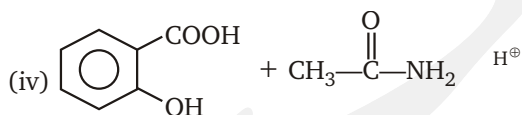
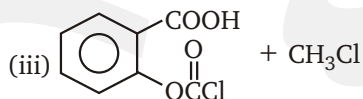
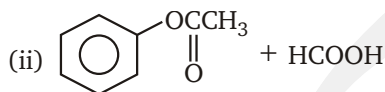
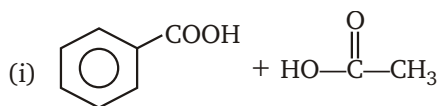
(d) II > I > III

3. Most reactive towards (acid catalysed dehydration) is:



4. Which of the following is the best method for preparing aspirin,





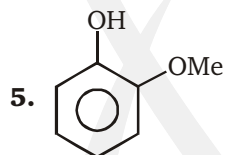
(a) i

(b) ii

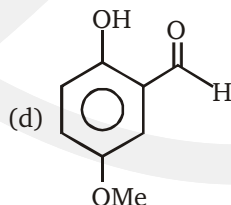
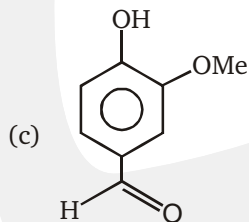
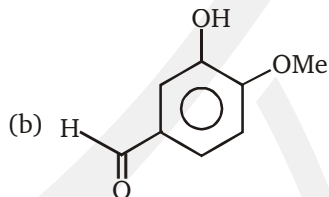
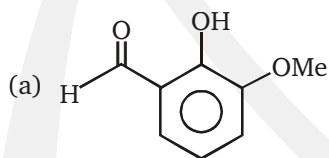
(c) iii

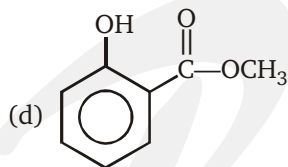
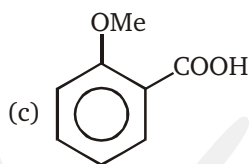
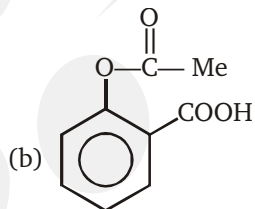
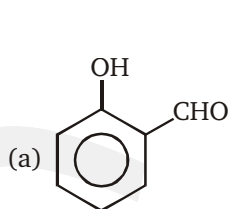
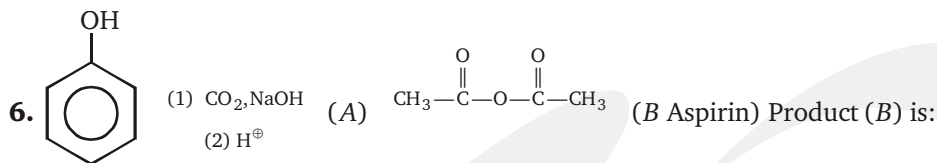
(d) iv

(e) v

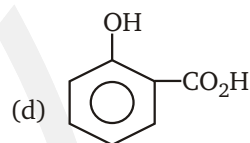
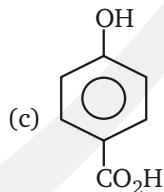
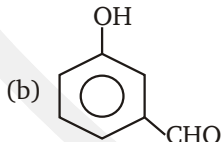
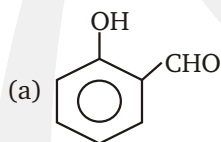
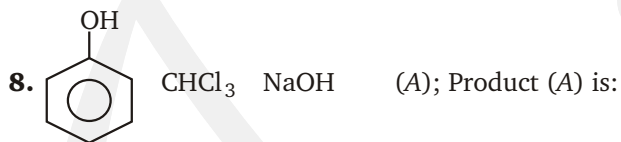
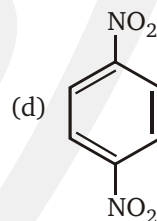
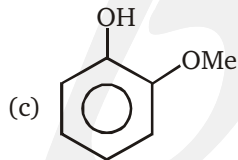
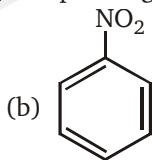
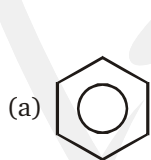
(i) $\text{KOH}, \text{CHCl}_3$ (ii) H_3O^+

'P', Major product (P) of the reaction is:

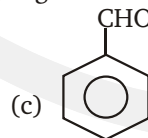
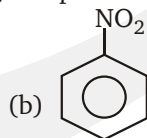
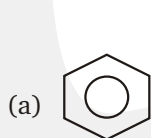




7. Which of the following compound gives Reimer-Tiemann reaction ?



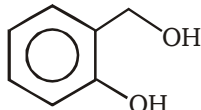
9. Which of the following compound will not undergo Reimer-Tiemann reaction?

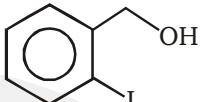
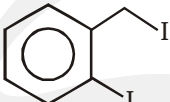
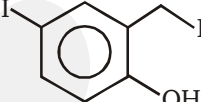
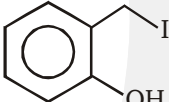


(d) All of these

10. Electrophile in Reimer-Tiemann reaction is:

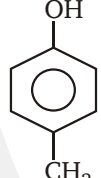
- (a) $\ominus\text{CCl}_3$ (b) CH_2Cl_2 (c) CHCl_3 (d) $:\text{CCl}_2$

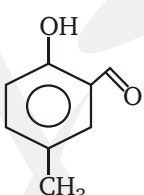
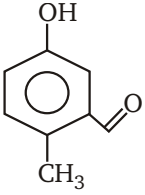
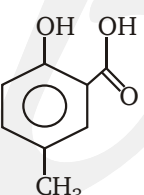
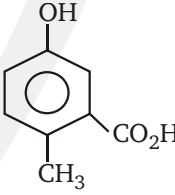
11.  $\xrightarrow[\text{excess}]{\text{HI}}$ 'P'; Major product 'P' is:

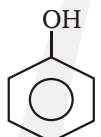
- (a)  (b)  (c)  (d) 

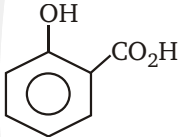
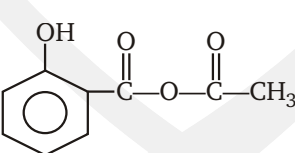
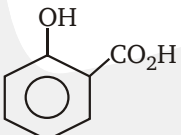
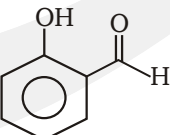
12. In Reimer-Tiemann reaction molecular weight of phenol increases by:

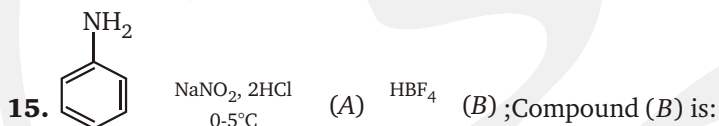
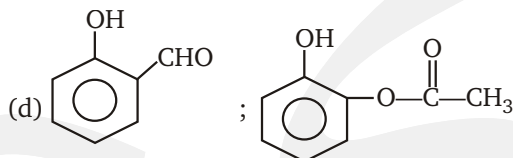
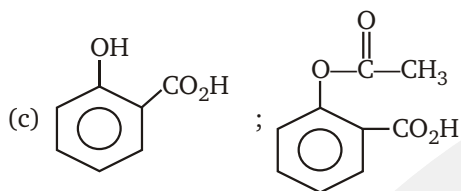
- (a) 28 (b) 29 (c) 30 (d) 31

13.  $\xrightarrow[\text{2H}^+]{\text{CHCl}_3, \text{NaOH}}$ (A); Product (A) is:

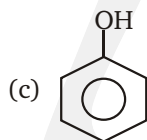
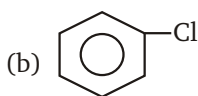
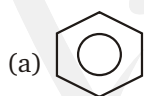
- (a)  (b)  (c)  (d) 

14.  $\xrightarrow[\text{2H}^+]{\text{CO}_2, \text{NaOH}}$ (A) $\text{CH}_3\text{--}\overset{\text{O}}{\parallel}\text{C--O--}\overset{\text{O}}{\parallel}\text{C--CH}_3$ (B); Products (A) and (B) respectively are:

- (a)  ; 
 (b)  ; 



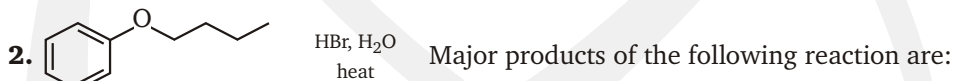
- (a) Phenol (b) Chlorobenzene (c) Fluorobenzene (d) Aniline



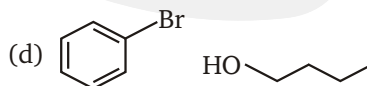
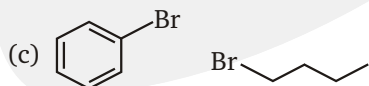
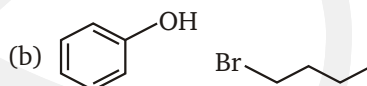
[C] Ethers

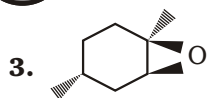


- (a) MeOH and Ph—I (b) MeI and Ph—CH₂—OH
(c) MeOH and Ph—CH₂—I (d) MeI and Ph—OH



- (a) recovered starting material (no reaction)

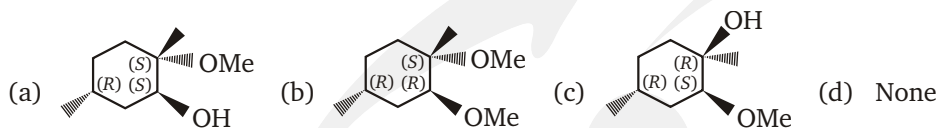




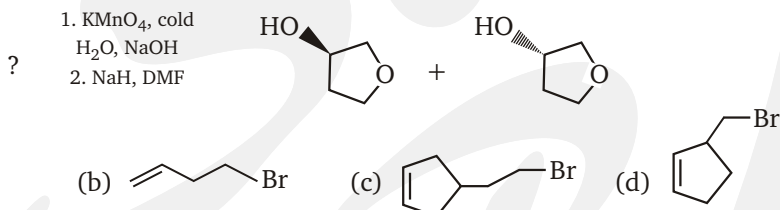
Optically active



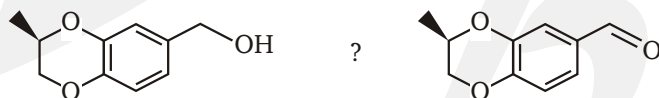
? Major product of the following reaction is:



4. What starting material is needed to synthesize the following compounds?

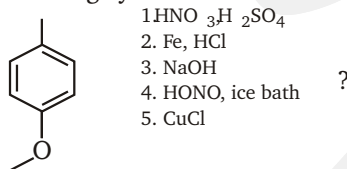


5. What reagents are needed to accomplish the following transformation?

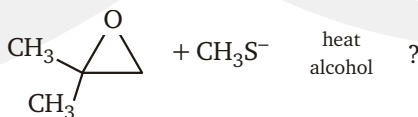


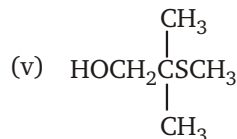
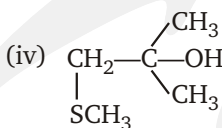
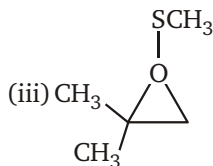
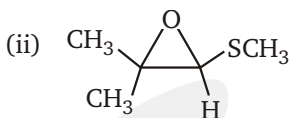
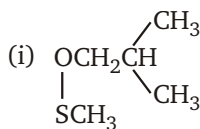
- (a) $\text{KMnO}_4, \text{NaOH}, \text{H}_2\text{O}, \text{heat}$
- (b) 1. $\text{LiAlH}_4, \text{Et}_2\text{O}$; 2. $\text{H}_2\text{O}, \text{H}_2\text{SO}_4$
- (c) $\text{CrO}_3, \text{HCl}, \text{pyridine}$
- (d) $\text{H}_2\text{CrO}_4, \text{acetone}, 35^\circ\text{C}$

6. What is the product of the following synthesis?



7. What is the major product of the following reaction ?





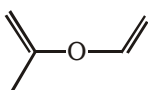
(a) i

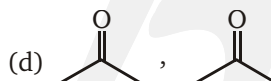
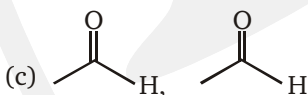
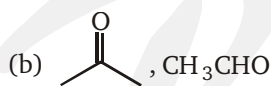
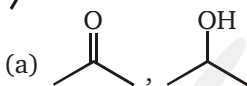
(b) ii

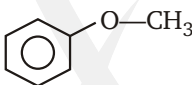
(c) iii

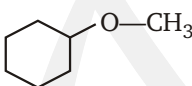
(d) iv

(e) v

8.  H_3O^+ ; Product of this reaction is:



9.  Conc. HI x mole of HI is consumed

 Conc. HI y mole of HI is consumed

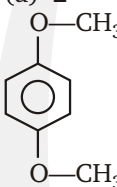
sum of (x y ?):

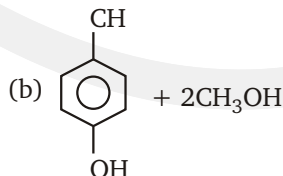
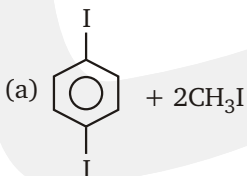
(a) 2

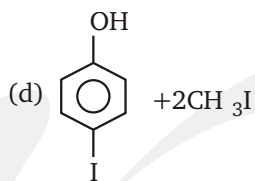
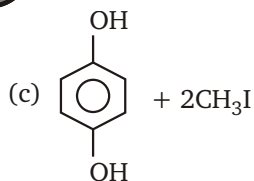
(b) 3

(c) 4

(d) 5

10.  Conc. HI (excess) Products ; Products of this reaction are:





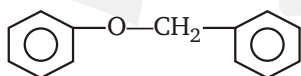
11. $\text{CH}_3 - \text{CH}_2 - \text{O} - \text{CH}_2 - \text{CH}_3$ $\xrightarrow[\text{(excess)}]{\text{Conc. HI}}$?; Major product of the reaction is:

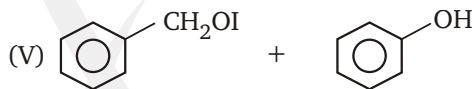
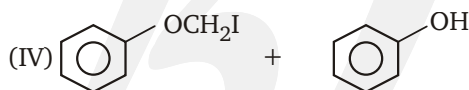
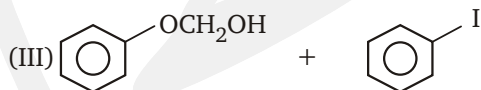
(a) 2EtOH

(b) 2EtI

(c) EtOH, EtI

(d) $2\text{H}_2\text{C} = \text{CH}_2$

12.  $\xrightarrow[\text{heat}]{\text{HI}}$?; Major products of this reaction are:



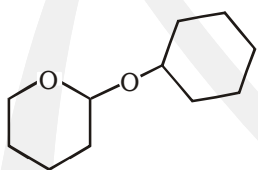
(a) I

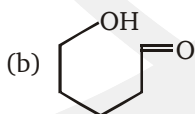
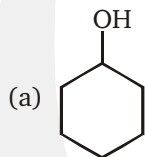
(b) II

(c) III

(d) IV

(e) V

13.  $\xrightarrow{\text{H}_3\text{O}^+}$ Product of the reaction is:



(c) Both (a) and (b) (d) No reaction

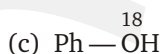
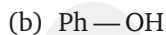
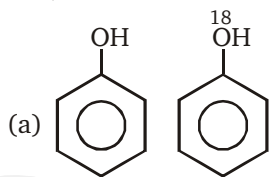
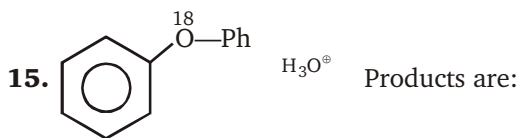
14. The reagent which will be suitable to distinguish 1-methoxy-3-methyl-2-butene from isomeric 4-methyl-3-penten-1-ol is:

(a) bromine in chloroform

(b) alkaline potassium permanganate

(c) ammoniacal silver nitrate

(d) sodium metal suspended in hexane



16. Compare the reactivities of ethers given in two pairs, towards hydrolysis :



(c)

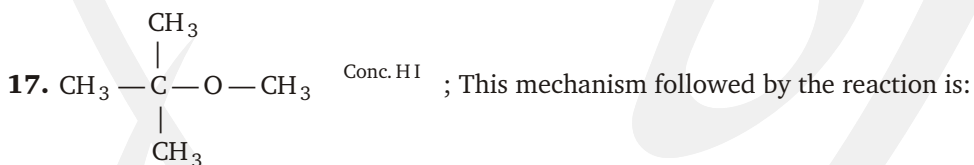
(d)

(a) 1 a, 2 c

(b) 1 b, 2 c

(c) 1 b, 2 d

(d) 1 a, 2 d



(a) $\text{S}_{\text{N}}1$

(b) $\text{S}_{\text{N}}2$

(c) SN AR

(d) NGP

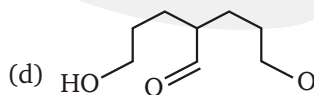
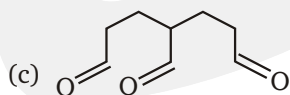
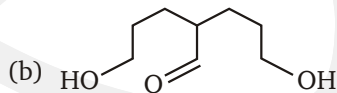
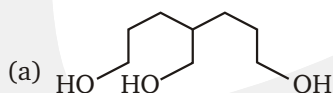


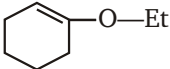
(a) $2\text{CH}_3\text{CHO}$

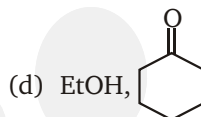
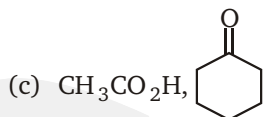
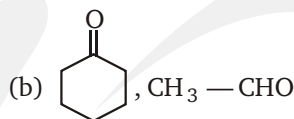
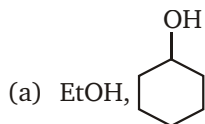
(b) 3HCHO

(c) $3\text{HCO}_2\text{H}$

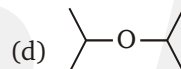
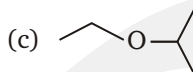
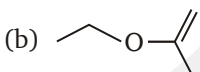
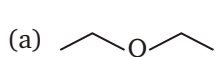
(d) $3\text{CH}_3\text{OH}$



20.  H_3O^+ ; Products obtained in this reaction are:



21. Which of the following is most reactive towards hydrolysis ?



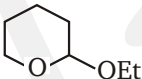
22. $\text{CH}_3 - \text{O} - \text{CH}_3$ $\xrightarrow[\text{H}_2\text{O}]{\text{HO}^-}$; Product of the reaction is:

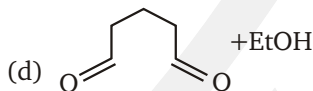
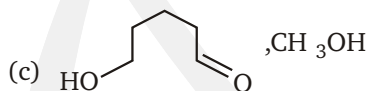
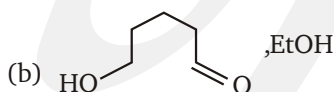
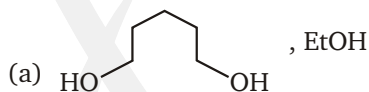
(a) No reaction

(b) $2\text{CH}_3\text{OH}$

(c) 2CH_4

(d) $\text{Et}-\text{H}$

23.  H_3O^+ ; Product of the reaction is:



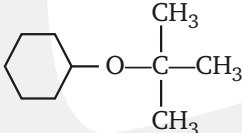
24. $\text{Ph}-\text{O}-\text{CH}_3$ $\xrightarrow{\text{Conc. HI}}$; Products of the reaction is:

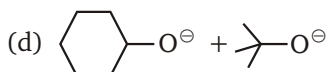
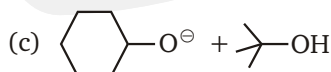
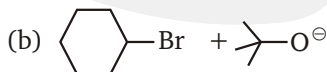
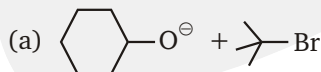
(a) $\text{Ph}-\text{OH}$, CH_3-I

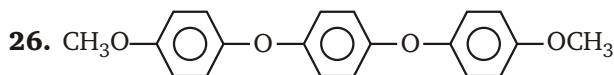
(b) $\text{Ph}-\text{I}$, CH_3OH

(c) $\text{Ph}-\text{I}$, CH_3-I

(d) $\text{Ph}-\text{OH}$, CH_3I

25.  , This ether is best prepared by:

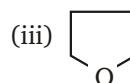
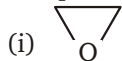




How many moles of HI consumed in above reaction ?

- (a) 1 (b) 2 (c) 3 (d) 4

27. Compare rate of reaction with OH :



- (a) $i > ii > iii$ (b) $i > iii > ii$ (c) $iii > ii > i$ (d) $ii > i > iii$

28. $\text{PhMgBr} + \text{Et}-\text{O}-\text{Et}$; Product is:

(diethyl ether)

- (a) $\text{Ph}-\text{OEt}$ (b) $\text{Ph}-\text{OH}$ (c) $\text{Ph}-\text{Et}$ (d) no reaction

Answers

[A] Alcohols

1.	(b)	2.	(d)	3.	(b)	4.	(a)	5.	(d)	6.	(d)	7.	(c)	8.	(b)	9.	(b)	10.	(b)
11.	(d)	12.	(d)	13.	(a)	14.	(c)	15.	(a)	16.	(d)	17.	(d)	18.	(d)	19.	(b)	20.	(b)
21.	(d)	22.	(b)	23.	(c)	24.	(b)	25.	(d)	26.	(b)	27.	(b)	28.	(b)	29.	(a)	30.	(d)
31.	(c)	32.	(b)	33.	(a)	34.	(b)	35.	(a)	36.	(c)	37.	(b)	38.	(c)	39.	(d)	40.	(b)
41.	(e)	42.	(a)	43.	(b)	44.	(b)	45.	(d)	46.	(b)	47.	(c)	48.	(c)	49.	(c)	50.	(b)
51.	(c)	52.	(b)	53.	(b)	54.	(b)	55.	(a)	56.	(b)	57.	(d)	58.	(a)	59.	(a)	60.	(d)
61.	(d)	62.	(d)	63.	(a)	64.	(c)	65.	(c)	66.	(b)	67.	(c)	68.	(b)	69.	(b)		

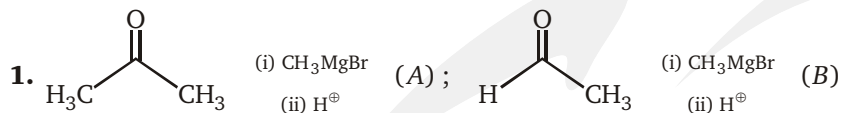
[B] Phenols

1.	(e)	2.	(b)	3.	(c)	4.	(e)	5.	(c)	6.	(b)	7.	(c)	8.	(a)	9.	(d)	10.	(c)
11.	(b)	12.	(b)	13.	(b)	14.	(a)	15.	(c)	16.	(b)								

[C] Ethers

1.	(c)	2.	(b)	3.	(a)	4.	(b)	5.	(d)	6.	(c)	7.	(d)	8.	(b)	9.	(b)	10.	(c)
11.	(b)	12.	(b)	13.	(c)	14.	(d)	15.	(a)	16.	(c)	17.	(a)	18.	(b)	19.	(b)	20.	(b)
21.	(b)	22.	(a)	23.	(b)	24.	(a)	25.	(b)	26.	(b)	27.	(a)	28.	(d)				

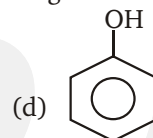
One or More than One Answers is/are Correct



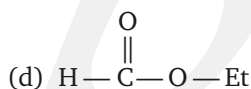
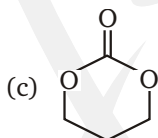
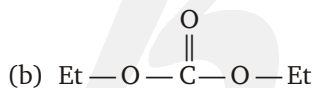
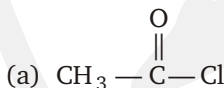
Which of the following reagent can be used to differentiate between A and B ?

- (a) FeCl_3 (b) Active metal Na (c) Lucas reagent (d) $\text{K}_2\text{Cr}_2\text{O}_7$

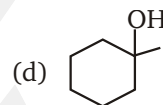
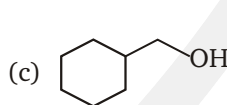
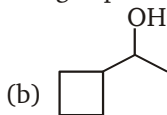
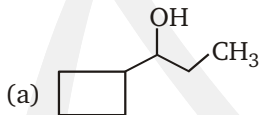
2. Which of the following compounds gives benzene when reacts with PhMgBr ?



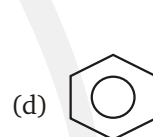
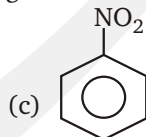
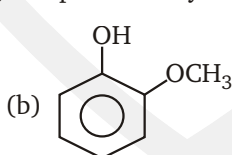
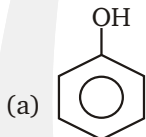
3. Which of the following compounds when reacts with PhMgBr (excess) followed by acidification will yield 3° alcohol ?



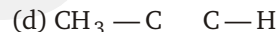
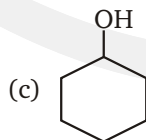
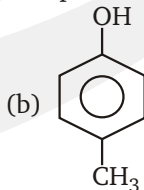
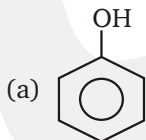
4. In which of the reaction ring expansion take place when protic acid is added in it ?



5. Which of the following compounds may undergo Reimer-Tiemann reaction ?



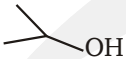
6. Which of the following compounds do/does not react with NaHCO_3 ?

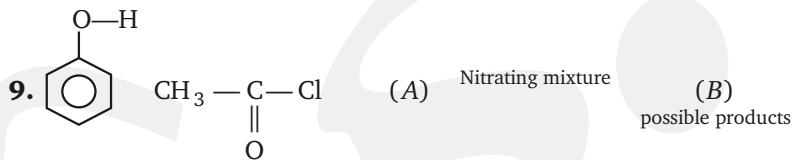


7. Which of the following compounds reacts with Na?

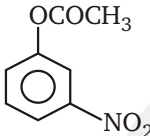
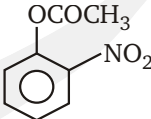
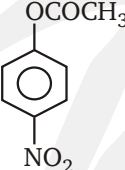
- (a) CH_3OH (b) $\text{CH}_3 - \text{C} \begin{array}{l} \diagup \text{CH} \\ \diagdown \end{array}$ (c) $\text{Ph} - \text{OH}$ (d) 

8. Which of the following is/are weaker acid than methanol?

- (a) $\text{CH}_3 - \text{C} \begin{array}{l} \diagup \text{CH} \\ \diagdown \end{array}$ (b)  (c)  (d) $\text{CH}_3 - \text{O}^+ \begin{array}{l} \diagup \text{H} \\ \diagdown \text{H} \end{array}$



Possible structure(s) of (B) is/are:

- (a)  (b)  (c)  (d) All of these

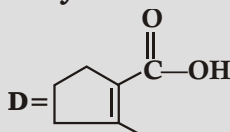
Answers

1.	(c, d)	2.	(a, c, d)	3.	(a, b, c)	4.	(a, b)	5.	(a, b)	6.	(a, b, c, d)
7.	(a, b, c)	8.	(a, b, c)	9.	(b, c)						

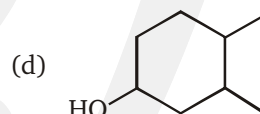
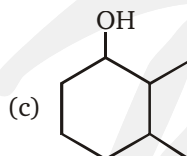
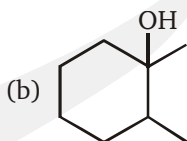
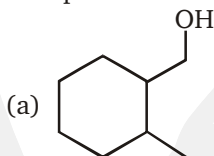
Comprehension Based Problems

Passage-1:

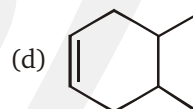
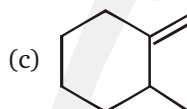
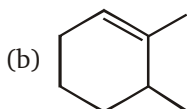
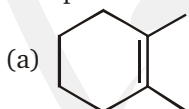
Compound (A) tertiary alcohol when it reacts with conc. H_2SO_4 gives alkene (B). B on O_3/Zn followed by reaction with dil. KOH and heating gives (C). (C) on reaction with NaOH followed by acidification gives D.



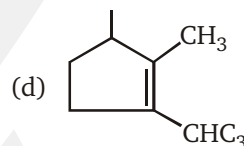
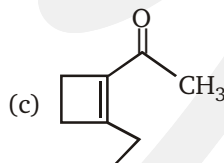
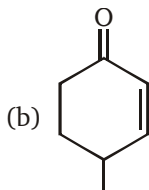
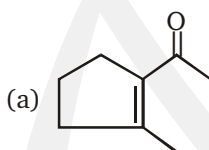
1. Compound A is:



2. Compound B is:

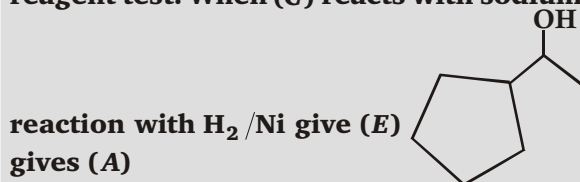


3. Compound C is:

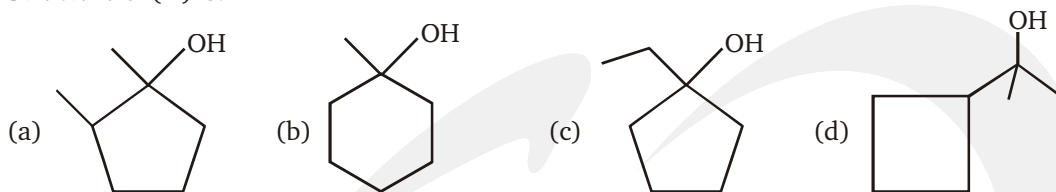


Passage-2:

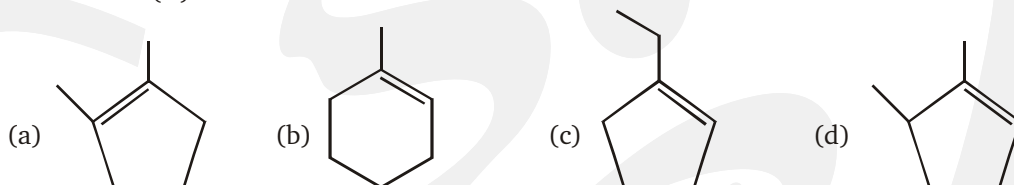
A tertiary alcohol (A) $\text{C}_7\text{H}_{14}\text{O}$ on reaction with conc. H_2SO_4 gives (B) C_7H_{12} . (B) on ozonolysis give (C) $\text{C}_7\text{H}_{12}\text{O}_2$. (C) gives both Haloform and Tollen's reagent test. When (C) reacts with sodium methoxide gives (D) $\text{C}_7\text{H}_{10}\text{O}$. (D) on



1. Structure of (A) is:



2. Structure of (B) is:

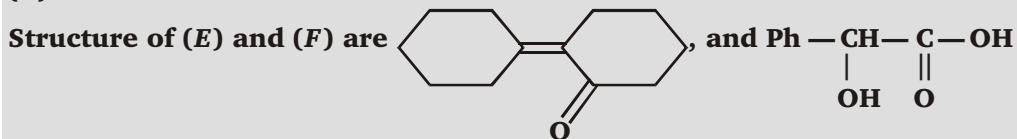


3. Which statement is true about D ?

- (a) Formed from Cannizaro reaction
- (b) Formed from Perkin reaction
- (c) Formed from cross Aldol reaction
- (d) Formed from intramolecular Aldol reaction

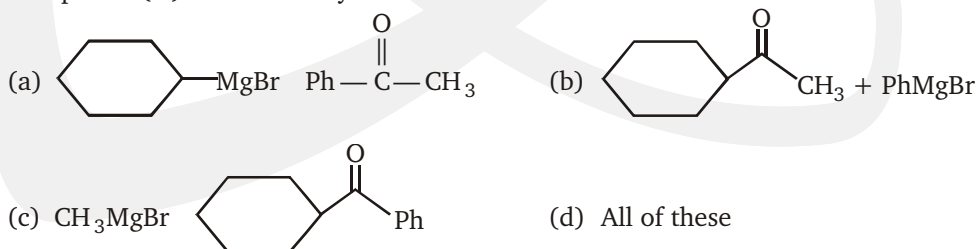
Passage-3:

A tertiary alcohol (A) on acid catalysed dehydration gives (B). Which on ozonolysis gives (C) and (D). C upon reaction with KOH gives (E). D upon reaction with SeO_2 followed by reaction with conc. KOH and acidification give (F).



respectively.

1. Compound (A) is obtained by:



(a)  (b)  (c)  (d) 

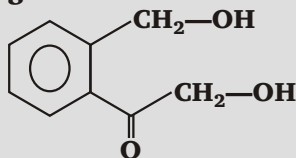
(a) , $\text{Ph}-\text{CHO}$

(b) , $\text{Ph}-\text{C}(=\text{O})-\text{CH}_3$

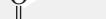
(c) , $\text{Ph}-\text{CH}_2-\text{CHO}$

(d) All of these

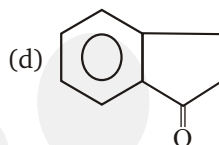
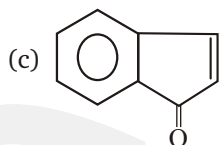
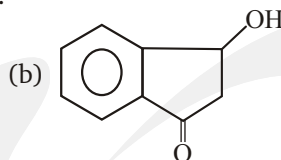
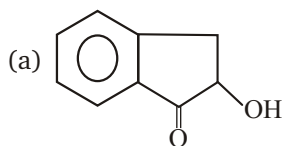
An alcohol P, which gives positive iodoform test, undergoes reaction with 2 moles of PCC (pyridinium chloro chromate) and gives compound Q. Compound Q on reaction with dil. KOH followed by heating gives compound R. Ozonolysis of R leads to formation of a tricarbonyl compound S, which on reaction with LiAlH_4 gives compound T. The structure of compound T is as given below.



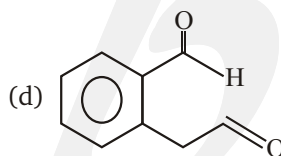
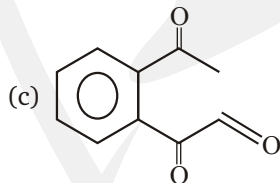
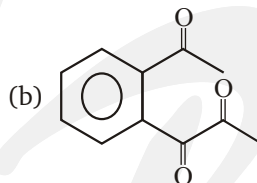
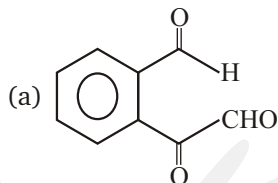
P 2 moles of PCC **Q** (i)OH⁻
 (ii)D **R** (i) O₃
 (ii) Zn,H₂O **S** LiAlH₄ **T**

(a)  (b)  (c)  (d) 

2. The structure of the carbonyl compound **R** is :

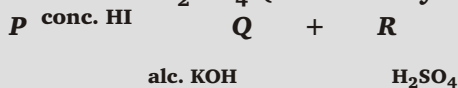


3. The structure of the carbonyl compound **S** is :

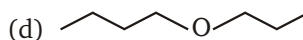
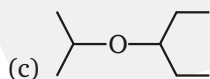
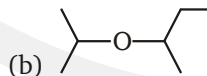
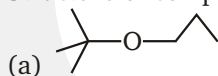


Passage-5:

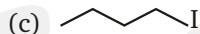
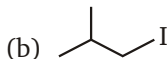
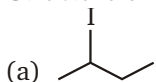
A organic compound **P** on reaction with concentrated **HI** gives products **Q** and **R**. Both compound **Q** and **R** gives alkene on reaction with respectively alcoholic **KOH** and concentrated **H₂SO₄** (followed by heating).



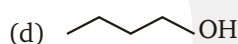
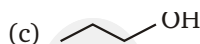
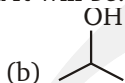
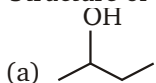
1. Structure of compound **P** will be:



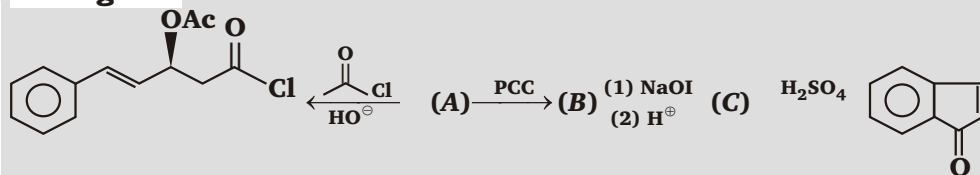
2. Structure of compound Q will be:



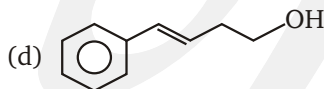
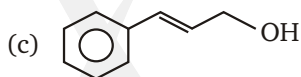
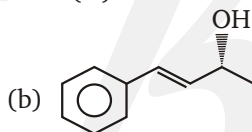
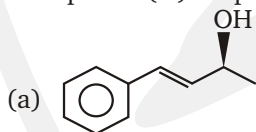
3. Structure of compound R will be:



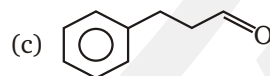
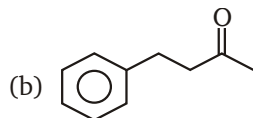
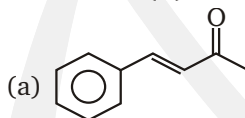
Passage-6:



1. Compound (A) is optically active so the structure of (A) is:

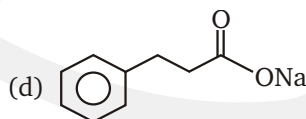
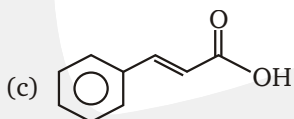
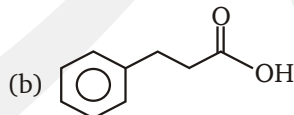
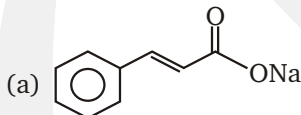


2. Structure of (B) is:



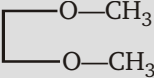
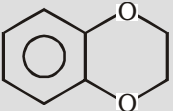
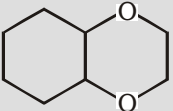
(d) (a) and (b) both

3. Structure of (C) is:

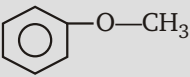
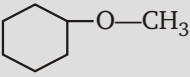
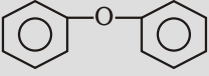
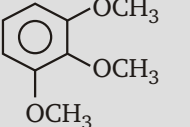


Matching Type Problems

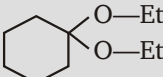
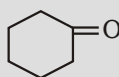
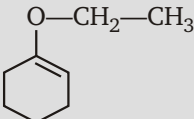
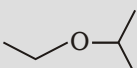
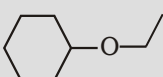
1. Match the column:

Column-I		Column-II (Moles of HI Consumed)	
(a)	 HI	(p)	2
(b)	 HI	(q)	3
(c)	 HI	(r)	5
(d)	Et — O — Et HI	(s)	6

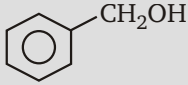
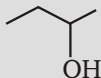
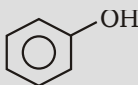
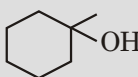
2. Match the column:

Column-I (Reaction)		Column-II (Moles of HI consumed)	
(a)	 Conc. HI	(p)	No reaction
(b)	 Conc. HI	(q)	1
(c)	 Conc. HI	(r)	2
(d)	 Conc. HI	(s)	3

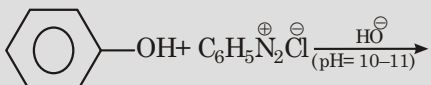
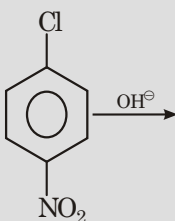
3. Match the column:

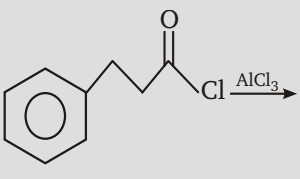
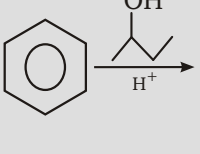
Column-I (Reaction)			Column-II (Product)	
(a)		H_3O^+	(p)	
(b)		H_3O^+	(q)	EtOH
(c)		H_3O^+	(r)	
(d)		H_3O^+	(s)	CH_3OH

4. Match the column:

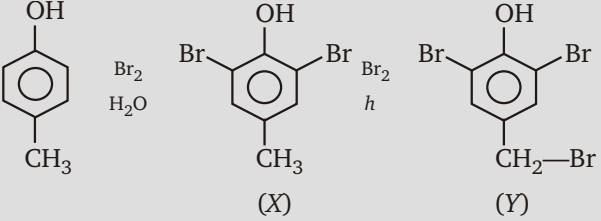
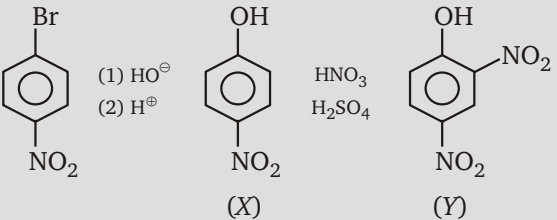
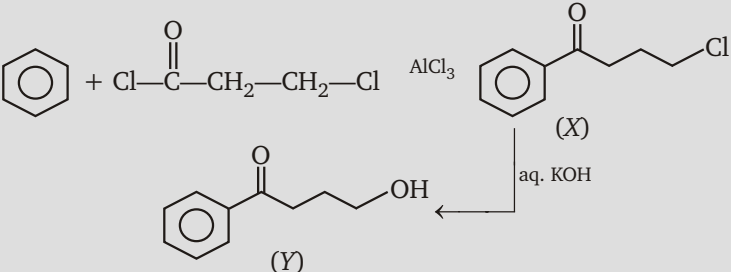
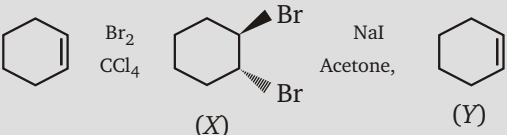
Column-I		Column-II	
(a)		(p)	Nucleophilic substitution reaction with HBr
(b)		(q)	Elimination reaction with H_2SO_4
(c)		(r)	Geometrical isomer will be formed when reaction with H_2SO_4
(d)		(s)	No reaction with H_2SO_4

5. Match the reactions in Column-I with appropriate types of steps/reactive intermediate involved in these reactions as given in Column-II:

Column-I		Column-II	
(a)		(p)	Nucleophilic aromatic substitution
(b)		(q)	Electrophilic aromatic substitution

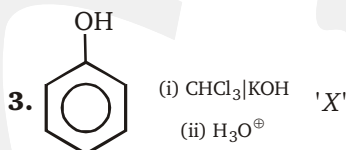
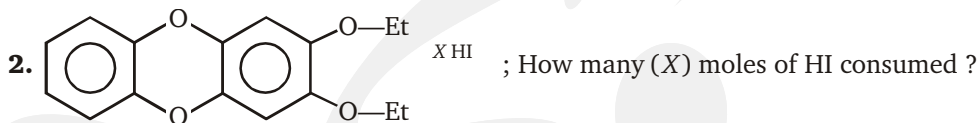
(c)		(r)	Carbocation intermediate
(d)		(s)	Carbanion intermediate
		(t)	Coupling reaction

6. Match the reactions with appropriate types of reactions and intermediates involved in formation of (X) and (Y).

	Column-I		Column-II
(a)		(p)	Nucleophilic substitution reaction
(b)		(q)	Electrophilic substitution reaction
(c)		(r)	Electrophilic addition reaction
(d)		(s)	Free-radical substitution
		(t)	Elimination reaction

Multi Concept Subjective Problems

1. How many isomers of $C_4H_{10}O$ are possible which can undergo oxidation reaction with $K_2Cr_2O_7$ (including stereoisomer) ?

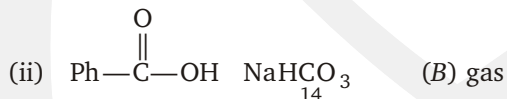
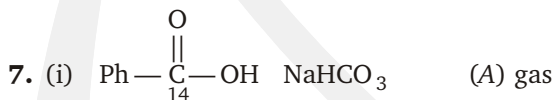


Identify the molecular weight of the product 'X' give your answer in this way if molecular weight is 121 means $1 + 2 + 1 = 4$ answer possible.

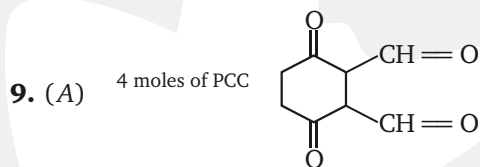
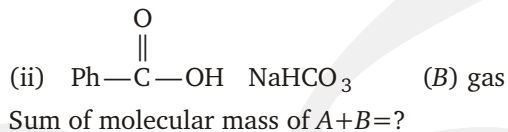
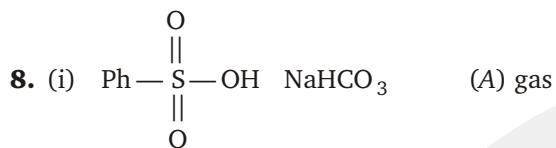
4. (i) $Ph-OH \xrightarrow{Na}$ (A) gas
(ii) $CH_3OH \xrightarrow{Na}$ (B) gas
Sum of molecular mass of gas $(A+B=?)$

5. (i) $Ph-OH \xrightarrow{K}$ (A) gas
(ii) $Ph-OH \xrightarrow{KH}$ (B) gas
Sum of molecular mass of $A+B=?$

6. (i) $Ph-OH \xrightarrow{Na}$ (A) gas
(ii) $Ph-OH \xrightarrow{NaH}$ (B) gas
(iii) $Ph-OH \xrightarrow{NaNH_2}$ (C) gas
Sum of molecular mass of $A+B+C=?$



Sum of molecular mass of $A+B=?$



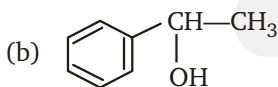
Maximum number of moles of Ac_2O consumed by reactant (A) is:

Unified Conceptual Problems

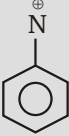
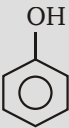
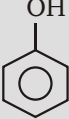
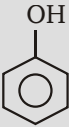
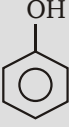
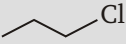
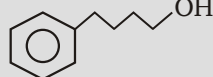
1. Which of the following alcohols, when heated with H_2SO_4 , will undergo dehydration more rapidly? Explain.



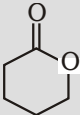
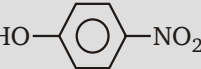
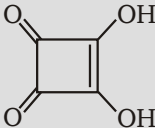
or



2. Identify product, Intermediate & Type of reaction (Substitution, Elimination, Addition)

S.No.	Reaction	Product	Intermediate	Type of reaction (S/E/A)
(i)	 M.A. M.A. mixed acid			
(ii)	  AlCl_3			
(iii)	 $\text{NCl}^\ominus$ H_2O (boil)			
(iv)	 CHCl_3 KOH			
(v)	 $(1) \text{CO}_2, \text{NaOH}$ $(2) \text{H}$			
(vi)	 $\text{Ph}-\text{N} \quad \text{N}$ $\text{HO}^\ominus$			
(vii)	 Br_2 H_2O			
(viii)	 +  AlCl_3			
(ix)	 H			

3. This question gives you idea about solubility of compound is Na, NaOH, NaHCO_3 .

S.No.	Compound	Evolve H_2 gas with Na metal	React with aq. NaOH	React with NaHCO_3
1.	$\text{Ph}-\text{CO}_2\text{H}$			
2.	$\text{Ph}-\text{OH}$			
3.	$\text{Ph}-\text{CH}_2-\text{OH}$			
4.	$\text{Ph}-\text{CH}_2-\text{CO}_2\text{H}$			
5.	$\text{Ph}-\text{CH}_2-\text{OCH}_3$			
6.	$\text{Ph}-\text{CH}_2-\text{CH}_2-\text{OH}$			
7.	$\text{Ph}-\text{SO}_3\text{H}$			
8.	$\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH}$			
9.	CH_3-CH_3			
10.	$\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{O}-\text{CH}_3$			
11.				
12.	$\text{CH}_3-\text{C} \quad \text{N}$			
13.	$\text{CH}_3-\text{C} \quad \text{CH}$			
14.				
15.				



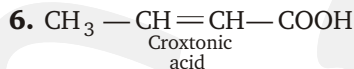
Cyclobutanone

1. LAH, ether
2. $\text{H}_3\text{O}^{\oplus}$



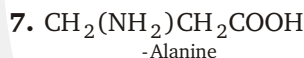
Palmitic acid

1. LAH, ether
2. $\text{H}_3\text{O}^{\oplus}$



Crotonic acid

1. LAH, ether
2. $\text{H}_3\text{O}^{\oplus}$



-Alanine

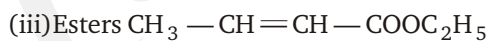
1. LAH, ether
2. $\text{H}_3\text{O}^{\oplus}$



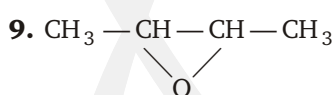
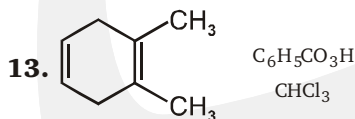
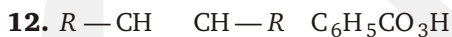
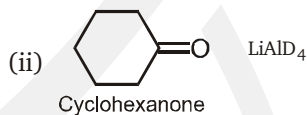
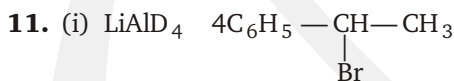
1. LAH, ether
2. $\text{H}_3\text{O}^{\oplus}$

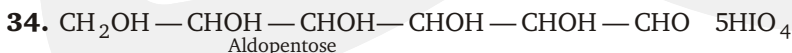
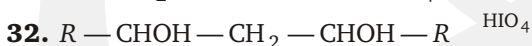
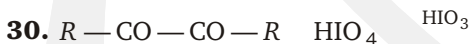
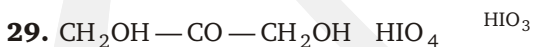
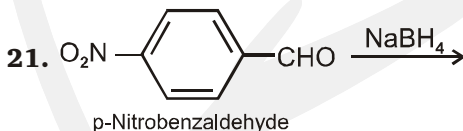
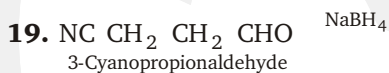
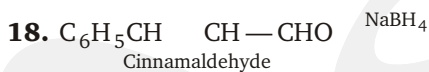
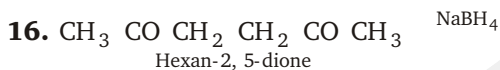
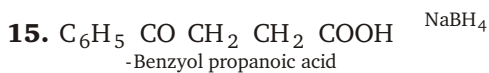


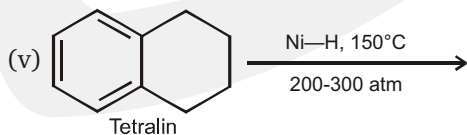
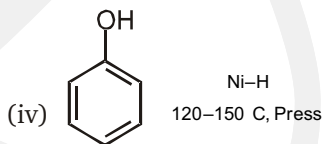
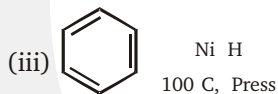
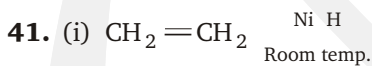
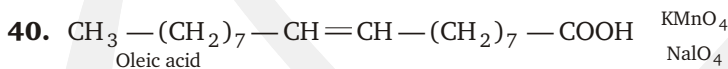
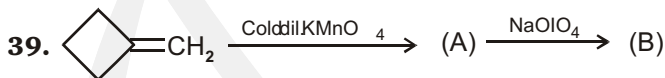
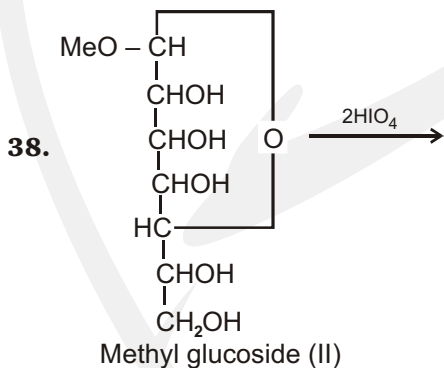
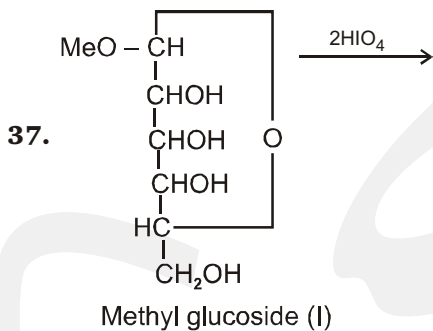
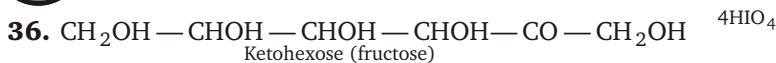
1. LAH, ether
2. $\text{H}_3\text{O}^{\oplus}$



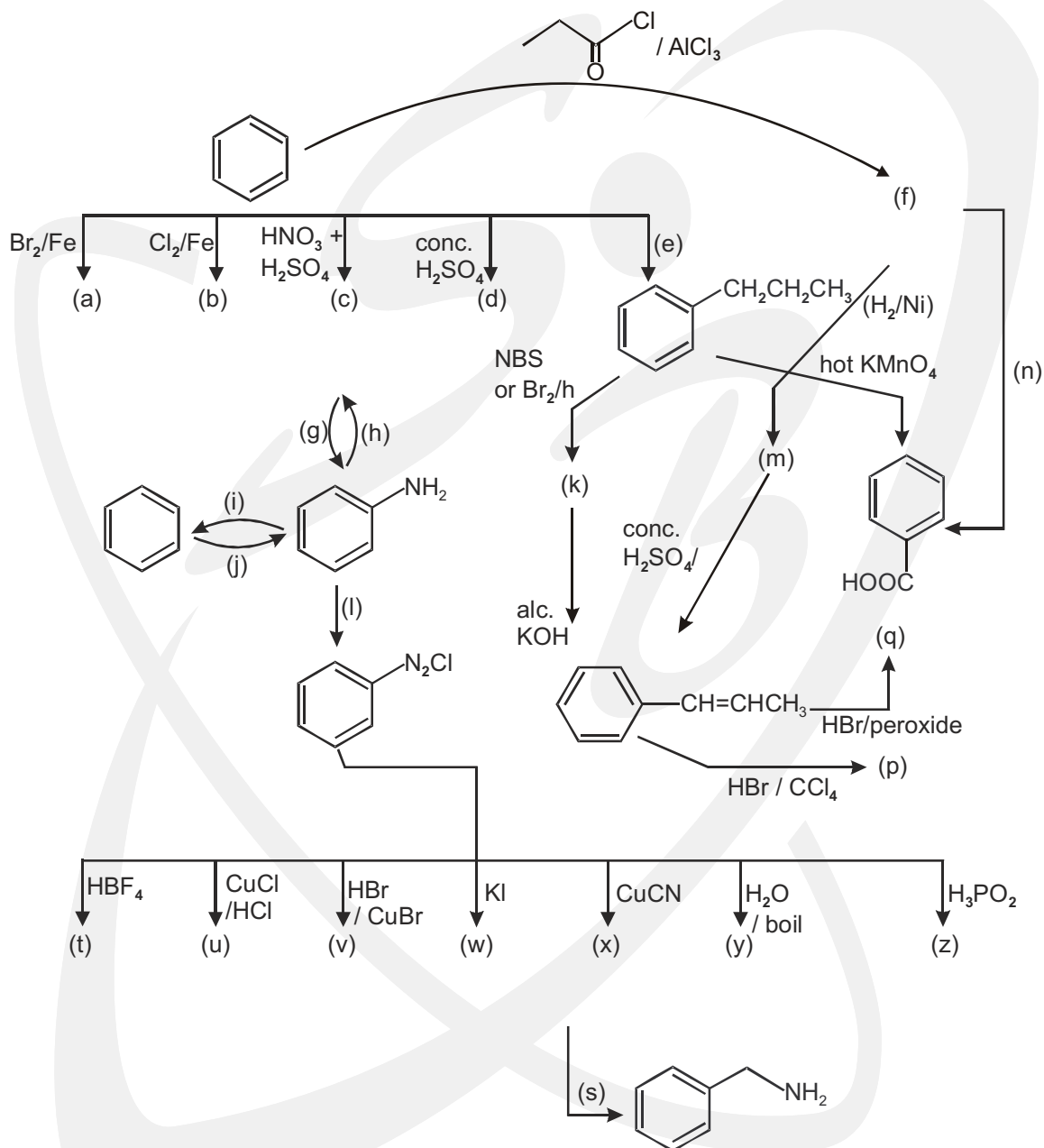
1. LAH, ether
2. $\text{H}_3\text{O}^{\oplus}$

LAH
Ether

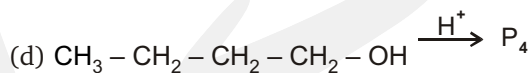
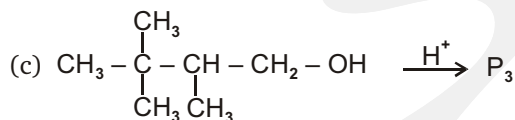
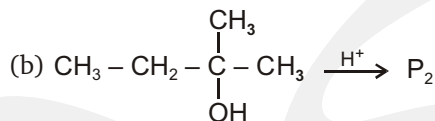
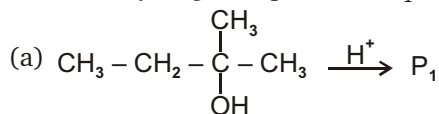




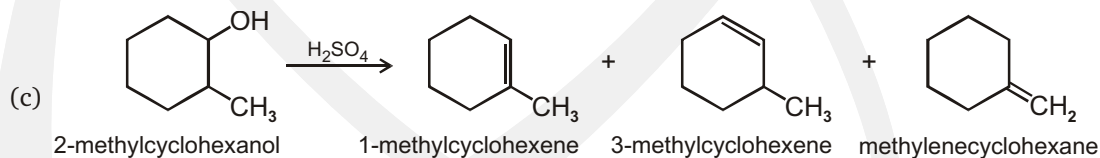
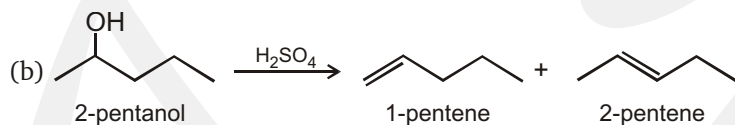
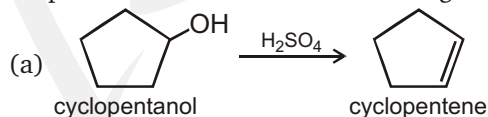
42. Provide necessary reagents in the following road-map :



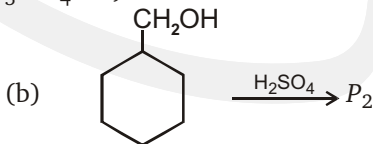
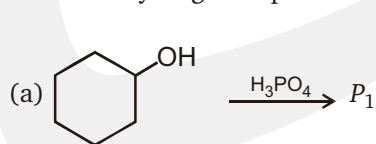
43. Sum of β -hydrogen in products (P_1 P_2 P_3 P_4 ?).

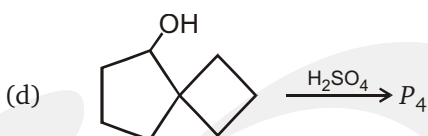
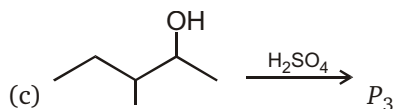


44. Propose mechanisms for the following reactions.

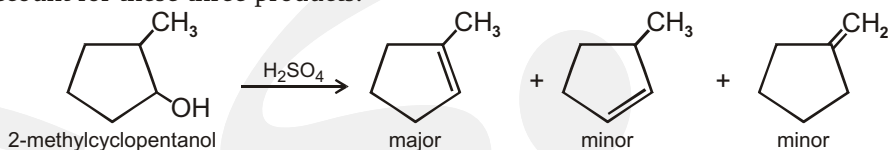


45. Sum of β -hydrogen in products (P_1 P_2 P_3 P_4 ?)

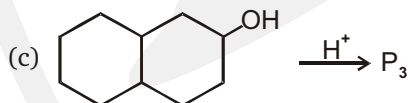
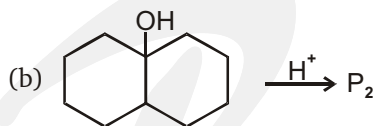
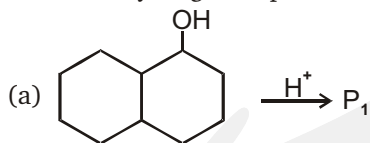




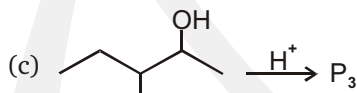
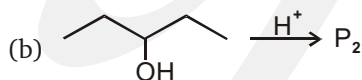
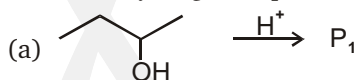
46. Dehydration of 2-methylcyclopentanol gives a mixture of three alkenes. Propose mechanisms to account for these three products.



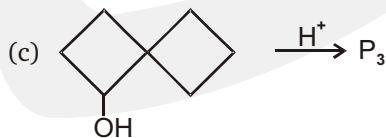
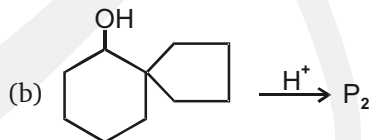
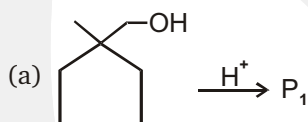
47. Sum of β -hydrogen in products (P_1 P_2 P_3 ?)



48. Sum of β -hydrogen in products (P_1 P_2 P_3 ?)



49. Sum of β -hydrogen in products (P_1 P_2 P_3 ?)



Answers

Comprehension Based Problems:

Passage-1:	1.	(b)	2.	(a)	3.	(a)				
Passage-2:	1.	(b)	2.	(b)	3.	(c)				
Passage-3:	1.	(d)	2.	(b)	3.	(b)				
Passage-4:	1.	(b)	2.	(c)	3.	(a)				
Passage-5:	1.	(a)	2.	(d)	3.	(c)				
Passage-6:	1.	(a)	2.	(a)	3.	(c)				

Matching Type Problems:

- (a) r; (b) q; (c) s; (d) p
- (a) q; (b) r; (c) p; (d) s
- (a) p, q; (b) p, q; (c) q, r; (d) q
- (a) p; (b) p, q, r; (c) s; (d) p, q
- (a) q, s, t; (b) p, s; (c) q, r; (d) q, r
- (a) q, s; (b) p, q; (c) p, q; (d) r, t

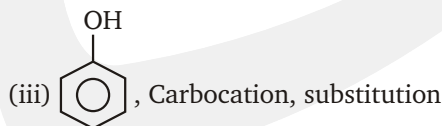
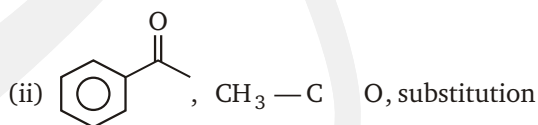
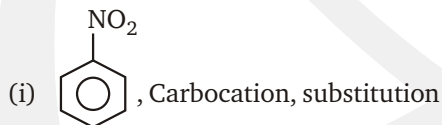
Multi Concept Subjective Problems:

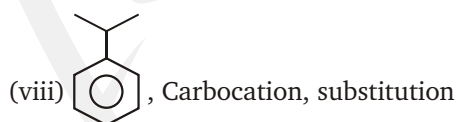
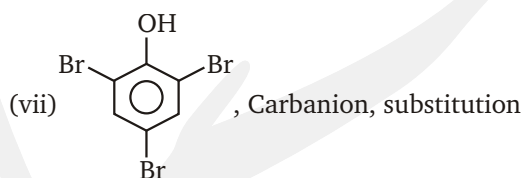
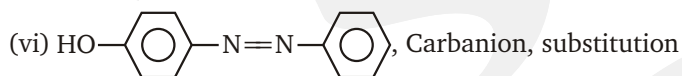
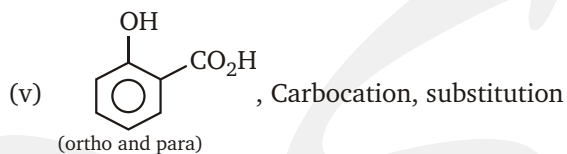
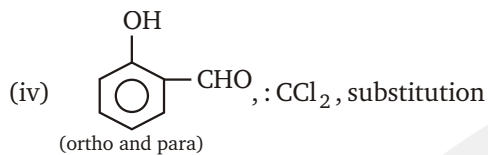
1.	5	2.	2	3.	5	4.	4	5.	4	6.	21	7.	90	8.	88	9.	4		
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Unified Conceptual Problems:

- (b) will undergo dehydration more rapidly because it produces a secondary benzylic carbocation, which is more stable than secondary carbocation due to resonance.

2.

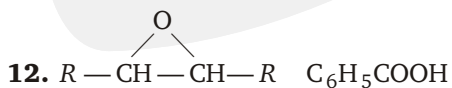
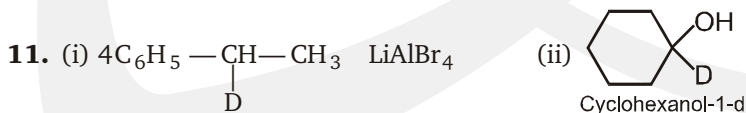
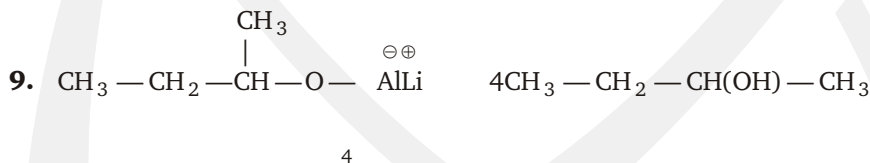
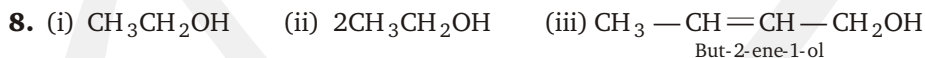
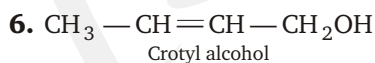
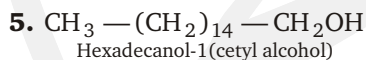
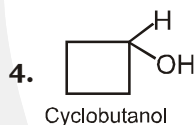


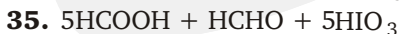
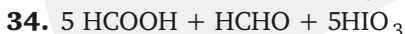
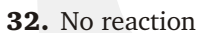
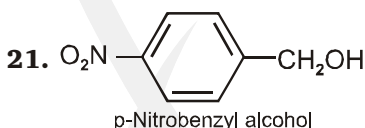
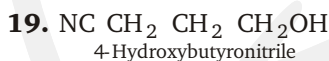
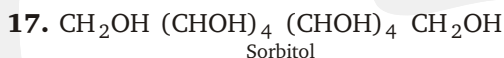
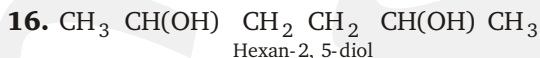
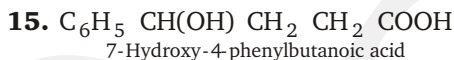
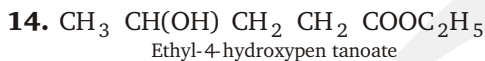
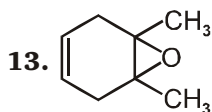


3.

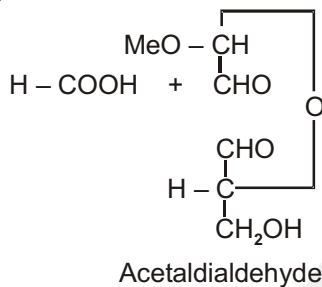
	Evolve H_2 gas with Na metal	React with aq. NaOH	React with NaHCO_3
1.	✓	✓	✓
2.	✓	✓	✗
3.	✓	✗	✗
4.	✓	✓	✓
5.	✗	✗	✗
6.	✓	✗	✗
7.	✓	✓	✓
8.	✓	✓	✓

9.	X	X	X
10.	X	✓	X
11.	X	✓	✓
12.	X	✓	X
13.	✓	X	X
14.	✓	✓	✓
15.	✓	✓	✓

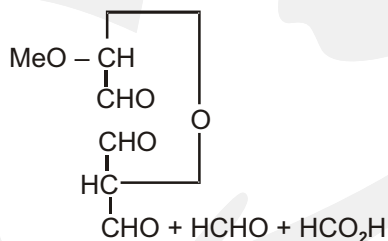




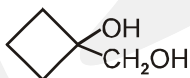
37.



38.



39. (A)



(B)

40. $\text{CH}_3-(\text{CH}_2)_7-\text{CHO}$
n-Nonaldehyde $\text{OHC}-(\text{CH}_2)_7-\text{COOH}$
Half aldehyde of azelaic acid41. (i) CH_3-CH_3

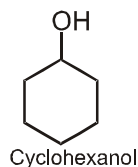
(ii)

 CH_3-CH_3

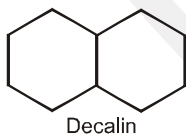
(iii)



(iv)

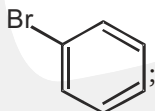


(v)

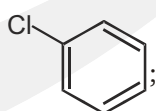


42.

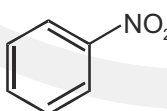
a



b

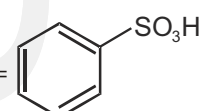


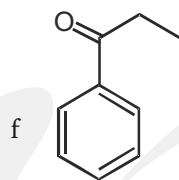
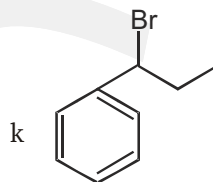
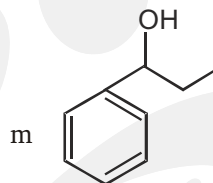
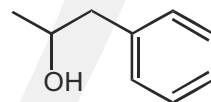
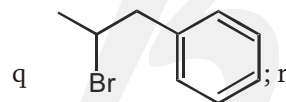
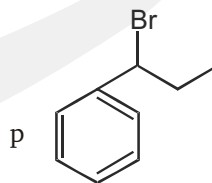
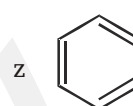
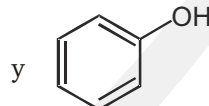
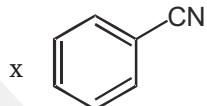
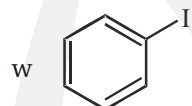
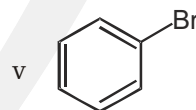
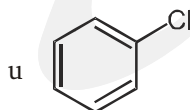
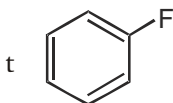
c



;

d



e $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}/\text{FeCl}_3$ g Sn/HCl h $\text{CF}_3\text{CO}_3\text{H}$;i $\text{NaNO}_2/\text{HCl}/\text{H}_3\text{PO}_2$ j (1) $\text{Cl}_2/\text{AlCl}_3$
(2) NaNH_2 l NaNO_2/HCl ;n hot KMnO_4 o hot KMnO_4 ;s LAH or H_2/Ni ;

43. 9 9 11 6 35

45. 4 7 8 8 27

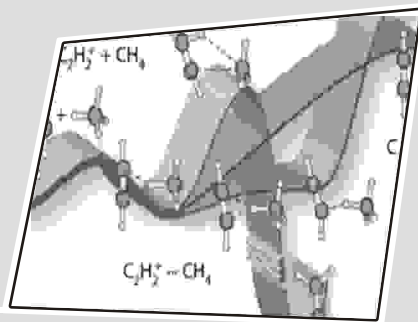
47. 8 8 4 20

48. 6 5 8 19

49. 7 8 8 23

5

Aldehydes, Ketones and Carboxylic Acids

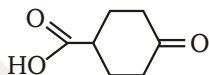


Only One Answer is Correct

[A] Aldehydes and Ketones

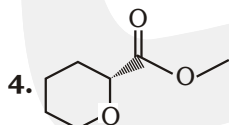
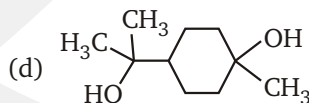
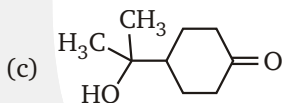
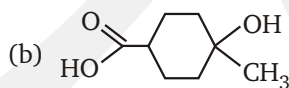
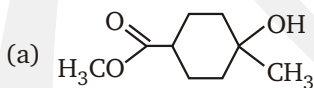
Methods of Preparation

- Predict which of the following compounds will have a lower boiling point than 1-butanal.
(a) 1-butanol (b) 2-butanol (c) 1-butene (d) butanoic acid
- Predict the product of the following reaction.
PhCH₂OH PCC (pyridinium chlorochromate/methylene chloride)
(a) benzophenone (b) benzoic acid (c) benzaldehyde (d) benzyl chloride
- Give the expected product of the following reaction.



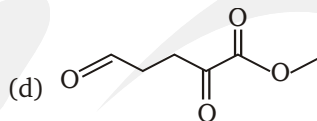
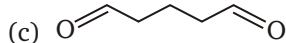
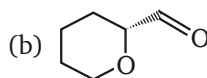
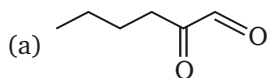
1. Excess MeMgBr
2. H₂O neutralize

?



1. DIBAL-H, hexane, -78°C
2. H₂O

? Major product of the reaction is:

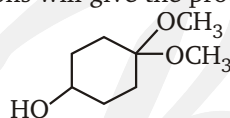
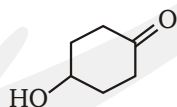


5. Di-isobutylaluminium hydride (DIBAL-H) can be used to carry out which of the following conversions ?

- (a) ester to aldehyde (b) carboxylic acid to ester
(c) 2° alcohol to ketone (d) aldehyde to carboxylic acid (e) ester to ketone

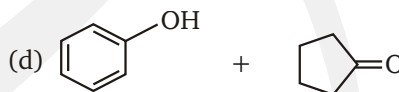
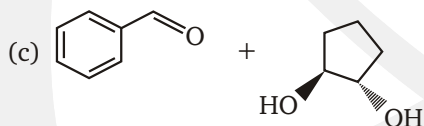
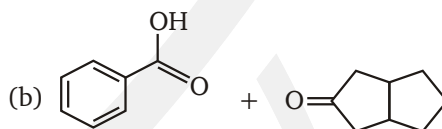
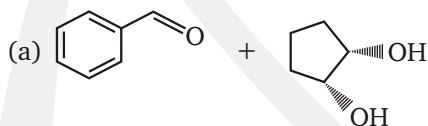
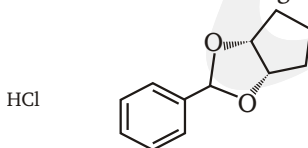
Properties of Carbonyl Compounds

1. Which of the following sets of reaction conditions will give the product indicated?

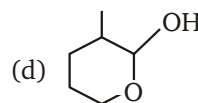
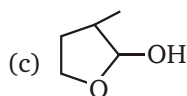
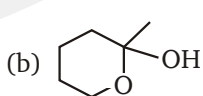
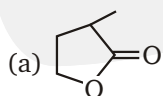


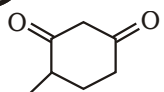
- (a) $(\text{CH}_3\text{O})_2\text{Ag}$, heat (b) CH_3OH , $\text{HO}^\ominus$, separate water
(c) (1) CH_3ONa , (2) MeI (d) CH_3OH , $\text{H}^\oplus$, separate water

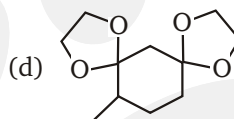
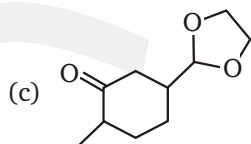
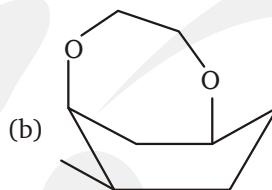
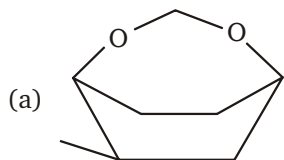
2. What are the starting materials needed for the following reaction?

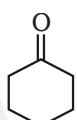


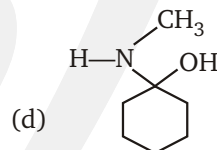
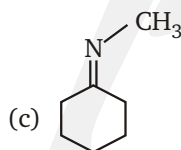
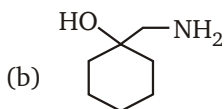
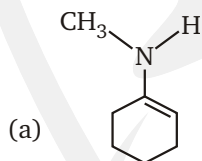
3. HCl ? ; Product of the reaction is:



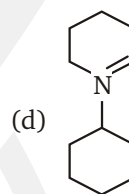
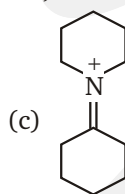
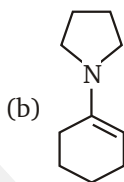
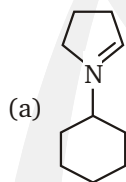
4.  $\text{HOCH}_2\text{CH}_2\text{OH}, \text{H}^+$? ; Product of the reaction is:

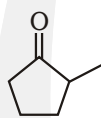


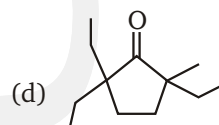
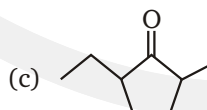
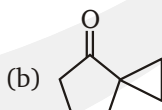
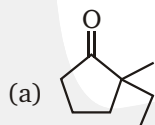
5.  CH_3NH_2
pH 4.5 ? ; Product of the reaction is:



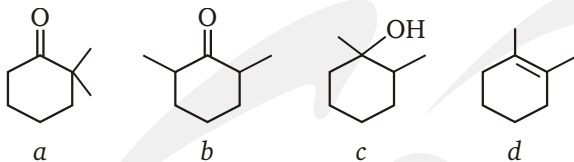
6. Cyclohexanone + Pyrrolidine + Tosic acid ? ; Product of the reaction is:



7.  1. LDA
2. $\text{CH}_3\text{CH}_2\text{I}$? ; Product of the reaction is:

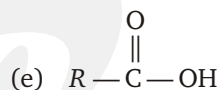
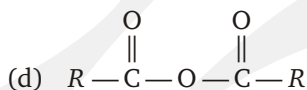
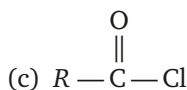
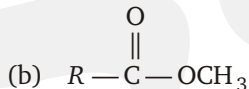
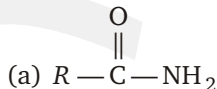


8. Which compound is formed when the enolate of 2-methylcyclohexanone, formed from lithium diisopropylamide (LDA), reacts with methyl iodide?

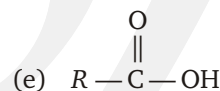
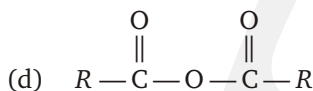
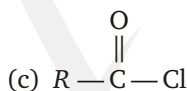
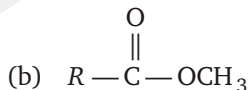
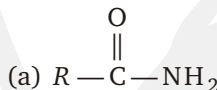


- (a) a (b) b (c) c (d) d

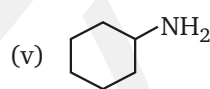
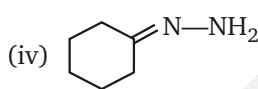
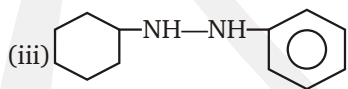
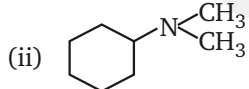
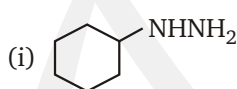
9. Which of the following compounds is most reactive toward nucleophilic acyl substitution ?



10. Which of the following compounds is the least reactive toward nucleophilic acyl substitution ?

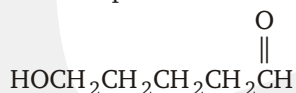


11. Which of the following compounds is a hydrazone ?

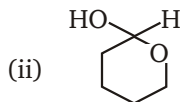
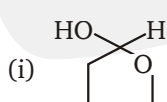


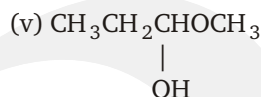
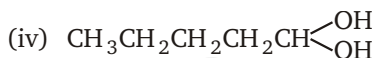
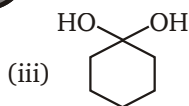
- (a) i (b) ii (c) iii (d) iv (e) v

12. The compound :



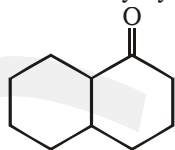
can form a hemiacetal by reacting with itself in solution. What is the structure of this hemiacetal?



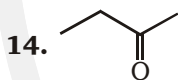


- (a) i (b) ii (c) iii (d) iv (e) v

13. How many hydrogens are replaced by deuterium in the following reaction?



- (a) 1 (b) 2 (c) 3 (d) 4



- (i) $x\text{PCl}_5$
 (ii) y -moles NaNH_2

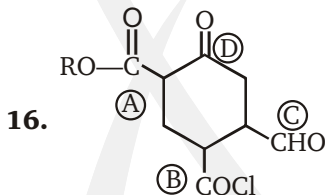


Sum of $(x \ y \ ?)$.

- (a) 2 (b) 3 (c) 4 (d) 5

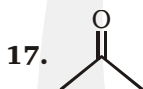
15. PhMgBr act as a nucleophile with:

- (a) CH_3SH (b) $\text{CH}_3 - \text{C} \quad \text{CH}$ (c) $\text{CH}_3 - \overset{\text{O}}{\parallel} \text{C} - \text{OH}$ (d) $\text{H} - \overset{\text{O}}{\parallel} \text{C} - \text{H}$



Find correct order of the reactivity of carbonyl groups with LiAlH_4 is:

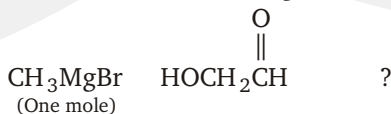
- (a) $\text{A} > \text{B} > \text{C} > \text{D}$ (b) $\text{B} > \text{C} > \text{D} > \text{A}$ ☐
 (c) $\text{D} > \text{C} > \text{B} > \text{A}$ ☐ (d) $\text{B} > \text{D} > \text{C} > \text{A}$ ☐

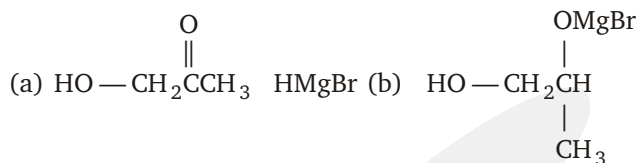


; Which type of mechanism is followed this reaction?

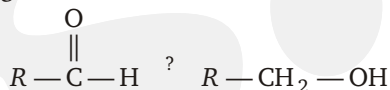
- (a) Electrophilic addition (b) Nucleophilic addition
 (c) Nucleophilic substitution (d) Electrophilic substitution

18. What is/are the major product(s) of the following reaction ?





19. Which of the following reagents does not form an alcohol from an aldehyde as shown below ?

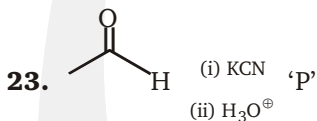
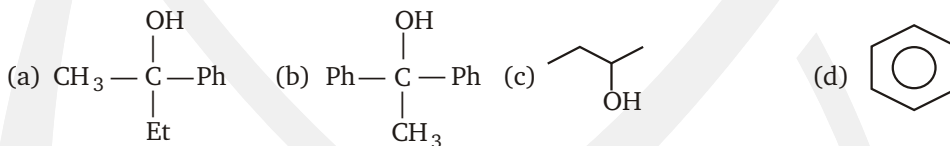
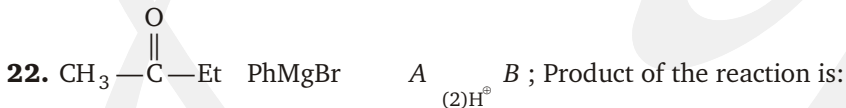
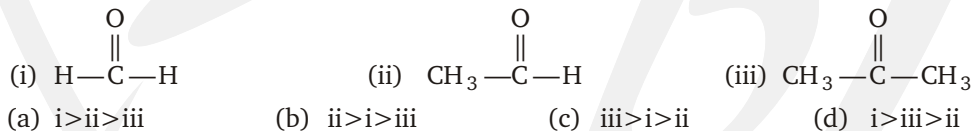


- (a) H_2/Pt (b) $1. \text{NaBH}_4/2. \text{H}_3\text{O}^+$
(c) $1. \text{LiAlH}_4/2. \text{H}_3\text{O}^+$ (d) $1. \text{Ag}_2\text{O}, \text{NH}_3/2. \text{H}_3\text{O}^+$ (e) $\text{H}_2/\text{Raney Ni}$

20. Which of the following compound undergo nucleophilic addition with RMgBr ?



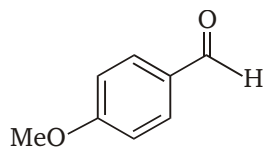
21. Compare rate of reaction of following with RMgBr :



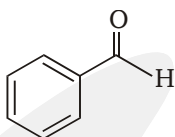
Find out the incorrect statement about the above given reaction:

- (a) It is a nucleophilic addition reaction
(b) Product obtained is present in racemic mixture
(c) If the hydrolysis of obtained product give α -hydroxy acid
(d) It is an electrophilic addition reaction

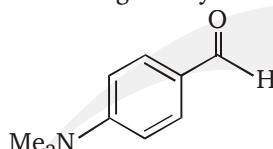
24. The K_{eq} values for the HCN addition reaction with the following aldehydes are in the order:



(i)



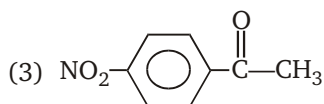
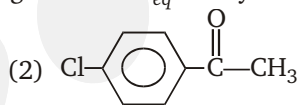
(ii)



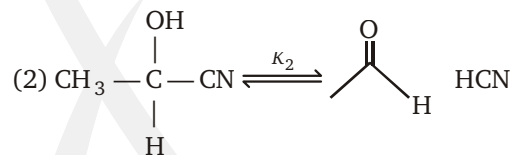
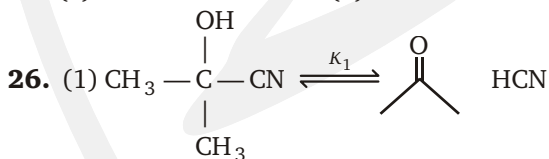
(iii)

- (a) $i > ii > iii$ (b) $ii > iii > i$ (c) $iii > i > ii$ (d) $ii > i > iii$

25. Arrange the following compounds in decreasing order of K_{eq} for hydrate formation:



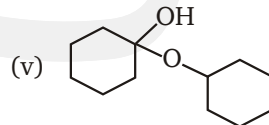
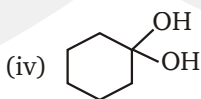
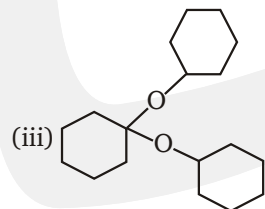
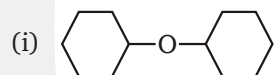
- (a) $3 > 2 > 1 > 4$ (b) $2 > 1 > 3 > 4$ (c) $3 > 1 > 2 > 4$ (d) $4 > 2 > 1 > 3$



Relation between K_1 and K_2 is:

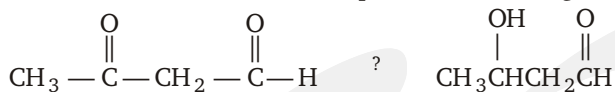
- (a) $K_1 > K_2$ (b) $K_1 < K_2$ (c) $K_2 > K_1$ (d) $K_1 < K_2 < 1$

27. ? Major product of the reaction is:



- (a) i (b) ii (c) iii (d) iv (e) v

28. What sequence of reactions would best accomplish the following conversion ?

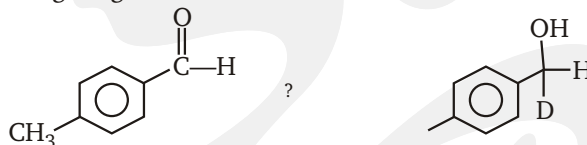


- (a) $\text{NaBH}_4/\text{H}_3\text{O}$ (b) $\text{HOCH}_2\text{CH}_2\text{OH}/\text{H}$; NaBH_4 ; $\text{H}/\text{H}_2\text{O}$
 (c) $\text{KMnO}_4/\text{HO}^-$; heat (d) $\text{LiAlH}_4/\text{H}_3\text{O}$ (e) H_2/Pt ; PCC

29. Which of the following compounds will give a positive iodoform test ?

- (a) propanal (b) 2-pentanone
 (c) 3-pentanone (d) benzophenone (e) cyclohexanone

30. Which of the following reagents is best used for the conversion shown below ?



- (a) 1. $\text{NaBH}_4/2.$ D_3O (b) 1. $\text{NaBD}_4/2.$ H_3O
 (c) 1. $\text{LiAlH}_4/2.$ D_3O (d) 1. $\text{H}_2/\text{Pt}/2.$ D_2O (e) 1. $\text{LiAlH}_4/2.$ H_3O

31. $\text{Ph} - \text{C}(\text{CH}_3) = \text{N}(\text{OH}) \xrightarrow{\text{H}_2\text{SO}_4} \text{Product (A)}$; Product (A) is:

- (a) $\text{Ph} - \overset{\overset{\text{O}}{\parallel}}{\text{C}} - \text{NH} - \text{CH}_3$ (b) $\text{CH}_3 - \overset{\overset{\text{O}}{\parallel}}{\text{C}} - \text{NH} - \text{Ph}$
 (c) $\text{CH}_3 - \overset{\overset{\text{O}}{\parallel}}{\text{C}} - \text{NH}_2$ (d) $\text{CH}_3 - \overset{\overset{\text{O}}{\parallel}}{\text{C}} - \text{NH} - \text{CH}_3$

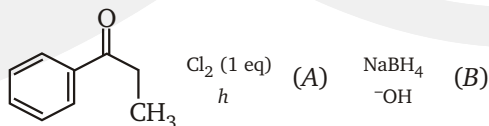
32. Benzaldehyde is more reactive than..... towards nucleophilic attack.

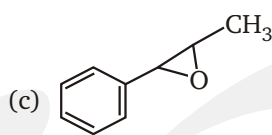
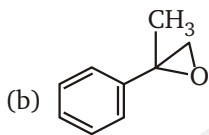
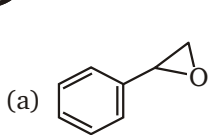
- (a) $\text{H} - \overset{\overset{\text{O}}{\parallel}}{\text{C}} - \text{H}$ (b) $\text{CH}_3 - \overset{\overset{\text{O}}{\parallel}}{\text{C}} - \text{H}$ (c) $\text{Ph} - \overset{\overset{\text{O}}{\parallel}}{\text{C}} - \text{Ph}$ (d) $\text{NO}_2 - \text{C}_6\text{H}_4 - \text{CHO}$

33. $\text{Cyclohexyl-CHO} \xrightarrow{\text{H}^+} \text{Product is :}$

- (a) cyclic acetal (b) cyclic hemiacetal
 (c) open chain acetal (d) open chain hemiacetal

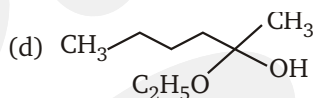
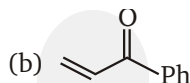
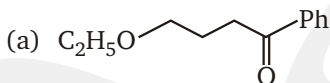
34. The end product of the reaction is:





(d) None of these

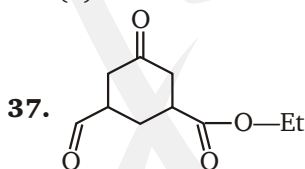
35. Alcoholic KOH (C), C is:



36. Rank the acid derivatives with respect to their reactivity with water.

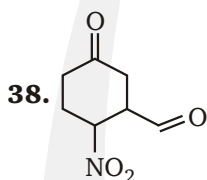
Hint: Most reactive > least reactive

- (a) acid halide > ester > acid anhydride > amide
 (b) acid anhydride > amide > acid halide > ester
 (c) amide > ester > acid anhydride > acid halide
 (d) acid halide > acid anhydride > ester > amide



Number of functional group reduced by NaBH_4 and LiAlH_4 respectively are:

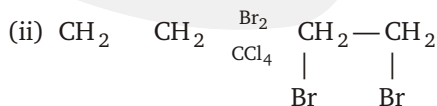
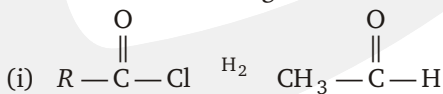
- (a) 2, 2 (b) 2, 3 (c) 3, 3 (d) 1, 3

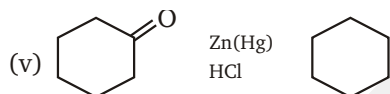
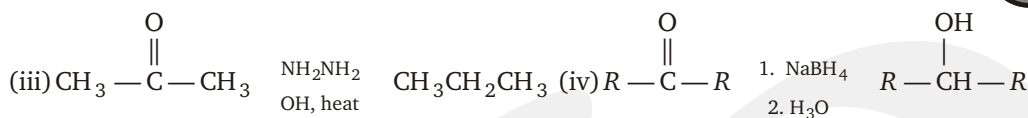


Number of functional groups that are reduced by LiAlH_4 and NaBH_4 is:

- (a) 3, 0 respectively (b) 3, 1 respectively (c) 3, 2 respectively (d) 2, 2 respectively

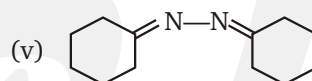
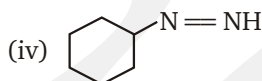
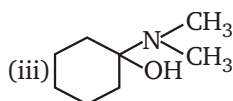
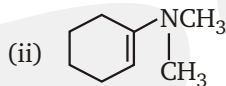
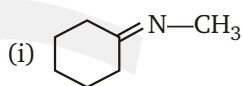
39. Which of the following is not a reduction reaction ?





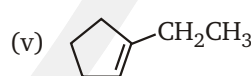
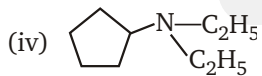
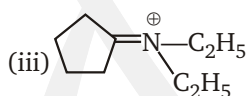
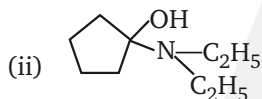
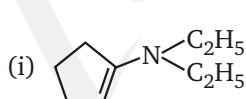
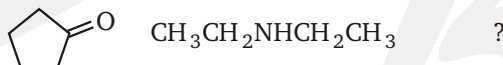
- (a) i (b) ii (c) iii (d) iv (e) v

40. Which of the following is an enamine ?



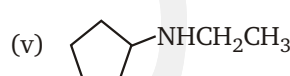
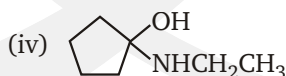
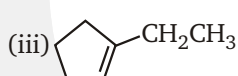
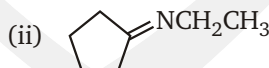
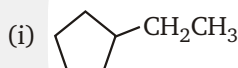
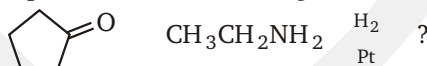
- (a) i (b) ii (c) iii (d) iv (e) v

41. What is the major organic product of the following reaction ?



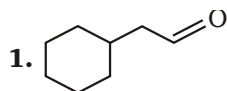
- (a) i (b) ii (c) iii (d) iv (e) v

42. What is the major organic product of the following reaction ?



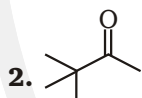
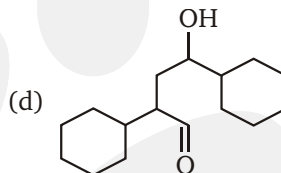
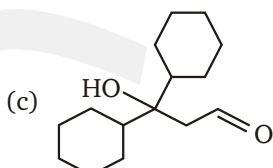
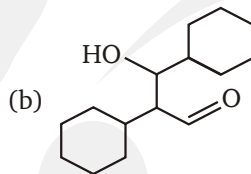
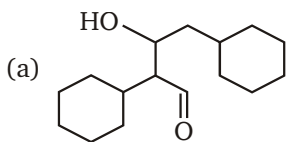
- (a) i (b) ii (c) iii (d) iv (e) v

Aldol and Cannizzaro Reactions



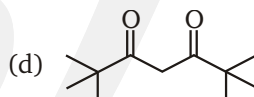
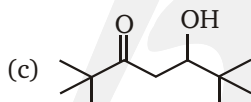
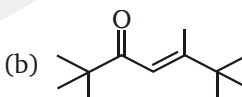
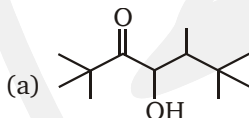
10%NaOH, $2O$
5°C

? Product of the reaction is :

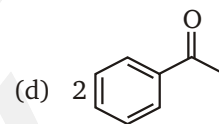
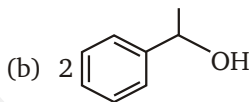
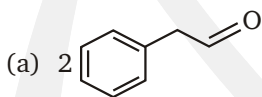
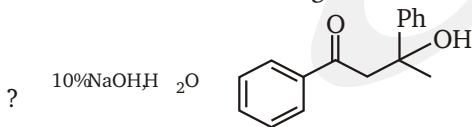


HCl, H_2O

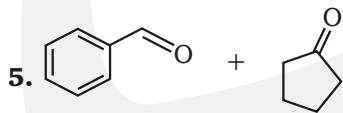
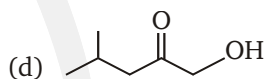
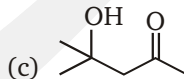
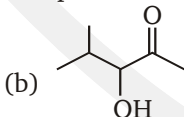
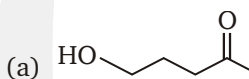
? Product of the reaction is :



3. What is the starting material used in the following reaction?

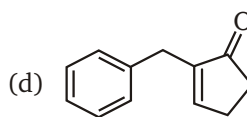
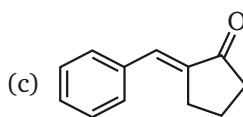
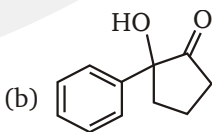
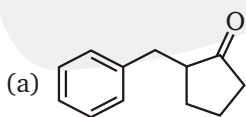


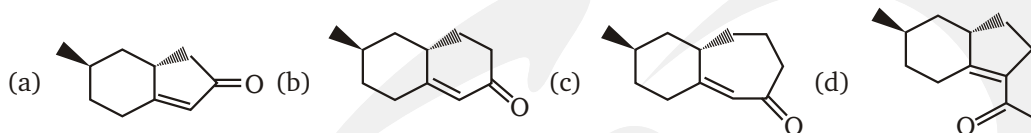
4. Which of the following compounds is the product of an aldol condensation reaction?



NaOH, $2O$
100°C

? Product of the reaction is :

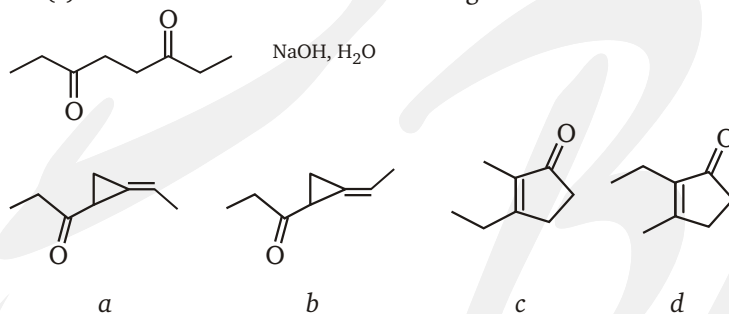




7. Which of the following is not the product of an intramolecular aldol condensation?

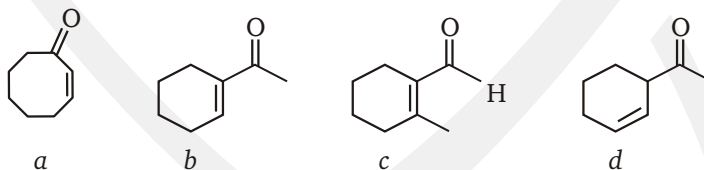
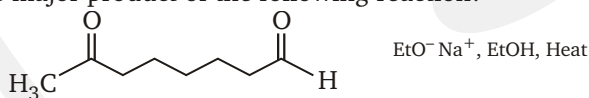


8. Which product(s) should be formed in the following reaction?

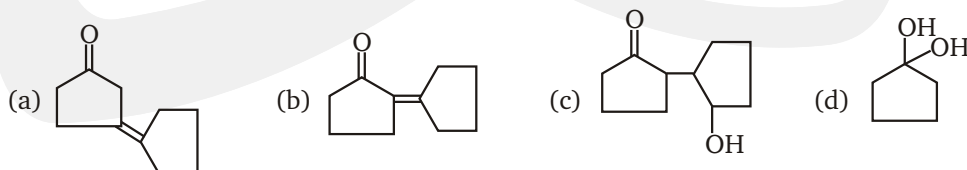
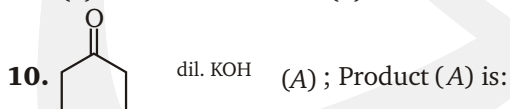


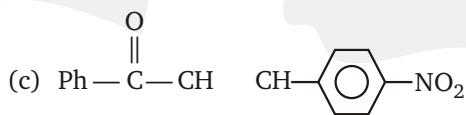
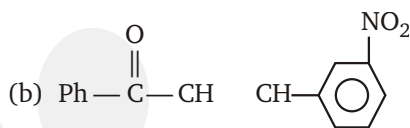
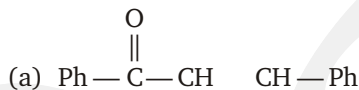
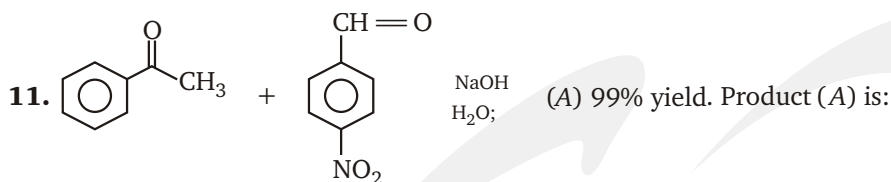
(a) a and b (b) a, b, and c (c) c and d (d) c only

9. Which is the major product of the following reaction?

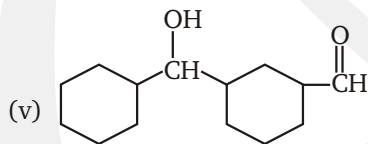
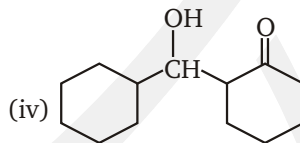
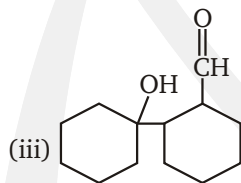
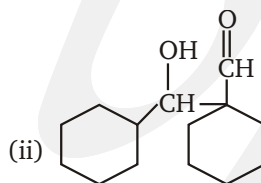
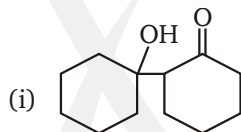
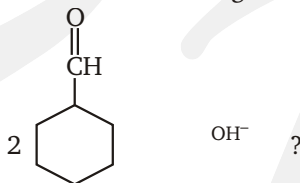


(a) a (b) b (c) c (d) d





12. What is the major organic product of the following aldol addition ?



(a) i

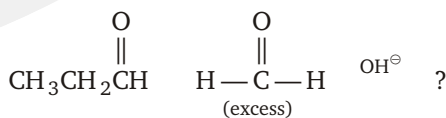
(b) ii

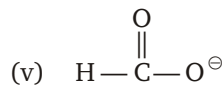
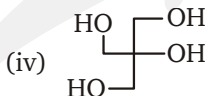
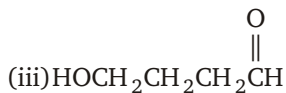
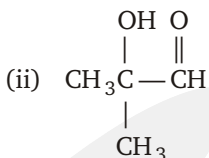
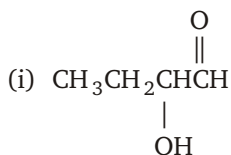
(c) iii

(d) iv

(e) v

13. What is the major organic product of the following addition ?





(a) i

(b) ii

(c) iii

(d) iv
base/

(e) v

14. $\text{Ph}-\text{CH}=\text{CH}-\text{CHO} + \text{CH}_3-\text{CH}=\text{CH}-\text{CHO}$ (A) aldol-condensation product. How many geometrical isomer of (A) will be formed?

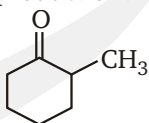
(a) 4

(b) 6

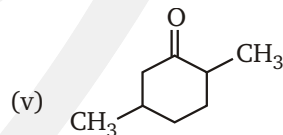
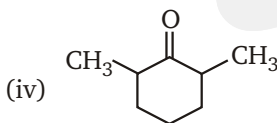
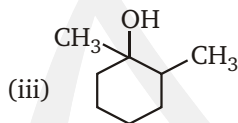
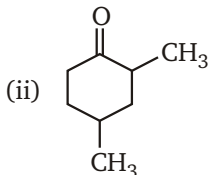
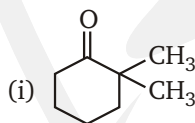
(c) 8

(d) 16

15. What is the major organic product of the following reaction?



1. LDA/THF
2. CH_3I ?



(a) i

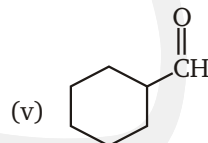
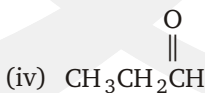
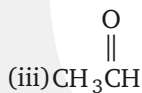
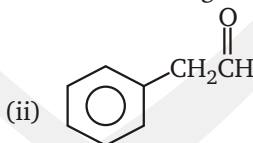
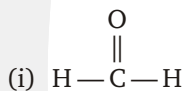
(b) ii

(c) iii

(d) iv

(e) v

16. Which of the following aldehydes does not undergo an aldol addition when mixed with base?



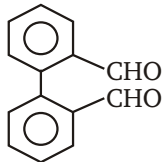
(a) i

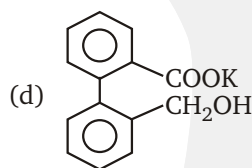
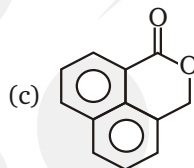
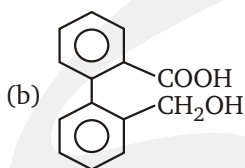
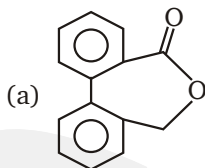
(b) ii

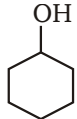
(c) iii

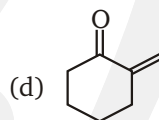
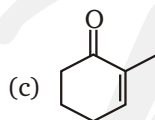
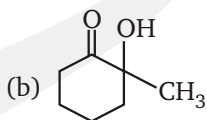
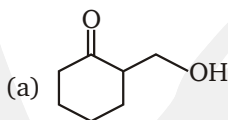
(d) iv

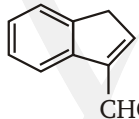
(e) v

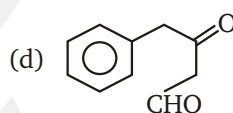
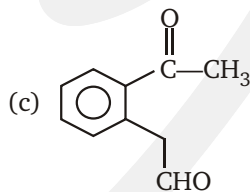
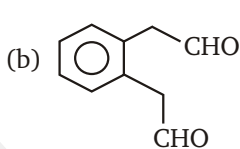
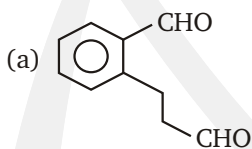
17.  $\xrightarrow{\text{conc. KOH}}$ (A) $\xrightarrow{\text{H}^+}$ (B) ; Structure of (B) is:
cyclic product

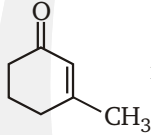


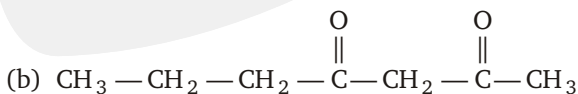
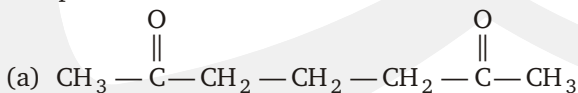
18.  $\xrightarrow{\text{PCC}}$ (A) + $\text{HCHO} \xrightarrow{\text{dil. KOH}}$ (B) ; Structure of (B) is:

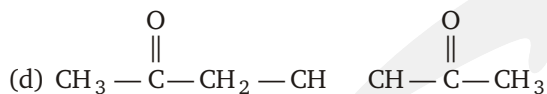
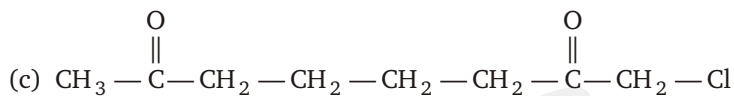


19.  can be synthesised from which of the following in presence of base:

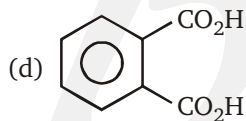
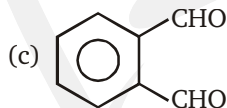
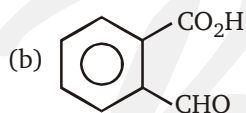
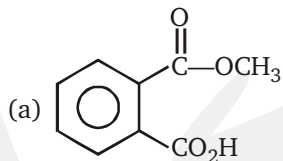
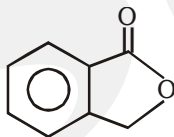


20.  is final product when one of following is allowed to react with base. Select the compound.

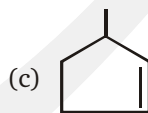
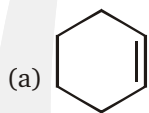
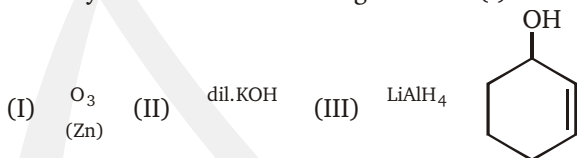




- 21.** Which of the following reactant on reaction with conc. NaOH followed by acidification gives the following lactone as the only product ?



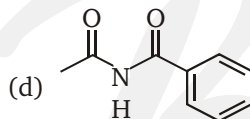
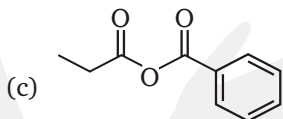
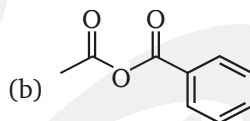
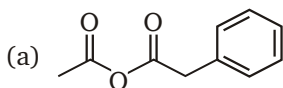
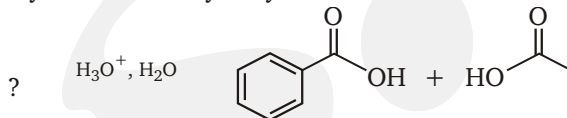
- 22.** Identify which of the following reactant (I) is best for the following conversion ?



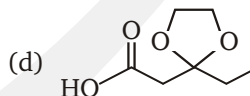
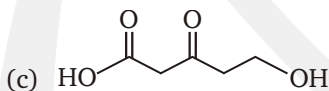
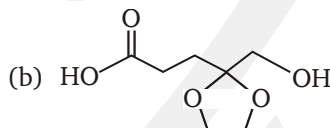
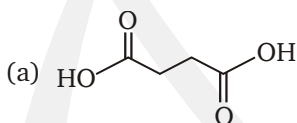
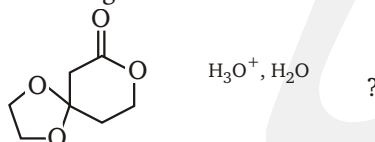
[B] Carboxylic Acids and Their Derivatives

Esterification

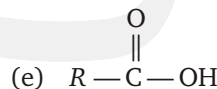
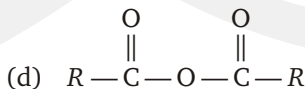
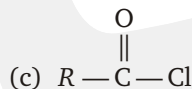
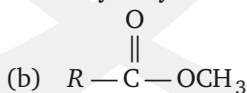
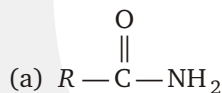
1. What is the acid anhydride that is hydrolyzed?



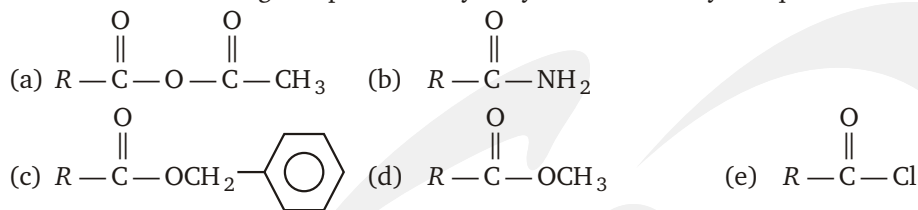
2. What is the product of the following reaction?



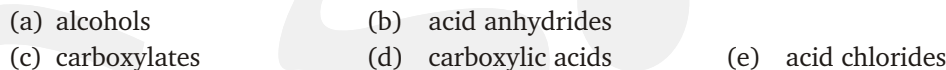
3. Which of the following compounds is hydrolyzed most rapidly in aqueous NaOH ?



4. Which of the following compounds is hydrolyzed most slowly in aqueous NaOH ?



5. Esters and amides are most easily made by nucleophilic acyl substitution reaction on:



6. 4-Oxobutanoic acid is reduced with Na-borohydride and the product is treated with aqueous acid. The final product is:



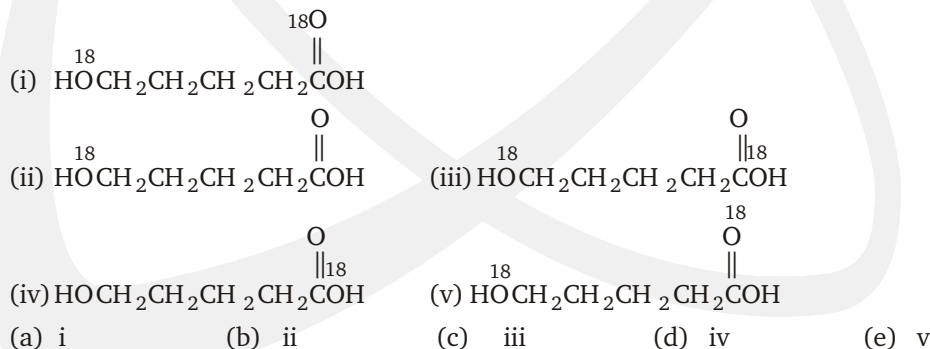
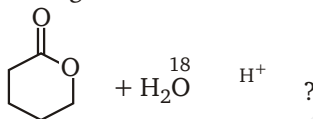
7. The hydrolysis of esters in base is called :



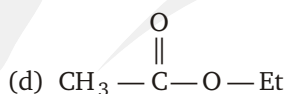
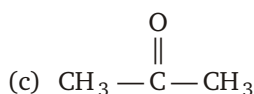
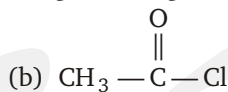
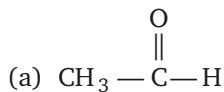
8. Which of the following conditions will drive the equilibrium of the Fischer esterification towards ester formation ?



9. What is the product of the following reaction ?



10. Which of the following is most reactive toward Grignard reagent ?



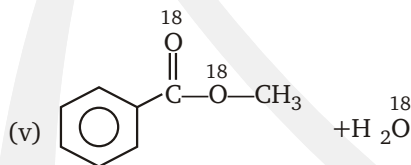
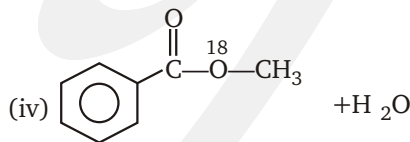
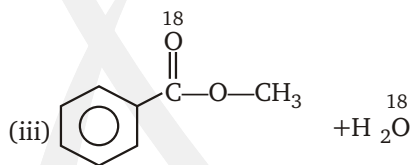
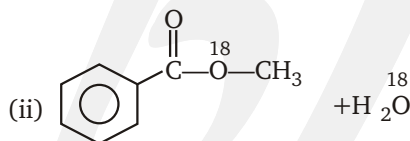
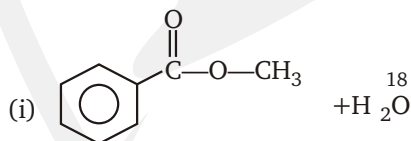
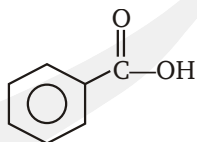
11. Typically, amides will hydrolyze under conditions than esters.

- (a) milder (b) more dilute
(c) more vigorous (d) less vigorous (e) more saline

12. Which of the following esters undergoes hydrolysis in basic medium most easily ?

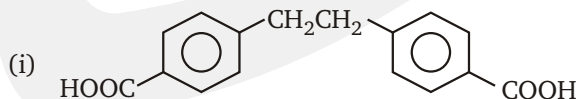
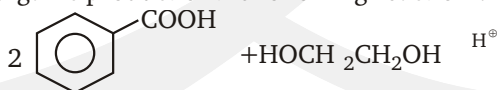
- (a) $\text{CH}_3\text{CH}_2\text{CO}_2\text{CH}_3$ (b) $(\text{CH}_3)_2\text{CHCH}_2\text{CO}_2\text{CH}_3$
(c) $\text{C}_6\text{H}_5\text{O}_2\text{CCH}_3$ (d) $p\text{-CH}_3\text{C}_6\text{H}_4\text{O}_2\text{CCH}_3$ (e) $p\text{-NO}_2\text{C}_6\text{H}_4\text{O}_2\text{CCH}_3$

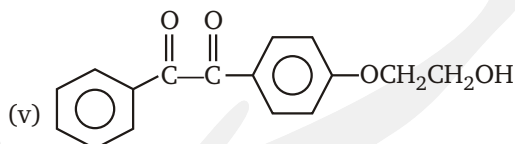
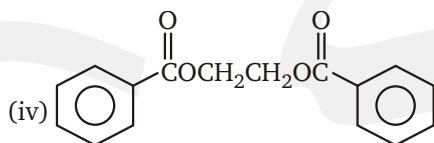
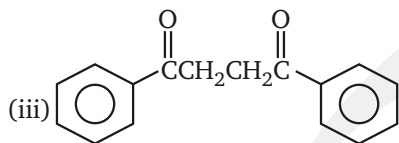
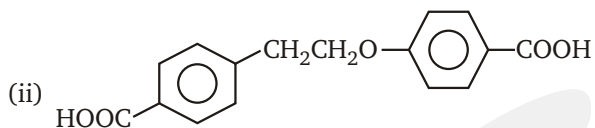
13. What is the product obtained from the following reaction ?



- (a) i (b) ii (c) iii (d) iv (e) v

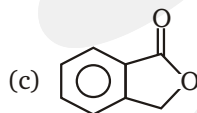
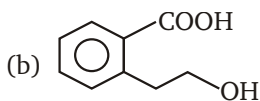
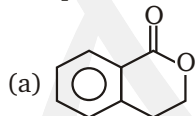
14. What is the major organic product of the following reaction ?



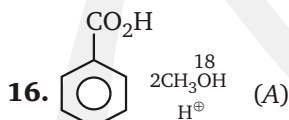


15. (a) i (b) ii NaBH_4 (A) (c) iii H^+ (B) Cyclic

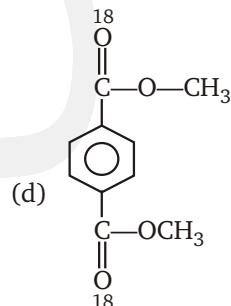
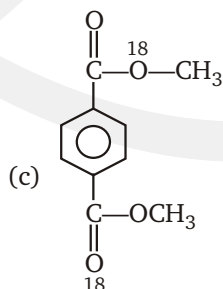
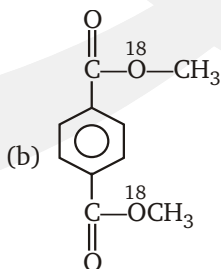
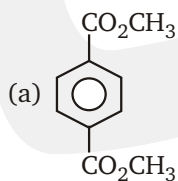
Compound (B) is:



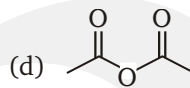
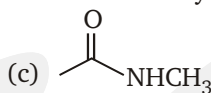
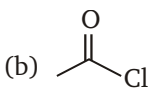
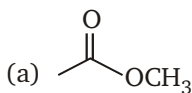
(d) (a) and (c) both



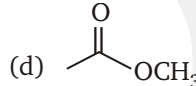
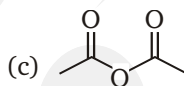
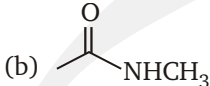
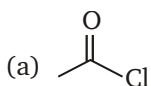
Product (A) is:



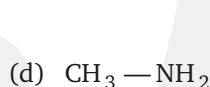
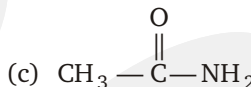
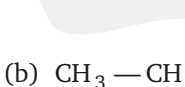
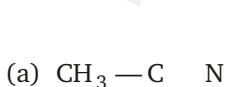
17. Which of the following carboxylic acid derivatives is most easily hydrolyzed?



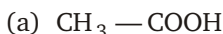
18. Which of the following carboxylic acid derivatives is most difficult to hydrolyze?



19. Which of the following compounds will not undergo hydrolysis? (H_3O^+)



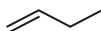
20. $\text{C}_2\text{H}_5\text{OH}$ PI_3 A KCN B Hydrolysis (C), (C) is:



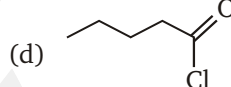
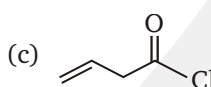
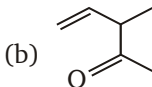
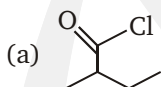
Carboxylic Acids and Their Derivatives

1. What is the product of the following synthesis?

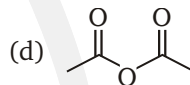
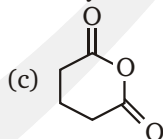
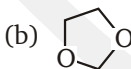
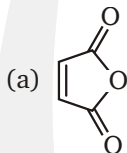
1. HBr, peroxide
2. Mg, Et_2O
3. CO_2 , Et_2O
4. H_3O^+ , H_2O
5. SOCl_2 , pyridine



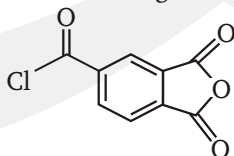
?



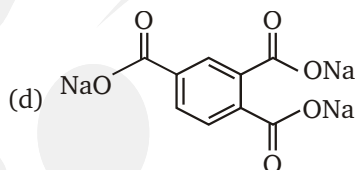
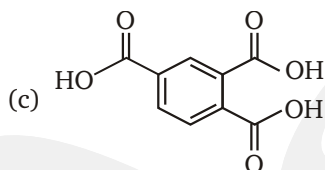
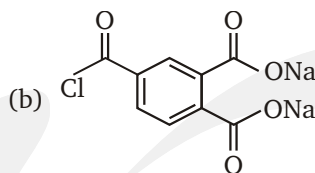
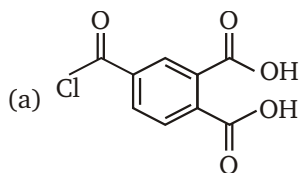
2. Which of the following compounds is not an acid anhydride?



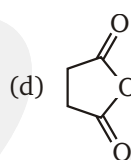
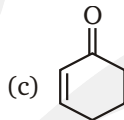
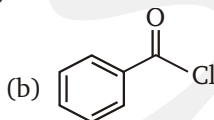
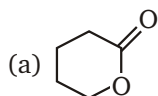
3. What is the product of the following reaction?



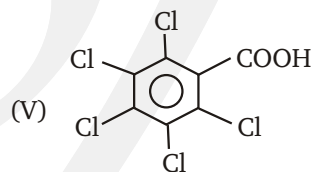
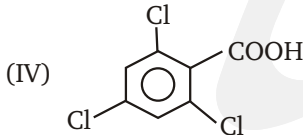
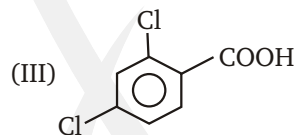
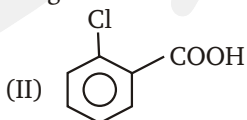
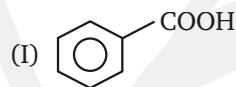
NaOH , H_2O ?



4. Which of the following is not an acid derivative?



5. Which of the following is the strongest acid?



(a) I

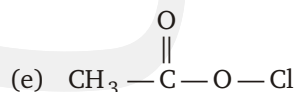
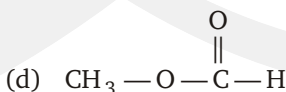
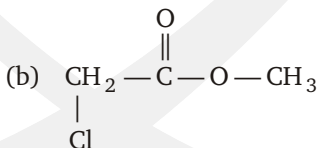
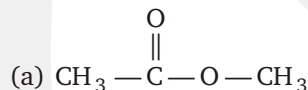
(b) II

(c) III

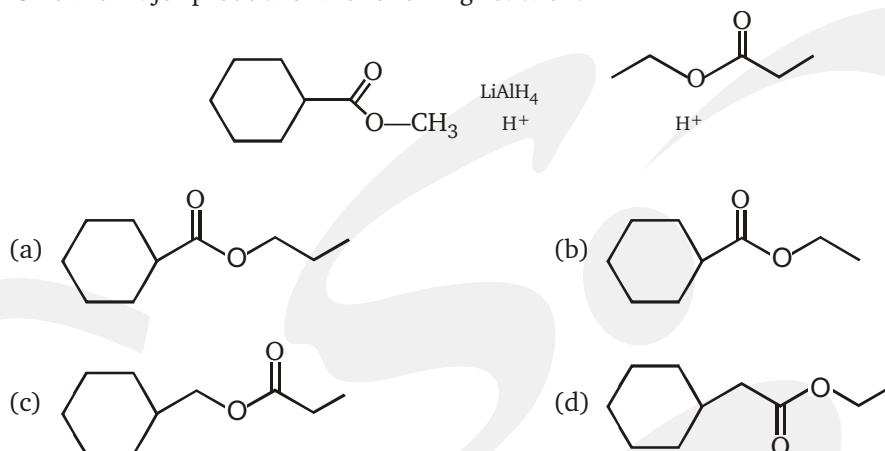
(d) IV

(e) V

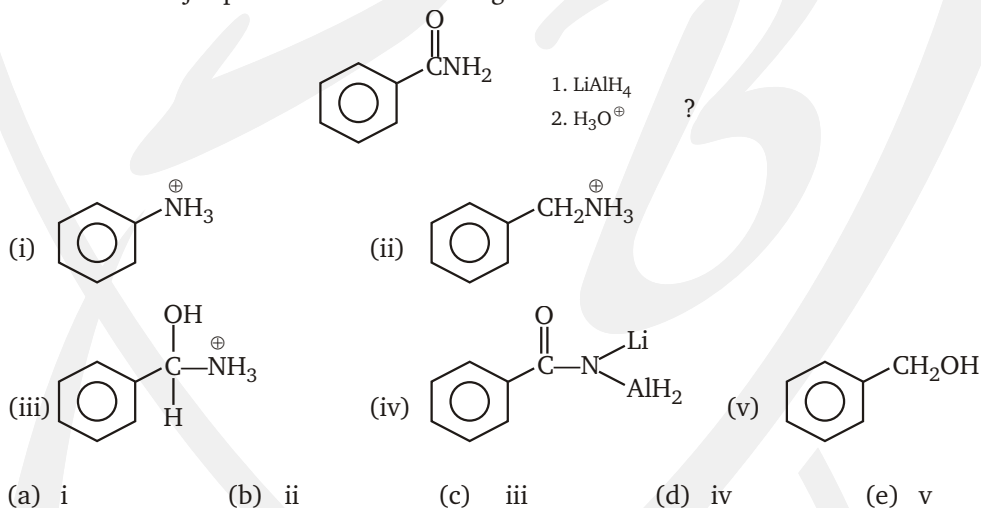
6. What is the major organic product obtained from the following sequence of reactions ?

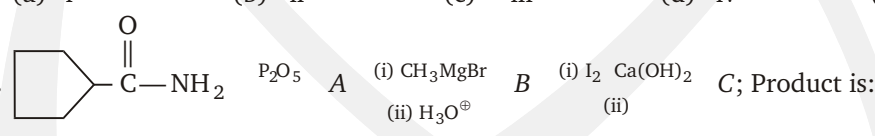


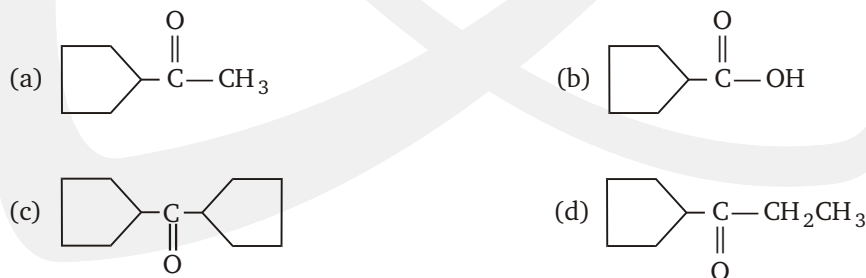
7. Give the major product of the following reaction:

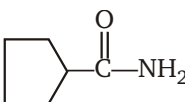


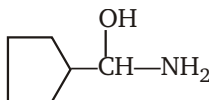
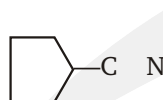
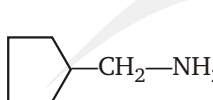
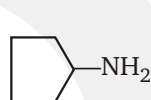
8. What is the major product of the following reaction ?



9.  Product is:



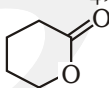
10.  $\xrightarrow{\text{P}_4\text{O}_{10}}$ (A) ; Product (A) is :

- (a)  (b)  (c)  (d) 

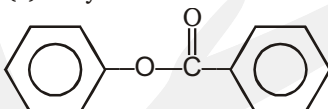
11. Which of the following reagents would reduce carboxylic acids and esters into alcohols ?

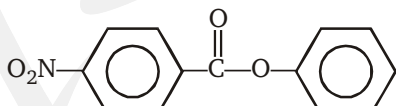
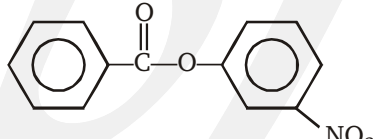
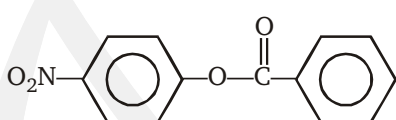
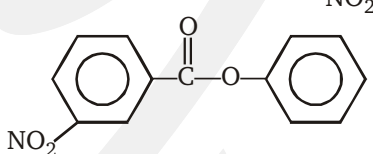
- (a) $\text{H}_2/\text{Raney Ni}$ (b) $1. \text{LiAlH}_4/2. \text{H}_3\text{O}^+$
(c) Na/NH_3 (d) $1. \text{NaBH}_4/2. \text{H}_3\text{O}^+$ (e) $\text{Zn}(\text{Hg})/\text{H}^+$

12. When the lactone below is treated with LiAlH_4 , the major organic product is



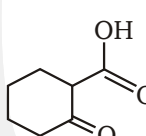
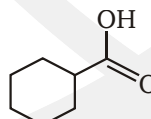
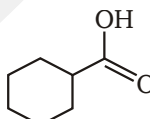
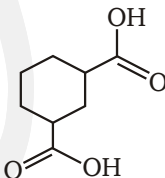
- (a) a cyclic ether (b) a cyclic alcohol
(c) a cyclic ketone (d) an acyclic aldehyde (e) an acyclic diol

13.  $\xrightarrow[\text{H}_2\text{SO}_4]{\text{HNO}_3}$ (A) ; Major product (A) is:

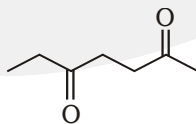
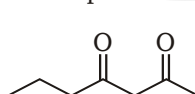
- (a)  (b) 
(c)  (d) 

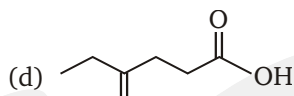
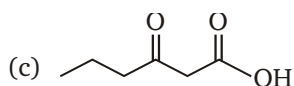
Decarboxylation

1. Which of the following compounds loses carbon dioxide upon heating to 100°C ?

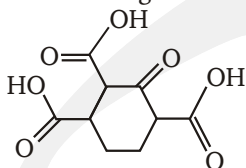
- (a)  (b)  (c)  (d) 

2. Which of the following compounds loses carbon dioxide upon heating to 100°C ?

- (a)  (b) 

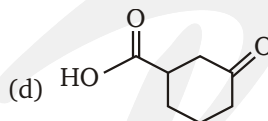
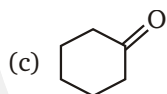
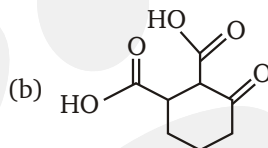
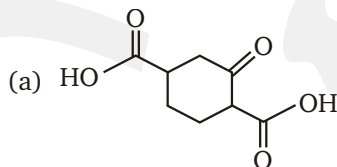


3. What is the product of the following reaction?

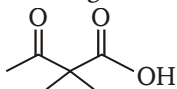


Heat

?



4. Predict the product of the following reaction.



Heat

?

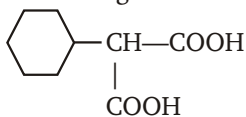
(a) 3-methyl-2-butanone

(b) 3,3-dimethylmalonic acid

(c) propane

(d) 3,3-dimethyl-2,4-pentadione

5. Predict the product of the following reaction.



Heat

?

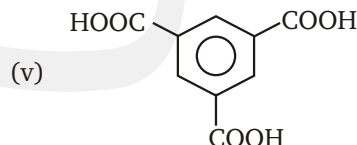
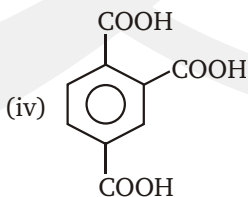
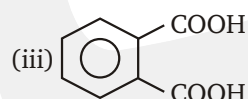
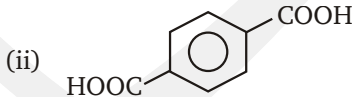
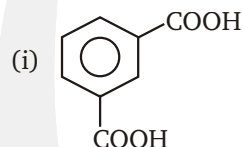
(a) toluene

(b) benzoic acid

(c) phenylacetic acid

(d) cyclohexylacetic acid

6. Which of the following is phthalic acid ?



(a) i

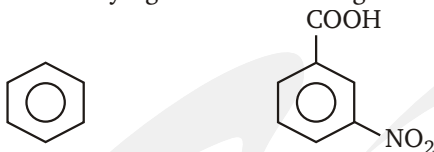
(b) ii

(c) iii

(d) iv

(e) v

7. Which is the best method for carrying out the following reaction ?



- (a) $\text{COOH}; \text{HNO}_3 / \text{H}_2\text{SO}_4$
 (b) $\text{CH}_3\text{Cl} / \text{AlCl}_3; \text{HNO}_3 / \text{H}_2\text{SO}_4; \text{KMnO}_4 / \text{H}^+$, heat
 (c) $\text{CH}_3\text{Cl} / \text{AlCl}_3; \text{KMnO}_4 / \text{H}^+$, heat; $\text{HNO}_3 / \text{H}_2\text{SO}_4$
 (d) $\text{HNO}_3 / \text{H}_2\text{SO}_4; \text{CH}_3\text{Cl} / \text{AlCl}_3; \text{KMnO}_4 / \text{H}^+$, heat
 (e) $\text{HNO}_3 / \text{H}_2\text{SO}_4; \text{COOH}$

8. $\text{Ph}-\overset{\text{O}}{\parallel}{\text{C}}-\text{NH}_2 \xrightarrow[\text{P}_2\text{O}_5]{\text{POCl}_3} (\text{A})$; Product (A) is:

- (a) $\text{Ph}-\text{NH}_2$ (b) $\text{Ph}-\text{CH}_2-\text{NH}_2$ (c) $\text{Ph}-\overset{\text{OH}}{\underset{|}{\text{CH}}}-\text{NH}_2$ (d) $\text{Ph}-\text{C}\equiv\text{N}$

9. $\text{Ph}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_2-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH}$ (A) (B) (i) CH_3MgBr (ii) H (C)
 (gas)

Product (C) is:

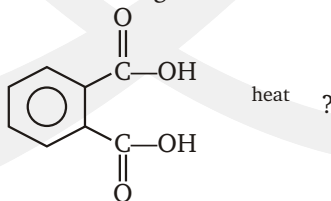
- (a) $\text{CH}_3-\text{CO}_2\text{H}$ (b) $^{14}\text{CH}_3\text{CO}_2\text{H}$ (c) $^{14}\text{CH}_3\text{CO}_2\text{H}$ (d) $^{14}\text{CH}_3\text{CO}_2\text{H}$

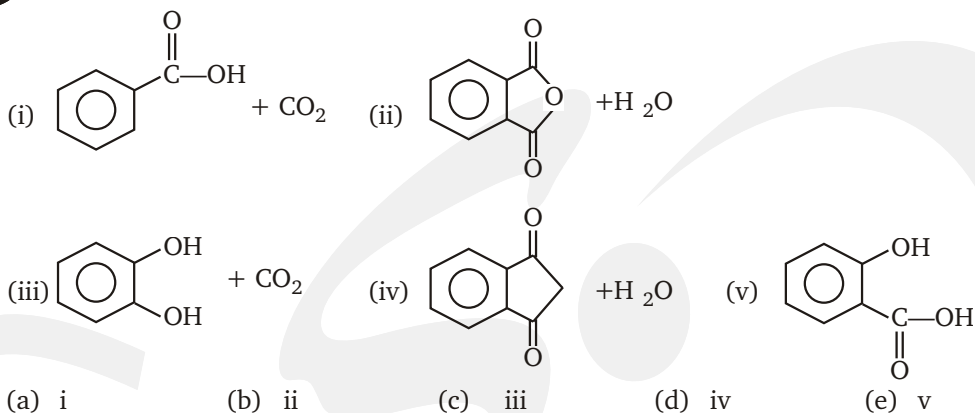
10. $\text{Ph}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH}$ NaHCO_3 (A) PhMgBr (^{2}H) (B)
 Gas

Product (B) is:

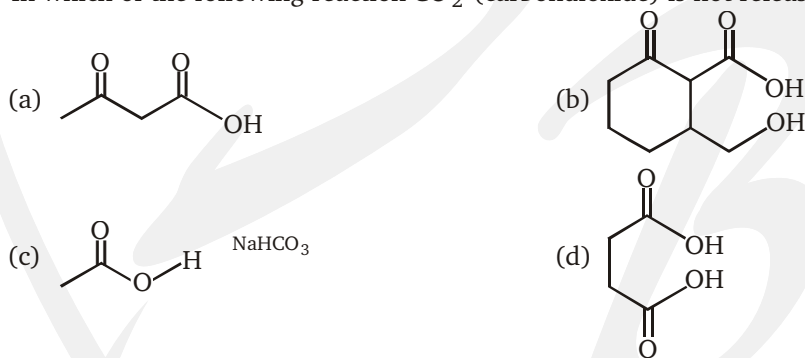
- (a) $\text{Ph}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH}$ (b) $\text{Ph}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH}$ (c) $\text{Ph}-\text{OH}$ (d) $\text{Ph}-\text{CH}_3$

11. What is the major product of the following reaction ?

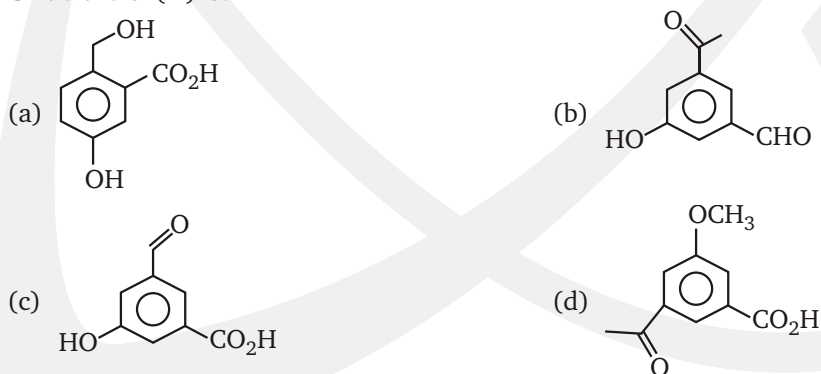




12. In which of the following reaction CO₂ (carbondioxide) is not released ?



13. Compound (A) evolves CO₂ gas with NaHCO₃,
Compound (A) evolves H₂ gas with Na,
Compound (A) gives Iodoform test,
Structure of (A) is:



Answers**[A] Aldehydes and Ketones****Methods of Preparation**

1.	(c)	2.	(c)	3.	(b)	4.	(b)	5.	(a)										
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Properties of Carbonyl Compounds

1.	(b)	2.	(a)	3.	(b)	4.	(b)	5.	(c)	6.	(b)	7.	(c)	8.	(b)	9.	(c)	10.	(a)
11.	(d)	12.	(b)	13.	(c)	14.	(b)	15.	(d)	16.	(b)	17.	(b)	18.	(c)	19.	(d)	20.	(a)
21.	(a)	22.	(a)	23.	(d)	24.	(d)	25.	(a)	26.	(b)	27.	(c)	28.	(b)	29.	(b)	30.	(b)
31.	(a)	32.	(c)	33.	(b)	34.	(c)	35.	(c)	36.	(d)	37.	(b)	38.	(c)	39.	(b)	40.	(b)
41.	(a)	42.	(e)																

Aldol and Cannizaro Reactions

1.	(a)	2.	(a)	3.	(d)	4.	(c)	5.	(c)	6.	(d)	7.	(c)	8.	(d)	9.	(c)	10.	(b)
11.	(a)	12.	(b)	13.	(d)	14.	(c)	15.	(a)	16.	(a)	17.	(a)	18.	(a)	19.	(b)	20.	(a)
21.	(c)	22.	(b)																

[B] Carboxylic Acids and Their Derivatives**Esterification**

1.	(b)	2.	(c)	3.	(c)	4.	(b)	5.	(e)	6.	(a)	7.	(e)	8.	(e)	9.	(d)	10.	(b)
11.	(c)	12.	(e)	13.	(d)	14.	(d)	15.	(a)	16.	(b)	17.	(b)	18.	(b)	19.	(d)	20.	(b)

Carboxylic Acids and Their Derivatives

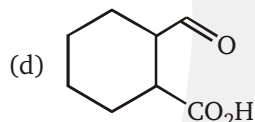
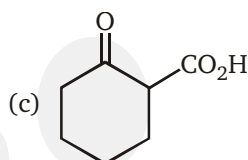
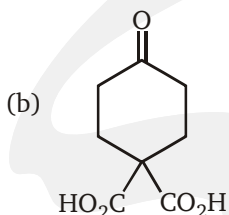
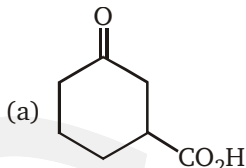
1.	(a)	2.	(b)	3.	(d)	4.	(c)	5.	(e)	6.	(a)	7.	(c)	8.	(b)	9.	(c)	10.	(b)
11.	(b)	12.	(e)	13.	(c)														

Decarboxylation

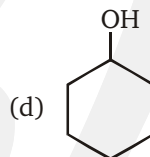
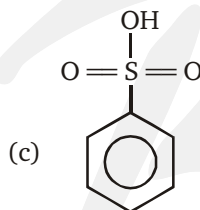
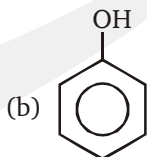
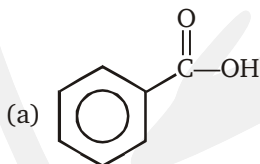
1.	(a)	2.	(c)	3.	(d)	4.	(a)	5.	(d)	6.	(c)	7.	(d)	8.	(d)	9.	(a)	10.	(b)
11.	(b)	12.	(d)	13.	(d)														

One or More than One Answers is/are Correct

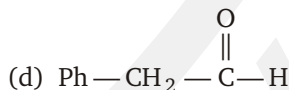
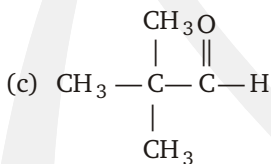
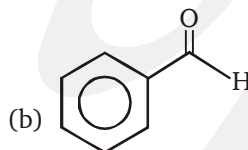
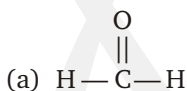
1. Which of the following compound easily undergo decarboxylation on heating ?



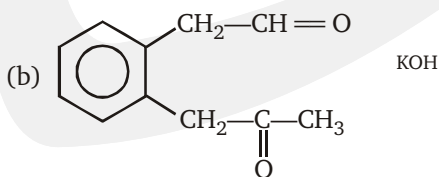
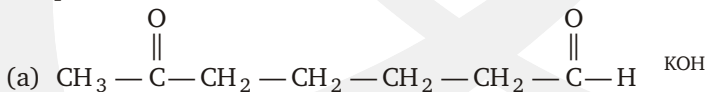
2. Which of the following compound reacts with NaHCO_3 to evolve CO_2 gas ?

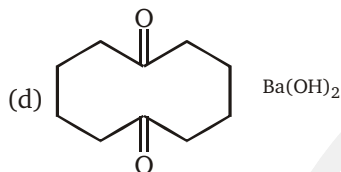
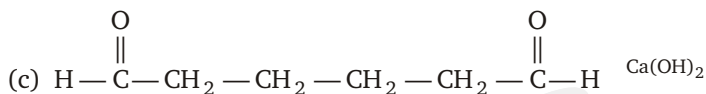


3. Which of the following compound undergo Cannizzaro reaction ?

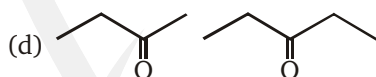
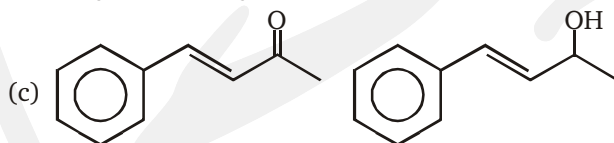
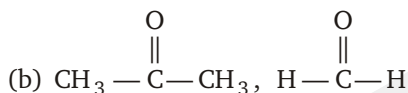
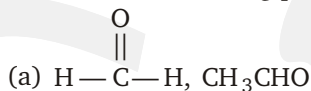


4. In which of the following intramolecular aldol condensation reaction more than two products are possible ?

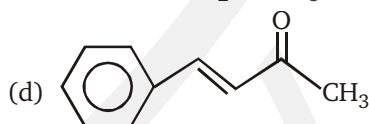
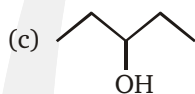
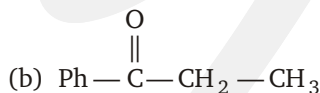
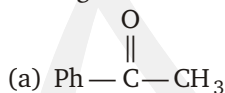




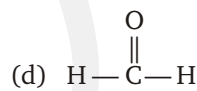
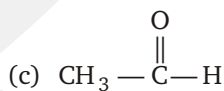
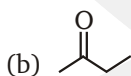
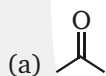
5. Which of the following pair is differentiated by iodoform test ?



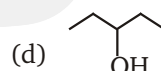
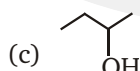
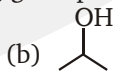
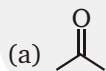
6. Among the following which compound gives haloform test ?



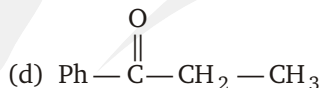
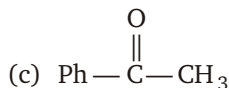
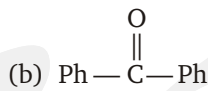
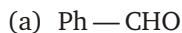
7. Which of the following gives positive test with 2, 4-DNP ?



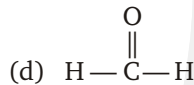
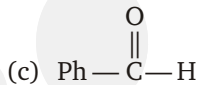
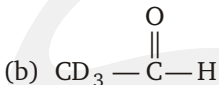
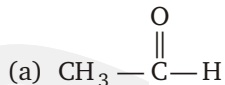
8. Which of the following gives positive iodoform test?



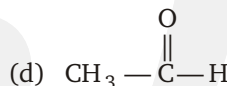
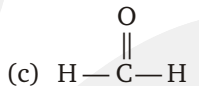
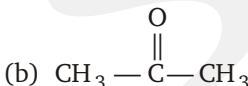
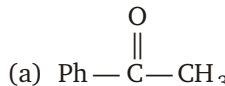
9. Which of the following gives negative iodoform test?



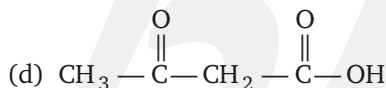
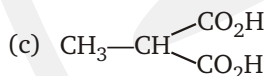
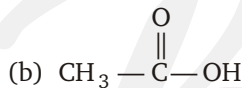
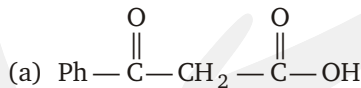
10. Which of the following compounds undergo self aldol?



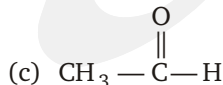
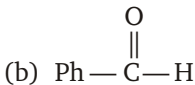
11. Which of the following compounds give geometrical isomers when reacts with NH_2-OH ?



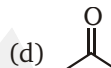
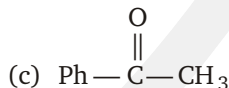
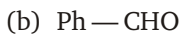
12. Which of the following compound easily undergo decarboxylation on heating?



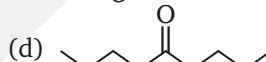
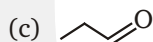
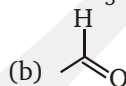
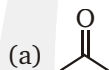
13. Which of the following compounds will undergo cannizaro reaction?



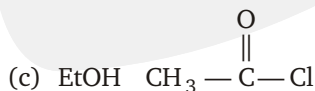
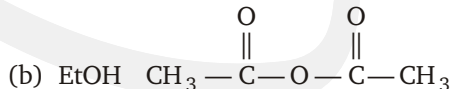
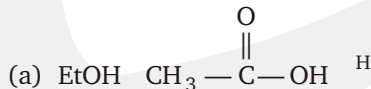
14. Which of the following gives positive Tollen's test?



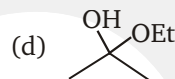
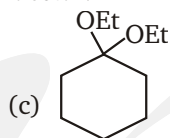
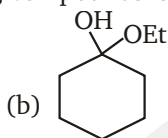
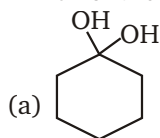
15. Which of the following compounds reacts with NaHSO_3 ?



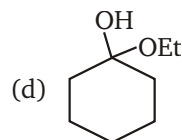
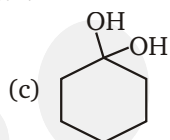
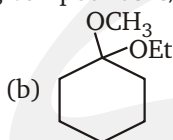
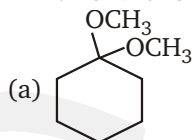
16. Which of the following reaction is esterification?



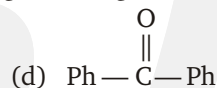
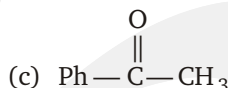
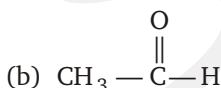
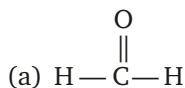
17. Which of the following compounds is/are hemiacetal?



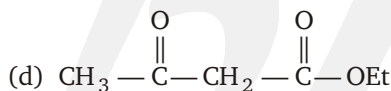
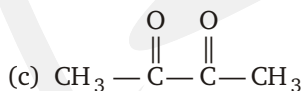
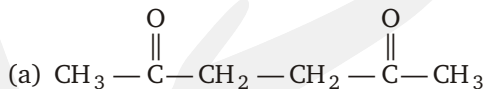
18. Which of the following compounds is/are acetal?



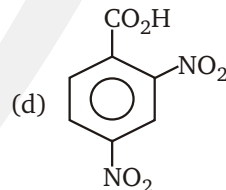
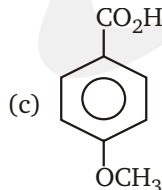
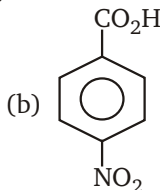
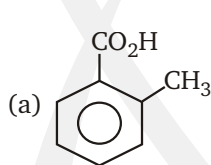
19. Which of the following is/are more reactive than acetone toward Grignard reagent?



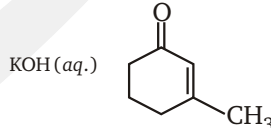
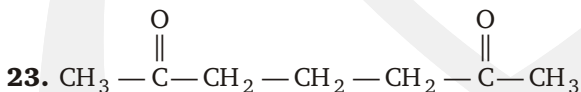
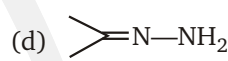
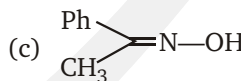
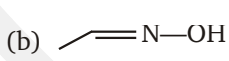
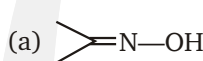
20. Which of the following compounds has active methylene group?



21. Which of the following acids is/are stronger acid than benzoic acid?



22. Which of the following is/are ketoxime ?



Correct statement regarding given reaction is:

(a) It is a dehydration reaction

(b) Carbanion is intermediate

(c) One of the step is nucleophilic addition

(d) One of the step is electrophilic addition

24. In which of the following reaction aldehyde will form ?

(a) Stephen's reduction $R-\text{C}\equiv\text{N} \xrightarrow{\text{SnCl}_2, \text{HCl}} R-\text{CHO} \xrightarrow{\text{H}_3\text{O}^+}$

(b) Etard reaction $\text{C}_6\text{H}_5\text{CH}_3 \xrightarrow[\text{H}_2\text{O}]{\text{CrO}_2\text{Cl}_2}$

(c) Gattermain's formylation $\text{C}_6\text{H}_6 \xrightarrow[\text{HCl}]{\text{CO, AlCl}_3}$

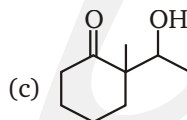
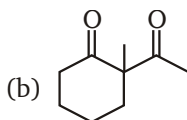
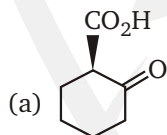
(d) $\text{Ph}-\text{CH}=\text{N}-\text{CH}_3 \xrightarrow{\text{H}_3\text{O}^+}$

25. $\text{CH}_3\text{CH}=\text{CHCH}_3 \xrightarrow[\text{Zn}]{\text{O}_3}$ (A) (B)

Compound (A) and (B) is differentiated by

(a) NaOI (b) Tollens reagent (c) Fehling solution (d) Na-metal

26. (A) $\xrightarrow[\text{H}^+]{\text{NaOI}}$ (B) $\text{C}_6\text{H}_{11}\text{O} + \text{CO}_2$; Structure of (A) is:



(d) None of these

Answers

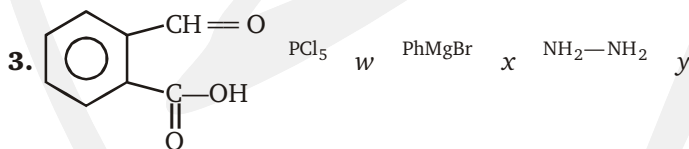
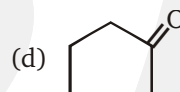
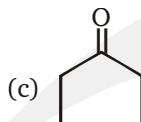
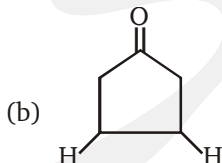
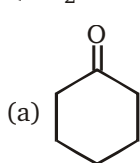
1.	(b, c)	2.	(a, c)	3.	(a, b, c)	4.	(a, b)	5.	(a, b, d)	6.	(a, d)
7.	(a, b, c, d)	8.	(a, b, c)	9.	(a, b, d)	10.	(a, b)	11.	(a, d)	12.	(a, c, d)
13.	(a, b, d)	14.	(a, b)	15.	(a, b, c)	16.	(a, b, c)	17.	(b, d)	18.	(a, b)
19.	(a, b)	20.	(b, d)	21.	(a, b, d)	22.	(a, c)	23.	(a, b, c)	24.	(a, b, c, d)
25.	(b, c)	26.	(b, c)								

Comprehension Based Problems

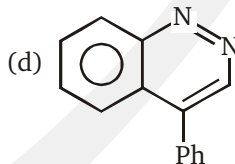
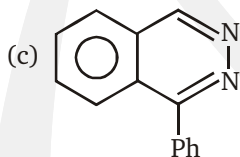
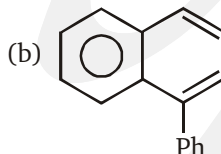
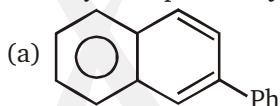
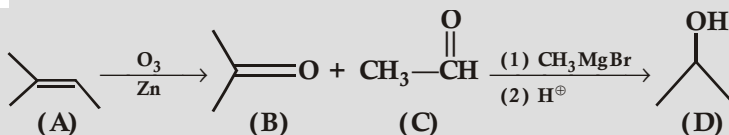
Passage-1:

Carbonyl group show characteristic reaction of nucleophilic addition in which nucleophile attack in rate determining step on carbonyl carbon.

- Which pH is suitable for oxime formation ?
(a) 4 5 (b) 1 2 (c) 8 10 (d) 12 14
- Which of the reactant show geometrical isomerism when it react with hydrazine ($\text{NH}_2 - \text{NH}_2$)?



Identify final product 'y' of the above given reaction:

**Passage-2:**

- Compound (B) and (C) is differentiated by:
(a) Iodoform test (b) 2, 4-DNP (c) Tollen's reagent (d) All

2. Which compound give negative iodoform test ?

- (a) A (b) B (c) C (d) D

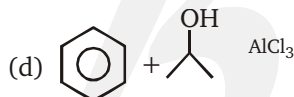
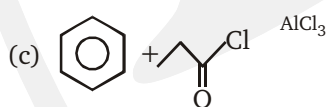
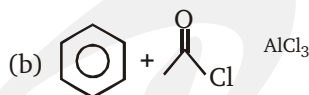
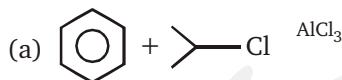
3. When (C) reacts with hydroxylamine then formed product is:

- (a) $\text{CH}_2 = \text{N} - \text{OH}$ (b) $\text{CH}_3 - \text{CH} = \text{N} - \text{OH}$
 (c) $\text{CH}_3 - \text{CH} = \text{N} - \text{NH}_2$ (d)

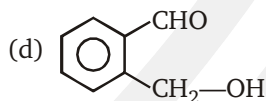
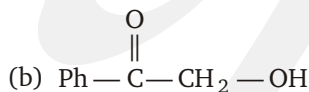
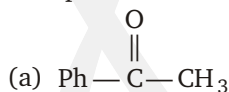
Passage-3:



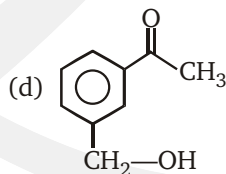
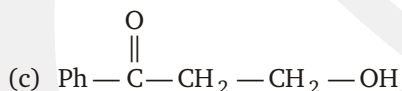
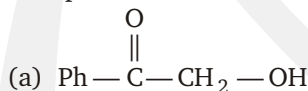
1. Compound A and B is:

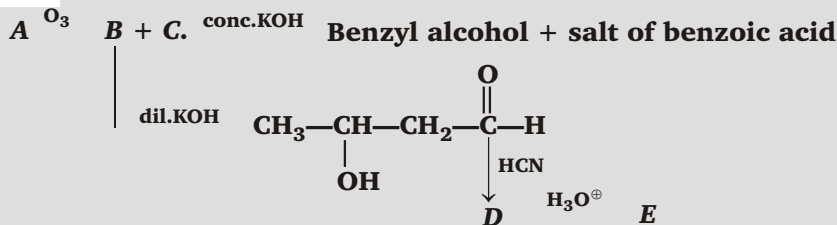


2. Compound C is:

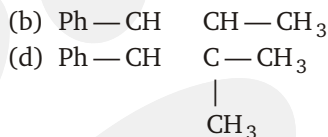
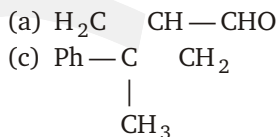


3. Compound D is:



Passage-4:

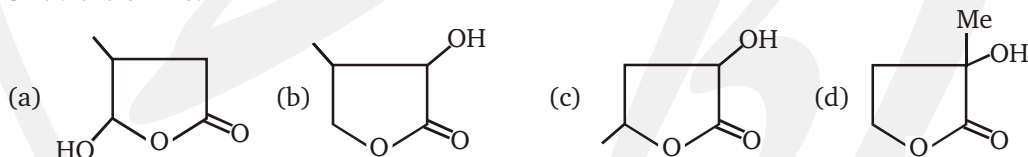
1. Structure of A is:



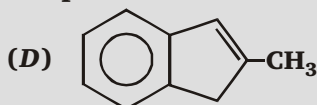
2. Structure of (B) and (C) differentiated by:



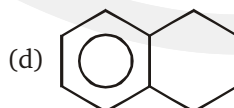
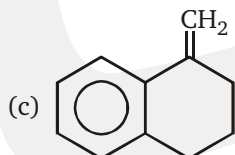
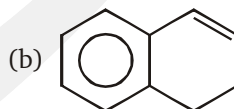
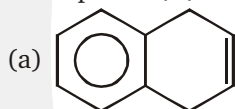
3. Structure of E is:

**Passage-5:**

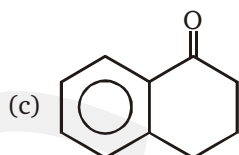
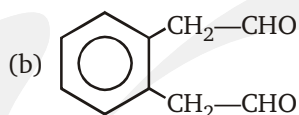
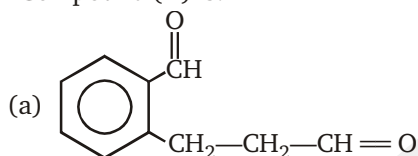
A compound (A) gives positive test with Br_2/CCl_4 . When (A) undergoes reductive ozonolysis product (B) is obtained. When (B) reacts with dil. KOH followed by heating (C) will be obtained (C) gives positive Tollens test. When (C) reacts with $\text{NH}_2 - \text{NH}_2/\text{KOH}$ product (D) will be obtained.



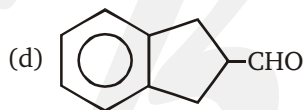
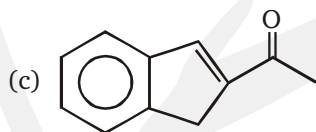
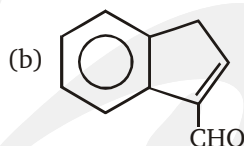
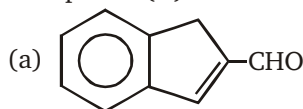
1. Compound (A) is:



2. Compound (B) is:



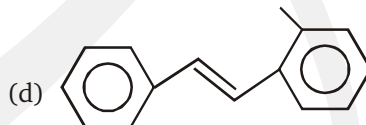
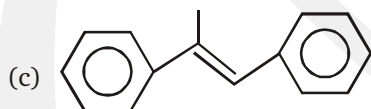
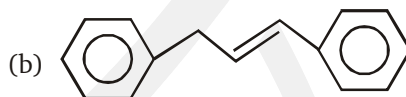
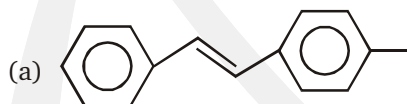
3. Compound (C) is:



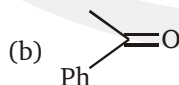
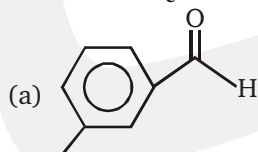
Passage-6:

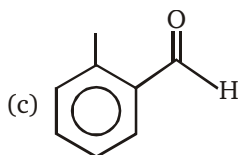
An organic compound 'P' ($C_{15}H_{14}$) on ozonolysis in the presence of Zn give Q (C_8H_8O) and R (C_7H_6O). R either react with tollens's reagent as well as conc. KOH following by acidic medium give same product S. Q reacts with dil. KOH to give T product. Q also give test with $I_2 / ^-OH$ and obtained product is S.

1. Structure of 'P' is:

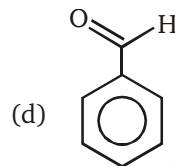
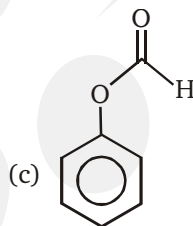
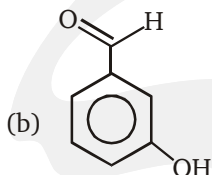
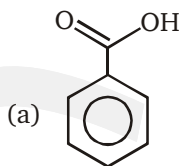


2. Structure of Q is:





3. Structure of 'S' is:

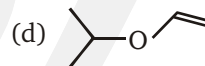
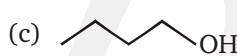
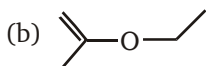
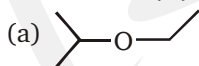


Passage-7:

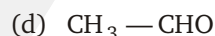
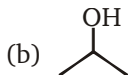
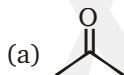


(B) give positive iodoform.

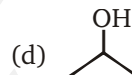
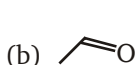
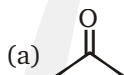
1. Structure of (A) is:



2. Structure of (B) is:



3. Structure of (C) is:



Passage-8:

Compound (A) gives $\frac{3}{2}$ mole of H_2 gas with Na.

Compound (A) reacts with NaHCO_3

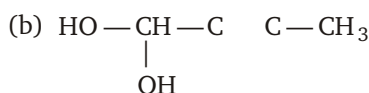
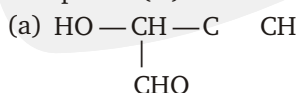
Compound (B) reacts with amm. AgNO_3

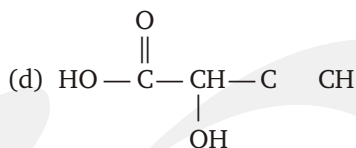
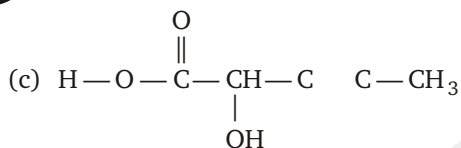
Compound (B) gives 1-mole of H_2 gas with Na

Compound (C) highly reactive towards Na

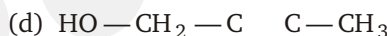
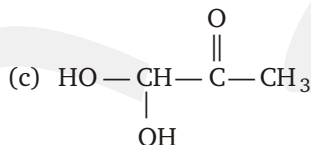
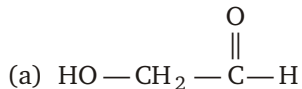
Compound (D) does not reacts with Na.

1. Compound (A) is:

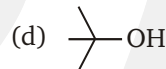
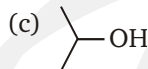
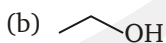




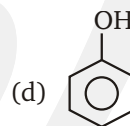
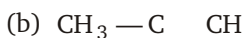
2. Compound (B) is:



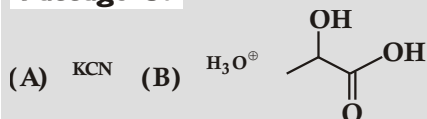
3. Compound (C) is:



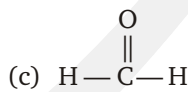
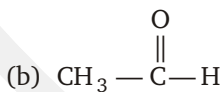
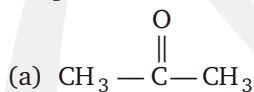
4. Compound (D) is:



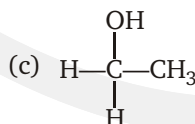
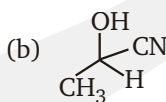
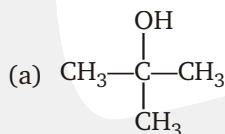
Passage-9:



1. Compound (A) is:



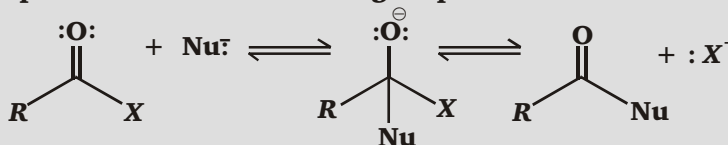
2. Compound (B) is:



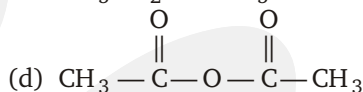
Passage-10:

Acid derivatives (RCOX) undergo nucleophilic reactions at the carbonyl group. The rate of such a reaction depends on the group X .

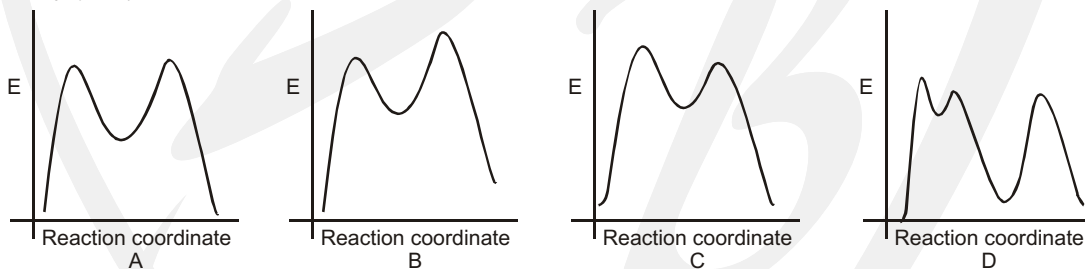
The reaction proceeds via the following steps.



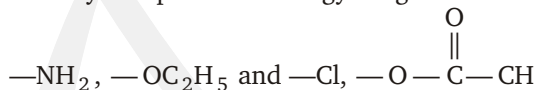
1. The compound which would undergo nucleophilic substitution fastest would be:



2. Based on the above mechanism, following four probable potential energy diagrams can be drawn.



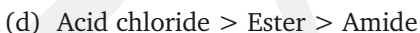
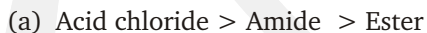
Identify the potential energy diagrams for the best and the poorest leaving groups among



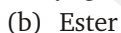
Best ☐

Poorest ☐

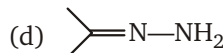
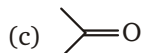
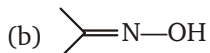
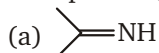
3. The correct order of resonance stabilization is:



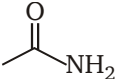
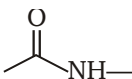
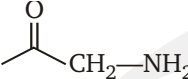
4. Oxygen atom of the carbonyl group is most basic in:

**Passage-11:**

1. Compound (A) is:



2. Compound (B) is:

- (a)  (b)  (c)  (d) None of these

3. Compound (C) is:

- (a) EtOH (b) $\text{CH}_3 - \text{CO}_2\text{H}$ (c)  (d) $\text{CH}_3 - \text{CH}_3$

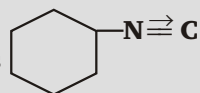
Passage-12:

(A) NH_3

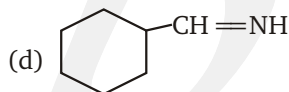
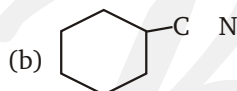
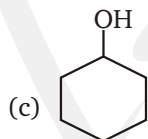
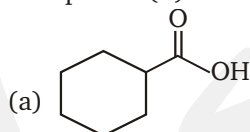
(B) $\frac{\text{Br}_2}{\text{KOH}}$

(C) $\frac{\text{CHCl}_3}{\text{KOH}}$

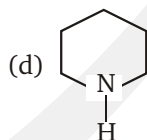
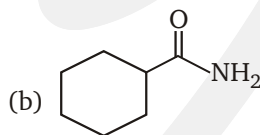
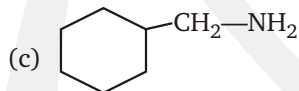
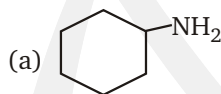
(D) Compound (D) is



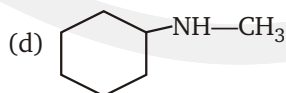
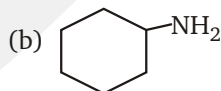
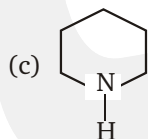
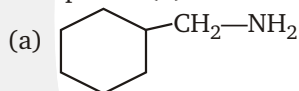
1. Compound (A) is:



2. Compound (B) is:

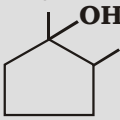


3. Compound (C) is:

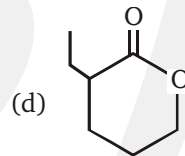
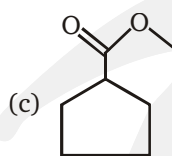
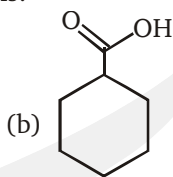
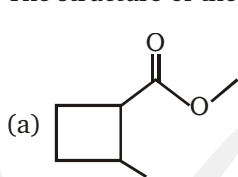


Passage-13:

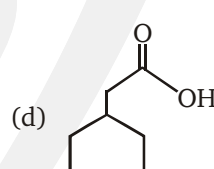
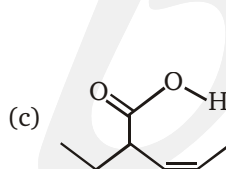
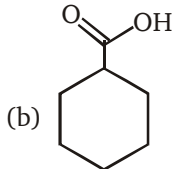
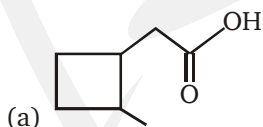
A derivative of acid (A), $C_7H_{12}O_2$ on reduction with $LiAlH_4$ (lithium aluminium halide) gives (B) on reaction with $SOCl_2$ gives (C) $C_6H_{11}Cl$. (C) on reaction with $Mg/ether$ and CO_2 following by H_3O^+ gives (D) $C_7H_{12}O_2$. (D) on reacts with ammonia and after dehydration with P_2O_5 gives (E) $C_7H_{11}N$. (E) after reduction with LAH gives $C_7H_{15}N$ (F). (F) on reaction with $NaNO_2/HCl$

gives (G). G is  and (F) on reaction with $CHCl_3/KOH$ gives foul like smell.

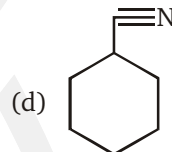
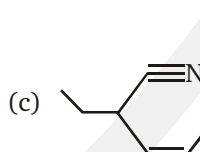
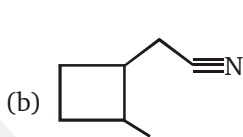
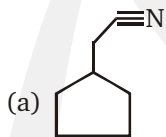
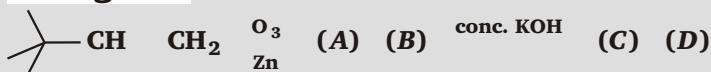
1. The structure of the A is:



2. The structure of D is:



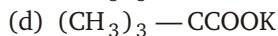
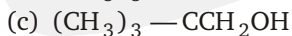
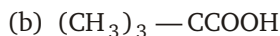
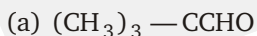
3. The structure of E is:

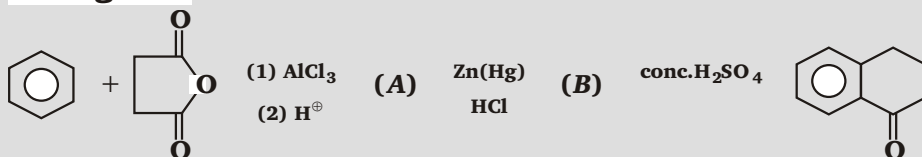
**Passage-14:**

1. Structure of (C) is:

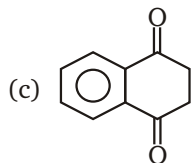
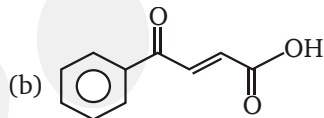
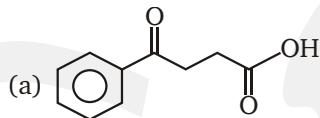


2. Structure of (D) is:



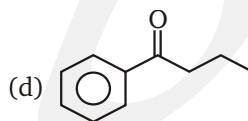
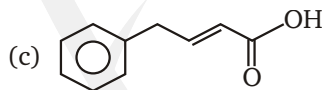
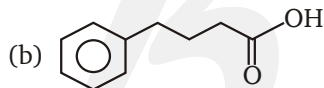
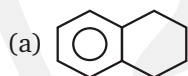
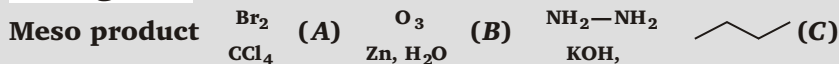
Passage-15:

1. Structure of (A) is:



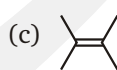
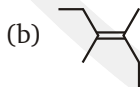
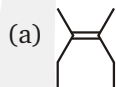
(d) None of these

2. Structure of (B) is:

**Passage-16:**

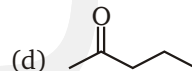
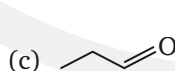
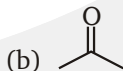
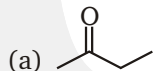
Compound (B) gives positive iodoform test

1. Structure of (A) is:



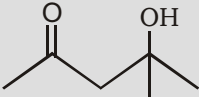
(d) (a) and (b) both

2. Structure of (B) is:

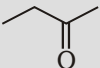
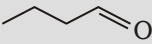
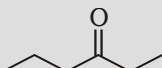
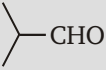


Matching Type Problems

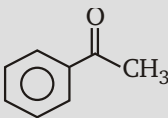
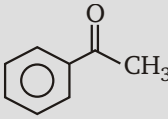
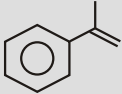
1. Match the column:

Column-I		Column-II	
(a)	(A) conc. KOH $\text{Ph}-\text{CH}_2\text{OH}$ $\text{Ph}-\text{CO}_2\text{K}^{\ominus\oplus}$	(p)	A gives positive Tollen's test
(b)	(A) dil. KOH $\text{C}_3\text{H}_6\text{O}$ 	(q)	A gives positive Fehling test
(c)	(A) $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{O}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3$ $\text{Ph}-\text{CH}$ $\text{CH}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH}$ (1) $\text{CH}_3\text{CO}_2\text{Na}^{\ominus\oplus}$ (2) $\text{H}_3\text{O}^{\oplus}$	(r)	A gives positive iodoform test
(d)	(A) $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH}$ $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{O}-\text{CH}_2-\text{CH}_3$ H ,	(s)	A gives positive 2-4-DNP test
		(t)	A when reacts with $\text{R}-\text{NH}_2$ then formed product is Schiff-base

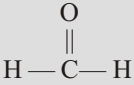
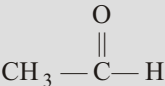
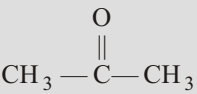
2. Match the column:

Column-I		Column-II	
(a)		(p)	Reacts with NaBH_4 to given secondary alcohol
(b)		(q)	Silver mirror react with $\text{Ag}(\text{NH}_3)_2^1$
(c)		(r)	Isomer of But-2-en-1-ol
(d)		(s)	When reacts with PhMgBr tertiary alcohol will form.

3. Match the column:

Column-I		Column-II	
(a)	 $(1) \text{CH}_3\text{MgBr}$ $(2) \text{H}^+$	(p)	Nucleophilic addition reaction
(b)	 $(1) \text{LiAlH}_4$ $(2) \text{H}^+$	(q)	Electrophilic addition reaction
(c)	 $\text{H}_3\text{O}^{\oplus}$	(r)	Formation of 3° alcohol takes place
(d)	 HBr H_2O	(s)	Formation of 2° alcohol takes place.

4. Match the column:

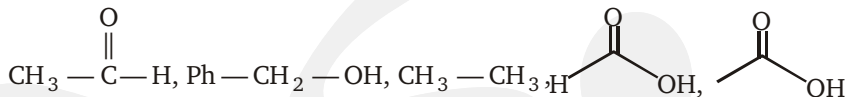
Column-I		Column-II	
(a)	PhMgBr (x) $(2) \text{H}^{\oplus}$ 1° alcohol	(p)	
(b)	PhMgBr (y) $(2) \text{H}^{\oplus}$ 2° alcohol	(q)	
(c)	PhMgBr (z) $(2) \text{H}^{\oplus}$ 3° alcohol	(r)	
(d)	PhMgBr (p) $(2) \text{H}^{\oplus}$ 	(s)	EtOH

Multi Concept Subjective Problems

1. (x) isomer of carbonyl compound of $C_5H_{10}O$ give positive iodoform test (including stereoisomer) ?

2. How many compound give positive Tollen's test ?

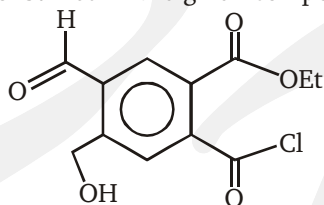
D-Glucose, D-Fructose,



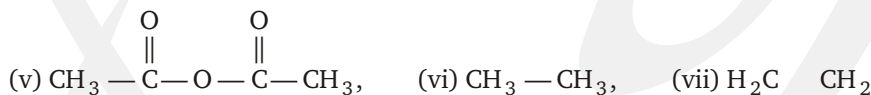
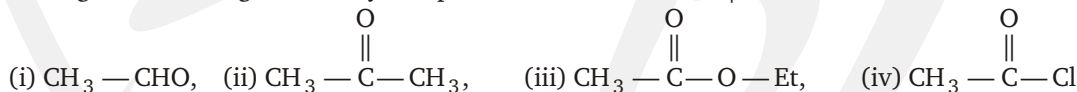
3. How many isomer $C_4H_{10}O$ give positive Iodoform test ?

(Excluding stereoisomer)

4. How many moles of $RMgX$ consumed in the given compound ?

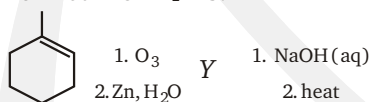


5. Among the following how many compound reacts with $NaBH_4$?



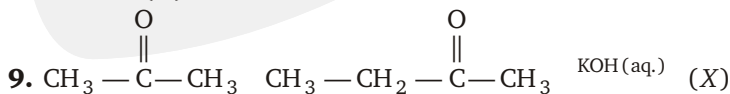
6. $HC \equiv CH \xrightarrow[H_2SO_4]{HgSO_4}$ (A) $\xrightarrow[NaOH]{dil.}$ (B); total number of 'C' atom in the product B is:

7. In the scheme given below, the total number of intramolecular aldol condensation products formed from 'Y' is:



X moles of $HCHO$ consumed.

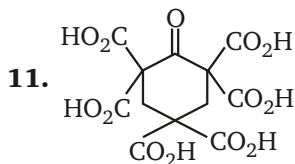
Value of (X) will be:



X number of aldol condensation product (including stereoisomer).

Find out the value of (X)

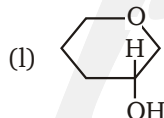
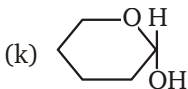
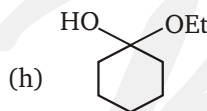
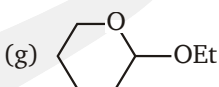
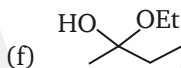
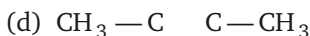
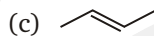
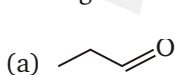
10. How many isomers of $C_4H_{10}O$ give positive iodoform test ? (include stereoisomer)



(X) moles of CO_2 evolved during given reaction?

So the value of (X) is:

12. Among the following compounds how many gives positive Tollen's test?



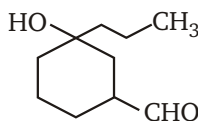
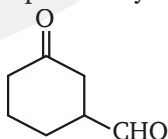
13. How many isomers having molecular formula $C_7H_{14}O$ give cannizaro reaction?

Unified Conceptual Problems

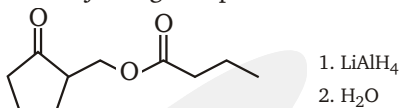


Find the value of x
where $x = \frac{\text{molecular weight of (B)}}{2}$

2. Propose a sequence of steps to carry out the following conversion:



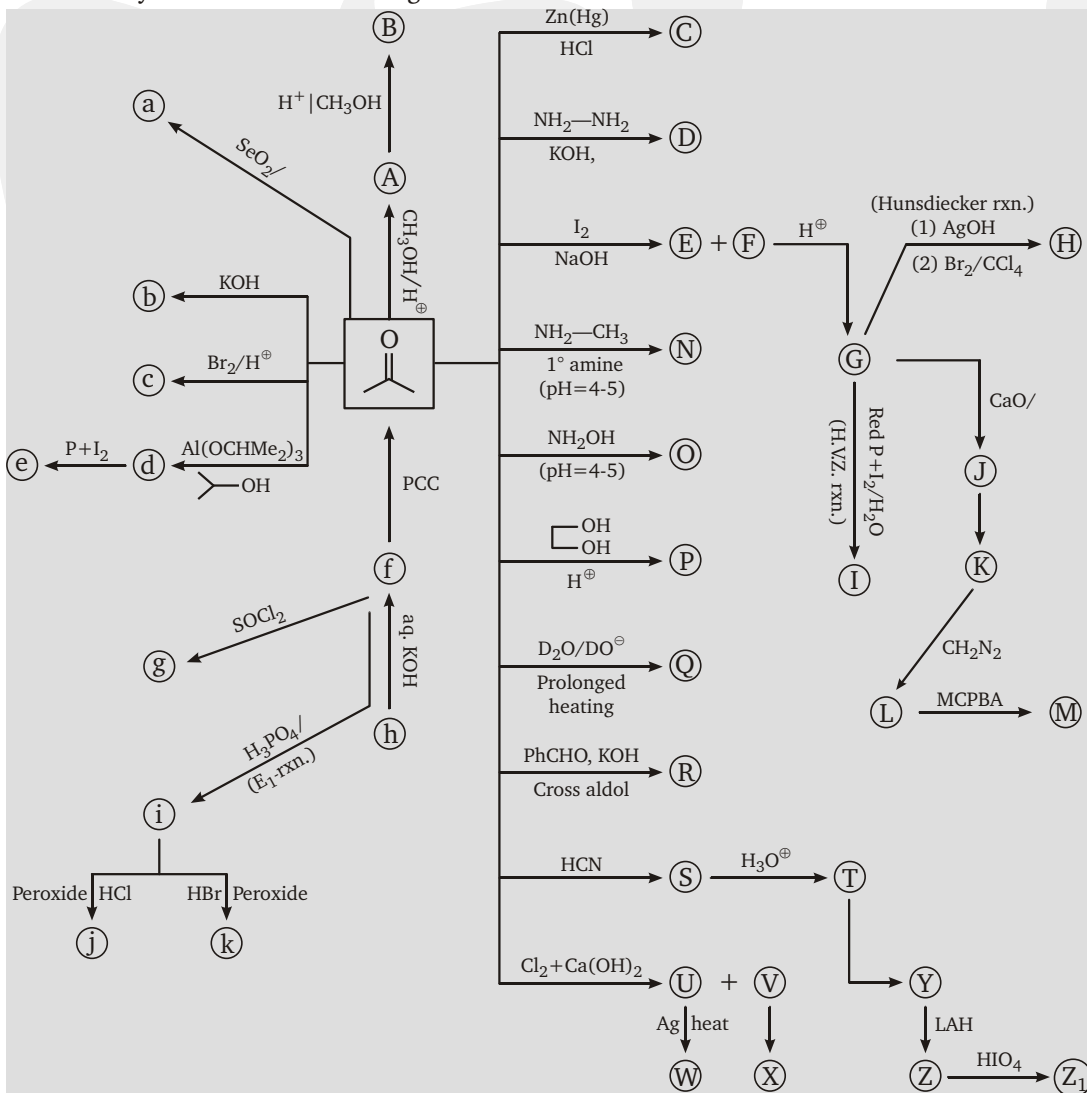
3. Provide the structure of the major organic product of the following reaction sequence .



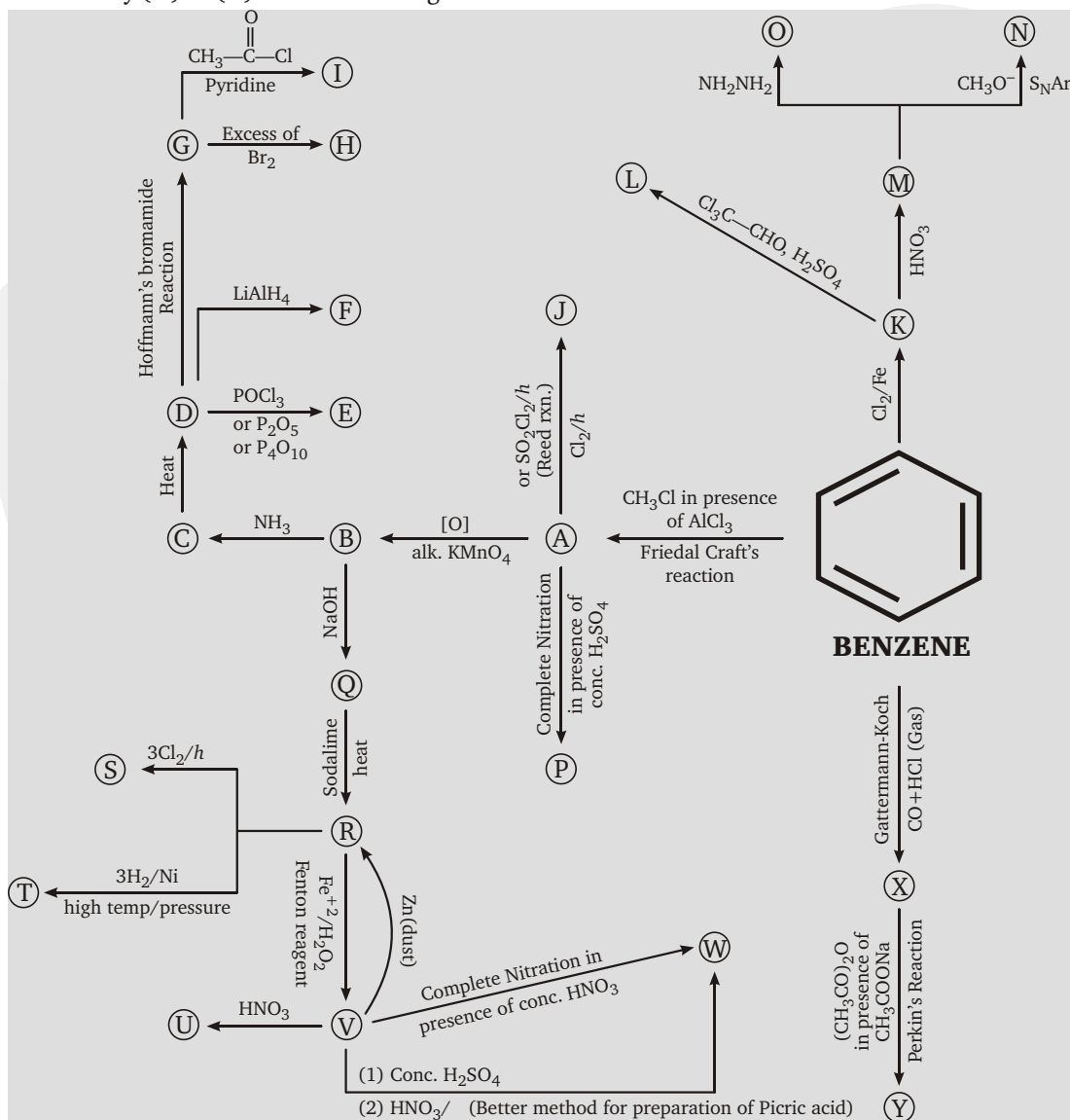
4. $\text{Ph}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH} \xrightarrow{\text{NaHCO}_3}$ (A) gas; $\text{Ph}-\text{OH} \xrightarrow{\text{Na}}$ (B) gas

Sum of molecular mass of gas (A B ?)

5. Identify A to Z in the following:



6. Identify (A) to (Y) in the following:



7.

S.No.	Compound	Reaction with aq. NaOH	Reaction with H_3O^+	Positive test with Tollen's reagent
1.				
2.				
3.				
4.	$\text{CH}_3 - \text{C} \quad \text{N}$			
5.	$\text{R} - \text{O} - \text{R}$			
6.	$\text{CH}_3 - \text{CH}_2 - \text{OH}$			
7.	$\text{Ph} - \text{O} - \text{Ph}$			
8.	$\text{Ph} - \text{O} - \text{CH}_3$			
9.				
10.				
11.				
12.				
13.	$\text{CH}_3 - \text{CHO}$			
14.	$\text{Ph} - \text{COCH}_3$			
15.				

Answers

Comprehension Based Problems:

Passage-1:	1.	(a)	2.	(d)	3.	(c)				
Passage-2:	1.	(c)	2.	(a)	3.	(b)				
Passage-3:	1.	(b)	2.	(a)	3.	(c)				
Passage-4:	1.	(a)	2.	(d)	3.	(c)				
Passage-5:	1.	(b)	2.	(a)	3.	(a)				
Passage-6:	1.	(c)	2.	(b)	3.	(a)				
Passage-7:	1.	(b)	2.	(c)	3.	(a)				
Passage-8:	1.	(d)	2.	(b)	3.	(a)	4.	(c)		
Passage-9:	1.	(b)	2.	(b)						
Passage-10:	1.	(c)	2.	$-\text{Cl}, -\text{NH}_2$	3.	(c)	4.	(a)		
Passage-11:	1.	(b)	2.	(b)	3.	(b)				
Passage-12:	1.	(a)	2.	(b)	3.	(b)				
Passage-13:	1.	(a)	2.	(a)	3.	(b)				
Passage-14:	1.	(b)	2.	(c)						
Passage-15:	1.	(a)	2.	(b)						
Passage-16:	1.	(b)	2.	(a)						

Matching Type Problems:

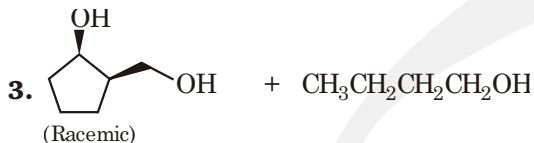
1. (a) p,s,t; (b) r,s,t; (c) p,s,t; (d) r
2. (a) p, r, s; (b) q, r; (c) p, s; (d) q, r
3. (a) p, r; (b) p, s; (c) q, s; (d) q, r
4. (a) p; (b) q; (c) r; (d) d s

Multi Concept Subjective Problems:

1.	2	2.	4	3.	1	4.	6	5.	3	6.	4	7.	3	8.	6	9.	9	10.	2
11.	5	12.	³ (a,e,k)	13.	3														

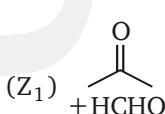
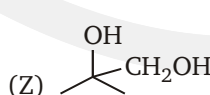
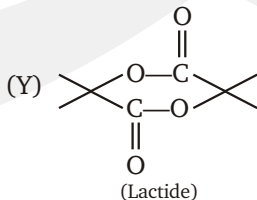
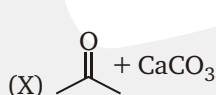
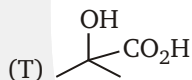
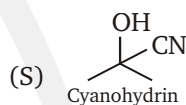
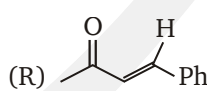
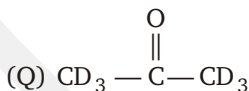
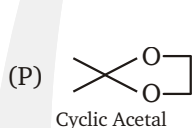
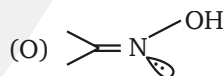
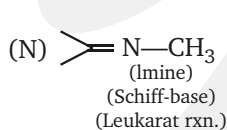
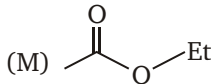
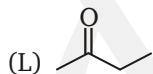
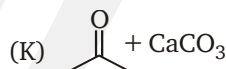
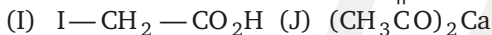
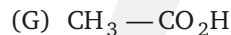
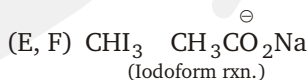
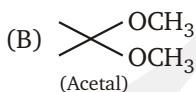
Unified Conceptual Problems

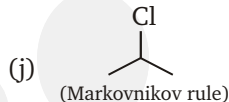
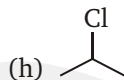
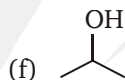
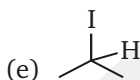
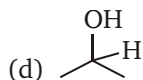
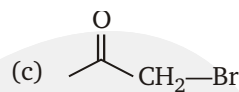
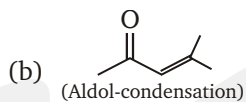
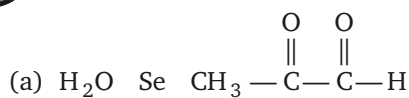
1. 67

2. (i) $\text{HOCH}_2\text{CH}_2\text{OH}$, H (ii) $\text{CH}_3\text{CH}_2\text{CH}_2\text{MgBr}$ (iii) H_3O^+ 

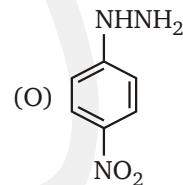
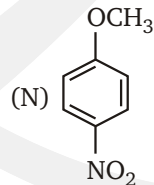
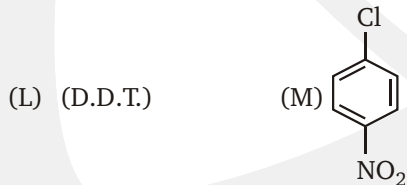
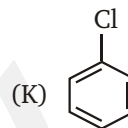
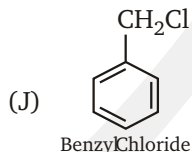
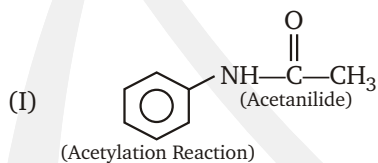
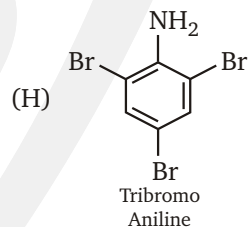
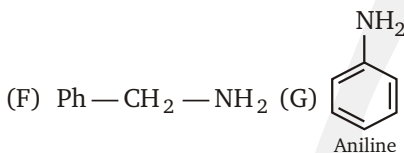
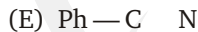
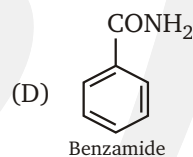
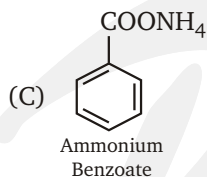
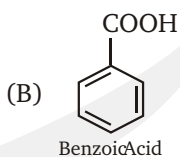
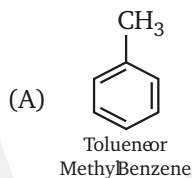
4. 46 2 48

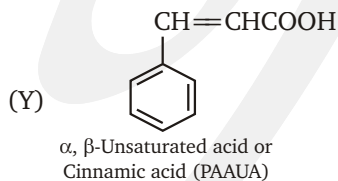
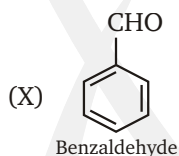
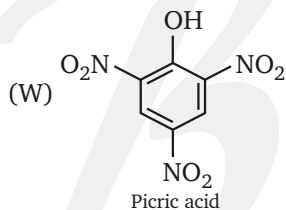
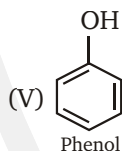
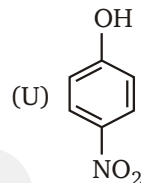
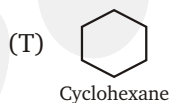
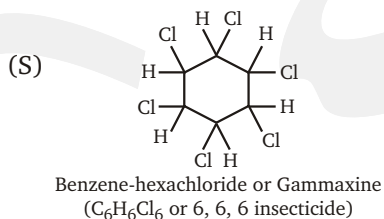
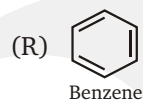
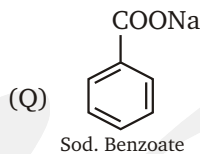
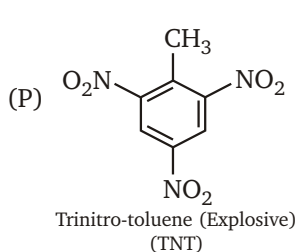
5.





6.





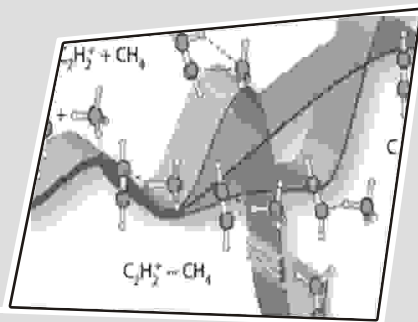
7.

	Product when react with aqueous NaOH	Product when react with H_3O^+	Positive test with Tollen's reagent
1.	✓	✓	✓
2.	✗	✓	✗
3.	✓	✓	✗
4.	✓	✓	✗
5.	✗	✓	✗
6.	✗	✗	✗

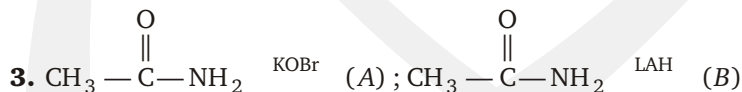
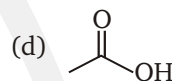
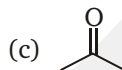
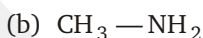
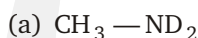
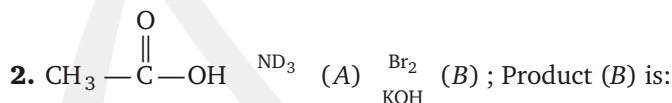
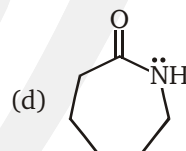
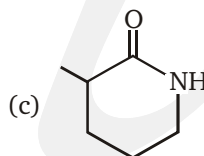
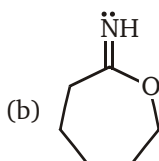
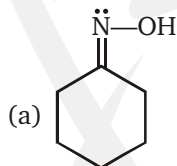
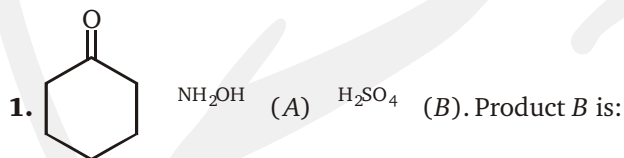
7.	X	X	X
8.	X	✓	X
9.	X	✓	X
10.	✓	✓	✓
11.	✓	✓	✓
12.	✓	✓	X
13.	✓	✓	✓
14.	✓	✓	X
15.	✓	✓	X

6

Organic Compounds Containing Nitrogen



Only One Answer is Correct



Relation between (A) and (B) is:

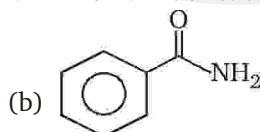
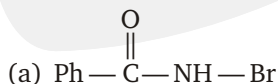
(a) Chain isomer

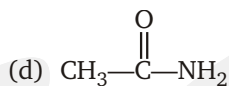
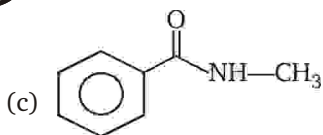
(b) Homologues

(c) Functional isomer

(d) Identical

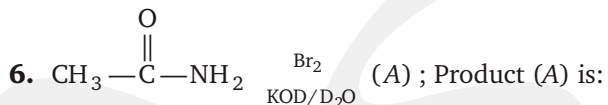
4. Which of the following will not undergo Hoffmann bromamide reaction?



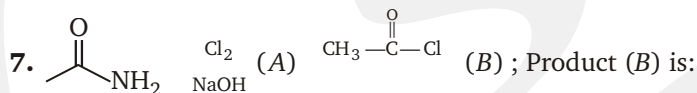


5. Number of moles of Br_2 consumed in Hoffmann bromamide reaction is:

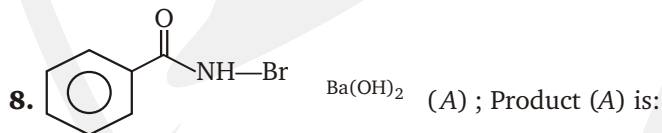
- (a) 1 (b) 2 (c) 3 (d) 4

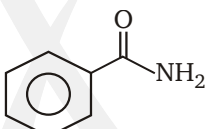
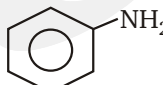
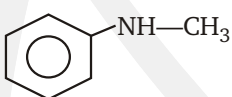
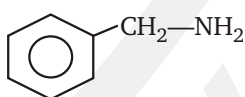


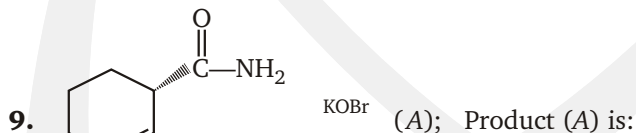
- (a) CH_3-NH_2 (b) CH_3-ND_2 (c) CH_3-OH (d) $\text{CH}_3-\text{O}-\text{D}$

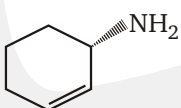
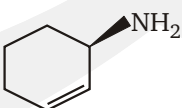
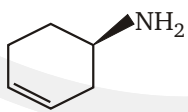
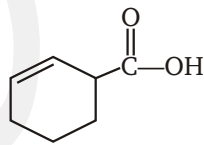


- (a) $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{NH}_2$ (b) $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{NH}-\text{Et}$ (c) $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{NH}-\text{CH}_3$ (d) $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{NH}-\text{CH}_2\text{CH}_3$



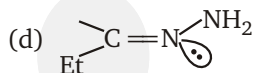
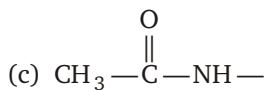
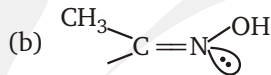
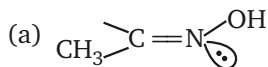
- (a)  (b) 
 (c)  (d) 



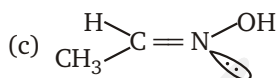
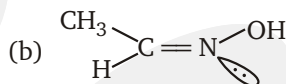
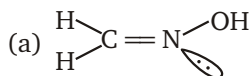
- (a)  (b)  (c)  (d) 

O

10. (A) $\xrightarrow{\text{H}_2\text{SO}_4}$ $-\text{C}-\text{NH}-\text{CH}_3$ [Ph] ; Reactant (A) is:

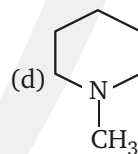
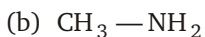
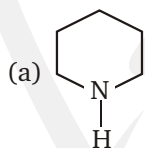


11. Which of the following is syn aldoxime ?



(d) All of these

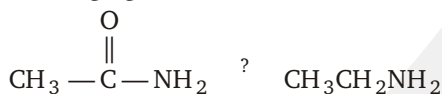
12. Which of the following give positive isocyanide test?



13. An isocyanide on reduction with hydrogen in the presence of platinum gives:

(a) amide (b) primary amine (c) secondary amine (d) alcohol

14. Which of the following reducing agents is best used in the reaction shown below ?



(a) $\text{H}_2/\text{Raney Ni}$

(b) 1. $\text{LiAlH}_4/2. \text{H}_3\text{O}^+$

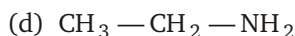
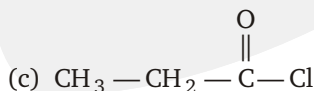
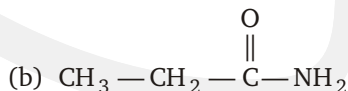
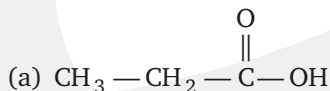
(c) Na/NH_3

(d) 1. $\text{NaBH}_4/2. \text{H}_3\text{O}^+$

(e) $\text{Zn}(\text{Hg})/\text{H}^+$

15. (A) $\xrightarrow[\text{CHCl}_3]{\text{KOH}}$ (B) $\xrightarrow[\text{H}^+]{\text{LiAlH}_4}$ $\text{CH}_3\text{CH}_2\text{NHCH}_3$

Identify compound A:

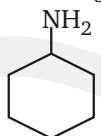


16. A nitrile can be made by dehydrating an amide. However, for this reaction to occur, the amide must be:

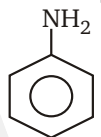
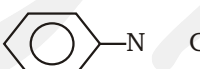
- (a) primary (b) secondary (c) tertiary (d) N-methylated (e) part of a lactam

17. $\text{CH}_3 - \text{CH}_2 - \text{NH}_2$ CHCl_3 KOH (A) ; Product (A) is:

- (a) $\text{CH}_3 - \text{CH}_2 - \text{C} \equiv \text{N}$ (b) $\text{CH}_3 - \text{CH}_2 - \text{N} = \text{C}$
(c) $\text{CH}_3 - \text{N} = \text{C}$ (d) N_2

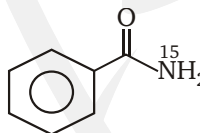
18.  $x \text{ HO}^\ominus$ isocyanide. Reagent x is:

- (a) CH_4 (b) CCl_4 (c) CHCl_3 (d) CH_2Cl_2

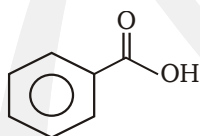
19.  CHCl_3 $x\text{KOH}$  C

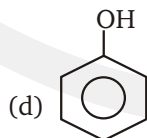
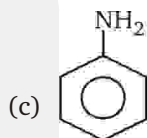
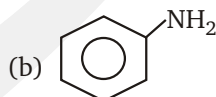
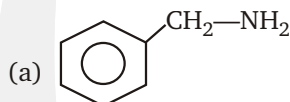
Moles of KOH used in the reaction are :

- (a) 1 (b) 2 (c) 3 (d) 4

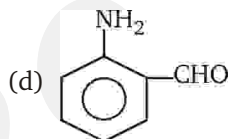
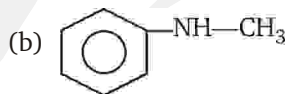
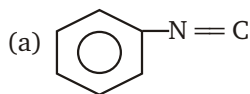
20.  Br_2 KOH (A) ; Product (A) is:

- (a) $\text{Ph} - \text{NH}_2$ (b) $\text{Ph} - \text{CH}_2 - \text{NH}_2$ (c) $\text{Ph} - \text{CO}_2\text{H}$ (d) $\text{Ph} - \text{CH}_2 - \text{OH}$

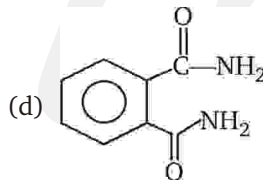
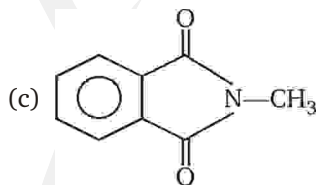
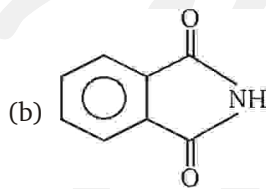
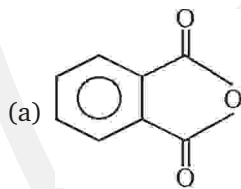
21.  NH_3 (A) Br_2 KOH (B) ; Product (B) is:



22. c1ccccc1C(=O)Cl $\xrightarrow[\text{KOH}]{\text{NH}_3}$ (A) $\xrightarrow[\text{KOH}]{\text{Br}_2}$ (B) $\xrightarrow[\text{KOH}]{\text{CHCl}_3}$ (C) ; Product (C) is:



23. (A) $\xrightarrow[\text{KOH}]{\text{Br}_2}$ Nc1ccccc1C(=O)[O-] ; Reactant (A) is:



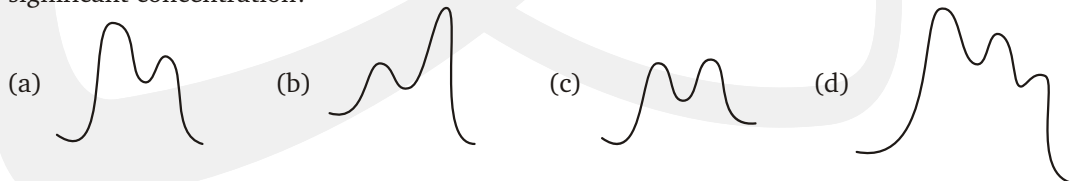
24. c1ccccc1C(=O)N $\xrightarrow[\text{(Partial hydrolysis)}]{\text{H}_3\text{O}^+}$ (A) $\xrightarrow[\text{KOH}]{\text{Br}_2}$ (B) ; Product (B) is:



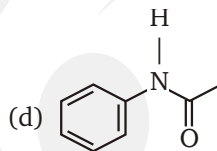
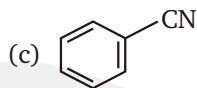
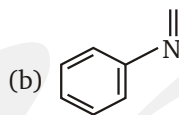
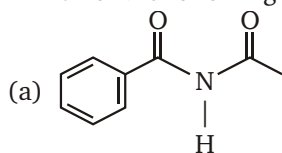
25. Which one of the following compounds when heated with KOH and *p*-amine gives carbylamine test?



26. Which energy diagram corresponds to a reaction in which an intermediate would build up to a significant concentration?

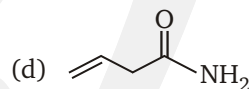
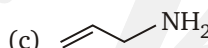
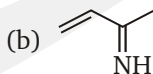
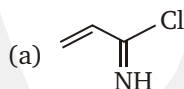
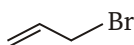


27. Which of the following compounds is an imine?

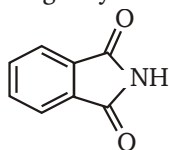


28. What is the product of the following reactions?

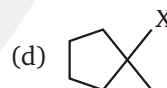
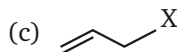
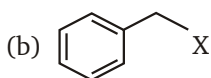
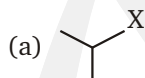
1. Mg, Et₂O
2. CO₂, Et₂O
3. H₃O⁺, H₂O
4. SOCl₂, pyridine
5. NH₃ (Solvent)



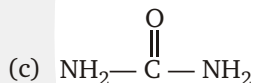
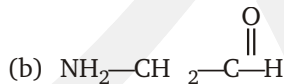
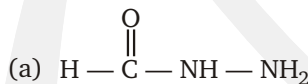
29. Which of the following alkyl halides is not a reasonable substrate for the Gabriel synthesis?



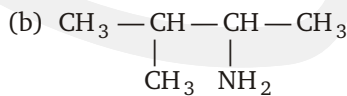
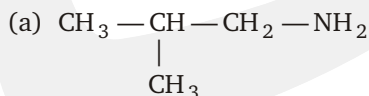
1. KOH
 2. RX
 3. NH₂NH₂, EtOH, reflux
- R—NH₂

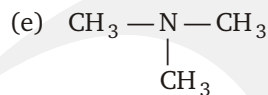
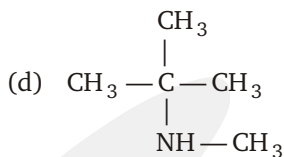
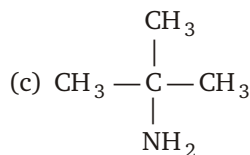


30. What is the molecular formula of urea ?

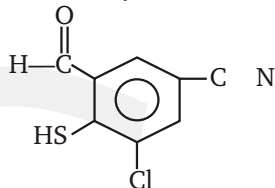


31. Which of the following is a tertiary amine?





32. How many moles of Grignard reacts with following compound ?



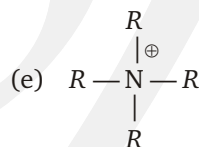
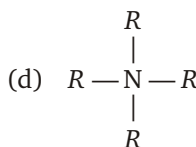
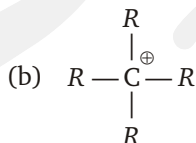
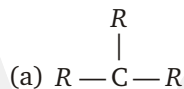
(a) 2

(b) 3

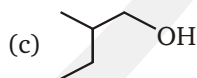
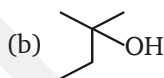
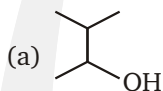
(c) 4

(d) 5

33. Which of the following is a quaternary ammonium ion ?

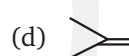
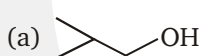


34. $\xrightarrow{\text{HNO}_2}$ (A) ; Product (A) is:
(Major)

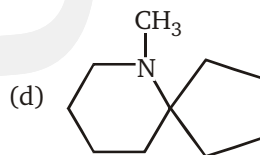
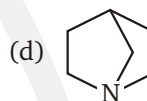
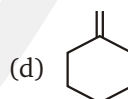
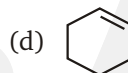
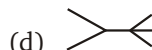
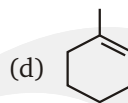


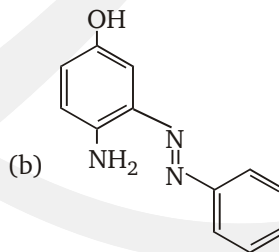
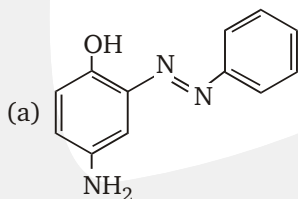
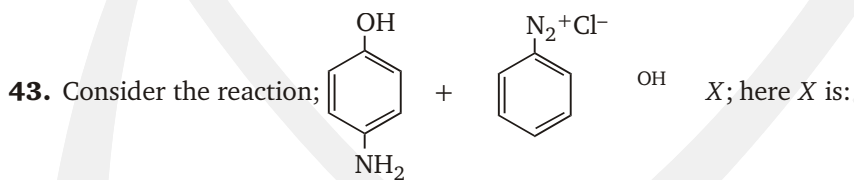
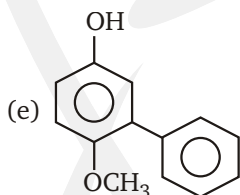
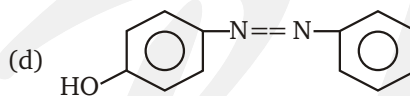
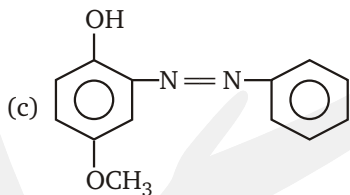
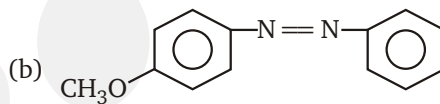
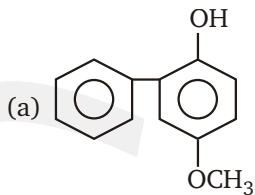
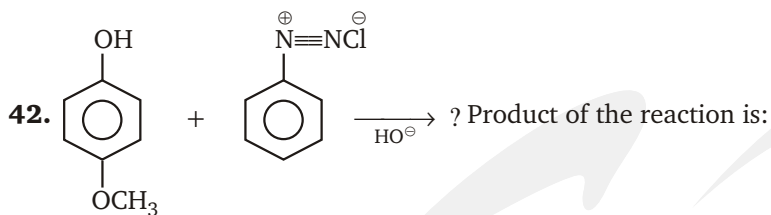
(d) All of these

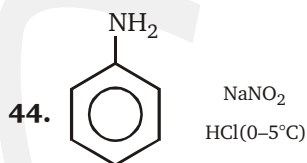
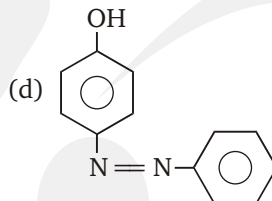
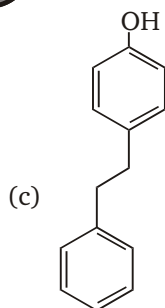
35. $\text{CH}_3 - \underset{\text{CH}_3}{\text{CH}} - \text{CH}_2 - \text{NH}_2 \xrightarrow{\text{HNO}_2}$ (A) ; Major product (A) is:



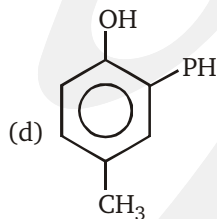
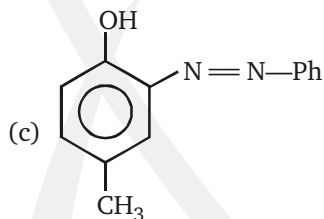
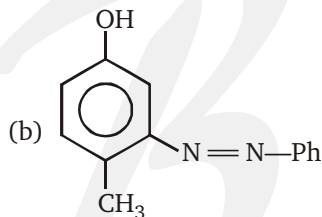
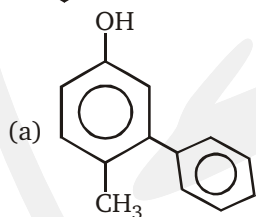
36. $\xrightarrow{\text{HNO}_2}$ (A) ; Product (A) is :



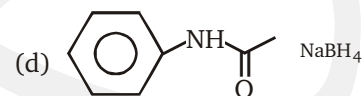
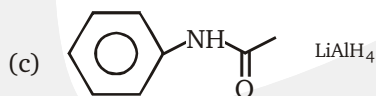
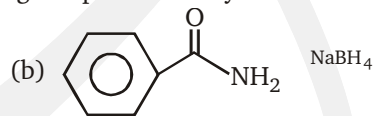
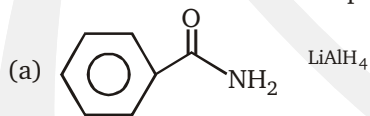




(B), Product (B) is:

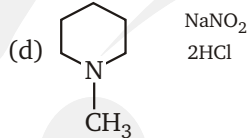
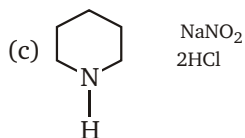
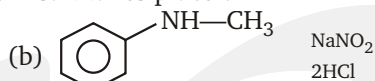
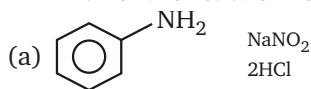


45. Find out a reaction in which product obtained gives positive isocyanide test

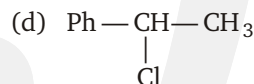
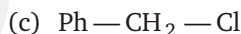
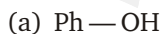


- (a) $\text{Ph}-\text{CH}_2-\text{OH}$ (b) $\text{Ph}-\text{OH}$ (c) $\text{Ph}-\text{CH}=\text{O}$ (d) $\text{Ph}-\text{CO}_2\text{H}$

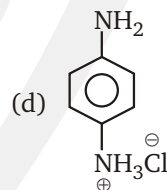
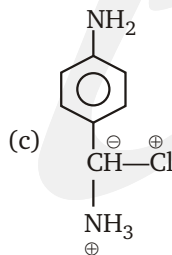
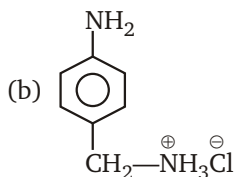
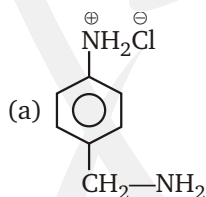
47. In which of the reaction formation of Diazonium salt takes place ?



48. $\text{Ph}-\text{CH}_2-\text{NH}_2 \xrightarrow[2\text{HCl}]{\text{NaNO}_2}$ (A)(Major) ; Product (A) is:



49.  $\xrightarrow[1\text{-mole}]{\text{H}-\text{Cl}}$; Product formed in this reaction is:

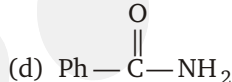
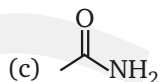
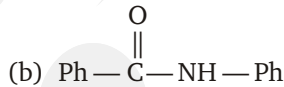
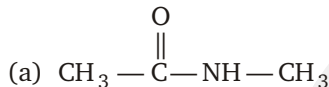


Answers

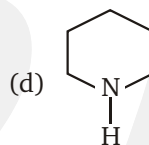
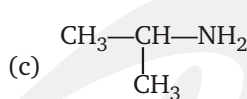
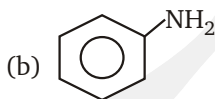
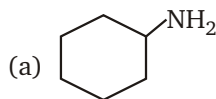
1.	(d)	2.	(b)	3.	(b)	4.	(c)	5.	(a)	6.	(b)	7.	(c)	8.	(b)	9.	(a)	10.	(a)
11.	(c)	12.	(b)	13.	(c)	14.	(b)	15.	(d)	16.	(a)	17.	(b)	18.	(c)	19.	(c)	20.	(a)
21.	(b)	22.	(a)	23.	(b)	24.	(c)	25.	(a)	26.	(b)	27.	(b)	28.	(d)	29.	(d)	30.	(c)
31.	(e)	32.	(b)	33.	(e)	34.	(b)	35.	(c)	36.	(a)	37.	(b)	38.	(a)	39.	(b)	40.	(d)
41.	(b)	42.	(c)	43.	(a)	44.	(c)	45.	(a)	46.	(a)	47.	(a)	48.	(b)	49.	(b)		

One or More than One Answers is/are Correct

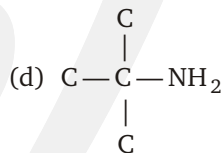
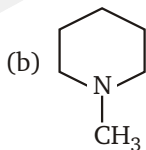
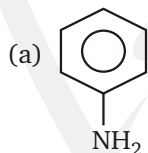
1. Which of the following compounds will not undergo Hoffmann bromamide reaction?



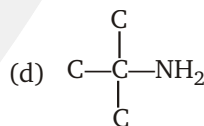
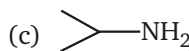
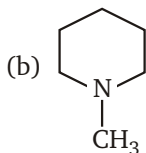
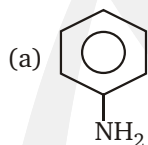
2. Which of the following gives positive carbylamine reaction?



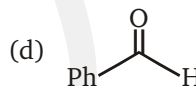
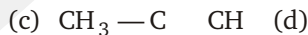
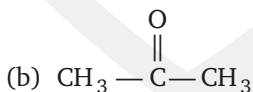
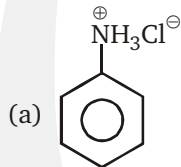
3. Which of the following gives positive carbylamine reaction?



4. Which of the following is/are primary amine?



5. Which of the following compound reacts with ammonical AgNO_3 ?



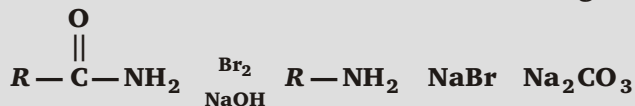
Answers

1.	(a, b)	2.	(a, b, c)	3.	(a, c, d)	4.	(a, c, d)	5.	(a, c, d)		
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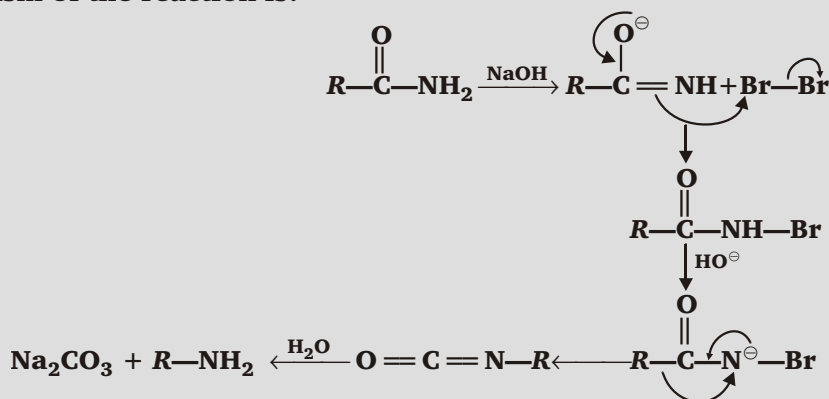
Comprehension Based Problems

Passage-1:

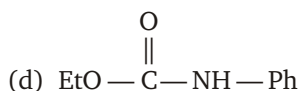
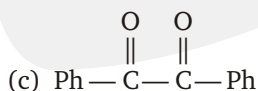
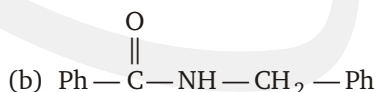
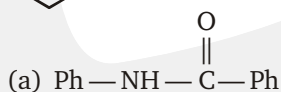
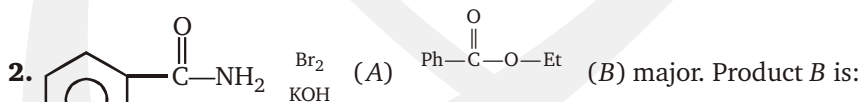
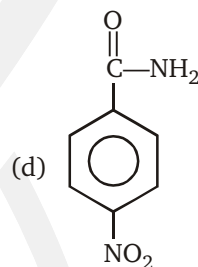
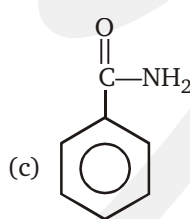
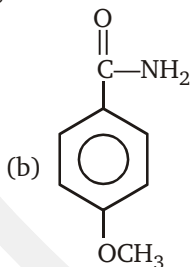
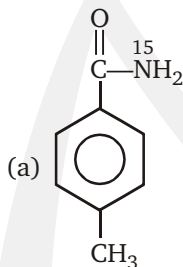
Hoffmann bromamide reaction involves conversion of a carboxylic acid amide into an amine with a loss of one carbon atom on treatment with aqueous sodium hypobromite. Thus Hoffmann result in shortening of a carbon chain.



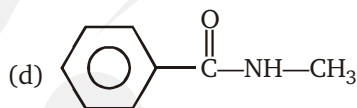
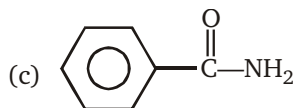
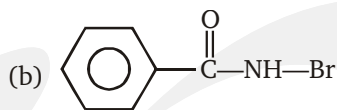
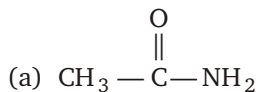
Mechanism of the reaction is:



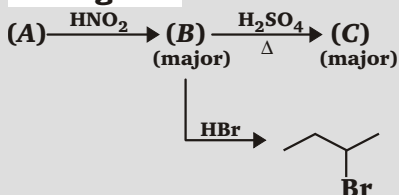
1. Which of the following most reactive towards Hoffmann bromamide reaction ?



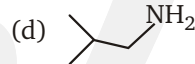
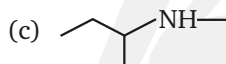
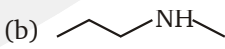
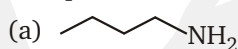
3. Which of the following will not give Hoffmann bromamide reaction ?



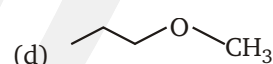
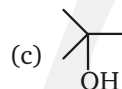
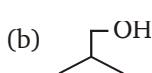
Passage-2:



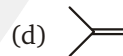
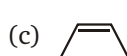
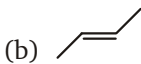
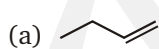
1. Compound (A) is:



2. Compound (B) is:



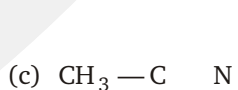
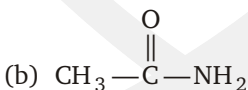
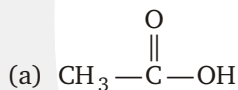
3. Compound (C) is:



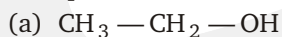
Passage-3:

$(A) \xrightarrow{\text{LiAlH}_4} (B)$; $(C) \xrightarrow{\text{LiAlH}_4} (D)$
(A) and (C) are isomers while (B) and (D) are isomers.

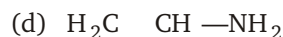
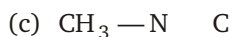
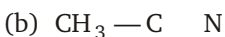
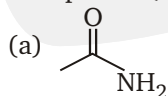
1. Compound (A) is:



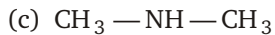
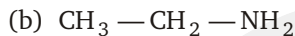
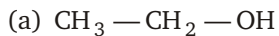
2. Compound (B) is:



3. Compound (C) is:



4. Compound (D) is:



5. Which of the following gives positive isocyanide test?

(a) A

(b) B

(c) C

(d) D

Passage-4:

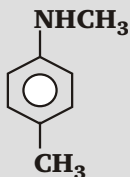
(A)



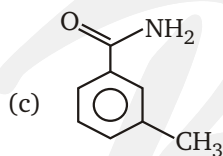
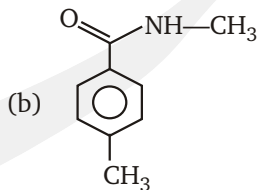
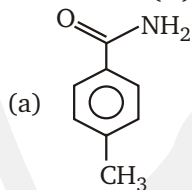
(B)



(C)

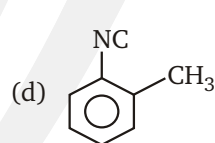
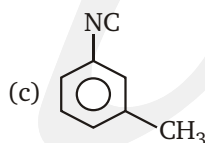
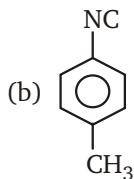
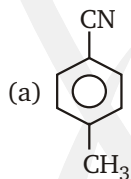


1. Structure of (A) is:



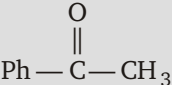
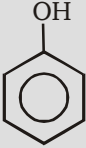
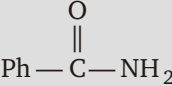
(d) none of these

2. Structure of (C) is:

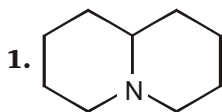


Matching Type Problems

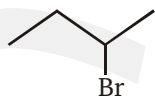
1. Which of the compound react with the following reagent ?

Column-I		Column-II	
(a)		(p)	Positive isocyanide test
(b)		(q)	Undergo Reimer Teimann reaction
(c)		(r)	Hoffmann bromamide reaction
(d)		(s)	2,4 DNP (Positive test)
		(t)	Reacts with CH_3MgBr and evolve CH_4 gas.

Multi Concept Subjective Problems



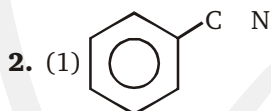
x is total number of HEM (Hoffman Exhaustive Methylation and elimination) to remove nitrogen from given compound.



alc. KOH

y is total number of possible product (including stereoisomer)

Sum of x y ?

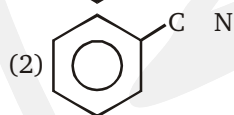


H_3O^+

(A)

NaHCO_3

(B) gas



LiAlH_4

(C)

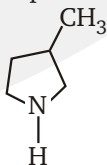
HNO_2

(D) gas

Find out the sum of molecular mass of gas $\frac{B + D}{12}$?

Unified Conceptual Problems

1. Provide the major organic product(s) of the reaction below.

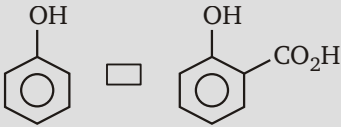
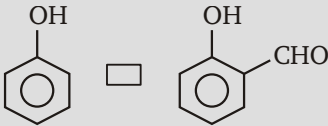
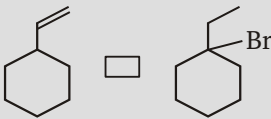
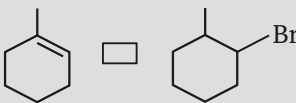


1. CH_3I (excess), K_2CO_3

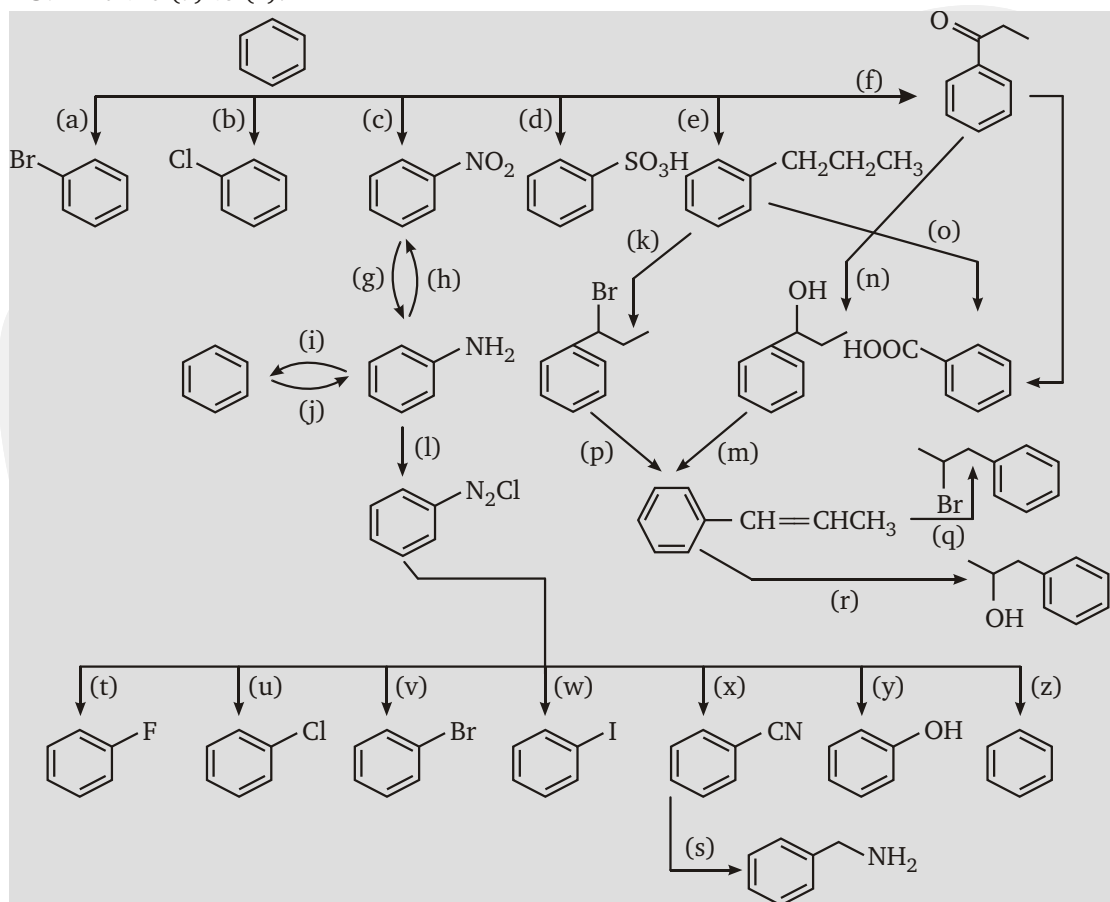
2. Ag_2O , H_2O

3. Heat

2. Find missing reagent & Name of reaction:

S.No.	Reaction	Missing Reagent	Name of reaction
(i)	$R-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{NH}_2 \quad \square \quad R-\text{NH}_2$		
(ii)	$\text{Ph}-\text{NH}_2 \quad \square \quad \text{Ph}-\text{N} \quad \text{C}$		
(iii)	 $\text{C}_6\text{H}_5\text{OH} \quad \square \quad \text{C}_6\text{H}_4(\text{OH})(\text{CO}_2\text{H})$		
(iv)	 $\text{C}_6\text{H}_5\text{OH} \quad \square \quad \text{C}_6\text{H}_4(\text{OH})(\text{CHO})$		
(v)	$\text{Ph}-\text{C}(\text{CH}_3)=\text{N}-\text{OH} \quad \square \quad \text{Ph}-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{NH}-\text{CH}_3$		
(vi)	$\text{Ph}-\text{C} \quad \text{CH} \quad \square \quad \text{Ph}-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{CH}_3$		
(vii)	 $\text{C}_6\text{H}_{11} \quad \square \quad \text{C}_6\text{H}_{11}\text{Br}$		
(viii)	 $\text{C}_7\text{H}_{12} \quad \square \quad \text{C}_7\text{H}_{13}\text{Br}$		

3. Find the (a) to (z).



Answers

Comprehension Based Problems:

Passage-1:	1.	(b)	2.	(a)	3.	(d)				
Passage-2:	1.	(a)	2.	(a)	3.	(b)				
Passage-3:	1.	(c)	2.	(b)	3.	(c)	4.	(c)	5.	(b)
Passage-4:	1.	(a)	2.	(b)						

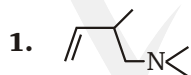
Matching Type Problems:

1. (a) p, t; (b) s; (c) q, t; (d) r, t

Multi Concept Subjective Problems:

1.	6	2.	6																
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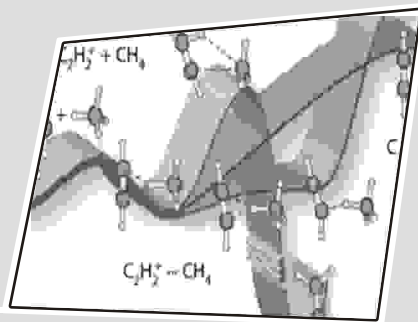
Unified Conceptual Problems



2.

- (i) Br_2 / NaOH ; Hoffmann's Bromamide
 - (ii) CHCl_3 / KOH ; Isocyanide Test/Hoffmann's Carbyl amine test
 - (iii) CO_2 / KOH ; Kolbe schmidt
 - (iv) CHCl_3 / KOH ; Reimer-Tiemann reaction
 - (v) $\text{H} / \text{PCl}_5 / \text{SO}_3$; Beckmann rearrangement
 - (vi) HgSO_4 / $\text{dil. H}_2\text{SO}_4$; Kucherov's reaction
 - (vii) HBr ; Markovnikov's addition
 - (viii) HBr ; Peroxide; Anti Markovnikov's addition
3. (a) Br_2 / Fe ; (b) Cl_2 / Fe ; (c) HNO_3 / H_2SO_4 ; (d) conc. H_2SO_4 ;
 (e) $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl} / \text{FeCl}_3$; (f) $\text{CH}_3\text{CH}_2\text{CH}_2\text{COCl} / \text{AlCl}_3$;
 (g) Sn / HCl ; (h) $\text{CF}_3\text{CO}_3\text{H}$; (i) $\text{NaNO}_2 / \text{HCl}$, H_3PO_2 ; (j) $\text{Cl}_2 / \text{AlCl}_3$, NaNH_2 ;
 (k) NBS ; (l) $\text{NaNO}_2 / \text{HCl}$; (m) conc. H_2SO_4 ; (n) $\text{NaBH}_4 / \text{LAH}(\text{H}_2 / \text{Ni})$;
 (o) hot KMnO_4 ; (p) alc. KOH ; (q) HBr /peroxide; (r) $\text{BH}_3 : \text{THF} / \text{H}_2\text{O}_2$;
 (s) LAH ; (t) HBF_4 ; (u) CuCl / HCl ; (v) HBr / CuBr ; (w) KI ; (x) CuCN ;
 (y) $\text{H}_2\text{O} / \text{boil}$; (z) H_3PO_2

Biomolecules

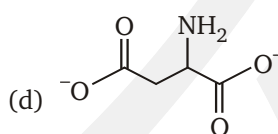
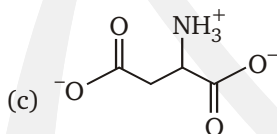
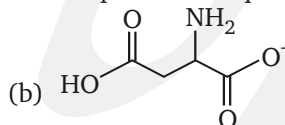
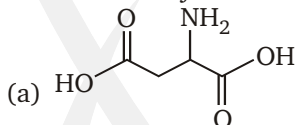


Only One Answer is Correct

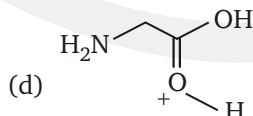
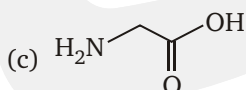
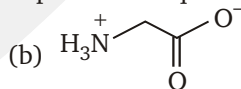
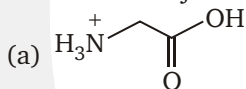
1. Which of the following is the only naturally occurring amino acid with the *R* configuration at the (-carbonyl stereogenic centers?

- (a) serine (b) arginine (c) cysteine (d) proline

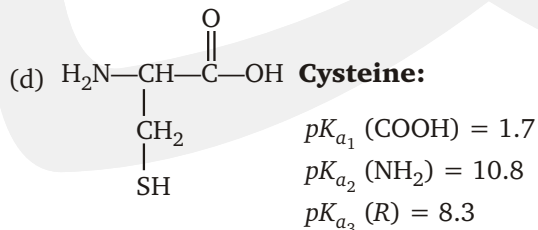
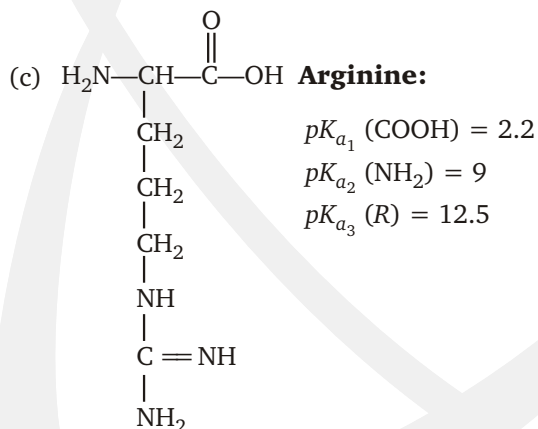
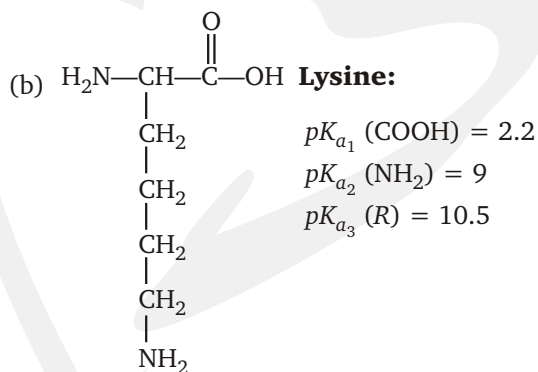
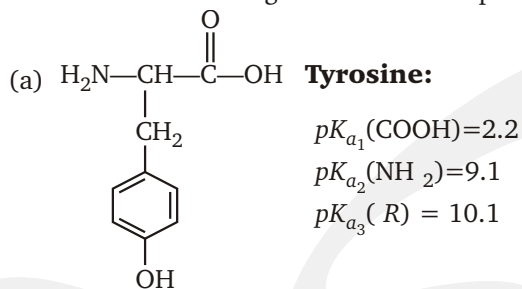
2. What is the major structure of aspartic acid present at pH = 7 in aqueous solution?



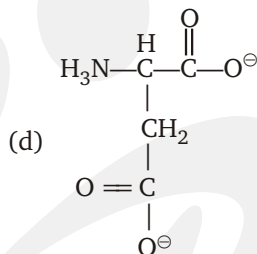
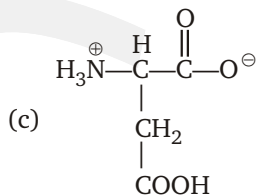
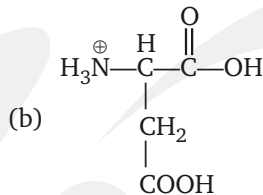
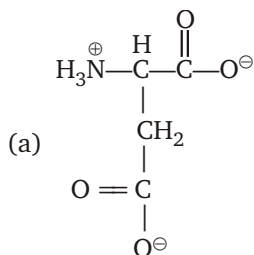
3. What is the major structure of glycine present at pH = 1 in aqueous solution?



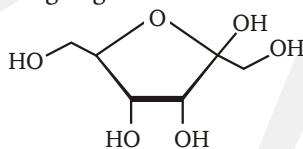
4. Which of the following amino acids has $pI \sim 9.8$?



5. Which of the following structures is called a Zwitter ion?



6. Which descriptors fit the following sugar best?



(a) ketose, furanose,

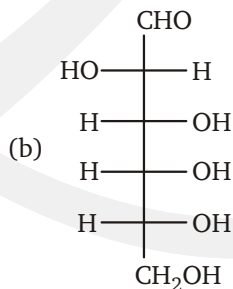
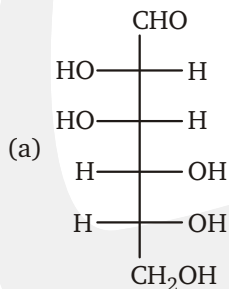
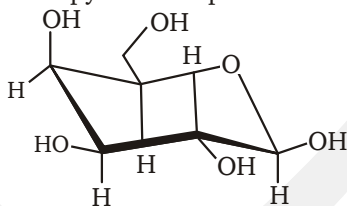
(b) ketose, furanose,

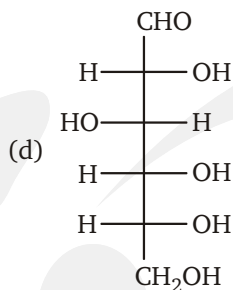
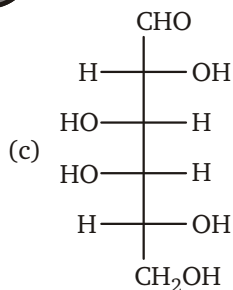
(c) aldose, pyranose,

(d) aldose, pyranose,

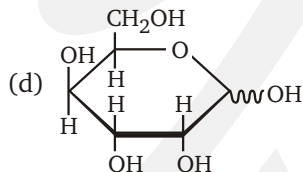
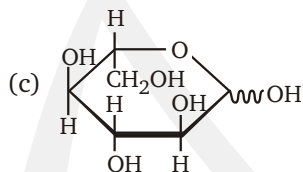
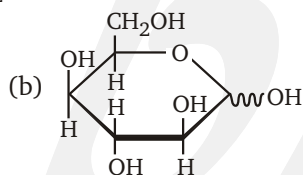
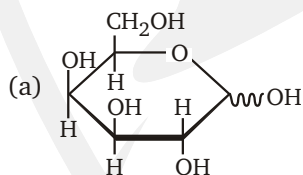
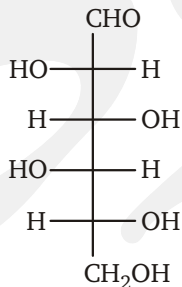
7. Which of the following is the correct Fischer representation of D-galactose?

Hint: The structure of α -D-galactopyranose is provided.





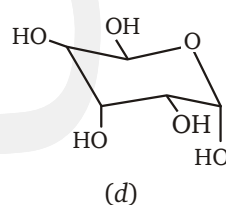
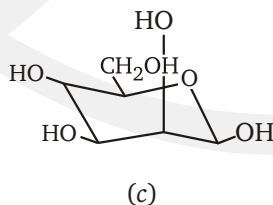
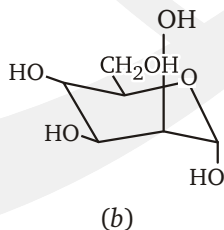
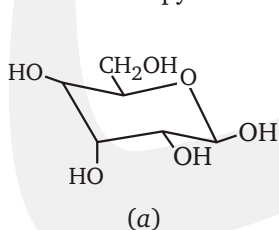
8. Which of the following structures is the correct Haworth representation of D-idose?



9. Which of the following compounds does not undergo mutarotation?

- (a) glucose (b) sucrose (c) ribose (d) fructose

10. D-Mannose is epimeric with D-glucose at C_2 . Which of the following structures represents -D-mannopyranose?



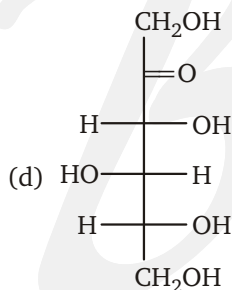
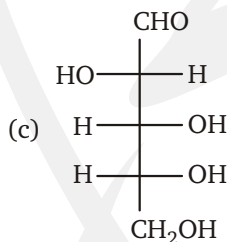
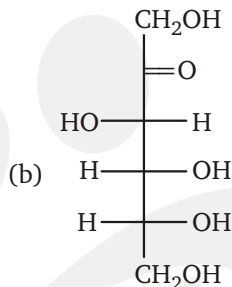
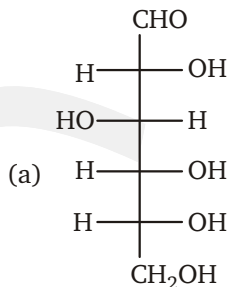
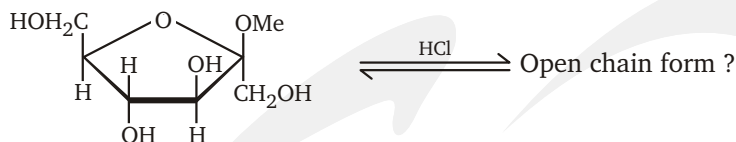
(a) a

(b) b

(c) c

(d) d

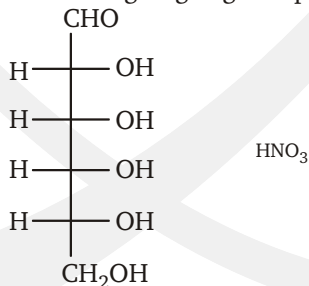
11. A glycoside is shown below. What is the structure of the carbohydrate from which it was formed?

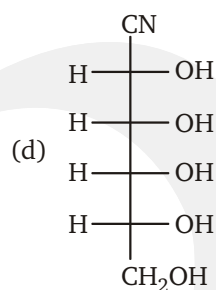
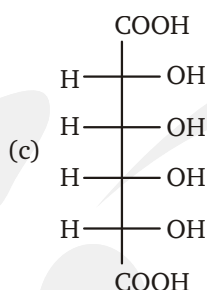
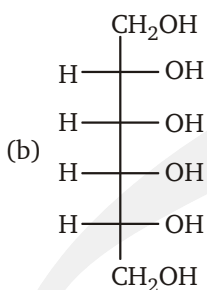
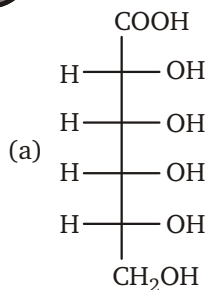


12. A glycoside is the carbohydrate form of an

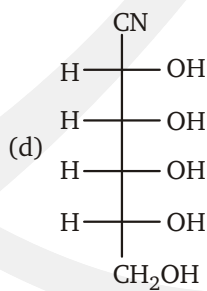
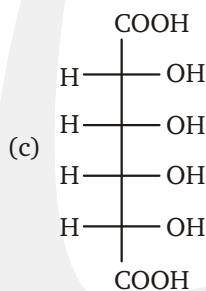
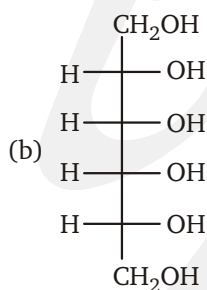
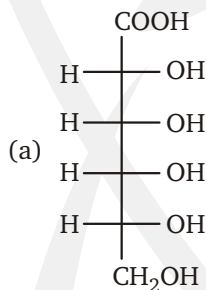
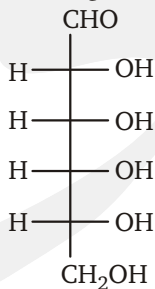
- (a) ether (b) acetal
(c) aglycone (d) alcohol

13. Which of the following structures is not going to give a positive test with Benedict's reagent?





14. Predict the product of the following reaction.



15. Reduction of hexose A (molecular formula $\text{C}_6\text{H}_{12}\text{O}_6$) with sodium borohydride gives compounds B and C. Compound B is optically inactive, whereas compound C is optically active. Which of the following is compound A?

(a) D-fructose

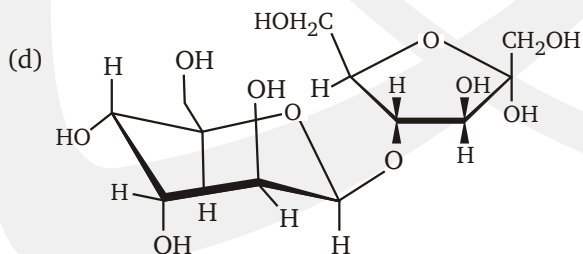
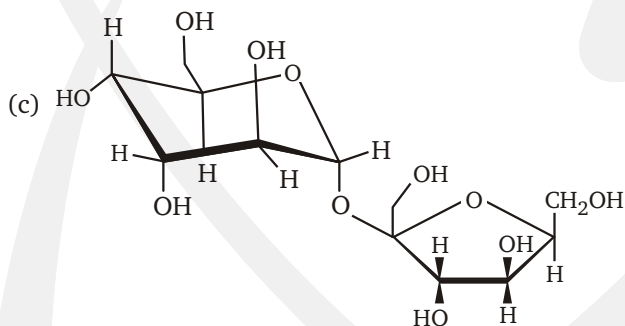
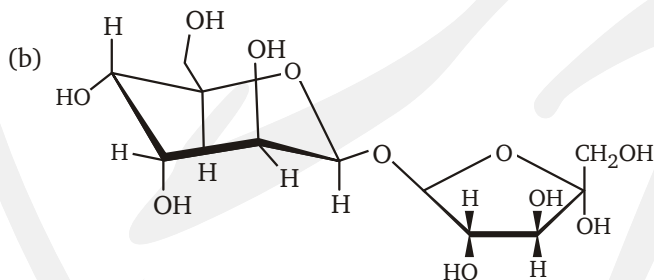
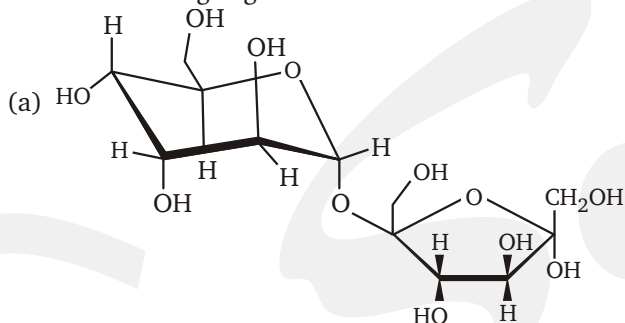
(b) D-glucose

(c) D-mannose

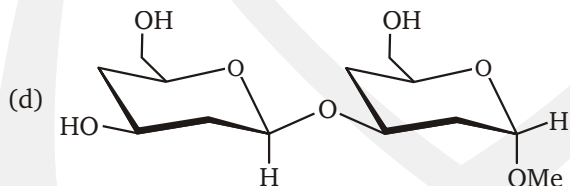
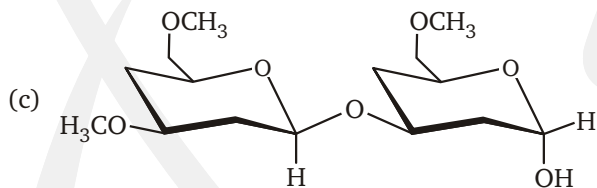
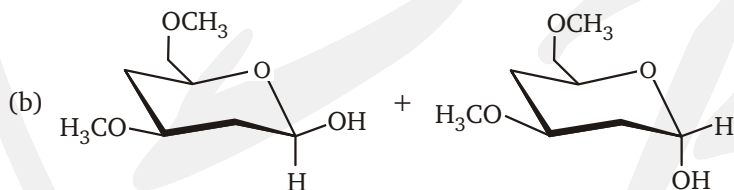
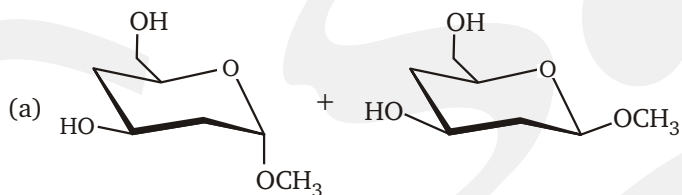
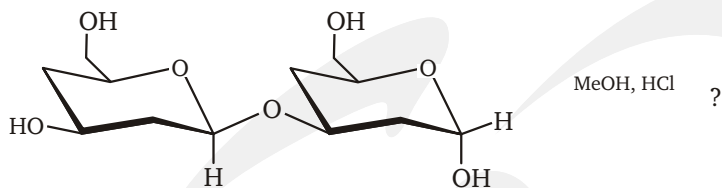
(d) D-galactose

16. Which of the following disaccharides fulfills the two requirements stated below?

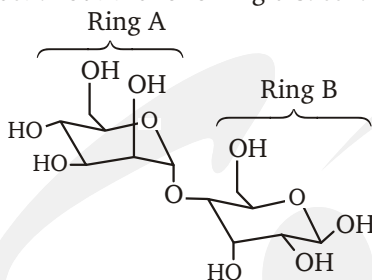
1. α -(1, 2) glycosidic linkage
2. Not a reducing sugar



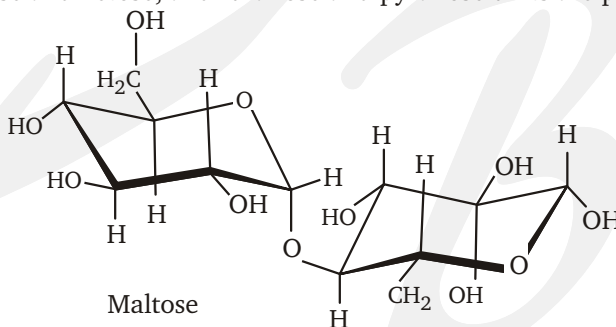
17. What is the product of the following reaction?



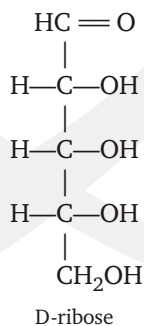
18. Which statements are correct about the following disaccharide?



- (a) Ring A is an acetal in the configuration; ring B is a hemiacetal in the configuration.
 (b) Ring A is an acetal in the configuration; ring B is a hemiacetal in the configuration.
 (c) Ring A is a hemiacetal in the configuration; ring B is an acetal in the configuration.
 (d) Ring A is a hemiacetal in the configuration; ring B is an acetal in the configuration.
19. How many aldose and ketose, and furanose and pyranose units are present in maltose?

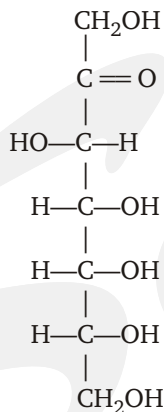


- (a) 0 aldose, 2 ketose, 0 furanose, 2 pyranose
 (b) 1 aldose, 1 ketose, 1 furanose, 1 pyranose
 (c) 2 aldose, 0 ketose, 0 furanose, 2 pyranose
 (d) 2 aldose, 0 ketose, 1 furanose, 2 pyranose
20. Given structure is :



- (a) Aldopentose (b) Aldohehexose (c) Ketopentose (d) Aldotetrose

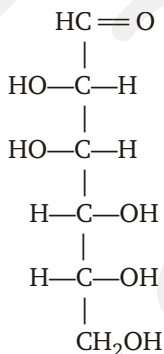
21. Given structure is :



D-sedoheptulose

- (a) Aldopentose (b) Aldoheptose (c) Ketoheptose (d) Ketohexose

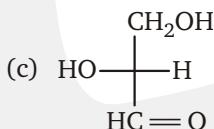
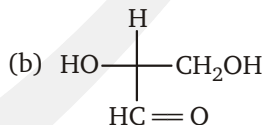
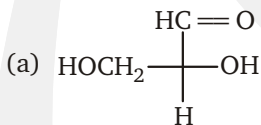
22. Given structure is :



D-mannose

- (a) Aldopentose (b) Ketohexose
 (c) Ketoheptose (d) Aldoheptose

23. Which of the following is D-glyceraldehyde ?



(d) Both (a) and (b)

24. D-Glucose $\xrightarrow[\text{H}_2\text{O}]{\text{Br}_2}$

- (a) Aldonic acid (b) Adaric acid (c) Alditol (d) Tartaric acid

25. D-Glucose will form same osazone with:

- (a) D-Mannose (b) D-Fructose (c) D-Allose (d) Both (a) and (b)

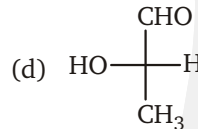
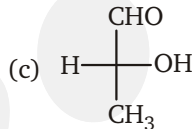
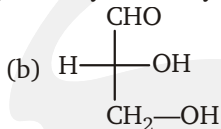
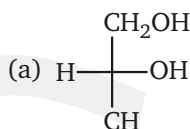
26. Relation between D-Glucose and D-Fructose is:

- (a) C₂-epimer (b) C₃-epimer (c) Functional isomer (d) Positional isomer

27. Which of following is Oligosaccharides ?

- (a) Glucose (b) Fructose (c) Sucrose (d) Starch

28. Which of the following is D-Glyceraldehyde ?



29. C₂-epimer of D-Glucose is:

- (a) D-Glucose (b) D-Allose (c) D-Altrose (d) D-Mannose

30. C₃-epimer of D-Glucose is:

- (a) D-Glucose (b) D-Allose (c) D-Altrose (d) D-Mannose

31. Which of the following carbohydrates is Aldohexose ?

- (a) D-Mannose (b) D-Glucose
(c) D-Fructose (d) Both (a) and (b)

32. D-Glucose $\xrightarrow[\text{H}_2\text{O}]{\text{Br}_2}$; Product is known as:

- (a) D-Glucitol (b) D-Gluconic acid
(c) D-Glucaric acid (d) Tartaric acid

33. D-Glucose $\xrightleftharpoons{\text{HO}^\ominus}$ D-Fructose $\xrightleftharpoons{\text{HO}^\ominus}$ (X); (X) is:

- (a) D-Allose (b) D-Altrose (c) D-Mannose (d) D-Glucose

34. D-Glucose $\xrightarrow{5\text{HIO}_4}$; Product is:

- (a) 4HCO₂H, HCHO (b) 5HCO₂H, HCHO
(c) 4HCO₂H, CO₂, HCHO (d) 5HCHO, HCO₂H

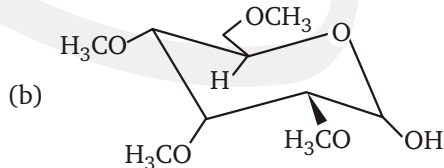
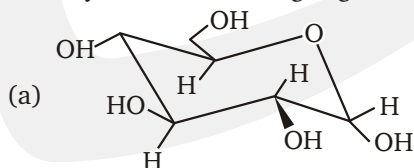
35. Which of the following pairs give same Osazone ?

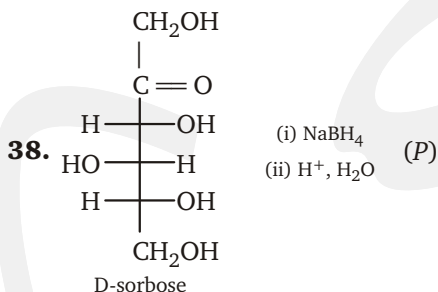
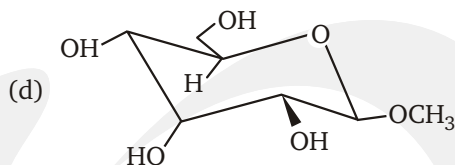
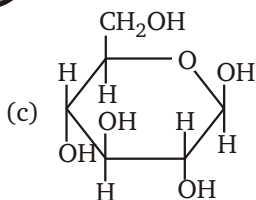
- (a) D-Glucose, Fructose (b) D-Glucose, D-Mannose
(c) D-Allose, D-Altrose (d) All of these

36. -D-Glucose and -D-Glucose are:

- (a) Epimers (b) Anomers
(c) Enantiomers (d) Acetals

37. Identify the non-reducing sugar :



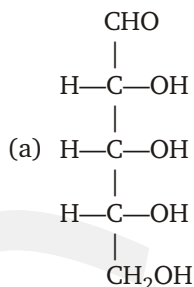


Number of stereoisomer of product (P) is:

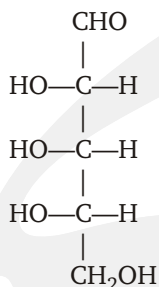
- (a) 4 (b) 10 (c) 12 (d) 16
39. The main structural features of proteins is:
(a) Ester linkage (b) Ether linkage (c) Peptide linkage (d) All of these
40. Which compound can exist in a dipolar (Zwitter ion) structure ?
(a) $\text{C}_6\text{H}_5\text{CH}_2\text{CH}(\text{N}^+\text{CH}_3)\text{COOH}$ (b) $(\text{CH}_3)_2\text{CHCH}(\text{NH}_2)\text{COOH}$
(c) $\text{C}_6\text{H}_5\text{CONHCH}_2\text{COOH}$ (d) $\text{HOOCCH}_2\text{CH}_2\text{COCOOH}$
41. Vitamin C is chemically :
(a) Ascorbic acid (b) Citric acid (c) Aspirin (d) Aspartic acid
42. In an amino acid, the carboxylic group ionises at pK_{a_1} 2.34 and ammonium ion at pK_{a_2} 9.60. The isoelectric point of the amino acid is at pH :
(a) 5.97 (b) 2.34 (c) 9.60 (d) 6.97
43. Which statement is incorrect about peptide bond ?
(a) C—N bond length in proteins is longer than usual bond length of C—N bond
(b) Spectroscopic analysis shows planar structure of $\text{—}\overset{\text{O}}{\underset{\parallel}{\text{C}}}\text{—NH—}$ bond
(c) C—N bond length in proteins is smaller than usual bond length of C—N bond
(d) None of these
44. A triglyceride can have how many different acyl groups ?
(a) 3 (b) 2 (c) 1 (d) 4
45. Glycosidic linkage is present in:
(a) Sucrose (b) Maltose (c) Lactose (d) All of these
46. Which of the following amino acid has lowest isoelectric point ?
(a) Lysine (b) Aspartic acid (c) Glycine (d) Alanine

47. Protein on hydrolysis gives:
(a) -amino acid (b) -amino acid (c) -amino acid (d) All of these
48. Which of the following amino acid is optically inactive ?
(a) Glycine (b) Alanine (c) Valine (d) Leucine
49. Peptide linkage is present in:
(a) Protein (b) Nylon-66 (c) Sucrose (d) Both (a) and (b)
50. Product obtained by hydrolysis of lactose are:
(a) Glucose and fructose (b) Glucose and mannose
(c) Mannose and fructose (d) Galactose and Glucose
51. Which of the following is amino acid ?
(a) $\text{NH}_2 - \text{CH}_2 - \text{CO}_2\text{H}$ (b) $\text{NH}_2 - \text{CH}_2 - \text{CH}_2 - \text{CO}_2\text{H}$
(c) $\text{NH}_2 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{CO}_2\text{H}$ (d) All of these
52. Which of the following is structure of α -D-Glucopyranose ?
- (a)
- (b)
- (c)
- (d)
53. Maximum number of monosaccharide units present in oligosaccharides is:
(a) 8 (b) 15 (c) 10 (d) 40
54. Which one of the following is non-reducing sugar ?
(a) Glucose (b) Arabinose (c) Fructose (d) Sucrose
55. Which one of the following is reducing sugar ?
(a) Starch (b) Cellulose (c) Glycogen (d) Fructose
56. In aqueous solution glucose exist in how many isomeric forms ?
(a) Two (b) Sixteen (c) Three (d) Four
57. Reducing property of monosaccharide is due to the presence of:
(a) —OH group (b) Keto group
(c) Acetal group (d) Anomeric hydroxyl group (a Hemiacetal)
58. Which one of the following carbohydrates will show mutarotation ?
(a) Sucrose (b) Maltose (c) Lactose (d) Both (b) and (c)
59. Glucose and mannose are:
(a) Anomers (b) Positional isomer
(c) Functional isomers (d) Epimers

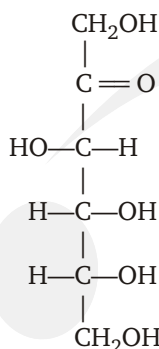
60. Which of the following pairs are epimers ?



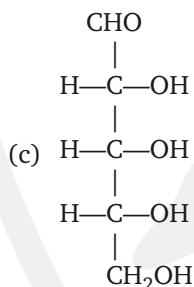
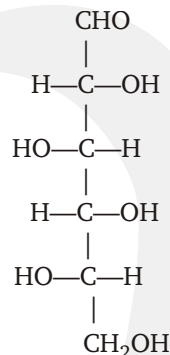
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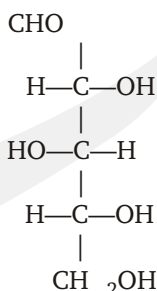
(b)



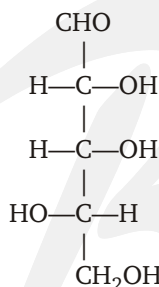
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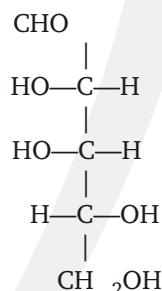
and



(d)



and



61. Glucose molecule reacts with 'X' number of molecule of phenylhydrazine to yield osazone. The value of X is:

- (a) Three (b) Two (c) One (d) Four

62. Which of the following reagents cannot distinguish between glucose and fructose ?

- (a) Tollens' reagent (b) Fehling's solution (c) Benedict's solution (d) All of these

63. Cellulose is a polymer of:

- (a) Glucose (b) Fructose (c) Ribose (d) Sucrose

64. Which of the following is an example of Ketohehexose ?

- (a) Mannose (b) Galactose (c) Maltose (d) Fructose

65. When glucose reacts with bromine water, the main product is :

- (a) Acetic acid (b) Saccharic acid (c) Glyceraldehyde (d) Gluconic acid

66. All monosaccharideTollens' reagent :

- (a) oxidises (b) reduces (c) condense with (d) add to

67. The number of chiral centres in (+) glucose is:

- (a) 4 (b) 3 (c) 2 (d) 1

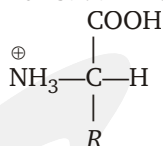
68. Protein is polymer of:

- (a) Amino acid (b) -amino acid (c) -amino acid (d) -amino acid

69. Amino acid may be:

- (a) Neutral (b) Basic (c) Acidic (d) Any one of these

70. The given structure of α -amino acid will exist at which pH ?



- (a) 7 (b) 14 (c) 0 (d) 12

71. Peptide linkage is:



72. Complete hydrolysis of cellulose gives:

- (a) L-glucose (b) D-fructose (c) D-ribose (d) D-glucose

73. Number of chiral carbons in α -D-(+)-glucose is:

- (a) six (b) three (c) four (d) five

74. The pair of compounds in which both the compounds give positive test with Tollens' reagent is:

- (a) glucose and sucrose (b) fructose and sucrose
(c) acetophenone and hexanal (d) glucose and fructose

75. D-Glucose and L-Glucose are:

- (a) Enantiomers (b) Diastereomers (c) Epimers (d) Anomers

76. Identify the pair of Epimers:

- (a) D-glucose and D-fructose (b) D-glucose and L-glucose
(c) D-glucose and D-mannose (d) D-glucose and D-gulose

77. The number of stereogenic centres in α -D-Glucose are:

- (a) 4 (b) 5 (c) 3 (d) 2

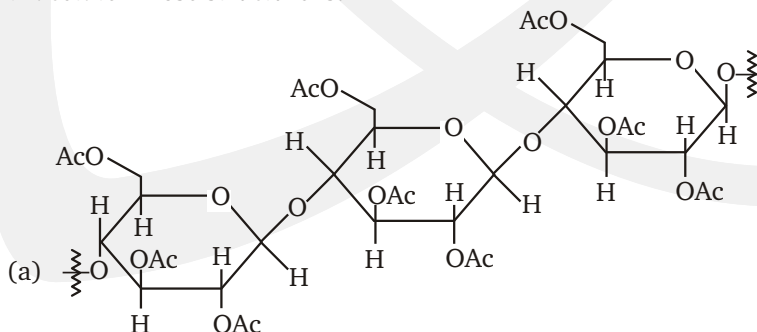
78. Which of the following pairs give positive Tollens' test ?

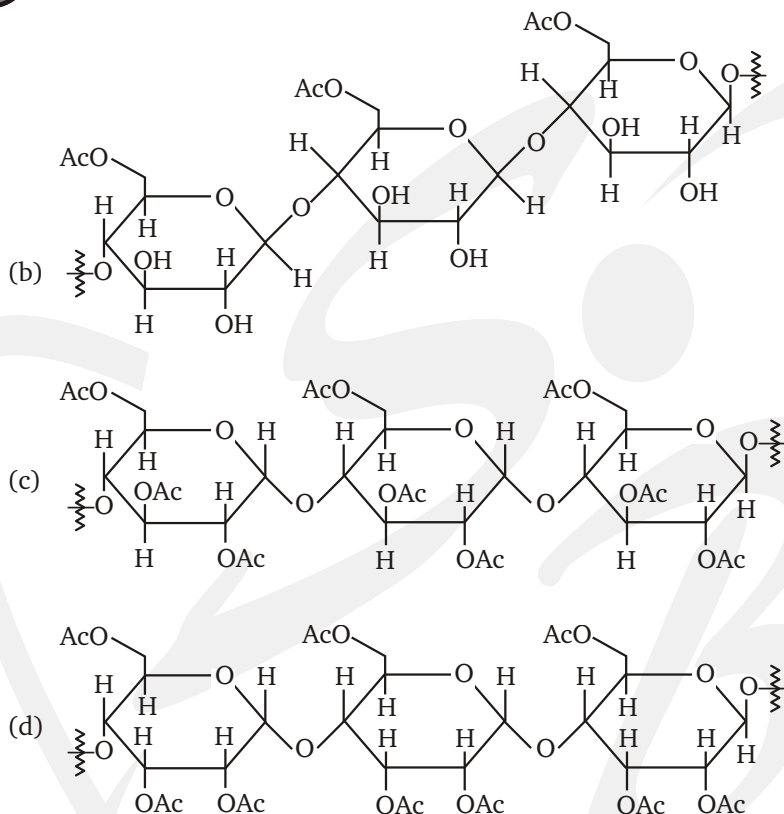
- (a) Glucose, sucrose (b) Glucose, fructose
(c) Hexanal, acetophenone (d) Fructose, sucrose

79. The two forms of D-glucopyranose obtained from the solution of D-glucose are called :

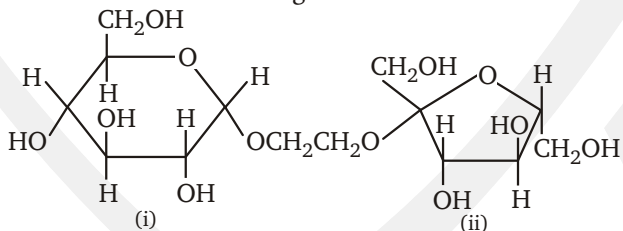
- (a) isomer (b) anomer (c) epimer (d) enantiomer

80. Cellulose upon acetylation with excess acetic anhydride/ H_2SO_4 (catalytic) gives cellulose triacetate whose structure is:



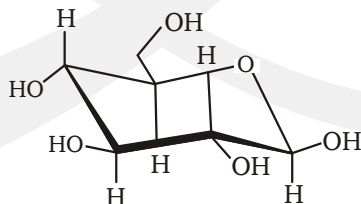


81. The correct statement about the following disaccharide is:



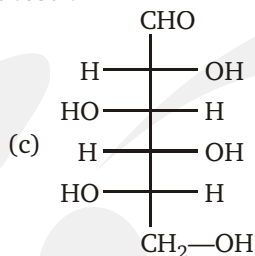
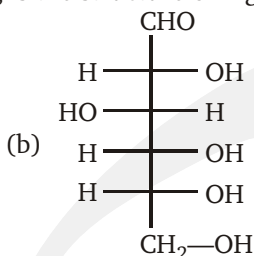
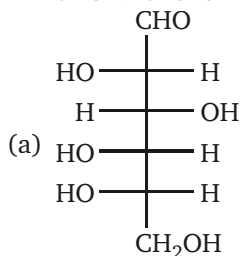
- (a) Ring (i) is pyranose with α -glycosidic link (b) Ring (i) is furanose with α -glycoside link
 (c) Ring (ii) is furanose with α -glycoside link (d) Ring (ii) is pyranose with α -glycosidic link

82. The following carbohydrate is:



- (a) a ketohexose (b) an aldohexose (c) an α -furanose (d) an α -pyranose

83. Which of the following is the structure of L-glucose ?



(d) None of these

84.
$$\begin{array}{c} \text{HC-OH} \\ || \\ \text{C-OH} \\ | \\ \text{HO} - \text{C} - \text{H} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{CH}_2\text{OH} \end{array}$$
, the given enol form is:

(a) D-glucose

(b) D-mannose

(c) D-fructose

(d) All of these

85. Component of starch is:

(a) Amylose

(b) Allose

(c) Fructose

(d) Mannose

86. A carbohydrate composed of three to ten sugar molecules is called a (n) :

(a) single carbohydrate

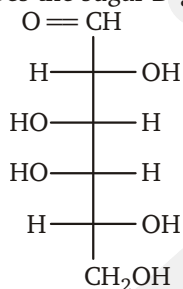
(b) disaccharide

(c) oligosaccharide

(d) polysaccharide

(e) monosaccharid

87. Which of the following best describes the sugar D-galactose,



(a) D-aldohexose

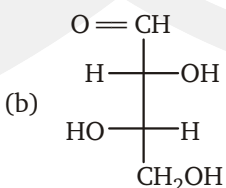
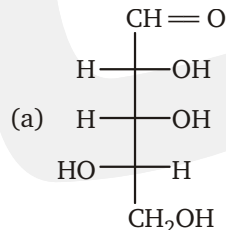
(b) D-ketohexose

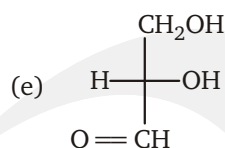
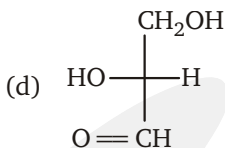
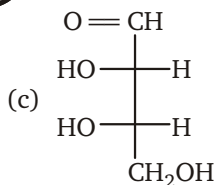
(c) L-aldohexose

(d) L-ketohexose

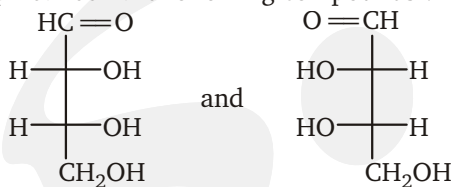
(e) D-aldopentose

88. Which of the following compounds is a D-sugar ?





89. What is the relationship between the following compounds ?

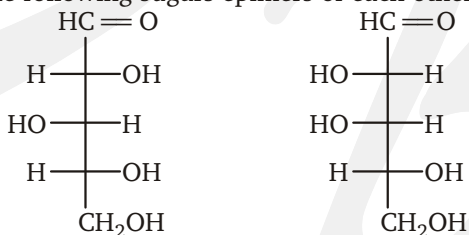


D-erythrose

L-erythrose

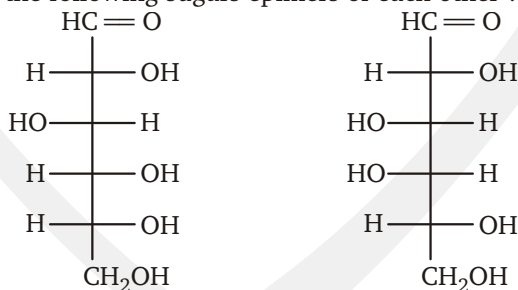
- (a) conformational isomers (b) constitutional isomers
 (c) identical (d) enantiomers (e) diastereomers

90. At which carbon are the following sugars epimers of each other ?



- (a) C-1 (b) C-2 (c) C-3 (d) C-4 (e) C-5

91. At which carbon are the following sugars epimers of each other ?



D-Glucose

D-Galactose

- (a) C-1 (b) C-2 (c) C-3 (d) C-4 (e) C-5

92. How many chirality centers are there in an aldohexose ?

- (a) 2 (b) 3 (c) 4 (d) 5 (e) 6

93. How many chirality centers are there in a 2-ketohexose ?

- (a) 2 (b) 3 (c) 4 (d) 5 (e) 6

94. How many stereoisomers are possible for a 2-ketohexose ?

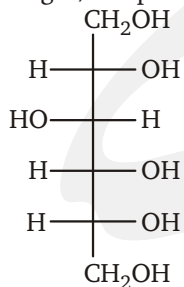
- (a) 2 (b) 4 (c) 8 (d) 16 (e) 32

95. Which of the following sugars yields the same alditol upon reduction with NaBH_4 as does D-glucose ?

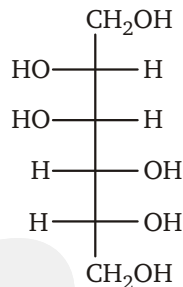
- (a) L-glucose (b) D-mannose
(c) L-gulose (d) D-arabinose (e) none of these

96. Which of the following is/are product(s) obtained from the reduction of D-fructose ?

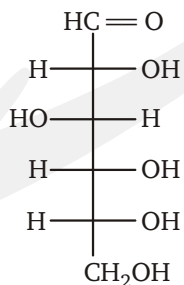
(a) D-Glucitol



(b) D-Mannitol



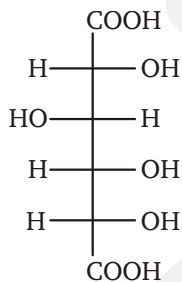
(c) D-Glucose



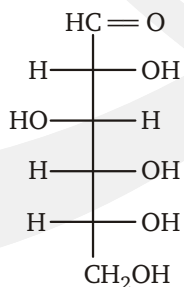
(d) (a) and (b)

(e) (a) and (c)

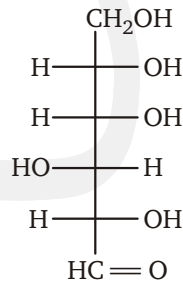
97. Which of the following monosaccharides will form D-glucaric acid upon oxidation with nitric acid ?

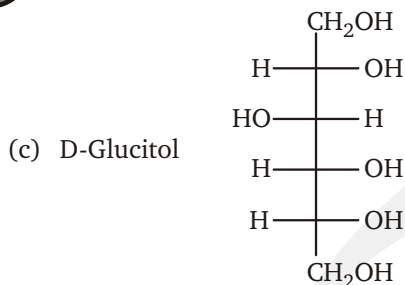


(a) D-Glucose



(b) L-Glucose





(d) (a) and (b)

(e) (a) and (c)

98. Reduction of a 2-ketohexose with NaBH_4 yields:

(a) a single aldohexose

(b) a mixture of acetals

(c) a mixture of alditols

(d) a mixture of cyclic hemiacetals

(e) a single pyranose

99. When two aldohexoses yield the same osazone upon treatment with phenylhydrazine, then they are:

(a) enantiomeric

(b) epimeric at C-2

(c) epimeric at C-3

(d) epimeric at C-5

(e) both necessarily D-sugars

100. Which of the following pairs of monosaccharides will form the same osazone when reacted with phenylhydrazine ?

(a) D-glucose and L-glucose

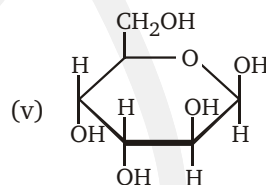
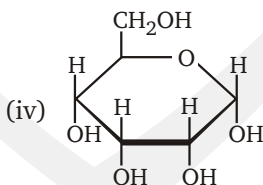
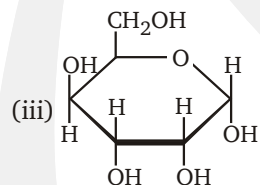
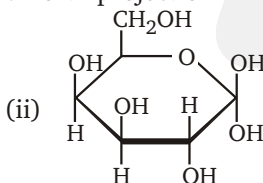
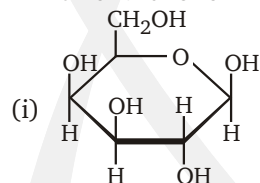
(b) D-glucose and D-gulose

(c) D-glucose and D-fructose

(d) D-glucose and D-allose

(e) D-glucose and D-galactose

101. Which of the following is the Haworth projection -D-galactopyranose ?



(a) i

(b) ii

(c) iii

(d) iv

(e) v

102. When a monosaccharide reacts to give the pyranose form from its open-chain form, how many distinct pyranose forms are possible ?

(a) 1

(b) 2

- (c) 2^n , where n is the number of carbons present
(d) $4n - 2$, where n is the number of carbons present
(e) 4

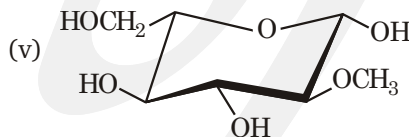
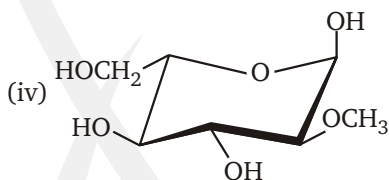
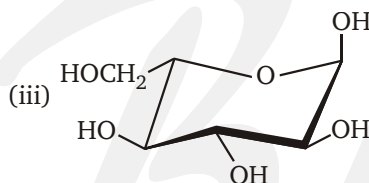
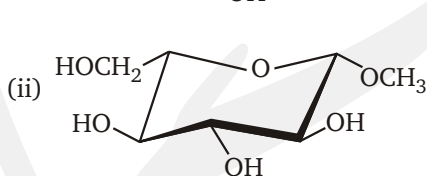
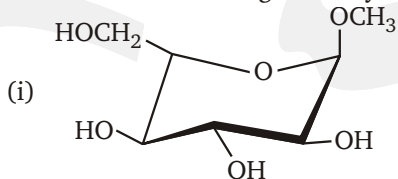
103. Anomers of D-glucopyranose differ in their stereochemistry at:

- (a) C-1 (b) C-2 (c) C-3 (d) C-4 (e) C-5

104. A pyranose with the hydroxyl group on the anomeric carbon pointing up in the Haworth structure is designated:

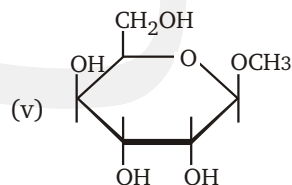
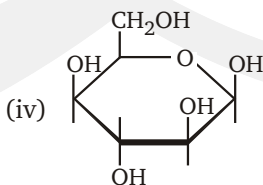
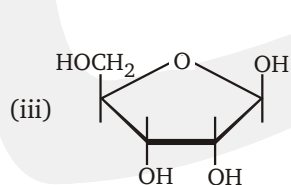
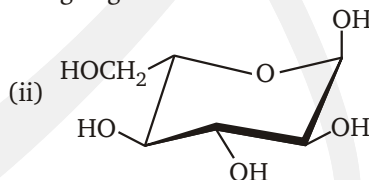
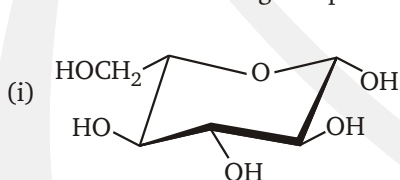
- (a) a (b) b (c) (d) (e) f

105. Which of the following is methyl- α -D-glucoside ?



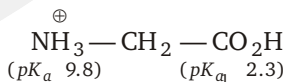
- (a) i (b) ii (c) iii (d) iv (e) v

106. Which of the following compounds is a nonreducing sugar ?



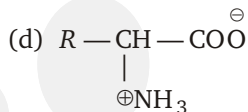
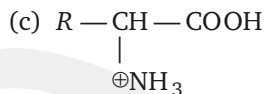
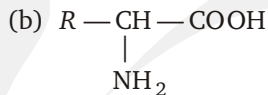
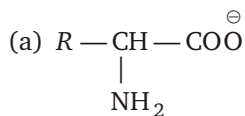
- (a) i (b) ii (c) iii (d) iv (e) v

- 107.** Which of the following statements regarding methyl- α -D-glucopyranoside is correct ?
 (a) This glycoside is a reducing sugar
 (b) This glycoside undergoes mutarotation in aqueous base
 (c) This glycoside will undergoes no reaction with treated with excess $\text{CH}_3\text{I}/\text{Ag}_2\text{O}$
 (d) This glycoside will be hydrolyzed to the cyclic hemiacetal in dilute aqueous acid
 (e) This glycoside will readily undergoes Killani-Fischer chain elongation
- 108.** Which of the following would give a positive Tollen's test ?
 (a) α -D-glucopyranose
 (b) methyl α -D-glucopyranoside
 (c) sucrose
 (d) methyl α -D-ribofuranoside
 (e) none of the above
- 109.** How many different proteins containing 100 amino acid can be made from the 20 naturally occurring amino acids ?
 (a) 100^{20} (b) 20^{100} (c) 100×20 (d) None
- 110.** The maximum concentration of zwitter ion of given amino acid will be at

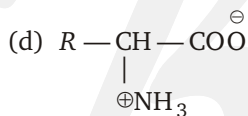
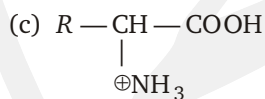
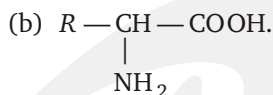
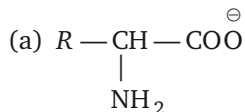


- (a) 1 (b) 11 (c) 6 (app.) (d) 5 (app.)
- 111.** Which of the following molecules is the skeletal structure of an amino acid ?
- (a) $\text{R}-\text{C}(\text{ONH}_3^+)-\text{CH}-\text{COO}^-$ (b) $\text{R}-\text{CH}(\text{NH}_3^+)-\text{CH}_2-\text{COO}^-$
- (c) $\text{R}-\text{CH}(\text{NH}_3^+)-\text{COO}^-$ (d) $\text{R}-\text{CH}(\text{NH}_3^+)-\text{C}(=\text{O})-\text{COO}^-$ (e) $\text{R}-\text{CH}(\text{COO}^-)-\text{C}(=\text{O})-\text{NH}_3^+$
- 112.** Which of the following is an *L*-amino acid ?
- (i) $\text{H}_3\text{N}^+-\text{C}(\text{H})_2-\text{COO}^-$ (ii) $\text{H}_3\text{N}^+-\text{C}(\text{H})(\text{R})-\text{COO}^-$
- (iii) $\text{H}-\text{C}(\text{R})(\text{NH}_3^+)-\text{COO}^-$ (iv) $\text{H}_3\text{N}^+-\text{C}(\text{H})(\text{R})-\text{COO}^-$ (v) $\text{R}-\text{C}(\text{H})(\text{NH}_3^+)-\text{COO}^-$
- (a) i (b) ii (c) iii (d) iv (e) v

113. Which of the following best represents the structure of an amino acid in basic solution (pH 11)?



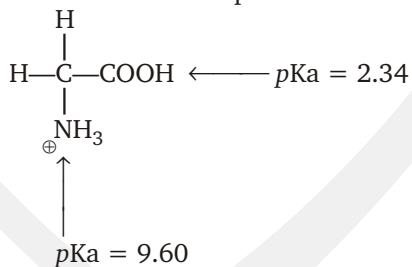
114. Which of the following is a Zwitter ion ?



115. Which of the following amino acids has the highest isoelectric point ?

- (a) lysine (b) aspartic acid
(c) glycine (d) proline (e) phenylalanine

116. What is the pI of glycine ? The structure and pKa values are shown below.

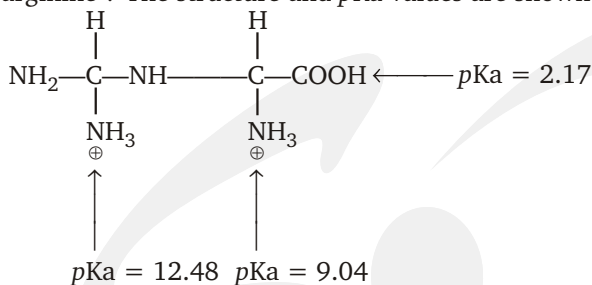


- (a) 7.26 (b) 5.97 (c) 3.63 (d) 7.50 (e) 11.94

117. Which of the following amino acids has its isoelectric point at the lowest pH ?

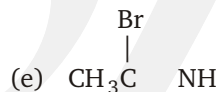
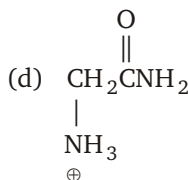
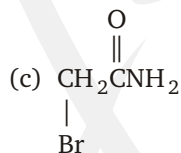
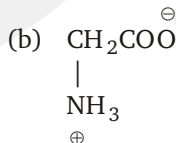
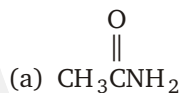
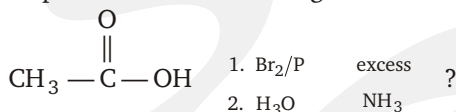
- (a) glutamic acid (b) lysine
(c) valine (d) glycine (e) methionine

118. What is the pI of arginine ? The structure and pKa values are shown below.



- (a) 10.76 (b) 7.90 (c) 5.61 (d) 7.33 (e) 9.67

119. What is the major organic product of the following reaction ?



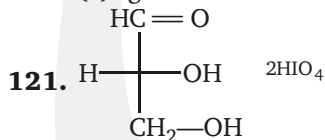
120. What is the name of the amino acid produced when propanoic acid is subjected to the following sequence of reagents :

1. $\text{PBr}_3, \text{Br}_2$;
2. H_2O
3. $\text{NH}_3, ?$

- (a) alanine
 (c) glutamic acid

- (b) aspartic acid
 (d) valine

(e) asparagine

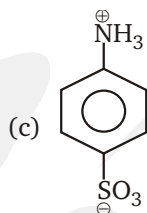
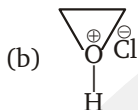
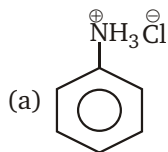


Product formed is:

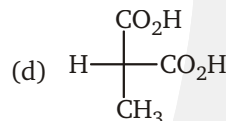
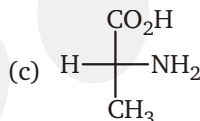
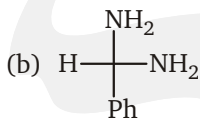
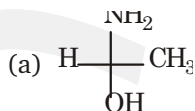
- (a) $2\text{HCHO}, \text{HCO}_2\text{H}$
 (c) CO_2, HCHO

- (b) $\text{HCHO}, 2\text{HCO}_2\text{H}$
 (d) $\text{CO}_2, \text{HCO}_2\text{H}$

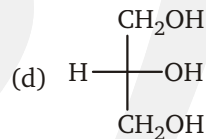
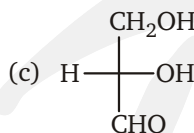
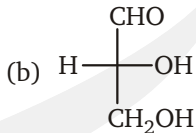
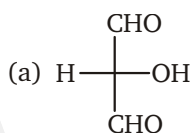
122. Which of the following is Zwitter ion ?



123. Which of the following compound can form Zwitter ion?



124. Which of the following is *D*-carbohydrate ?



125. Which of the following pair will form same osazone ?

(a) D-Glucose, D-Allose

(b) D-Glucose, D-Altrose

(c) D-Alose, D-Altrose

(d) D-Fructose, D-Galactose

126. Total number of stereoisomers possible for ketohexose are :

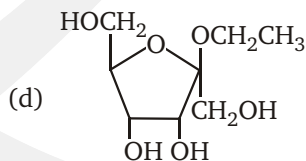
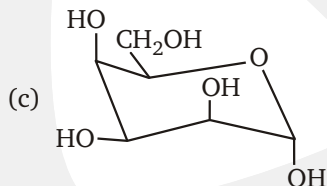
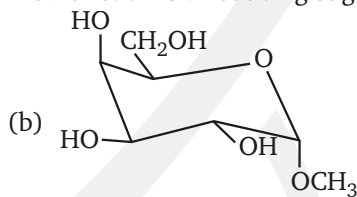
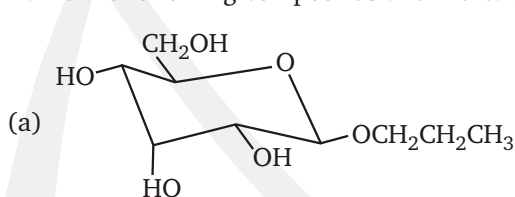
(a) 4

(b) 8

(c) 16

(d) 32

127. Name the following compounds and indicate whether each is a reducing sugar:

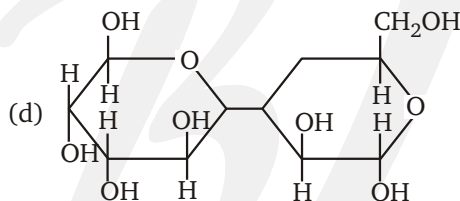
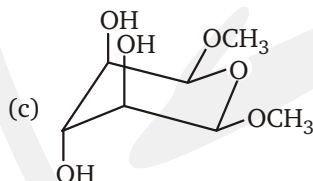
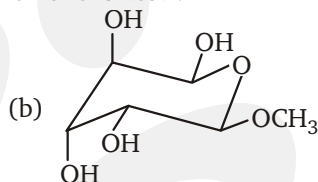
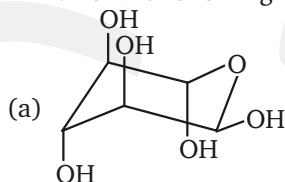


Answers

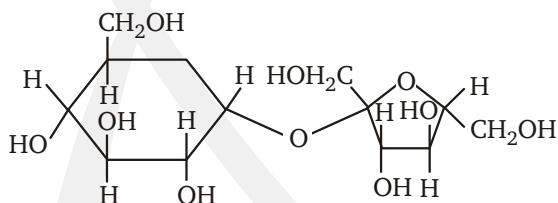
1.	(c)	2.	(c)	3.	(a)	4.	(b)	5.	(c)	6.	(b)	7.	(c)	8.	(b)	9.	(b)	10.	(c)
11.	(b)	12.	(b)	13.	(b)	14.	(c)	15.	(a)	16.	(c)	17.	(a)	18.	(a)	19.	(c)	20.	(a)
21.	(c)	22.	(d)	23.	(d)	24.	(a)	25.	(d)	26.	(c)	27.	(c)	28.	(b)	29.	(d)	30.	(c)
31.	(d)	32.	(b)	33.	(c)	34.	(b)	35.	(d)	36.	(b)	37.	(d)	38.	(b)	39.	(c)	40.	(b)
41.	(a)	42.	(a)	43.	(a)	44.	(a)	45.	(d)	46.	(b)	47.	(a)	48.	(a)	49.	(d)	50.	(d)
51.	(d)	52.	(b)	53.	(c)	54.	(d)	55.	(d)	56.	(c)	57.	(d)	58.	(d)	59.	(d)	60.	(c)
61.	(a)	62.	(d)	63.	(a)	64.	(d)	65.	(d)	66.	(b)	67.	(a)	68.	(b)	69.	(d)	70.	(c)
71.	(d)	72.	(d)	73.	(d)	74.	(d)	75.	(a)	76.	(c)	77.	(b)	78.	(b)	79.	(b)	80.	(a)
81.	(a)	82.	(b)	83.	(a)	84.	(d)	85.	(a)	86.	(c)	87.	(a)	88.	(d)	89.	(d)	90.	(b)
91.	(d)	92.	(c)	93.	(b)	94.	(c)	95.	(c)	96.	(d)	97.	(e)	98.	(c)	99.	(b)	100.	(c)
101.	(a)	102.	(b)	103.	(a)	104.	(d)	105.	(a)	106.	(e)	107.	(d)	108.	(a)	109.	(b)	110.	(c)
111.	(c)	112.	(b)	113.	(a)	114.	(d)	115.	(a)	116.	(b)	117.	(a)	118.	(a)	119.	(b)	120.	(a)
121.	(b)	122.	(c)	123.	(c)	124.	(b)	125.	(c)	126.	(b)	127.	(c)						

One or More than One Answers is/are Correct

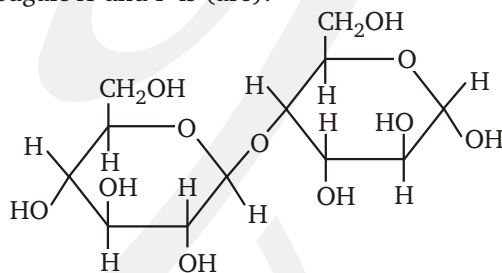
1. Which of the following pairs forms the same osazone with phenylhydrazine ?
 (a) D-Glucose and D-Fructose (b) D-Fructose and D-Mannose
 (c) D-Glucose and D-Mannose (d) D-Glucose and D-Galactose
2. Which of the following compounds give positive Tollens' test ?



3. The correct statement(s) about the following sugars X and Y is (are):



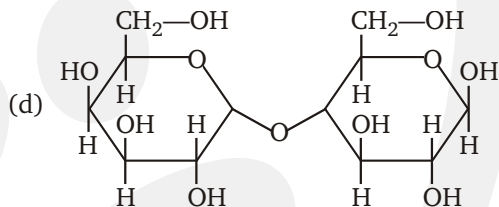
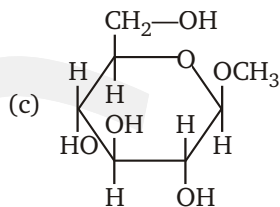
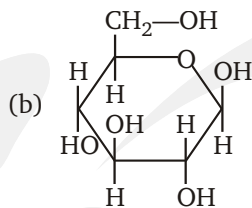
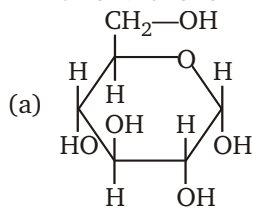
(X)



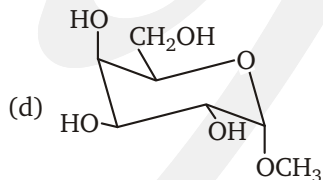
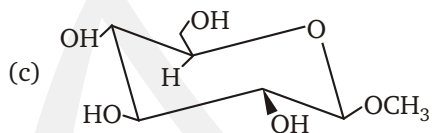
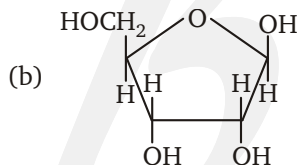
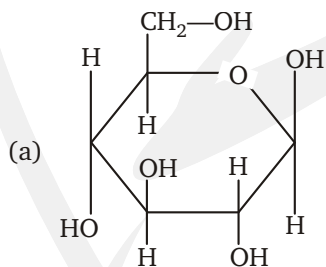
(Y)

- (a) X is a reducing sugar and Y is a non-reducing sugar
 (b) X is a non-reducing sugar and Y is a reducing sugar
 (c) The glucosidic linkages in X and Y are and , respectively
 (d) The glucosidic linkages in X and Y are and , respectively
4. Which of the following is reducing sugar ?
 (a) Sucrose (b) Maltose
 (c) Lactose (d) -D-Glucopyranose
5. In which of the following polymer there is presence of peptide linkage ?
 (a) Starch (b) Cellulose (c) Protein (d) Terylene

6. Which of the following is reducing sugar ?



7. Which of the following compound will show mutarotation ?



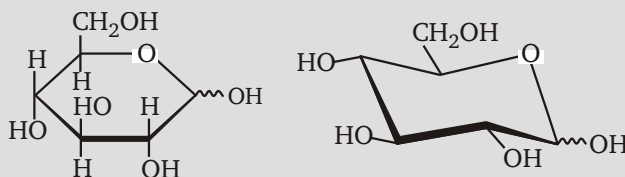
Answers

1.	(a, b, c)	2.	(a, b, d)	3.	(b, c)	4.	(b, c, d)	5.	(c)	6.	(a, b, d)
7.	(a, b)										

Comprehension Based Problems

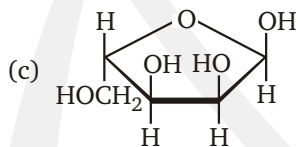
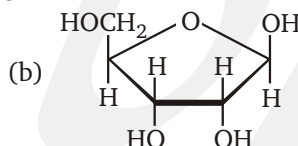
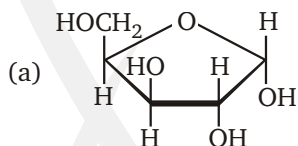
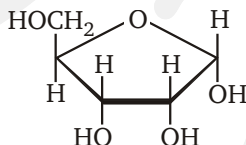
Passage-1:

It is convenient at times to represent the cyclic structures of a monosaccharide without specifying whether the configuration of the anomeric carbon atom is α or β . When we do this, we shall use formulas such as the following



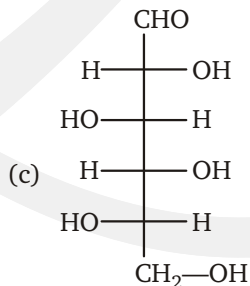
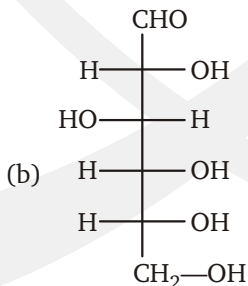
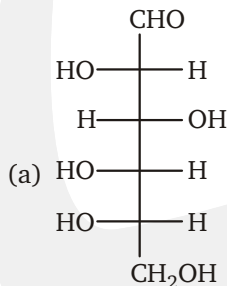
~ Indicates α or β (three-dimensional view not specified)

1. Which of the following represents the anomer of the compound shown ?

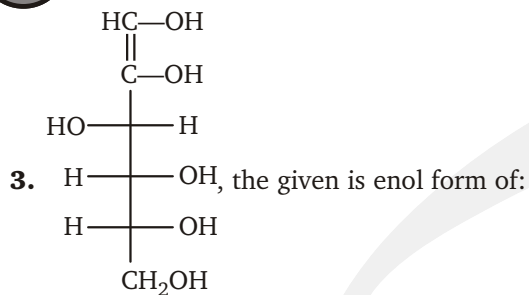


(d) None of these

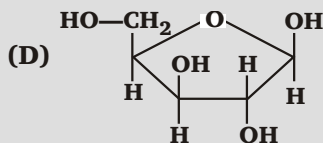
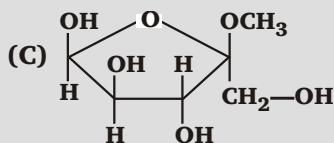
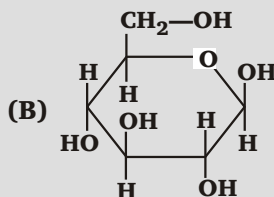
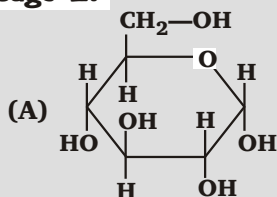
2. What is the structure of L-Glucose ?



(d) None of these



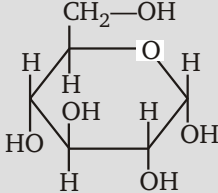
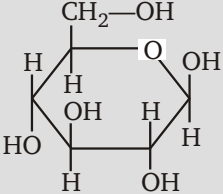
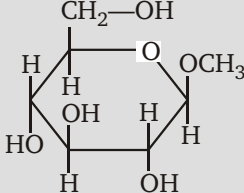
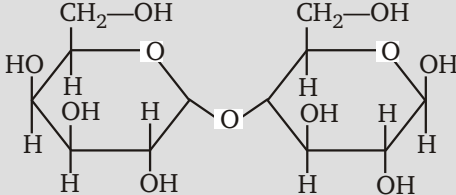
- (a) D-glucose (b) D-mannose
(c) D-fructose (d) All of these

Passage-2:


- Relation between (A) and (B) is:
 - Anomers
 - Positional
 - Functional isomer
 - Enantiomer
- Structure of α -D-Glucopyranose is:
 - (A)
 - (B)
 - (C)
 - (D)
- Which of compound not undergo mutarotation ?
 - (A)
 - (B)
 - (C)
 - None of these
- How many mole of acetic anhydride will consumed when it reacts with compound (A) ?
 - 3
 - 4
 - 5
 - 6

Matching Type Problems

1. Match the column:

Column-I		Column-II	
(a)		(p)	It will undergo osazone formation
		(q)	When undergo acetylation reaction with acetic anhydride molecular weight increases by 210
		(r)	It is reducing sugar
		(s)	It is known as -D-Glucopyranose
(d)		(t)	Non-reducing sugar

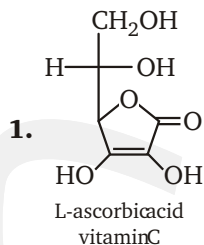
2. Match the column:

Column-I		Column-II	
(a) (b) (c) (d)		(p)	Reducing sugar
		(q)	Used as a monomer of cellulose
		(r)	Presence of Glycosidic linkage
		(s)	It will undergo mutarotation
		(t)	When reacts with acetic anhydride 4 moles will consumed

3. Match the column:

Column-I			Column-II (Reagent used is)	
(a)	Sucrose	$\text{H}_3\text{O}^{\oplus}$	(p)	Product is glucose + Fructose
(b)	Maltose	$\text{H}_3\text{O}^{\oplus}$	(q)	Product is 2 Glucose
(c)	Lactose	$\text{H}_3\text{O}^{\oplus}$	(r)	Product is Glucose + Galactose
			(s)	Oligo saccharides

Unified Conceptual Problems

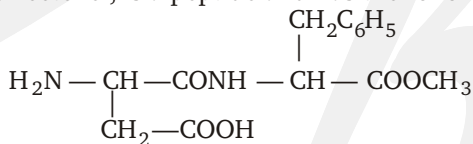


Explain why the C-3 OH group of vitamin C is more acidic than the C-2 OH group.

2. Give the structures of the products in the following reaction



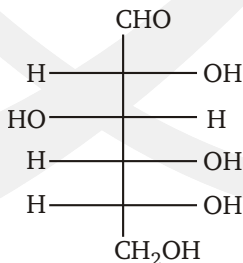
3. Aspartame, an artificial sweetener, is a peptide and has the following structure:



- Identify the four functional groups.
 - Write the Zwitter ionic structure.
 - Write the structure of the amino acids obtained from the hydrolysis of aspartame.
 - Which of the two amino acids is more hydrophobic?
4. Write the structure of alanine at pH 2 and pH 10.
5. Following two amino acids lysine and glutamine form dipeptide linkage. What are two possible dipeptides ?

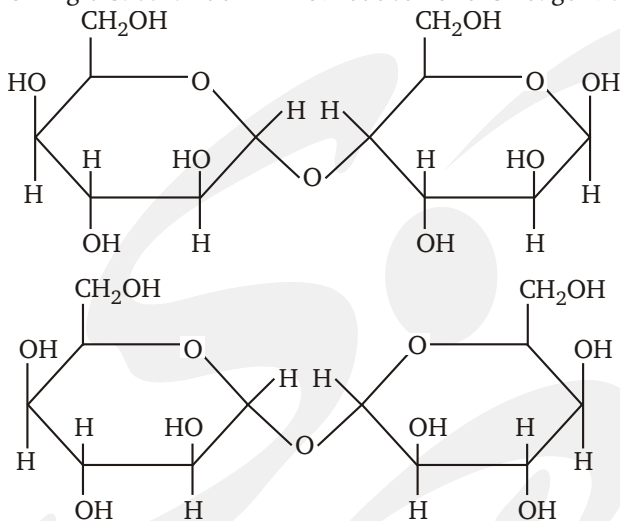


6. The structure of D-glucose is as follows :

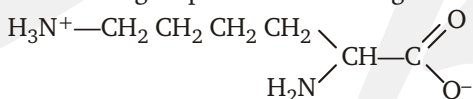


- Draw the structure of L-glucose.
- Give the reaction of L-glucose with Tollens' reagent.

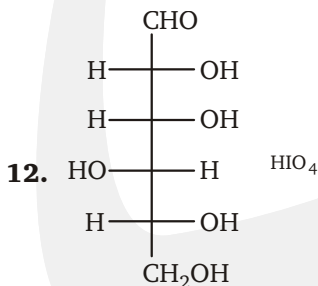
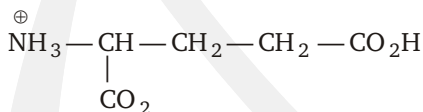
7. Which of the following disaccharide will not reduce Tollens' reagent ?



8. What is the total number of basic groups in the following form of lysine ?

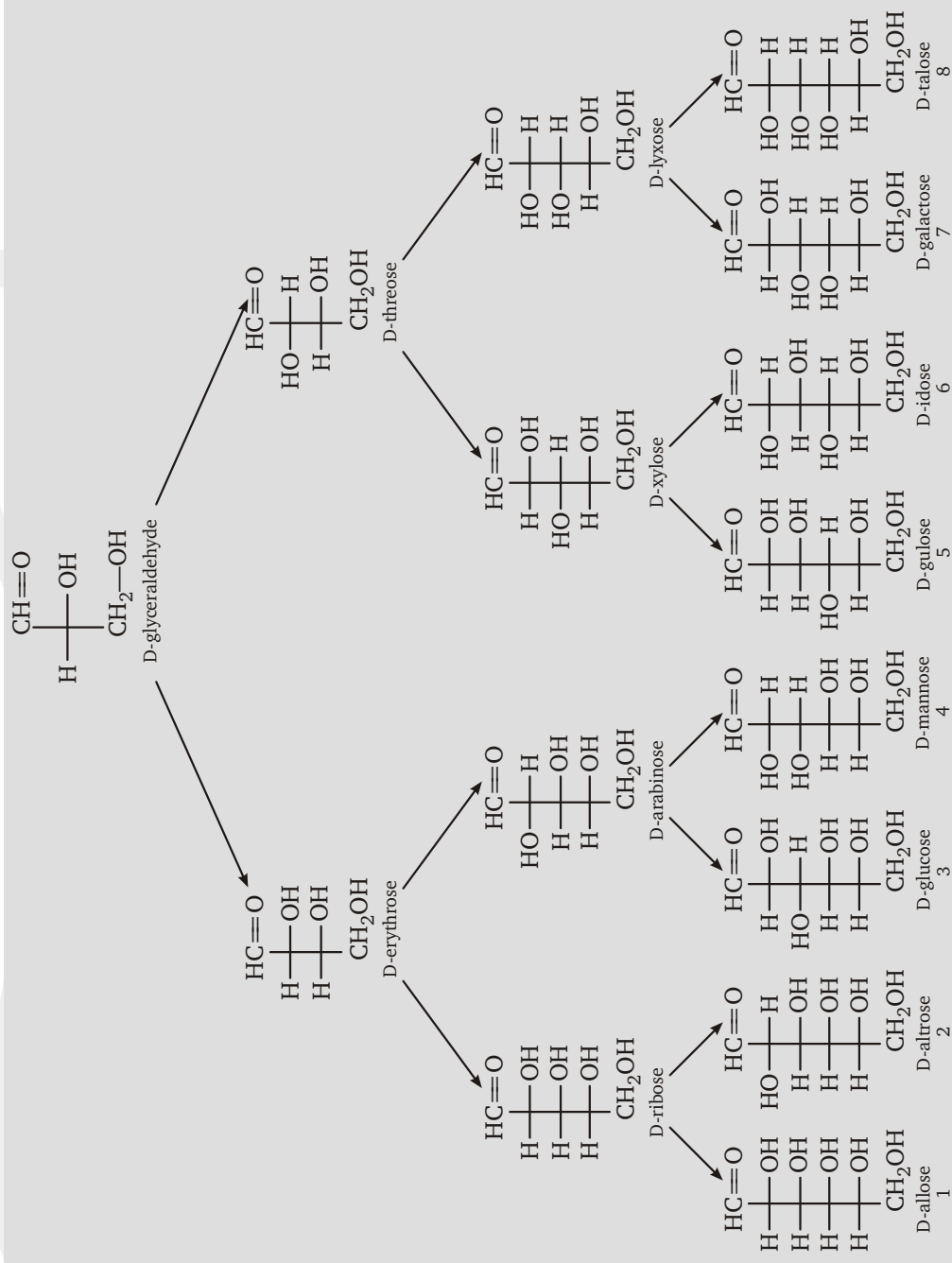


9. A decapeptide (Mol. wt. 796) on complete hydrolysis gives glycine (Mol. wt. 75), alanine and phenylalanine. Glycine contributes 47.0% to the total weight of the hydrolysed products. The number of glycine units present in the decapeptide is :
10. How many compound which is given below is isomer of D-Glucose ?
D-Mannose, D-Fructose, D-Gulose, D-Idose, D-Galactose, D-Arabinose, D-Ribose
11. How many acidic group is present in given amino acid ?



16-isomer of above compound when reacts with periodic acid (HIO_4)- How many mole of periodic acid will consumed ?

13. On the basis of given comprehension, answer the given questions.



- (i) Are D-erythrose and L-erythrose enantiomers or diastereomers?
 (ii) Are L-erythrose and L-threose enantiomers or diastereomers?
 (iii) Which sugar is the C-3 epimer of D-xylose?
 (iv) Which sugar is the C-5 epimer of D-allose?
 (v) Which sugar is the C-4 epimer of L-gulose?
 (vi) How many stereoisomers are possible for:
 (a) a 2-ketoheptose? (b) an aldoheptose? (c) a ketotriose?

Answers

Comprehension Based Problems:

Passage-1:	1.	(b)	2.	(a)	3.	(d)				
Passage-2:	1.	(a)	2.	(b)	3.	(d)	4.	(c)		

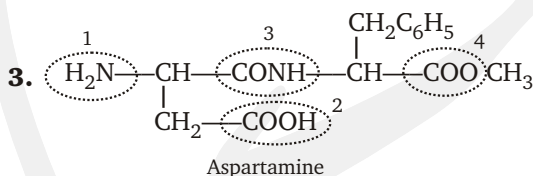
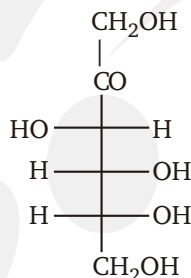
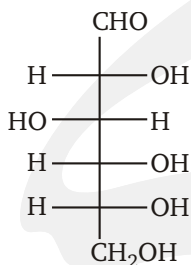
Matching Type Problems:

1. (a) p,q,r,s; (b) p,q,r; (c) t; (d) p,r
 2. (a) p,r,s; (b) p,r,s; (c) p,q,s; (d) t
 3. (a) p,s; (b) q,s; (c) r,s

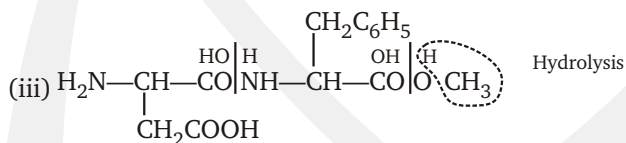
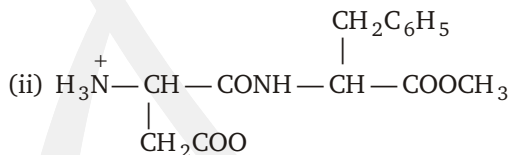
Unified Conceptual Problems:

1. Due to the stability of conjugate base.

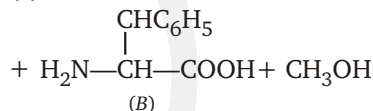
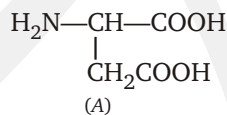
2. Sucrose $\xrightarrow[\text{H}_2\text{O}]{\text{H}}$ D-glucose + D-fructose



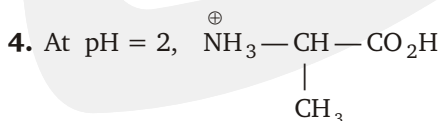
- (i) NH_2 — gp. (1) — amino gp.
 —COOH gp. (2) — carboxyl gp.
 —CONH gp. (3) — 2°- amide gp.
 —COO gp. (4) — ester gp.

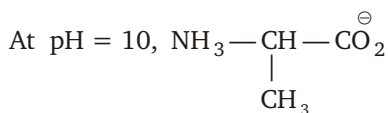


Hydrolysis

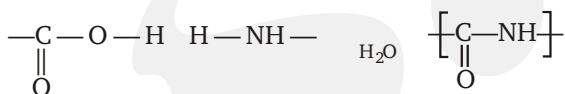


(iv) B is more hydrophobic due to presence of larger organic gp., benzyl gp. ($\text{C}_6\text{H}_5\text{CH}_2-$).

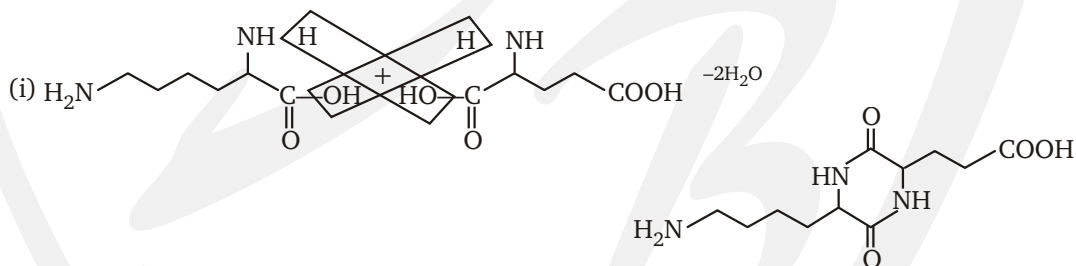




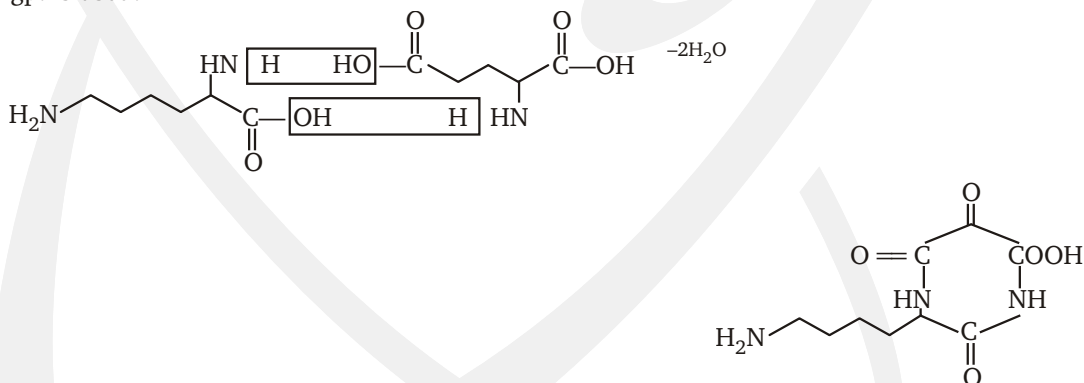
5. Peptide linkage is $-\text{NH}-\text{C}(=\text{O})-$ and it is formed by the condensation between $-\text{NH}_2$ group and $-\text{COOH}$ group as follows:



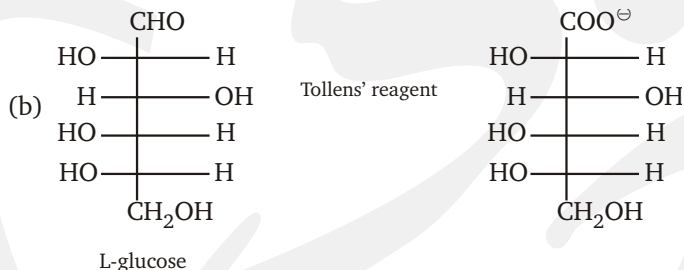
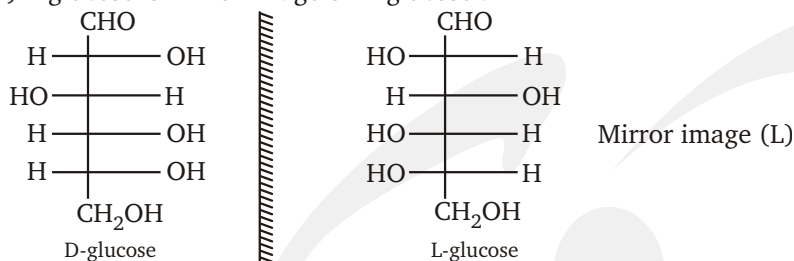
In given amino acids lysine and glutamine, the two possible dipeptides are formed as follows:



(ii) When $-\text{NH}_2$ of both amino acids is used but in one terminal $-\text{COOH}$ gp. lacking $-\text{NH}_2$ gp. is used.



6. (a) L-glucose is mirror image of D-glucose .



7. In structure (a), one ring has a free hemiacetal group, will hydrolyse into open chain in aqueous solution and therefore will reduce Tollen's reagent. Structure (b) has only acetal groups, will not hydrolyse in aqueous solution into open chain, will not reduce Tollens' reagent.

8. 2

9. 6

10. 5

11. 2

12. 80

13.

(i) Enantiomers

(ii) Diastereomers

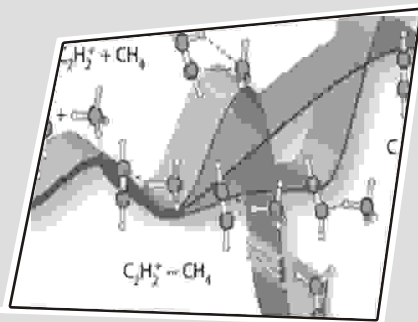
(iii) D-ribose

(iv) L-talose

(v) L-allose

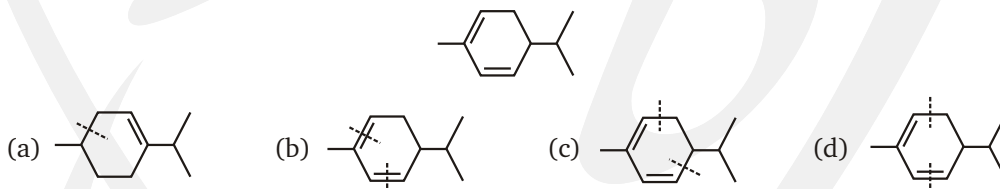
(vi) (a) 16, (b) 32, (c) 0

Polymers

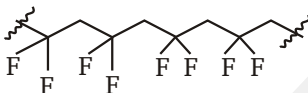


Only One Answer is Correct

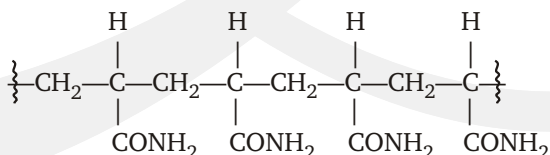
1. When a carbohydrate reacts with NaBH_4 , the product is an:
 (a) alditol (b) aldaric acid (c) aldonic acid (d) aglycone
2. Where are the isoprene subunits in the following terpene?

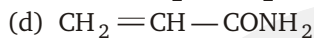
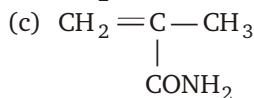
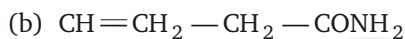
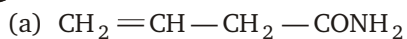


3. The following chain-growth polymer is made up of how many difluoroethylene monomer units?

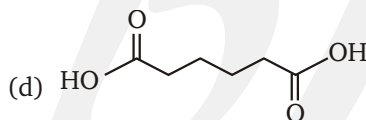
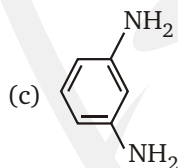
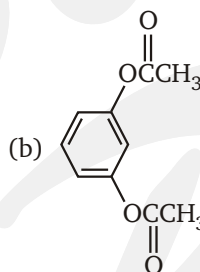
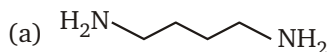
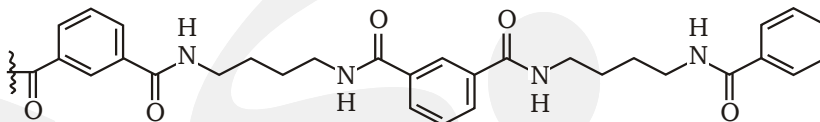


- (a) 1 (b) 2 (c) 3 (d) 4
4. Which of the following is the correct structure for the monomer that is used to make the following chain-growth polymer?

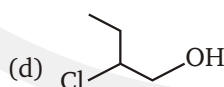
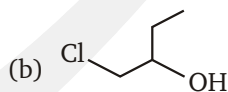
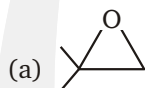
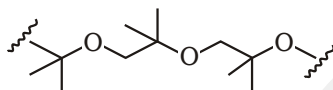




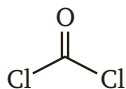
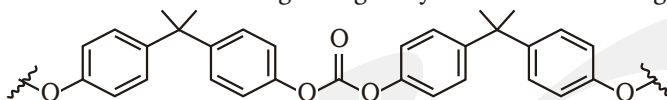
5. What is the structure of one of the compounds used in synthesis of the following step-growth polymer?



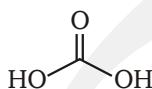
6. Draw the structure of the monomer used to produce this polymer through anionic polymerization.



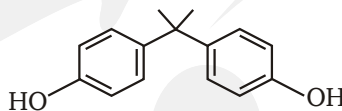
7. Which two monomers would react to give a good yield of the following polymer?



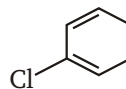
I



II



III



IV

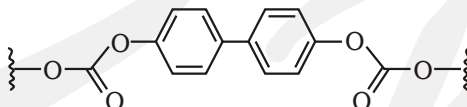
(a) I and III

(b) II and IV

(c) I and IV

(d) II and III

8. To which functional group class does the following polymer belong?



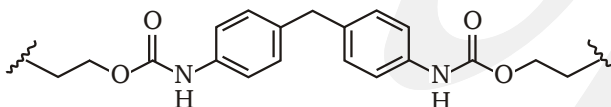
(a) Polyester

(b) Polycarbonate

(c) Polyurethane

(d) Polyamide

9. To which functional group class does the following polymer belong?



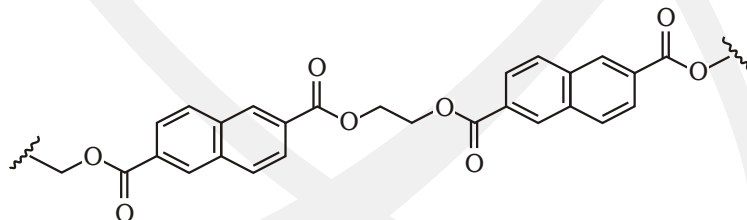
(a) Polyester

(b) Polycarbonate

(c) Polyurethane

(d) Polyamide

10. To which functional group class does the following polymer belong?



(a) Polyester

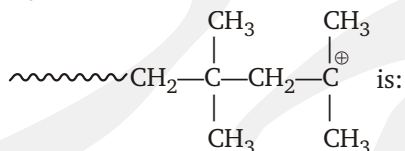
(b) Polycarbonate

(c) Polyurethane

(d) Polyamide

11. Which percentage of sulphur is used in the vulcanization of rubber ?
 (a) 3% (b) 5% (c) 30% (d) 55%
12. Reductive ozonolysis of polymer of 1, 3-butadiene will mainly form :
 (a) HCHO (b) $\begin{array}{c} \text{CHO} \\ | \\ \text{CHO} \end{array}$ (c) $\begin{array}{c} \text{CH}_2 - \text{CHO} \\ | \\ \text{CH}_2 - \text{CHO} \end{array}$ (d) $\text{CH}_3 - \text{CHO}$
13. Which one of the following is natural polymer ?
 (a) Starch (b) Nylon-6 (c) Teflon (d) Buna-S, SBR
14. Which of the following is homopolymer ?
 (a) Starch (b) Polystyrene (c) Orlon (d) All of these
15. Which of the following monomers can undergoes radical, cationic as well as anionic polymerisation with equal ease ?
 (a) $\begin{array}{c} \text{CH}_3 - \text{C} \quad \text{CH}_2 \\ | \\ \text{CH}_3 \end{array}$ (b) $\text{C}_6\text{H}_5 - \text{CH} \quad \text{CH}_2$
 (c) $\text{CH}_2 \quad \text{CH} - \text{CN}$ (d) $\text{CH}_2 \quad \text{CH}_2$
16. Ziegler-Natta catalyst is:
 (a) R_3Al (b) TiCl_4 (c) $\text{R}_3\text{Al} \quad \text{TiCl}_4$ (d) $\text{R}_3\text{B} \quad \text{TiCl}_2$
17. Monomer of Teflon is :
 (a) Monochloroethene (b) 1, 2-Difluoroethene
 (c) 1, 1, 2-Trifluoroethene (d) Tetrafluoroethene
18. Orlon is polymer of:
 (a) Styrene (b) $\text{CF}_2 \quad \text{CF}_2$ (c) Vinyl chloride (d) Acrylonitrile
19. Intermolecular force present in nylon-6,6 is:
 (a) van der Waals (b) Hydrogen bond
 (c) Dipole-Dipole interactions (d) Sulphide linkage
20. Nylon-6,6 is made by using:
 (a) Phenol (b) Benzaldehyde
 (c) Adipic acid (d) Succinic acid
21. Polymer which has amide linkage is:
 (a) Nylon-6,6 (b) Terylene (c) Teflon (d) Bakelite
22. Bakelite is prepared by the reaction between:
 (a) Phenol and formaldehyde
 (b) Ethylene glycol and dimethylphthalate
 (c) Urea and formaldehyde
 (d) Tetramethylene glycol and hexamethylene diamine
23. Natural rubber is a polymer of:
 (a) Butadiene (b) Ethyne (c) Styrene (d) Isoprene
24. Terylene is a condensation polymer of ethylene glycol and:
 (a) Benzoic acid (b) Phthalic acid
 (c) Salicylic acid (d) Terephthalic acid

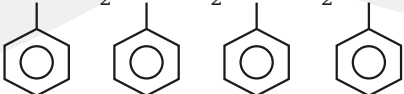
25. Cellulose acetate is a:
 (a) Natural polymer (b) Semisynthetic polymer
 (c) Synthetic polymer (d) Plasticiser
26. A condensation polymer among the following polymers is:
 (a) Teflon (b) Polystyrene (c) PVC (d) Dacron
27. In elastomer, intermolecular forces are:
 (a) Nil (b) Weak (c) Strong (d) Very strong
28. Which one of the following is a polyamide ?
 (a) Teflon (b) Nylon-6,6 (c) Terylene (d) Bakelite
29. Which of the following is fully fluorinated polymer ?
 (a) Neoprene (b) Teflon (c) Thiokol (d) PVC
30. The monomer of the polymer

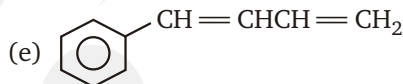
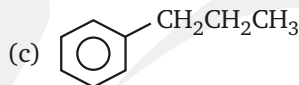
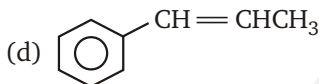
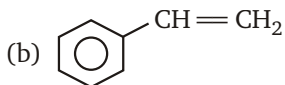
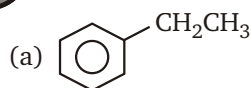


- (a) $\text{CH}_2=\text{C}\begin{matrix} \text{CH}_3 \\ \text{CH}_3 \end{matrix}$ (b) $\text{CH}_3-\text{CH}=\text{CH}-\text{CH}_3$
 (c) $\text{CH}_3-\text{CH}=\text{CH}_2$ (d) $\text{CH}_3-\text{C}(\text{CH}_3)=\text{C}(\text{CH}_3)-\text{CH}_3$
31. Buna-S is a polymer of:
 (a) butadiene only (b) butadiene and styrene
 (c) styrene only (d) butadiene and nitril
32. Among cellulose, polyvinyl chloride, nylon and natural rubber, the polymer in which the intermolecular force of attraction is the weakest, is :
 (a) polyvinyl chloride (b) natural rubber (c) nylon (d) cellulose
33. What is the structure of the monomer from which the following polymer is made ?

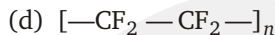
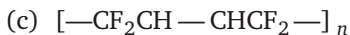
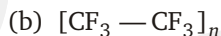
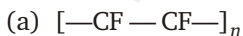
$$\begin{array}{cccccccc} \text{—CH—CH—CH—CH—CH—CH—CH—CH—} \\ | & | & | & | & | & | & | & | \\ \text{CH}_3 & \text{CH}_3 & \text{CH}_3 & \text{CH}_3 & \text{CH}_3 & \text{CH}_3 & \text{CH}_3 & \text{CH}_3 \end{array}$$

 (a) CH_3CH_3 (b) $\text{CH}_2=\text{CH}_2$
 (c) $\text{CH}_3\text{CH}=\text{CH}_2$ (d) $\text{CH}_2=\text{CHCH}=\text{CH}_2$ (e) $\text{CH}_3\text{CH}=\text{CHCH}_3$
34. What is the structure of the monomer from which the following polymer is made ?

$$\text{—CHCH}_2\text{—CHCH}_2\text{—CHCH}_2\text{—CHCH}_2\text{—}$$




35. Which of the following addition polymers results from the reaction below ?



36. Chain-growth polymerization proceeds by which of the following mechanisms ?

(a) radical polymerization

(b) cationic polymerization

(c) anionic polymerization

(d) (a) and (b)

(e) (a), (b) and (c)

37. Which of the following species can best serve as a radical initiator for radical polymerization ?

(a) ROH

(b) ROR

(c) ROOR

(d) RCOOR

(e) RCOOH

38. Which of the following species can best serve as an initiator for cationic polymerization ?

(a) ROOR

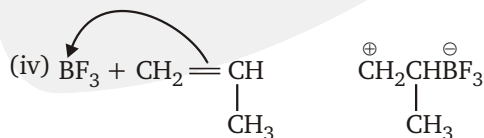
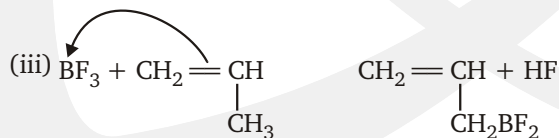
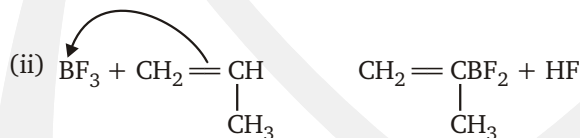
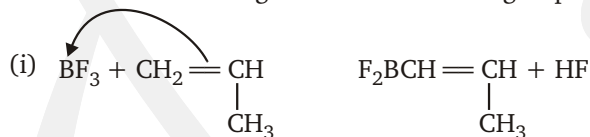
(b) ROH

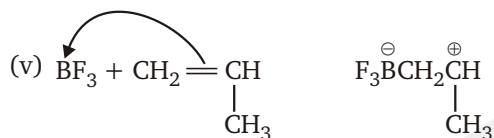
(c) ROR

(d) AlCl_3

(e) RCOOR

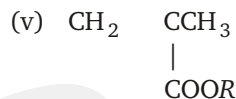
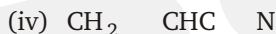
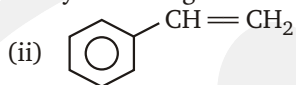
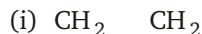
39. Which of the following is the chain-initiating step in the cationic polymerization of propene ?





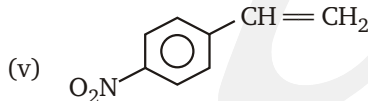
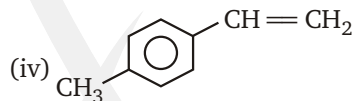
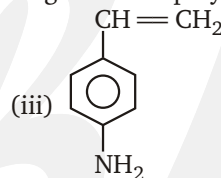
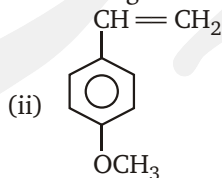
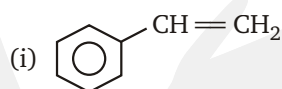
- (a) i (b) ii (c) iii (d) iv (e) v

40. Which of the following is more likely to undergo both radical and cationic polymerization ?



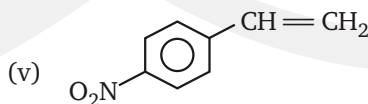
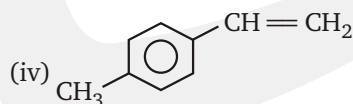
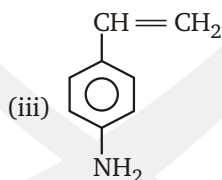
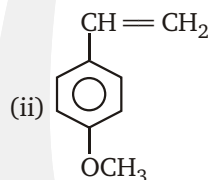
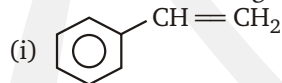
- (a) i (b) ii (c) iii (d) iv (e) v

41. Which of the following monomers has the greatest ability to undergo cationic polymerization ?



- (a) i (b) ii (c) iii (d) iv (e) v

42. Which of the following monomers has the greatest ability to undergo anionic polymerization ?

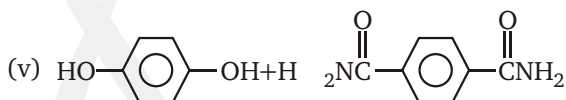
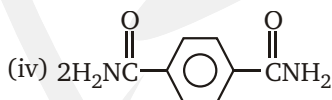
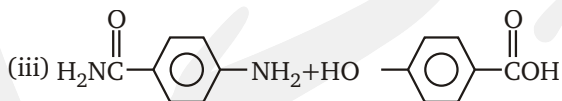
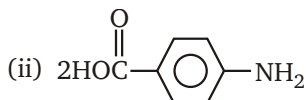
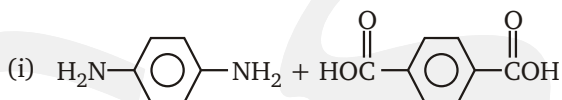


- (a) i (b) ii (c) iii (d) iv (e) v

43. Which of the following is a step-growth polymer ?

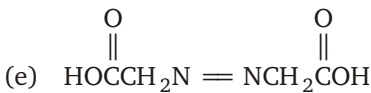
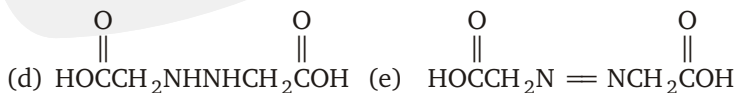
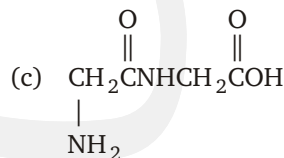
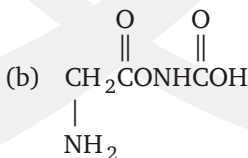
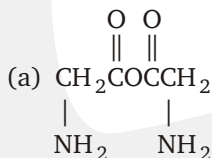
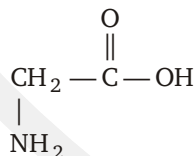
- (a) polyethylene (b) polyester
(c) polypropylene (d) polystyrene (e) plexiglass

44. What monomers formed the following polymer ?



- (a) i (b) ii (c) iii (d) iv (e) v

45. Which dimer (step-growth dimerization) will form from the following monomer ?



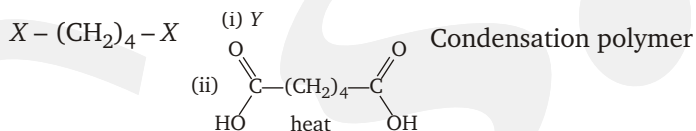
46. The joining together of many monomers to form a very large molecule is called ?
 (a) polymerization (b) monomerization
 (c) dimerization (d) hydroxylation (e) homogenization
47. Amide linkage is present in :
 (a) Nylon-6,6 (b) Bakelite (c) Sucrose (d) Starch
48. When phenol reacts with formaldehyde in the presence of acidic medium then obtained polymer will be:
 (a) Resin (b) Bakelite (c) Dacron (d) Teflon
49. Polymer of α -D-Glucopyranose is:
 (a) starch (b) cellulose (c) amylose (d) sucrose
50. Among cellulose, polyvinyl chloride, nylon and natural rubber, the polymer in which the intermolecular force of attraction is weakest is:
 (a) nylon (b) polyvinyl chloride (c) cellulose (d) natural rubber
51. Which of the following polymer, not having C—C bond formation during polymerisation.
 (a) Teflon (b) Bakelite (c) Natural rubber (d) Terylene

Answers

1.	(a)	2.	(c)	3.	(d)	4.	(d)	5.	(a)	6.	(a)	7.	(a)	8.	(b)	9.	(c)	10.	(a)
11.	(b)	12.	(c)	13.	(a)	14.	(d)	15.	(b)	16.	(c)	17.	(d)	18.	(d)	19.	(b)	20.	(c)
21.	(a)	22.	(a)	23.	(d)	24.	(d)	25.	(b)	26.	(d)	27.	(b)	28.	(b)	29.	(b)	30.	(a)
31.	(b)	32.	(b)	33.	(e)	34.	(b)	35.	(d)	36.	(e)	37.	(c)	38.	(d)	39.	(e)	40.	(b)
41.	(c)	42.	(e)	43.	(b)	44.	(a)	45.	(c)	46.	(a)	47.	(a)	48.	(b)	49.	(b)	50.	(d)
51.	(d)																		

One or More than One Answers is/are Correct

1. The correct functional group X and the reagent/ reaction conditions Y in the following scheme are:



- (a) $X = \text{COOCH}_3, Y = \text{H}_2/\text{Ni}/\text{heat}$ (b) $X = \text{CONH}_2, Y = \text{H}_2/\text{Ni}/\text{heat}$
 (c) $X = \text{CONH}_2, Y = \text{Br}_2/\text{NaOH}$ (d) $X = \text{CN}, Y = \text{H}_2/\text{Ni}/\text{heat}$
2. In which of the following polymer there is presence of amide linkage?
- (a) Starch (b) Cellulose
 (c) Protein (d) Nylon-6,6

Answers

1.	(c, d)	2.	(c, d)										
----	--------	----	--------	--	--	--	--	--	--	--	--	--	--

Matching Type Problems

1. Match the column:

Column-I		Column-II	
(a)	Nylon-6,6	(p)	Condensation polymerization
(b)	Styrene	(q)	Addition polymerization
(c)	Bakelite	(r)	Homo-polymer
(d)	Teflon	(s)	Co-polymer

2. Match the column:

Column-I		Column-II	
(a)	Terylene	(p)	Condensation polymerization
(b)	Styrene	(q)	Addition polymerization
(c)	Bakelite	(r)	Homo-polymer
(d)	Teflon	(s)	Co-polymer

3. Match the column:

Column-I		Column-II	
(a)	Cellulose	(p)	Natural polymer
(b)	Nylon-6,6	(q)	Synthetic polymer
(c)	Protein	(r)	Amide linkage
(d)	Sucrose	(s)	Glycoside linkage

4. Match the column:

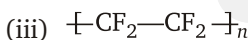
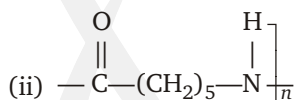
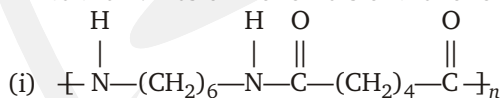
Column-I		Column-II	
(a)	Natural polymer	(p)	Nylon-6,6
(b)	Synthetic polymer	(q)	Protein
(c)	Amide-linkage	(r)	Cellulose
(d)	Glycosidic linkage	(s)	Starch

5. Match the column:

Column-I		Column-II	
(a)	Protein	(p)	Natural polymer
(b)	Starch	(q)	Synthetic polymer
(c)	Cellulose	(r)	Amide-linkage
(d)	Nylon-6,6	(s)	Glycosidic linkage

Unified Conceptual Problems

1. Write the names of monomers of the following polymers:



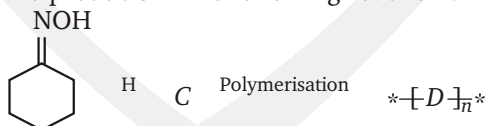
2. Classify the following as addition and condensation polymers: Terylene, Bakelite, Polyvinyl chloride, Polythene.

3. Arrange the following polymers in increasing order of their intermolecular forces.

(i) Nylon-6,6, Buna-S, Polythene

(ii) Nylon-6, Neoprene, Polyvinyl chloride

4. Give the structures of the products in the following reaction :



5. Write down the heterogeneous catalyst involved in the polymerization of ethylene.

6. Monomer A of a polymer on ozonolysis yields two moles of HCHO and one mole of CH_3COCHO .

(a) Deduce the structure of A.

(b) Write the structure of "all cis"- form of polymer of compound A.

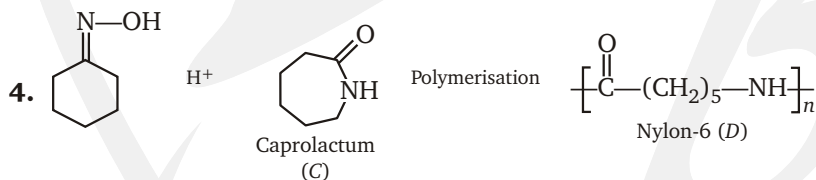
Answers

Matching Type Problems:

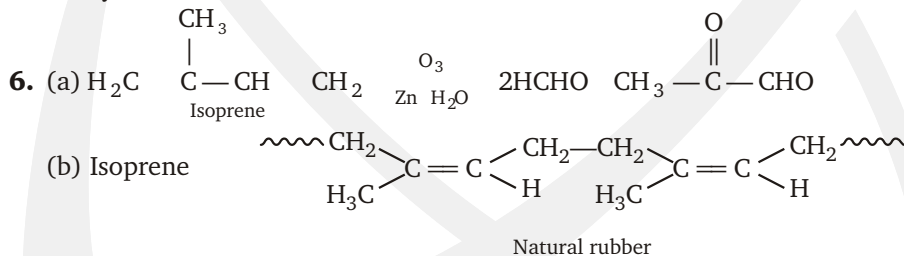
1. (a) p,s; (b) q,r; (c) p,s; (d) q,r
2. (a) p,s; (b) q,r; (c) p,s; (d) q,r
3. (a) p,s; (b) q,r; (c) p,r; (d) s
4. (a) q,r,s; (b) p; (c) p,q; (d) r,s
5. (a) p,r; (b) p,s; (c) p,s; (d) q,r

Unified Conceptual Problems:

1. (i) Hexamethylene diamine and adipic acid, (ii) Caprolactum, (iii) Tetrafluoroethene
2. Addition polymers: Polyvinyl chloride, Polythene
Condensation polymers: Terylene, Bakelite
3. in increasing order of intermolecular forces.
(i) Buna-S, Polythene, Nylon-6,6, (ii) Neoprene, polyvinyl chloride, Nylon-6.

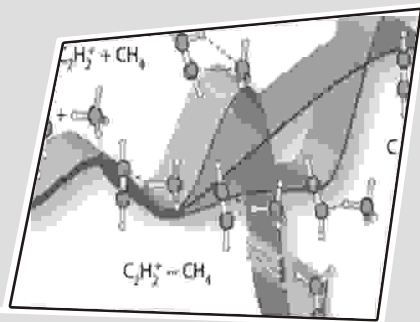


5. $n\text{CH}_2=\text{CH}_2 \xrightarrow[\text{Ziegler-Natta Catalyst}]{\text{R}_3\text{Al} \quad \text{TiCl}_4} \text{-(CH}_2\text{—CH}_2\text{)}_n$ Ziegler-Natta catalyst ($\text{R}_3\text{Al} \quad \text{TiCl}_4$) acts as heterogenous catalyst. While Wilkinsons' catalyst ($(\text{Ph}_3\text{P})_3\text{RhCl}$) acts as homogenous catalyst.



9

Grignard Reaction



Only One Answer is Correct

1. What are the major products of the following reaction?



1. EtMgBr, Et₂O
2. H₂O, H₃O⁺

?

Optically active

- (a) + (b) +
- (c) + (d) +

2. Which of the following compounds would not give *tert*-butyl alcohol when treated with excess methylmagnesium bromide?

- (a) Acetyl chloride (b) Acetaldehyde (c) Methyl acetate (d) Acetic anhydride

3. Which of the following reagents, when treated with phenylmagnesium bromide followed by acid workup, will yield 2-phenylethanol?

- (a) Ethanol (b) Oxirane (c) Acetaldehyde (d) Ethylene

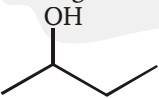
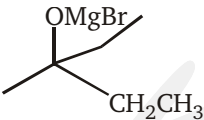
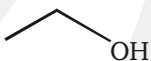
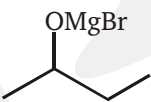
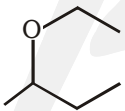
4. What is the major product of the reaction sequence shown ?



- (a) (CH₃)₂CHCH₂CH₂OH (b) (CH₃)₂CHCHOHCH₃
(c) CH₃CH₂CH₂CH₂CH₂OH (d) CH₃CH₂CH₂CHOHCH₃

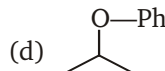
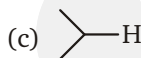
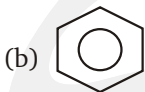
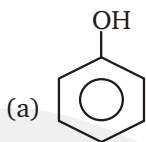
5. In which of the reaction CH₄ formation takes place ?

- (a) CH₃MgBr + Ph-CH=O (b) CH₃MgBr + CH₃-CH=O

- (c) CH_3MgBr $\text{CH}_3 - \overset{\text{O}}{\parallel} \text{C} - \text{CH}_3$ (d) CH_3MgBr EtOH
6. $\text{PhMgBr} + (x)$ ; (x) is:
- (a) CH_4 (b) $\text{CH}_3 - \overset{\text{CH}_3}{\underset{\text{CH}_3}{\text{C}}} - \text{OH}$ (c)  (d) ether
7. The product/s of the following reaction is/are:
-  $\text{CH}_3\text{CH}_2\text{MgBr}$ ether
- (a)  (b)  + 
- (c)  CH_3CH_3 (d) 
8. $\text{CH}_3 - \text{Br}$ $\overset{\text{Mg}}{\text{ether}}$ (A) $\text{Ph} - \overset{\text{O}}{\parallel} \text{C} - \text{H}$ $\overset{(2)}{\text{H}^+}$ (B); Product (B) is:
- (a) $\text{CH}_3 - \overset{\text{O}}{\parallel} \text{C} - \text{Ph}$ (b) $\text{CH}_3 - \overset{\text{OH}}{\text{CH}} - \text{Ph}$
- (c) $\text{CH}_3 - \underset{\text{Ph}}{\text{CH}} - \text{CH}_3$ (d) $\text{Ph} - \text{CH}_2 - \overset{\text{O}}{\parallel} \text{C} - \text{H}$
9. The product $\text{CH}_3 - \overset{\text{OH}}{\underset{\text{Ph}}{\text{C}}} - \text{Ph}$ is obtained in the reaction. (Assume acidification at the end of the reaction)
- (a) CH_3MgBr $\text{Ph} - \overset{\text{O}}{\parallel} \text{C} - \text{Ph}$ (b) PhMgBr $\text{CH}_3 - \overset{\text{O}}{\parallel} \text{C} - \text{Ph}$

(c) 2PhMgBr $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{O}-\text{Et}$ (d) All of these

10. Reaction with PhMgBr $\text{H}-\text{O}-\underset{\text{CH}_3}{\underset{|}{\text{CH}}}-\text{CH}_3$ gives:



11. Compare reactivity toward Grignard reagent :



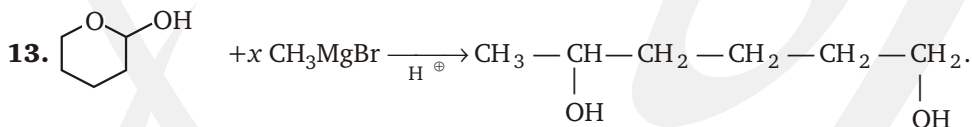
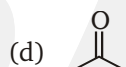
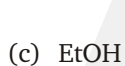
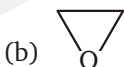
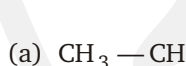
(a) $c > b > a$

(b) $b > c > a$

(c) $a > b > c$

(d) none of these

12. $\text{PhMgBr} + (x) \text{ } 2\text{H}^+ \text{ Ph}-\text{CH}_2-\text{CH}_2-\text{OH}$; Reactant (x) is:



The value of x is:

(a) 1

(b) 2

(c) 3

(d) 4

14. Compare rate of reaction of the following with $\text{OH}^- / \text{H}_2\text{O}$



(a)

(b)

(c)

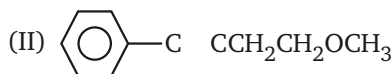
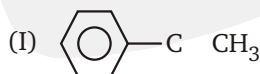
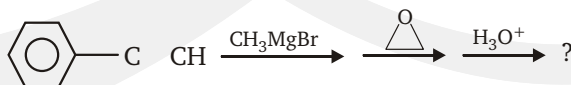
(a) $c > b > a$

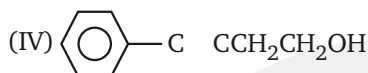
(b) $a > b > c$

(c) $b > a > c$

(d) $b > c > a$

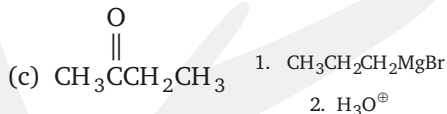
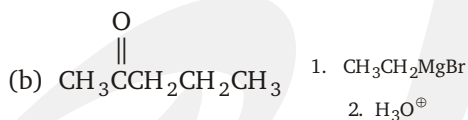
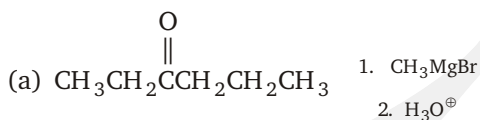
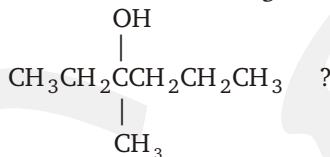
15. What is the product of the following reaction sequence?



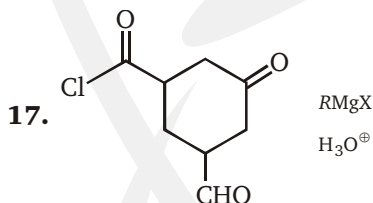


- (a) I (b) II (c) III (d) IV (e) V

16. Which of the following reactions can be used to prepare 3-methyl-3-hexanol,



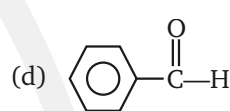
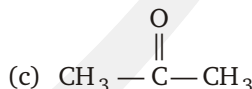
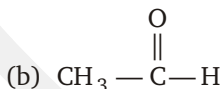
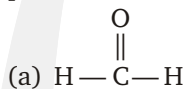
- (d) (a) and (b) (e) (a), (b) and (c)



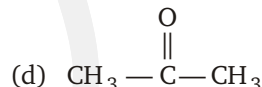
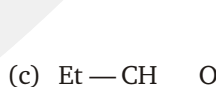
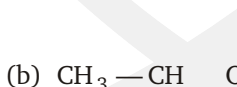
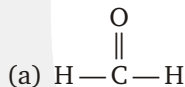
How many molecules of RMgX are consumed in the above given reaction ?

- (a) 2 (b) 4 (c) 5 (d) 6

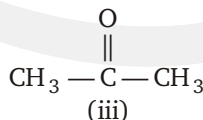
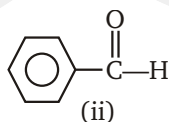
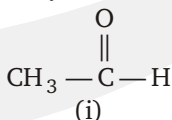
18. Which of the following compound when reacts with CH₃MgBr formation of 3° alcohol take place ?



19. Which of the following is least reactive toward Grignard reagent ?

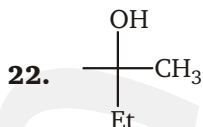
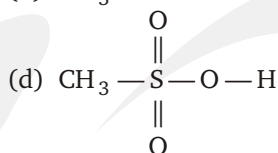
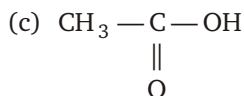


20. Compare reactivity toward Grignard reagent ?

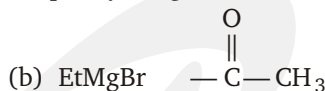
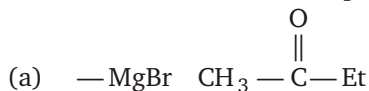


- (a) i > ii > iii (b) i > iii > ii (c) ii > i > iii (d) iii > ii > i

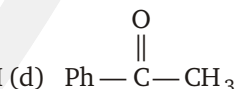
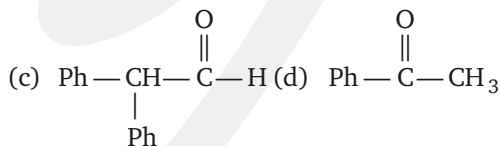
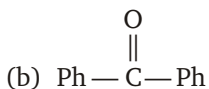
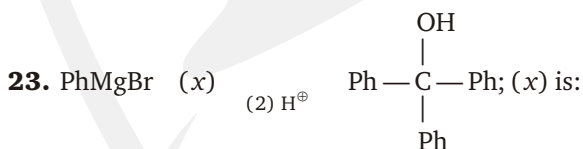
21. Which of the following is most reactive toward Grignard reagent ?



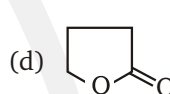
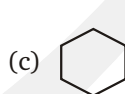
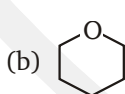
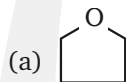
The above 3 alcohol can be prepared by [Ph (phenyl ring)] :



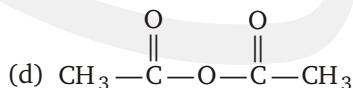
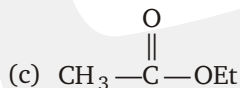
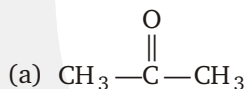
(d) All of these



24. Which of the following compound react with CH_3MgBr ?



25. PhMgBr act as a base with :



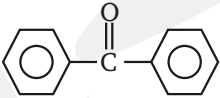
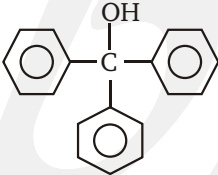
26. $\text{CH}_3 - \overset{\text{O}}{\underset{\text{14}}{\parallel}}\text{C} - \text{OH}$; the given carboxylic acid is prepared by:

- (a) $\text{CH}_3\text{MgBr} \xrightarrow{(2)\text{H}^+} \text{O} \quad \text{C} \quad \text{O}$ (b) $\text{CH}_3\text{MgBr} \xrightarrow{(2)\text{H}^+} \text{O} \quad \text{C}_{14} \quad \text{O}$
 (c) $\text{CH}_3\text{MgBr} \xrightarrow{\text{CH}_3\text{OH}} \text{14}$ (d) $\text{CH}_3\text{MgBr} \xrightarrow{\text{CH}_3\text{OH}} \text{14}$

27. When CH_3MgBr reacts with dry ice followed by acidification product will be :

- (a) $\text{CH}_3 - \overset{\text{O}}{\parallel} \text{C} - \text{OH}$ (b) $\text{CH}_3 - \text{CH}_2 - \text{OH}$
 (c) $\text{CH}_3 - \text{CH}_2 - \overset{\text{O}}{\parallel} \text{C} - \text{H}$ (d) CH_3OH

28. $\text{C}_6\text{H}_5\text{MgBr} + \text{Et-O-CO-O-Et} \xrightarrow{(\text{excess})} \text{H}^+ \text{ (B)} ; \text{ Final product (B) is :}$

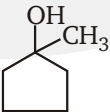
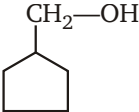
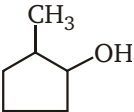
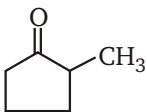
- (a)  (b) 
 (c) PhCH (d) $\text{Ph} - \underset{\text{OH}}{\text{CH}} - \text{Ph}$

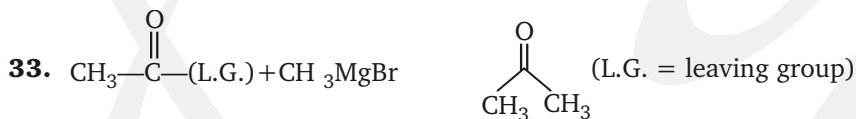
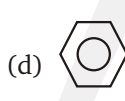
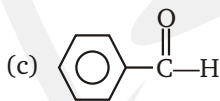
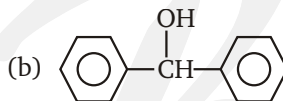
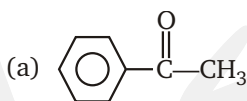
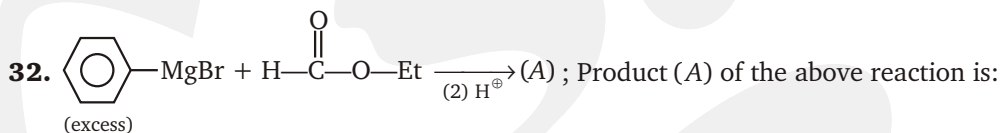
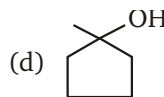
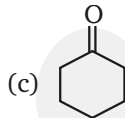
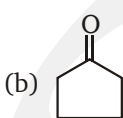
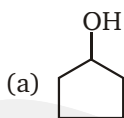
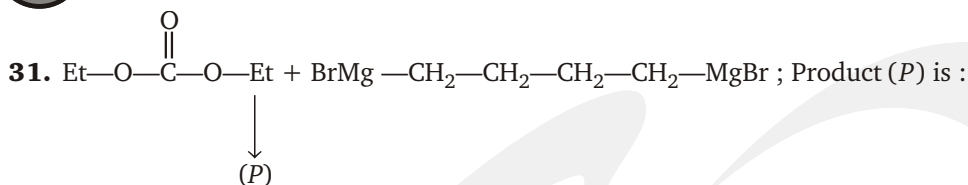
29. $\text{C}_6\text{H}_5\text{MgBr} + \text{Cl-CO-CH}_3 \xrightarrow{(\text{excess})} (2)\text{H}^+$

Final product of above reaction is:

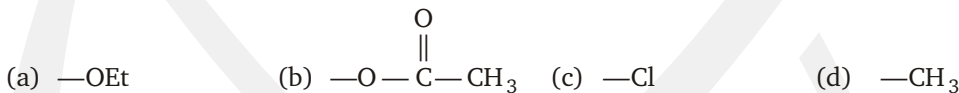
- (a) $\text{Ph} - \underset{\text{OH}}{\text{CH}} - \text{CH}_3$ (b) $\text{Ph} - \underset{\text{Ph}}{\underset{\text{OH}}{\text{C}}} - \text{CH}_3$ (c) $\text{Ph} - \overset{\text{O}}{\parallel} \text{C} - \text{CH}_3$ (d) Ph_3CH

30. $\text{CH}_3 - \overset{\text{O}}{\parallel} \text{C} - \text{O-Et} + \text{BrMg-CH}_2\text{-CH}_2\text{-CH}_2\text{-CH}_2\text{-MgBr} \xrightarrow{(2)\text{H}^+} \text{(A)} ; \text{ Product (A) is:}$

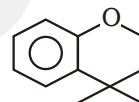
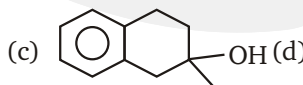
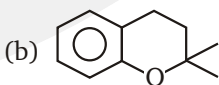
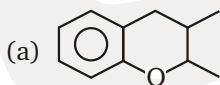
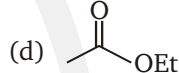
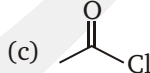
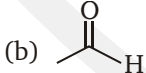
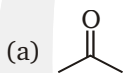
- (a)  (b) 
 (c)  (d) 

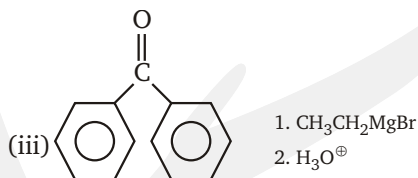


Rate of reaction will be maximum when leaving group is:



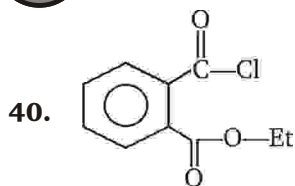
34. Most reactive towards Grignard reagent is :



CC(C1=CC=CC=C1)(O)C2=CC=CC=C2

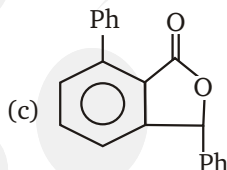
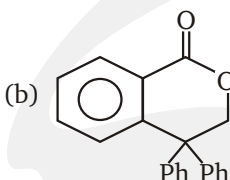
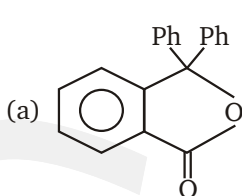
- CC(=O)C(=O)c1ccc(O)c(S)c1

- For More Material Join: @JEEAdvanced_2024

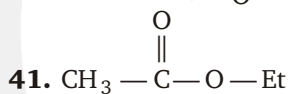


(1) 2PhMgBr
(2) H⁺

(A); Final product (A) is:



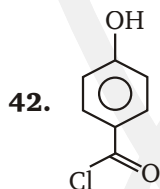
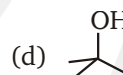
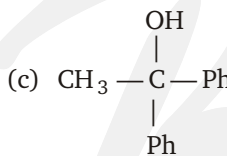
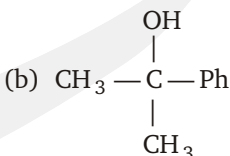
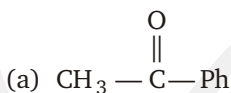
(d) None of these



PhMgBr
(excess)

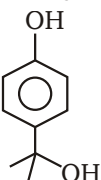
(2) H⁺

(A); Product (A) is:



+ xCH₃MgBr

(2) H⁺



; The value of x is:

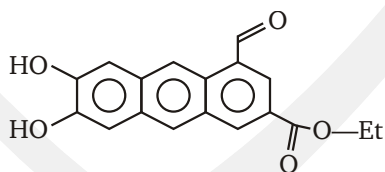
(a) 1

(b) 2

(c) 3

(d) 4

43. How many moles of Grignard reagent will consumed when following compound reacts with it?

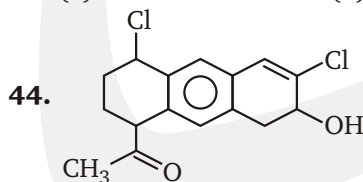


(a) 2

(b) 3

(c) 4

(d) 5



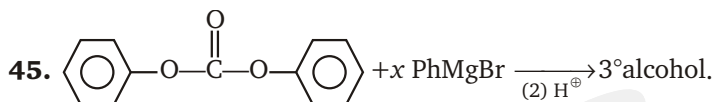
+ x CH₃MgBr. The value of x is:

(a) 1

(b) 2

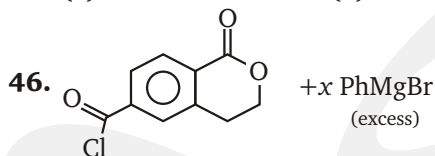
(c) 3

(d) 4



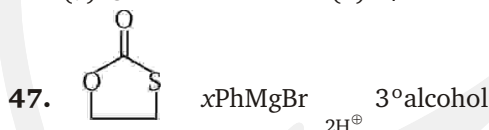
The value of x is:

- (a) 2 (b) 3 (c) 4 (d) 5



Moles of Grignard reagent consumed is:

- (a) 3 (b) 4 (c) 5 (d) 6

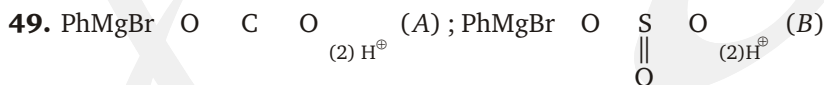


Value of x is:

- (a) 2 (b) 3 (c) 1 (d) 4

48. Which of the following is least reactive toward RMgBr?

- (a) $CH_3 - \overset{\overset{O}{\parallel}}{C} - Et$ (b) $CH_3 - \overset{\overset{O}{\parallel}}{C} - Ph$ (c) $Ph - \overset{\overset{O}{\parallel}}{C} - Ph$ (d) $Ph - \overset{\overset{O}{\parallel}}{C} - H$



Product (A) and (B) respectively are:

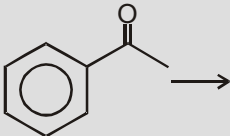
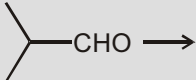
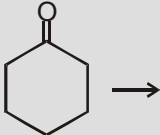
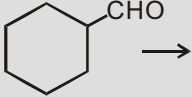
- (a) Ph — CO₂H, Ph — SO₂H (b) Ph — OH, Ph — SO₃H
(c) Ph — CO₂H, Ph — SO₃H (d) Ph — SO₃H, Ph — CO₂H

Answers

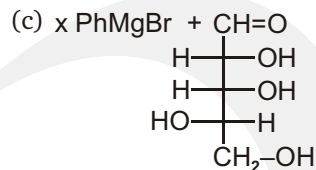
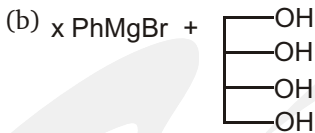
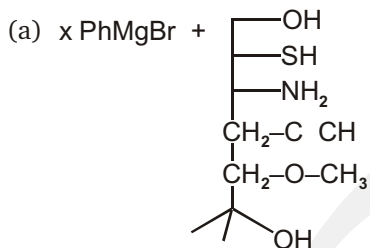
1.	(a)	2.	(b)	3.	(b)	4.	(a)	5.	(d)	6.	(b)	7.	(c)	8.	(b)	9.	(d)	10.	(b)
11.	(c)	12.	(b)	13.	(a)	14.	(b)	15.	(d)	16.	(e)	17.	(b)	18.	(c)	19.	(d)	20.	(b)
21.	(d)	22.	(d)	23.	(b)	24.	(d)	25.	(b)	26.	(b)	27.	(a)	28.	(b)	29.	(b)	30.	(a)
31.	(a)	32.	(b)	33.	(c)	34.	(c)	35.	(b)	36.	(e)	37.	(c)	38.	(c)	39.	(c)	40.	(a)
41.	(c)	42.	(c)	43.	(d)	44.	(c)	45.	(b)	46.	(b)	47.	(b)	48.	(c)	49.	(c)		

Unified Conceptual Problems

1.

S.No.	Reaction	Product
1.	PhMgBr 2-propanol	
2.	PhMgBr + Ph —CH ₂ — OH	
3.	PhMgBr CH ₃ — SH	
4.	PhMgBr CH ₃ — C — CH	
5.	PhMgBr + Acetic acid	
6.	PhMgBr + 	
7.	PhMgBr Ph —  — Ph	
8.	PhMgBr 	
9.	PhMgBr + 	
10.	PhMgBr + 	
11.	CH ₃ MgBr Butanone	
12.	CH ₃ MgBr 2-Pentanone	
13.	CH ₃ MgBr 3-Pentanone	

2. Find the value of x in each reaction ?

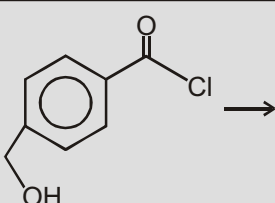


3. Identify product and compare rate of reaction ?

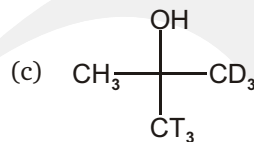
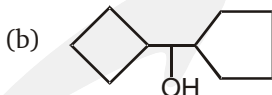
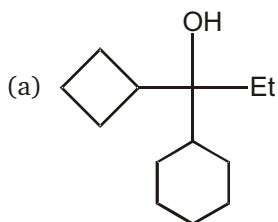
(a)	$\text{CH}_3\text{MgBr} + \text{CH}_3 - \overset{\text{O}}{\parallel} \text{C} - \text{H} \longrightarrow r_1$ $\text{CH}_3\text{MgBr} + \text{Ph} - \overset{\text{O}}{\parallel} \text{C} - \text{H} \longrightarrow r_2$ $\text{CH}_3\text{MgBr} + \text{CH}_3\text{CH}_2\text{COCH}_3 \longrightarrow r_3$	
(b)	$\text{CH}_3\text{MgBr} + \text{H} - \overset{\text{O}}{\parallel} \text{C} - \text{H} \longrightarrow r_1$ $\text{CH}_3\text{MgBr} + \text{CH}_3\text{CHO} \longrightarrow r_2$	
(c)	$\text{CH}_3\text{MgBr} + \text{C}_6\text{H}_5\text{CHO} \longrightarrow r_1$ $\text{CH}_3\text{MgBr} + \text{CH}_3\text{O-C}_6\text{H}_4\text{CHO} \longrightarrow r_2$ $\text{CH}_3\text{MgBr} + \text{NO}_2\text{-C}_6\text{H}_4\text{CHO} \longrightarrow r_3$	

(d)	$\text{PhMgBr} + \text{CH}_3\text{CH}_2\text{CHO} \rightarrow r_1$ $\text{PhMgBr} + \text{CH}_3\text{COCH}_3 \rightarrow r_2$ $\text{PhMgBr} + \text{CH}_3\text{CH}_2\text{COCH}_2\text{CH}_3 \rightarrow r_3$	
(e)	$\text{PhMgBr} + \text{CH}_3\text{CH}_2\text{CHO} \rightarrow r_1$ $\text{PhMgBr} + \text{Ph-CH=O} \rightarrow r_2$ $\text{PhMgBr} + \text{CH}_3\text{COCH}_2\text{CH}_3 \rightarrow r_3$ $\text{PhMgBr} + \text{C}_6\text{H}_5\text{COCH}_3 \rightarrow r_4$	
(f)	$\text{PhMgBr} + \text{epoxide} \rightarrow r_3$ $\text{PhMgBr} + \text{oxetane} \rightarrow r_3$ $\text{PhMgBr} + \text{tetrahydrofuran} \rightarrow r_3$	

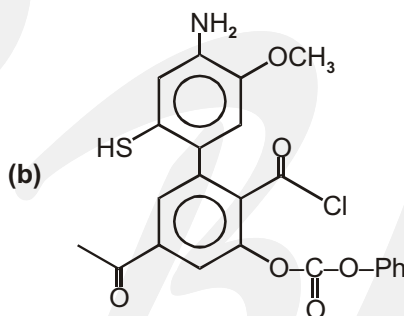
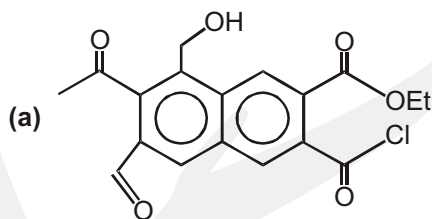
4. Write the product of following reactions.

S.No.	Reaction	Product
1.	PhMgBr (excess) + $\text{Ph}-\overset{\text{O}}{\parallel}{\text{C}}-\text{Cl}$	
2.	PhMgBr (excess) + $\text{Ph}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OEt}$	
3.	PhMgBr (excess) + $\text{Ph}-\overset{\text{O}}{\parallel}{\text{C}}-\text{NH}_2$	
4.	PhMgBr (excess) + $\text{EtO}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OEt}$	
5.	PhMgBr (excess) + $\text{Et}-\text{S}-\overset{\text{O}}{\parallel}{\text{C}}-\text{Cl}$	
6.	PhMgBr (excess) + $\text{Ph}-\text{C}\equiv\text{N}$	
7.	PhMgBr (excess) + 	
8.	PhMgBr + CH_3-Cl	
9.	PhMgBr + $\text{Br}-\text{Br}$	

5. Propose the all suitable routes to prepare given alcohol by the help of grignard reagent ?



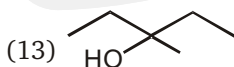
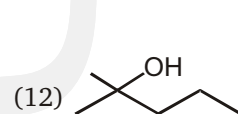
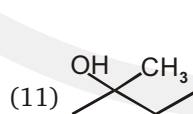
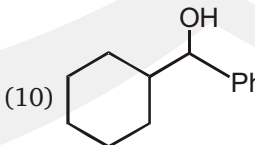
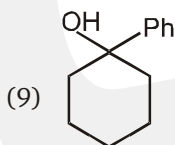
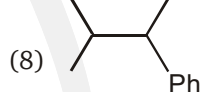
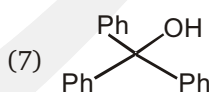
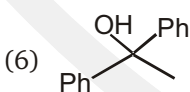
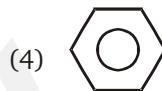
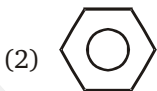
6. Find the value of x ? (x = mole of RMgX consumed).



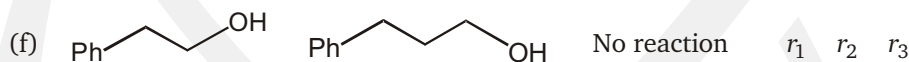
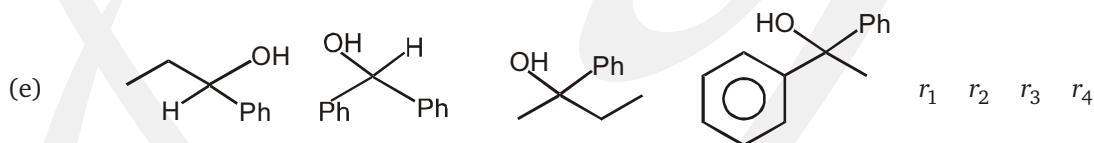
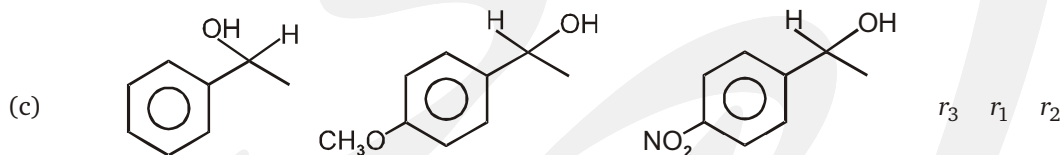
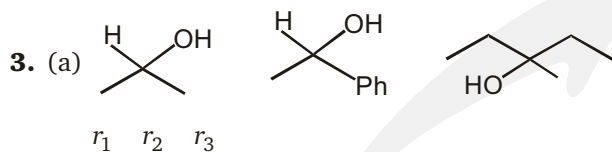
Answers

Unified Conceptual Problems:

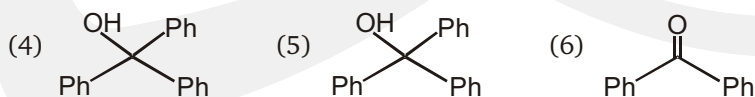
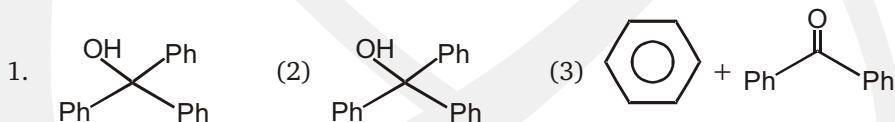
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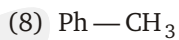
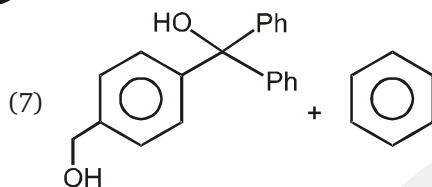


2. (a) x 5 (b) 4 (c) 5

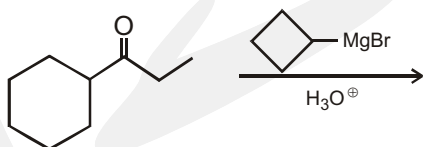
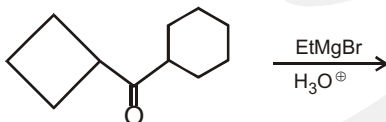
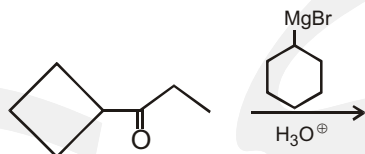


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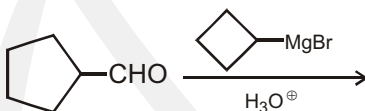
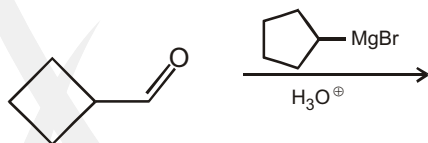




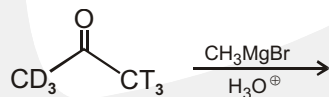
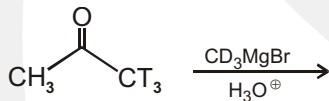
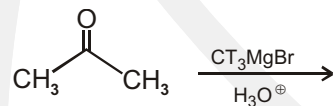
5. (a)



(b)



(c)

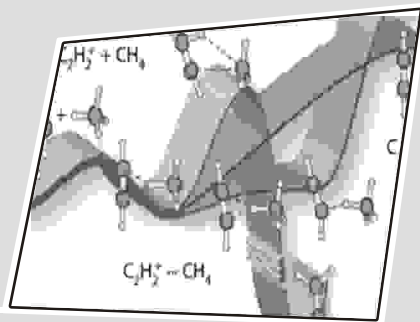


6. (a) 7

(b) 8

10

Practical Organic Chemistry



Only One Answer is Correct

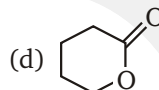
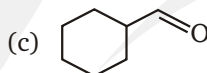
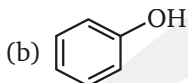
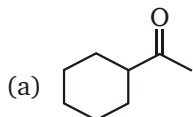
- Among the following which metal(s) is/are used to prepare sodium extract ?
 (a) Li (b) Na (c) K (d) all of these
- In organic analysis N is identified as:
 (a) CN (b) NO₂ (c) CNO (d) none of these
- The ammonia evolved from the treatment of 0.30g of an organic compound for the estimation of nitrogen was passed in 100 mL 0.1 M H₂SO₄. The excess of acid required 20 mL of 0.5M NaOH solution for complete neutralisation. The organic compound is :
 (a) Thio urea (b) Benzamide (c) Urea (d) Acetamide
- If a compound contains both N and S, the ions present in the Lassaigne's extract are:
 (a) CN S²⁻ (b) CNS (c) N³⁻ S²⁻ (d) CNS NO₂
- Before testing halogens by Lassaigne's extract, sodium extract is boiled with HNO₃ to:
 (a) make the medium acidic (b) remove CN⁻ or S²⁻ ions if present
 (c) suppress the dissociation of AgNO₃ (d) avoid co-precipitation
- Which of the following compounds interfere in the Beilsteins test for halogens ?
 (a) Urea (b) Thiourea (c) Pyridine (d) All of these
- Which halogen can't be tested by Beilstein test ?
 (a) F (b) Cl (c) Br (d) I
- There are several criteria of purity for organic compounds out of the following which is considered to be the best ?
 (a) Melting point (b) Mixed melting point
 (c) Microscopic examination (d) Colour
- Aniline has high boiling point, high vapour pressure at 100°C and is insoluble in water. Aniline is therefore, separated by:
 (a) Steam distillation (b) Simple distillation
 (c) Distillation at low pressure (d) Sublimation
- In paper chromatography:
 (a) Mobile phase is liquid and stationary phase is solid
 (b) Mobile phase is solid and stationary phase is liquid
 (c) Both phases are liquids
 (d) Both phases are solids

11. Absolute alcohol is prepared from rectified spirit by:
- (a) Fractional distillation
 - (b) Steam distillation
 - (c) Azeotropic distillation
 - (d) Vacuum distillation
12. Azeotropic mixture are:
- (a) Boils at different temperature
 - (b) Are mixture of solids
 - (c) Are constant boiling mixtures
 - (d) None of these
13. Anhydrous formic acid cannot be obtained from its aqueous solution by simple fractional distillation because:
- (a) Pure HCOOH is unstable
 - (b) HCOOH forms hydrogen bonds with water
 - (c) Boiling point of HCOOH is very close to that of water
 - (d) Constant boiling azeotropic mixture is formed with water
14. Glycerol boils at 290°C with slight decomposition, impure glycerine can be purified by:
- (a) Steam distillation
 - (b) Simple distillation
 - (c) Vacuum distillation
 - (d) Extraction with a solvent
15. Absolute alcohol cannot be obtained by simple fractional distillation because:
- (a) Pure $\text{C}_2\text{H}_5\text{OH}$ is unstable
 - (b) $\text{C}_2\text{H}_5\text{OH}$ forms hydrogen bonds with water
 - (c) Boiling point of $\text{C}_2\text{H}_5\text{OH}$ is very close to that of water
 - (d) Constant boiling azeotropic mixture is formed with water
16. A bottle containing two immiscible liquids is given to you. They may be separated by using a:
- (a) Fractionating column
 - (b) Separating funnel
 - (c) Vacuum distillation
 - (d) Steam distillation
17. The process of distillation involves all the following processes except:
- (a) Change of state
 - (b) Boiling
 - (c) Condensation
 - (d) Oxidation
18. Fractional crystallisation is carried out to separate such mixtures:
- (a) Organic solids mixed with inorganic solids
 - (b) Organic solids highly soluble in water
 - (c) Organic solids having small difference in their solubilities in suitable solvent
 - (d) Organic solids having great difference in their solubilities in suitable solvent
19. An organic substance from its aqueous solution can be separated by:
- (a) Distillation
 - (b) Extraction with solvent
 - (c) Steam distillation
 - (d) Fractional distillation
20. Two volatile and miscible liquids can be separated by fractional distillation into pure components under the condition:
- (a) They have low boiling points
 - (b) The difference in their boiling points is large
 - (c) The boiling points of the liquids are close to each other
 - (d) They do not form azeotropic mixture
21. For the separation of two miscible liquids which method is used ?
- (a) Chromatography
 - (b) Fractional column
 - (c) Fractional distillation
 - (d) Separating funnel
22. Which reagent is useful in separating benzoic acid from phenol?
- (a) Dil. HCl
 - (b) Dil. H_2SO_4
 - (c) 5% NaOH
 - (d) 5% NaHCO_3
23. The most suitable method for separation of 1:1 mixture of ortho and para nitrophenols is:
- (a) Distillation
 - (b) Crystallisation
 - (c) Sublimation
 - (d) Chromatography
24. In steam distillation, the vapour pressure of the volatile organic compound is:

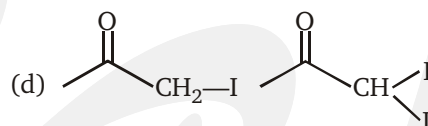
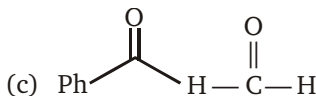
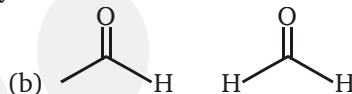
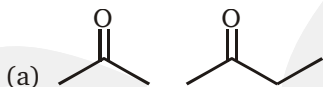
- (a) Equal to atmospheric pressure (b) Less than atmospheric pressure
(c) More than atmospheric pressure (d) None of these
25. Which of the following is used for removing moisture?
(a) Na_2CO_3 (b) silica gel (c) FeSO_4 (d) Potash alum
26. 1.0 g of an organic compound containing nitrogen on Kjeldahlising required 58.0 mL of $N/5$ H_2SO_4 for complete neutralization of ammonia. The percentage of nitrogen in the compound is about:
(a) 38.31 (b) 14.82 (c) 16.24 (d) 21.62
27. 0.099 g of an organic compound when heated with fuming nitric acid in the presence of silver nitrate in a Carius tube gave 0.287 g silver chloride. The percentage of chlorine in the compound is about:
(a) 28.64 (b) 71.72 (c) 35.45 (d) 64.23
28. In Lassaigne's test the organic compound is fused with sodium metal to:
(a) Ignite the compound (b) Oxidise the compound
(c) Convert the elements in ionic form (d) Reduce the compound
29. The fusion of nitrogen containing organic compounds with sodium results in the formation of:
(a) Sodium carbide (b) Sodium nitride (c) Sodium cyanide (d) Sodium carbonate
30. AgNO_3 solution does not give white ppt. with CCl_4 but with sodium extract of CCl_4 , AgNO_3 gives white precipitate because:
(a) CCl_4 is covalent compound
(b) CCl_4 is partially ionised
(c) Sodium extract of CCl_4 contains Cl^- ions and Na^+ ion
(d) Sodium extract of CCl_4 contains ClO_3^- ions
31. In Kjeldahl method function of CuSO_4 is:
(a) B.P. elevator (b) catalyst
(c) freezing point depressator (d) none of these
32. In the estimation of nitrogen by Duma's method 0.59 g of an organic compound gave 112 mL N_2 at NTP. The percentage of nitrogen in the compound is about:
(a) 23.7 (b) 11.8 (c) 20.1 (d) 47.5
33. 0.20 g of hydrocarbon on combustion gave 0.66 g CO_2 . The percentage of hydrogen in the hydrocarbon is about:
(a) 33 (b) 45 (c) 10 (d) 90
34. 0.58 g of a hydrocarbon on combustion gave 0.9 g water. The percentage of carbon in the hydrocarbon is about:
(a) 75.82 (b) 82.76 (c) 27.85 (d) 58.64
35. When sodium extract is prepared, generally the substance ignites is:
(a) Na (b) H_2 (c) Organic compound (d) O_2
36. The organic compound which does not give blue colour in Lassaigne's test is:
(a) Aniline (b) Glycine (c) Hydrazine (d) Urea
37. 0.2 g of an organic compound containing C, H and O, on combustion gave 0.147 g CO_2 and 0.12 g water. The percentage of oxygen in organic compound is:
(a) 73.29% (b) 78.45% (c) 83.23% (d) 89.50%

Miscellaneous Problems

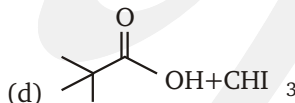
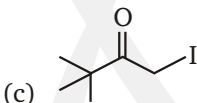
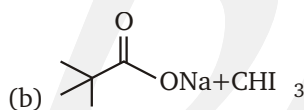
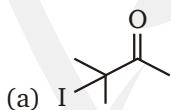
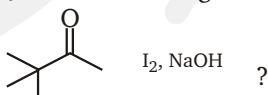
1. Which of the following structures will give a positive test with Tollens' reagent?



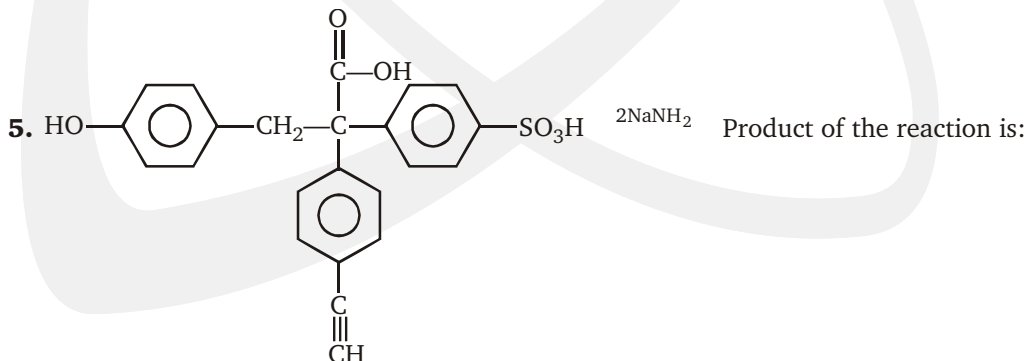
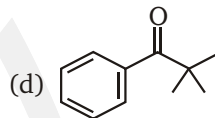
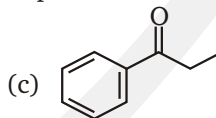
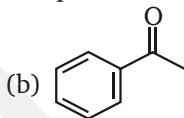
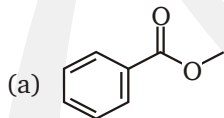
2. Which of the following pair is differentiated by iodoform test?

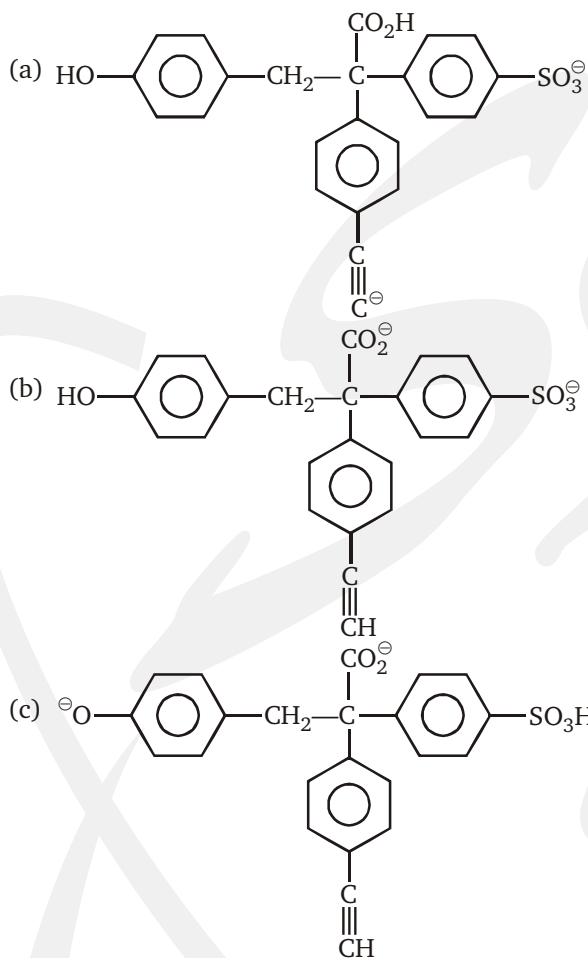


3. What is(are) the final product(s) of the following reaction?



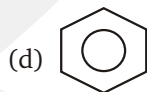
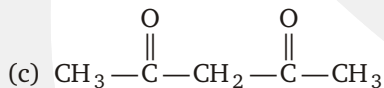
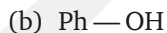
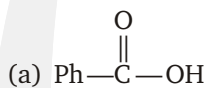
4. Which of the following compounds would give a positive iodoform test?



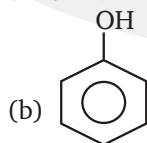
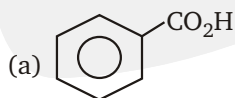


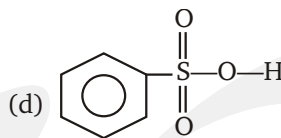
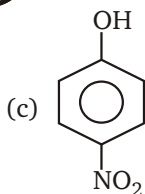
(d) None of these

6. Which of the following compound will not evolve H_2 gas with Na ?

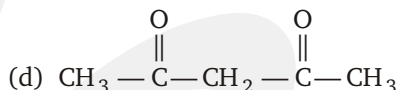
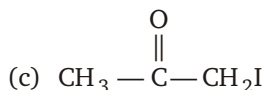
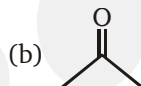
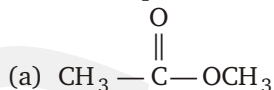


7. Which of the following compounds is most reactive with $NaHCO_3$?





8. Which compound does not give positive iodoform test ?



Answers

1.	(b)	2.	(a)	3.	(c)	4.	(b)	5.	(b)	6.	(d)	7.	(a)	8.	(b)	9.	(a)	10.	(a)
11.	(c)	12.	(c)	13.	(c)	14.	(c)	15.	(d)	16.	(b)	17.	(d)	18.	(c)	19.	(b)	20.	(d)
21.	(c)	22.	(d)	23.	(a)	24.	(b)	25.	(b)	26.	(c)	27.	(b)	28.	(c)	29.	(c)	30.	(c)
31.	(b)	32.	(a)	33.	(c)	34.	(b)	35.	(b)	36.	(c)	37.	(a)						

Miscellaneous Problems

1.	(c)	2.	(b)	3.	(b)	4.	(b)	5.	(b)	6.	(d)	7.	(d)	8.	(a)				
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One or More than One Answers is/are Correct

1. Among the following which compound can't be identified by Lassaigne's test ?

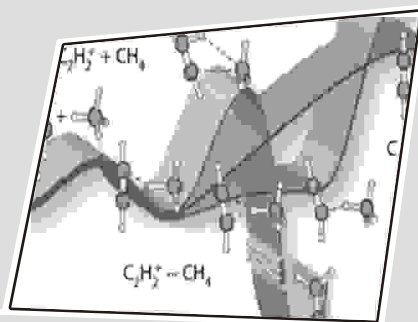
- (a) diazo compounds (b) NH_4NO_3 (c) NH_2NH_2 (d) NH_2OH

Answers

1.	(b, c, d)																		
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11

Chemistry in Everyday Life



Only One Answer is Correct

- Phenol is:
 - antimicrobial
 - antiseptic
 - disinfectant
 - all of these
- Select the correct statements:
 - 2-3% alcohol-water mixture is known as tincture of iodine.
 - Iodoform solution is antiseptic for wounds.
 - Boric acid solution is antiseptic for eyes.
 - All of these
- Select the correct statement:
 - Enzymes are proteins in nature and are biological catalysts
 - Proteins which help in communication system are called receptors
 - Polar molecules across the cell membrane are called carrier proteins
 - All are correct
- Some drugs interact with enzymes and make them biologically inactive. Such drugs are called:
 - enzyme promoters
 - enzyme inhibitors
 - allogens
 - all of these
- Drugs which do not bind to the enzyme's active site, but to different site. This different site is called :
 - inhibition site
 - competitive site
 - allosteric site
 - none of these
- Chemicals that are responsible for communication of message between neurons and muscles are known as:
 - messengers
 - allogens
 - antagonists
 - receptors
- Select the correct statement:
 - Antagonists are the substances which inhibit the natural function of receptor site.
 - Agonists are the drugs that mimic the natural messenger by switching on the receptor
 - Substrates may bind to active site of enzyme through a variety of interactions such as ionic bonding, H-bonding, van der Waals' interaction or dipole-dipole interaction.
 - All of these
- Among the following which is not antacid ?
 - Histamine
 - Cimetidine
 - Ranitidine
 - All are antacids
- Histamine stimulates the secretion of pepsin and hydrochloric acid and antihistamines are used to prevent the action of histamines. Among the following which are antihistamines ?
 - Ranitidine
 - Terfenadine
 - Brompheniramine
 - All of these

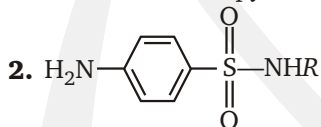
10. Among the following, which is not a tranquilizer ?
 (a) Noradrenaline (b) Chlordiazepoxide (c) Meprobamate
 (d) Equanil (e) All are tranquilizers
11. Iproniazid and phenelzine are:
 (a) antidepressants (b) tranquilizer (c) both (a) and (b) (d) none
12. Hypnotics are sleep producing drugs. Among the following select the hypnotics :
 (a) Barbiturates (b) Valium (c) Serotonin (d) All of these
13. Among the following analgesics, select the odd one:
 (a) codeine (b) morphine (c) aspirin (d) heroin
14. Morphine narcotics are obtained from:
 (a) opium (b) avena (c) datura (d) all of these
15. Morphine and heroin, both are narcotic analgesics. Heroin may be obtained from morphine by:
 (a) nitration (b) acetylation (c) chlorination (d) none of these

Answers

1.	(d)	2.	(d)	3.	(d)	4.	(b)	5.	(c)	6.	(a)	7.	(d)	8.	(a)	9.	(d)	10.	(e)
11.	(c)	12.	(d)	13.	(c)	14.	(a)	15.	(b)										

One or More than One Answers is/are Correct

1. Select the correct statements:
 (a) Drugs are chemicals of low molecular mass.
 (b) Drugs produce biological response.
 (c) Drugs which are used to treat diseases are called medicines.
 (d) Chemotherapy is the use of chemicals for therapeutic effect.



Above drug is an example of :

- (a) sulpha drug (b) sulphonamide (c) analgesic (d) tranquilizer
3. Select the correct statement:
 (a) Bactericidal antibiotics are those which kills the microbes.
 (b) Bacteriostatic antibiotics are those which inhibits the growth of microbes.
 (c) Ofloxacin is the bacteriostatic antibiotics.
 (d) penicillin is bactericidal

Answers

1.	(a, b, c, d)	2.	(a, b)	3.	(a, b, d)														
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